Internal Use Only

Service Manual

LG-D295

Date: Oct, 2014 / Issue 1.0

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1. INTRODUCTION

1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

Toll fraud, the unauthorized use of telecommunications system by an unauthorized part (for example, persons other than your company's employees, agents, subcontractors, or person working on your company's behalf) can result in substantial additional charges for your telecommunications services.
System users are responsible for the security of own system. There are may be risks of toll fraud associated with your telecommunications system. System users are responsible for programming and configuring the equipment to prevent unauthorized use. The manufacturer does not warrant that this product is immune from the above case but will prevent unauthorized use of common carrier telecommunication service of facilities accessed through or
connected to it. The manufacturer will not be responsible for any charges that result from such unauthorized use.

B. Incidence of Harm

If a telephone company determines that the equipment provided to customer is faulty and possibly causing harm or interruption in service to the telephone network, it should disconnect telephone service until repair can be done. A telephone company may temporarily disconnect service as long as repair is not done.

C. Changes in Service

A local telephone company may make changes in its communications facilities or procedure. If these changes could reasonably be expected to affect the use of the phones or compatibility with the net work, the telephone company is required to give advanced written notice to the user, allowing the user to take appropriate steps to maintain telephone service.

D. Maintenance Limitation

Maintenance limitations on the phones must be performed only by the manufacturer or its authorized agent. The user may not make any changes and/or repairs expect as specifically noted in this manual. Therefore, note that unauthorized alternations or repair may affect the regulatory status of the system and may void any remaining warranty.

E. Notice of Radiated Emission

This model complies with rules regarding radiation and radio frequency emission as defined by local regulatory agencies. In accordance with these agencies, you may be required to provide information such as the following to the end user.

F. Pictures

The pictures in this manual are for illustrative purposes only; your actual hardware may look slightly different.

G. Interference and Attenuation

A phone may interfere with sensitive laboratory equipment, medical equipment, etc. Interference from unsuppressed engines or electric motors may cause problems.

H. Electrostatic Sensitive Devices

ATTENTION

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign. Following information is ESD handling:

- Service personnel should ground themselves by using a wrist strap when exchange system boards.
- When repairs are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system boards or parts like EEPROM to the factory, use the protective package as described.

1.3 Specification

KEY FEATURES

- Android 4.4-(KitKat)
- 4.5"Touch Screen WVGA (480 x 800)
- 8MP AF with Flash
- 3G Mobile Hotspot

ENHANCED USER EXPERIENCE

- Android 4.4
- Proximity Sensor – locks the touch screen and buttons while talking on the phone
- Motion Sensor – Accelerometer
- USB Charging via Computer
- Airplane Mode (RF Off)

DESIGN

- Sleek and Stylish Design
- 4.5" WVGA LCD Screen
- Capacitive Touch Screen – Plastic with fast and accurate response
- Micro USB/Charging Port

CAMERA/VIDEO

- 8MP AF
- Camera Resolutions: 3264(H) x 2448(V) Pixels
- Image Editor: zoom, rotate, crop images
- Brightness, ISO, White Balance,* Color Effects, Timer
- Cheese shutter – take photo with voice command.
- Video Resolutions: 640 x 480 320 x 240 176 x 144 Pixels

MULTIMEDIA

- Music Player for MP3, WMA, and Unprotected AAC
- Stereo Sound via Headset or Bluetooth
- Music Library – Organized by Artists, Albums, Songs, and Playlists
- Repeat and Shuffle Music Playback Modes

BLUETOOTH®

- Version: 4.0
- Supported Profiles: headset, hands-free, object push, advanced audio distribution (stereo), audio/video remote control,file transfer, phone book access, message access, serial port
- Listen to Music with Optional Stereo Bluetooth Headset**
- Send User-Generated Pictures (JPEG) and Videos via Bluetooth Wireless Technology

*Depends on available memory.

**Accessories sold separately.

VOICE/AUDIO

- One-Touch Speakerphone*
- Speaker-Independent Voice Commands
- Voice-Activated Dialing
- Text to Speech
- MP3 Music Ringer Support

TOOLS & DATA

- Android Apps: Alarm/Clock, Browser, Calculator, Calendar (Corporate or Google), Camera, Contacts, Email, Gallery, Gmail, Google+, Maps, Messaging, Messenger, Music Player, Phone, Places, Play Books, Play Movies, Play Music, Play Store, Talk, Video Player, Voice Recorder, Youtube
- Navigation – spoken turn-by-turn directions showing real-time traffic
- Wi-Fi Connectivity – IEEE 802.11 b/g/n
- A-GPS Support for Enhanced Location Accuracy

SPECIFICATIONS

- Technology: GSM/WCMDA
- Frequencies: GSM 850,900,1800,1900
  WCDMA 2100, 900
- Data Transmission: WCMDA, GPRS
- Dimensions: 127.5X67.9X11.65(mm)
- Display: 4.5", WVGA (480 x 800) pixels
- Standard Battery: 1,900 [mAh]

ACCESSORIES

- Standard Battery
- Travel Adapter and USB Cable*
- Stereo Headset

2. PERFORMANCE

2.1 Product Name

LG-D295 : WCDMA900/2100+EGSM/GSM850/DCS/PCS(HSDPA 21Mbps HSUPA 5.76Mbps))

2.2 Supporting Standard

Item	Feature	Comment
Supporting Standard	WCDMA(FDD1,8)/EGSM/GSM850/DCS1800/PCS1900 with seamless handover Phase 2+(include AMR) SIM Toolkit : Class 1, 2, 3, C-E	
Frequency Range	WCDMA(FDD1) TX : 1920 – 1980 MHz WCDMA(FDD1) RX : 2110 – 2170 MHz WCDMA(FDD8) TX : 880 - 915 MHz WCDMA(FDD8) RX : 925 – 960 MHz EGSM TX : 880 – 915 MHz EGSM RX : 925 – 960 MHz GSM850 TX : 824 – 849 MHz GSM850 RX : 869 – 894 MHz GSM900 TX : 880 – 915 MHz GSM900 RX : 925 – 960 MHz DCS1800 TX :1710 – 1785 MHz DCS1800 RX : 1805 – 1880 MHz PCS1900 TX : 1850 – 1910 MHz PCS1900 RX : 1930 – 1990 MHz	
Application Standard		

2.3 Main Parts : GSM Solution

Item	Part Name	Comment
AP Chip	MSM8212	
CP Chip	MSM8212	

2.4 HW Features

List	Type / Spec.	
1. Phone Type	DOP Type	
2. Size	127.5X67.9X11.65 (mm)	
3. Weight	TBD (with battery)	
4. Battery	1,900mAh (typ.), 1,820mAh(min) (Li-Ion)	
5.Chipset	MSM8212	
6. Memory	4GB(eMMC) +1GB(LPDDR2)	
7. LCD	Size	4.5 inch
	Dot	480(H) X 800(V)
	Color	16.7M
	Display Type	TFT, Incell Touch
	NTSC	50% (Typ)
	Contrast Ratio	1:1000 (Typ)
	Pixel Pitch	TBD
	Color temperature	6000~10000K (TBD)
8. Main Camera (8M)	Type	BSI CIS Type
	Resolution	3264(H) X 2448(V) pixels.
	Frame Rate	TBD
	View Angle	71.2° ± 5% (Diagonal)
	Image Scaling Down	TBD
	Pixel x size	1.12um x1.12um

10. Audio	1) MIDI	MIDI
	2) Mono	18 X 10 X 3T Speaker
	3) Receiver	12 X 06 X 2.5T Receiver
11. Bluetooth	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
12. WLAN	standard	IEEE 802.11 b/g/n
	Throughput	Max 40Mbps (SDIO Driver performance)
	depend on environment	0 ~ 50m (depend on environment)
13. GPS	type	A-GPS

2.5 HW SPEC

1) WCDMA transmitter specification

Item	Spec.	Min	Typ.	Max	Unit
Transmit Frequency	B1(W2100): 1920 ~ 1980 MHz B8(W900) 880 - 915 MHz	-	-	-	MHz
Maximum Output Power	Power class Ⅲ	21.0	23.0	25.0	dBm/3.84MHz
Frequency Accuracy	-	-200	-	+200	Hz
Minimum Output Power	-	-	-	-50	dBm/3.84MHz
Occupied Bandwidth	-	-	-	4.6	MHz
Adjacent Channel Leakage Power Ratio	Δf = ± 5MHz	-	-	-33	dBm/3.84MHz
	Δf = ± 10MHz	-	-	-43	dBm/3.84MHz
Error Vector Magnitude(EVM)	-	-	-	17	%
Peak Code Domain Error	-	-	-	15	dBm/3.84MHz
Open Loop Output Power	Ior =-106.7dBm/3.84MHz	-46.7	-	-28.7	dBm/3.84MHz
	Ior =-65.7dBm/3.84MHz	-23.0	-	-5.0	dBm/3.84MHz
	Ior =-25.0dBm/3.84MHz	-0.0	-	14.0	dBm/3.84MHz
Transmit Off Power	-	-	-	-56	dBm/3.84MHz
Spectrum Emission Mask	Δf = 2.5~3.5MHz	-35-15 * (Δf-2.5)			dBc/30KHz
	Δf = 3.5~7.5MHz	-35-1 * (Δf-3.5)			dBc/1MHz

	Δf = 7.5~8.5MHz	-39 10 * (Δf-7.5)			dBc/1MHz
	Δf = 8.5~12.5MHz	-49			dBc/1MHz
Transmit Spurious Emission	9kHz ~ 150kHz			-36	dBm
	150kHz ~ 30MHz			-36	dBm
	30MHz ~ 1GHz			-36	dBm
	1GHz ~ 12.5GHz			-30	dBm

2) WCDMA receiver specification

Item	Specification
Receive Frequency	B1(W2100): 2110 – 2170 MHz B8(W900): 925 – 960 MHz
Reference Sensitivity Level <REF$\hat{I}_{or}$>	Band1 : BER < 0.001 when Îor = -106.7 dBm / 3.84 MHz Band8 : BER < 0.001 when Îor = -103.7 dBm / 3.84 MHz
Maximum Input Level	BER < 0.001 when Îor = -25 dBm / 3.84 MHz
Adjacent Channel Selectivity (ACS)	ACS > 33 dB where BER < 0.001 when Îor = <REF$\hat{I}_{or}$> + 14 dB / 3.84 MHz & Ioac = –52 dBm / 3.84 MHz @ ±5 MHz
Blocking Characteristic	BER < 0.001 when Îor = <REF$\hat{I}_{or}$> + 3 dB / 3.84 MHz & Iblocking = -56 dBm / 3.84 MHz @ Fuw(offset) = ±10 MHz or Iblocking = -44 dBm / 3.84 MHz @ Fuw(offset) = ±15 MHz
Spurious Response	BER < 0.001 when Îor = <REF$\hat{I}_{or}$> + 3 dB / 3.84 MHz & Iblocking = -44 dBm
Intermodulation	BER < 0.001 when Îor= <REF$\hat{I}_{or}$> + 3 dB / 3.84 MHz & Iouw1 = -46 dBm @ Fuw1(offset) = ±10 MHz & Iouw2 = -46 dBm / 3.84 MHz @ Fuw2(offset) = ±20 MHz
Spurious Emissions	< -57 dBm / 100 kHz BW @ 9 kHz ≤ f < 1 GHz < -47 dBm / 1 MHz BW @ 1 GHz ≤ f ≤ 12.75 GHz
Inner Loop Power Control In Uplink	Adjust output(TPC command) cmd 1dB 2dB 3dB +1 +0.5/1.5 +1/3 +1.5/4 0 -0.5/+0.5 -0.5/+0.5 -0.5/+0.5 -1 -0.5/-1.5 -1/-3 -1.5/-4 group(10equal command group) +1 +8/+12 +16/+24

3) HSDPA transmitter specification

	Item	Spec.		Min	Typ.	Max	Unit
Trandsmitter	5.2.A Maximum Output Power with HS-DPCCH	$1/15 \leq \beta c/\beta d \leq 12/15$ $13/15 \leq \beta c/\beta d \leq 15/8$ $15/7 \leq \beta c/\beta d \leq 15/0$		21.0 20 19	23.0 23.0 22.5	25.0 25 25	dBm/3.84MHz
	5.7A HS-DPCCH	MAX / 0dBm	Step 0	0 dBm +/- 1 dB			dB
			Step 1 – Step 0	6 dB +/- 2.3 dB			dB
			Step 2 – Step 3	1 dB +/- 0.6 dB			dB
			Step 5 – Step 4	0 dB +/- 0.6 dB			dB
			Step 6 – Step 7	5 dB +/- 2.3 dB			dB
	5.9A Spectrum Emission Mask with HS-DPCCH	Δf = 2.5~3.5MHz		-35-15 * (Δf-2.5)			dBc/30KHz
		Δf = 3.5~7.5MHz		-35-1 * (Δf-3.5)			dBc/1MHz
		Δf = 7.5~8.5MHz		-39 10 * (Δf-7.5)			dBc/1MHz
		Δf = 8.5~12.5MHz		-49			dBc/1MHz
	5.10A Adjacent Channel Leakage Power Ratio with HS-DPCCH	Δf = ± 5MHz		-	-	-33	dBm/3.84MHz
		Δf = ± 10MHz		-	-	-43	dBm/3.84MHz
	5.13.1A Error Vector Magnitude (EVM) with HS-DPCCH	MAX Power -20 dBm		-	-	17.5	%
Receiver	6.3A Maximum Input Level for HS-PDSCH Reception	Pin = -25dBm		-	-	10	BLER(%)

4) GSM receiver specification

Item	Spec.
Phase Error	Rms : 5° Peak : 20 °
Frequency Error	GSM : 0.1 ppm DCS/PCS : 0.1 ppm
EMC(Radiated Spurious Emission Disturbance)	GSM/DCS : < -28dBm
Transmitter Output power and Burst Timing	GSM : 5dBm – 33dBm ± 3dB DCS/PCS : 0dBm – 30dBm ± 3dB
Burst Timing	<3.69us
Spectrum due to modulation out to less than 1800kHz offset	200kHz : -36dBm 600kHz : -51dBm/-56dBm
Spectrum due to modulation out to larger than 1800kHz offset to the edge of the transmit band	GSM : 1800-3000kHz :< -63dBc(-46dBm) 3000kHz-6000kHz : <-65dBc(-46dBm) 6000kHz < : < -71dBc(-46dBm) DCS : 1800-3000kHz :< -65dBc(-51dBm) 6000kHz < : < -73dBc(-51dBm)
Spectrum due to switching transient	400kHz : -19dBm/-22dBm(5/0), -23dBm 600kHz : -21dBm/-24dBm(5/0), -26dBm
Reference Sensitivity – TCH/FS	Class II(RBER) : -105dBm(2.439%)
Usable receiver input level range	0.012(-15 - -40dBm)
Intermodulation rejection – Speech channels	± 800kHz, ± 1600kHz : -98dBm/-96dBm (2.439%)
AM Suppression - GSM : -31dBm - DCS : -29dBm	-98dBm/-96dBm (2.439%)
Timing Advance	± 0.5T

5) Bluetooth transmitter specification

Item	Spec.		Min	Typ.	Max	Unit
Output Power	DH5 PRBS9	Power (Class I)	0	-	20	dBm
Power Density	DH5 PRBS9	@100kHz RBW		-	+20	dBm
Power Control	DH1 PRBS9	Step size	2	-	8	dB
		Power(min)		-	4	dBm
Frequency Range	DH5 PRBS9	-30dBc @100kHz RBW	2400	-	2483.5	MHz
Tx Output Spectrum (20dB BW)	DH5 PRBS9	$\Delta f = f_H - f_L$		-	1	MHz
Tx Output Spectrum (Adjacent Output Power)	DH5 PRBS9	@M-N = 2		-	-20	dBm
		@M-N ≥ 3		-	-40	dBm
Modulation Index	DH5 11110000	$f1_{AVG}$	140	-	175	kHz
	DH5 10101010	$f2_{MAX}$	115	-		kHz
		$f2_{AVG}/f1_{AVG}$	0.8	-		
Initial Carrier Freq. Tolerance	DH5 PRBS9		-75	-	+75	kHz
Carrier Frequency Drift	DH1 10101010	Freq. Drift	-25	-	+25	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH3 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH5 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us

6) Bluetooth receiver specification

Item	Spec. BER=0.1%		Min	Typ.	Max	Unit
Sensitivity (Single Slot Packet)	DH1 PRBS9 Input power @BT_w	-		-	-70	dBm
Sensitivity (Multi Slot Packet)	DH5 PRBS9 Input power @BT_w	-		-	-70	dBm
C/I Performance*	DH1 PRBS9 Input power @BT_i	C/I@Co-Channel -60dBm @ BT		-	11	dB
		C/I @1MHz -60dBm@ BT		-	0	
		C/I @2MHz -60dBm@ BT		-	-30	
		C/I @≥3MHz -67dBm@ BT		-40	-37	
		C/I @Image -67dBm@ BT		-	-9	
		C/I @ Image±1MHz -67dBm@ BT		-	-20	
Blocking Performance	DH1 PRBS9 Input power @CW	30M ~2 GHz -67dBm@ BT	-10	-		dBm
		2 ~ 2.4 GHz -67dBm@ BT	-27	-		
		2.5 ~ 3 GHz -67dBm@ BT	-27	-		
		3 ~ 12.75 GHz -67dBm@ BT	-10	-		
Intermodulation Performance	DH1 PRBS9 Input power @CW_i	Δf = +5MHz (CW_i) Δf = +10MHz (BT_i) -64dBm@ BT_w	-39	-		dBm
		Δf = -5MHz (CW_i) Δf = -10MHz (BT_i) -64dBm@ BT_w	-39	-		dBm
Maximum Input Level	DH1 PRBS9 Input power @BT_w	-	-20	-		dBm

7) WLAN 802.11b transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm under (Europe), ≤ 30dBm under (USA)
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -30 @ fc-22MHz< f <fc-11MHz and fc+11MHz< f <fc+22MHz ≤ -50 @ f < fc-22MHz and f > fc+22MHz
Power ramp on/off time	≤ 2us
Carrier Suppression	≤ -15dB
Modulation Accuracy (Peak EVM)	≤ 35%
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	≤ -76dBm(1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER ≤ 8%
Rx Max input Sensitivity	≥ -10dBm(1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER ≤ 8%
Rx Adjacent Channel Rejection	≥ 35dB @FER ≤ 8%, interference input signal -70dBm@fc±25MHz(11Mbps)

8) WLAN 802.11g transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm under (Europe), ≤ 30dBm under (USA)
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -20 @ ±11MHz offset (9Mhz ~ 11MHz) ≤ -28 @ ±20MHz offset (11MHz ~ 20Mhz) ≤ -40 @ ±30MHz offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	≤ -5dB
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6Mbps, -81dBm@9Mbps, -79dBm@12Mbps -77dBm@18Mbps, -74dBm@24Mbps, -70dBm@36Mbps -66dBm@48Mbps, -65dBm@54Mbps
Rx Max input Sensitivity	≥ -20dBm(6,9,12,18,24,36,48,54Mbps) @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

9) WLAN 802.11n transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm under (Europe), ≤ 30dBm under (USA)
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -20 @ ±11MHz offset (9Mhz ~ 11MHz) ≤ -28 @ ±20MHz offset (11MHz ~ 20Mhz) ≤ -45 @ ±30MHz offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	≤ -5dB@6.5Mbps, ≤ -10dB@13Mbps, ≤ -13dB@19.5Mbps, ≤ -16dB@26Mbps, ≤ -19dB@39Mbps, ≤ -22dB@52Mbps, ≤ -25dB@58.5Mbps, ≤ -28dB@65Mbps
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps
Rx Max input Sensitivity	≥ -20dBm(6.5,13,19.5,26,39,52,58.5,65Mbps) @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

10) GPS receiver specification

Item	Spec.
Sensitivity Test	Success rate ≥ 60% @ -154 dBm

11) Current consumption

** Test condition for Standby current consumption should be as below.
: Measurement time 1hr, with Agilent & CMT JIG

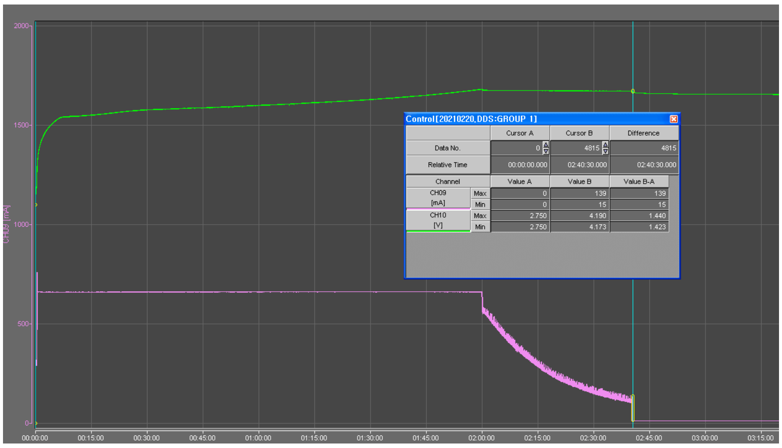
Item	Specification	
	WCDMA	GSM
1. Sleep Mode	Under 5mA	Under 5mA
2. Sleep : connector Ear jack	Under 6mA	Under 6mA
3. Current(Sleep & Idle AVG)	Under 8.5mA @ DRX 7	Under 8.5mA @ P.P 5
4. BT Connected idle	Under 10mA	Under 10mA
5. Talk Mode	- Max Pwr Under 600mA @ Avg - Tx 10dBm Under 300 mA @ Avg	GSM LVL 5 under 300 mA @ Avg
6. Power Off Mode	under 300uA	Under 300 uA
7. Backup Battery	N/A	N/A

12) RSSI indicator (Based on Cell power) _Updated08.11

BAR	WCDMA	GSM/DCS/PCS
5	Over -86 ±3dBm	Over -90 ±2dBm
5->4	-86± 3dBm	-90± 2dBm
4->3	-92± 3dBm	-96± 2dBm
3->2	-99± 3dBm	-98± 2dBm
2->1	-102± 3dBm	-102± 2dBm
1->0	-109± 3dBm	-104± 2dBm

13) Charging Time

Under 3hour 30min. (1900mAh battery, 700mA TA)



14) Phone Cut-off Volatage : 3.5V ± 0.1V

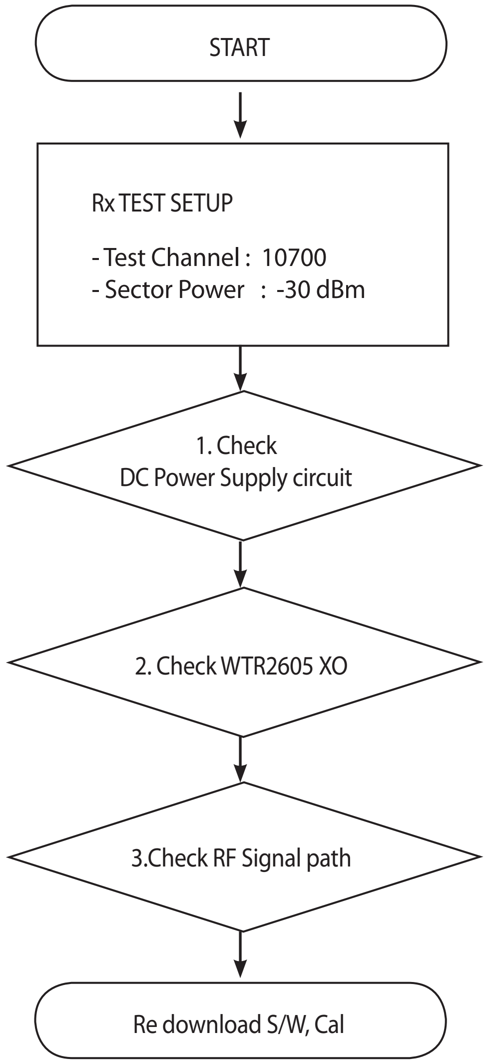
15) Battery indicator

Battery Bar	Specification	
BAR 20 (Full)	98% over	remain%
BAR 20 --> 19	98% ◊ 97%	
BAR 19 --> 18	93% ◊ 92%	
BAR 18 --> 17	88% ◊ 87%	
BAR 17 --> 16	83% ◊ 82%	
BAR 16 --> 15	78% ◊ 77%	
BAR 15 --> 14	73% ◊ 72%	
BAR 14 --> 13	68% ◊ 67%	
BAR 13 --> 12	63% ◊ 62%	
BAR 12 --> 11	58% ◊ 57%	
BAR 11 --> 10	53% ◊ 52%	
BAR 10 --> 9	48% ◊ 47%	
BAR 9 --> 8	43% ◊ 42%	
BAR 8 --> 7	38% ◊ 37%	
BAR 7 --> 6	33% ◊ 32%	
BAR 6 --> 5	28% ◊ 27%	
BAR 5 --> 4	23% ◊ 22%	
BAR 4 --> 3	16% ◊ 15%	
BAR 3 --> 2	13% ◊ 12%	
BAR 2 --> 1	8% ◊ 7%	
BAR 1 --> 0	3% ◊ 2%	
Low Battery Pop-up	16% →15% : One Time popup (No call) There is no pop-up until 5% if you delete pop-up	
Critical Low Battery Pop-up	6% → 5% : One Time popup (No call)	
Power Off	1%	

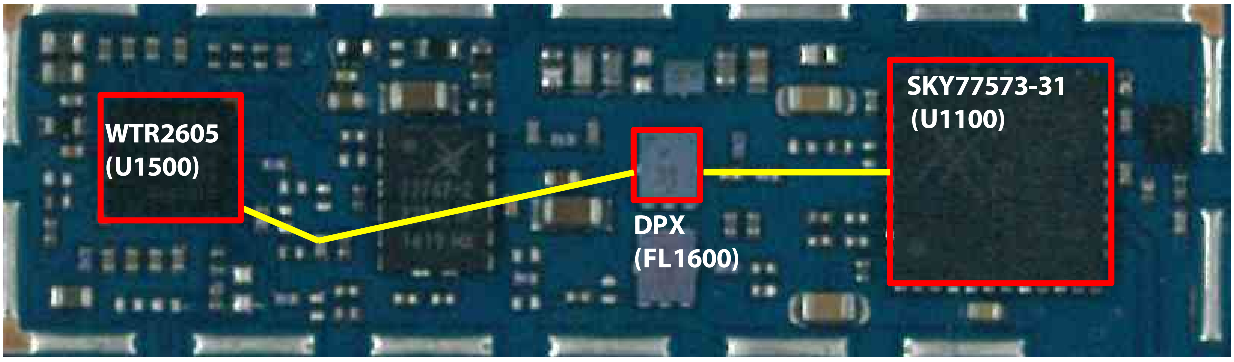
3. TROUBLE SHOOTING

3.1 WCDMA Rx Part

3.1.1 WCDMA_B1 Rx



* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form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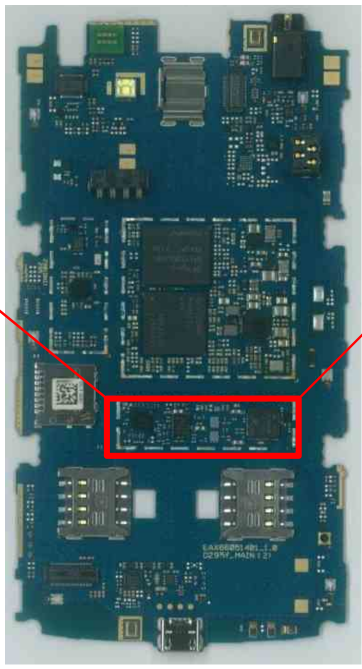
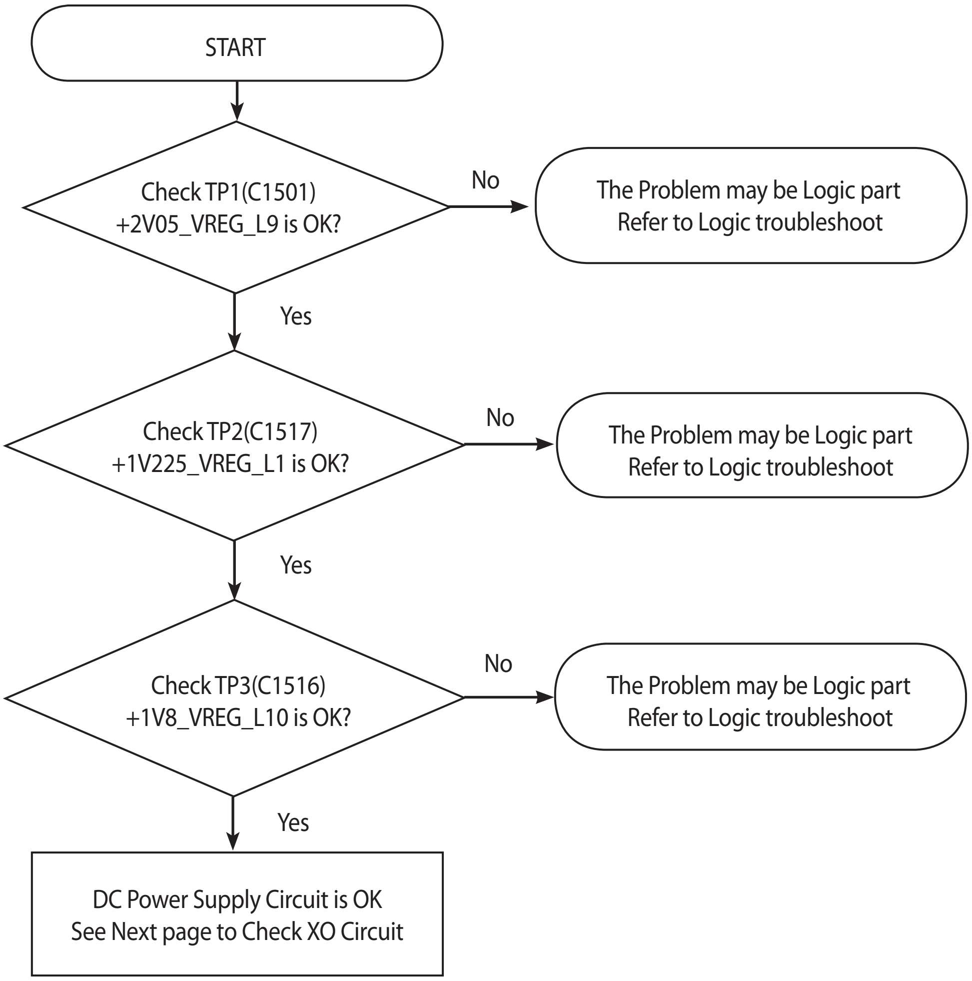
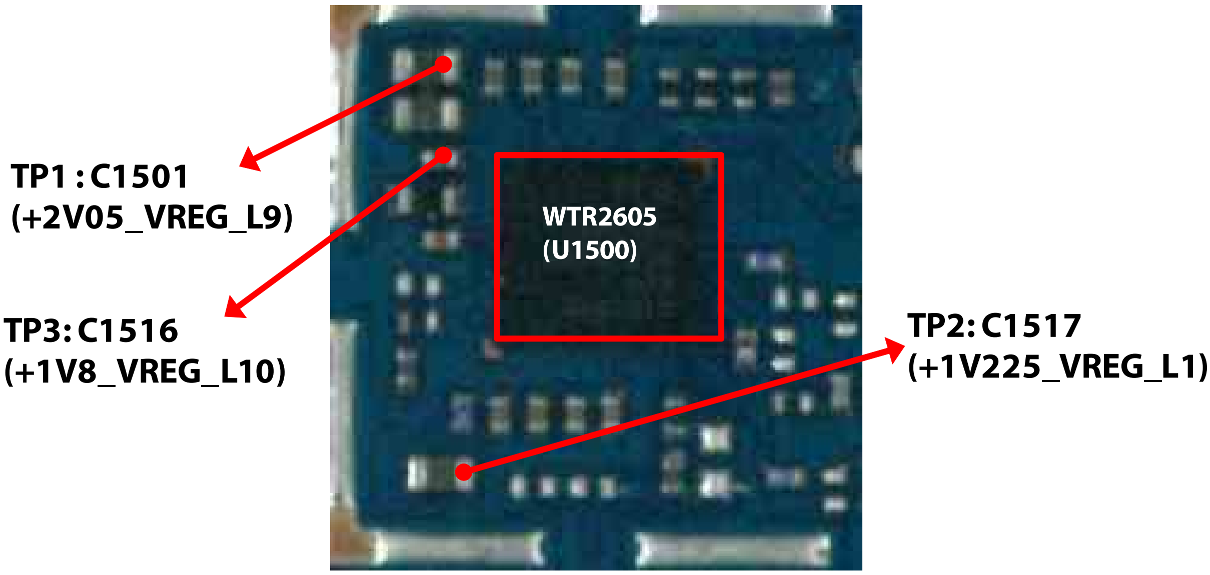
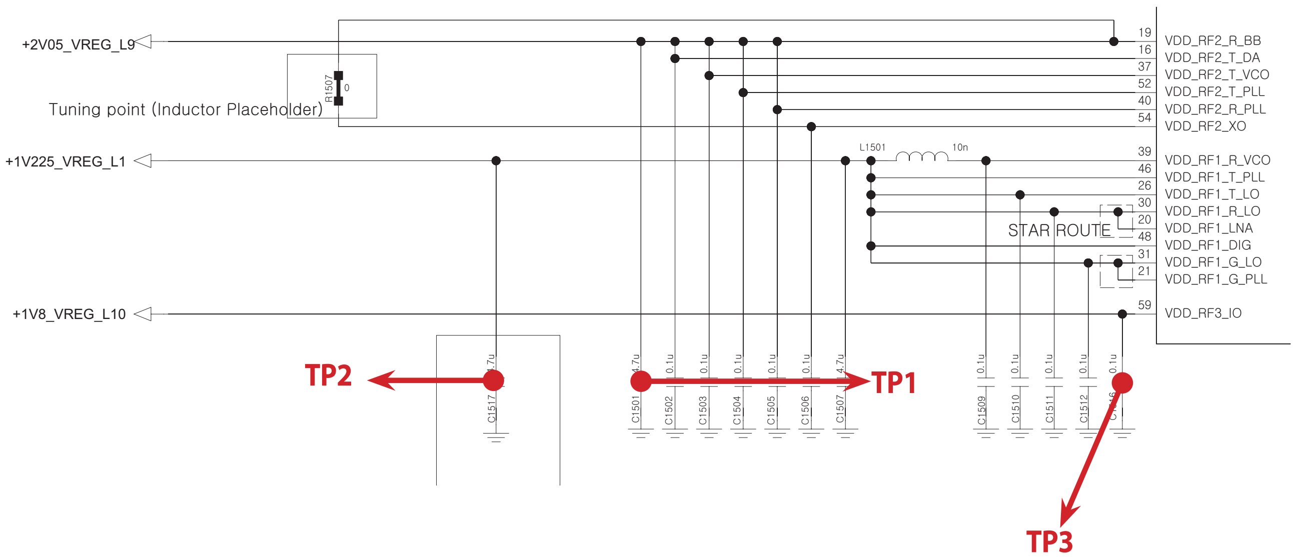
DPX
(FL1600)
SKY77573-31
(U1100)



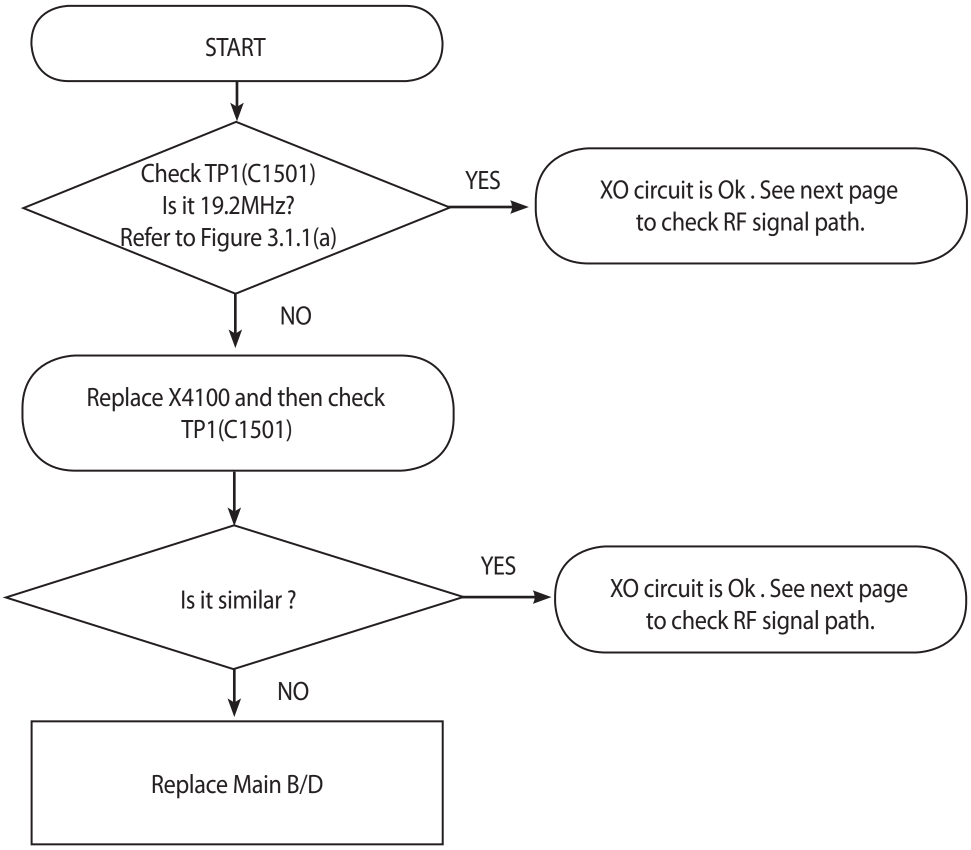
3.1.1.1 Checking DC Power supply circuit




TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)


+2V05_VREG_L9
Tuning point (Inductor Placeholder)
+1V225_VREG_L1
+1V8_VREG_L10
L1501
10n
STAR ROUTE
VDD_RF2_R_BB
VDD_RF2_T_DA
VDD_RF2_T_VCO
VDD_RF2_T_PLL
VDD_RF2_R_PLL
VDD_RF2_XO
VDD_RF1_R_VCO
VDD_RF1_T_PLL
VDD_RF1_T_LO
VDD_RF1_R_LO
VDD_RF1_LNA
VDD_RF1_DIG
VDD_RF1_G_LO
VDD_RF1_G_PLL
VDD_RF3_IO
TP2
TP1
TP3

3.1.1.2 Checking XO circuit



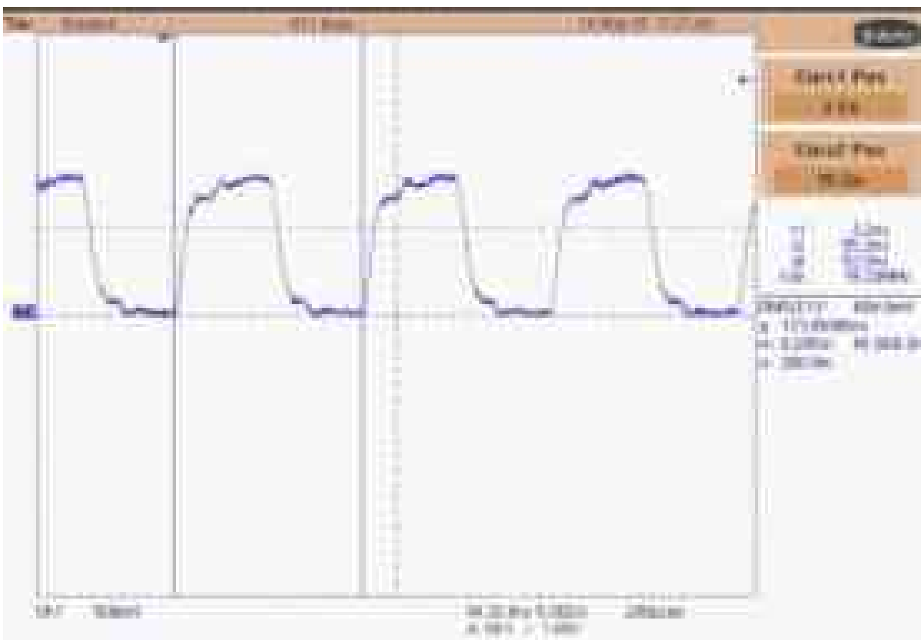
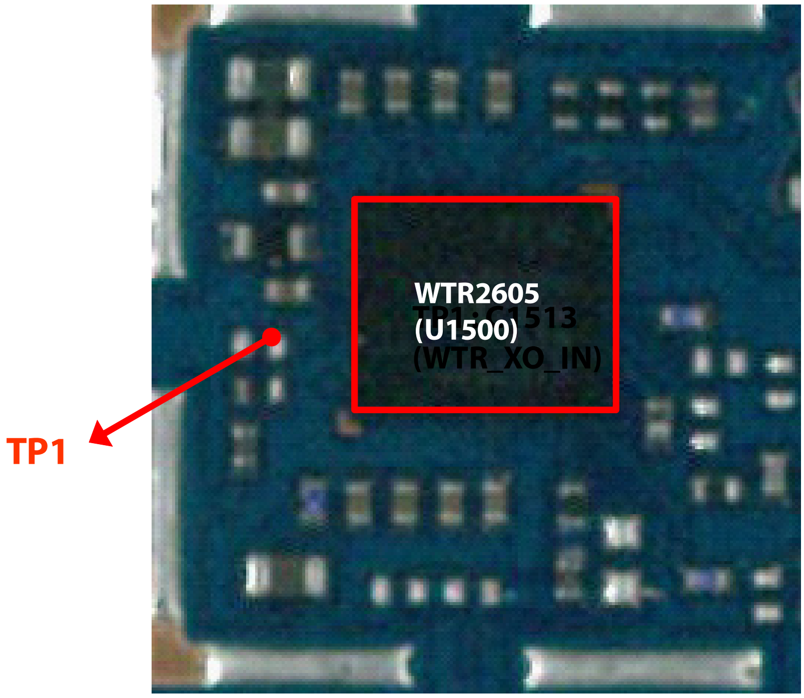
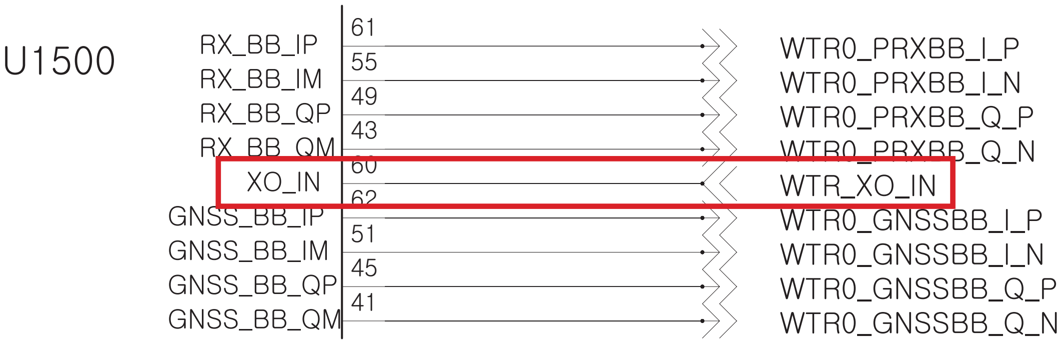
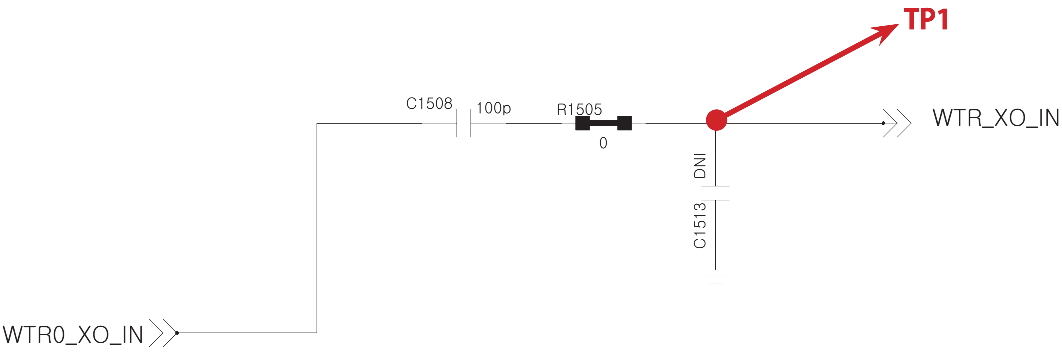


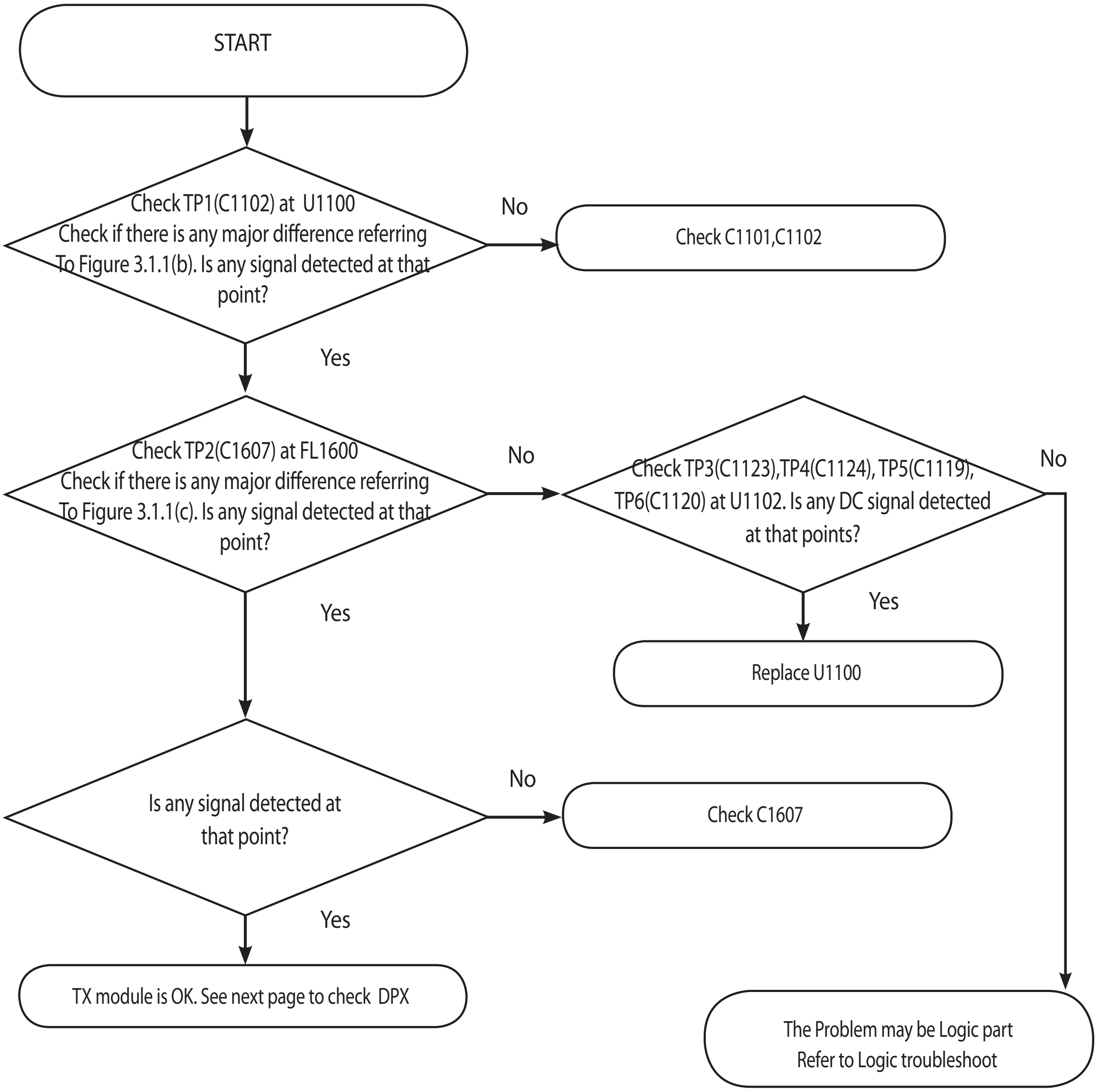
Figure 3.1.1 (a)

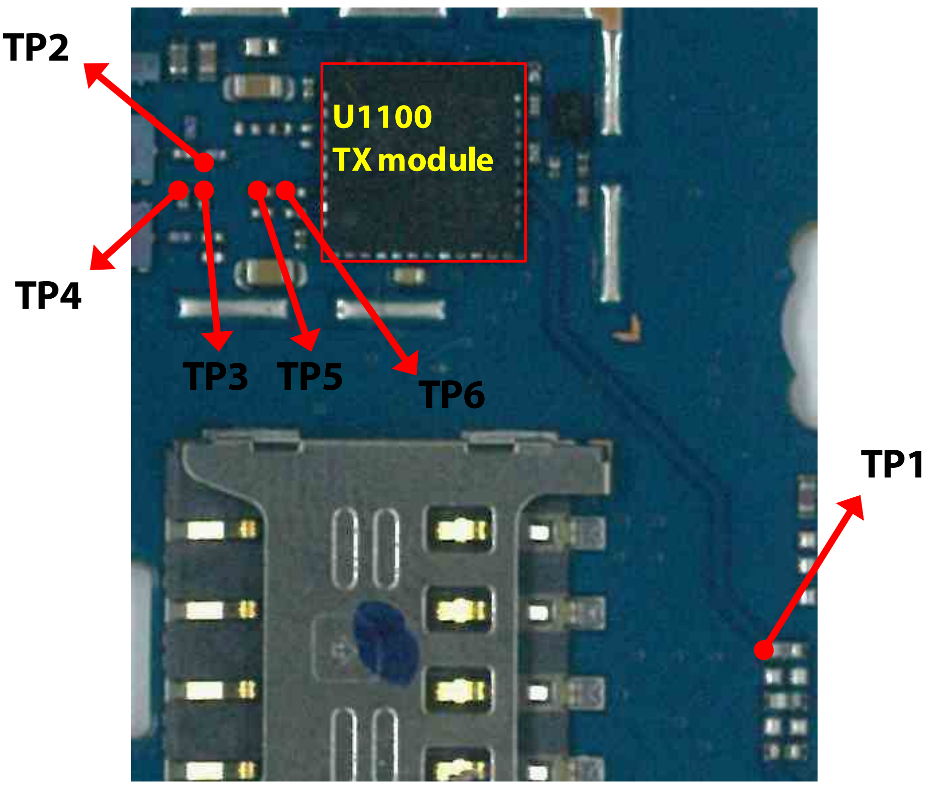

WTR2605
(U1500)
(WTR_XO_IN)
TP1

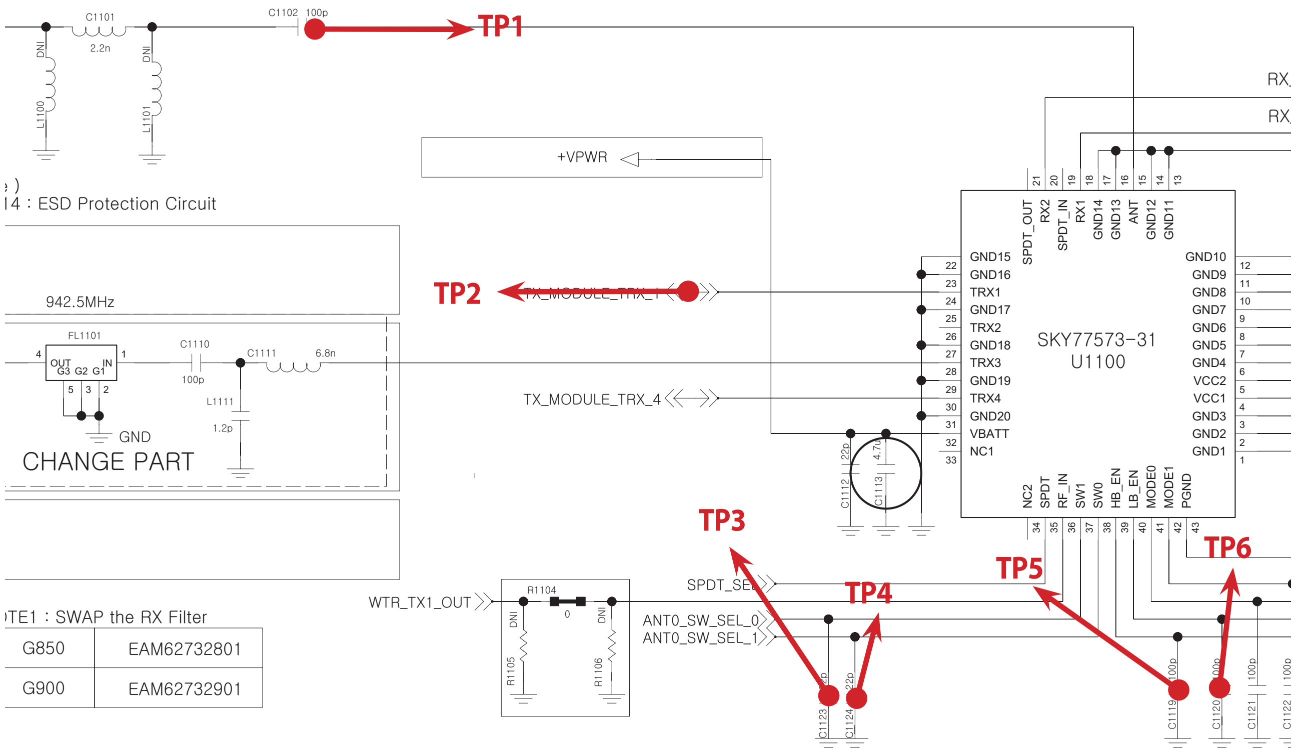

U1500
RX_BB_IP 61 WTR0_PRXBB_I_P
RX_BB_IM 55 WTR0_PRXBB_I_N
RX_BB_QP 49 WTR0_PRXBB_Q_P
RX_BB_QM 43 WTR0_PRXBB_Q_N
XO_IN 60 WTR_XO_IN
GNSS_BB_IP 62 WTR0_GNSSBB_I_P
GNSS_BB_IM 51 WTR0_GNSSBB_I_N
GNSS_BB_QP 45 WTR0_GNSSBB_Q_P
GNSS_BB_QM 41 WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
WTR_XO_IN
DNI
C1513
WTR0_XO_IN

3.1.1.3 Checking RF signal path (TX Module)







G850	EAM62732801
G900	EAM62732901

TP1

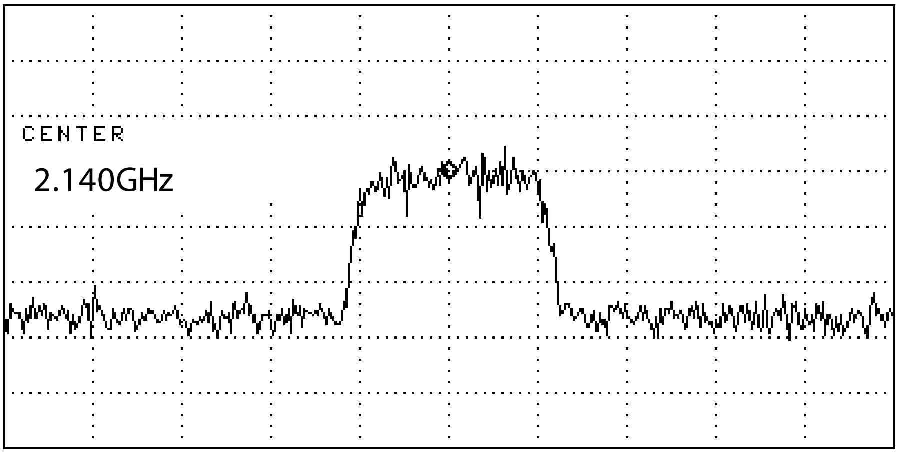


Figure 3.1.1 (b)

TP2

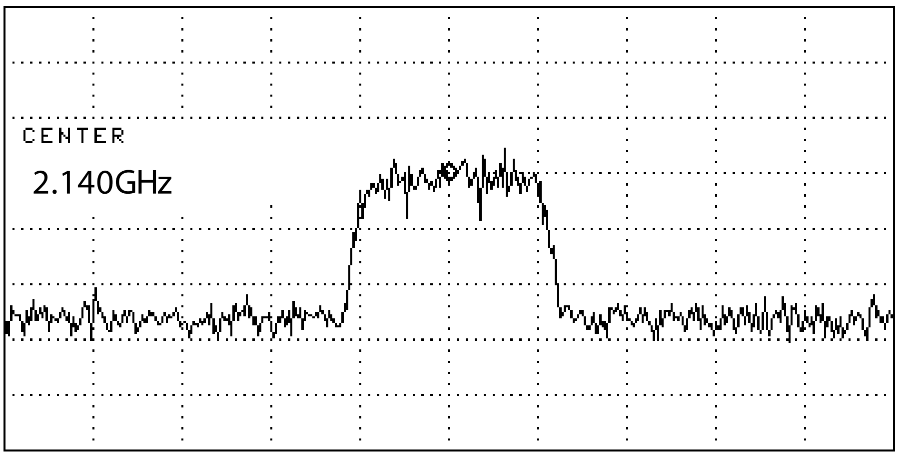
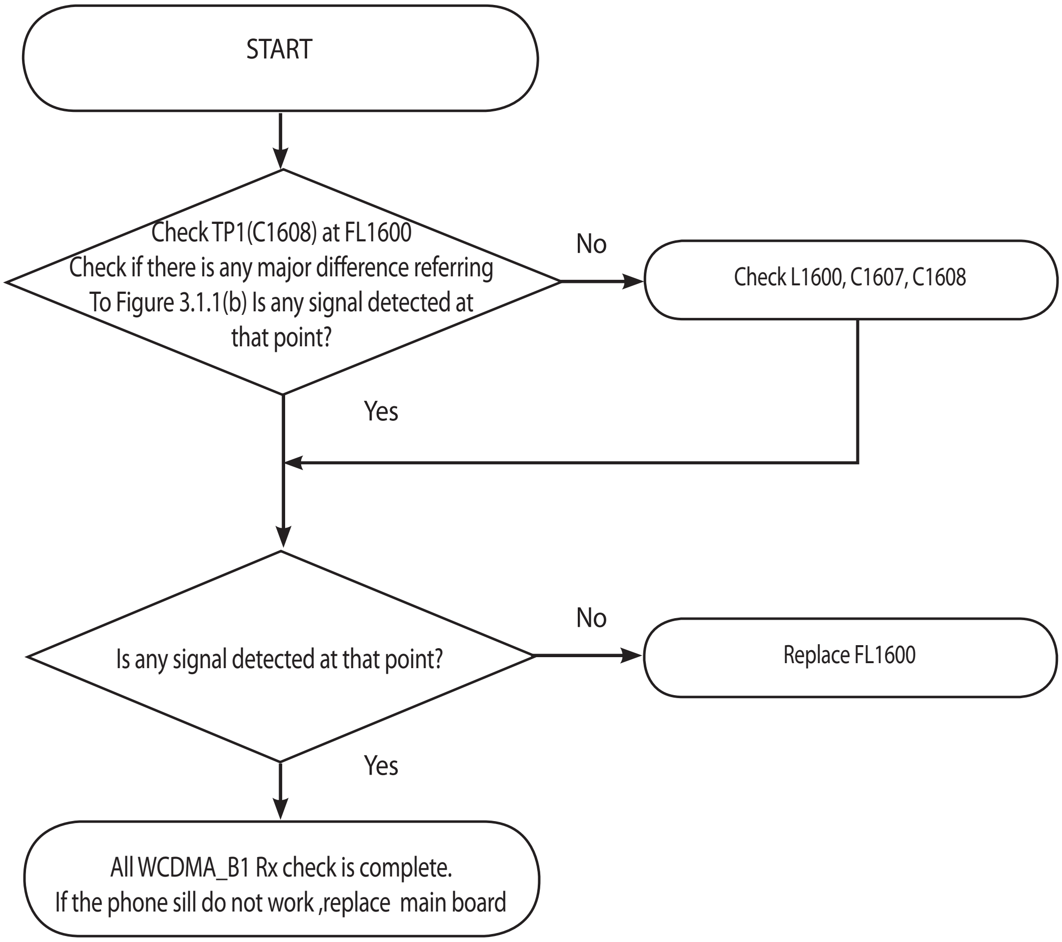
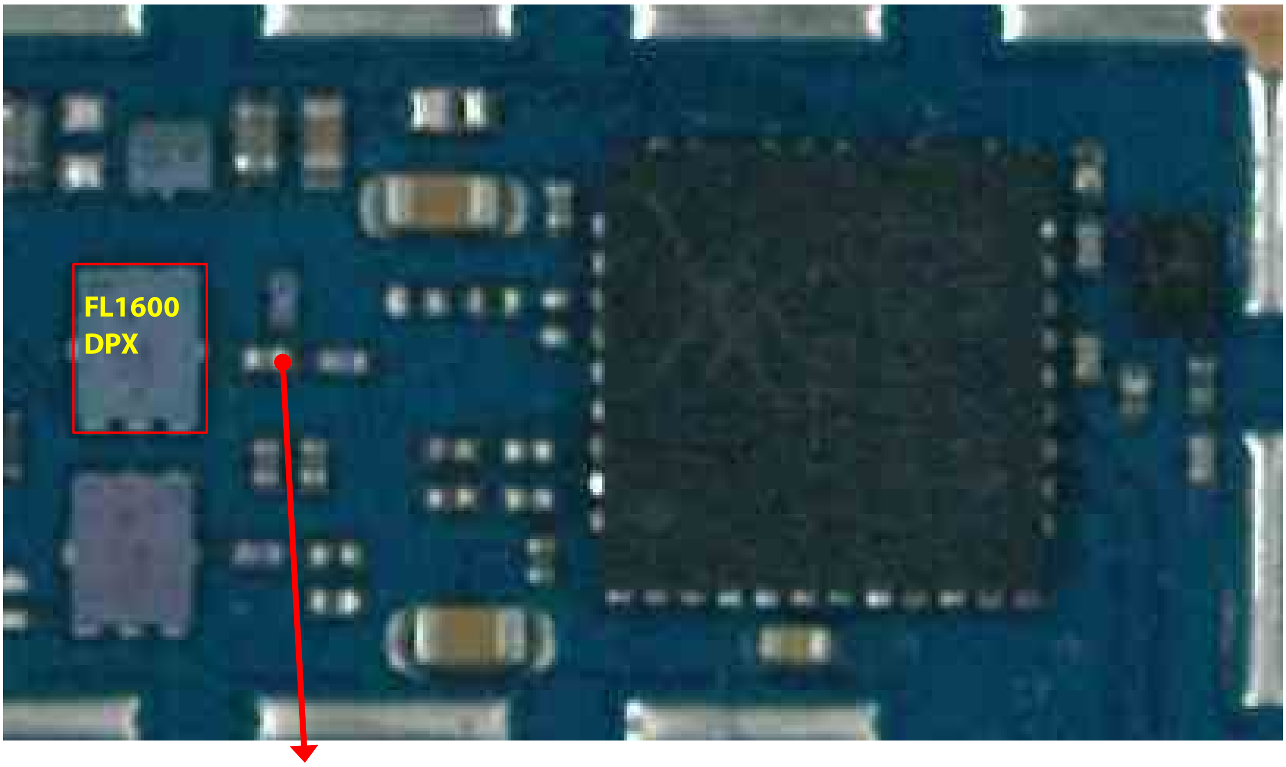


Figure 3.1.1 (c)

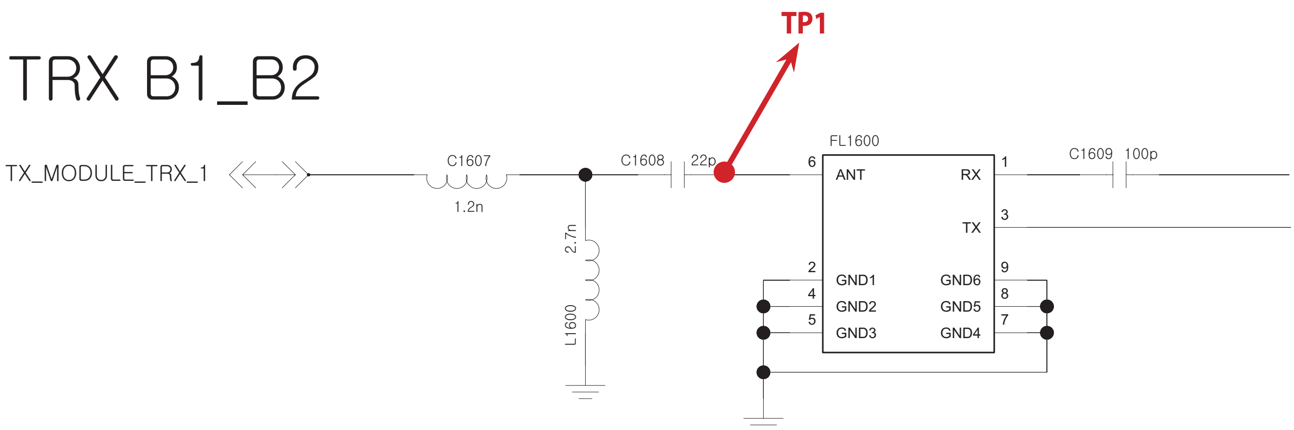
3.1.1.4 Checking RF signal path (Duplexer)




FL1600
DPX

TP1

TRX B1_B2


TP1
TX_MODULE_TRX_1
C1607
1.2n
L1600
2.7n
C1608
22p
FL1600
6
ANT
RX
1
3
TX
2
GND1
GND6
9
4
GND2
GND5
8
5
GND3
GND4
7
C1609
100p

TP1

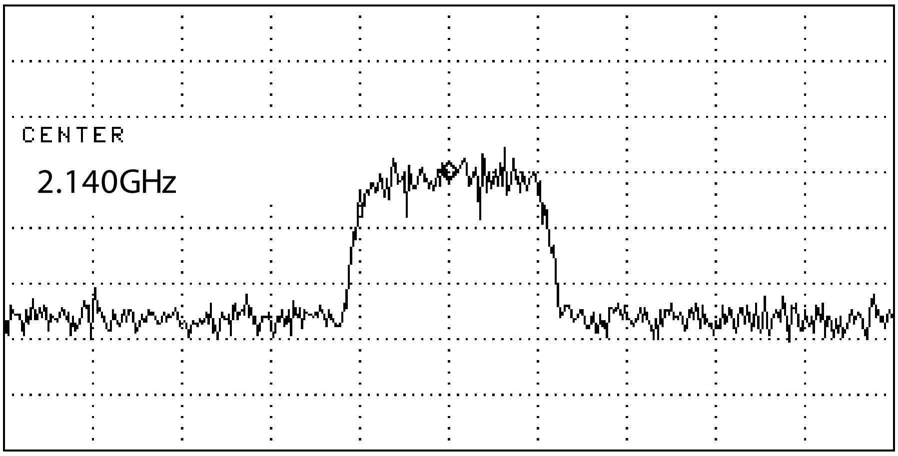
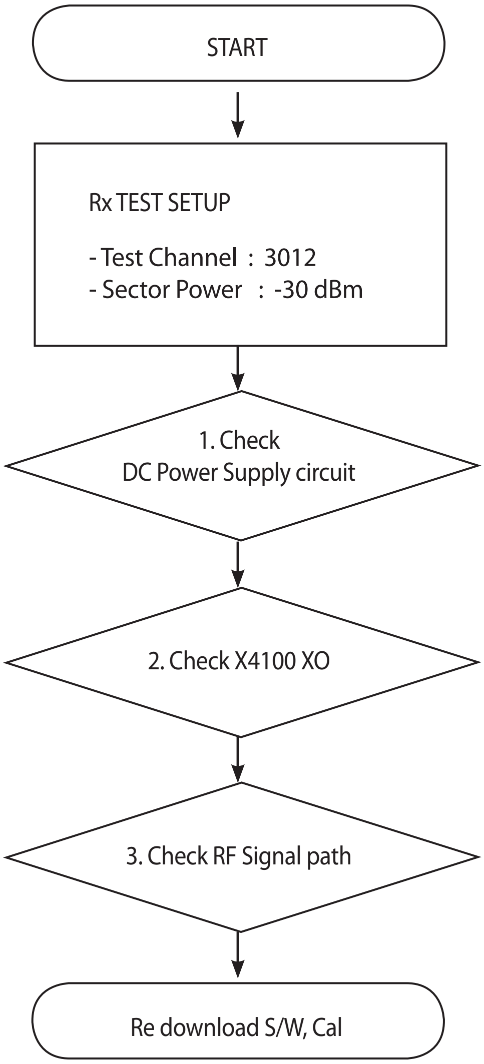
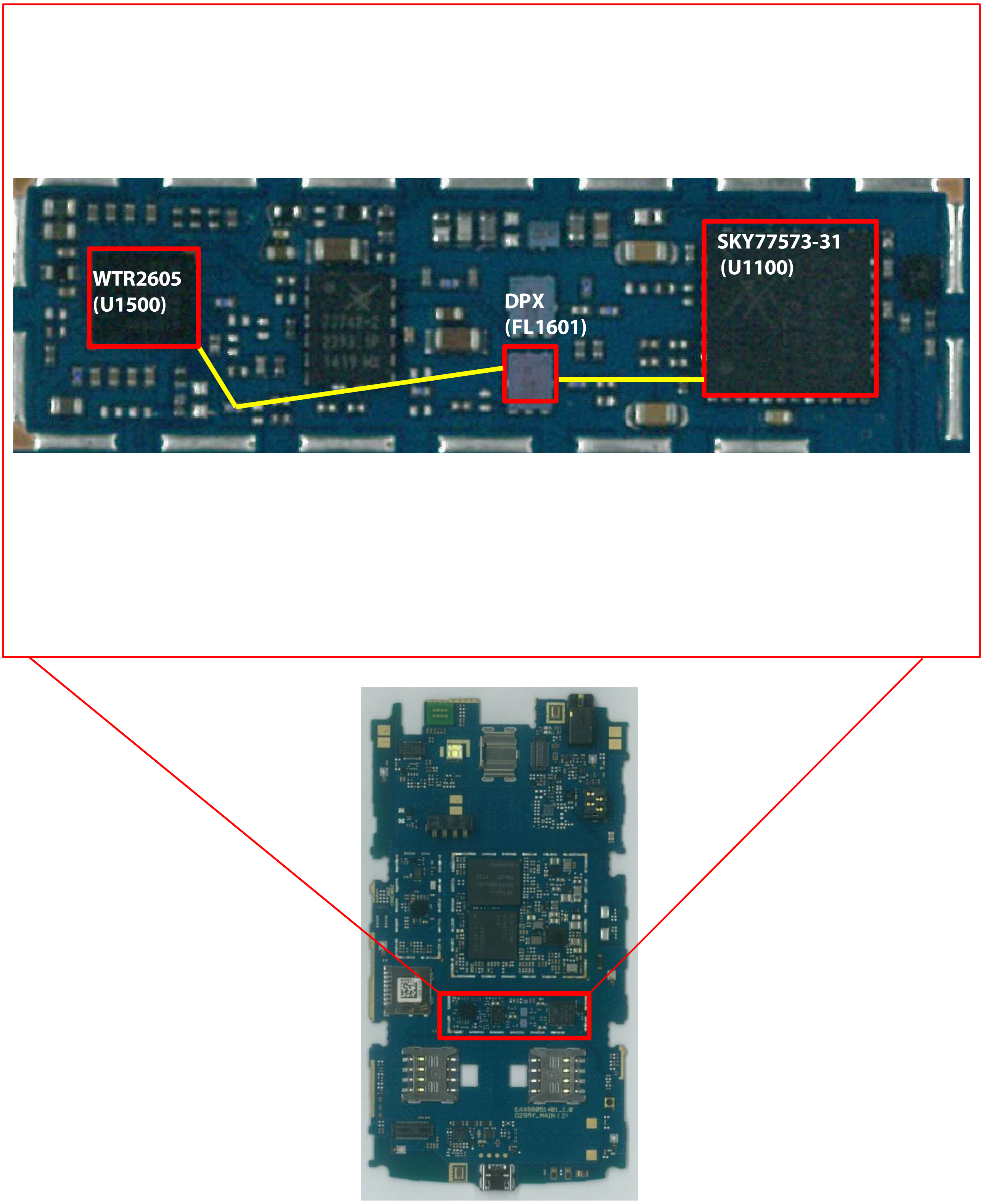


Figure 3.1.1 (b)

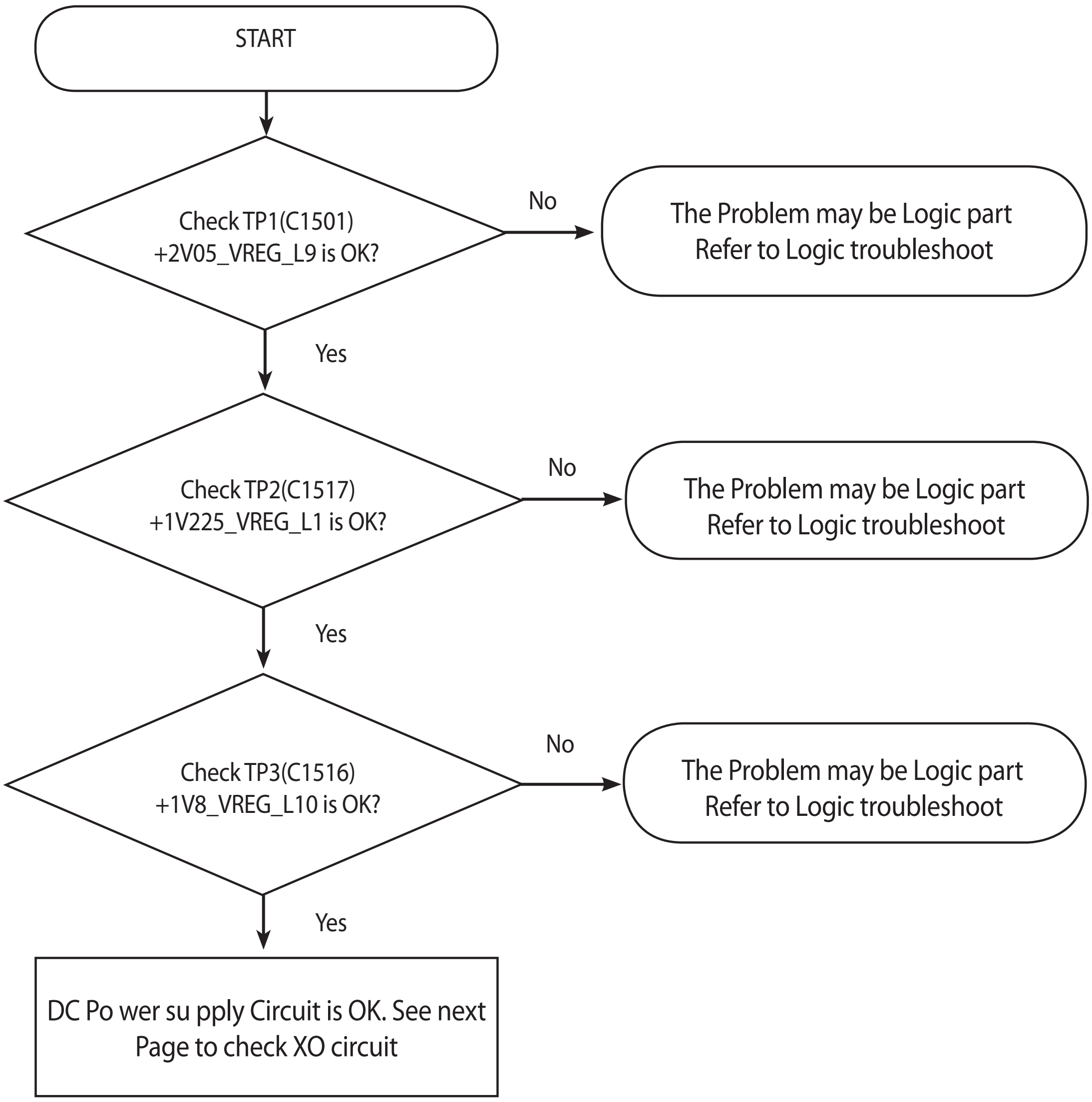
3.1.2 WCDMA_B8 Rx

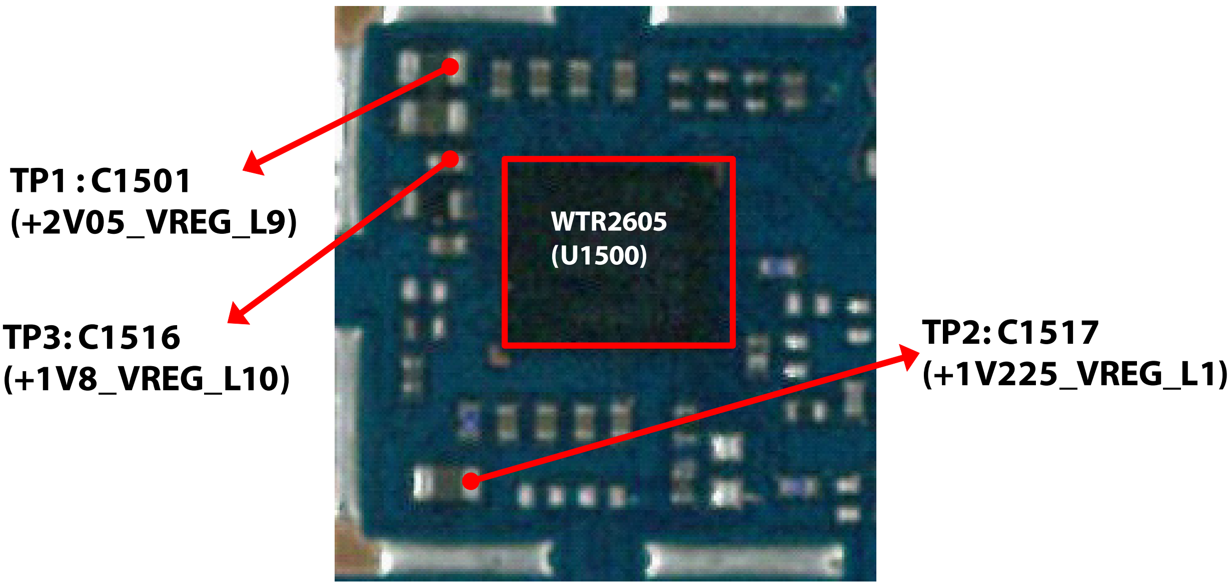


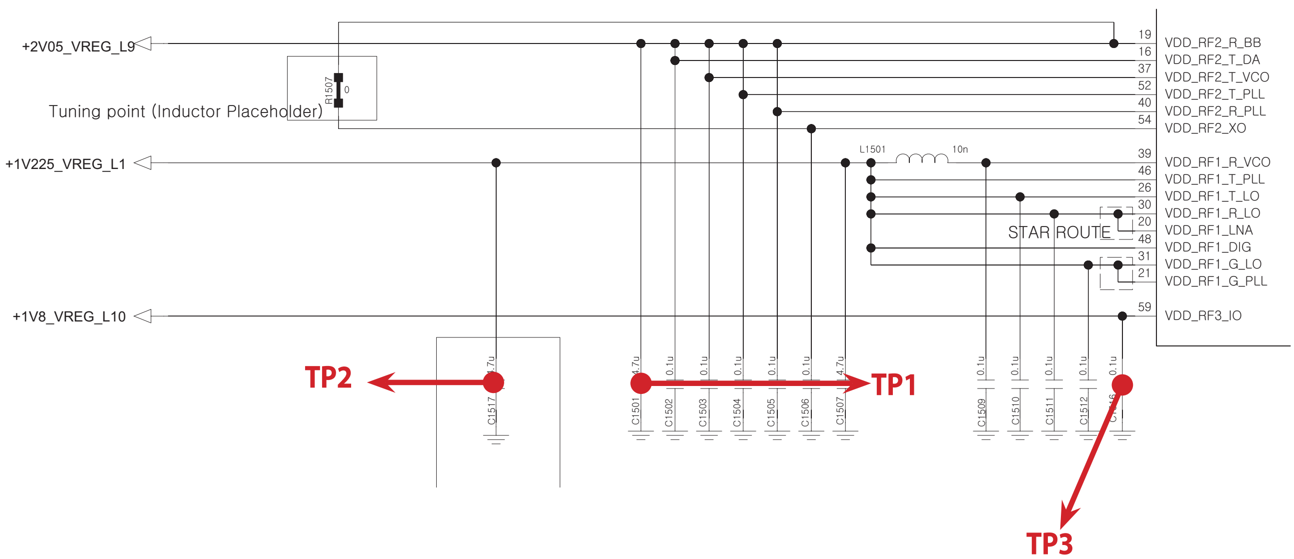
* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form


WTR2605
(U1500)
DPX
(FL1601)
SKY77573-31
(U1100)

3.1.2.1 Checking DC Power supply circuit




TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)


+2V05_VREG_L9
Tuning point (Inductor Placeholder)
R1507
+1V225_VREG_L1
+1V8_VREG_L10
L1501
10n
STAR ROUTE
VDD_RF2_R_BB
VDD_RF2_T_DA
VDD_RF2_T_VCO
VDD_RF2_T_PLL
VDD_RF2_R_PLL
VDD_RF2_XO
VDD_RF1_R_VCO
VDD_RF1_T_PLL
VDD_RF1_T_LO
VDD_RF1_R_LO
VDD_RF1_LNA
VDD_RF1_DIG
VDD_RF1_G_LO
VDD_RF1_G_PLL
VDD_RF3_IO
TP2
TP1
TP3
C1501
C1502
C1503
C1504
C1505
C1506
C1507
C1509
C1510
C1511
C1512

3.1.2.2 Checking XO circuit

START

Check TP1(C1501)
Is it 19.2MHz?
Refer to Figure 3.1.2(a)

Yes → XO circuit is Ok. See next page to check RF signal path

No → Replace X4100 and then check TP1(C1501)

Is it similar ?

Yes → XO circuit is Ok. See next page to check RF signal path.

No → Replace Main B/D

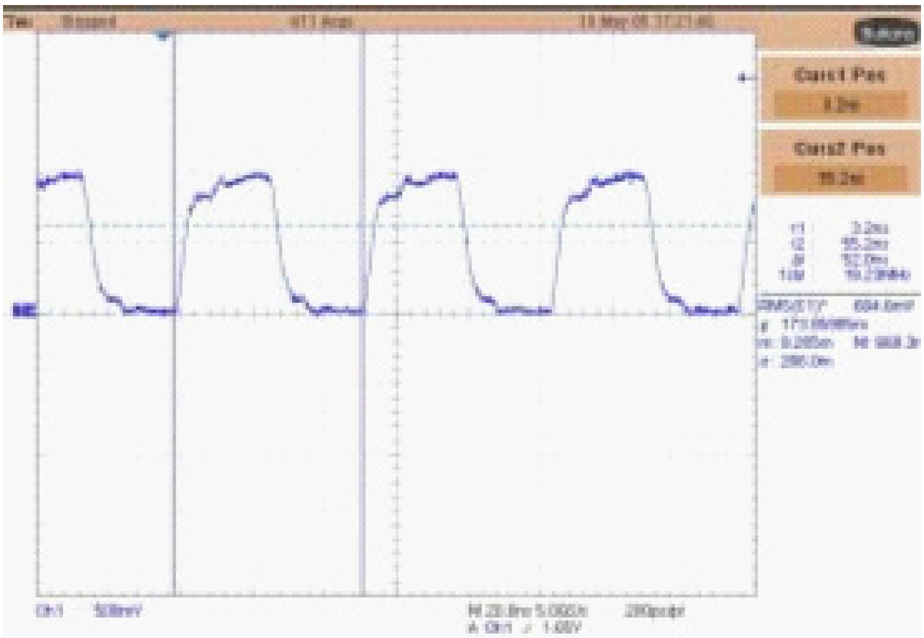
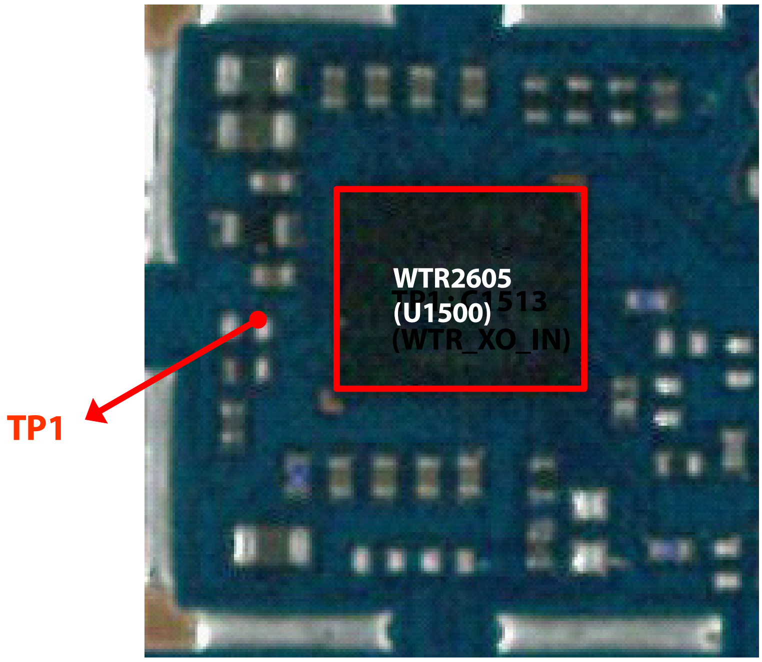
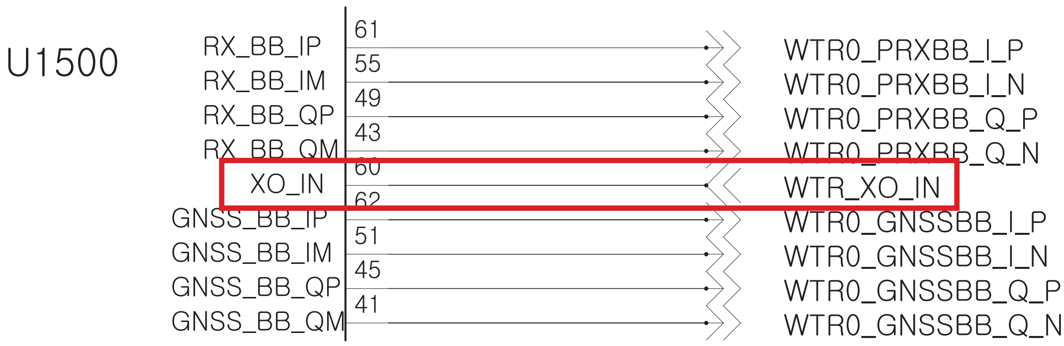
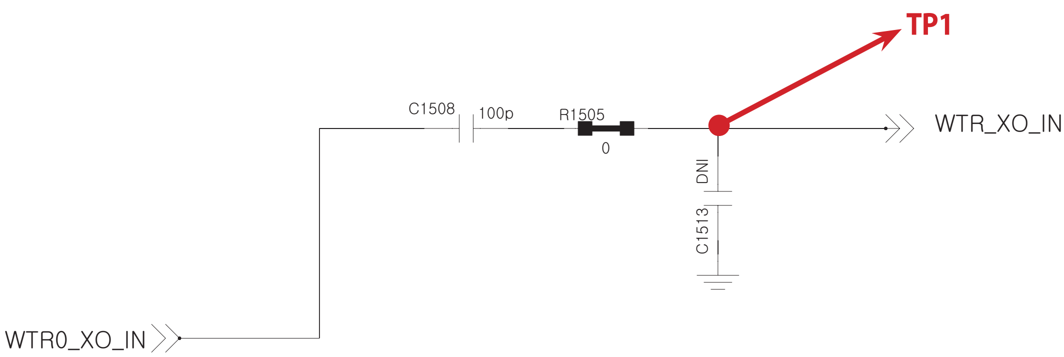


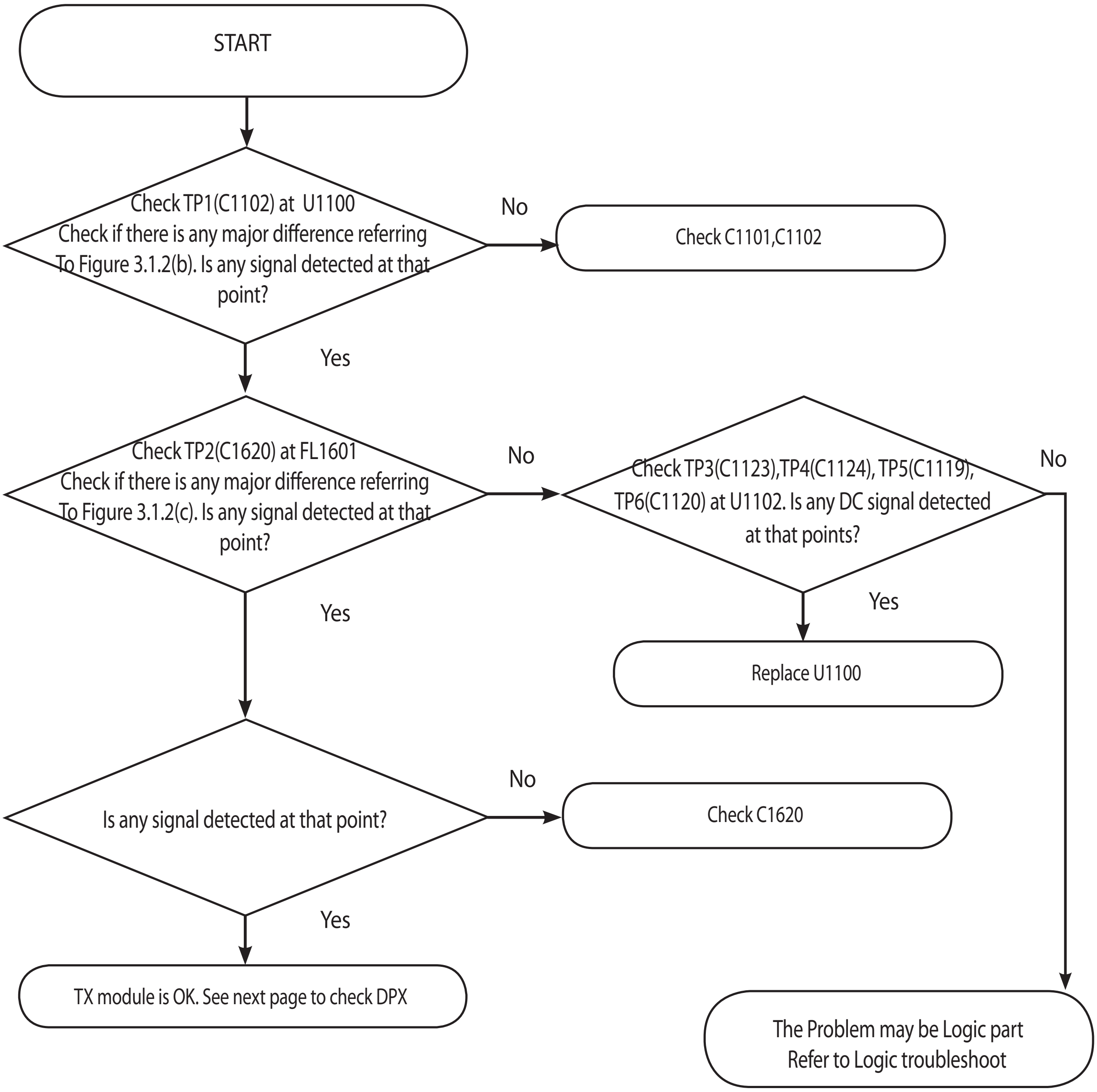
Figure 3.1.2 (a)

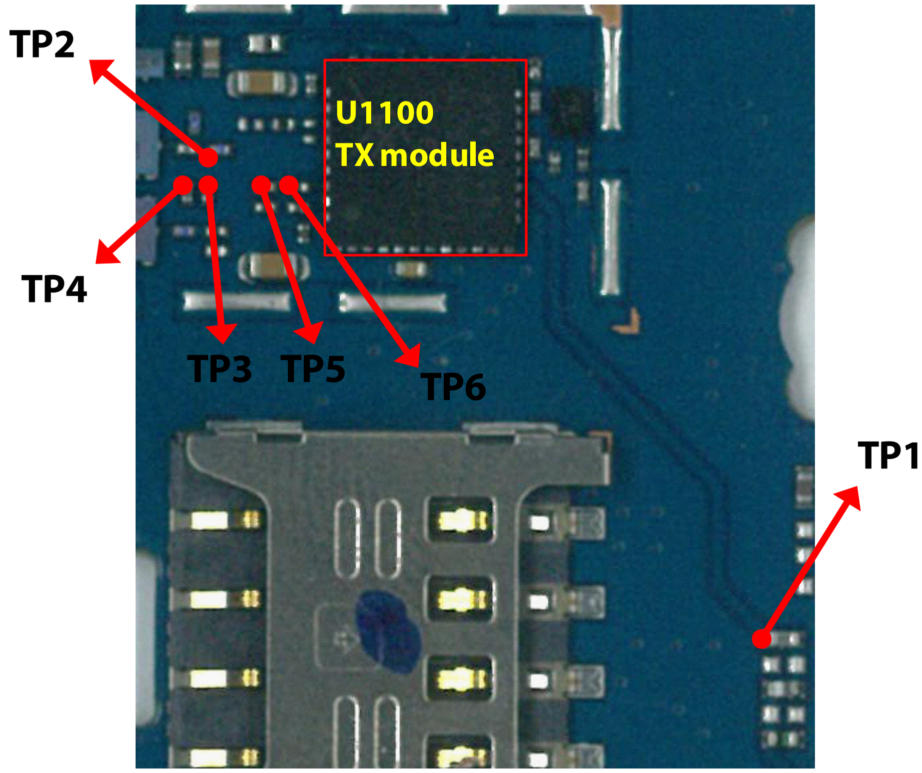

TP1
WTR2605
(U1500)
TP1 : C1513
(WTR_XO_IN)

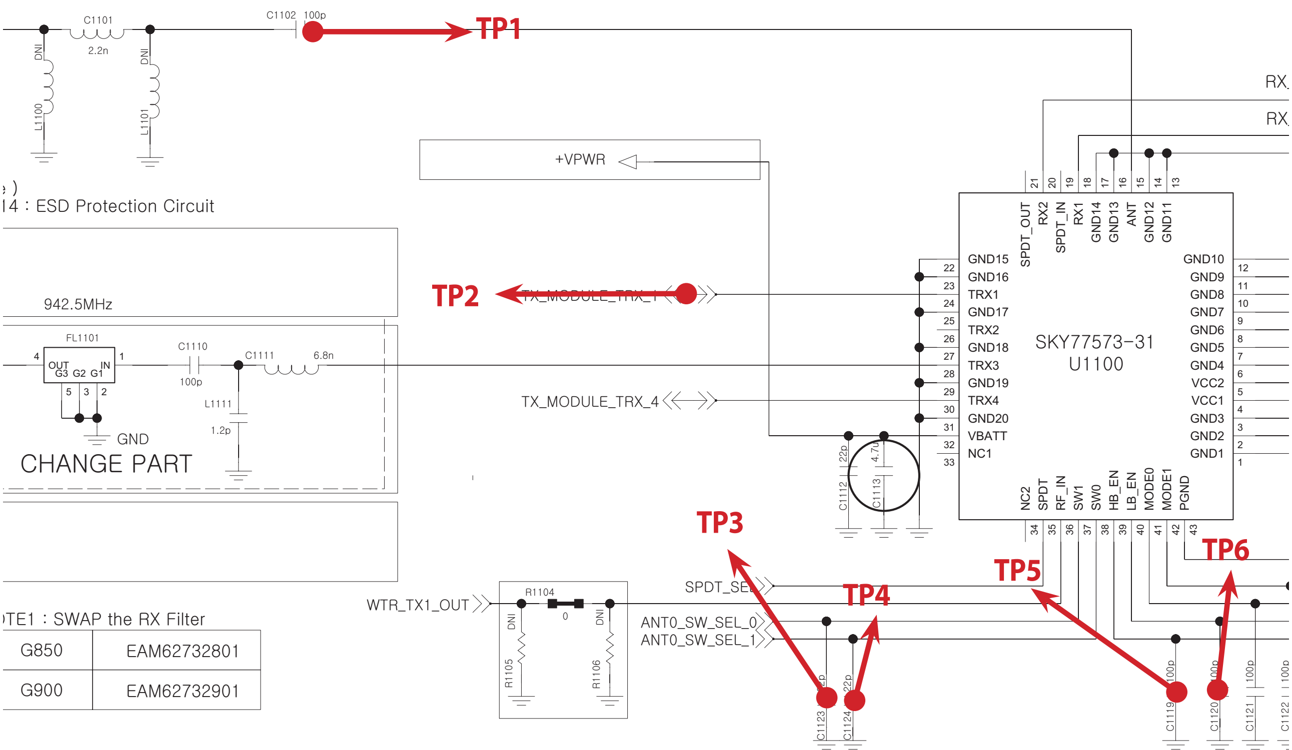

U1500
RX_BB_IP
RX_BB_IM
RX_BB_QP
RX_BB_QM
XO_IN
GNSS_BB_IP
GNSS_BB_IM
GNSS_BB_QP
GNSS_BB_QM
61
55
49
43
60
62
51
45
41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

3.1.2.3 Checking RF signal path (TX Module)




TP2
U1100
TX module
TP4
TP3
TP5
TP6
TP1


TP1
TP2
TP3
TP4
TP5
TP6
C1101
2.2n
C1102 100p
L1100
L1101
ESD Protection Circuit
+VPWR
942.5MHz
FL1101
C1110
100p
C1111
6.8n
L1111
1.2p
GND
CHANGE PART
TX_MODULE_TRX_4
SKY77573-31
U1100
WTR_TX1_OUT
R1104
R1105
R1106
SPDT_SEL
ANT0_SW_SEL_0
ANT0_SW_SEL_1
SWAP the RX Filter
G850
EAM62732801
G900
EAM62732901

TP1

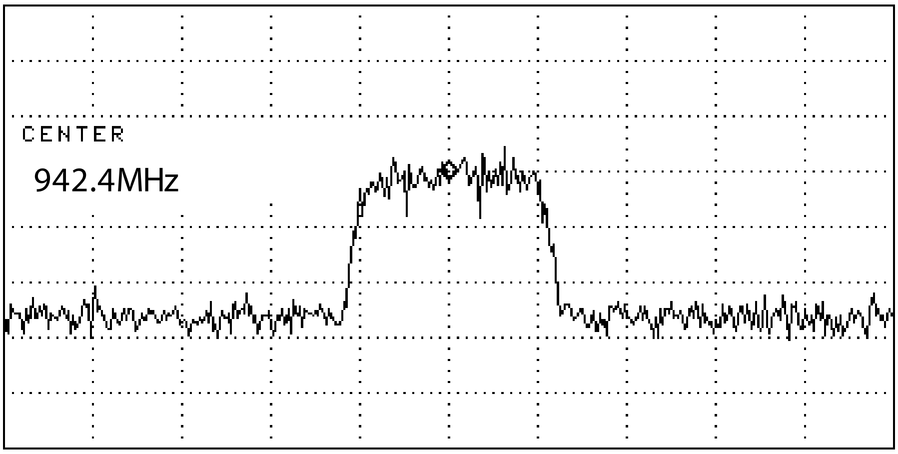


Figure 3.1.2 (b)

TP2

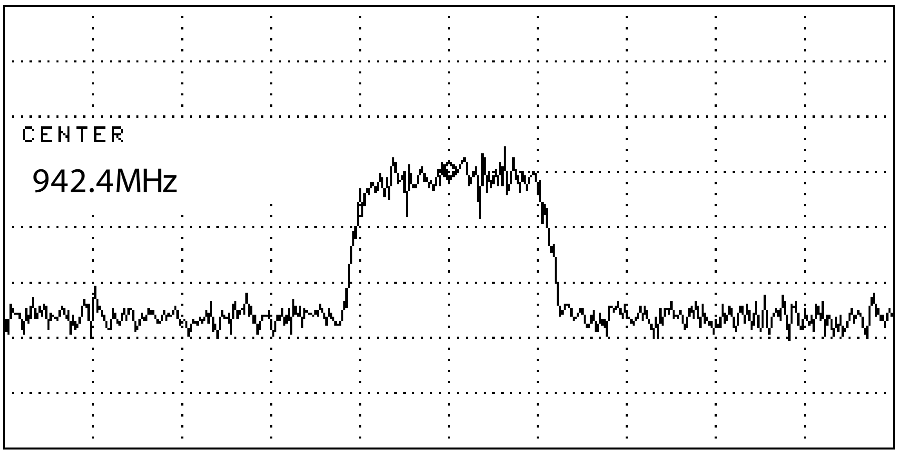
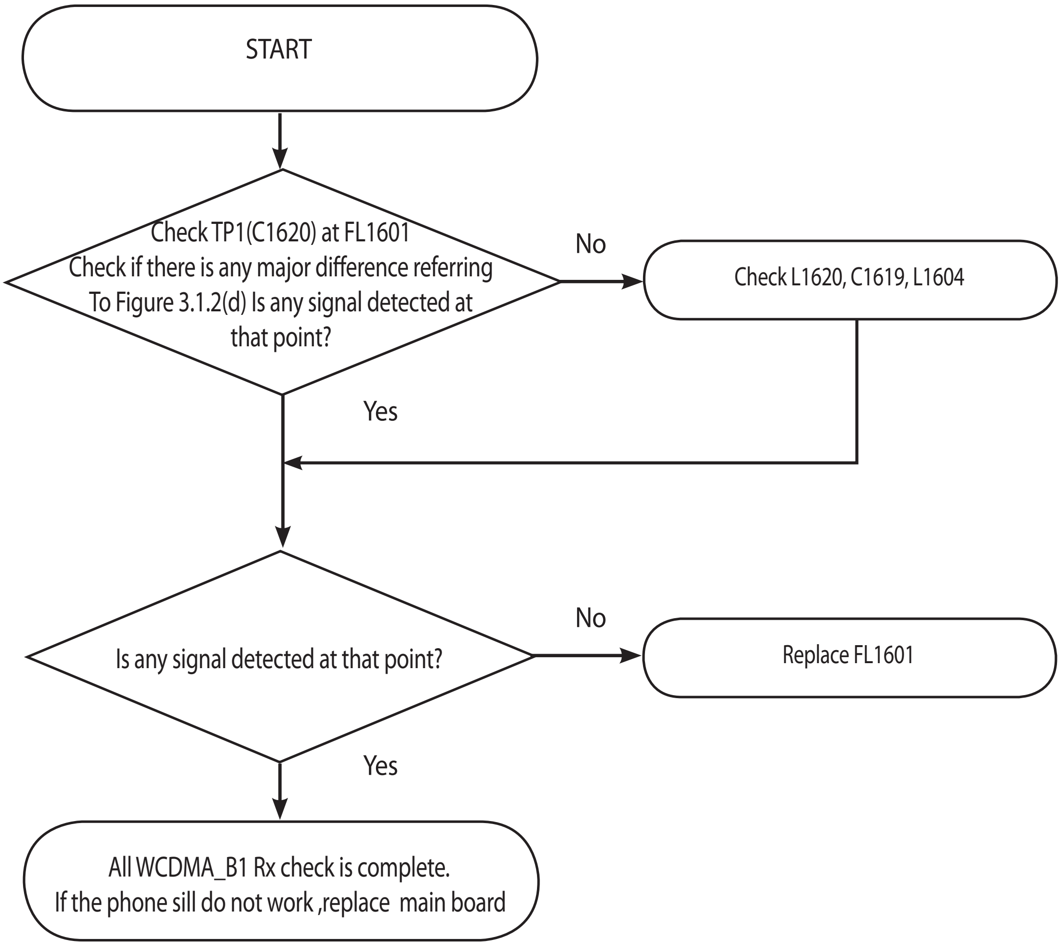


Figure 3.1.2 (c)

3.1.2.4 Checking RF signal path (Duplexer)



TP1

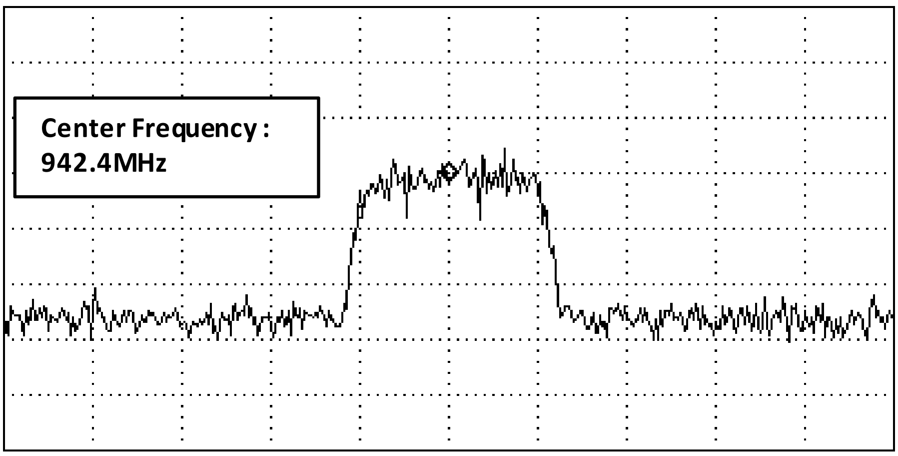
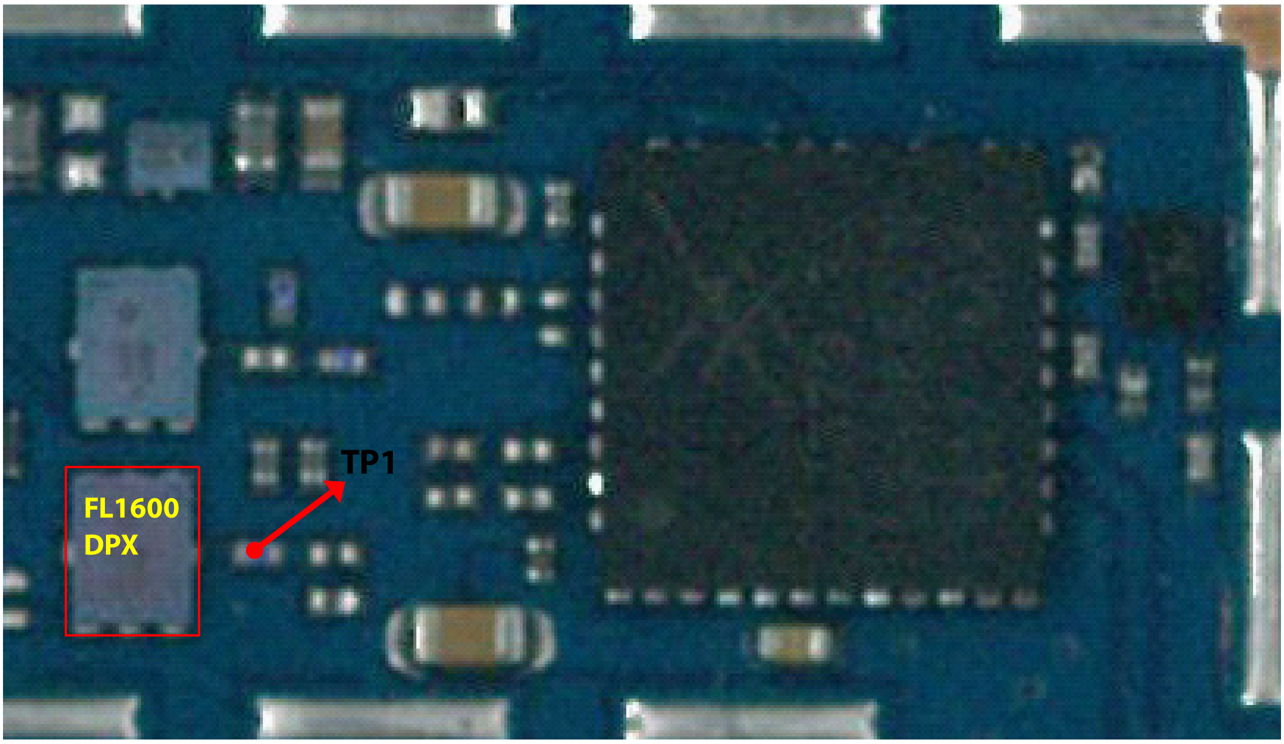
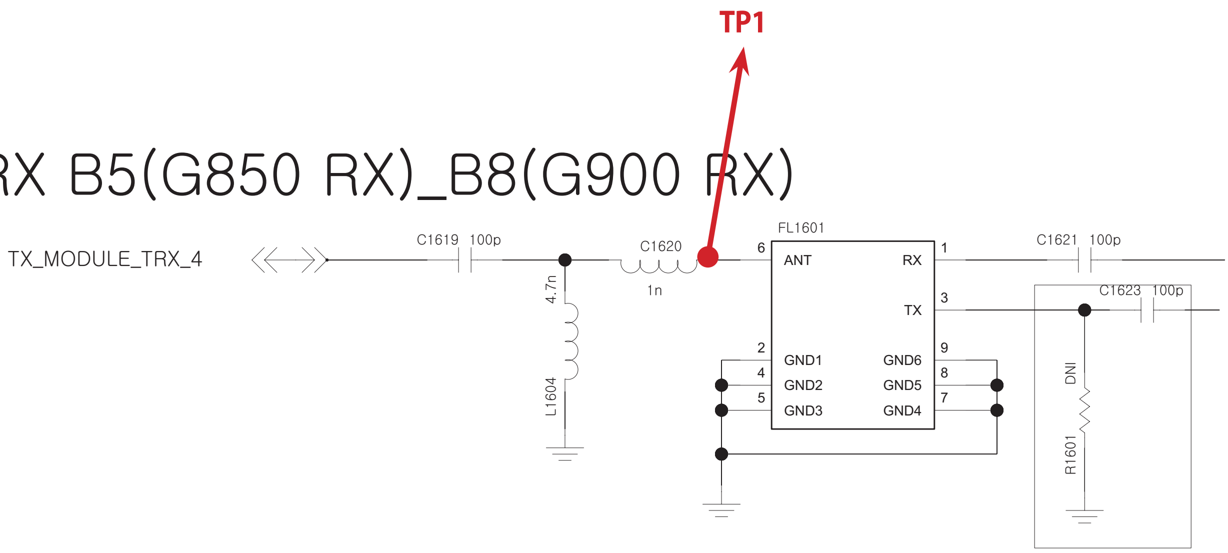


Figure 3.1.2 (d)

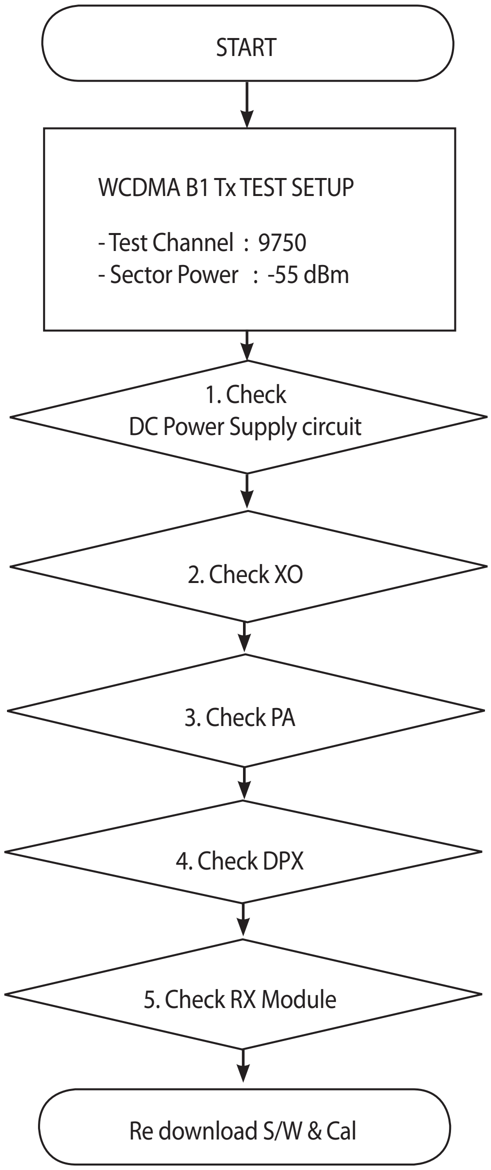


TRX B5(G850 RX)_B8(G900 RX)

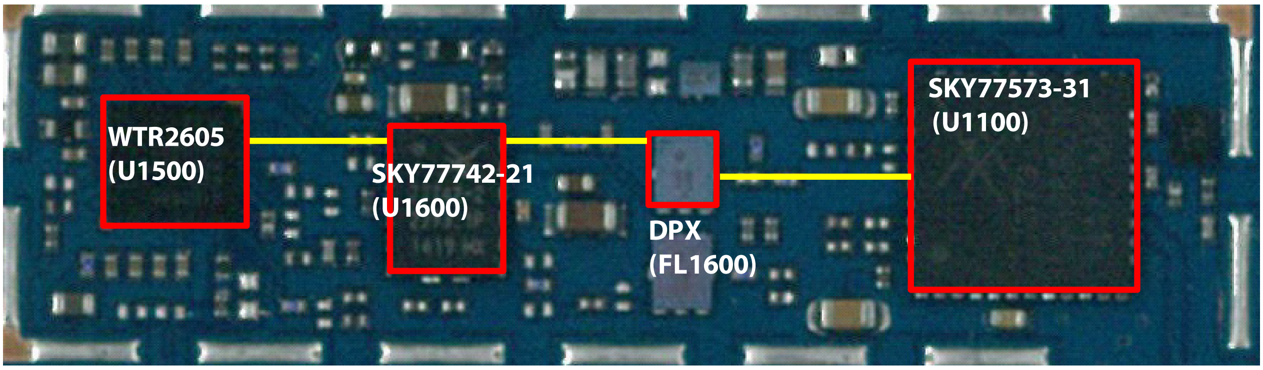


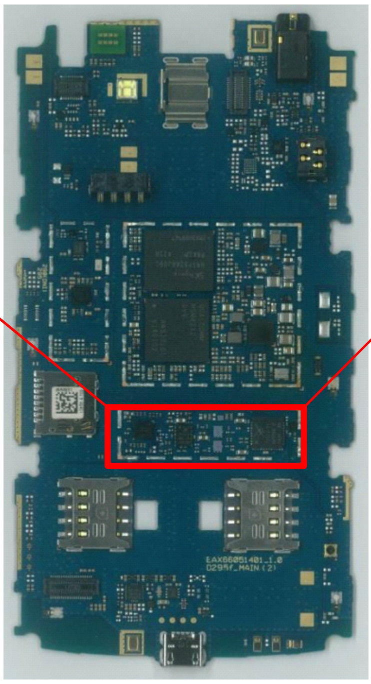
3.2 WCDMA Tx Part Trouble

3.2.1 WCDMA_B1 Tx

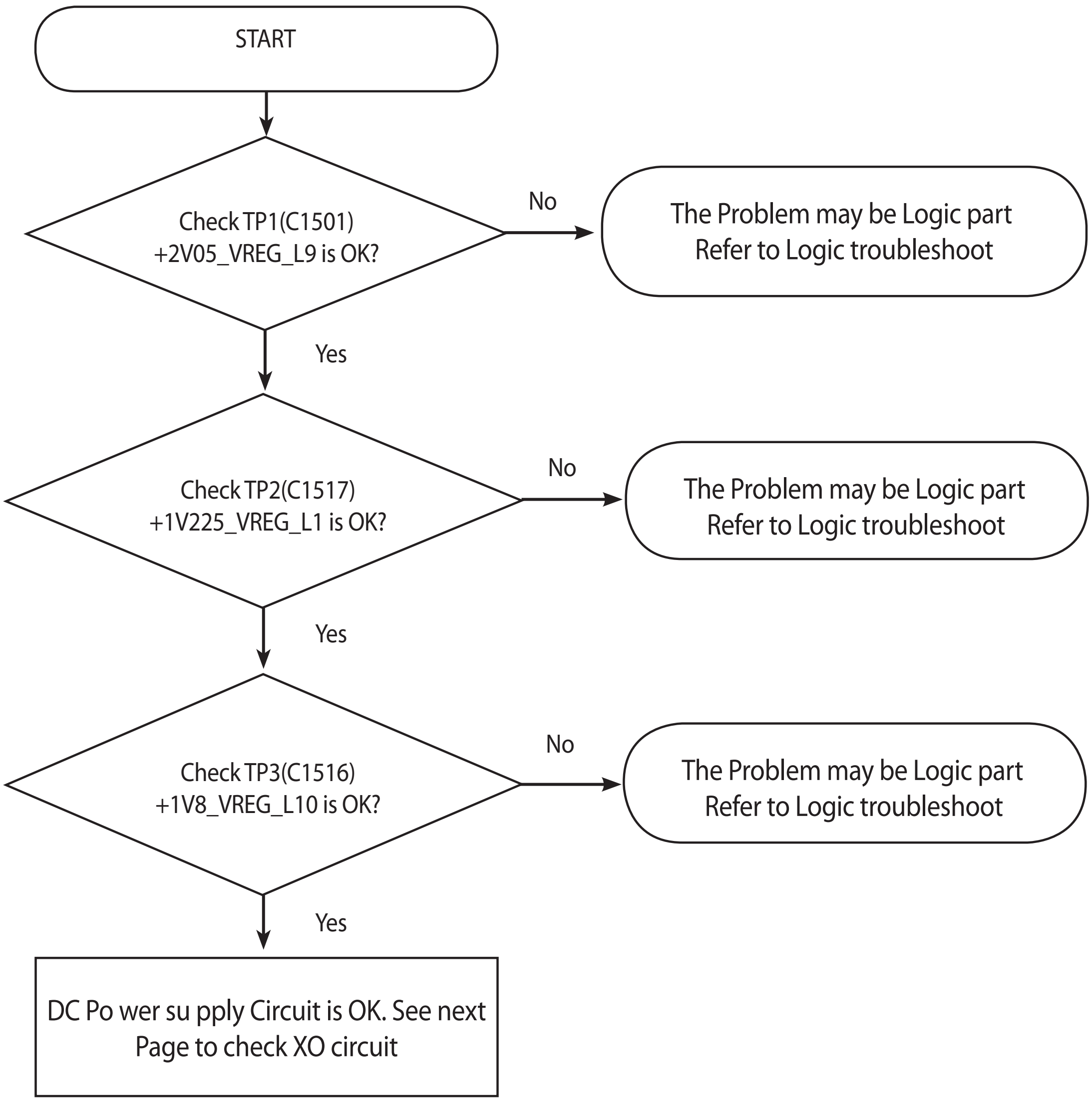


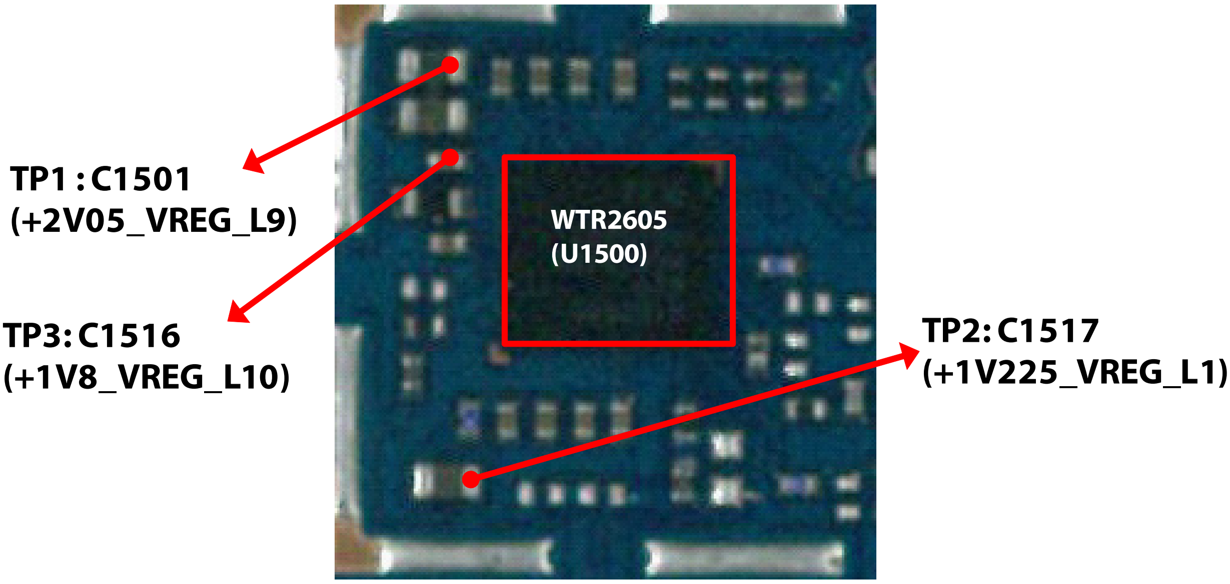
* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

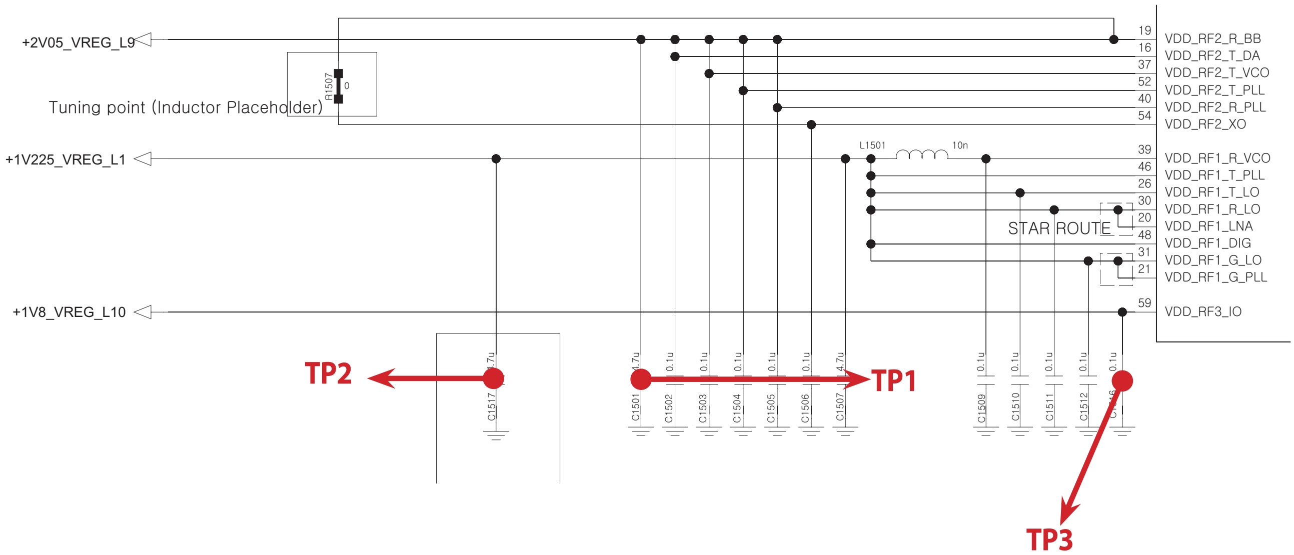

WTR2605
(U1500)
SKY77742-21
(U1600)
DPX
(FL1600)
SKY77573-31
(U1100)



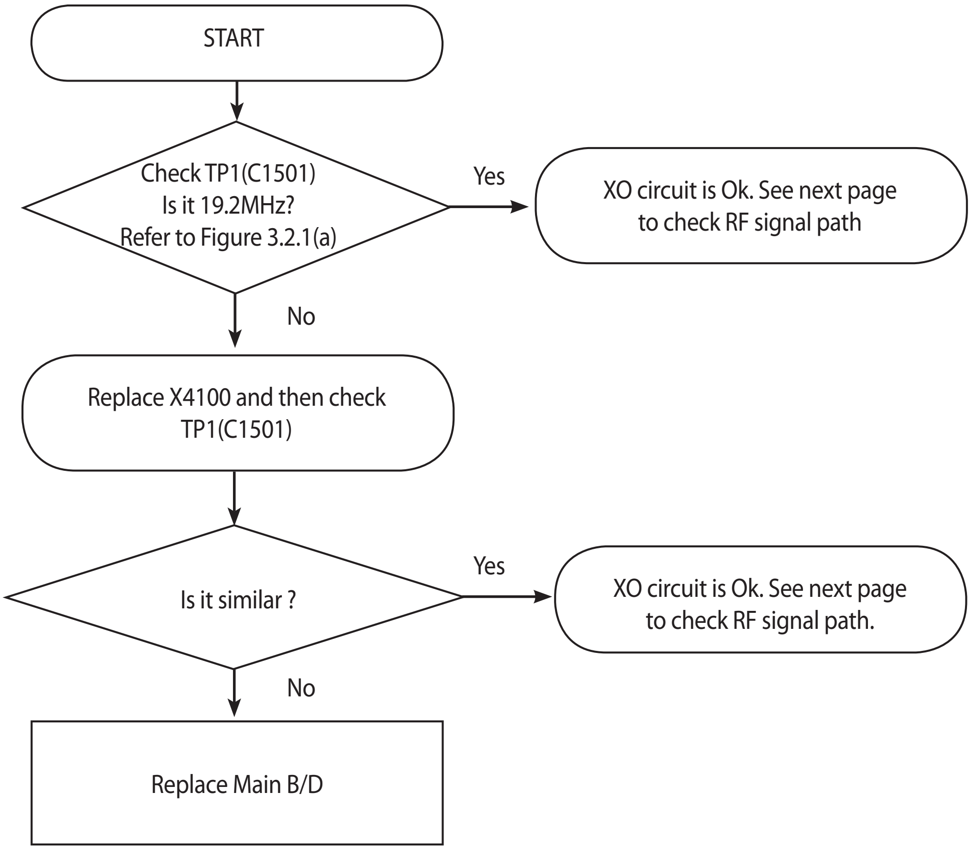
3.2.1.1 Checking DC Power supply circuit




TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)


+2V05_VREG_L9
Tuning point (Inductor Placeholder)
+1V225_VREG_L1
+1V8_VREG_L10
L1501
10n
STAR ROUTE
VDD_RF2_R_BB
VDD_RF2_T_DA
VDD_RF2_T_VCO
VDD_RF2_T_PLL
VDD_RF2_R_PLL
VDD_RF2_XO
VDD_RF1_R_VCO
VDD_RF1_T_PLL
VDD_RF1_T_LO
VDD_RF1_R_LO
VDD_RF1_LNA
VDD_RF1_DIG
VDD_RF1_G_LO
VDD_RF1_G_PLL
VDD_RF3_IO
TP2
TP1
TP3

3.2.1.2 Checking XO circuit



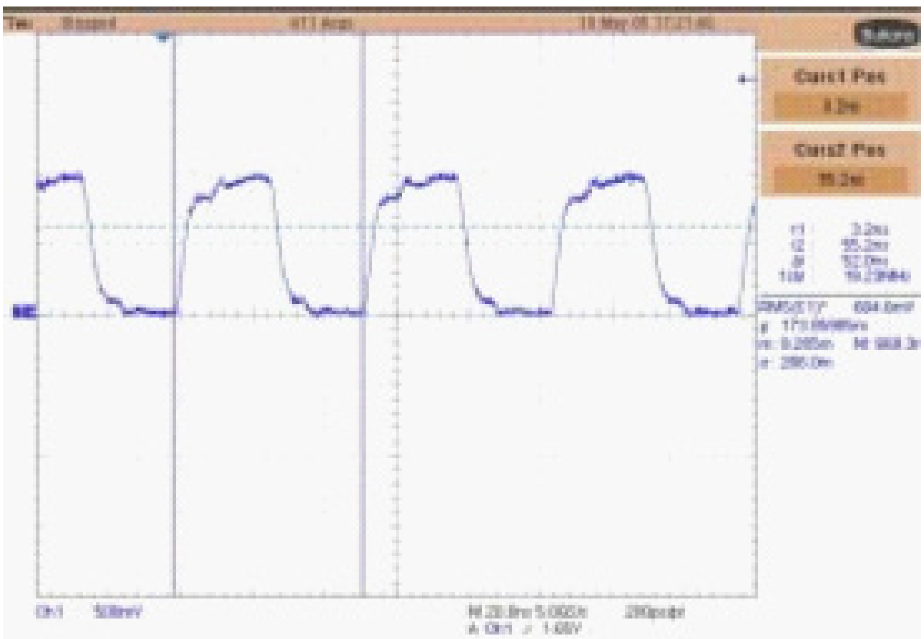
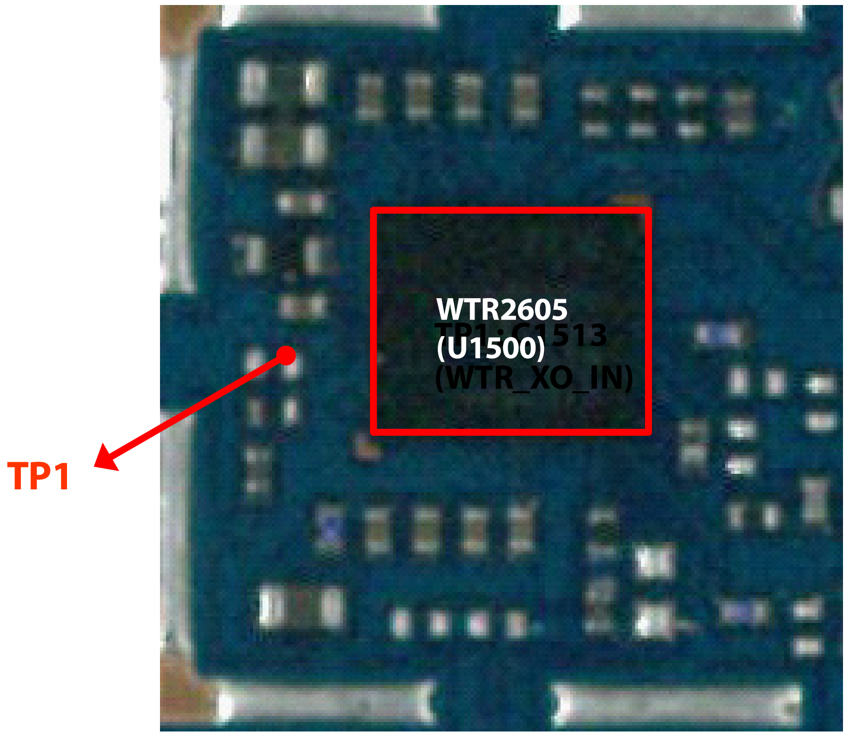
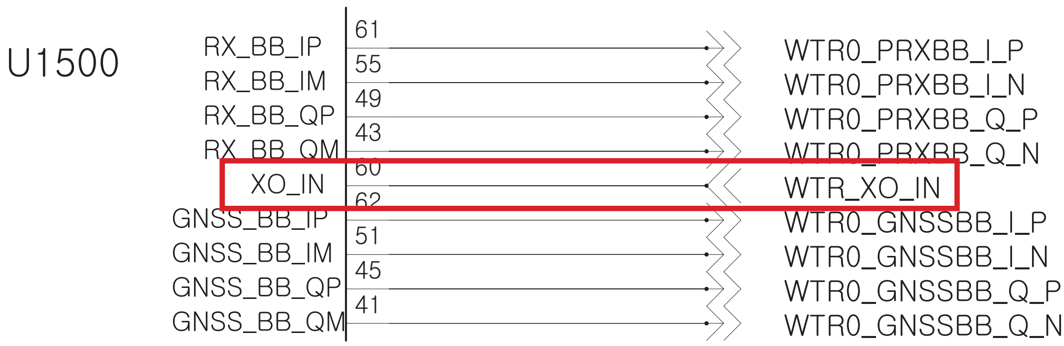
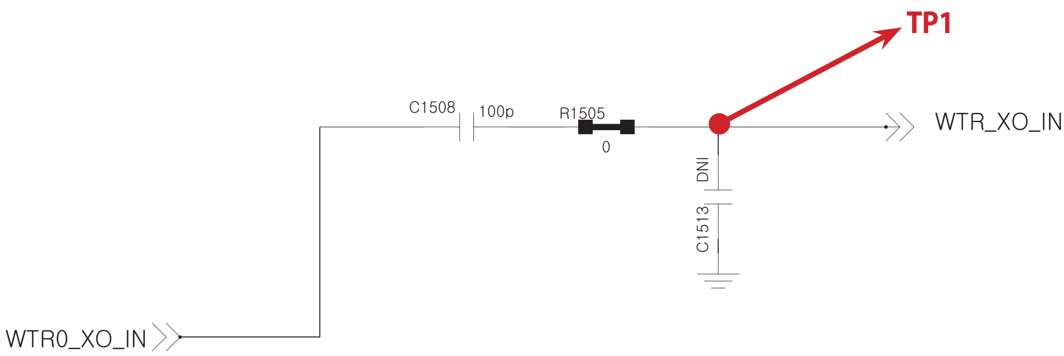


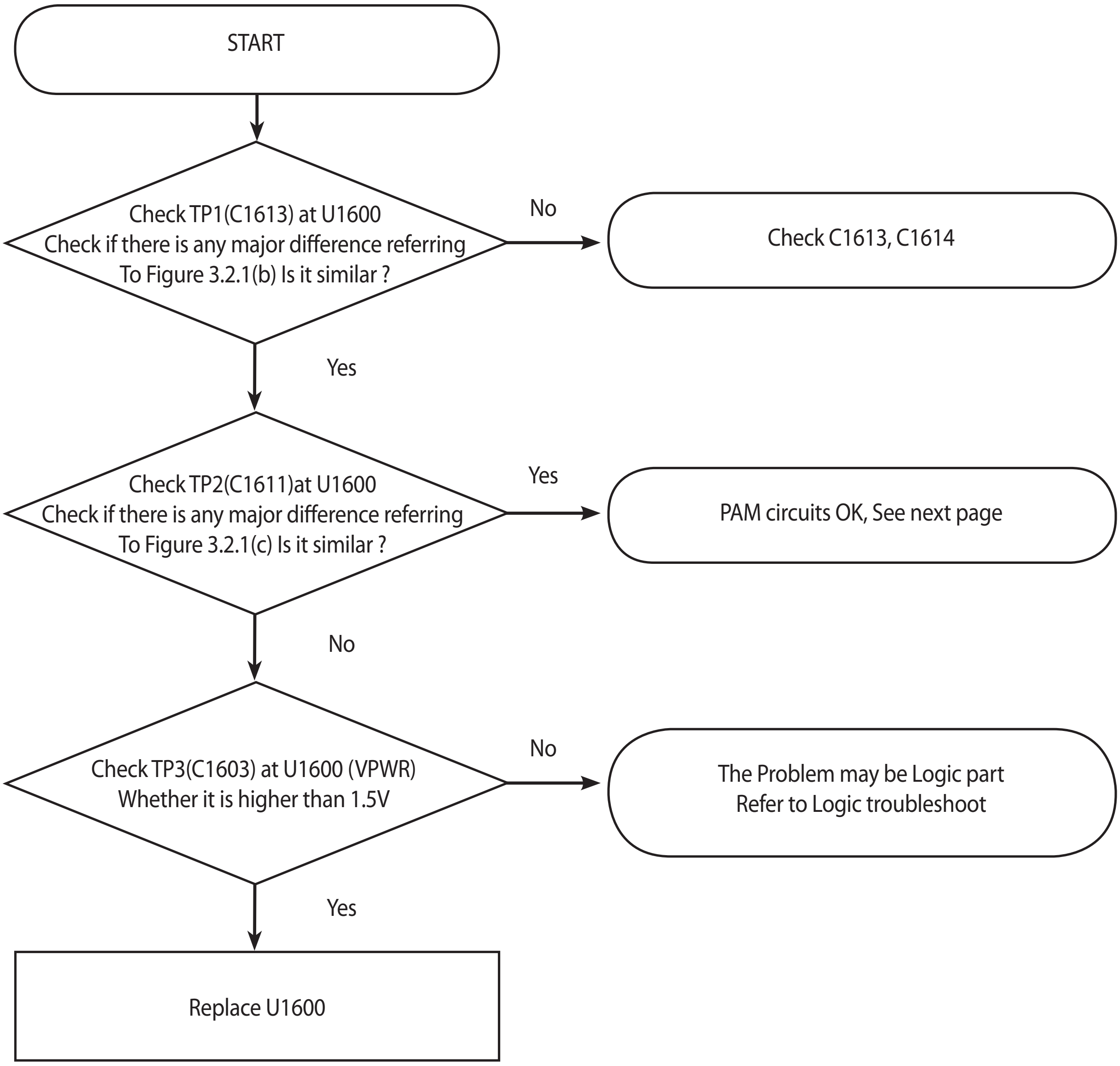
Figure 3.2.1 (a)

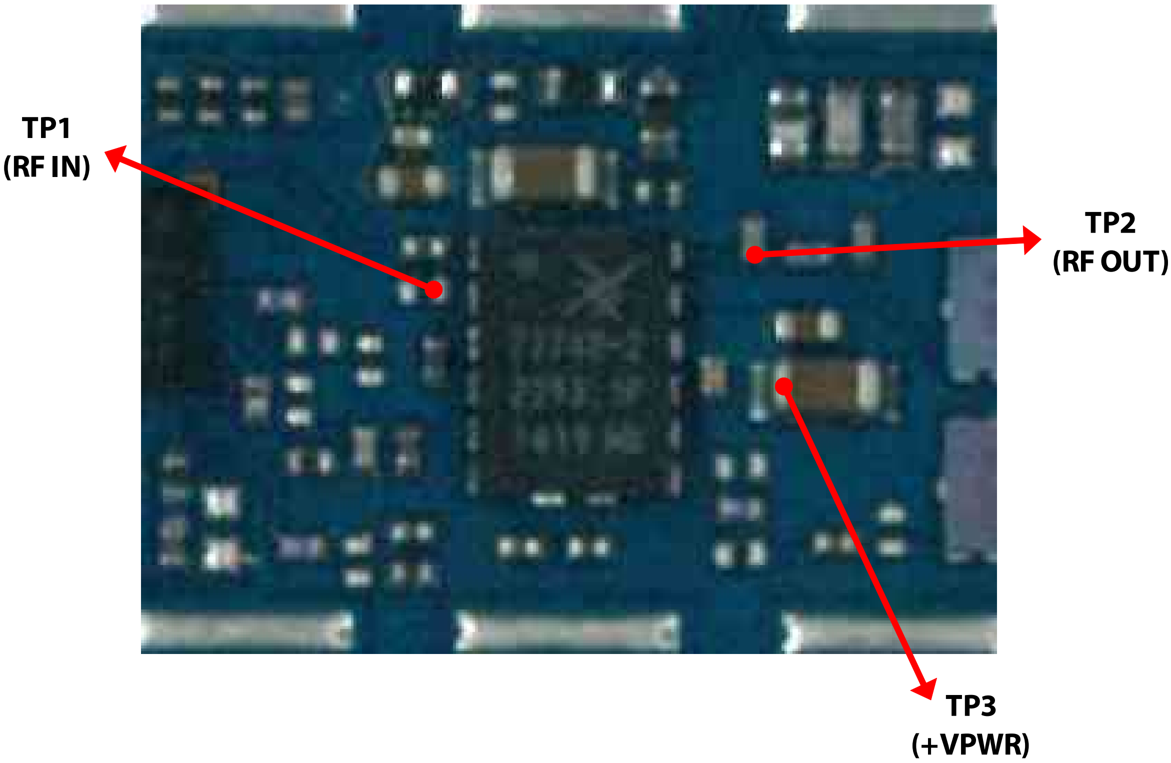

WTR2605
(U1500)
TP1 : C1513
(WTR_XO_IN)
TP1

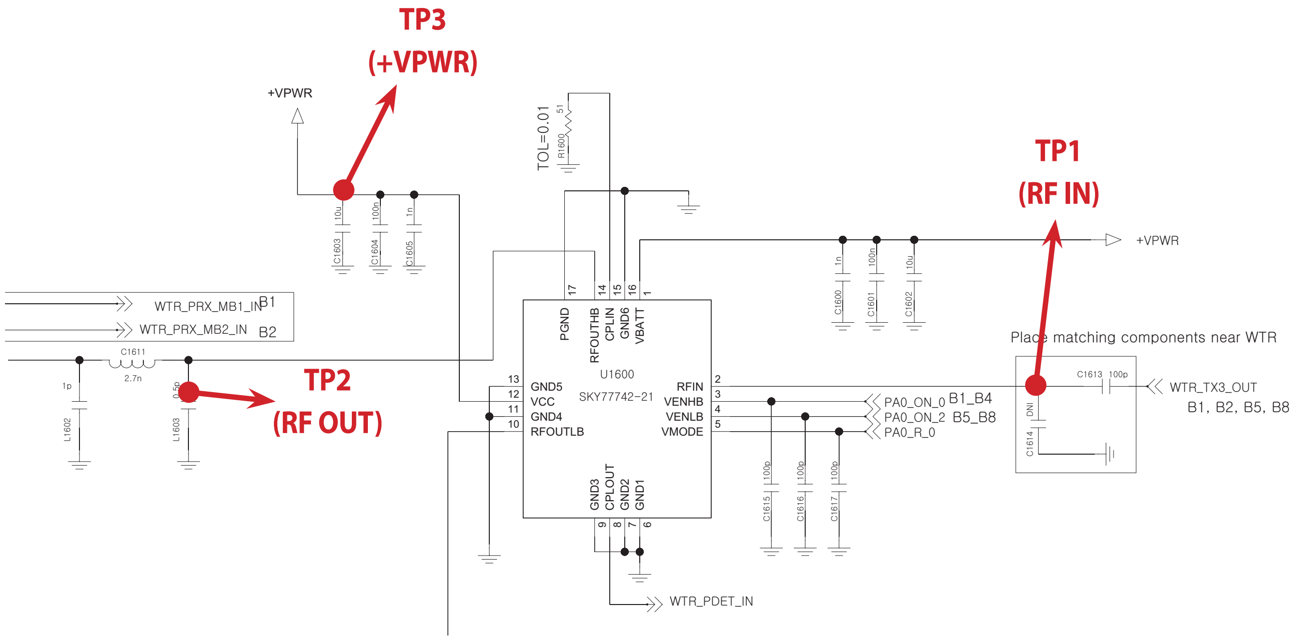

U1500
RX_BB_IP
RX_BB_IM
RX_BB_QP
RX_BB_QM
XO_IN
GNSS_BB_IP
GNSS_BB_IM
GNSS_BB_QP
GNSS_BB_QM
61
55
49
43
60
62
51
45
41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

3.2.1.3 Checking B1 TX PA




TP1
(RF IN)
TP2
(RF OUT)
TP3
(+VPWR)


TP3
(+VPWR)
TP1
(RF IN)
TP2
(RF OUT)
+VPWR
WTR_PRX_MB1_IN B1
WTR_PRX_MB2_IN B2
U1600
SKY77742-21
PGND
RFOUTHB
CPLIN
GND6
VBATT
GND5
VCC
GND4
RFOUTLB
RFIN
VENHB
VENLB
VMODE
GND3
CPLOUT
GND2
GND1
TOL=0.01
PA0_ON_0 B1_B4
PA0_ON_2 B5_B8
PA0_R_0
Place matching components near WTR
WTR_TX3_OUT
B1, B2, B5, B8
WTR_PDET_IN

TP1

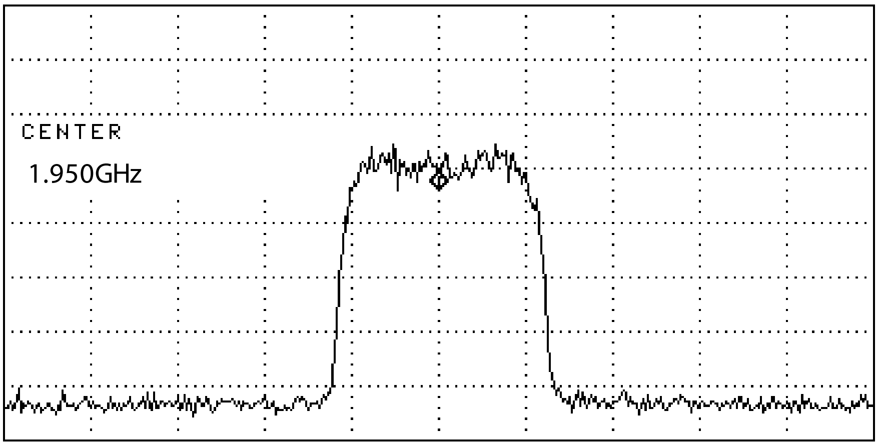


Figure 3.2.1 (b)

TP2

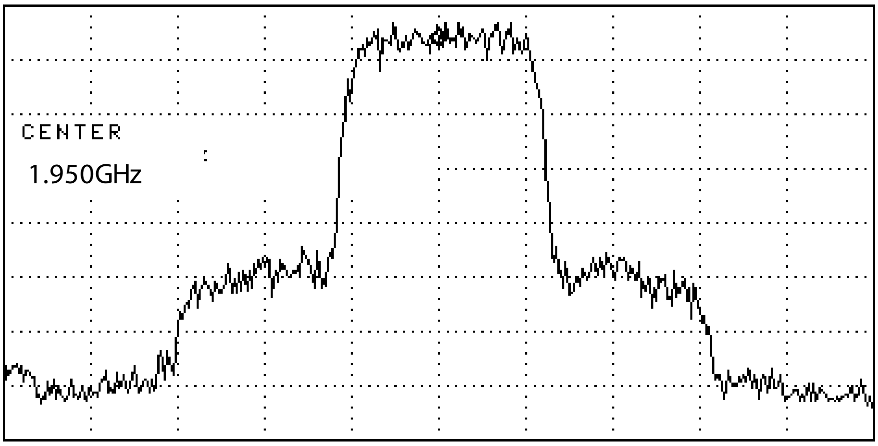
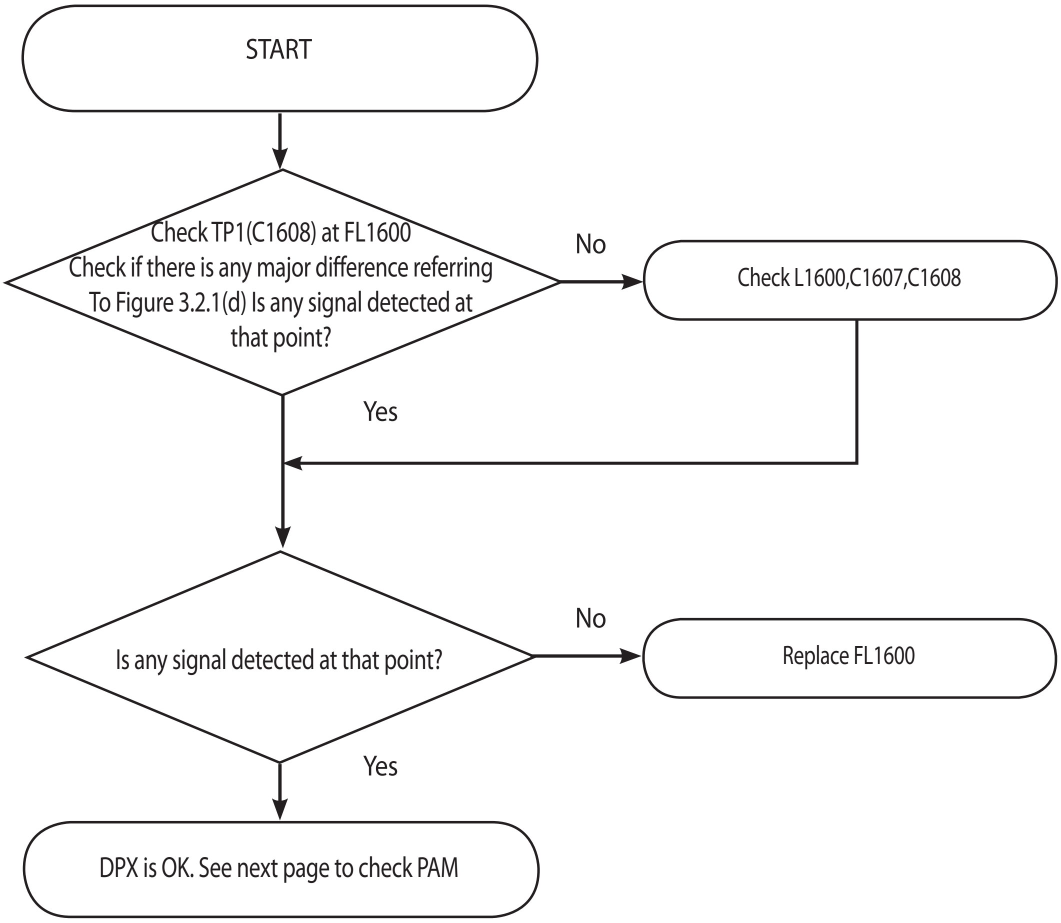


Figure 3.2.1 (c)

3.2.1.4 Checking RF signal path (DPX)



TP1

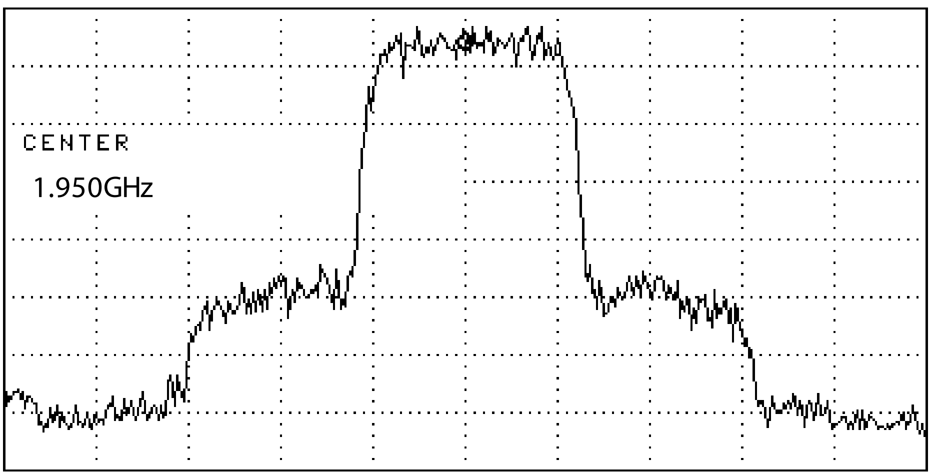
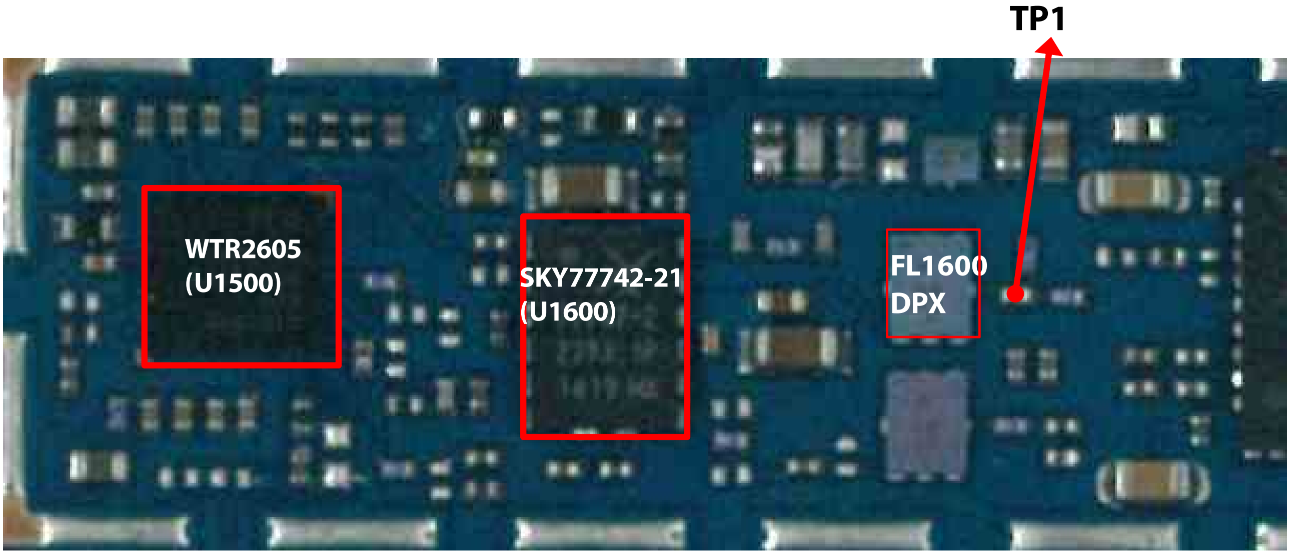
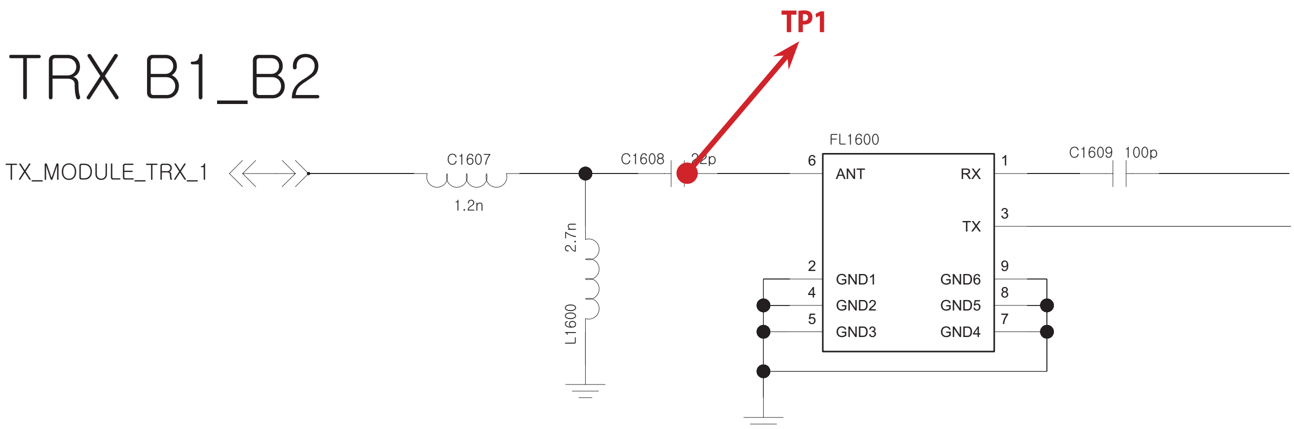


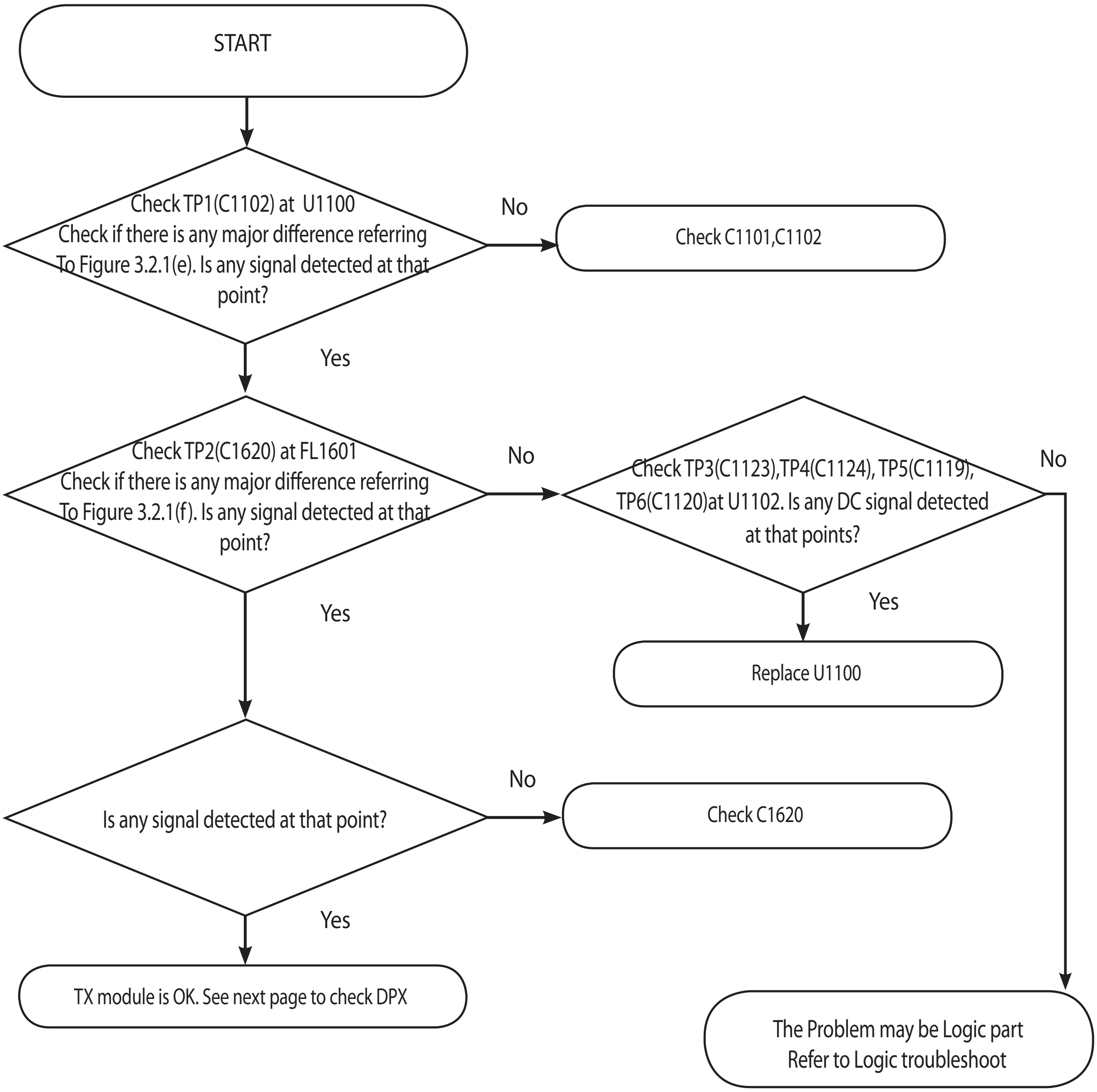
Figure 3.2.1 (d)

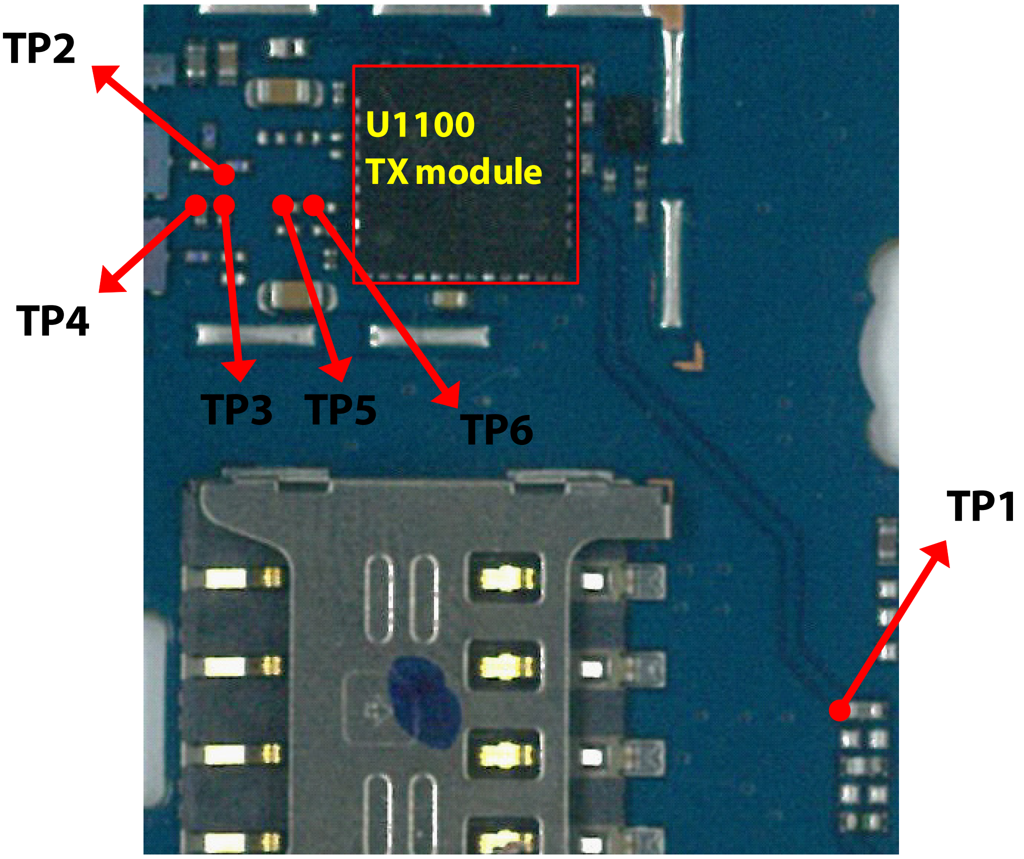

TP1
WTR2605
(U1500)
SKY77742-21
(U1600)
FL1600
DPX

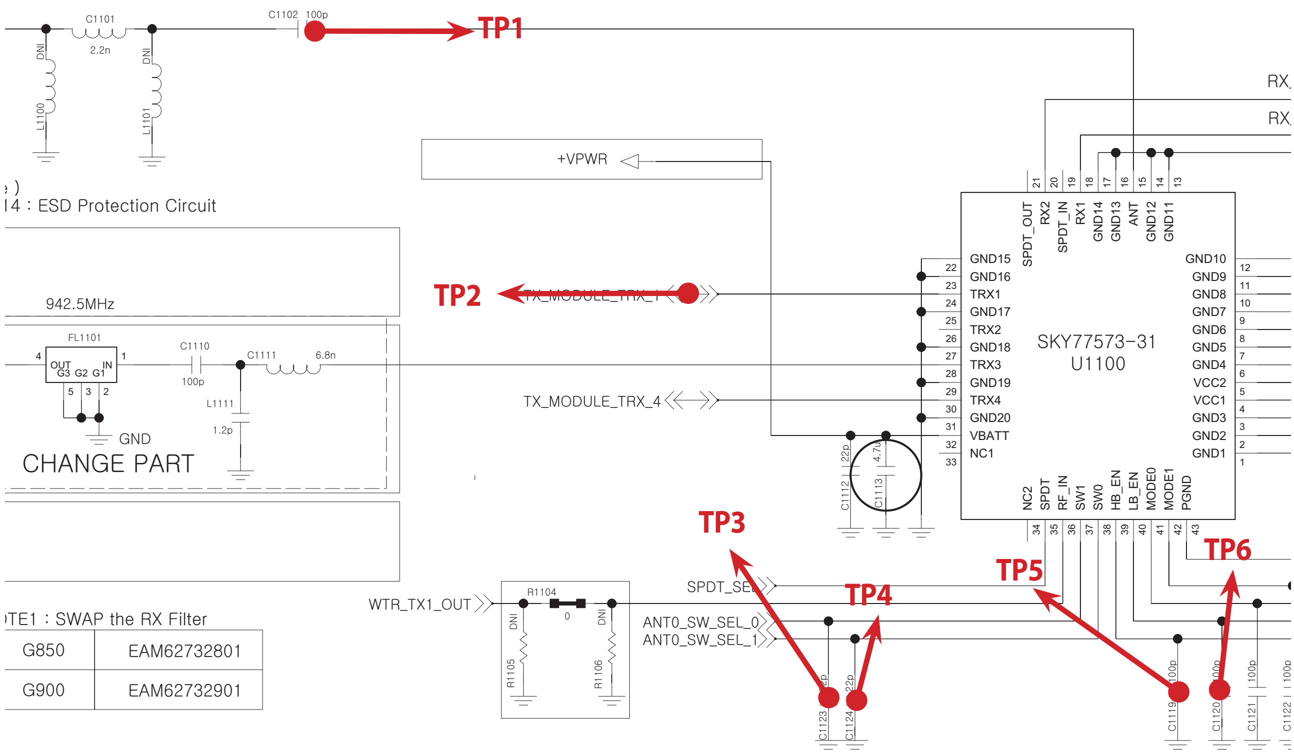
TRX B1_B2


TP1
TX_MODULE_TRX_1
C1607
1.2n
C1608
22p
2.7n
L1600
FL1600
ANT
RX
TX
GND1
GND2
GND3
GND6
GND5
GND4
6
1
3
2
4
5
9
8
7
C1609
100p

3.2.1.5 Checking RF signal path (TX Module)




TP2
U1100
TX module
TP4
TP3
TP5
TP6
TP1


C1101
2.2n
DNI
L1100
L1101
C1102 100p
TP1
RX
RX
+VPWR
14 : ESD Protection Circuit
942.5MHz
FL1101
OUT
IN
G3 G2 G1
C1110
100p
C1111
6.8n
L1111
1.2p
GND
CHANGE PART
TP2
TX_MODULE_TRX_1
TX_MODULE_TRX_4
SKY77573-31
U1100
SPDT_OUT
RX2
SPDT_IN
RX1
GND14
GND13
ANT
GND12
GND11
GND15
GND16
TRX1
GND17
TRX2
GND18
TRX3
GND19
TRX4
GND20
VBATT
NC1
GND10
GND9
GND8
GND7
GND6
GND5
GND4
VCC2
VCC1
GND3
GND2
GND1
NC2
SPDT
RF_IN
SW1
SW0
HB_EN
LB_EN
MODE0
MODE1
PGND
C1112
22p
C1113
4.7u
TP3
TP4
TP5
TP6
SPDT_SEL
WTR_TX1_OUT
R1104
0
R1105
R1106
DNI
ANT0_SW_SEL_0
ANT0_SW_SEL_1
C1123
C1124
22p
C1119
C1120
C1121
C1122
100p
TE1 : SWAP the RX Filter
G850
EAM62732801
G900
EAM62732901

TP1

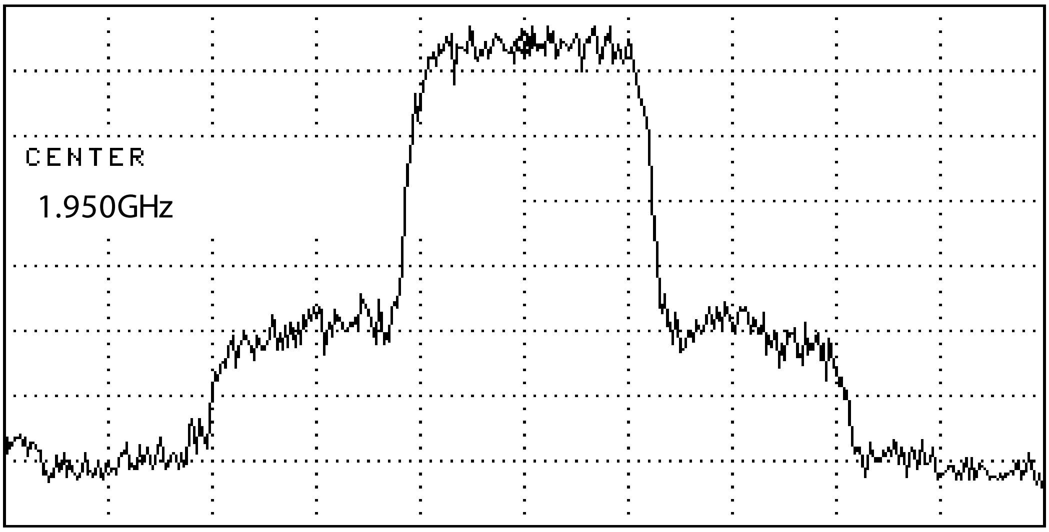


Figure 3.2.1 (e)

TP2

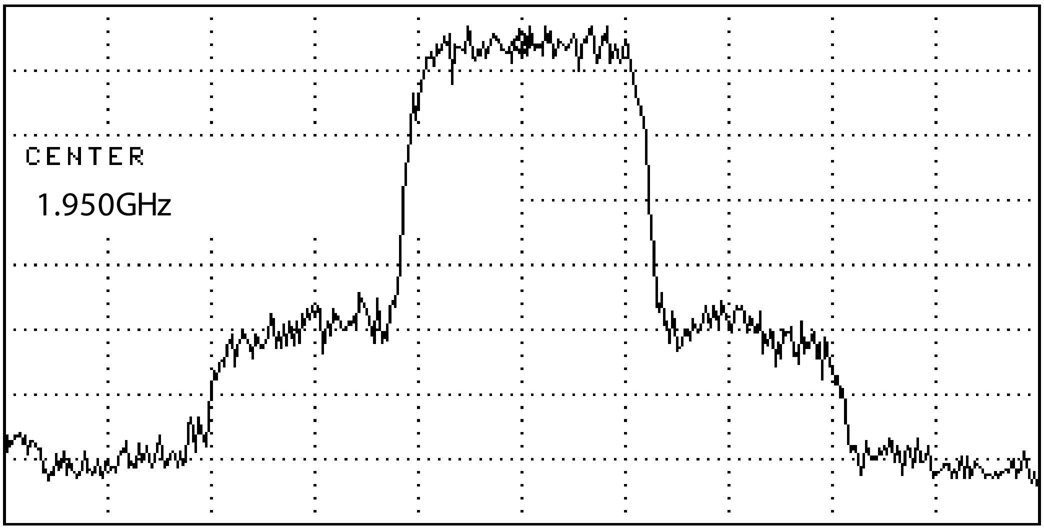
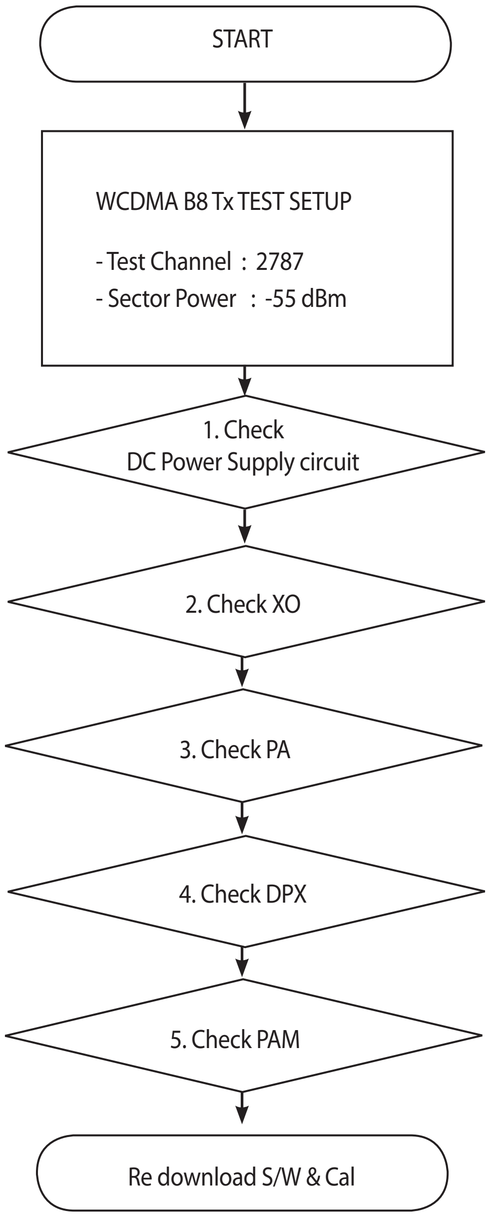


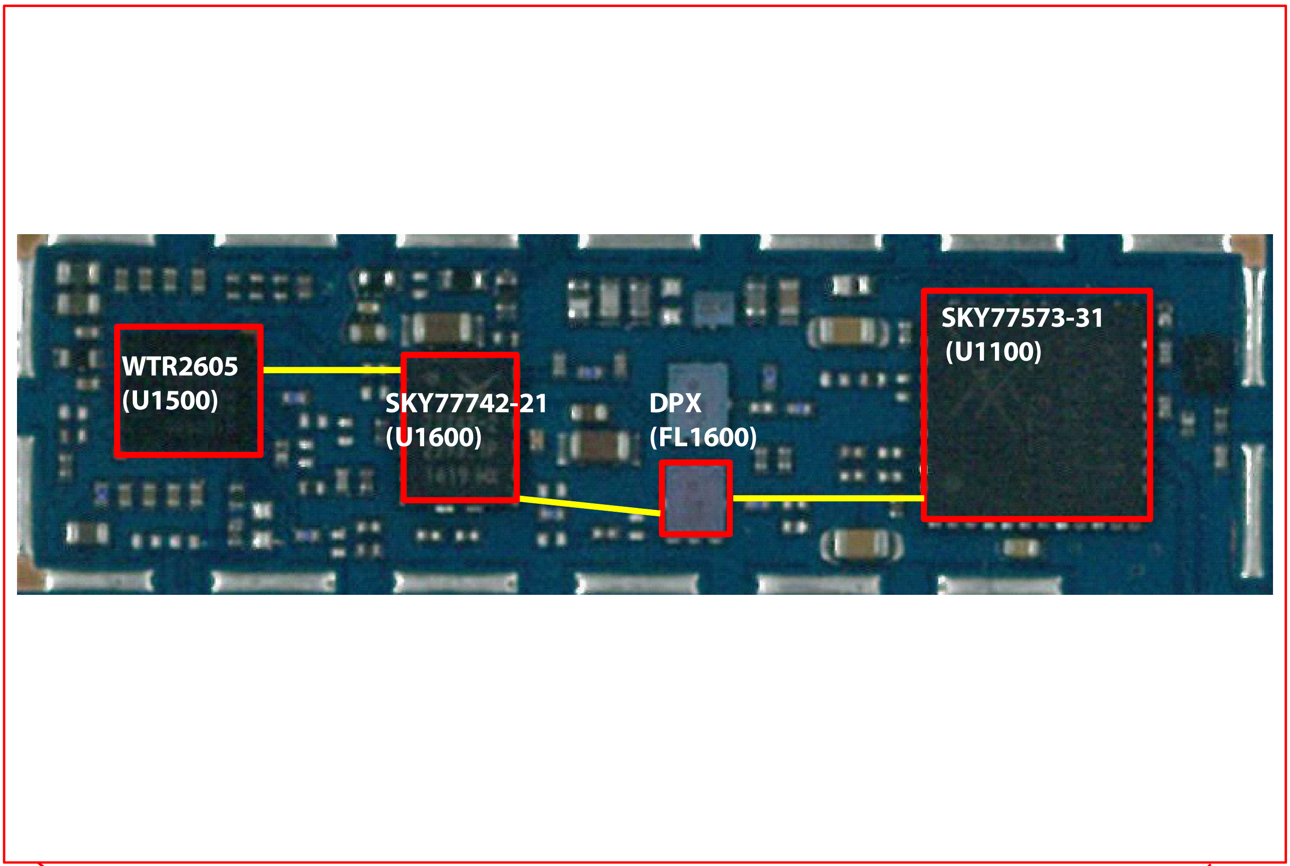
Figure 3.2.1 (f)

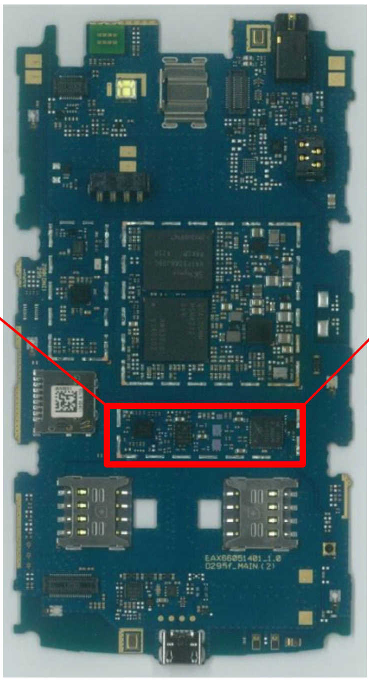
3.3 WCDMA Tx Part Trouble

3.3.1 WCDMA_B8 Tx

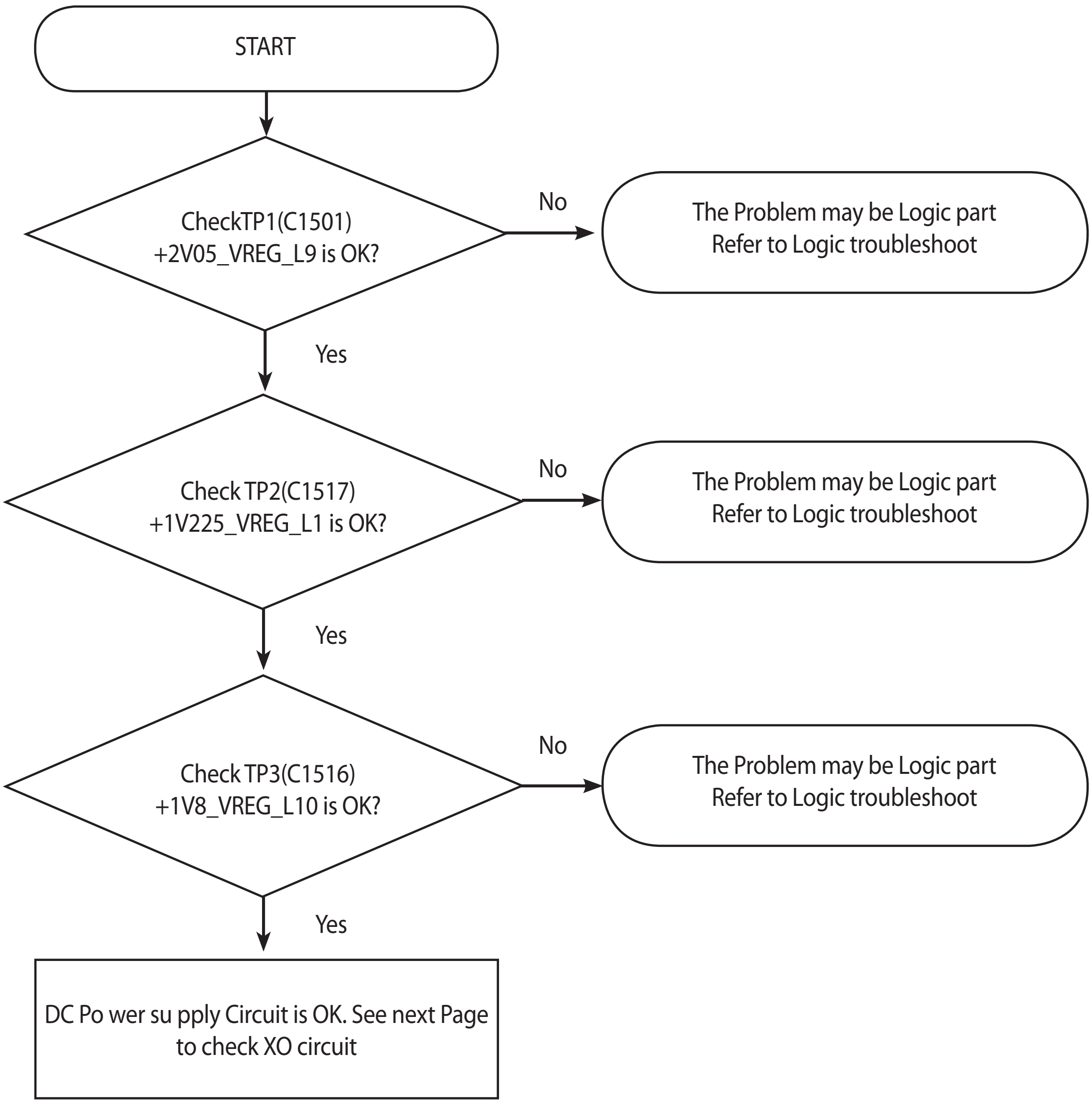


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form


WTR2605
(U1500)
SKY77742-21
(U1600)
DPX
(FL1600)
SKY77573-31
(U1100)

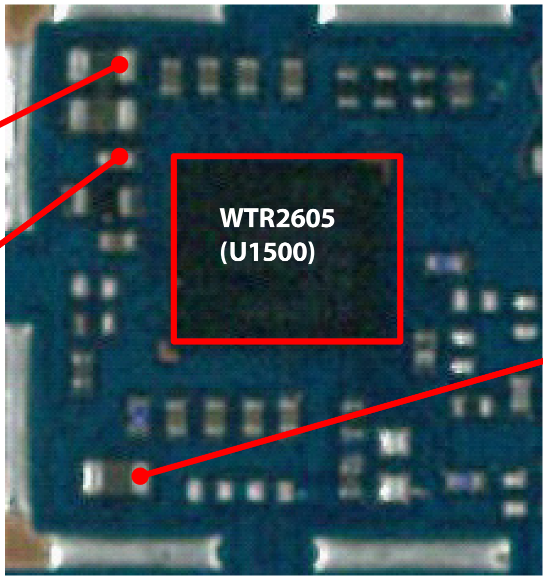


3.3.1.1 Checking DC Power supply circuit

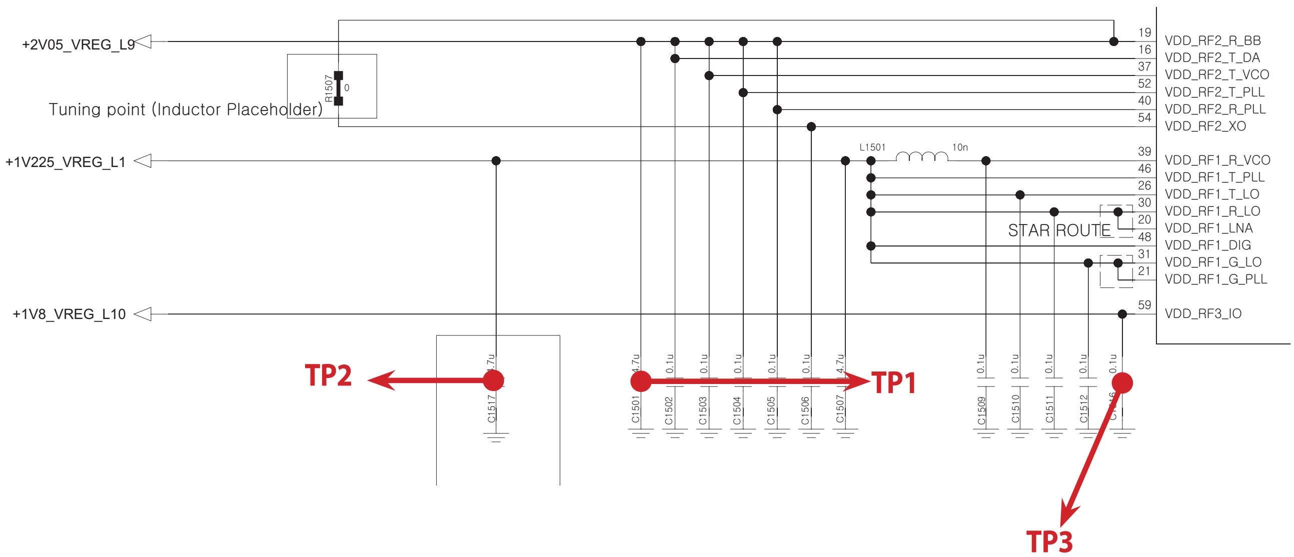


TP1 : C1501
(+2V05_VREG_L9)

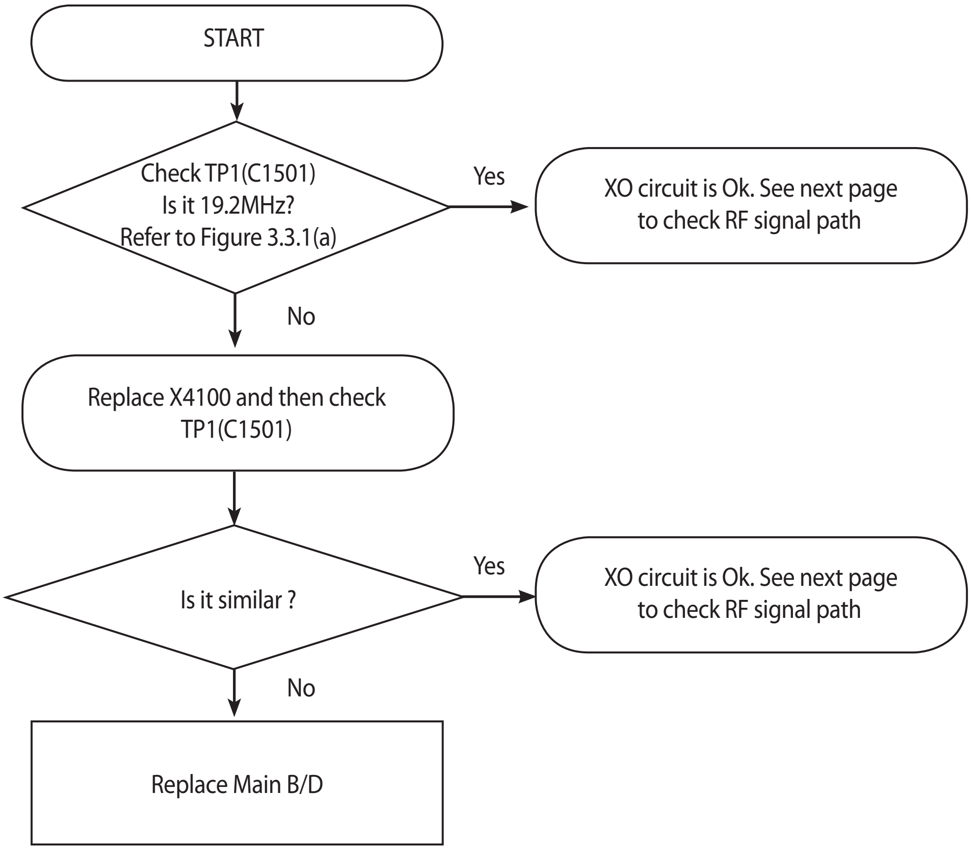
TP3: C1516
(+1V8_VREG_L10)



TP2: C1517
(+1V225_VREG_L1)



3.3.1.2 Checking XO circuit



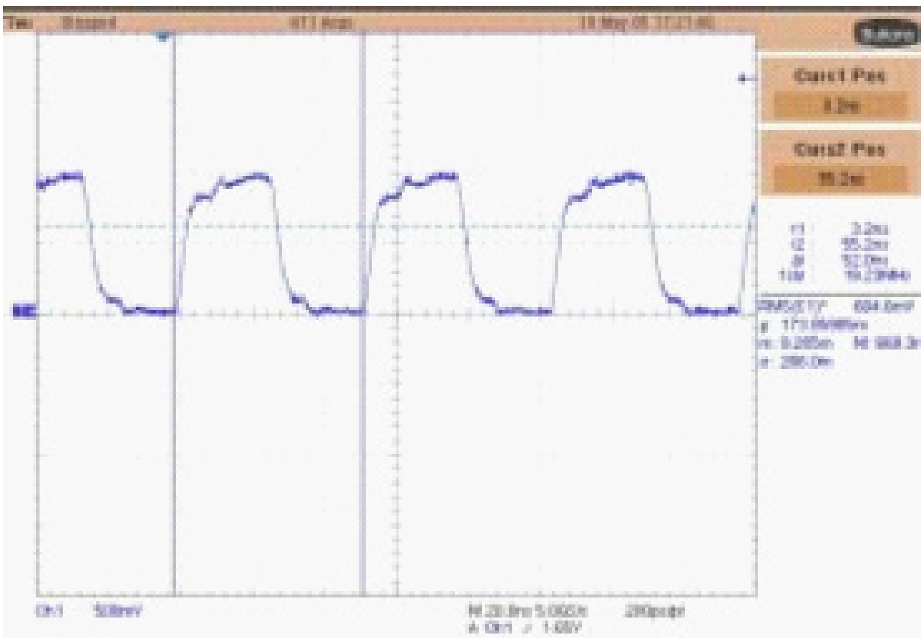
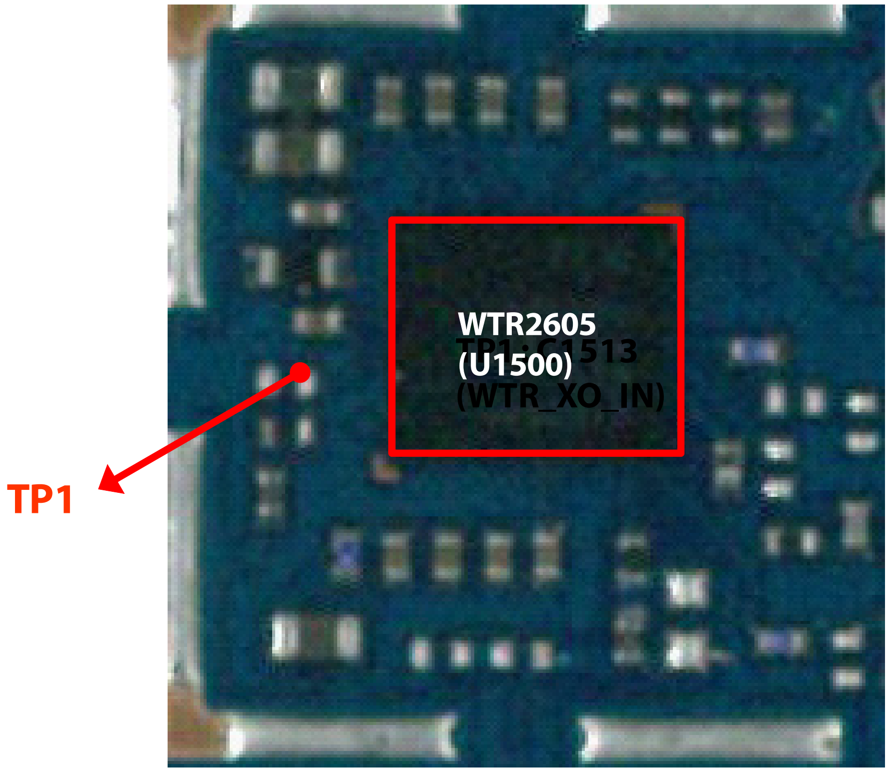
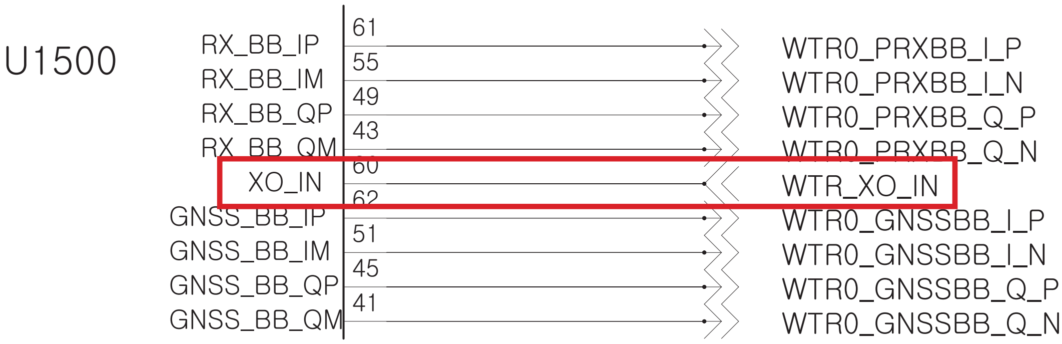
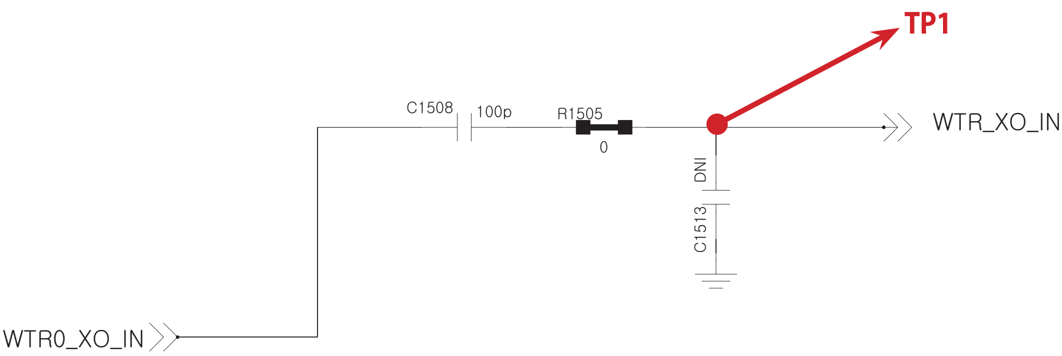


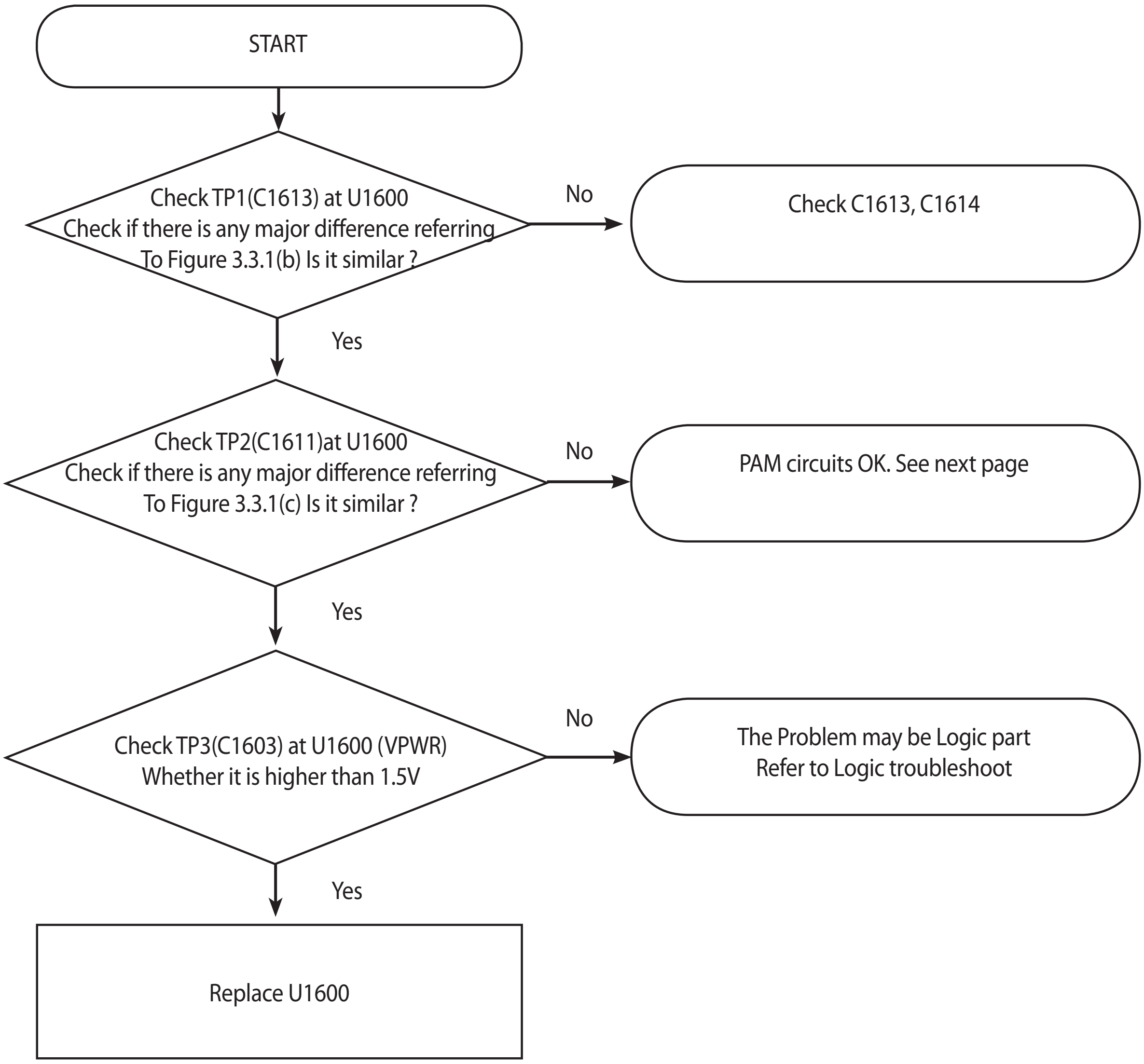
Figure 3.3.1 (a)

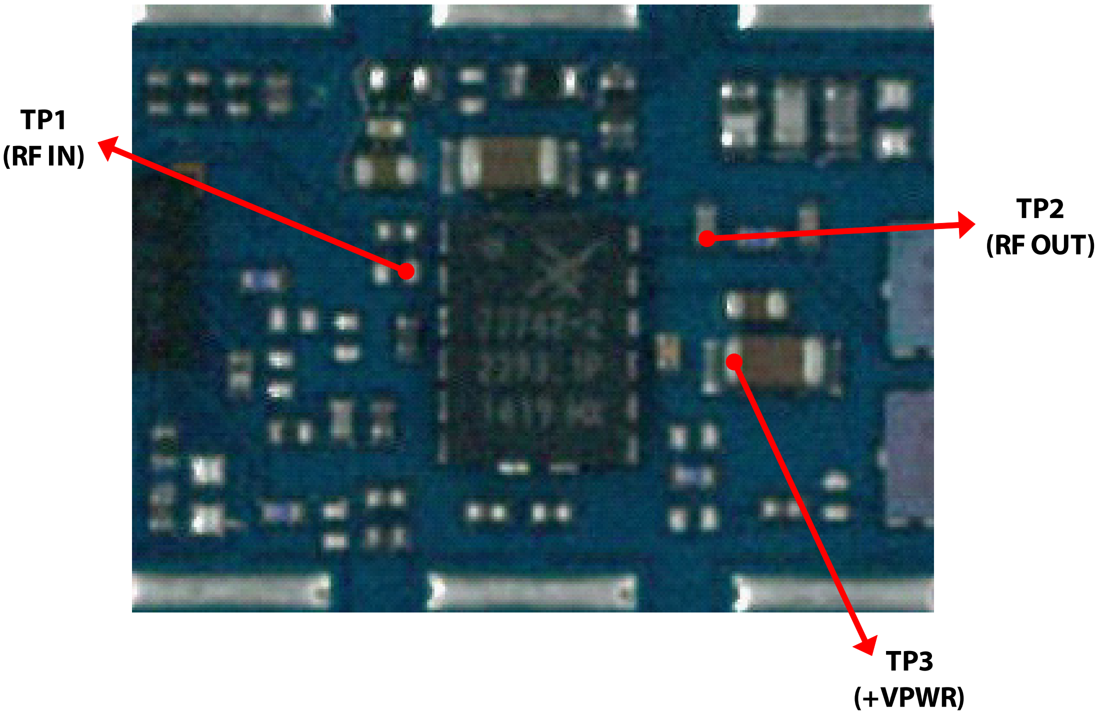

WTR2605
(U1500)
(WTR_XO_IN)
TP1

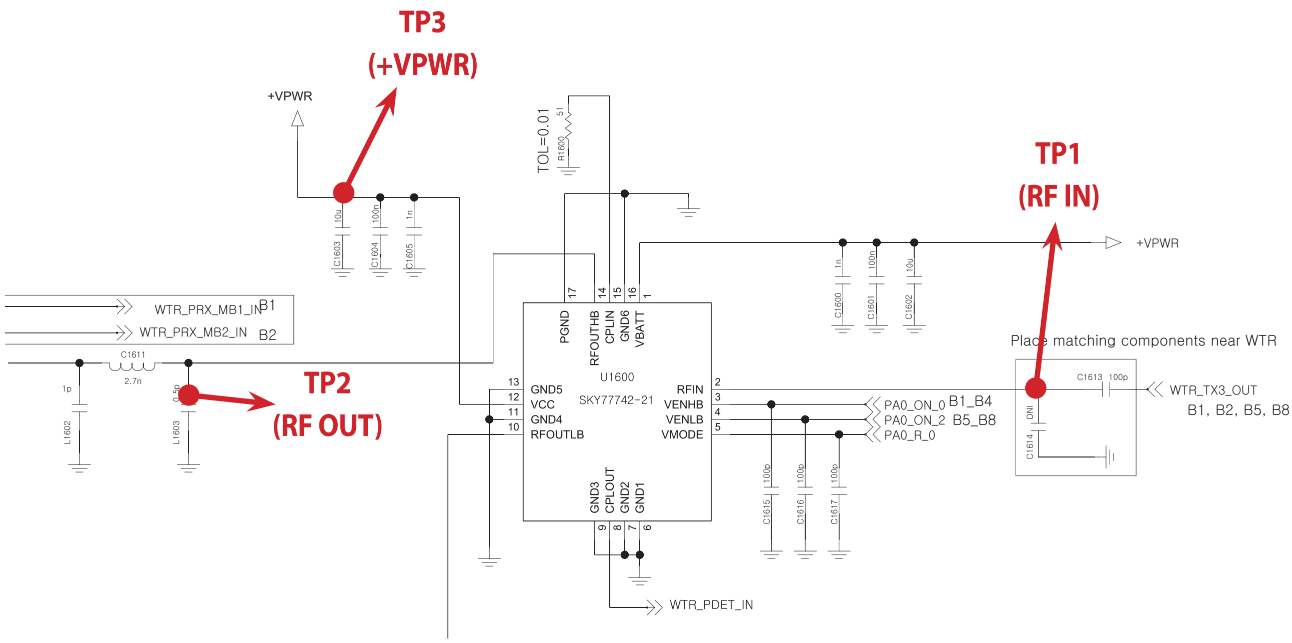

U1500
RX_BB_IP
RX_BB_IM
RX_BB_QP
RX_BB_QM
XO_IN
GNSS_BB_IP
GNSS_BB_IM
GNSS_BB_QP
GNSS_BB_QM
61
55
49
43
60
62
51
45
41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

3.3.1.3 Checking B8 TX PA




TP1
(RF IN)
TP2
(RF OUT)
TP3
(+VPWR)


TP3
(+VPWR)
+VPWR
TOL=0.01
TP1
(RF IN)
+VPWR
WTR_PRX_MB1_IN B1
WTR_PRX_MB2_IN B2
TP2
(RF OUT)
U1600
SKY77742-21
PGND
RFOUTHB
CPLIN
GND6
VBATT
GND5
VCC
GND4
RFOUTLB
RFIN
VENHB
VENLB
VMODE
GND3
CPLOUT
GND2
GND1
PA0_ON_0 B1_B4
PA0_ON_2 B5_B8
PA0_R_0
Place matching components near WTR
WTR_TX3_OUT
B1, B2, B5, B8
WTR_PDET_IN

TP1

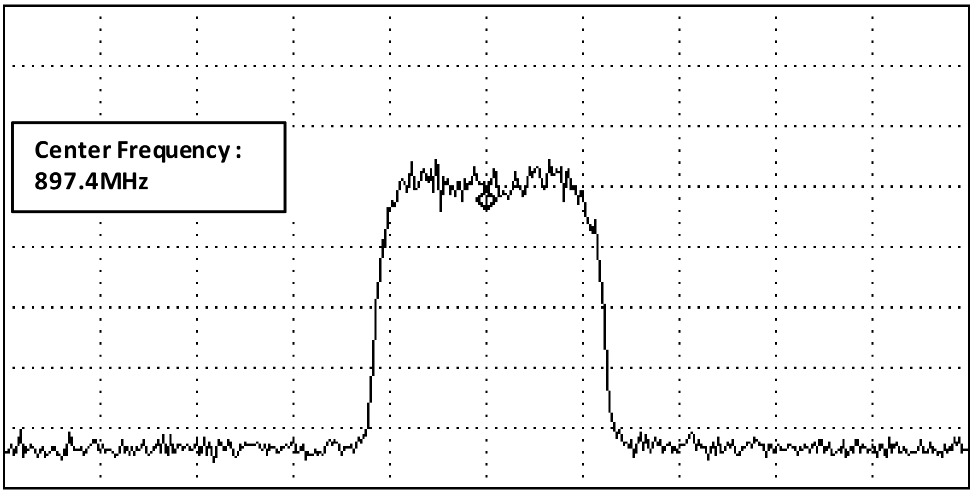


Figure 3.3.1 (b)

TP2

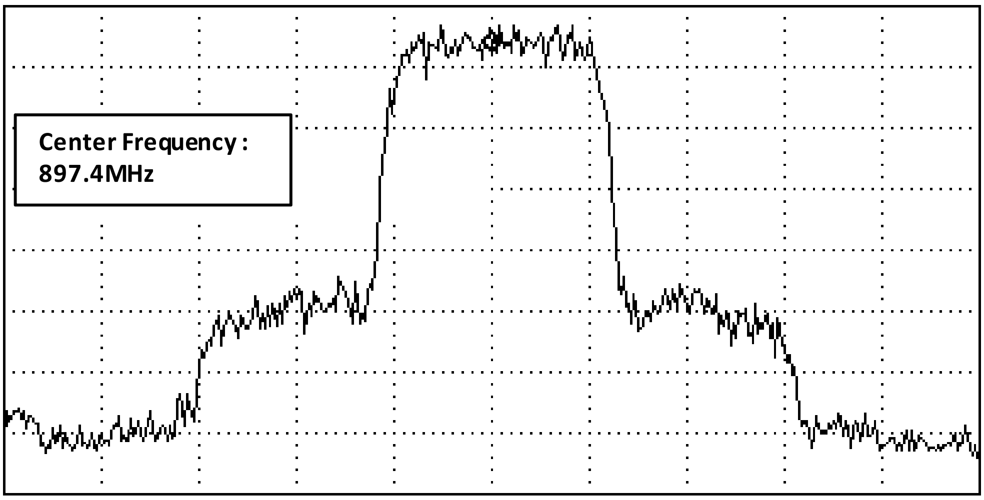
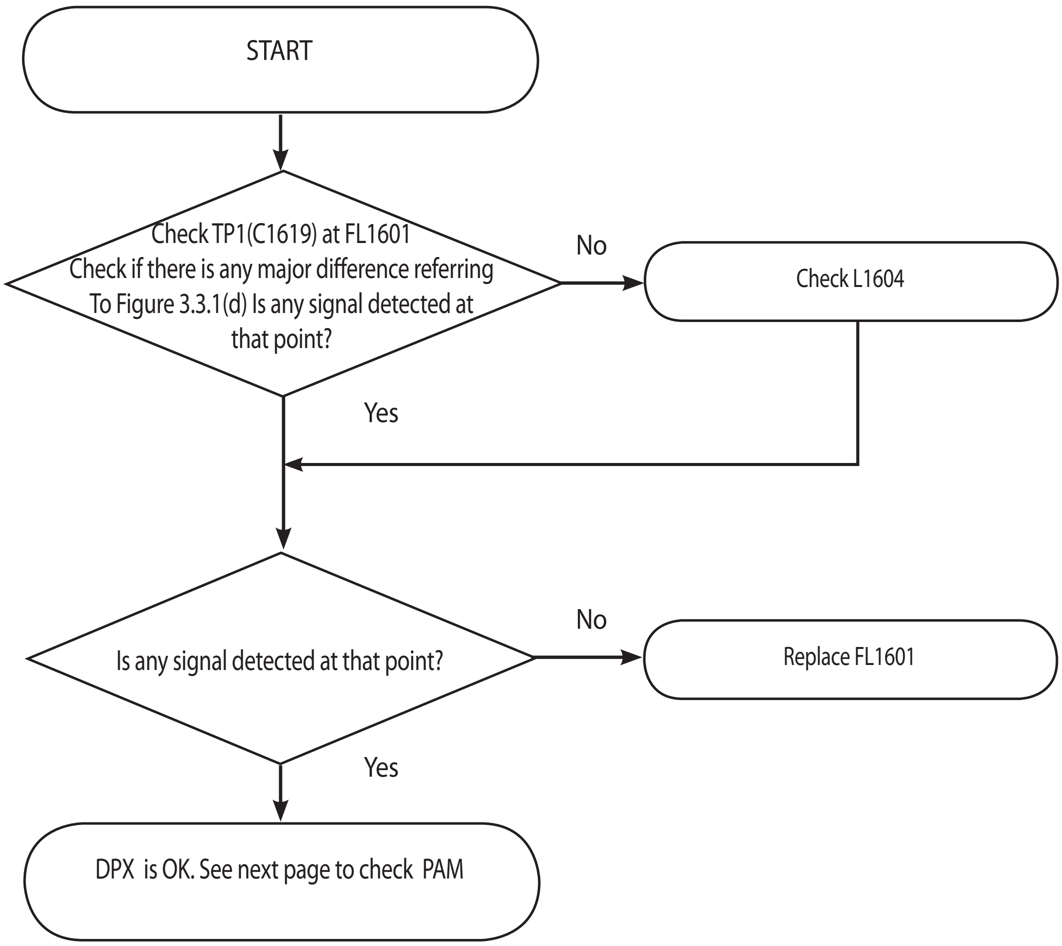


Figure 3.3.1 (C)

3.3.1.4 Checking RF signal path (DPX)



TP1

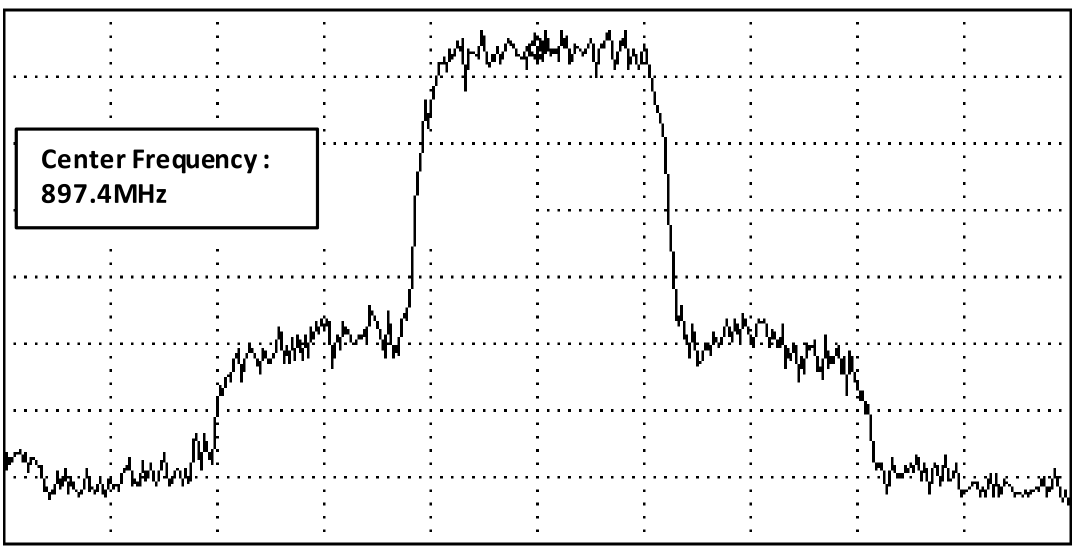
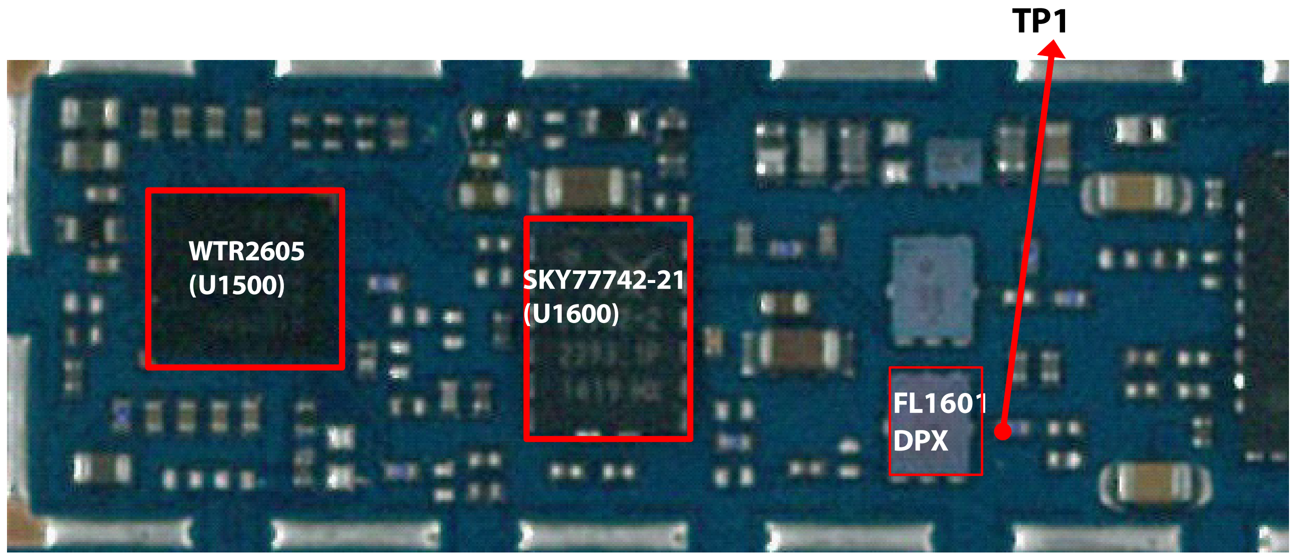
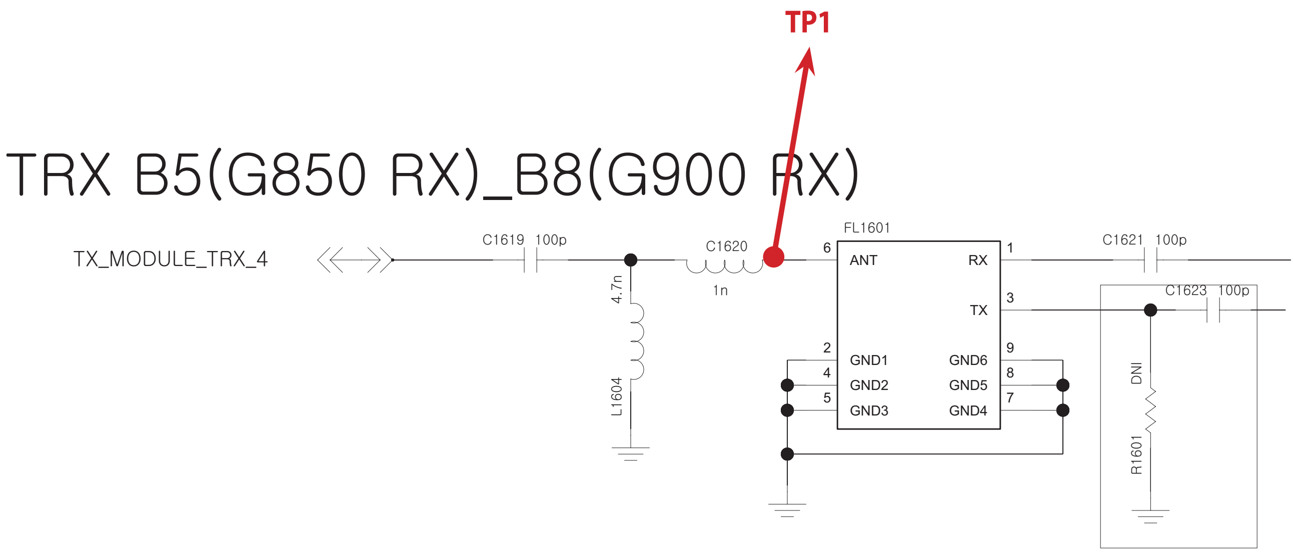
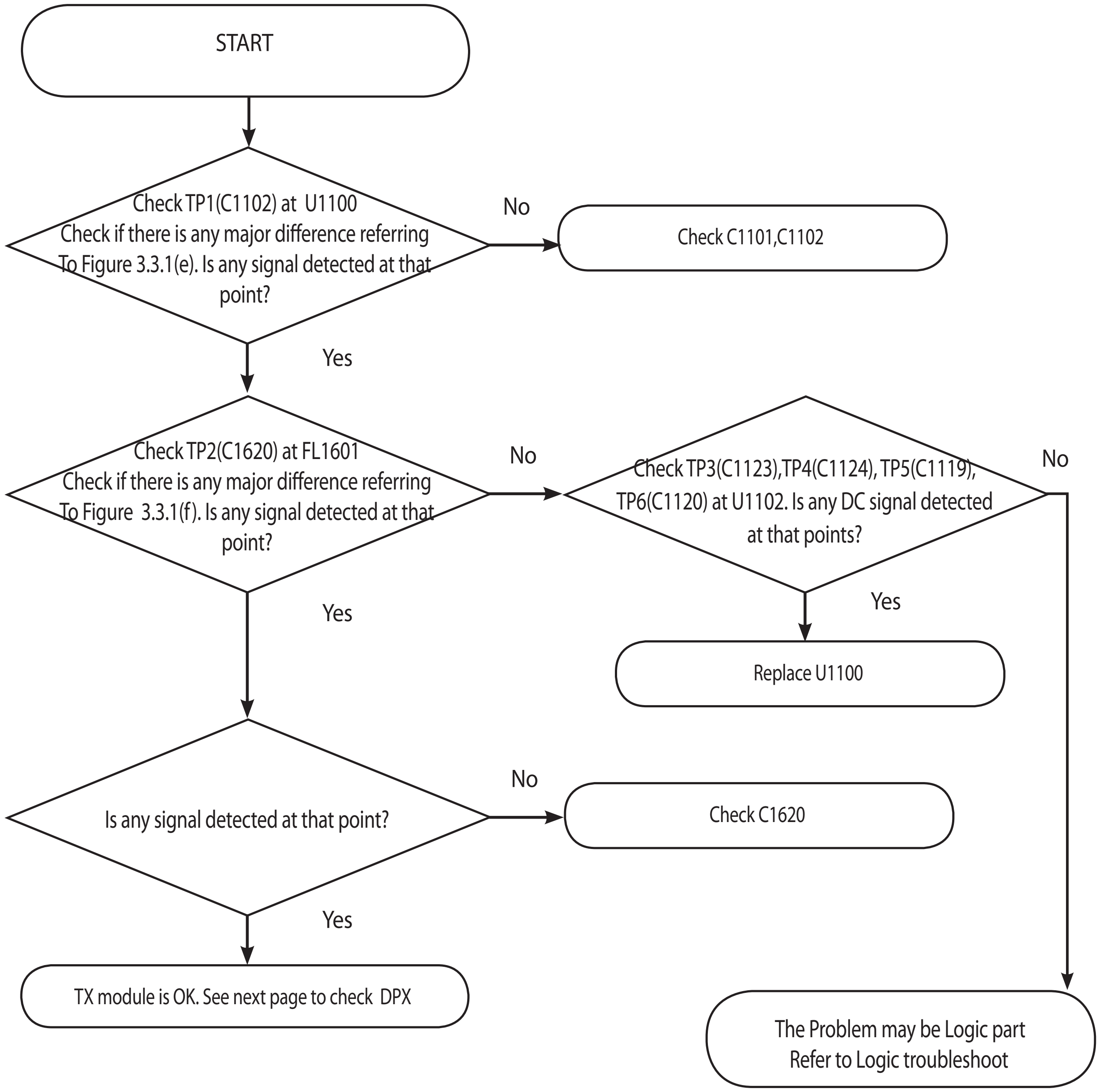


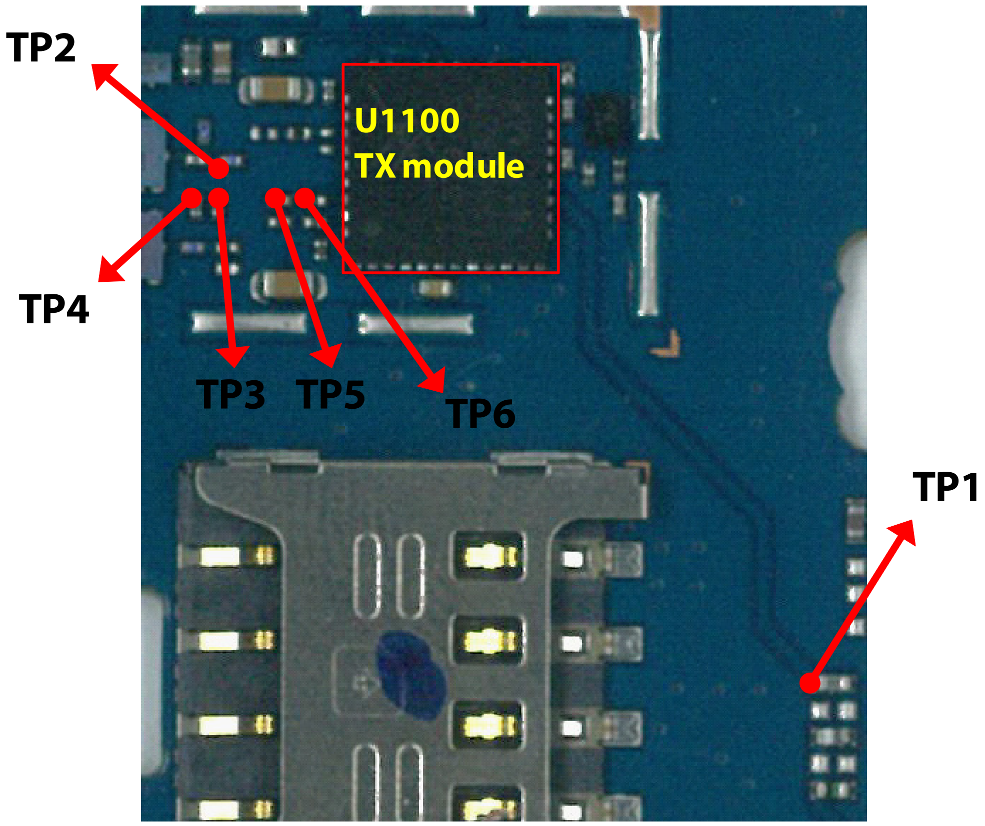
Figure 3.3.1 (d)

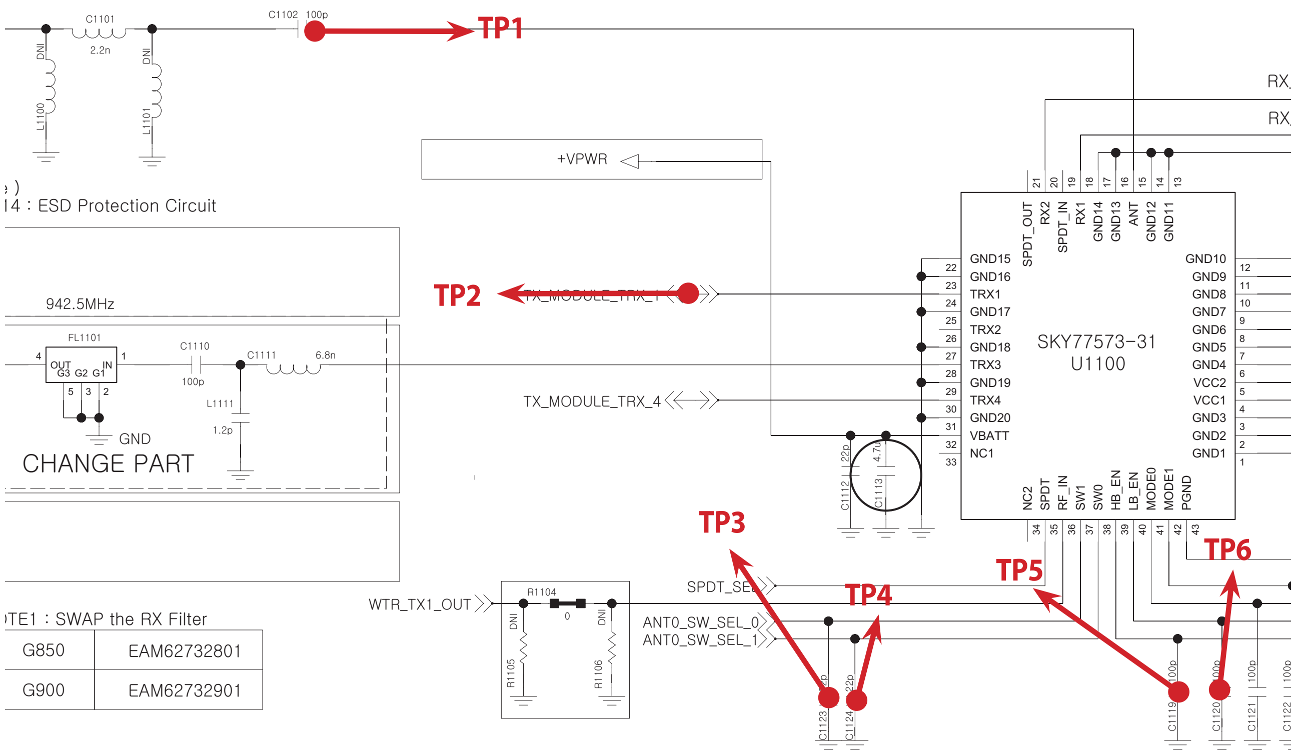

TP1
WTR2605
(U1500)
SKY77742-21
(U1600)
FL1601
DPX


TP1
TRX B5(G850 RX)_B8(G900 RX)
TX_MODULE_TRX_4
C1619 100p
C1620
1n
4.7n
L1604
FL1601
6
ANT
RX
1
TX
3
2
GND1
GND6
9
4
GND2
GND5
8
5
GND3
GND4
7
C1621 100p
C1623 100p
DNI
R1601

3.3.1.5 Checking RF signal path (TX Module)







TE1 : SWAP the RX Filter

G850	EAM62732801
G900	EAM62732901

TP1

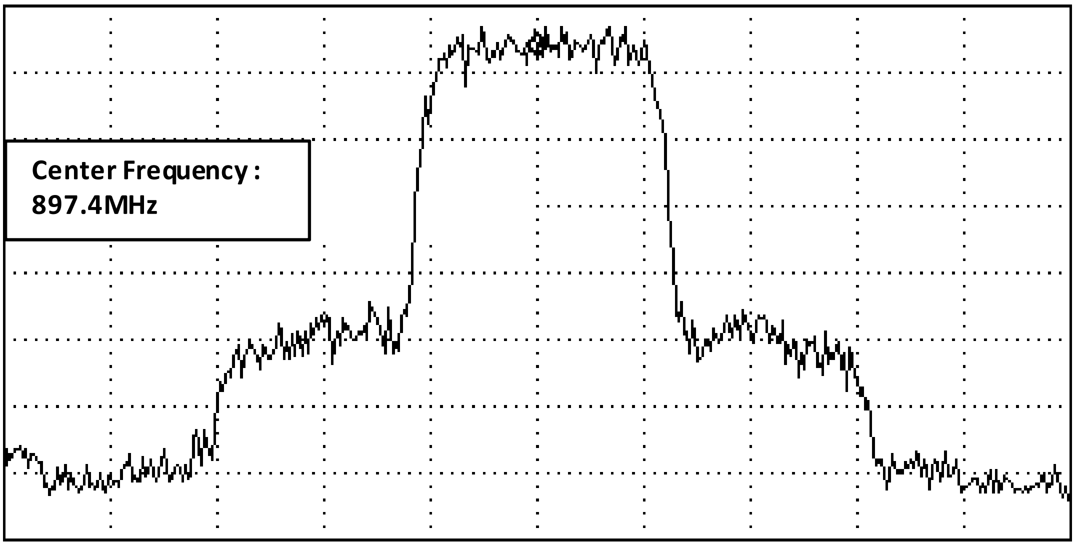


Figure 3.3.1 (e)

TP2

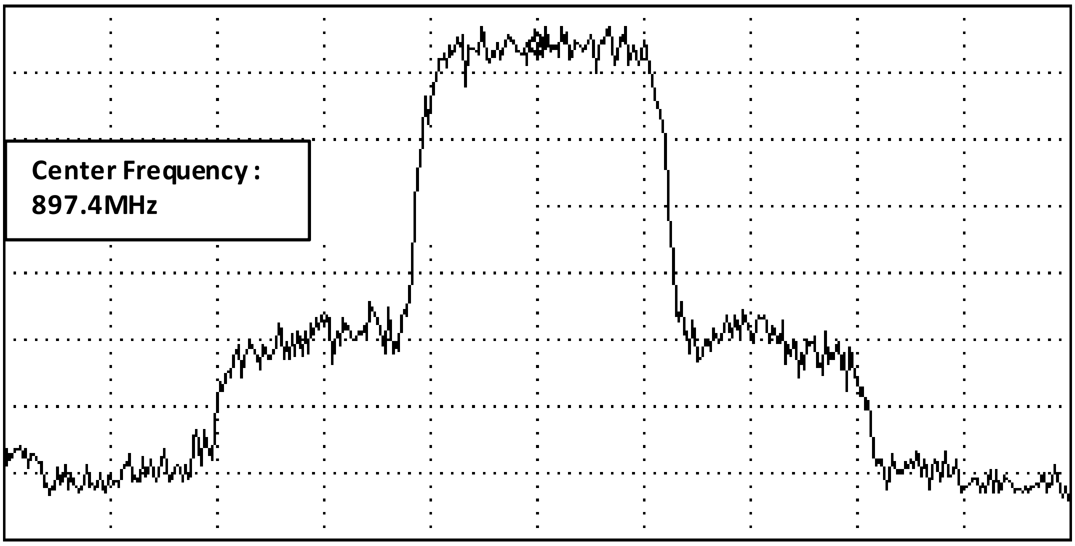
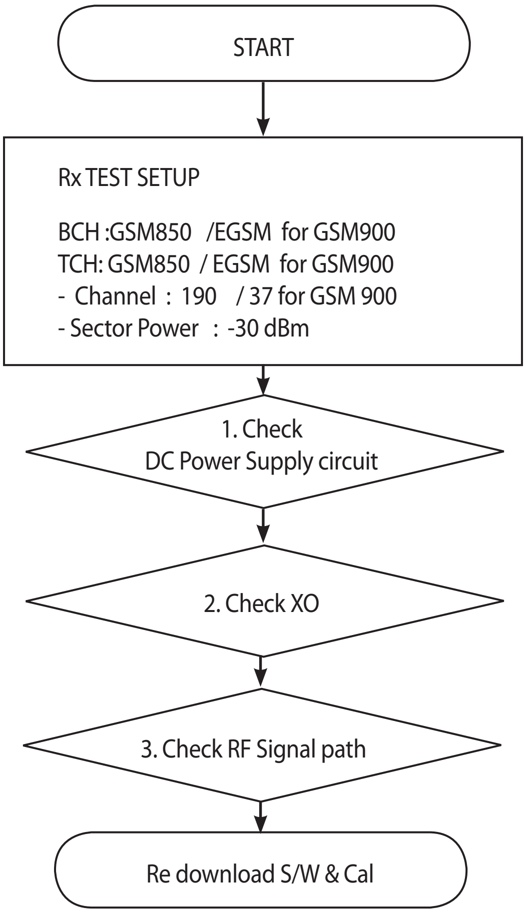


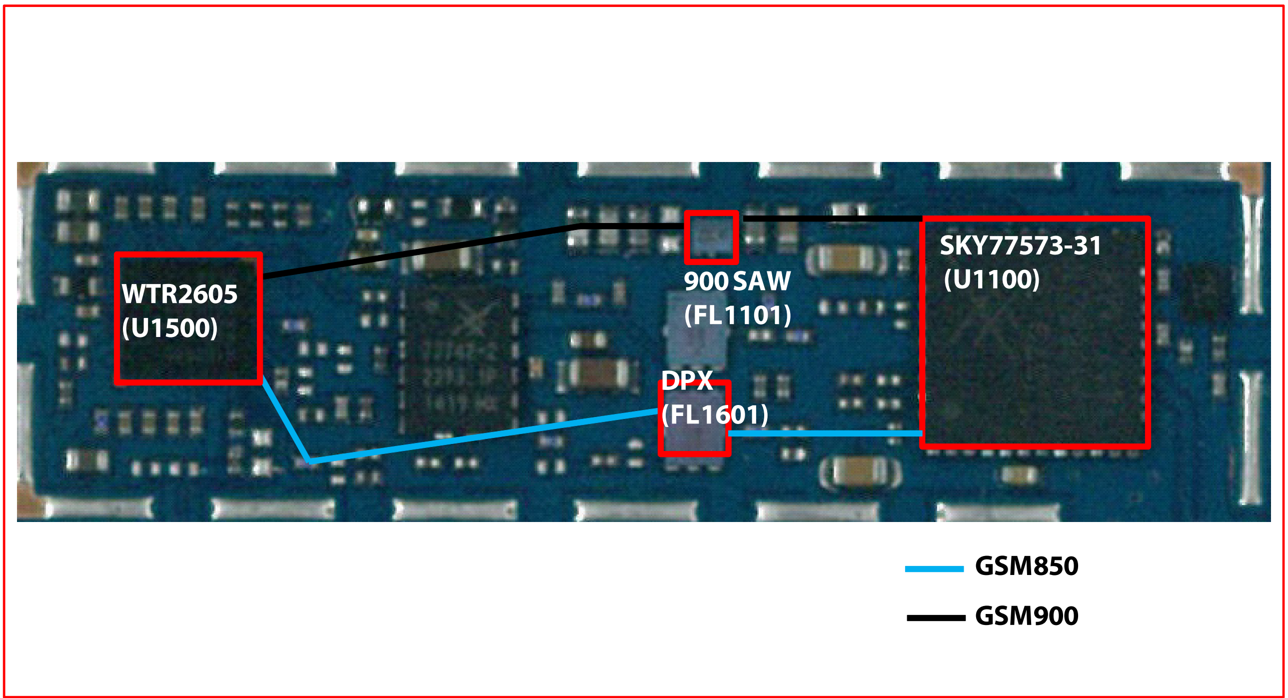
Figure 3.3.1 (f)

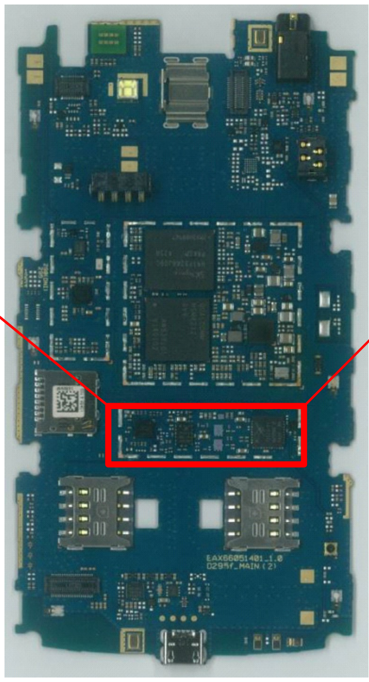
3.4 GSM Rx Part Troubleshooting

3.4.1 GSM850/900 Rx

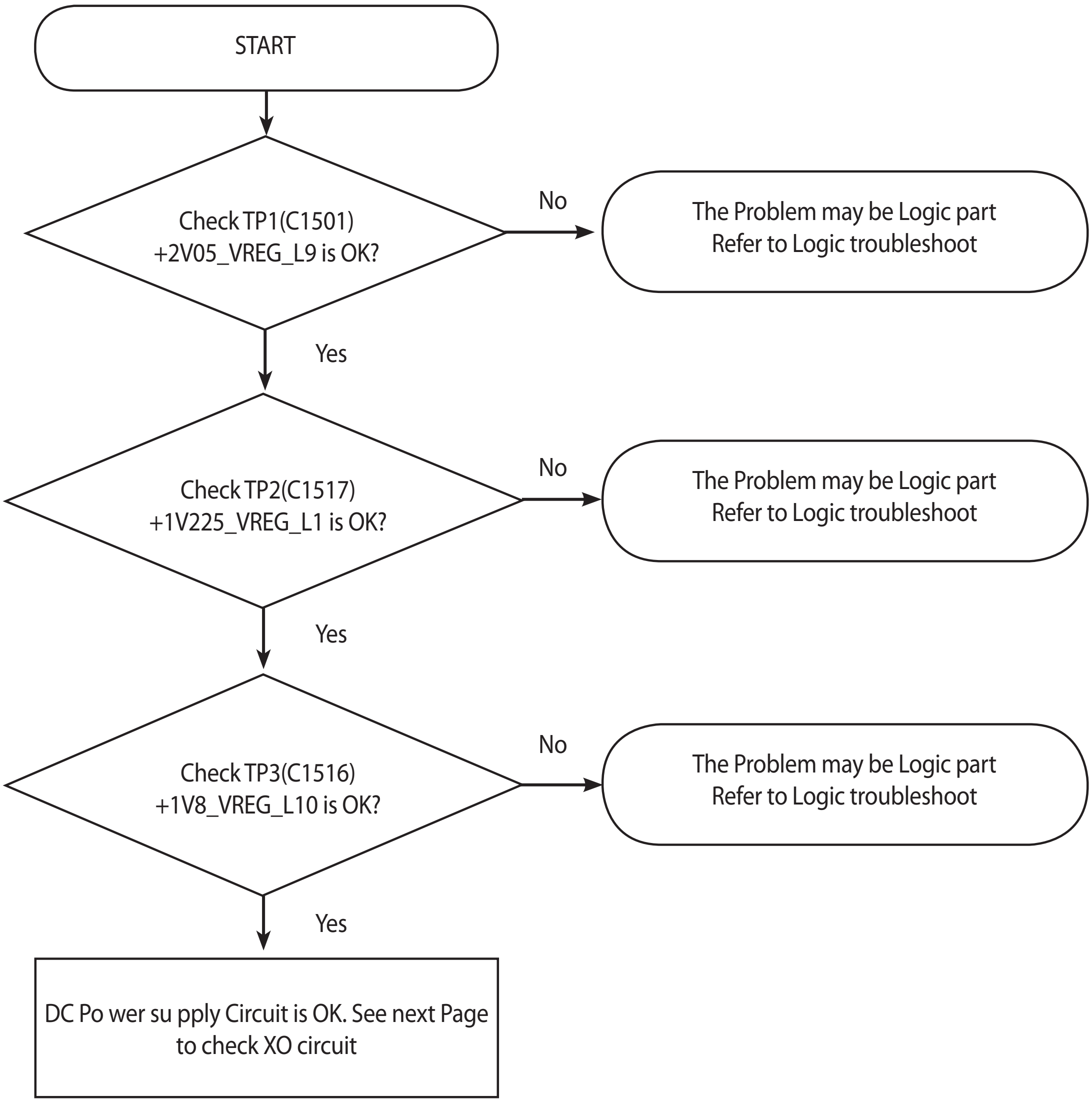


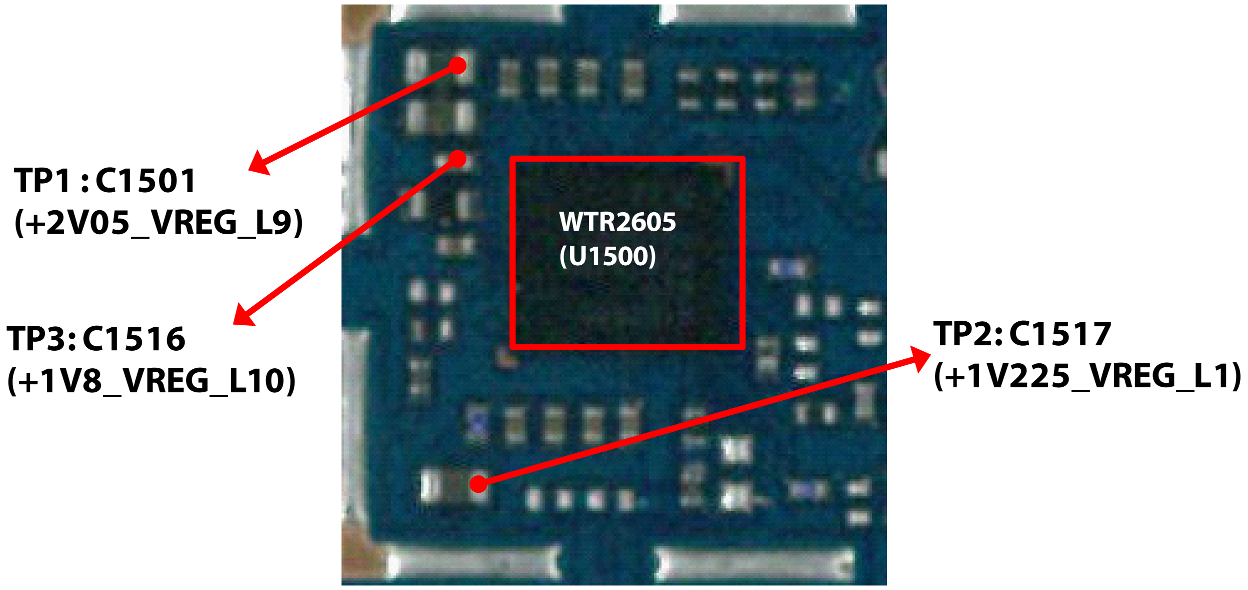
* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

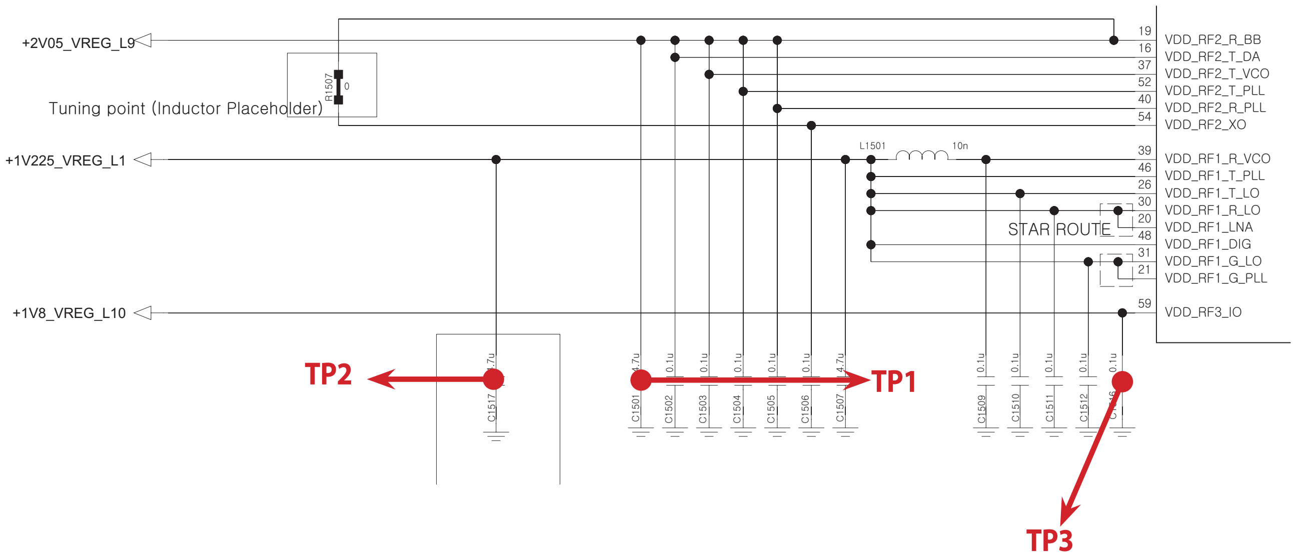

WTR2605
(U1500)
900 SAW
(FL1101)
DPX
(FL1601)
SKY77573-31
(U1100)
GSM850
GSM900



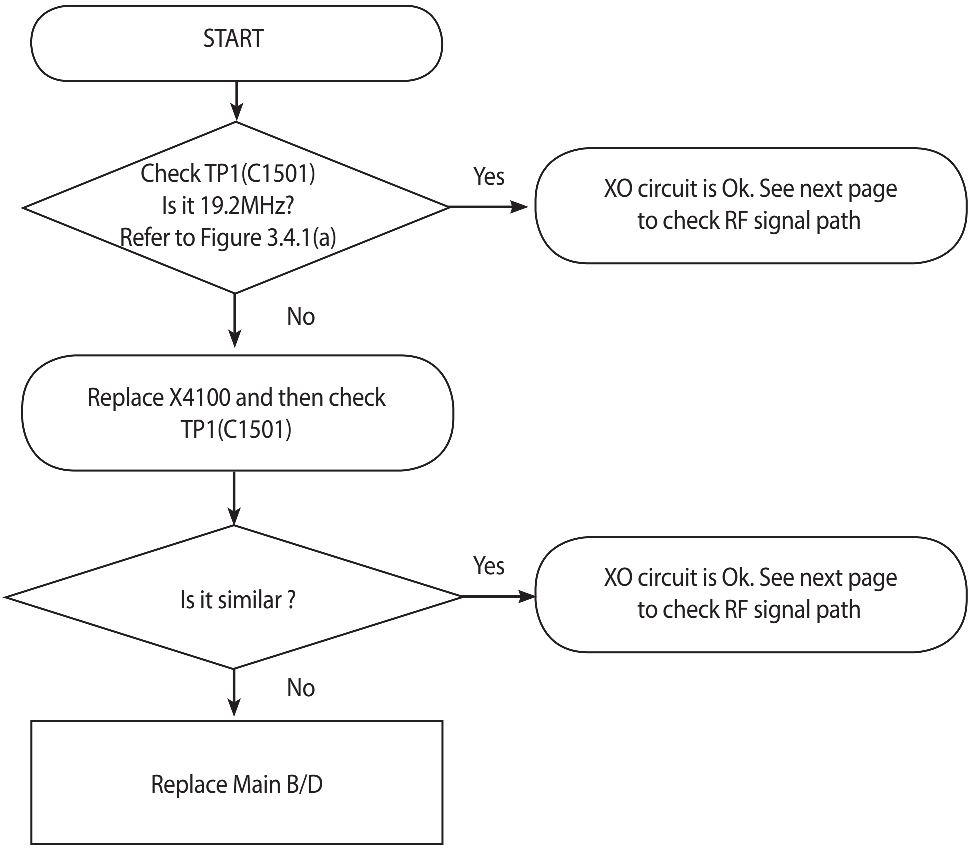
3.4.1.1 Checking DC Power supply circuit




TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)


+2V05_VREG_L9
Tuning point (Inductor Placeholder)
R1507
+1V225_VREG_L1
+1V8_VREG_L10
L1501
10n
STAR ROUTE
VDD_RF2_R_BB
VDD_RF2_T_DA
VDD_RF2_T_VCO
VDD_RF2_T_PLL
VDD_RF2_R_PLL
VDD_RF2_XO
VDD_RF1_R_VCO
VDD_RF1_T_PLL
VDD_RF1_T_LO
VDD_RF1_R_LO
VDD_RF1_LNA
VDD_RF1_DIG
VDD_RF1_G_LO
VDD_RF1_G_PLL
VDD_RF3_IO
TP2
TP1
TP3

3.4.1.2 Checking XO circuit



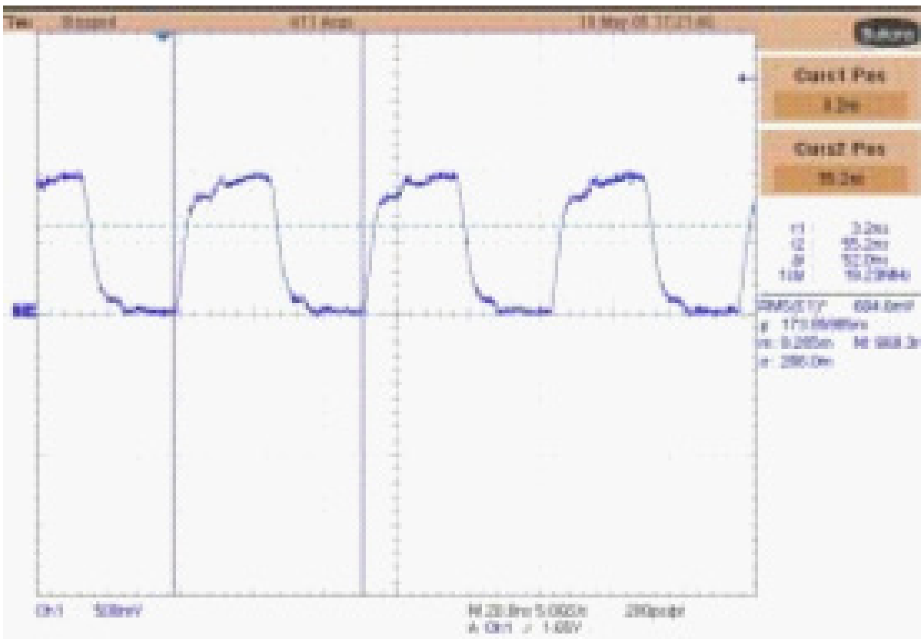
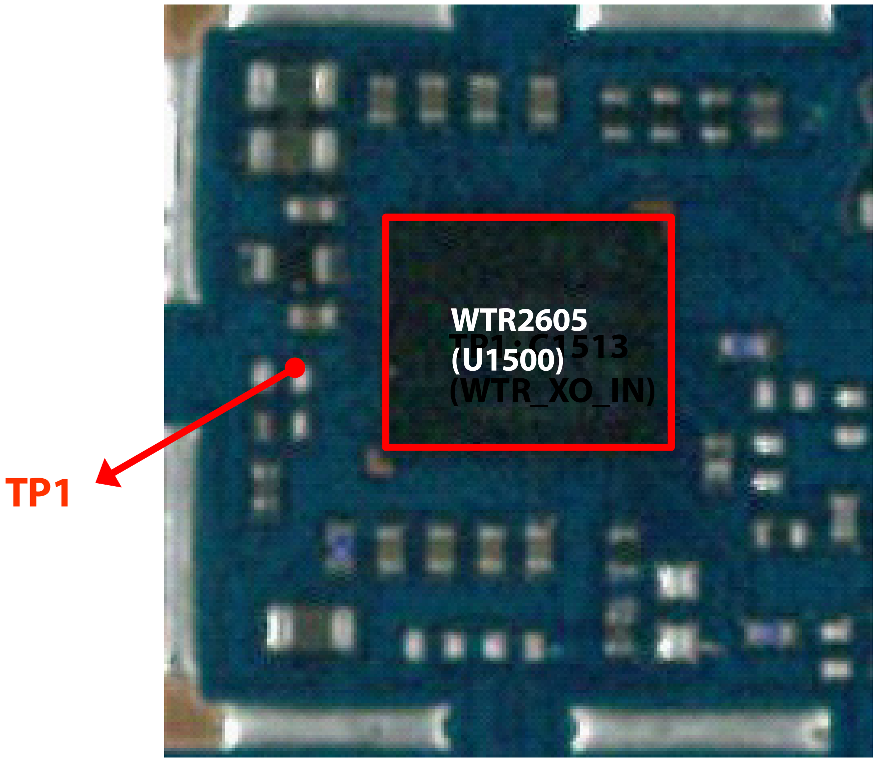
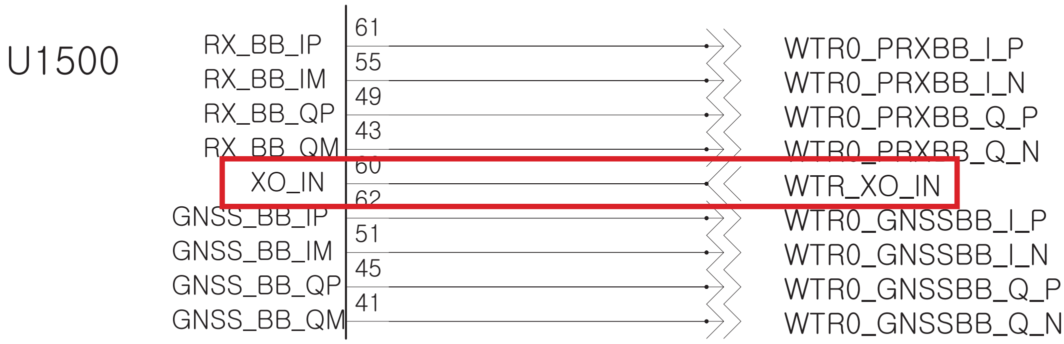
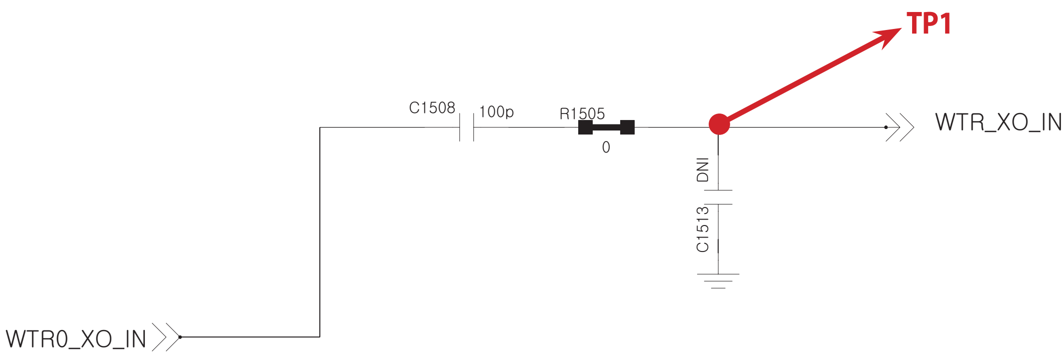


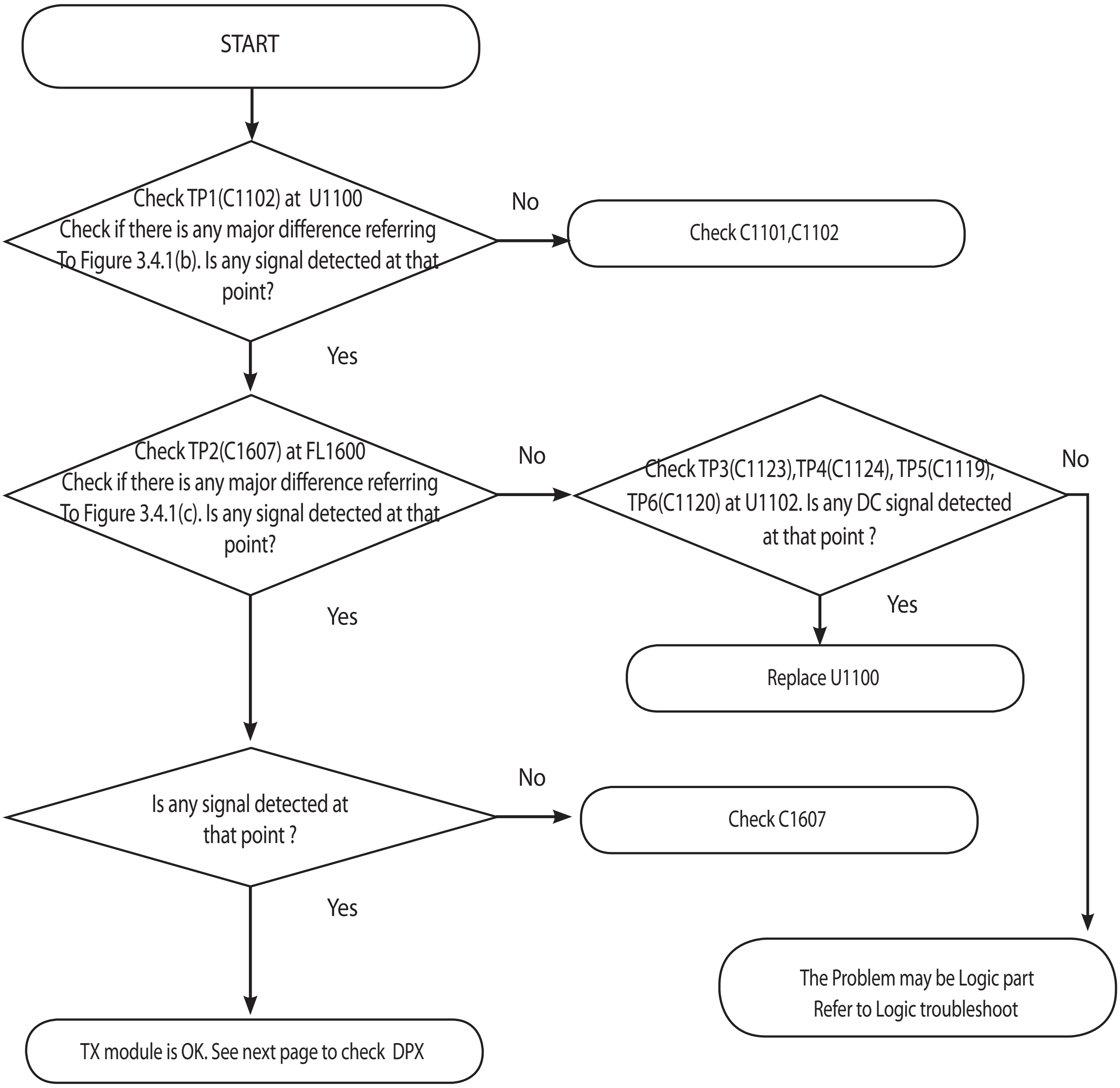
Figure 3.4.1 (a)

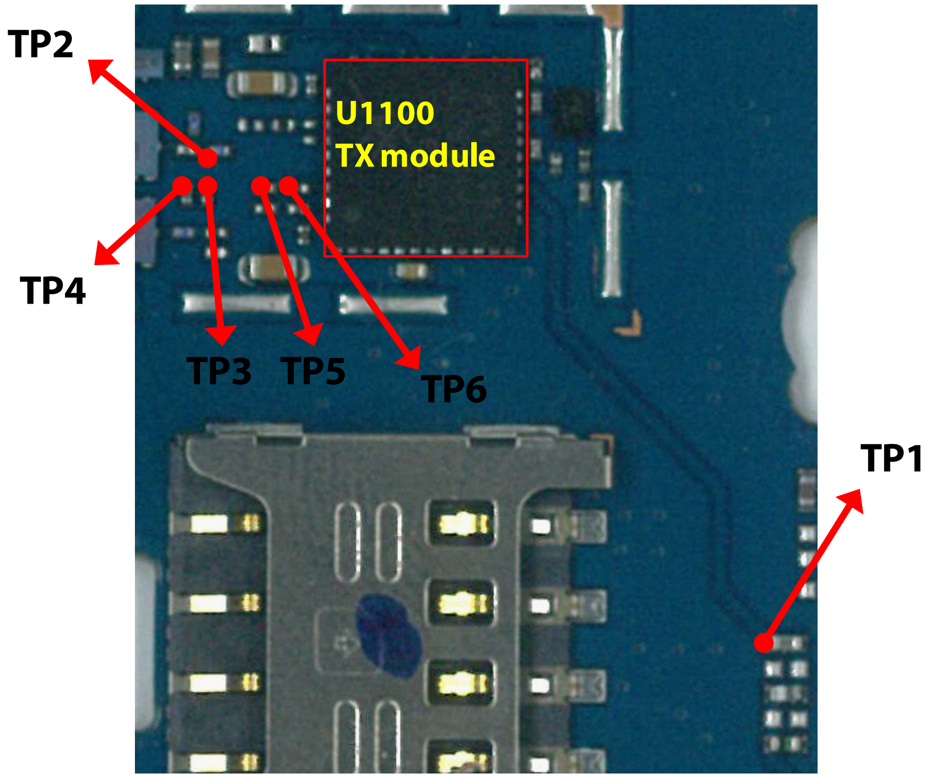

WTR2605
(U1500)
TP1 : C1513
(WTR_XO_IN)
TP1

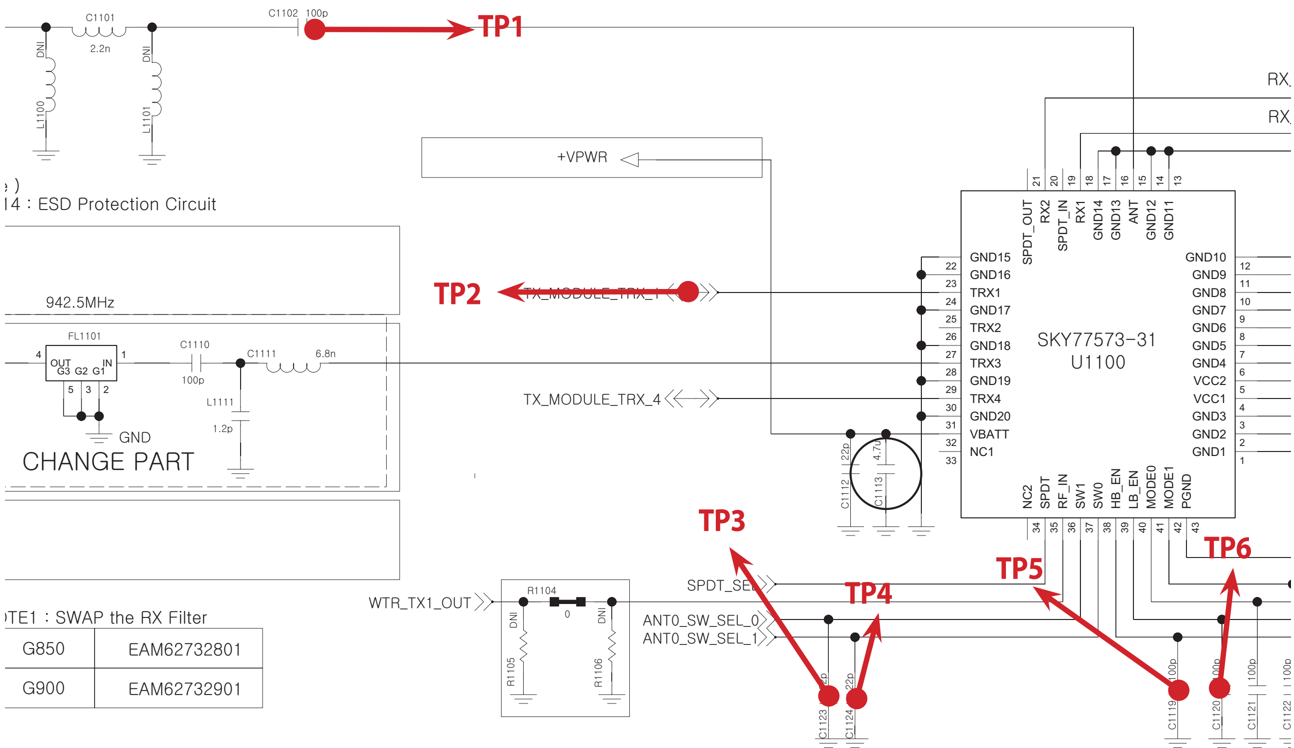

U1500
RX_BB_IP
RX_BB_IM
RX_BB_QP
RX_BB_QM
XO_IN
GNSS_BB_IP
GNSS_BB_IM
GNSS_BB_QP
GNSS_BB_QM
61
55
49
43
60
62
51
45
41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
WTR_XO_IN
DNI
C1513
WTR0_XO_IN

3.4.1.3 Checking RF signal path (TX Module)







G850	EAM62732801
G900	EAM62732901

TP1

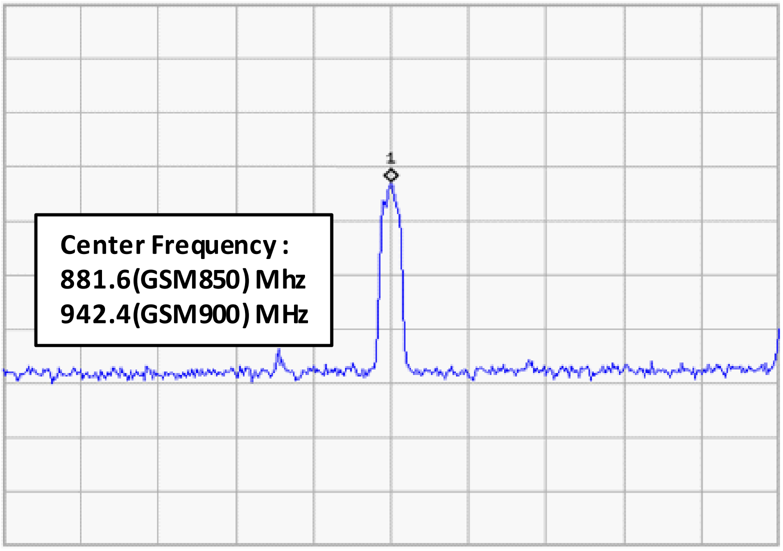


Figure 3.4.1 (b)

TP2

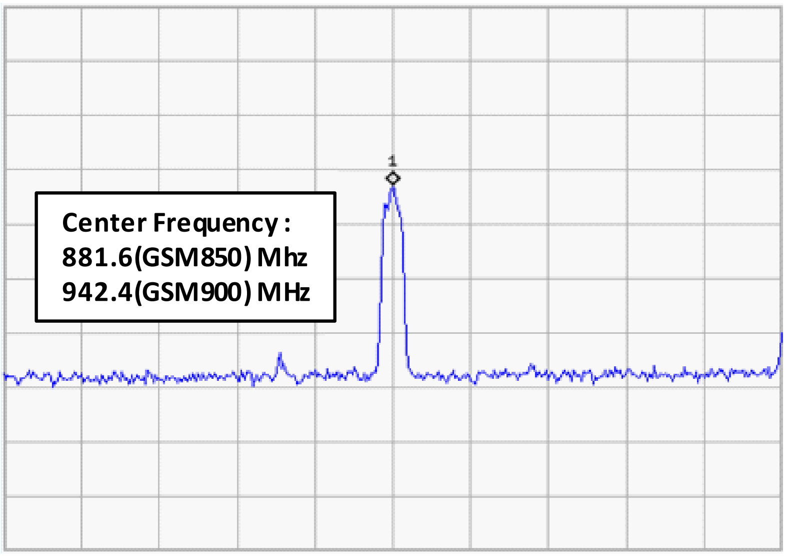
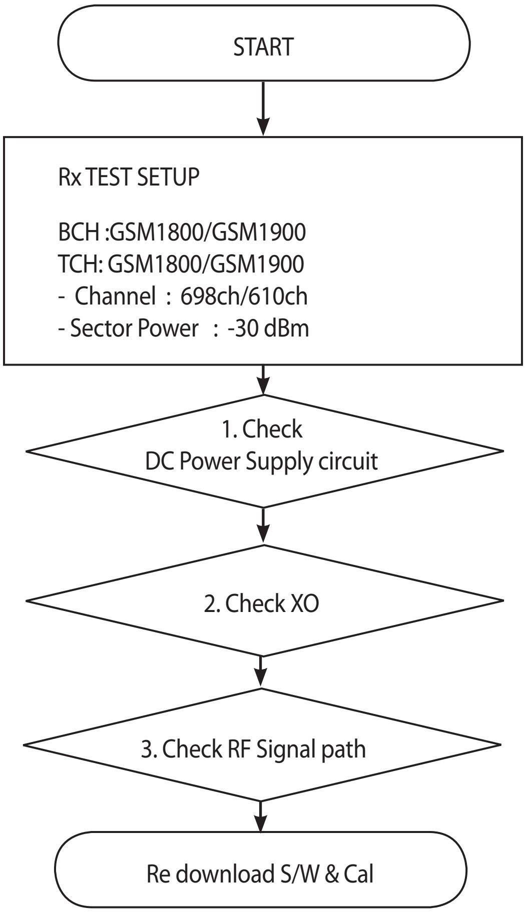


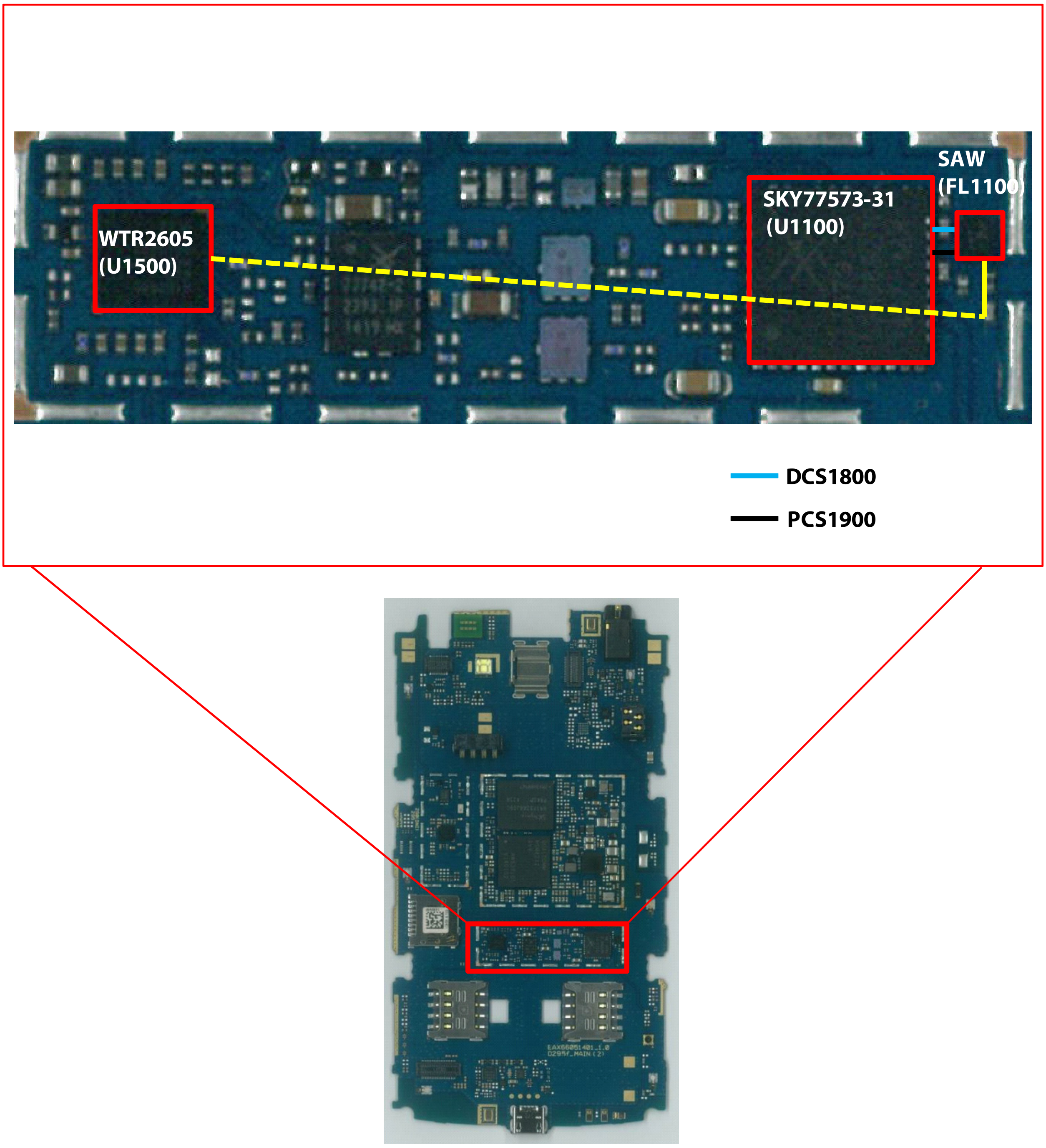
Figure 3.4.1 (C)

3.5 GSM Rx Part Troubleshooting

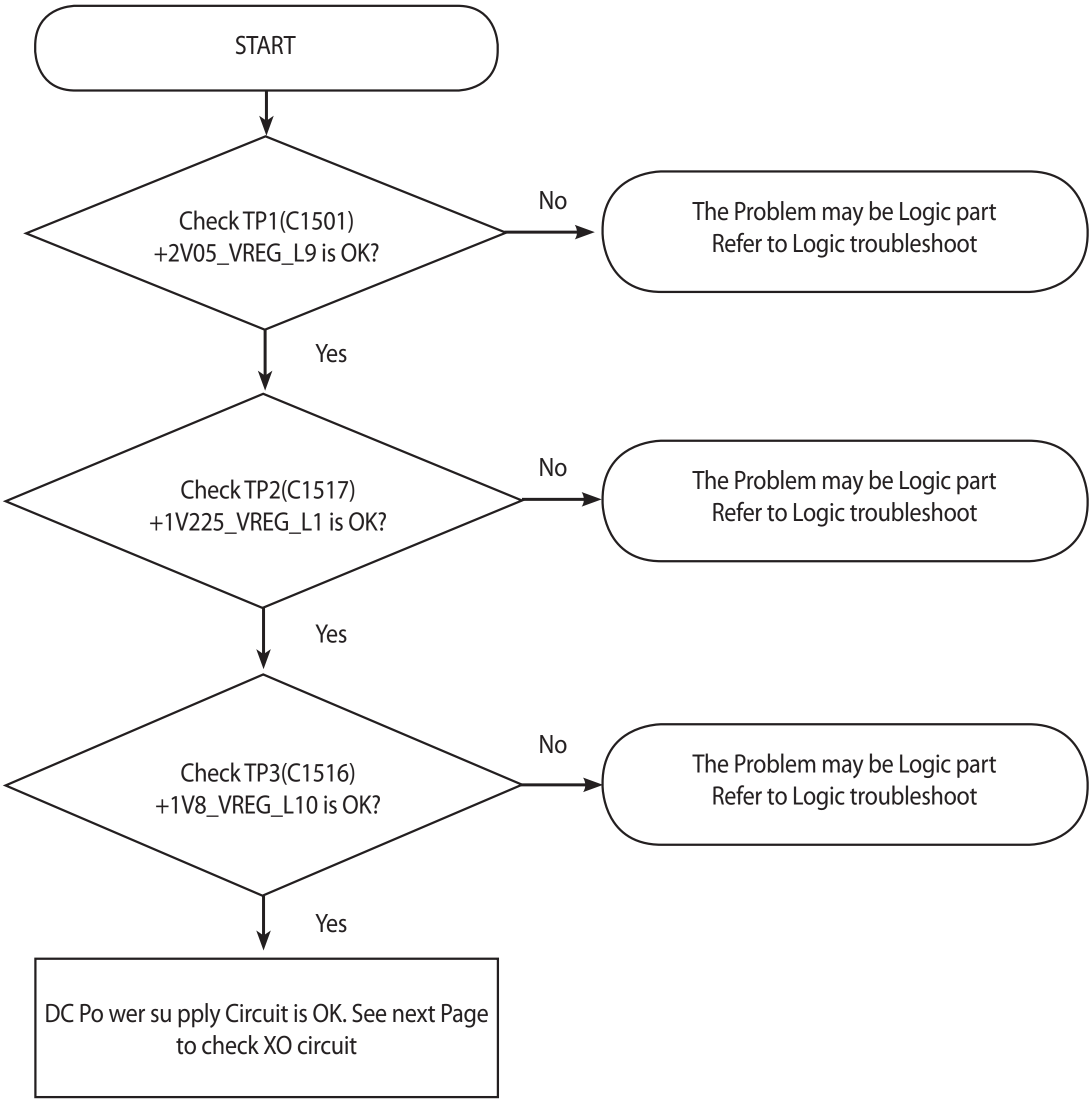
3.5.1 GSM1800/1900 Rx

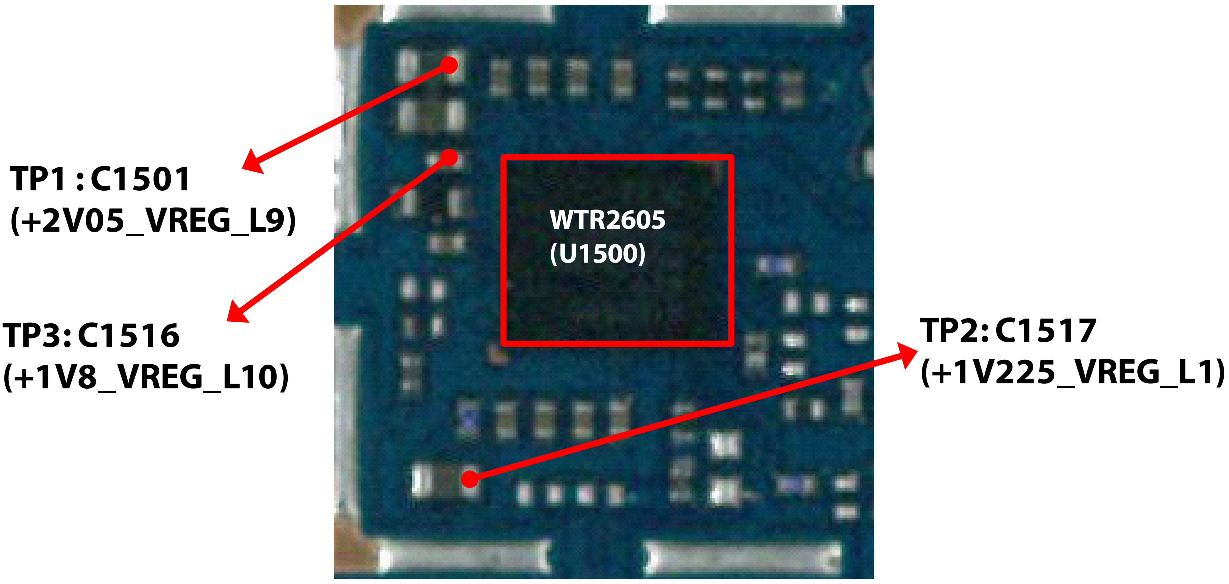


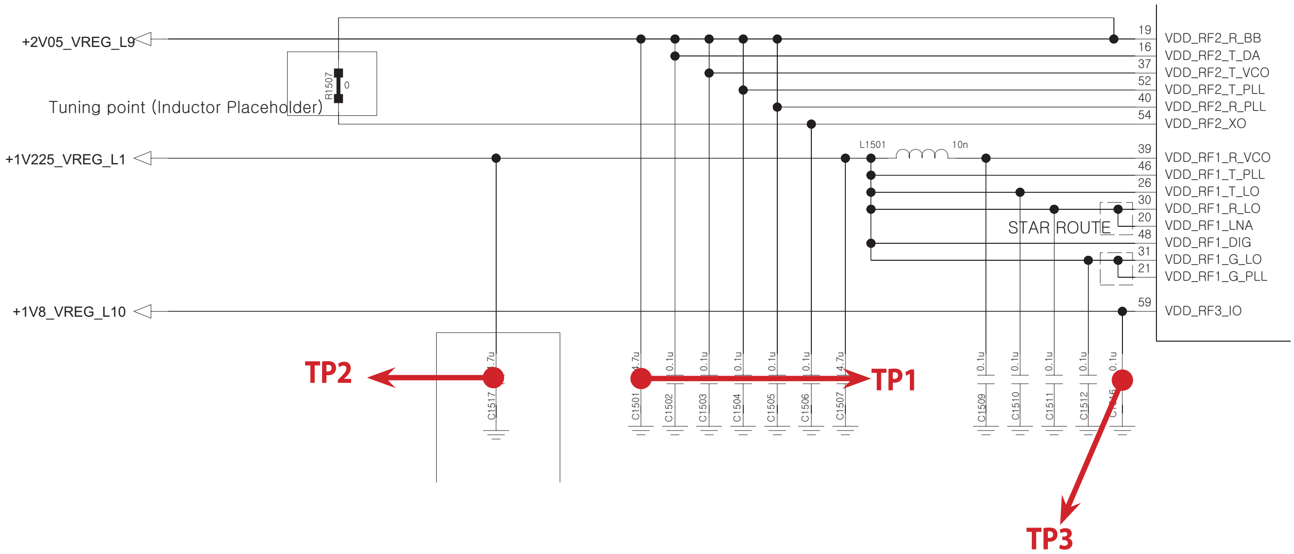
* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form


SAW
(FL1100)
SKY77573-31
(U1100)
WTR2605
(U1500)
DCS1800
PCS1900

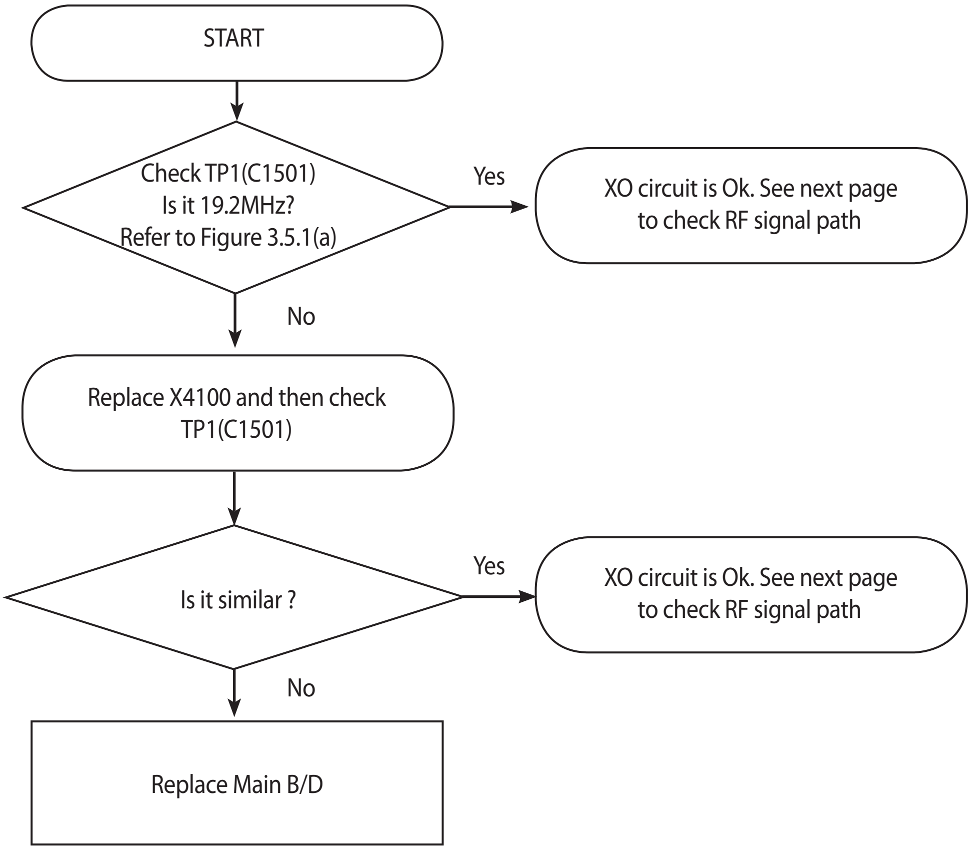
3.5.1.1 Checking DC Power supply circuit




TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)


+2V05_VREG_L9
Tuning point (Inductor Placeholder)
R1507
0
+1V225_VREG_L1
+1V8_VREG_L10
L1501
10n
STAR ROUTE
19 VDD_RF2_R_BB
16 VDD_RF2_T_DA
37 VDD_RF2_T_VCO
52 VDD_RF2_T_PLL
40 VDD_RF2_R_PLL
54 VDD_RF2_XO
39 VDD_RF1_R_VCO
46 VDD_RF1_T_PLL
26 VDD_RF1_T_LO
30 VDD_RF1_R_LO
20 VDD_RF1_LNA
48 VDD_RF1_DIG
31 VDD_RF1_G_LO
21 VDD_RF1_G_PLL
59 VDD_RF3_IO
C1517
C1501
4.7u
C1502
0.1u
C1503
0.1u
C1504
0.1u
C1505
0.1u
C1506
0.1u
C1507
4.7u
C1509
0.1u
C1510
0.1u
C1511
0.1u
C1512
0.1u
0.1u
TP2
TP1
TP3

3.5.1.2 Checking XO circuit



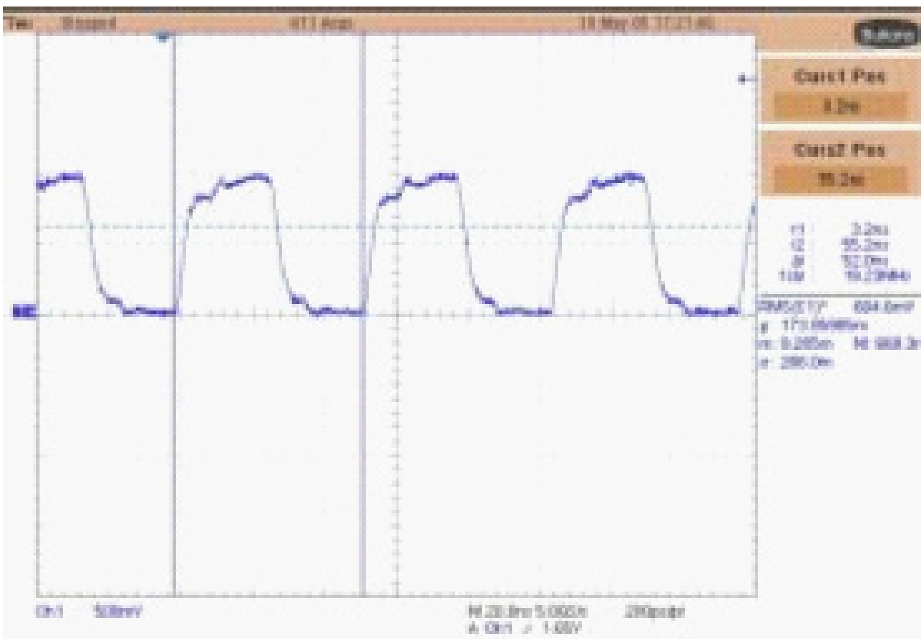
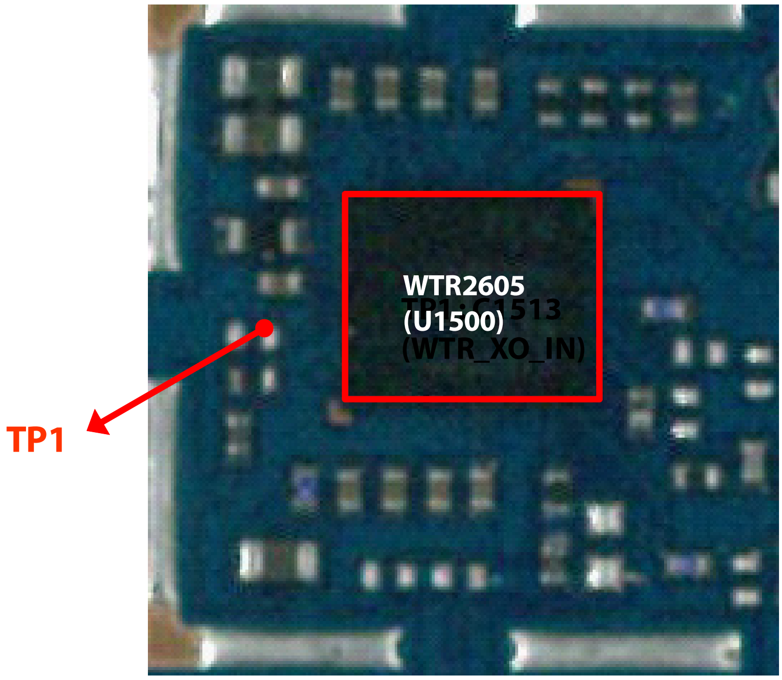
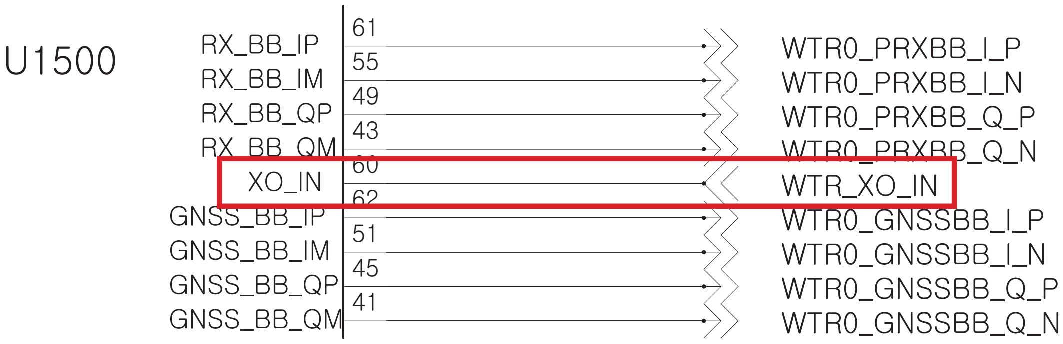
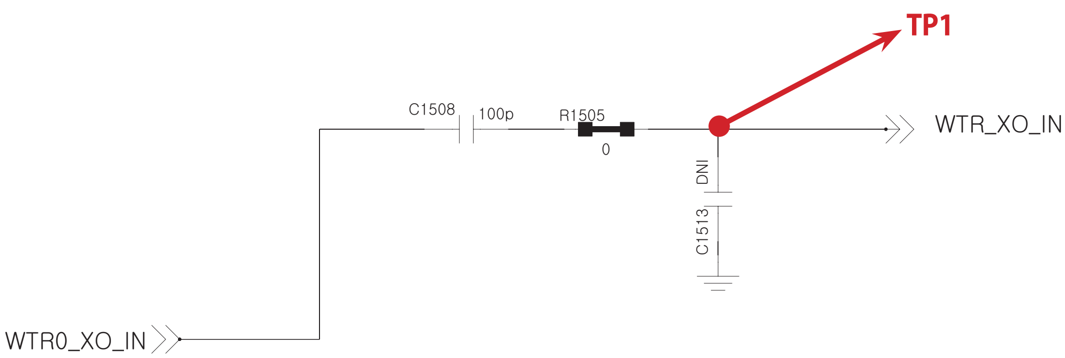


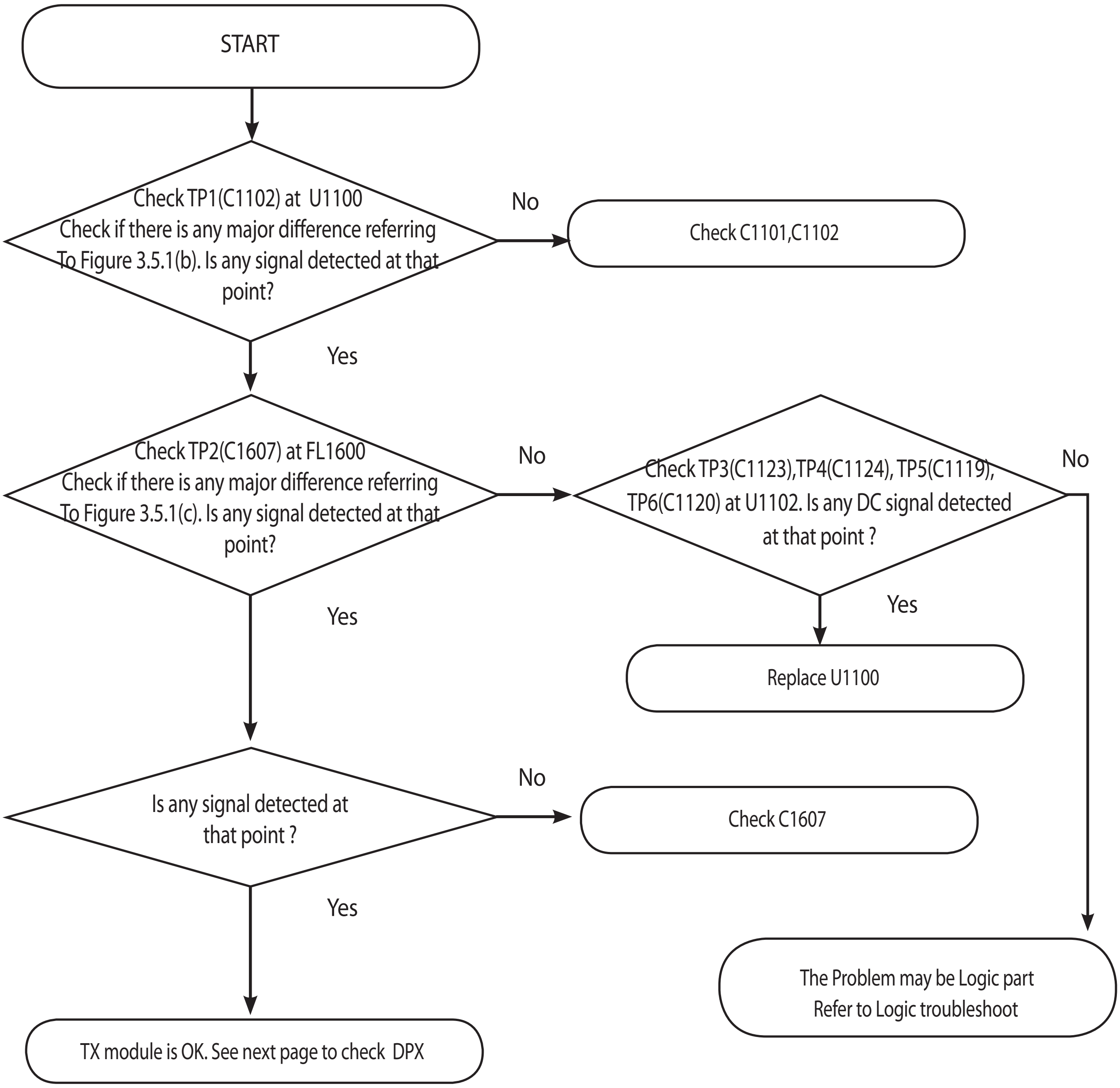
Figure 3.5.1 (a)

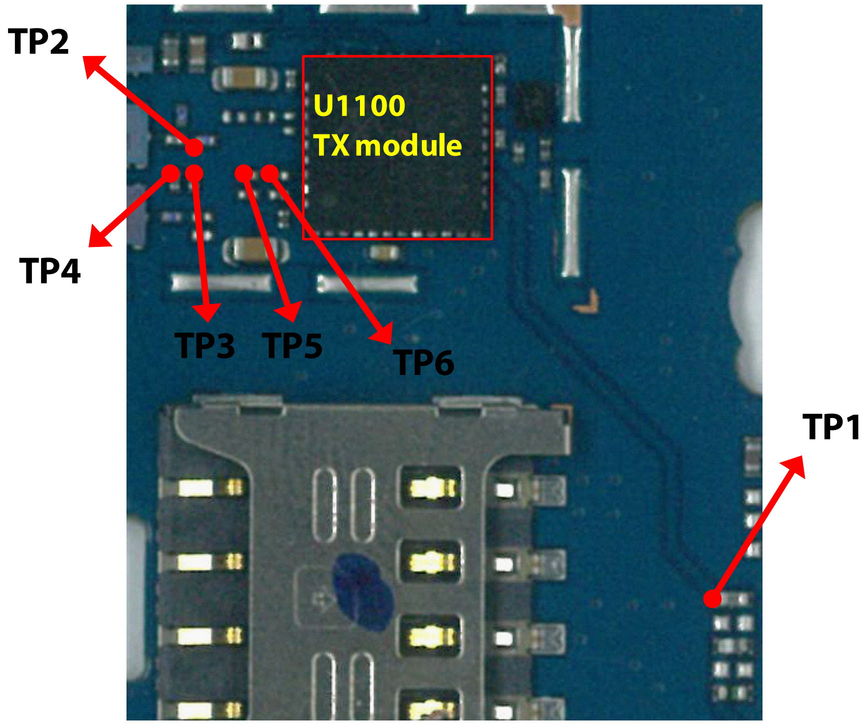

WTR2605
(U1500)
(WTR_XO_IN)
TP1

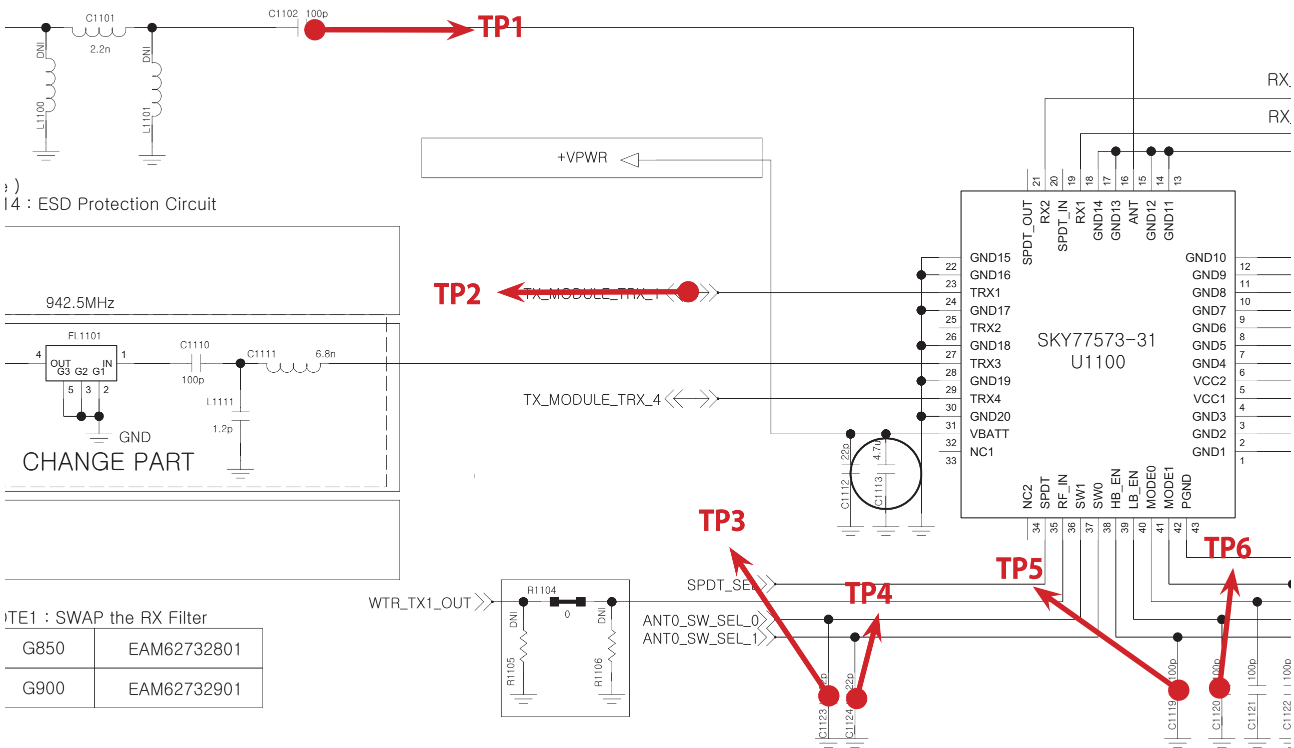

U1500
RX_BB_IP
RX_BB_IM
RX_BB_QP
RX_BB_QM
XO_IN
GNSS_BB_IP
GNSS_BB_IM
GNSS_BB_QP
GNSS_BB_QM
61
55
49
43
60
62
51
45
41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

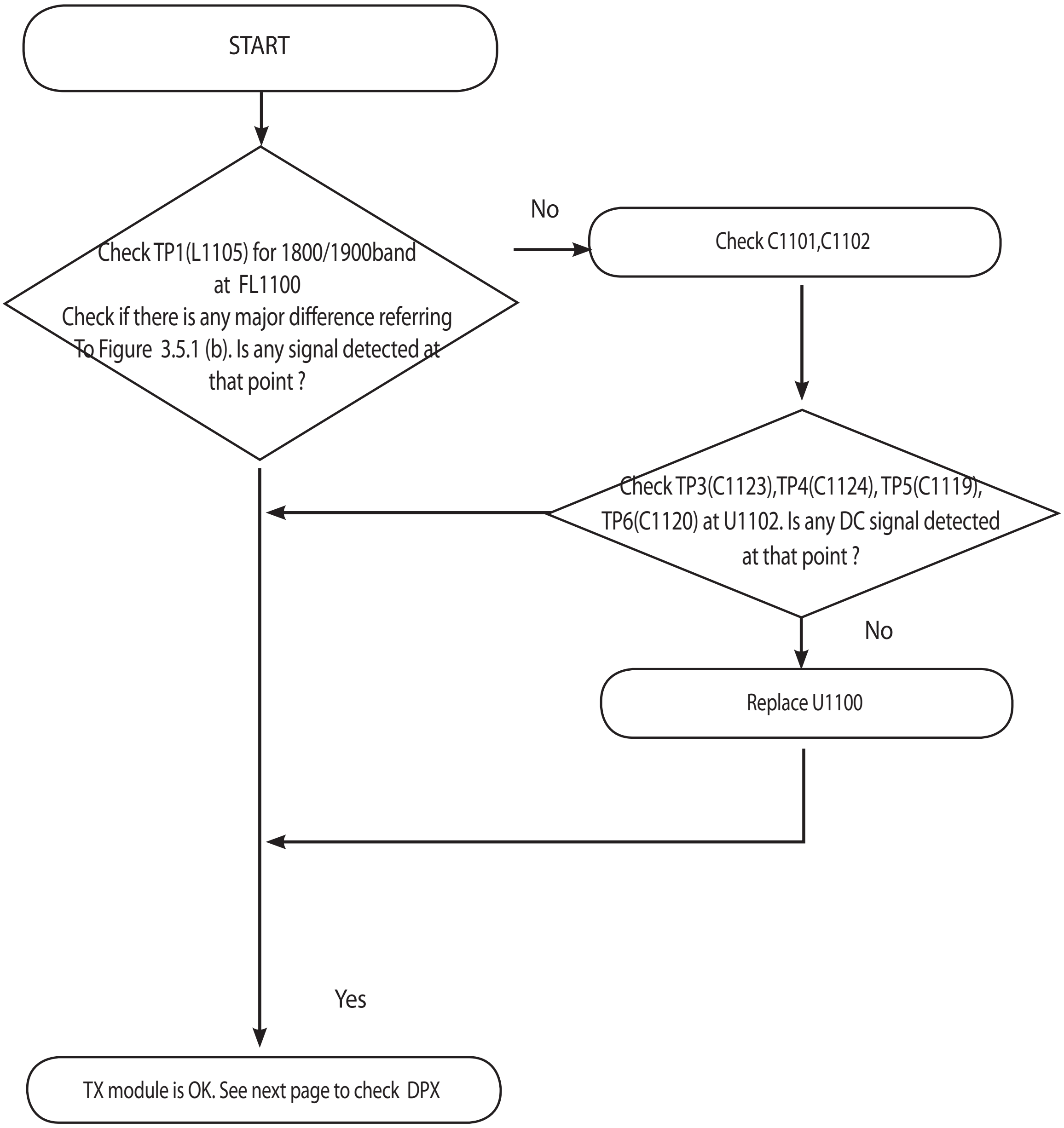
3.5.1.3 Checking RF signal path (TX Module)

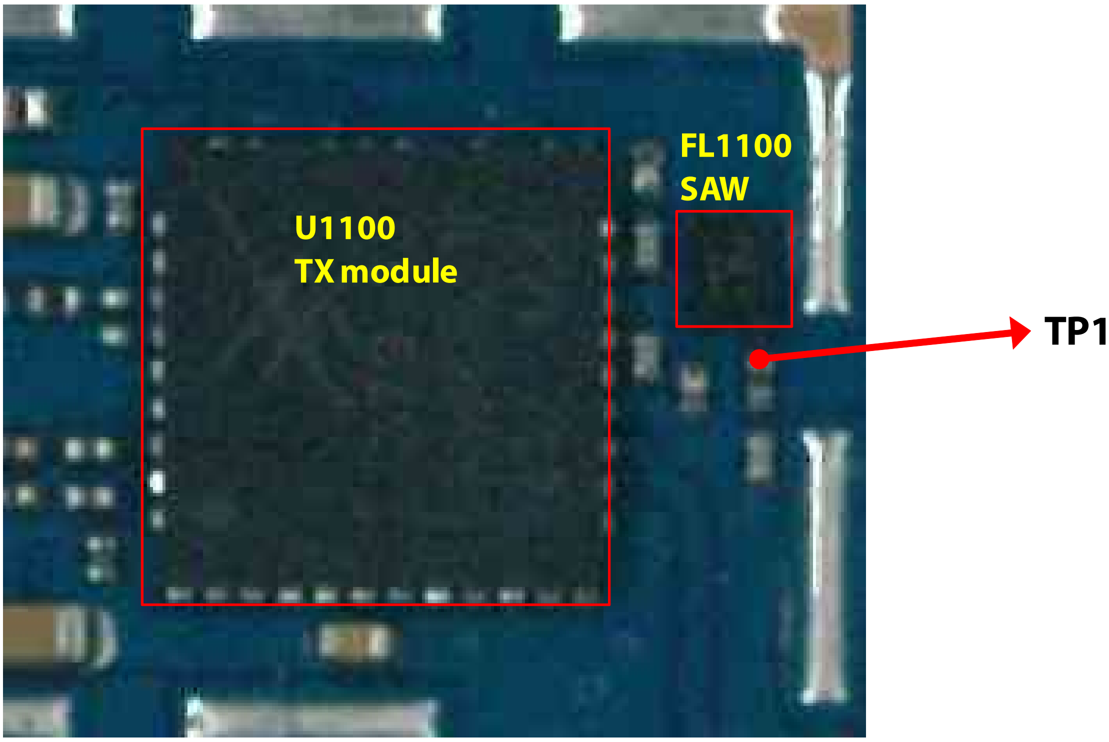


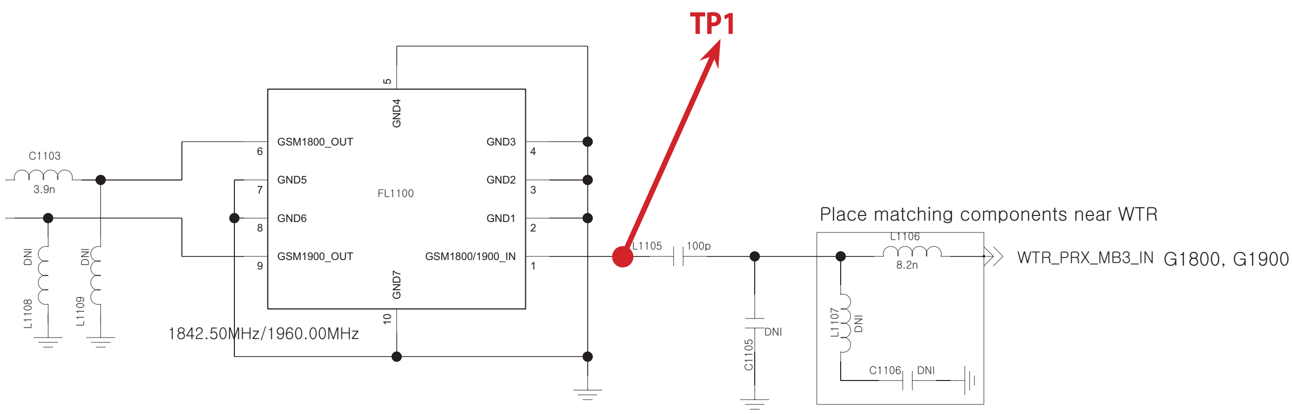

TP2
U1100
TX module
TP4
TP3
TP5
TP6
TP1


TP1
TP2
TP3
TP4
TP5
TP6
+VPWR
ESD Protection Circuit
942.5MHz
CHANGE PART
SWAP the RX Filter
G850 EAM62732801
G900 EAM62732901
TX_MODULE_TRX_4
WTR_TX1_OUT
SPDT_SEL
ANT0_SW_SEL_0
ANT0_SW_SEL_1
SKY77573-31
U1100

3.5.1.4 Checking RF Flow(SAW filter)




FL1100
SAW
U1100
TX module
TP1


TP1
C1103
3.9n
DNI
L1108
DNI
L1109
GSM1800_OUT
GND5
GND6
GSM1900_OUT
GND4
GND7
FL1100
GND3
GND2
GND1
GSM1800/1900_IN
1842.50MHz/1960.00MHz
L1105
100p
C1105
DNI
Place matching components near WTR
L1106
8.2n
L1107
DNI
C1106
DNI
WTR_PRX_MB3_IN
G1800, G1900

TP1

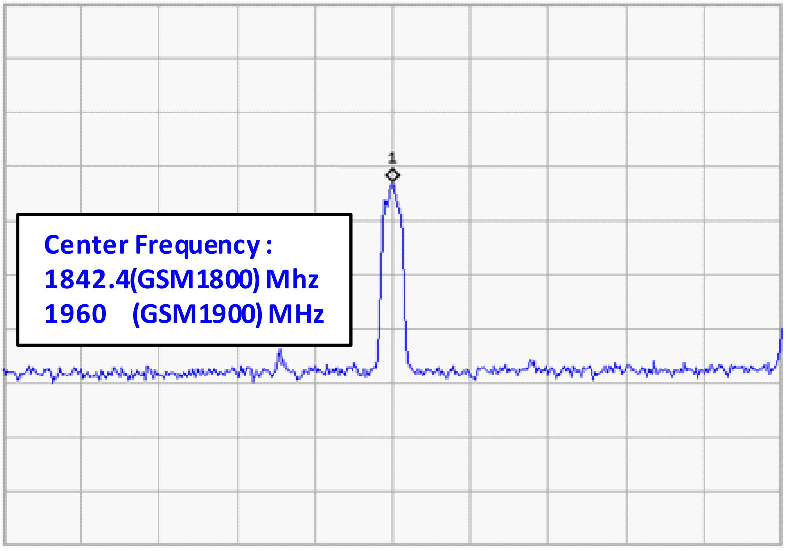


Figure 3.5.1 (b)

TP2

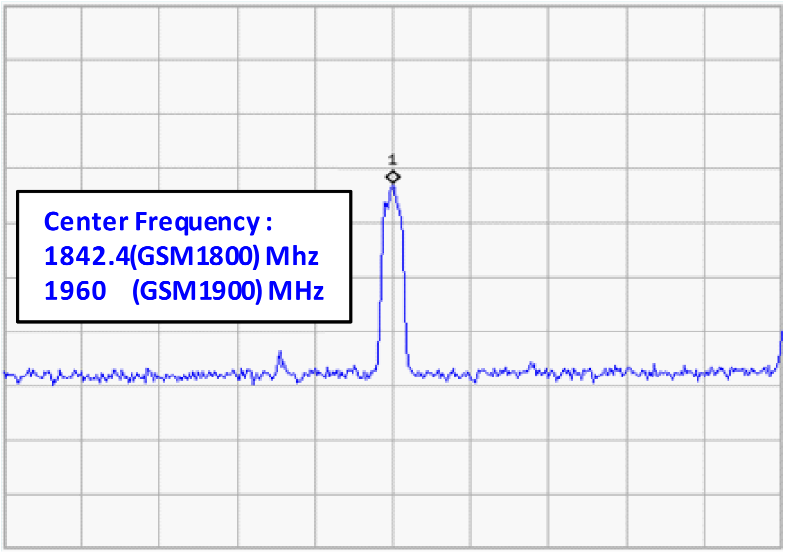
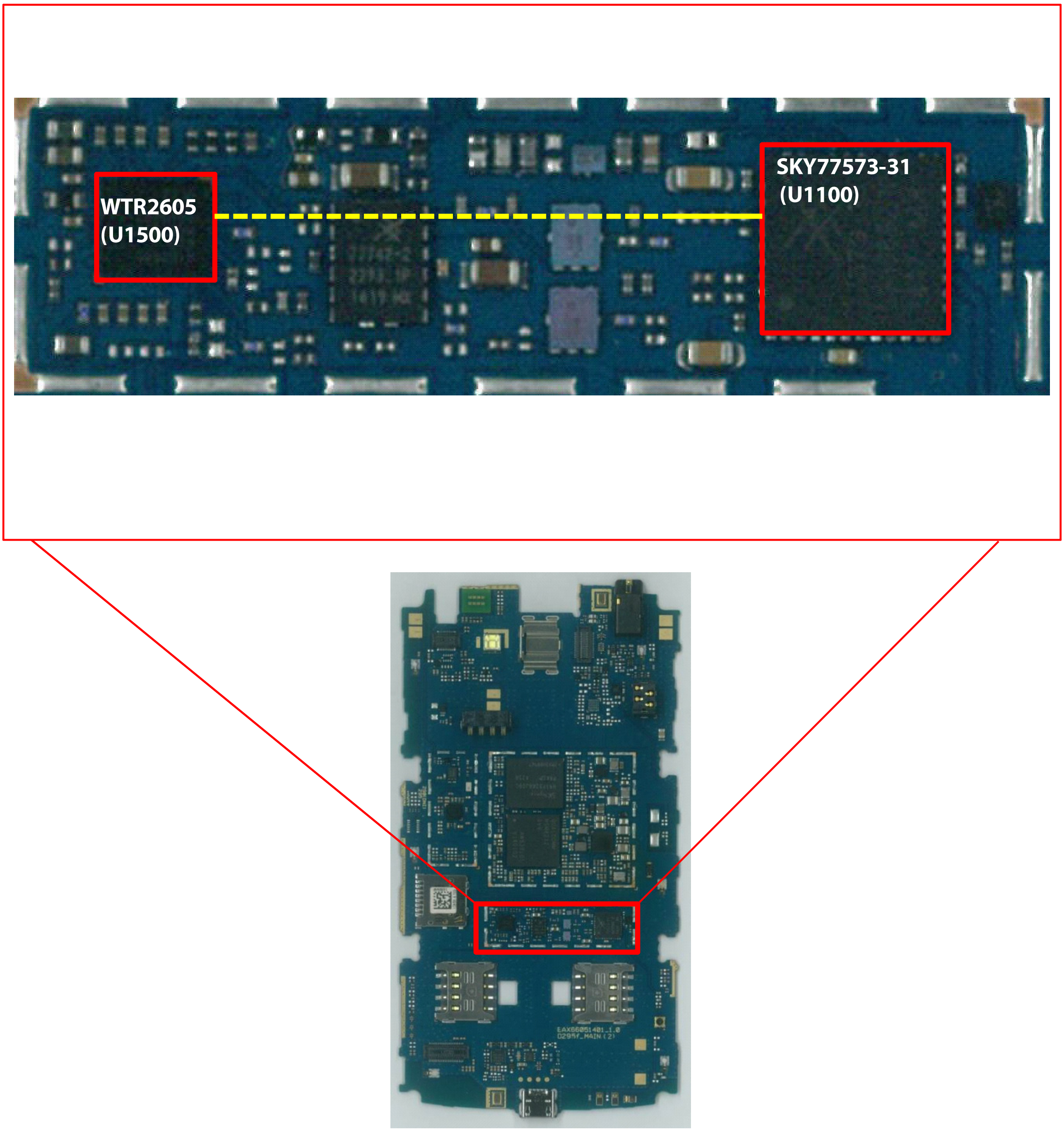


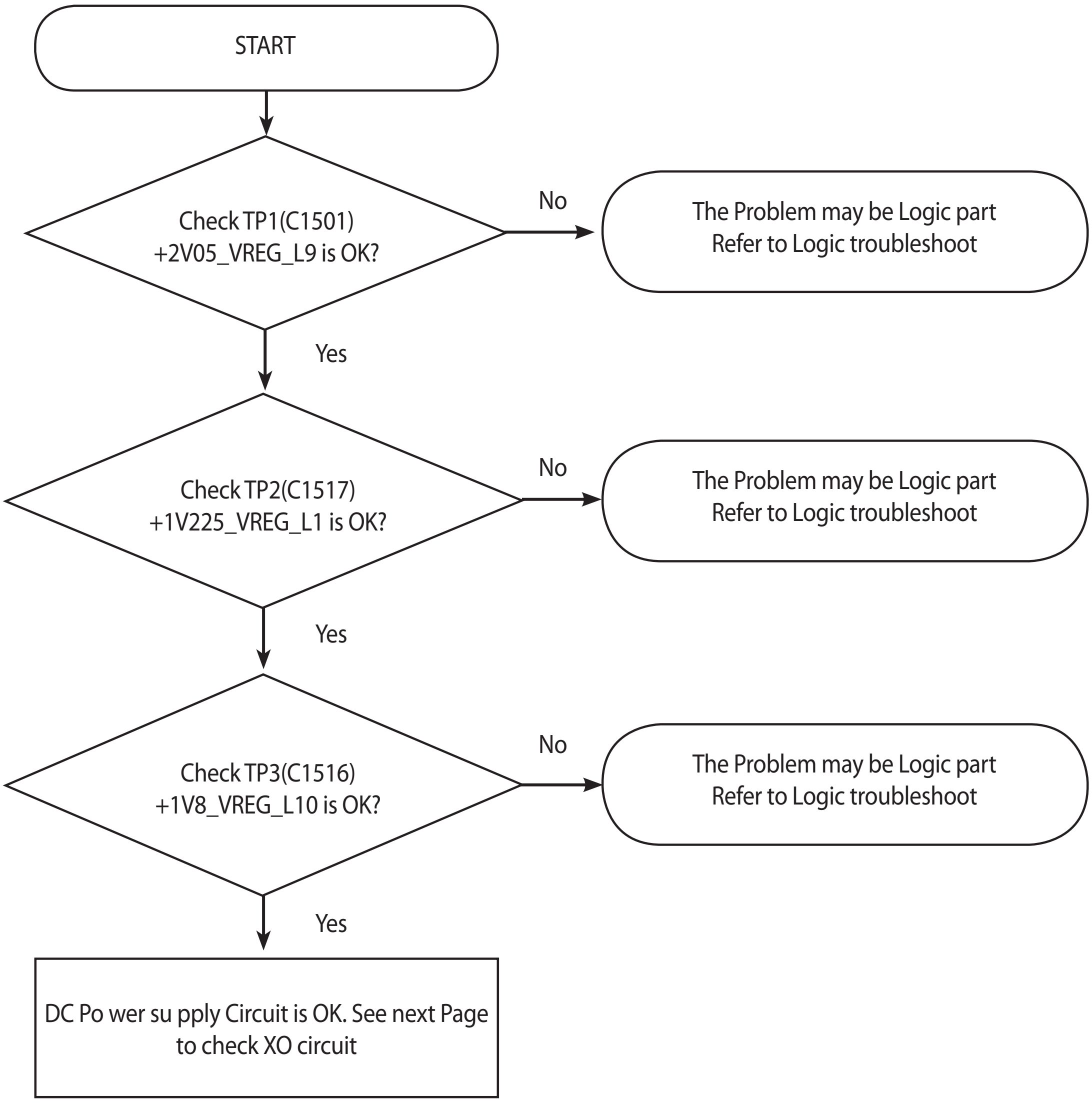
Figure 3.5.1 (c)

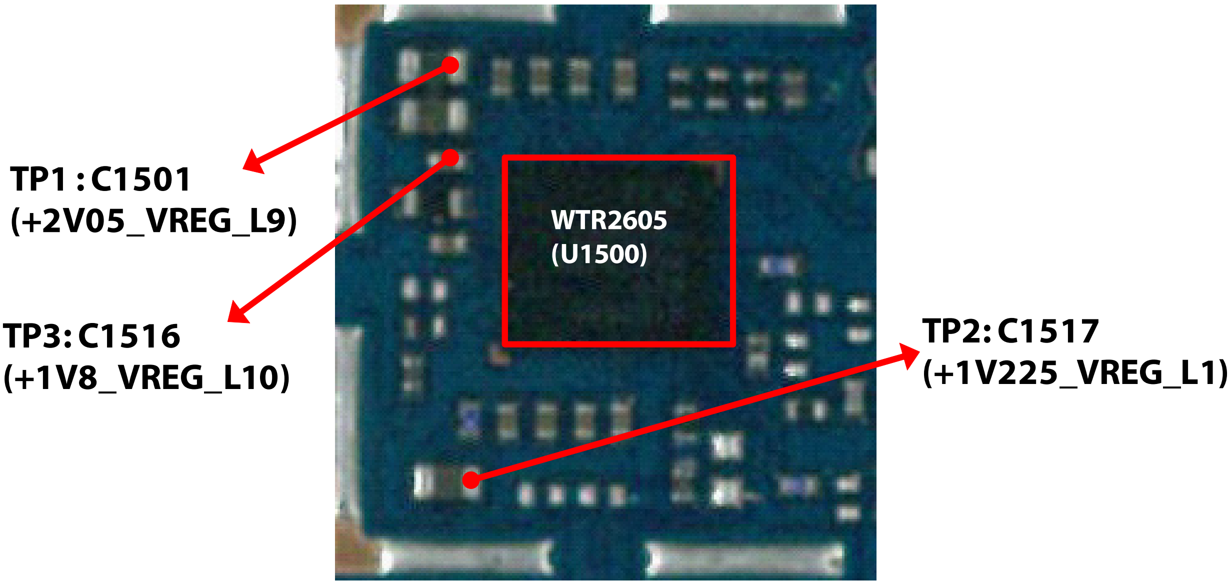
3.6 GSM Tx Part Troubleshooting

3.6.1 GSM850/900 Tx

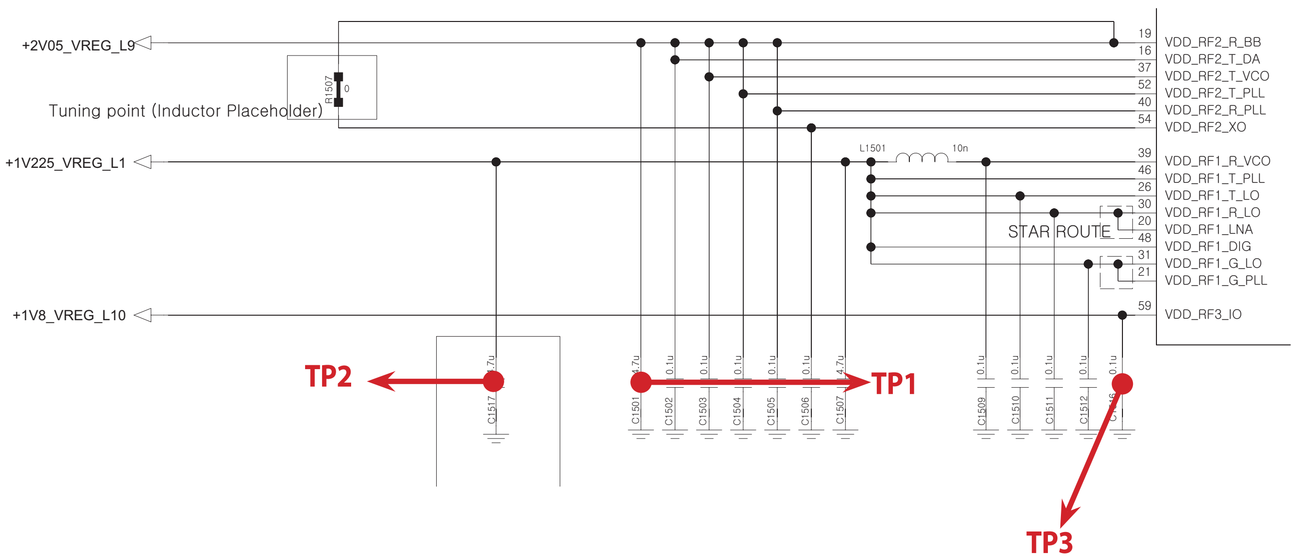


3.6.1.1 Checking DC Power supply circuit



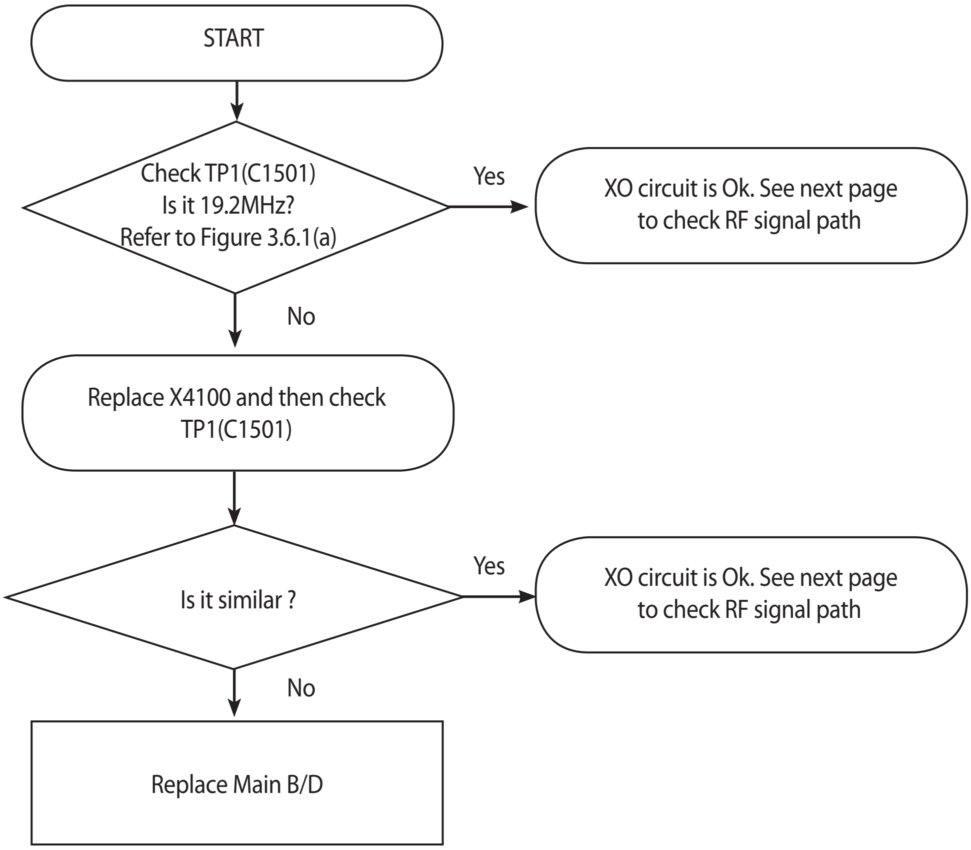


TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)



+2V05_VREG_L9
Tuning point (Inductor Placeholder)
R1507
+1V225_VREG_L1
L1501
10n
STAR ROUTE
+1V8_VREG_L10
VDD_RF2_R_BB
VDD_RF2_T_DA
VDD_RF2_T_VCO
VDD_RF2_T_PLL
VDD_RF2_R_PLL
VDD_RF2_XO
VDD_RF1_R_VCO
VDD_RF1_T_PLL
VDD_RF1_T_LO
VDD_RF1_R_LO
VDD_RF1_LNA
VDD_RF1_DIG
VDD_RF1_G_LO
VDD_RF1_G_PLL
VDD_RF3_IO
TP2
TP1
TP3

3.6.1.2 Checking XO circuit



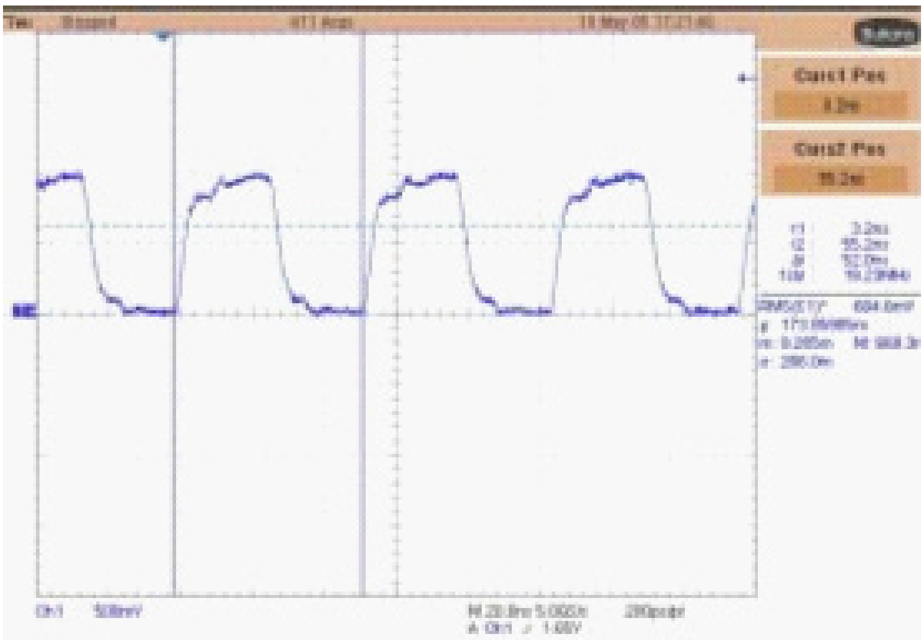
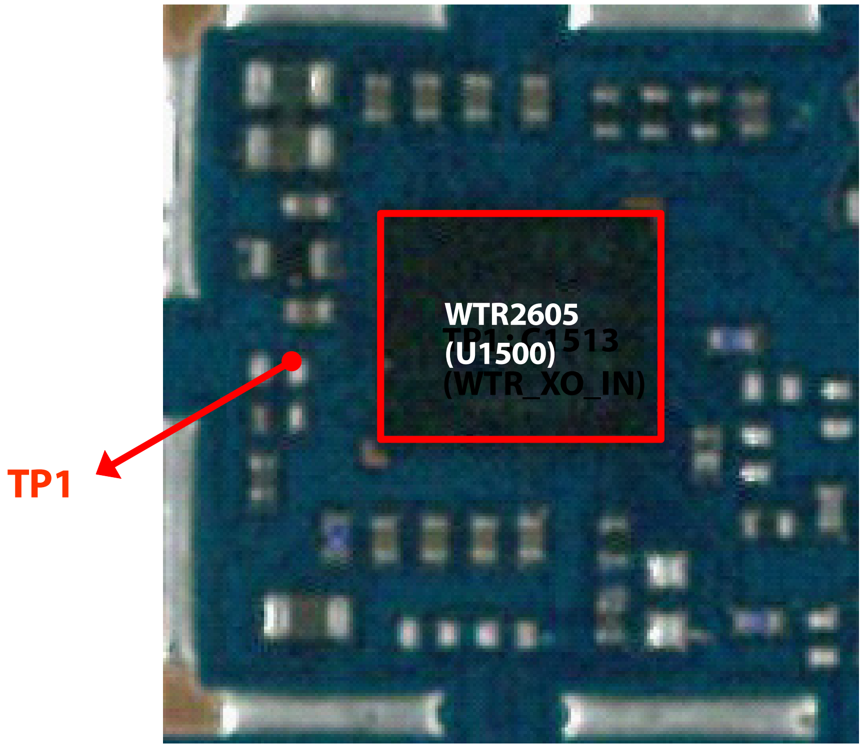
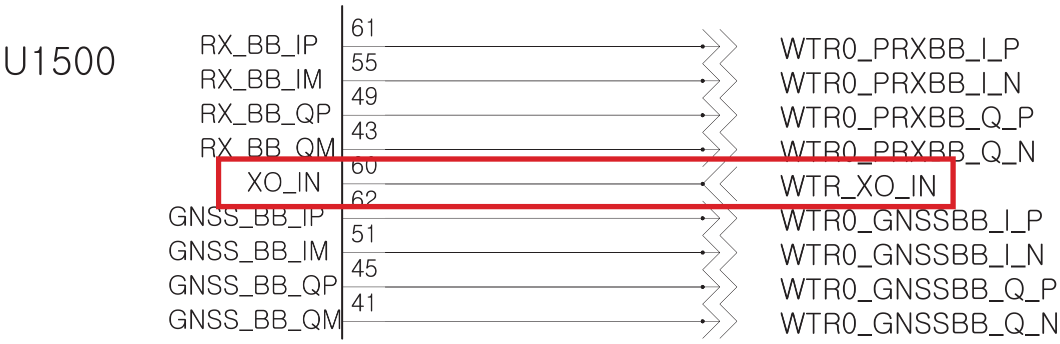
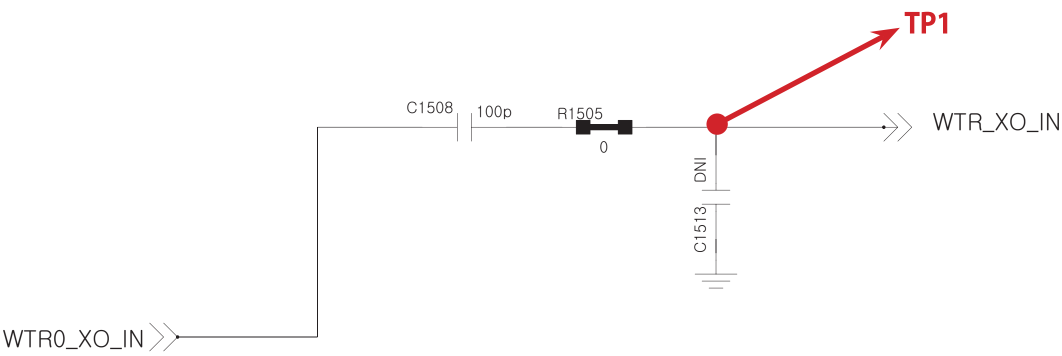


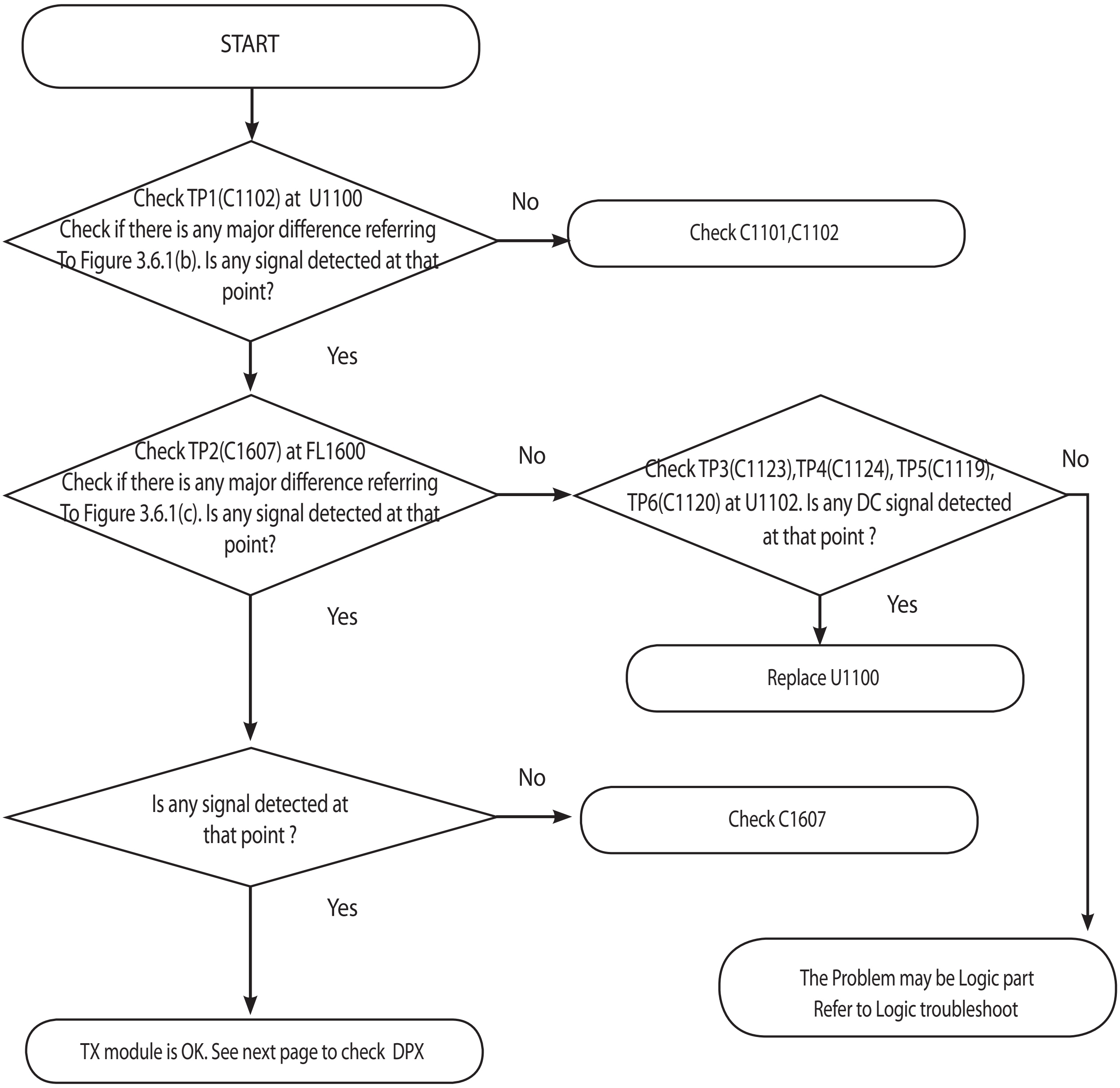
Figure 3.6.1 (a)

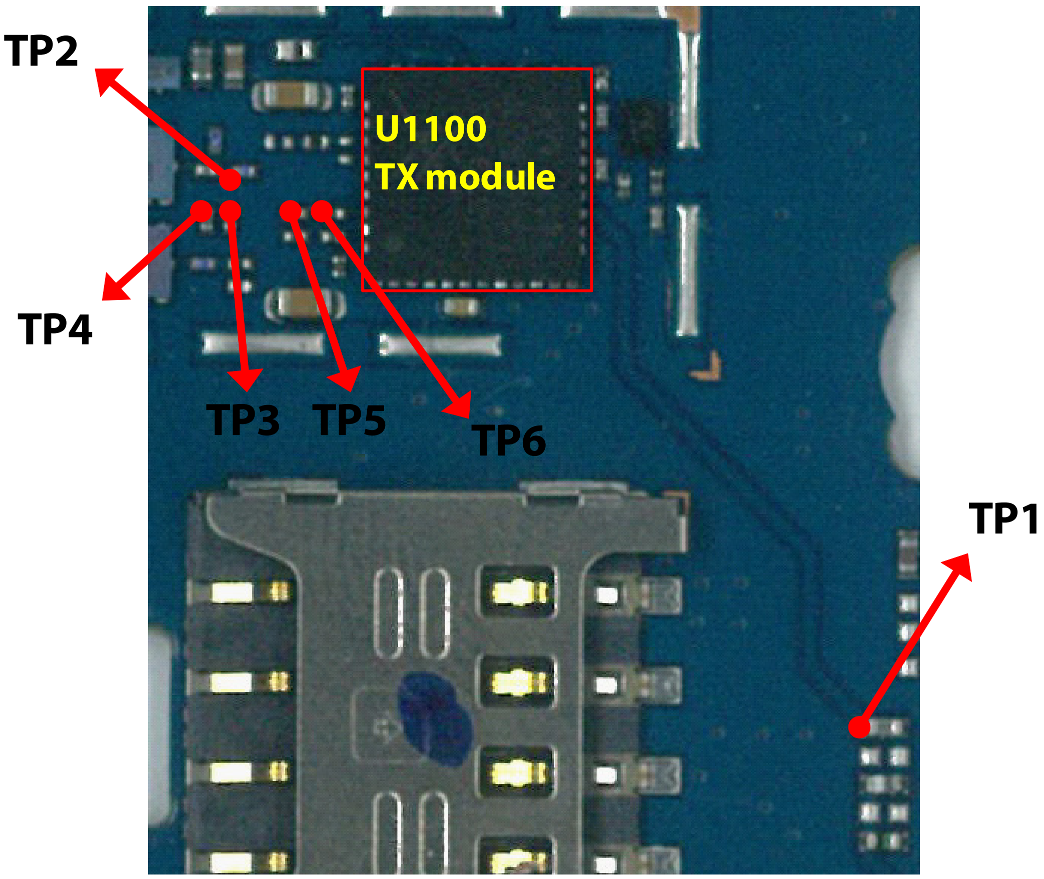

WTR2605
(U1500)
TP1 C1513
(WTR_XO_IN)
TP1

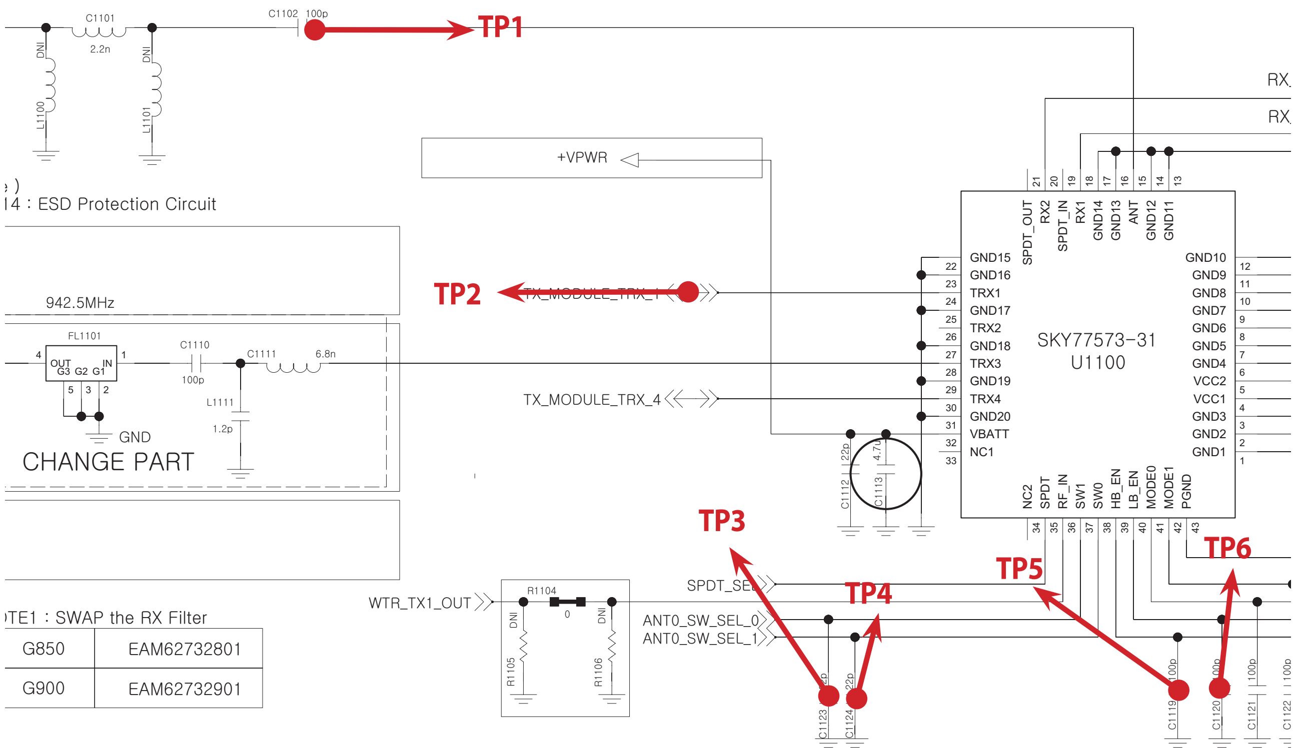

U1500
RX_BB_IP
RX_BB_IM
RX_BB_QP
RX_BB_QM
XO_IN
GNSS_BB_IP
GNSS_BB_IM
GNSS_BB_QP
GNSS_BB_QM
61
55
49
43
60
62
51
45
41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

3.6.1.3 Checking RF signal path (TX Module)







G850	EAM62732801
G900	EAM62732901

TP1

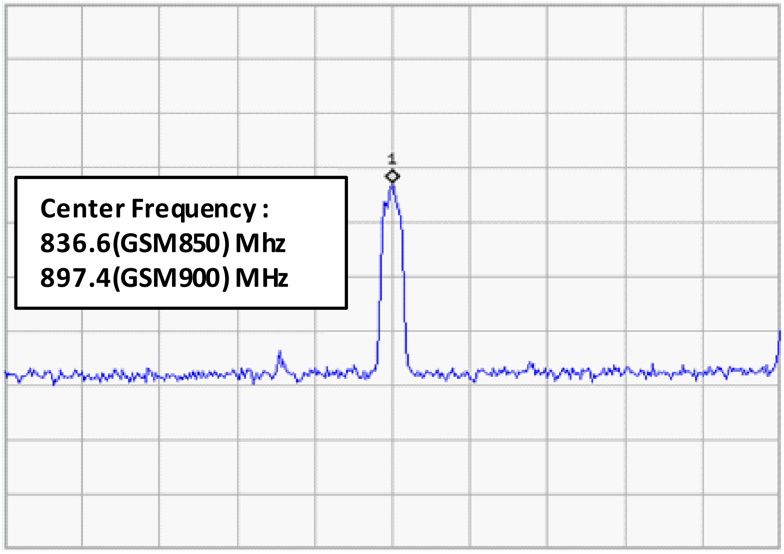


Figure 3.6.1 (b)

TP2

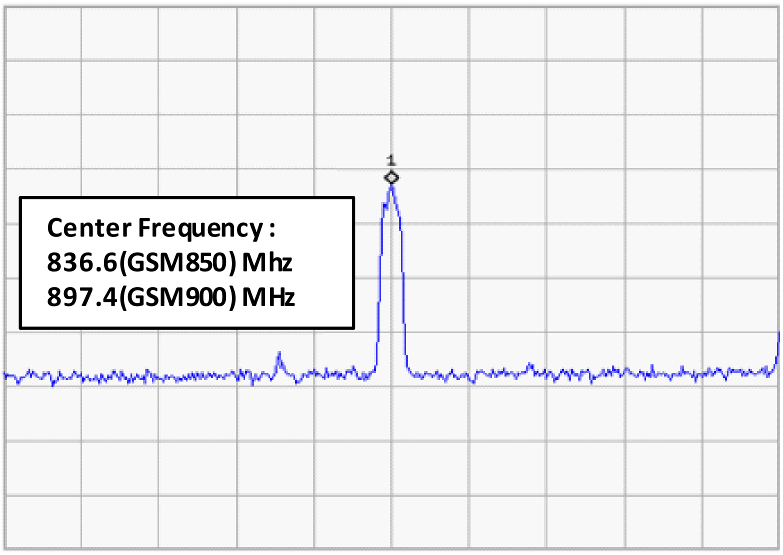
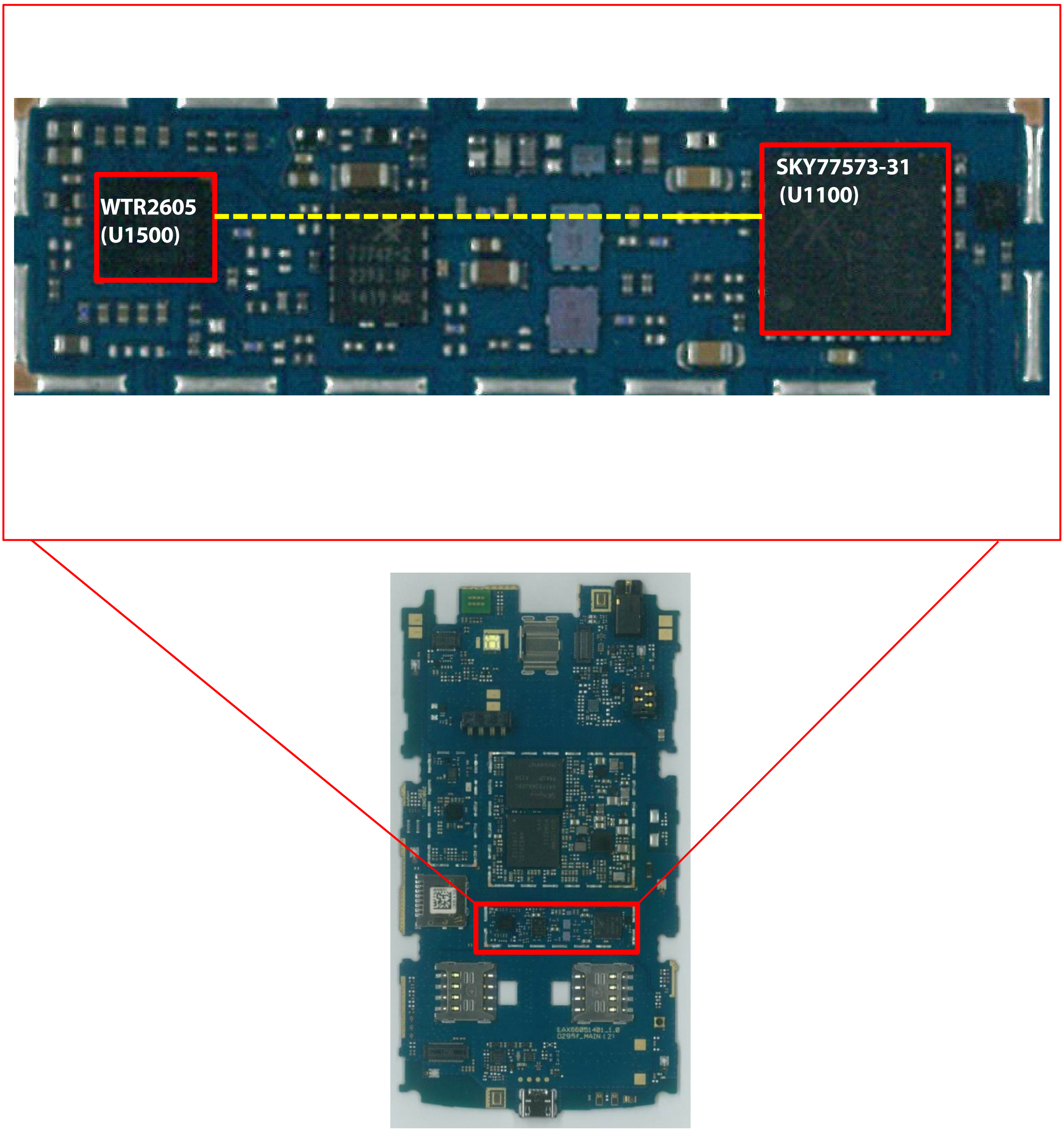


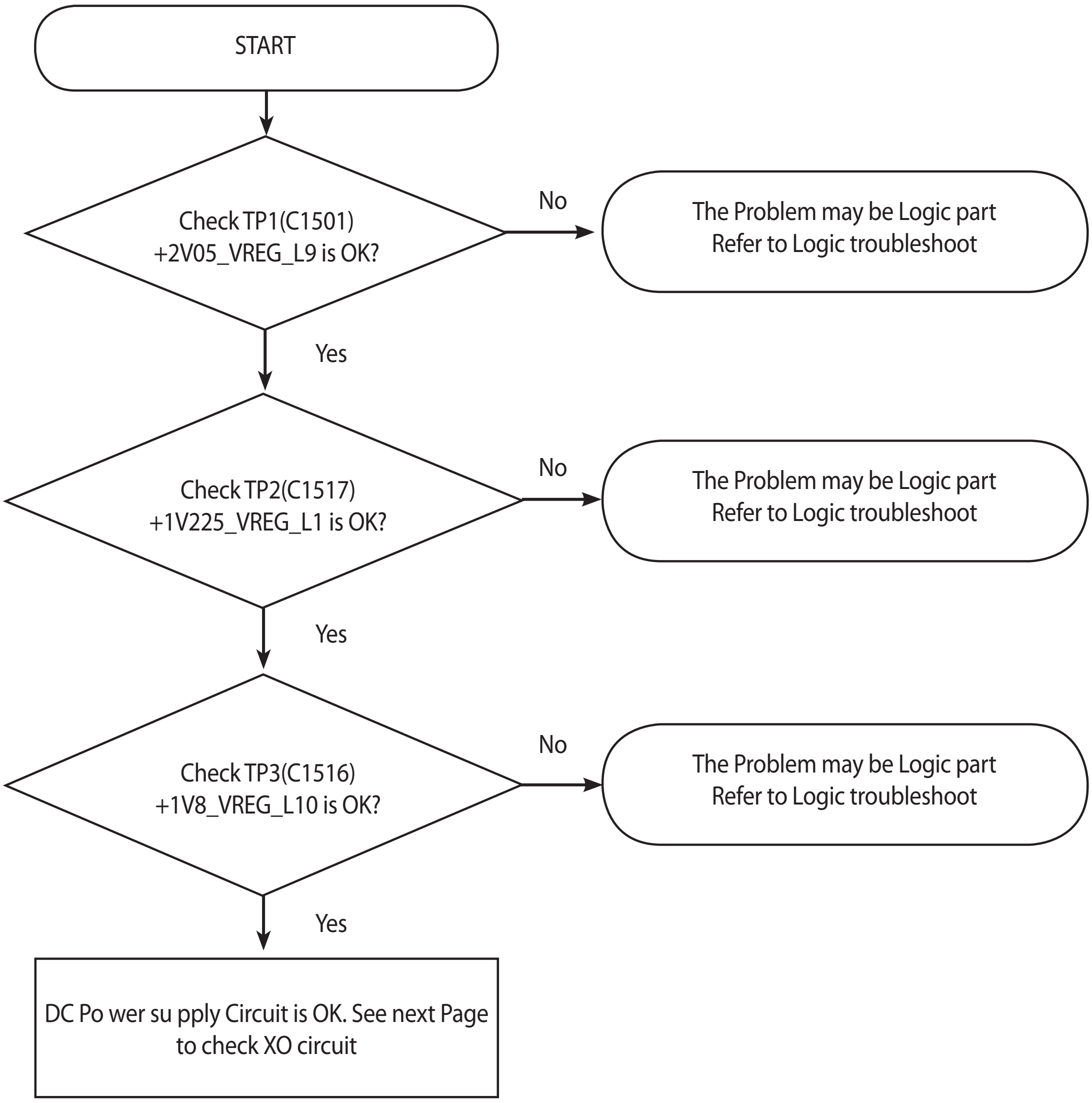
Figure 3.6.1 (c)

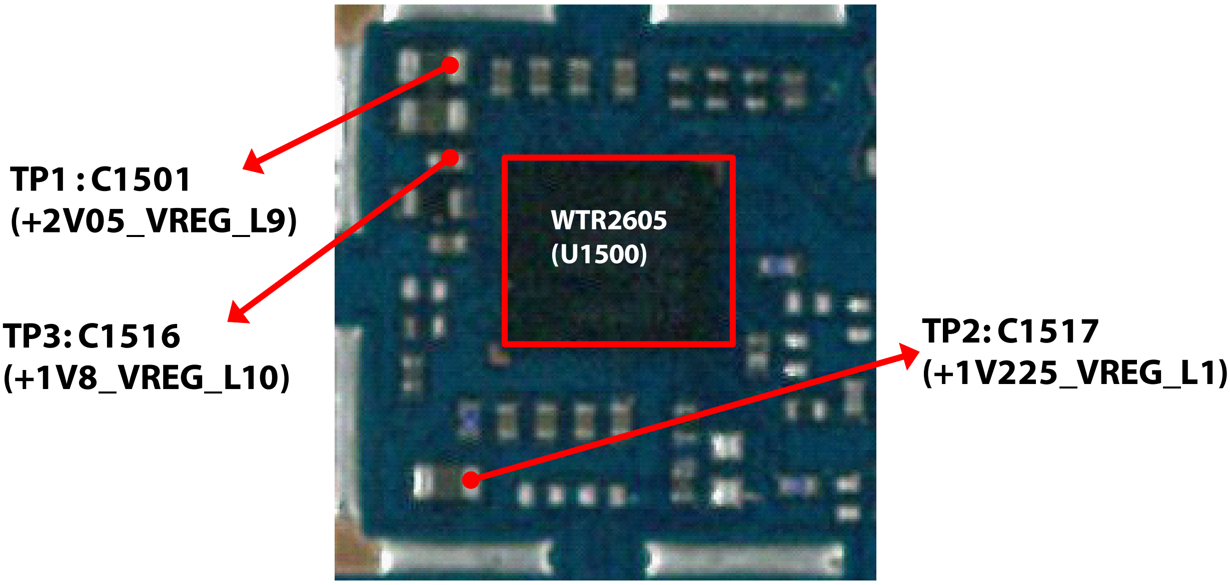
3.7 GSM Tx Part Troubleshooting

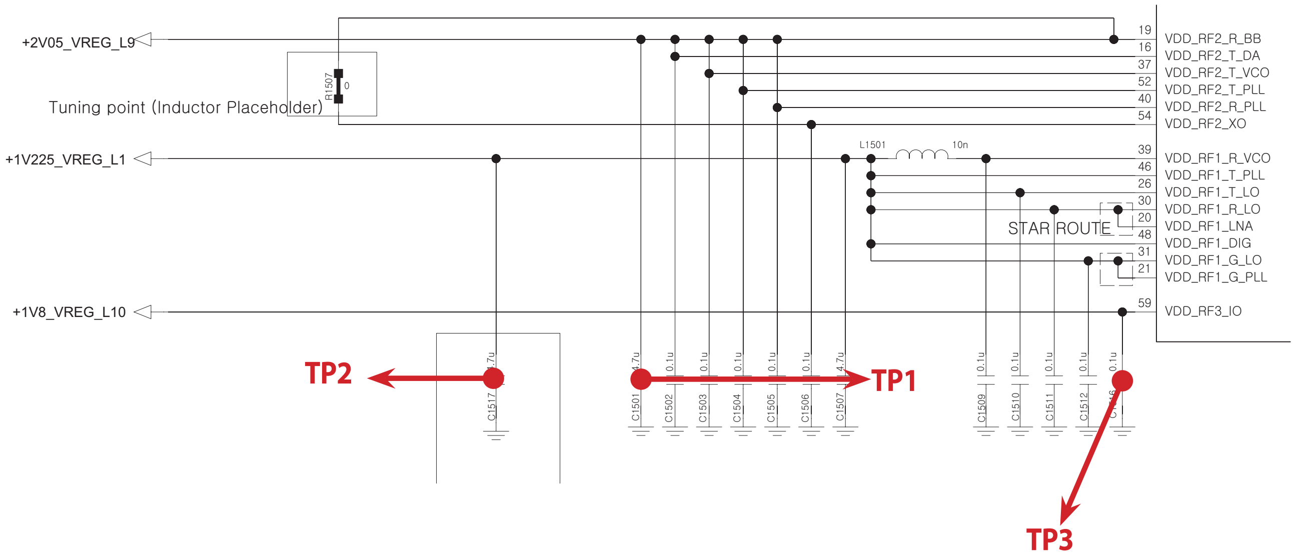
3.7.1 GSM1800/1900_Tx



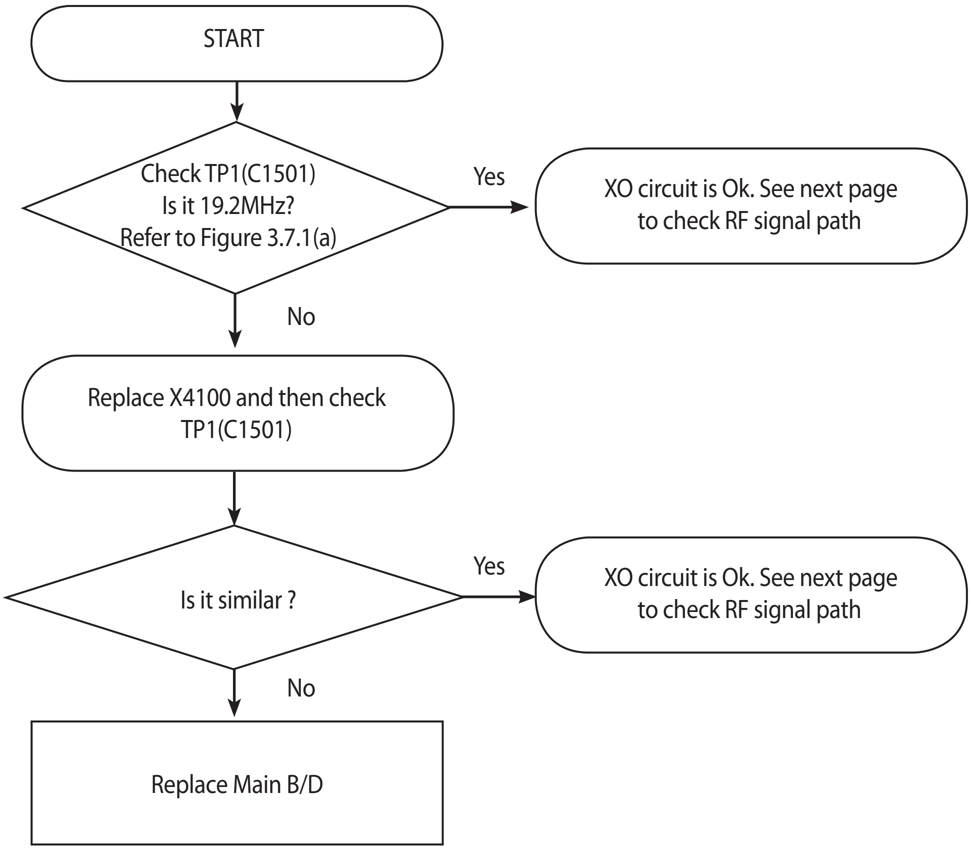
3.7.1.1 Checking DC Power supply circuit




TP1 : C1501
(+2V05_VREG_L9)
TP3: C1516
(+1V8_VREG_L10)
WTR2605
(U1500)
TP2: C1517
(+1V225_VREG_L1)


+2V05_VREG_L9
Tuning point (Inductor Placeholder)
+1V225_VREG_L1
+1V8_VREG_L10
L1501
10n
STAR ROUTE
VDD_RF2_R_BB
VDD_RF2_T_DA
VDD_RF2_T_VCO
VDD_RF2_T_PLL
VDD_RF2_R_PLL
VDD_RF2_XO
VDD_RF1_R_VCO
VDD_RF1_T_PLL
VDD_RF1_T_LO
VDD_RF1_R_LO
VDD_RF1_LNA
VDD_RF1_DIG
VDD_RF1_G_LO
VDD_RF1_G_PLL
VDD_RF3_IO
TP2
TP1
TP3

3.7.1.2 Checking XO circuit



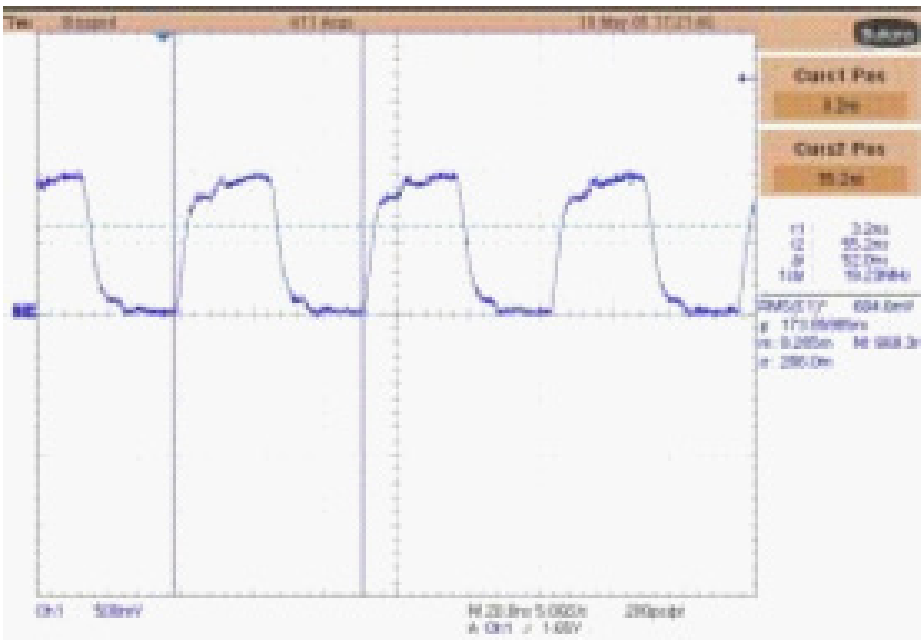
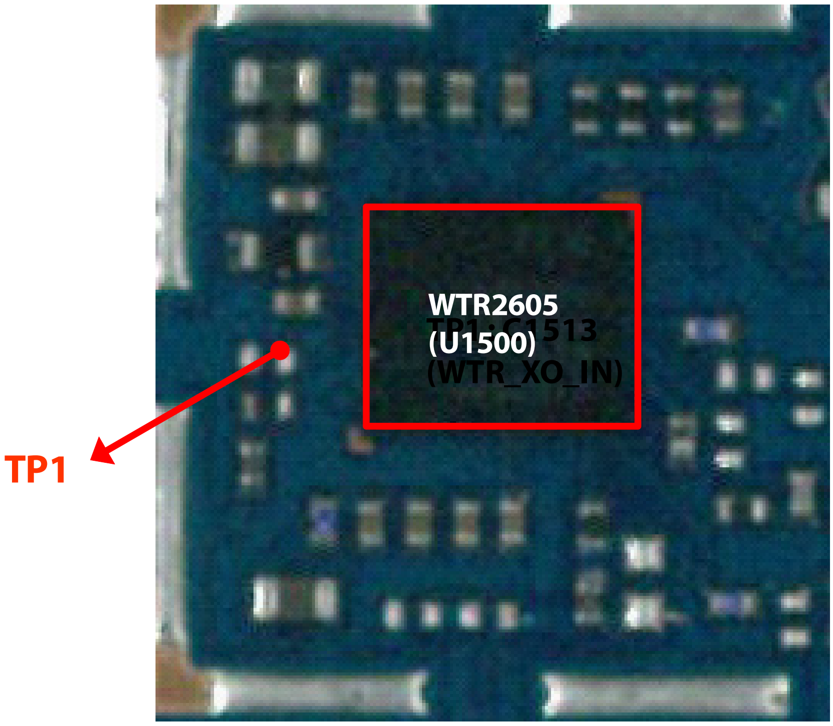
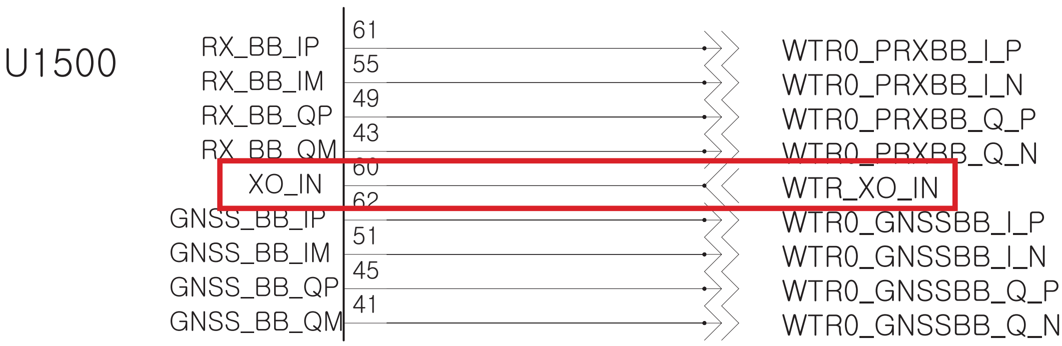
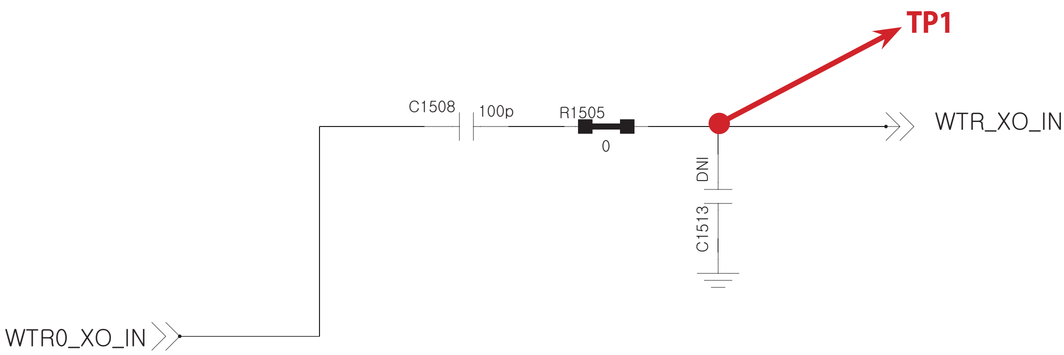


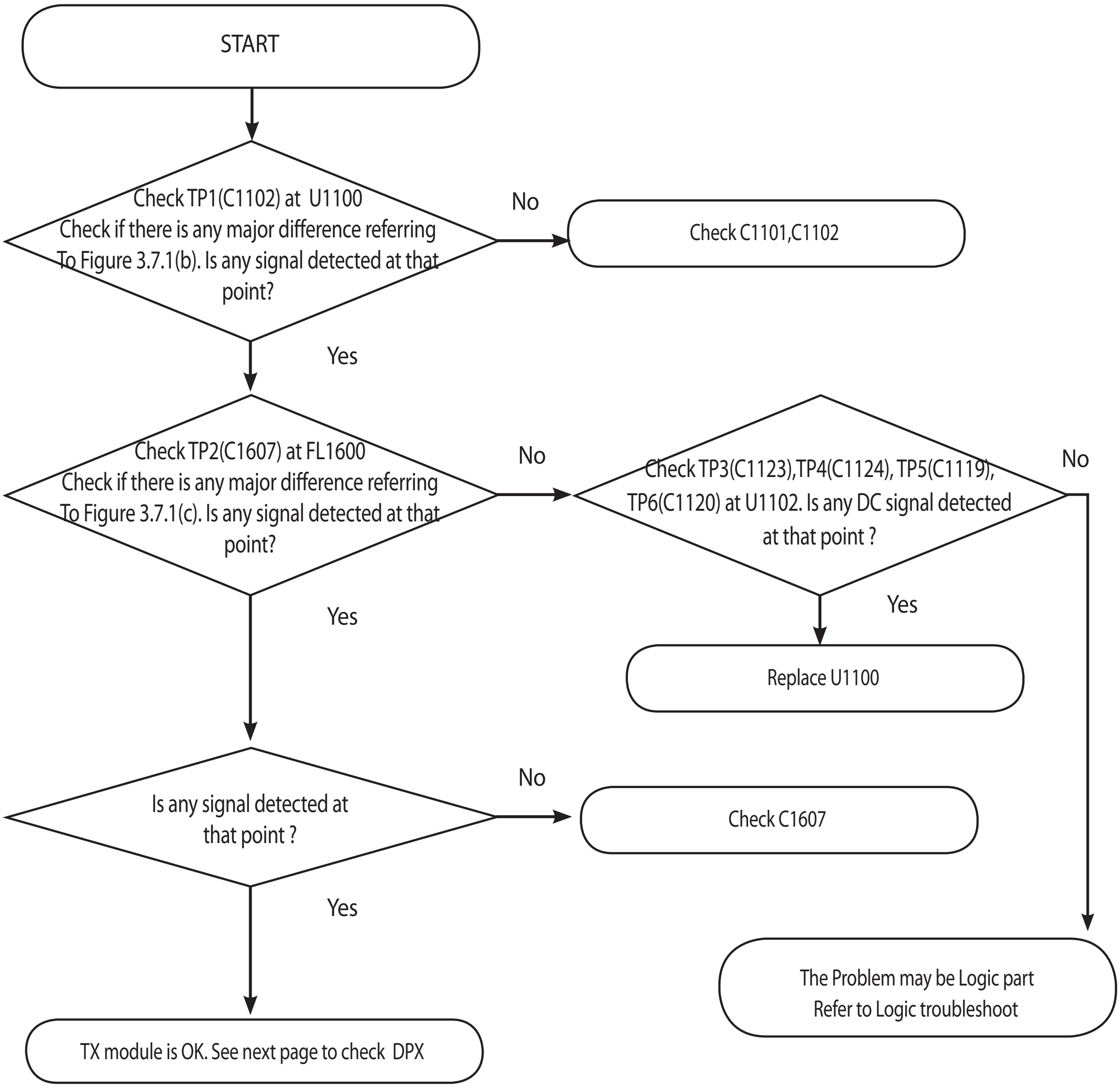
Figure 3.7.1 (a)

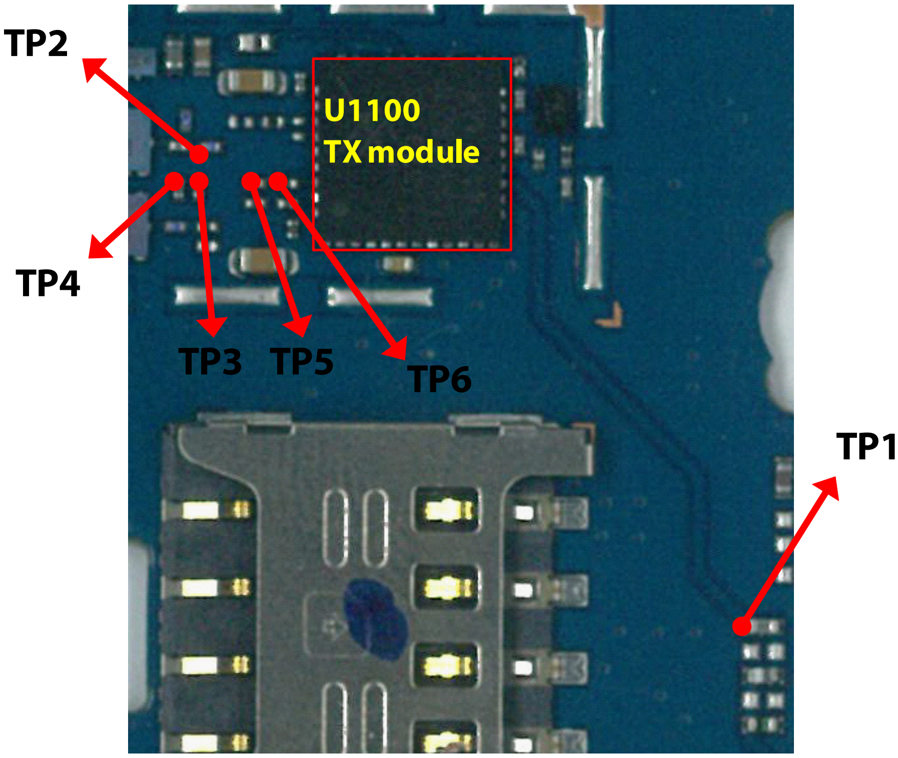

WTR2605
(U1500)
TP1 : C1513
(WTR_XO_IN)
TP1

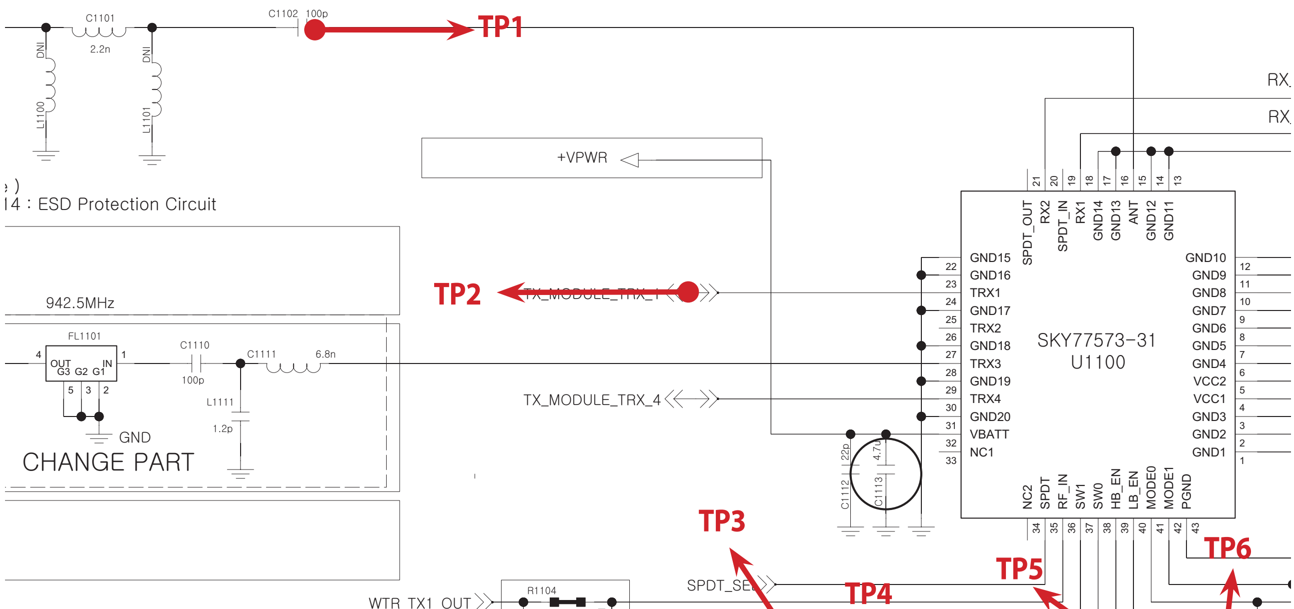

U1500
RX_BB_IP 61
RX_BB_IM 55
RX_BB_QP 49
RX_BB_QM 43
XO_IN 60
GNSS_BB_IP 62
GNSS_BB_IM 51
GNSS_BB_QP 45
GNSS_BB_QM 41
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_PRXBB_Q_N
WTR_XO_IN
WTR0_GNSSBB_I_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N


TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

3.7.1.3 Checking RF signal path (TX Module)







G850	EAM62732801
G900	EAM62732901

TP1

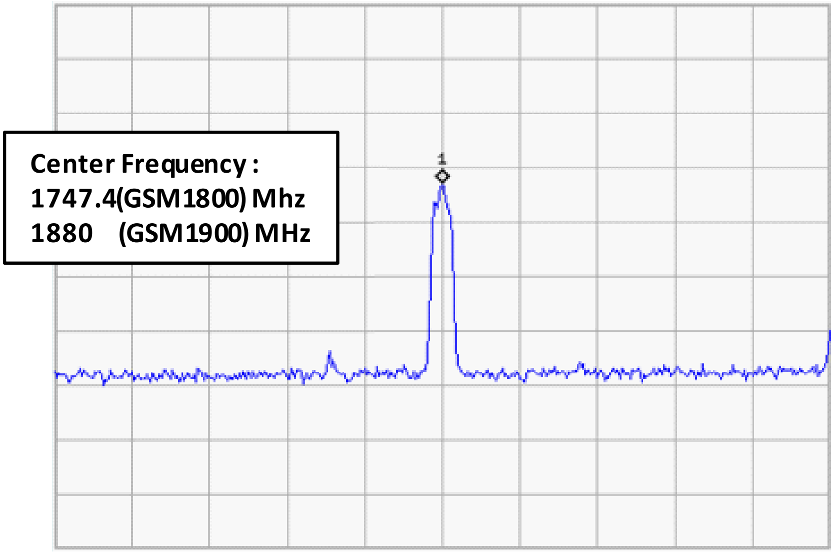


Figure 3.7.1 (b)

TP2

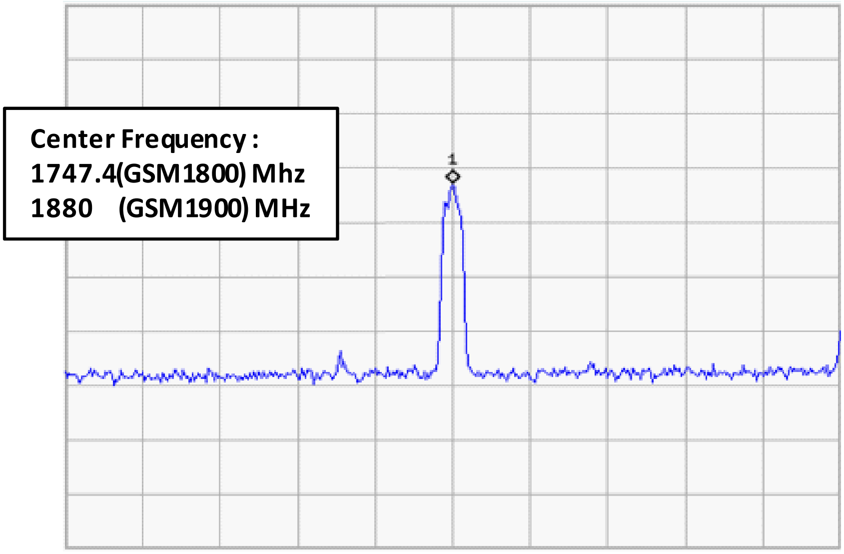
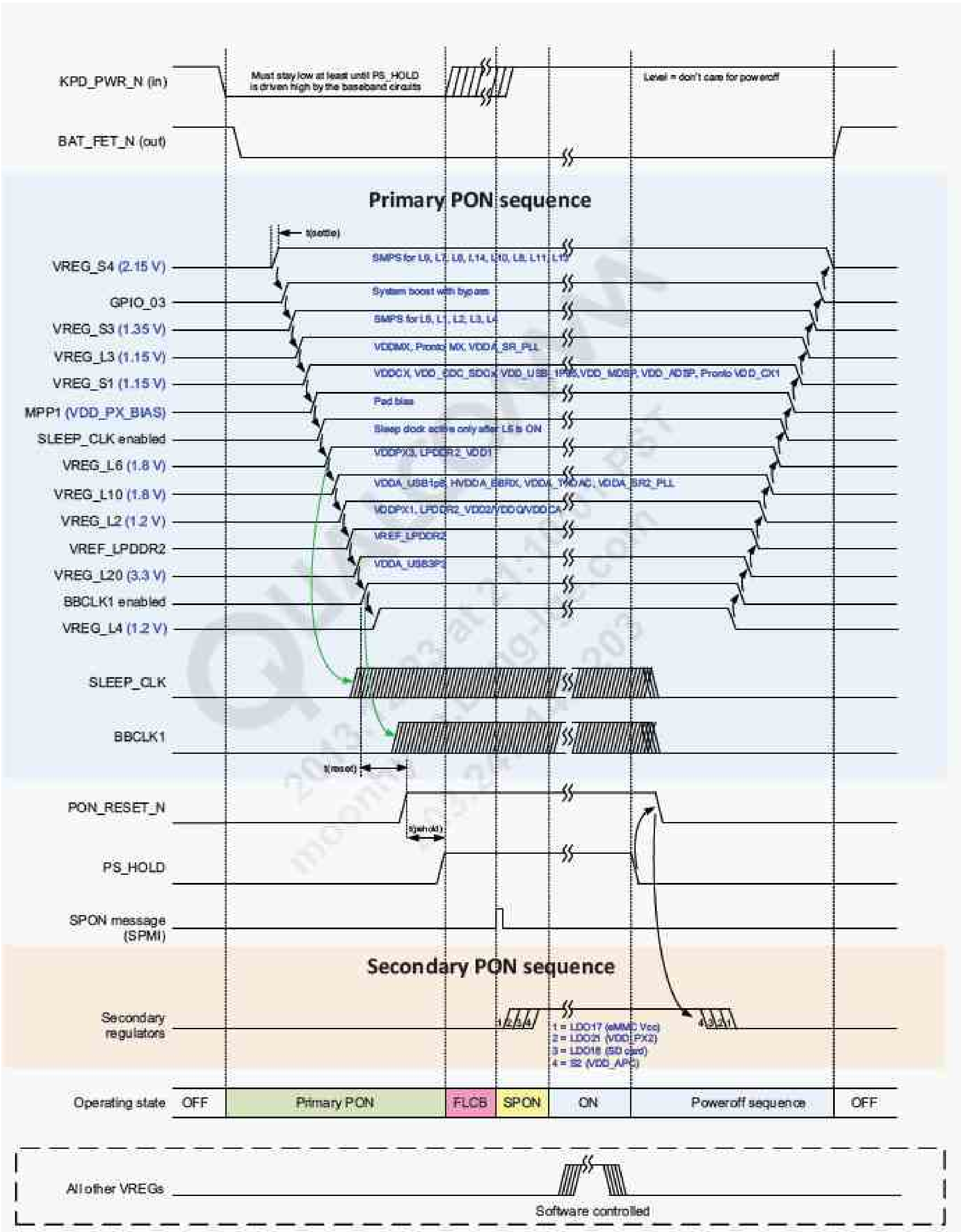


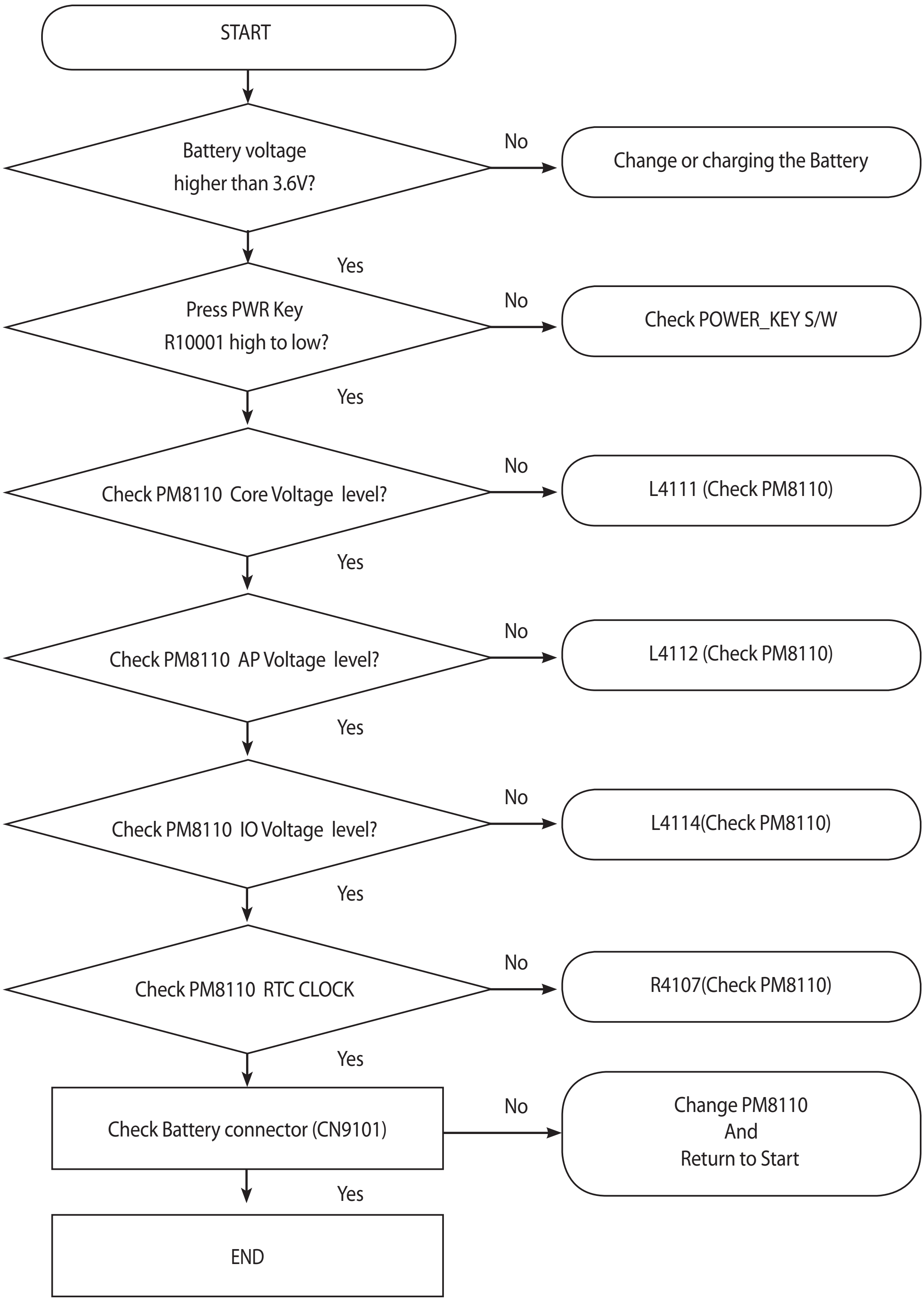
Figure 3.7.1 (c)

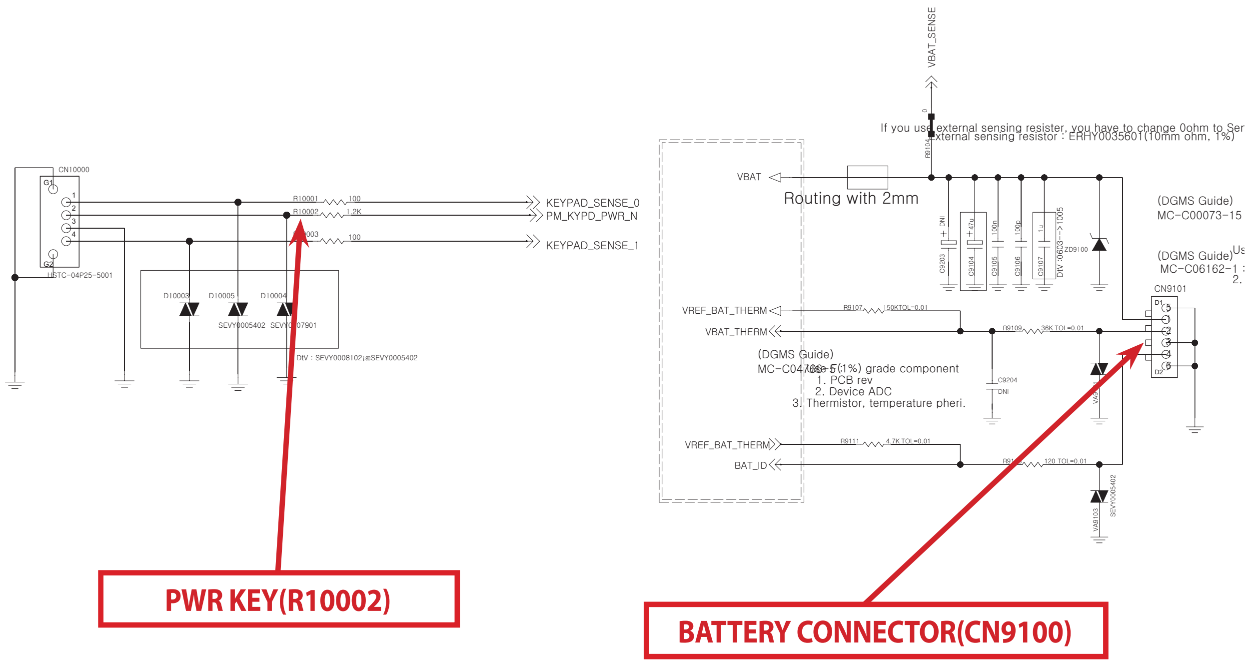
3.8. Power TS

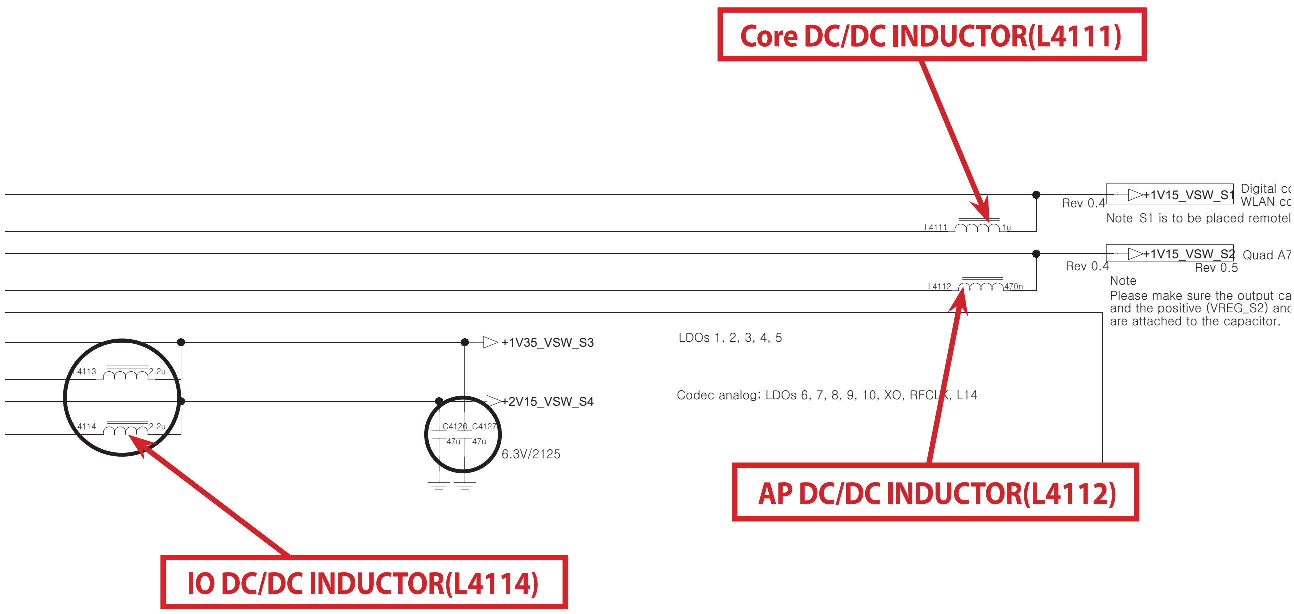
The main power source of D295 is provided by 1 chip which is PM8110

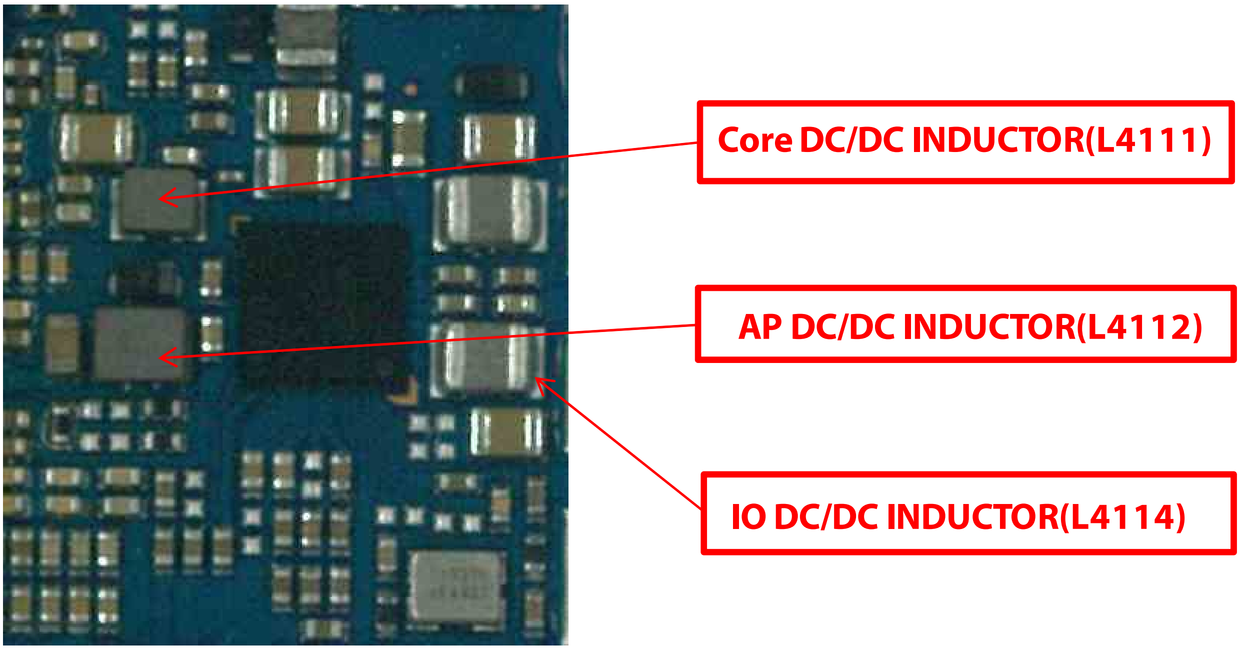
Power ON sequence of D295 is

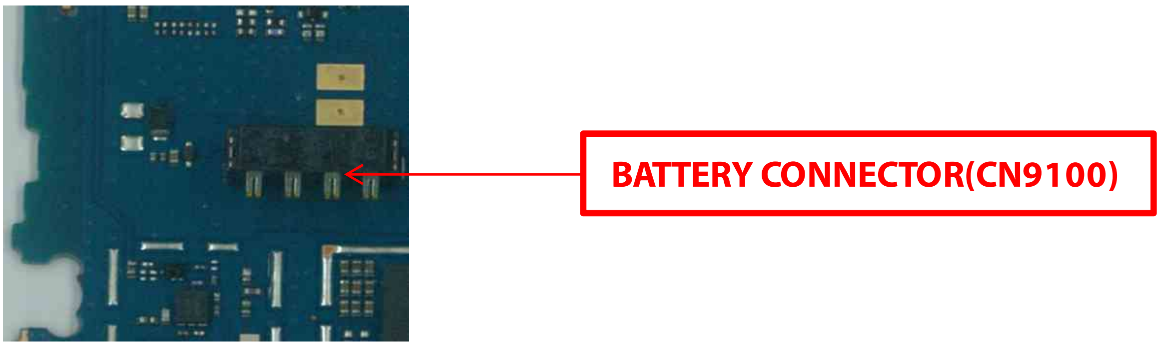


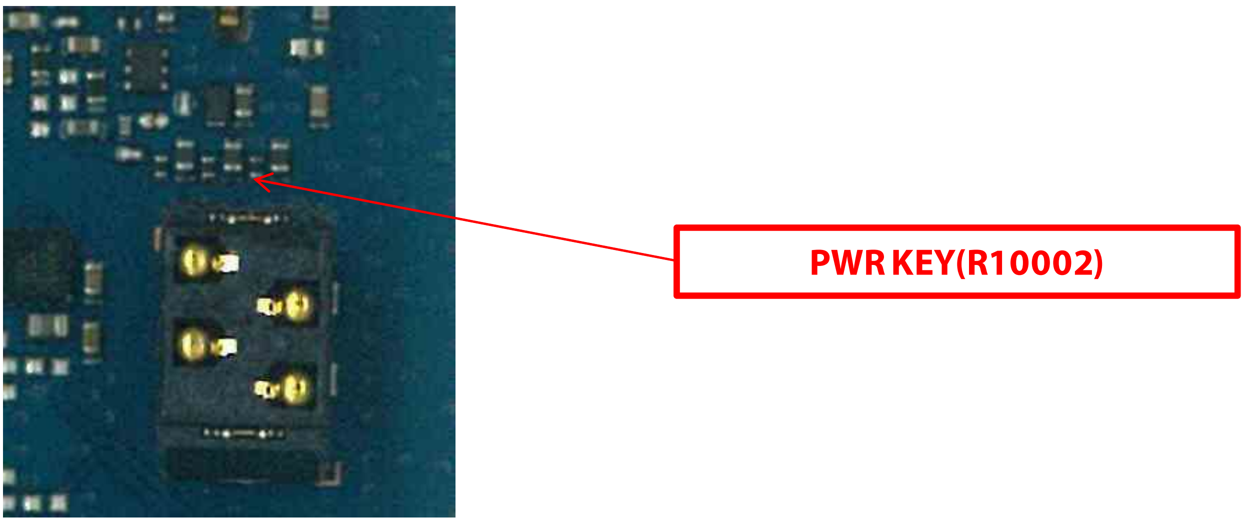

START
Battery voltage
higher than 3.6V?
No
Change or charging the Battery
Yes
Press PWR Key
R10001 high to low?
No
Check POWER_KEY S/W
Yes
Check PM8110 Core Voltage level?
No
L4111 (Check PM8110)
Yes
Check PM8110 AP Voltage level?
No
L4112 (Check PM8110)
Yes
Check PM8110 IO Voltage level?
No
L4114(Check PM8110)
Yes
Check PM8110 RTC CLOCK
No
R4107(Check PM8110)
Yes
Check Battery connector (CN9101)
No
Change PM8110
And
Return to Start
Yes
END


PWR KEY(R10002)
BATTERY CONNECTOR(CN9100)


Core DC/DC INDUCTOR(L4111)
AP DC/DC INDUCTOR(L4112)
IO DC/DC INDUCTOR(L4114)


Core DC/DC INDUCTOR(L4111)
AP DC/DC INDUCTOR(L4112)
IO DC/DC INDUCTOR(L4114)


BATTERY CONNECTOR(CN9100)

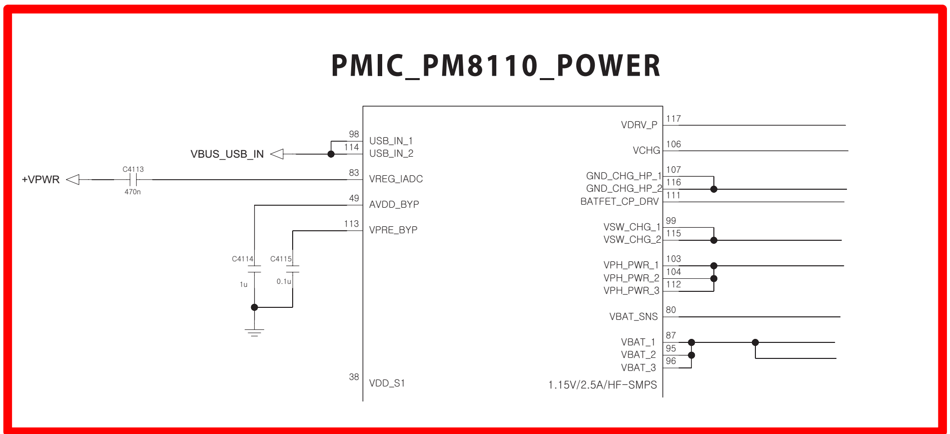

PWR KEY(R10002)

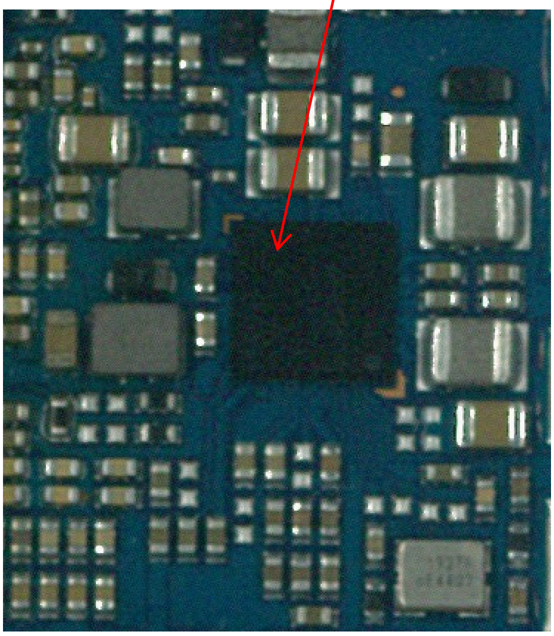
3.9 Charging TS

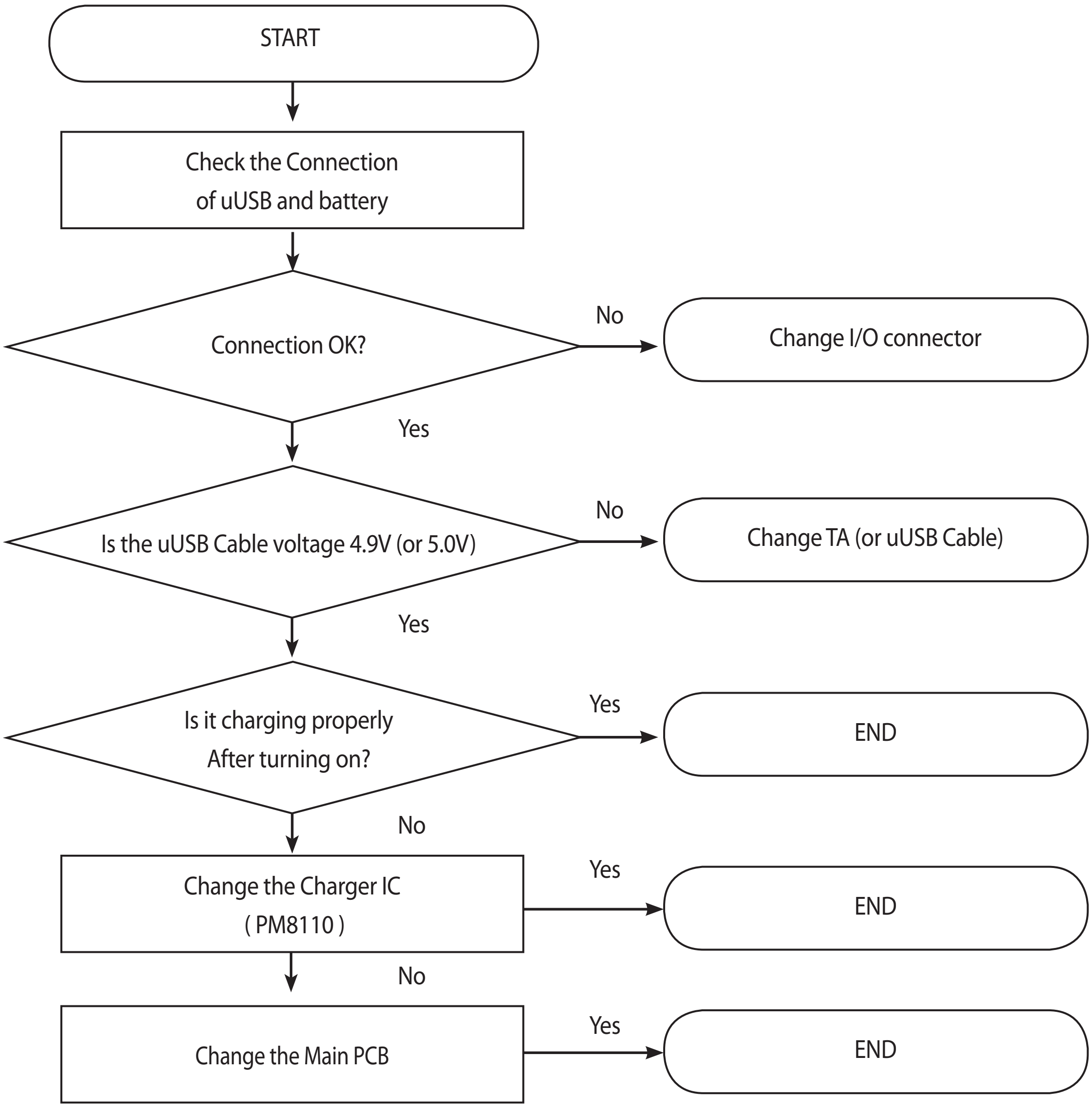
D295 uses PMIC(liner Charger) for charging control. The PMIC circuit includes the charging pass transistor detects the VDD and VBAT voltages, and its control driver. The PMIC maximum charging current is set around 700mA.

Charging Procedure
- Supply the VCHG use a TA or u-USB cable
- The PMIC control the charging current
- Charging current flows into the main battery

Troubleshooting Check Point
- Connection of the TA or u-USB cable
- Main battery

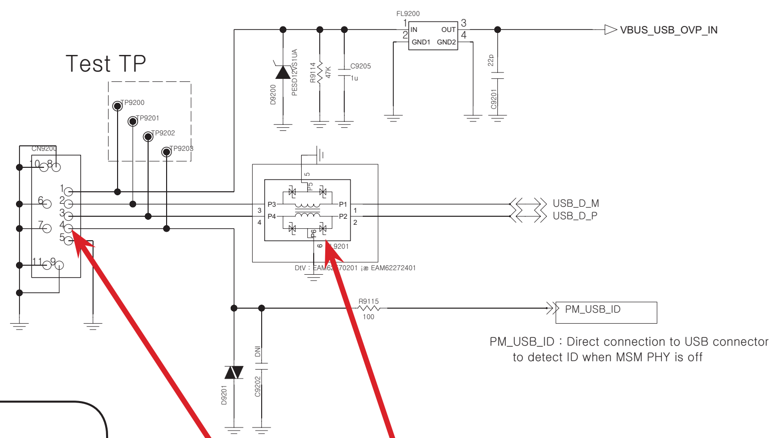





START
Check the Connection
of uUSB and battery
Connection OK?
No
Change I/O connector
Yes
Is the uUSB Cable voltage 4.9V (or 5.0V)
No
Change TA (or uUSB Cable)
Yes
Is it charging properly
After turning on?
Yes
END
No
Change the Charger IC
(PM8110)
Yes
END
No
Change the Main PCB
Yes
END

3.10. USB TS

< 9-2-1_IO_USB2.0 > Rev_0.5



START

Check the Connection of uUSB

Connection OK? — No → Change I/O connector(CN9200)

Yes

Main PCB Filter — No → Check Filter(FL9201)

Yes

Is the signal ok for USB+ & - ? — No → Change Filter(FL9201)

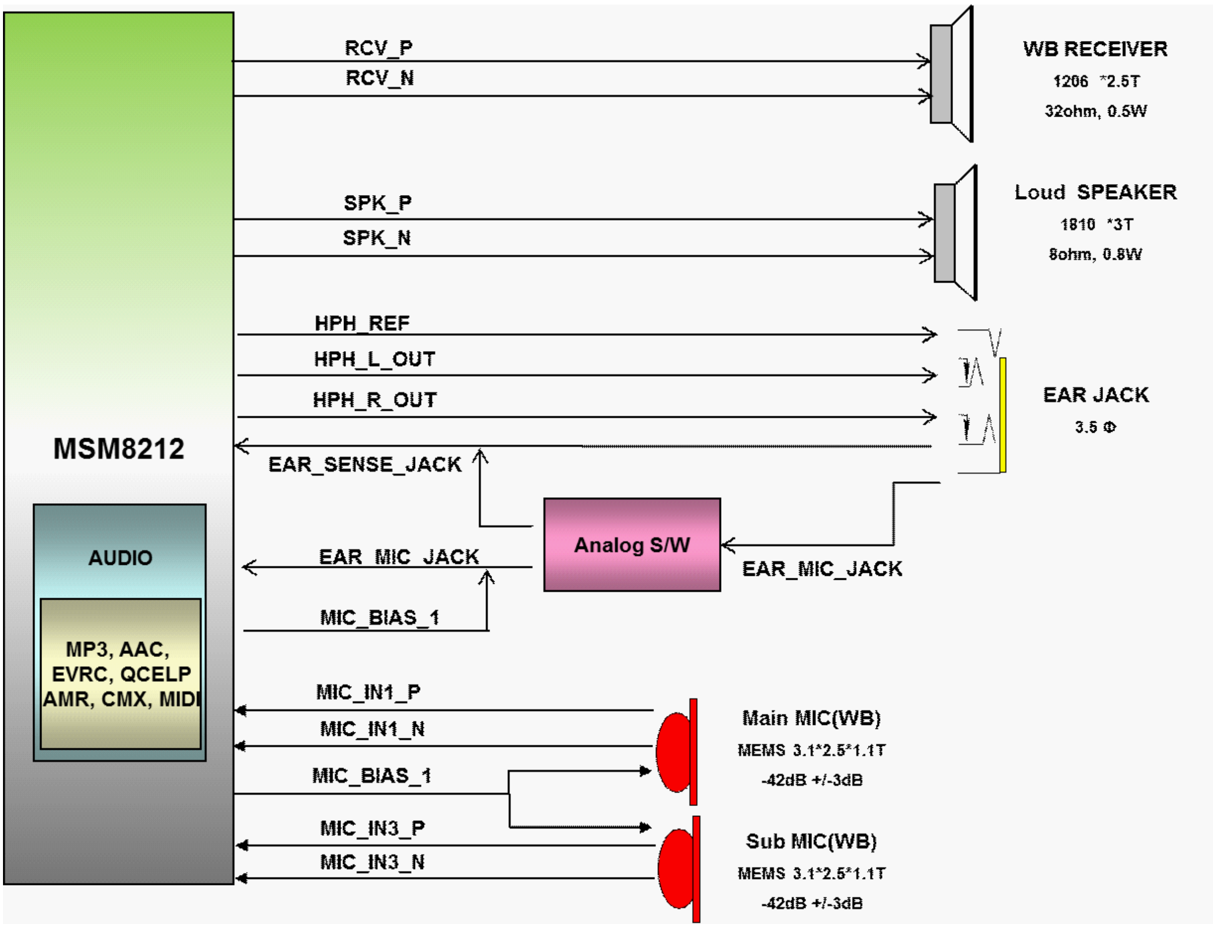
Yes

Change the Main PCB — Yes → END

3.11. Audio TS

3.11.1 Overview

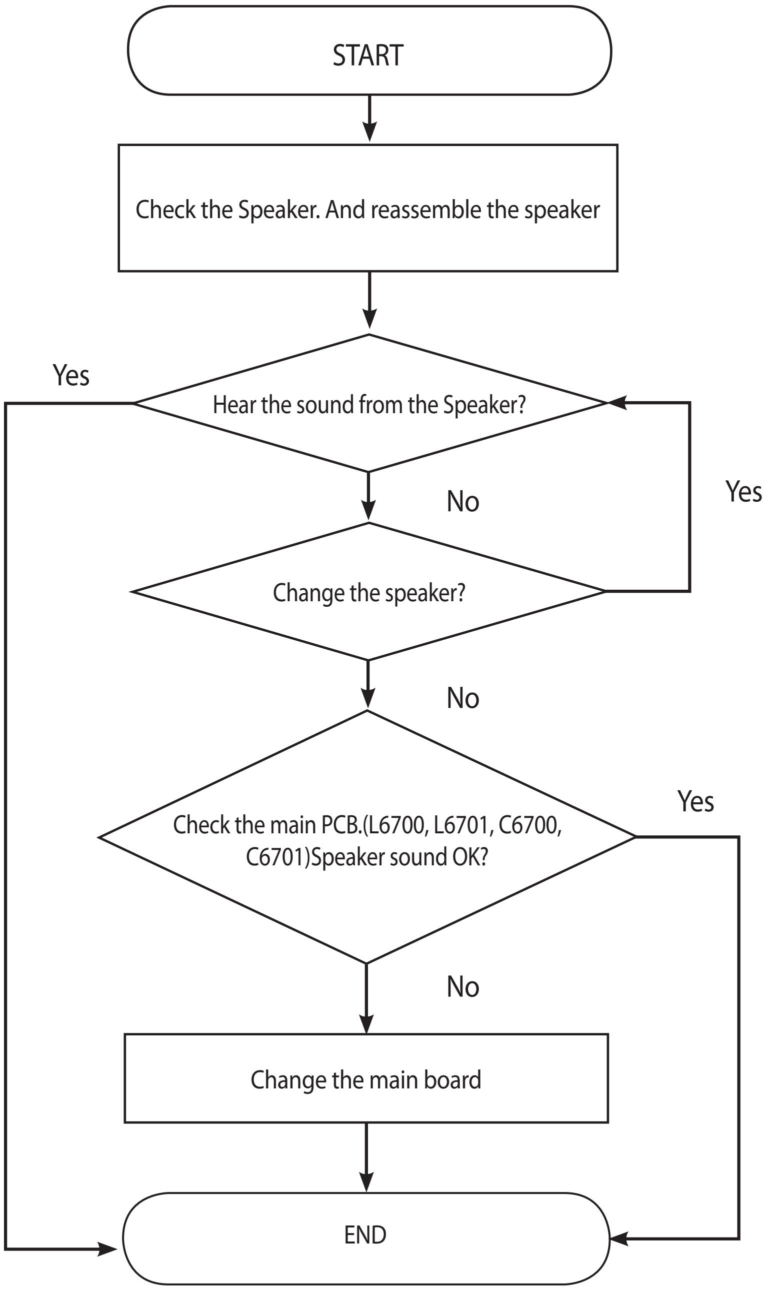
Audio Block Diagram

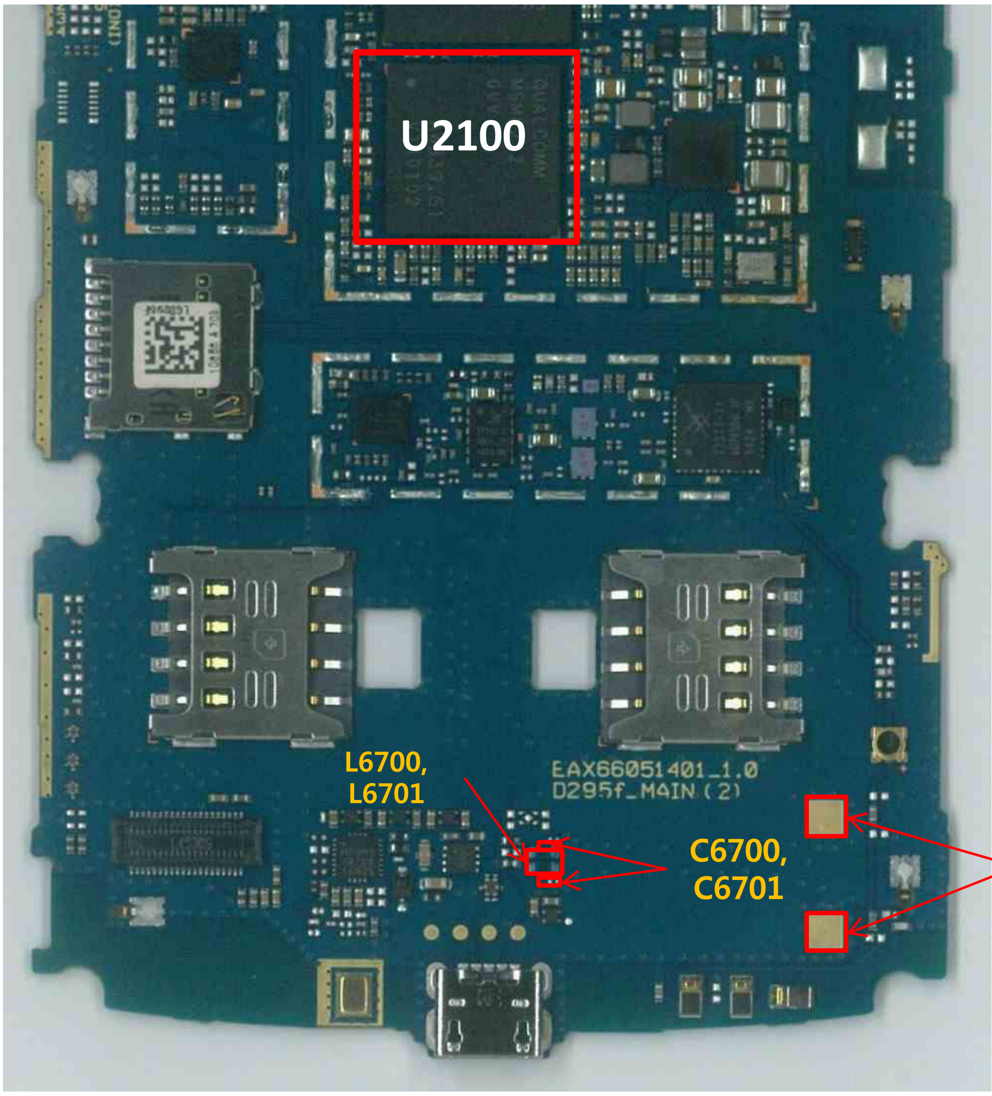


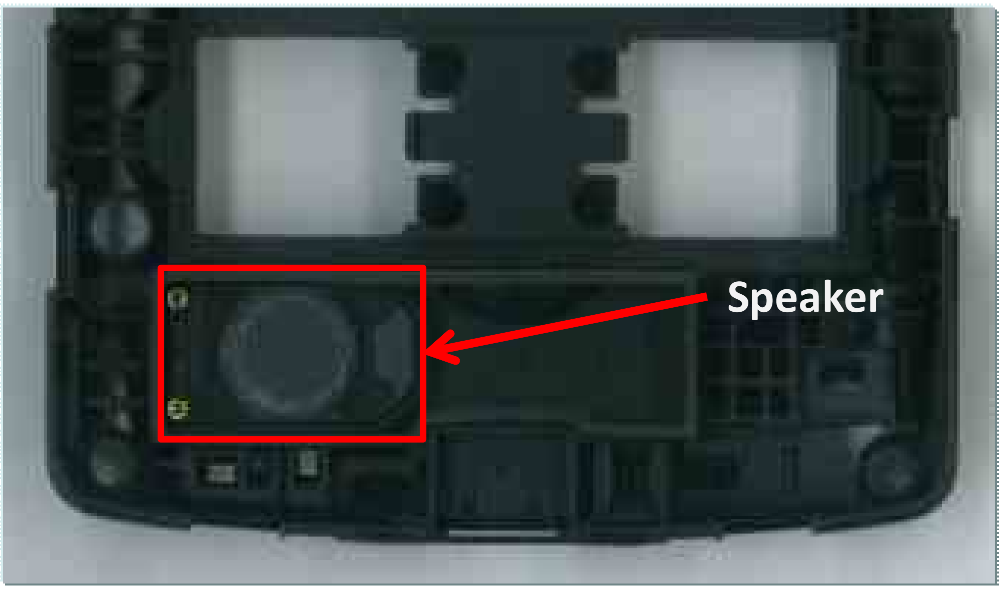
3.11.2 Speaker Trouble Shooting

It's trouble shooting guide for no sound case.

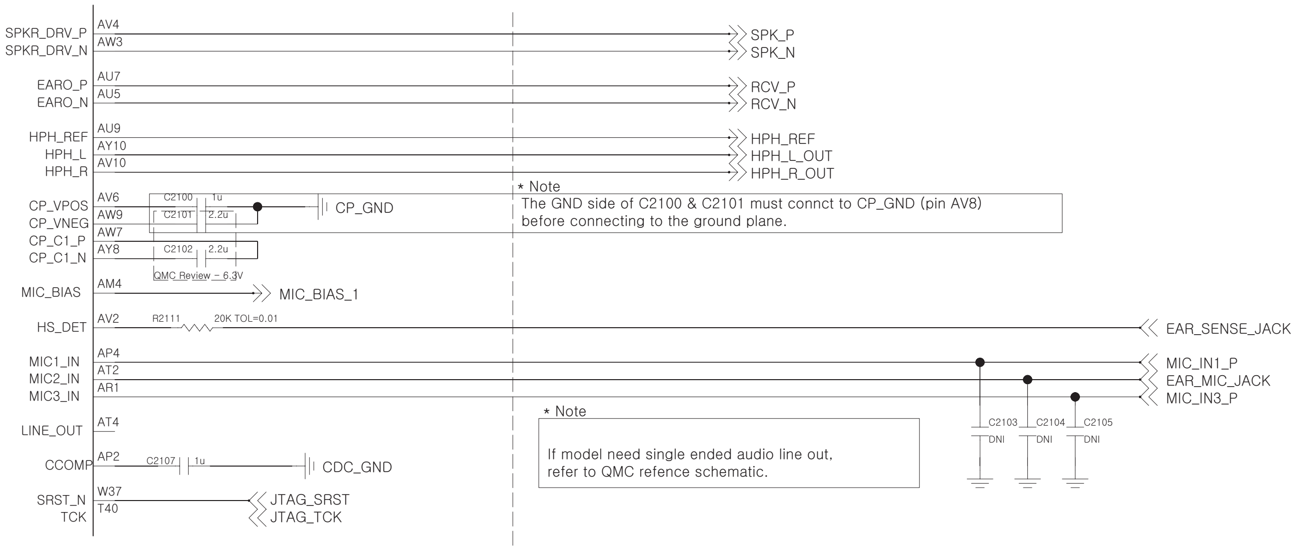
Speaker & Receiver control signals are generate by MSM8212 (U2100)If low level volume or breaking the sound, change the speaker



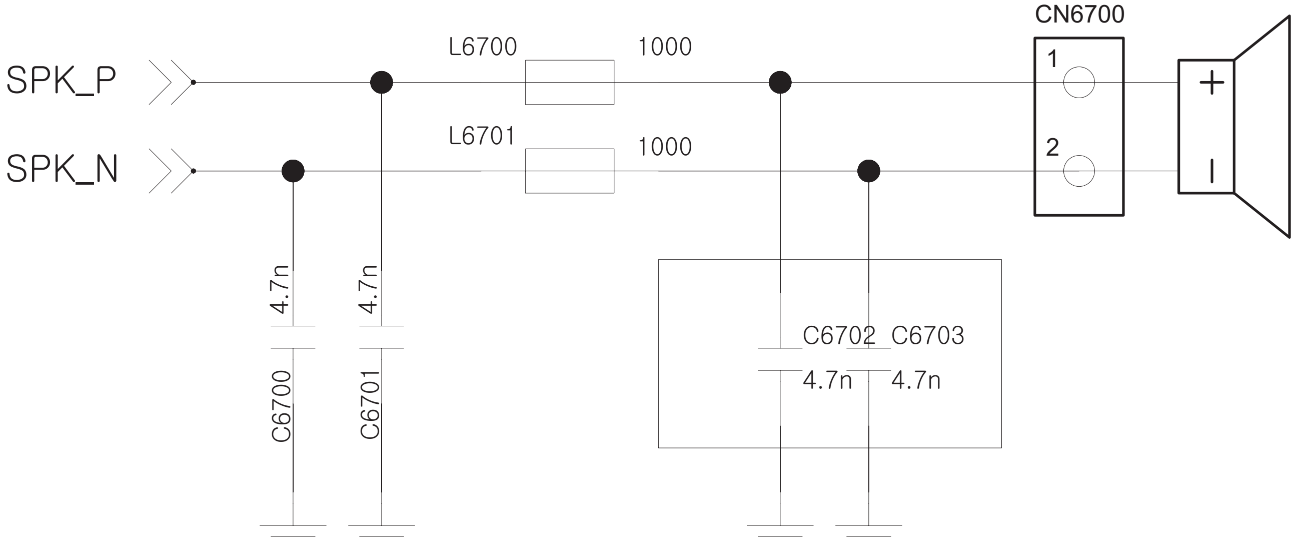

U2100
L6700,
L6701
EAX66051401_1.0
D295f_MAIN (2)
C6700,
C6701
Speaker PAD
(CN 6700)


Speaker

Main PCB



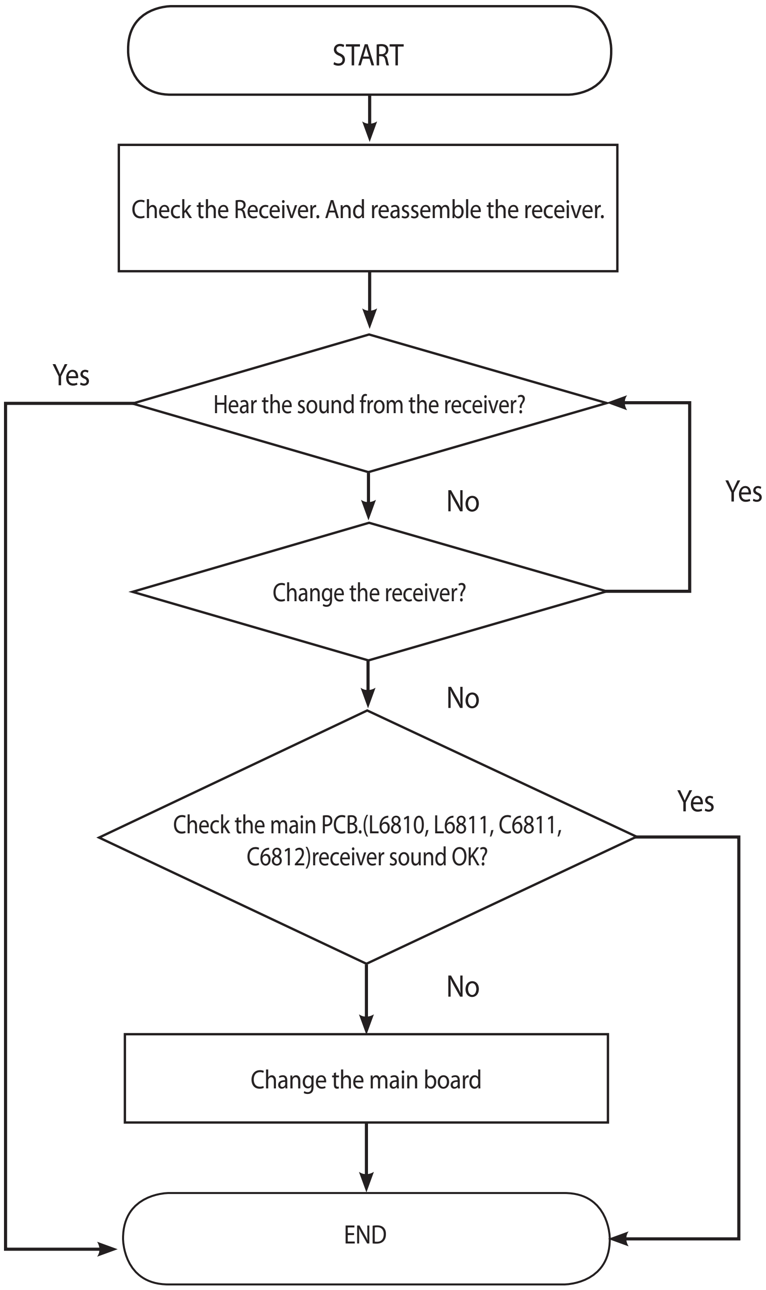
Speaker

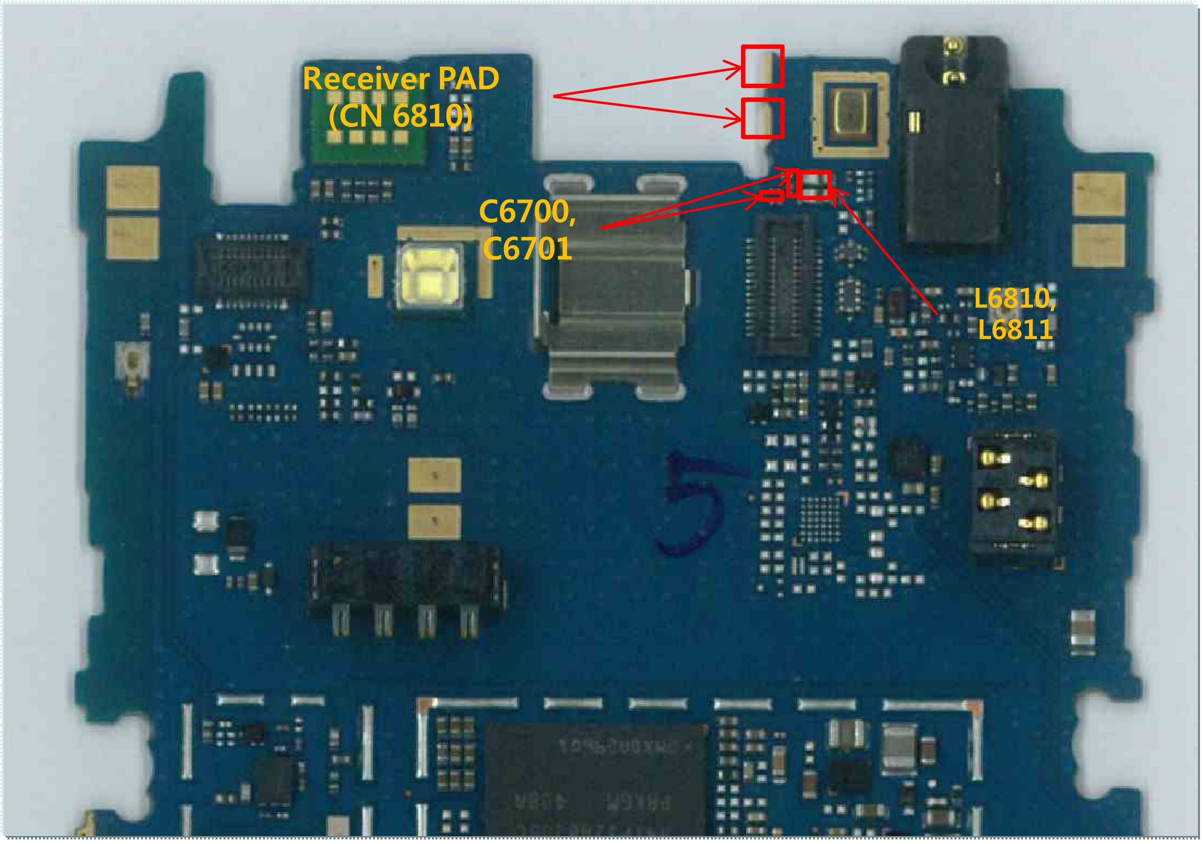


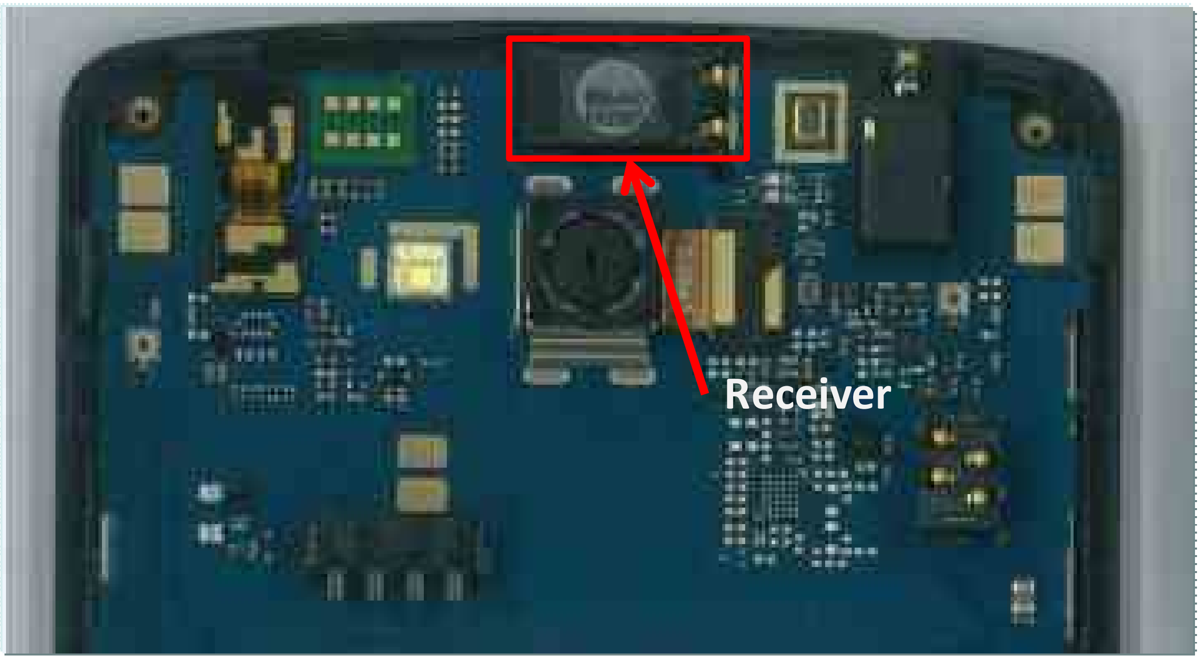
3.11.3 Receiver Trouble Shooting

It's trouble shooting guide for no sound case.

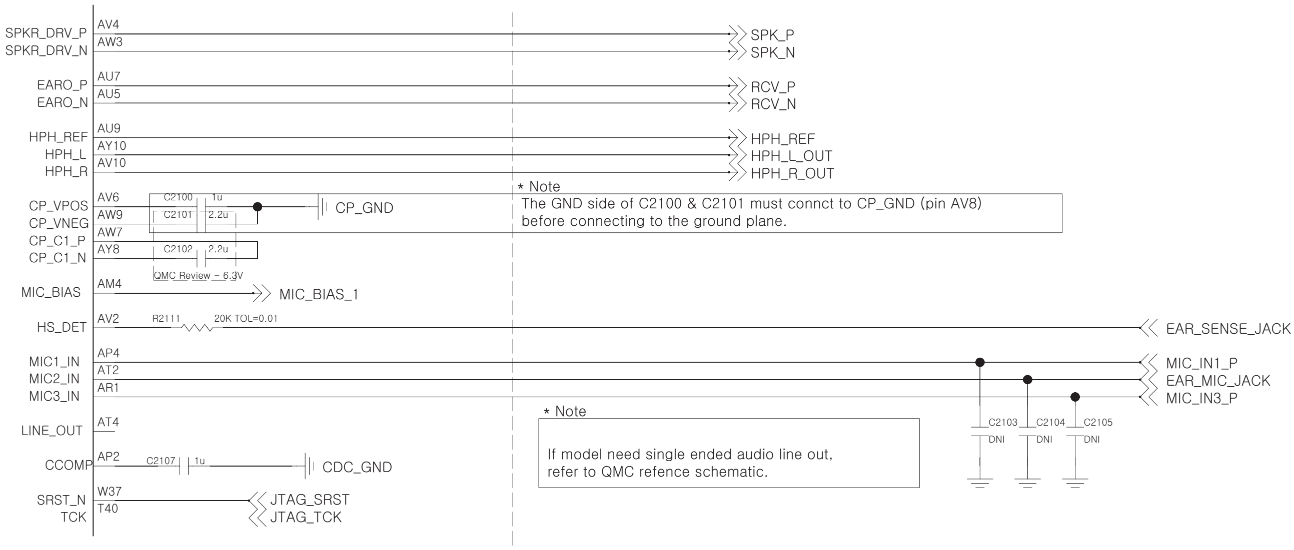
Speaker & Receiver control signals are generate by MSM8212 (U2100)If low level volume or breaking the sound, change the Receiver



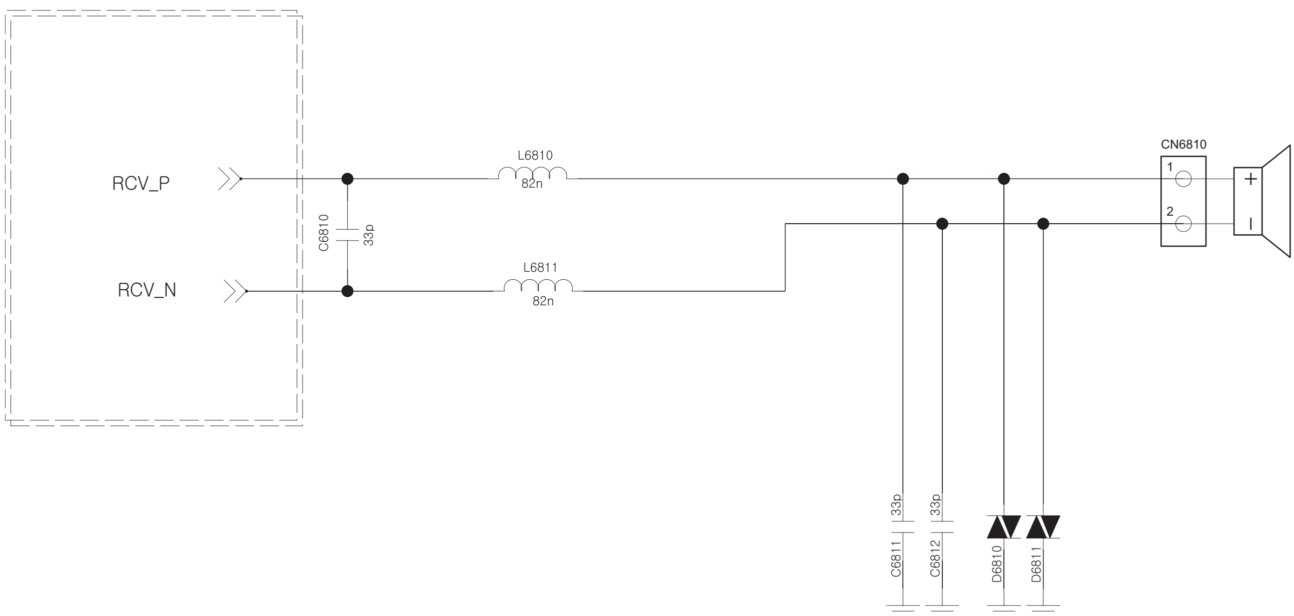

Receiver PAD
(CN 6810)
C6700,
C6701
L6810,
L6811


Receiver

Main PCB

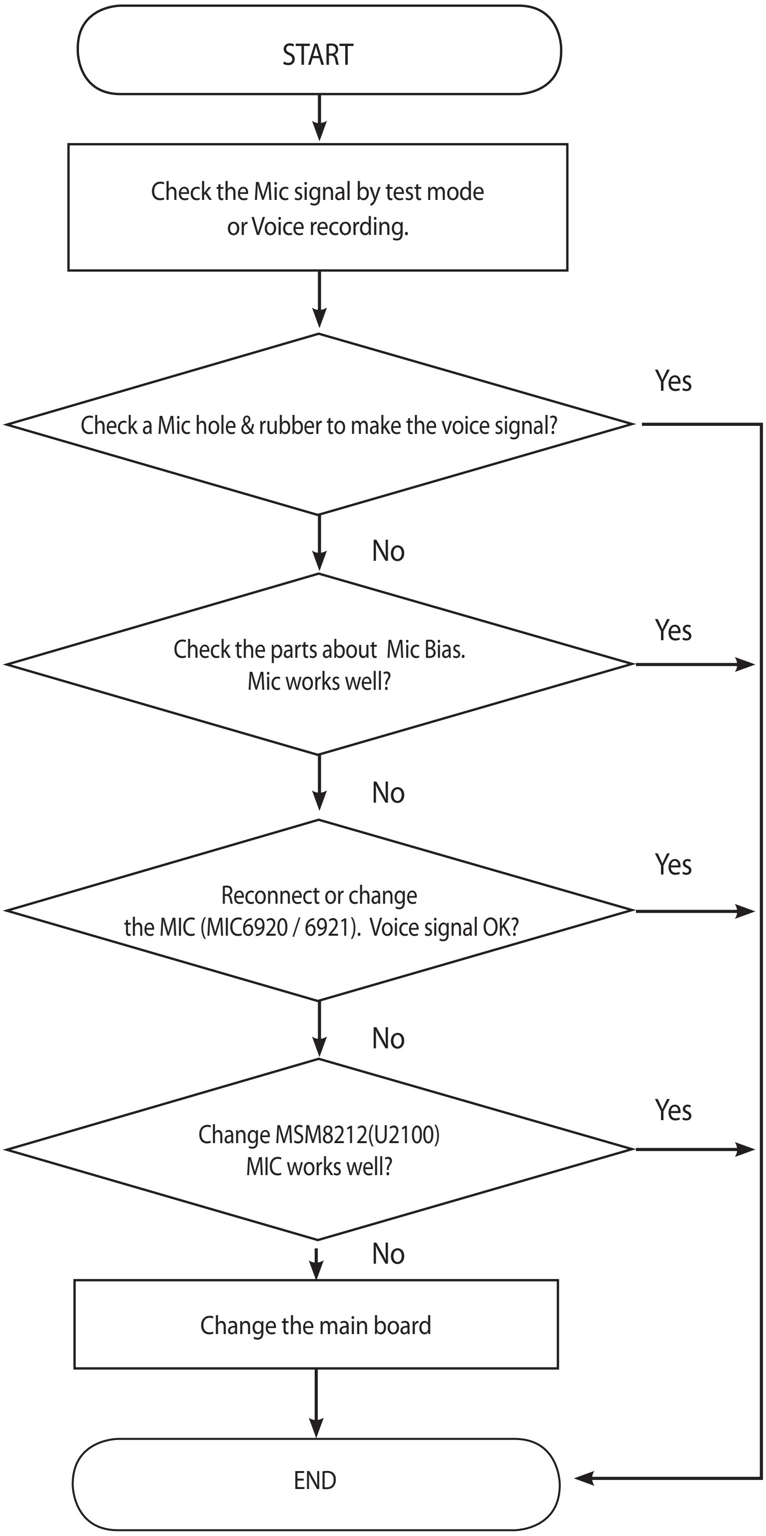


Receiver

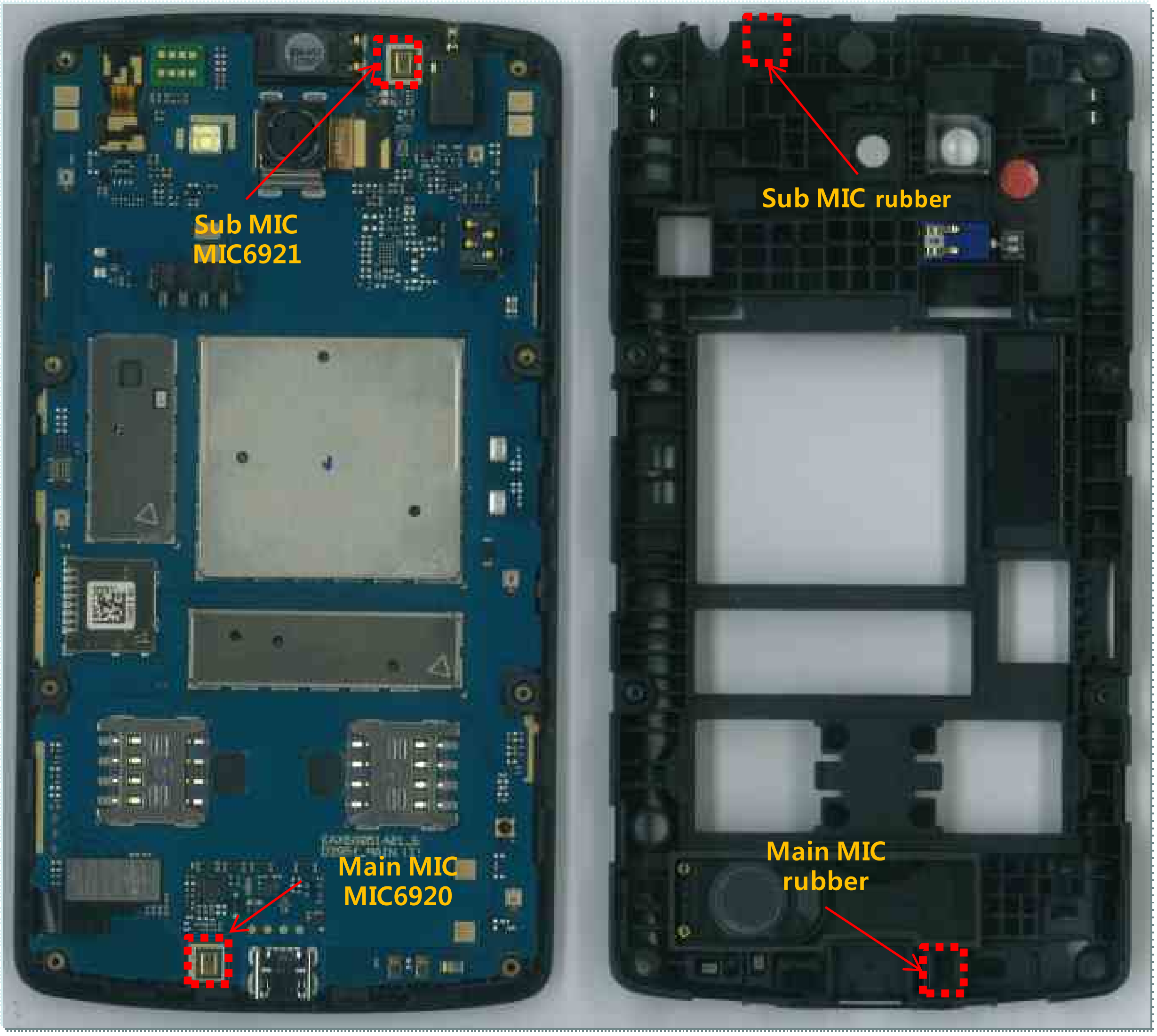


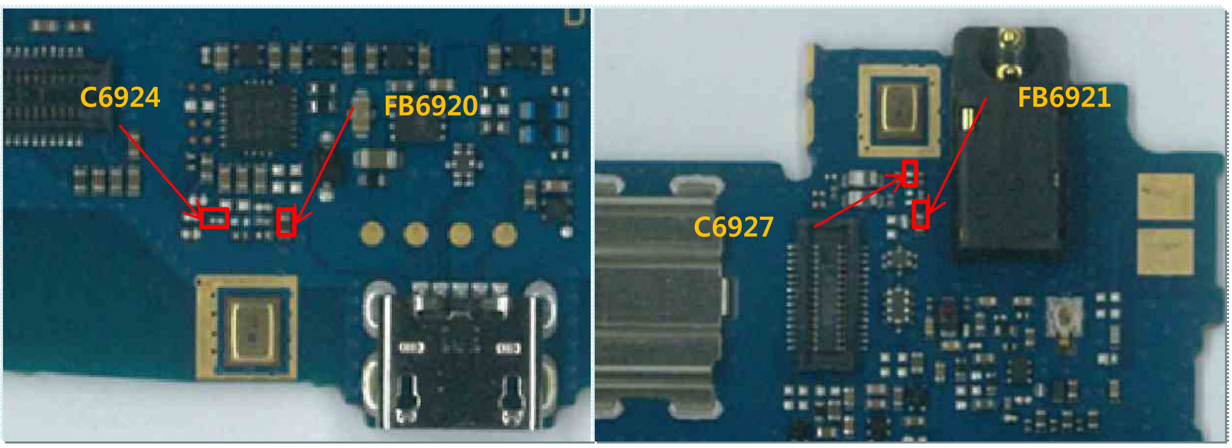
3.11.4 Main MIC / Sub MIC Trouble Shooting

It's operating voice call(including speaker phone), voice recording and camcorder recording.



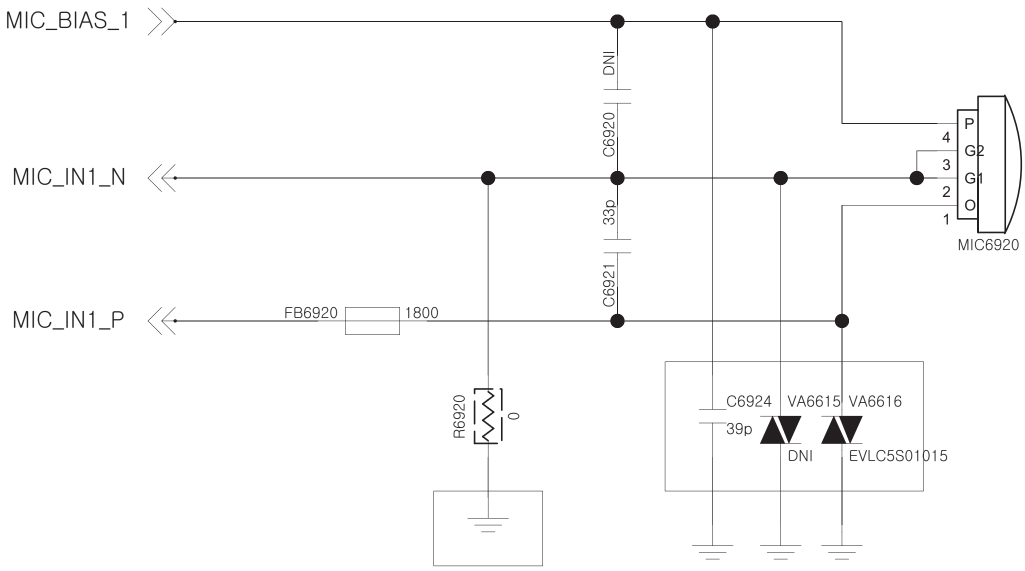
Check a Main Mic hole.



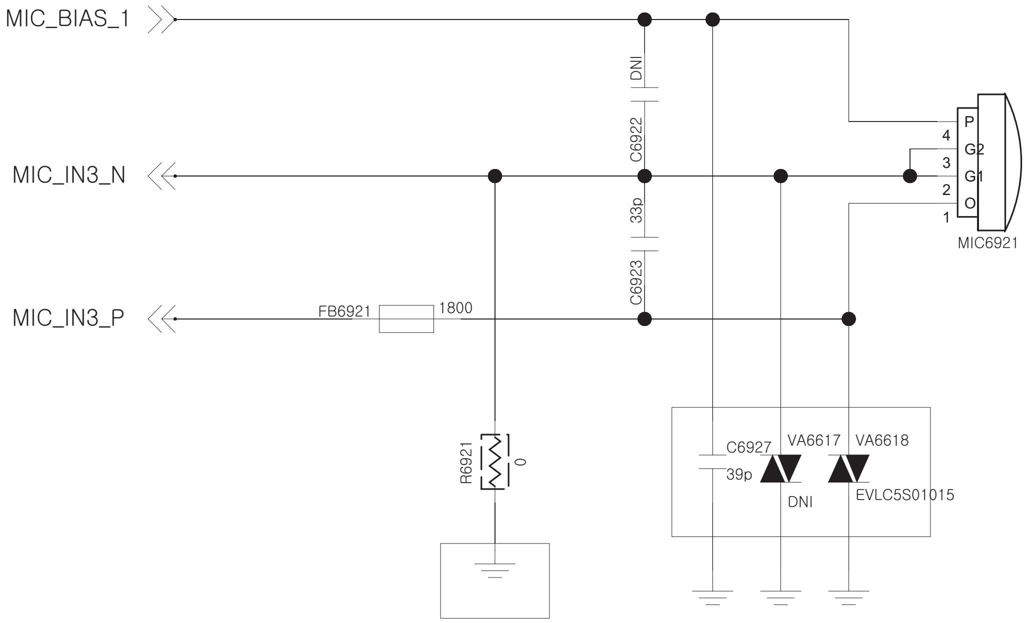


Check the MIC bias/output level

Main MIC



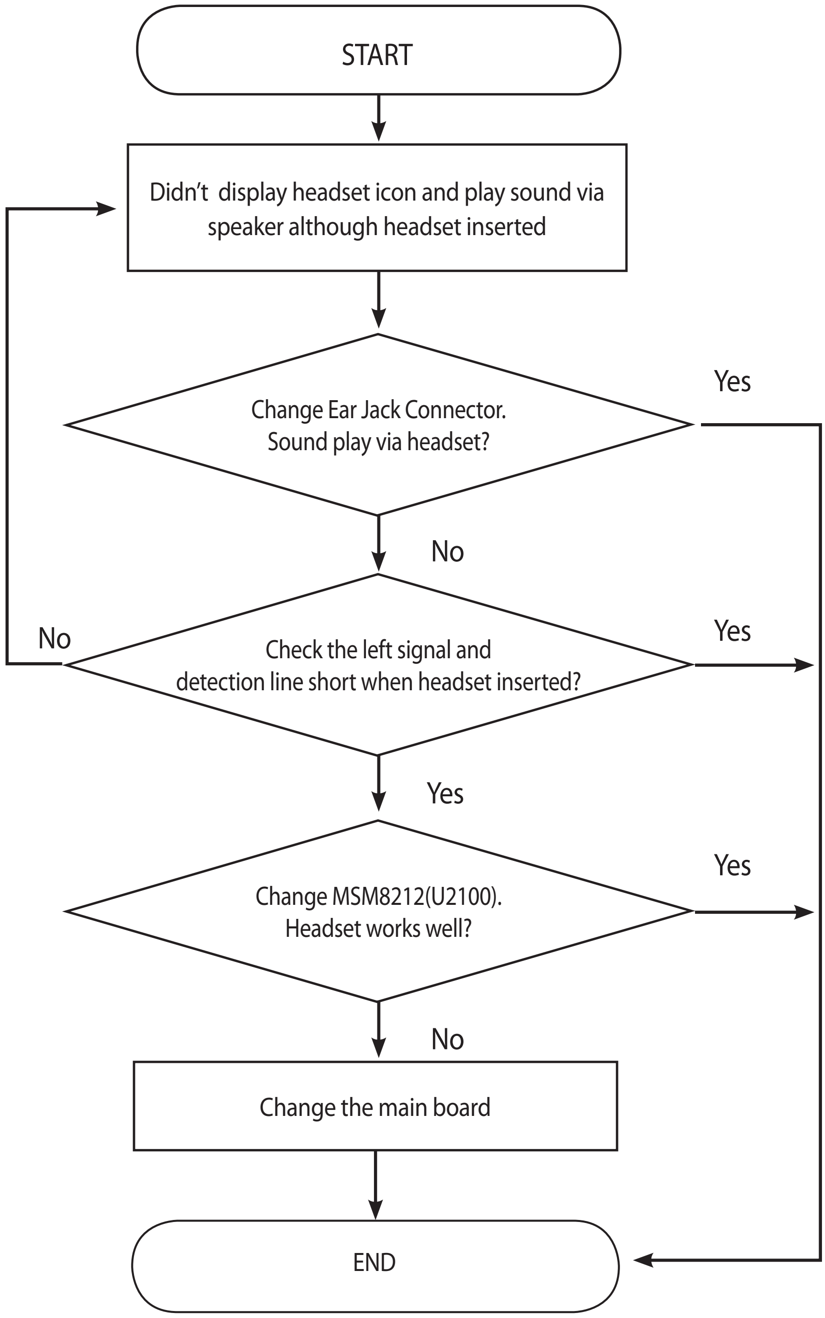
Sub MIC

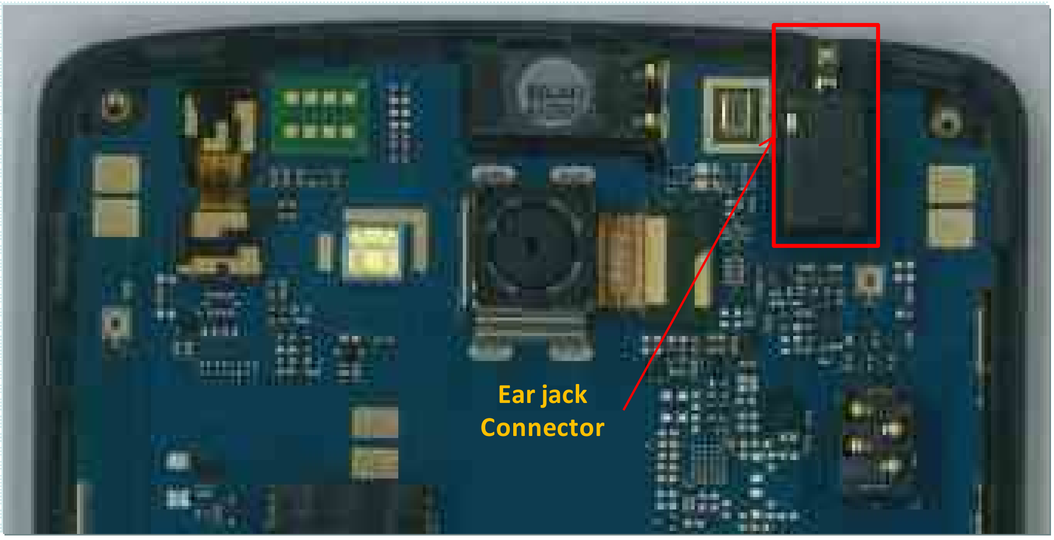


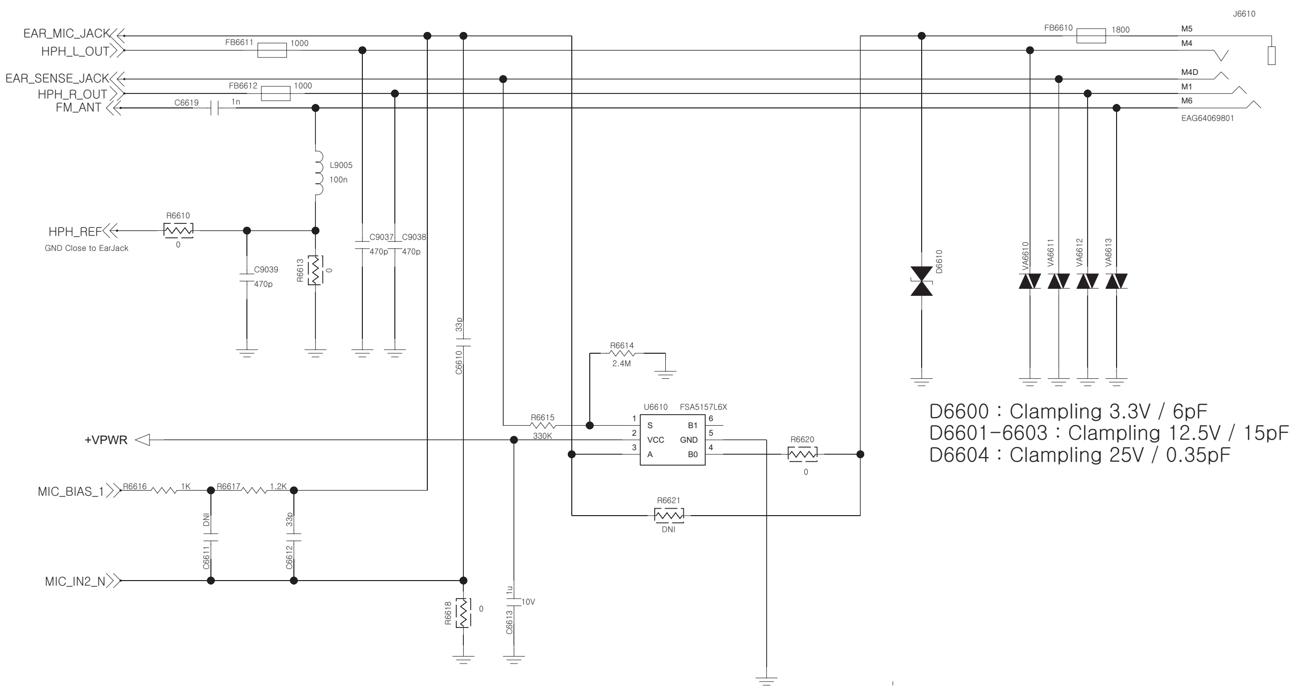
3.11.5. Ear-Mic Trouble Shooting

Ear MIC control signals are generate by MSM8212(U2100)

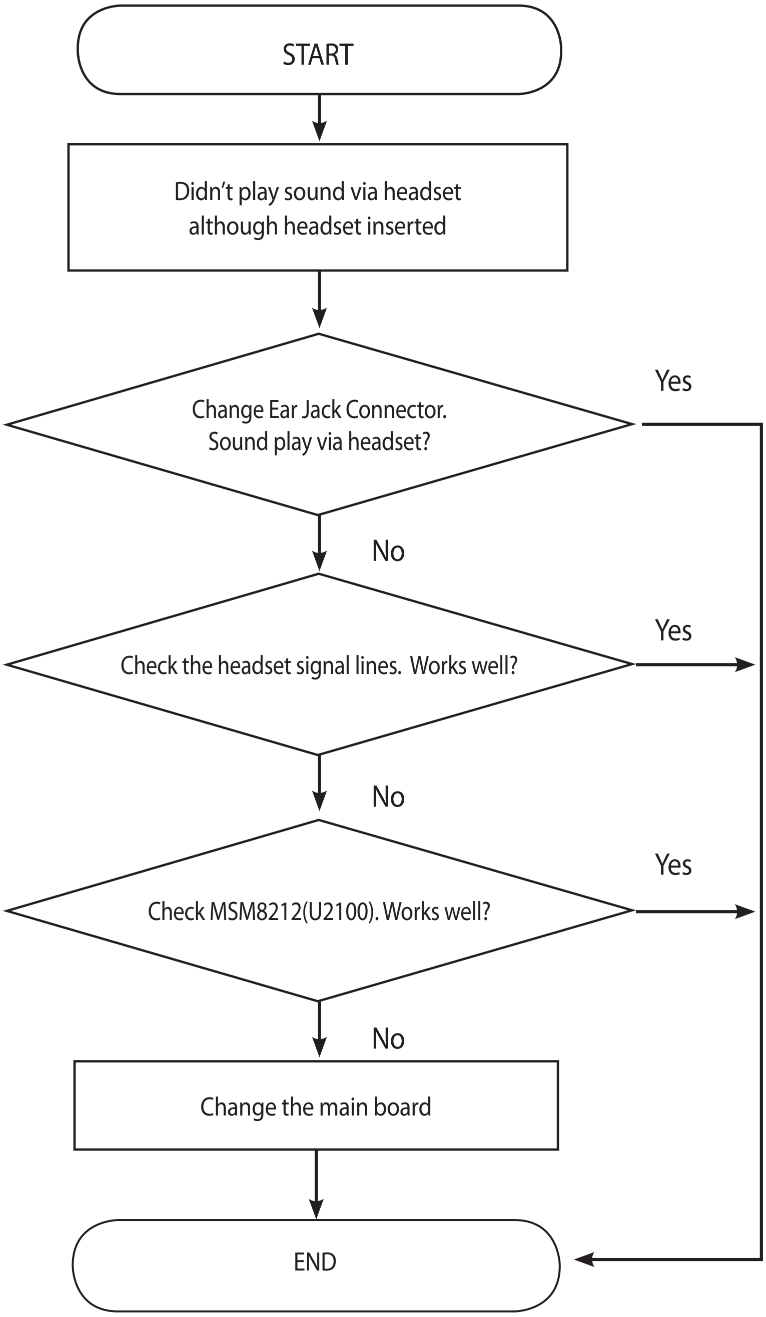
Case A. Disable detected headset insert.

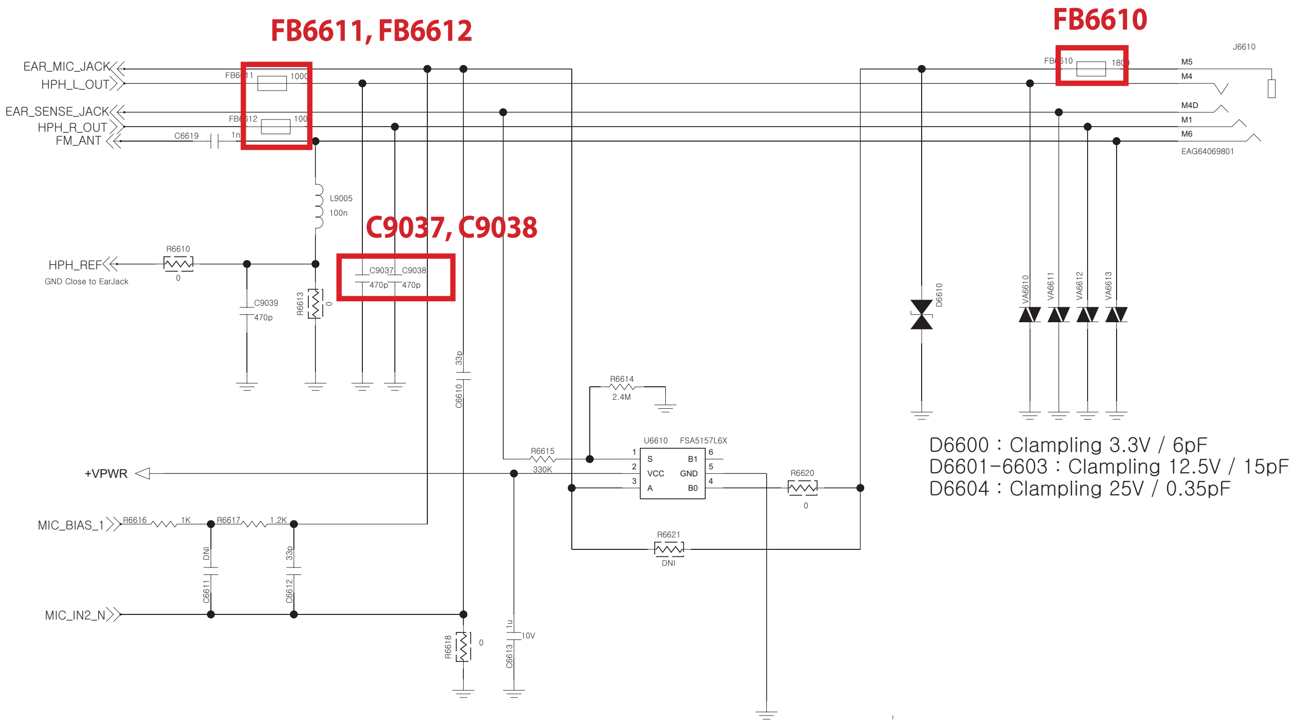


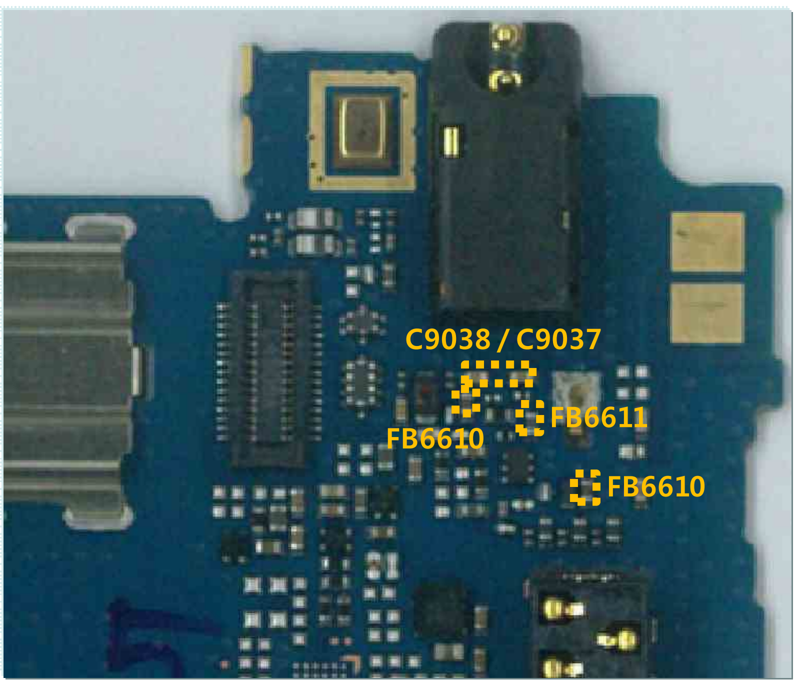

Ear jack
Connector


EAR_MIC_JACK
HPH_L_OUT
EAR_SENSE_JACK
HPH_R_OUT
FM_ANT
HPH_REF
GND Close to EarJack
+VPWR
MIC_BIAS_1
MIC_IN2_N
U6610 FSA5157L6X
J6610
EAG64069801
D6600 : Clampling 3.3V / 6pF
D6601~6603 : Clampling 12.5V / 15pF
D6604 : Clampling 25V / 0.35pF

Case B. No Sound from Headset







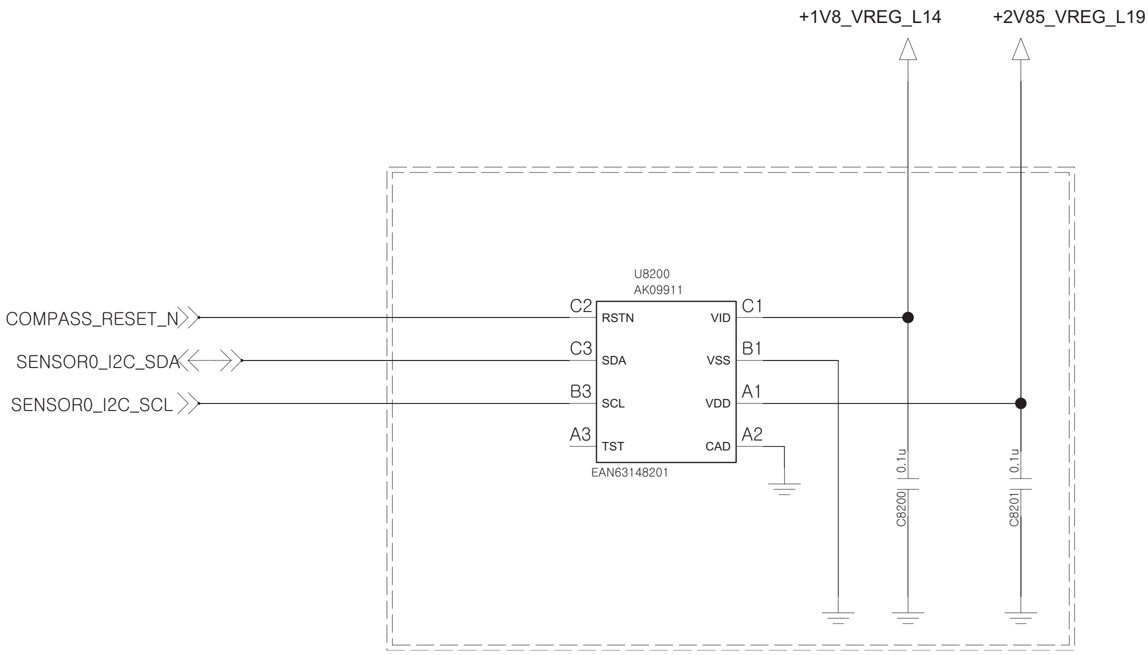
Check the headset MIC bias. (FB6610)

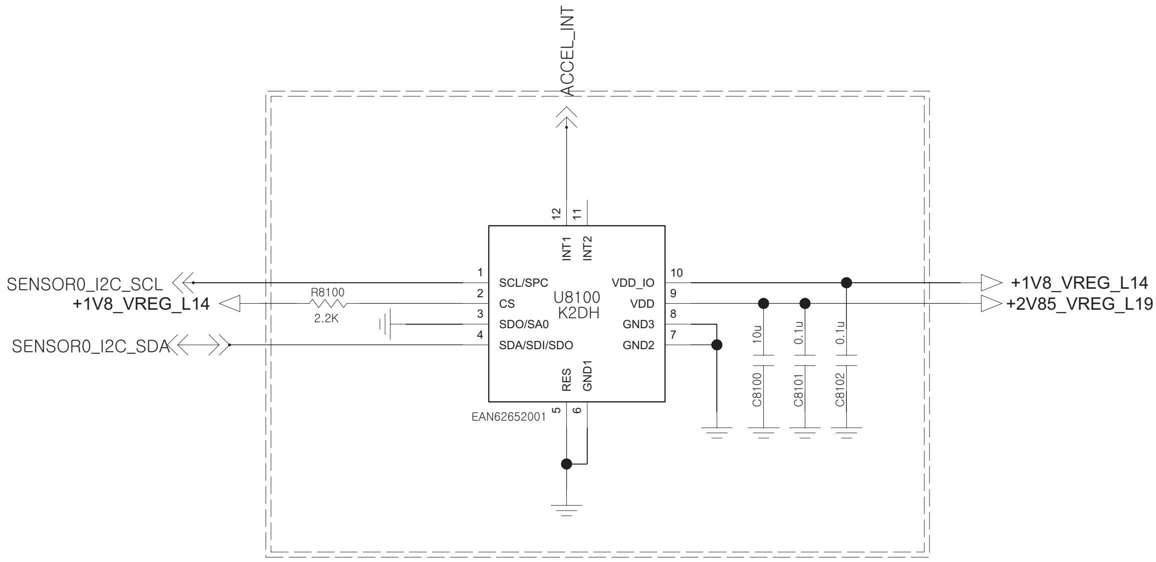
In case of you can't check the headset signal lines (FB6611, FB6612)

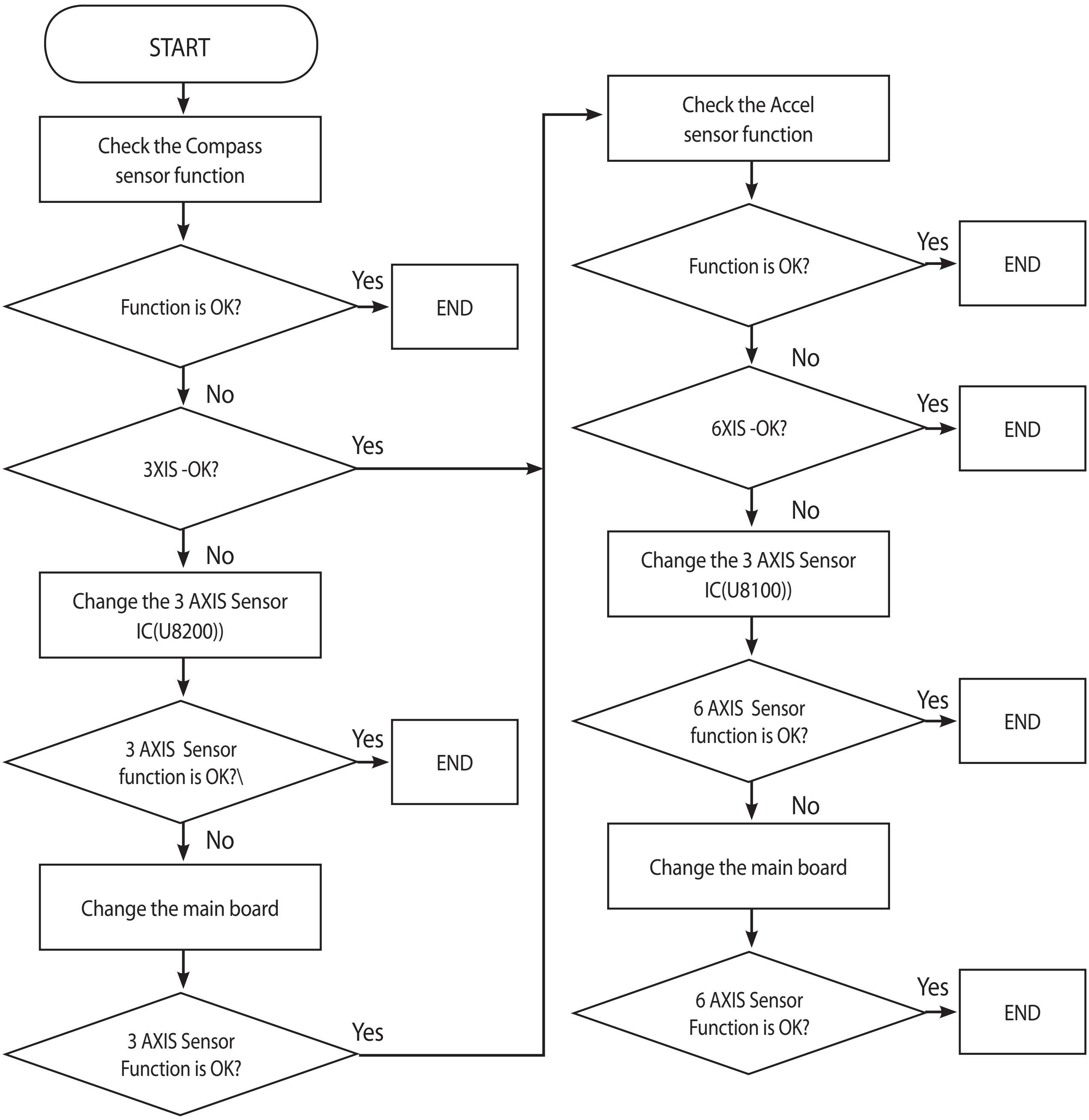
Can heard sound but sound is abnormal, check C6922, C6923

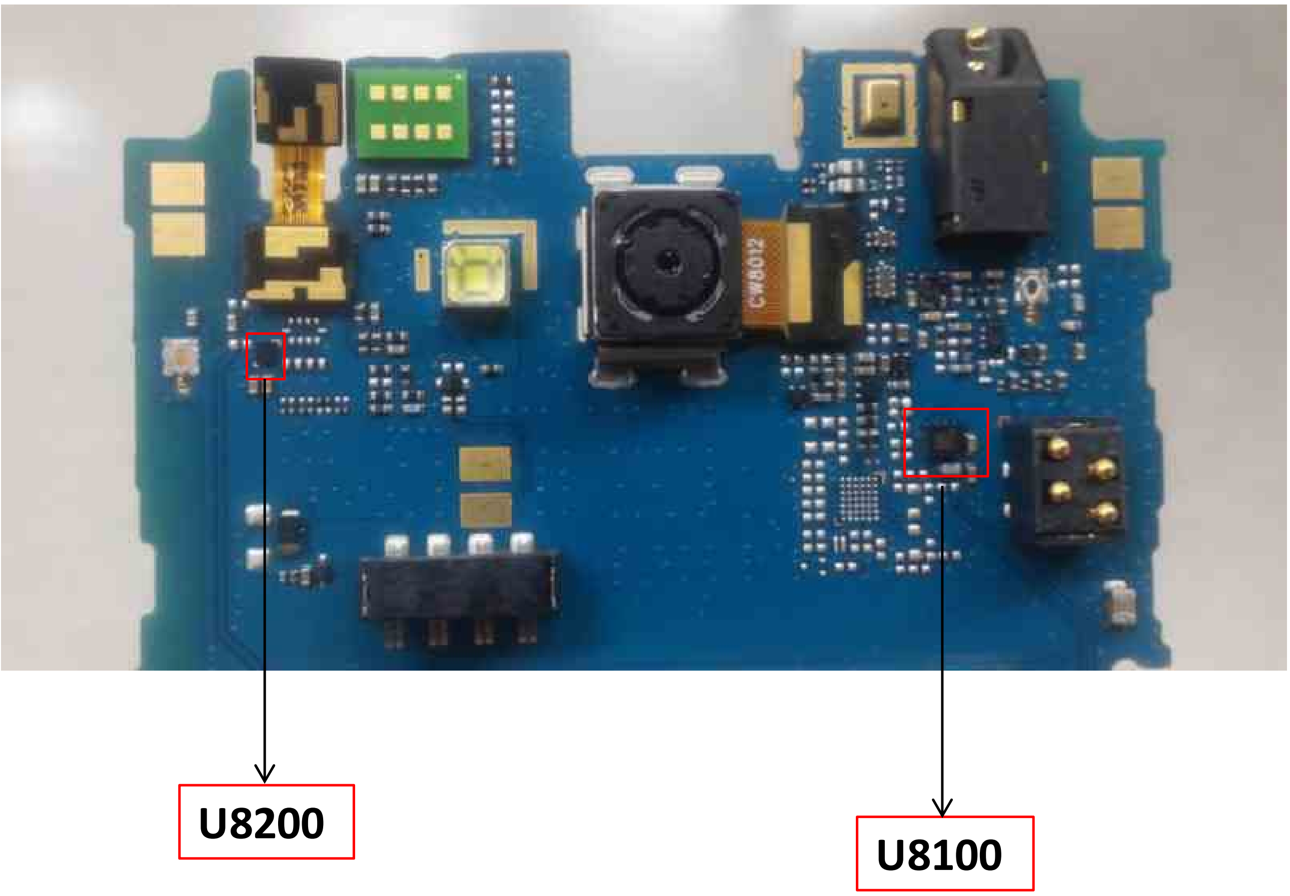
3.12. AXIS Sensor on/off TS

The **6 AXIS** sensor is calibrated by the sensor data using SW algorithm.



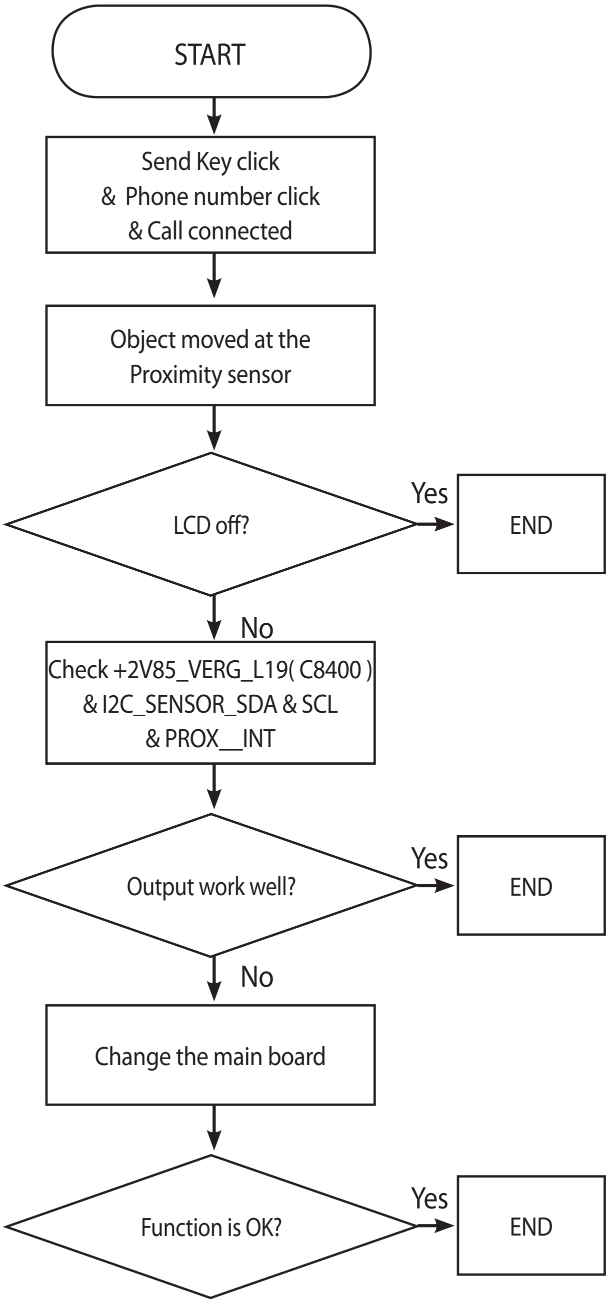


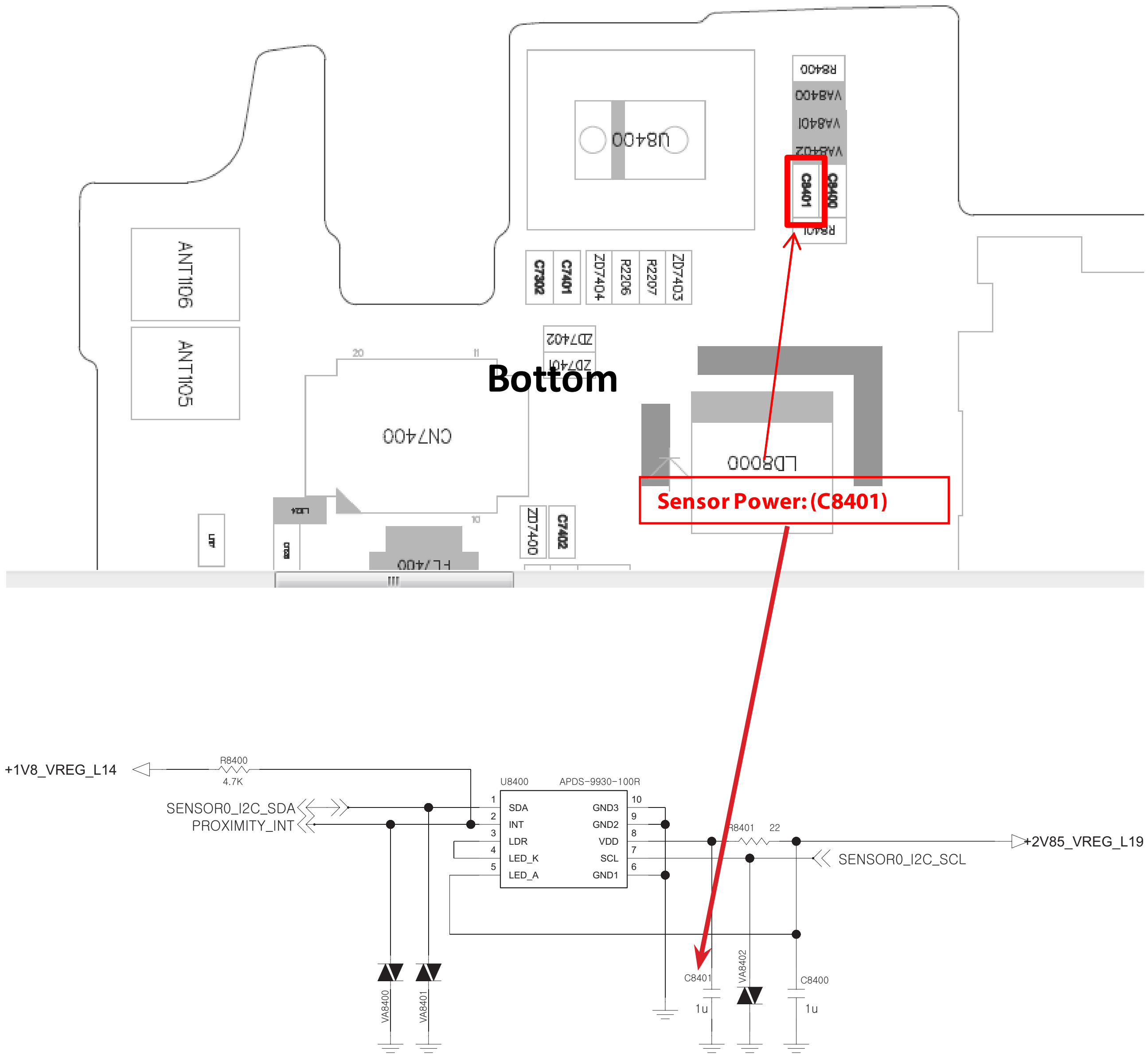

START
Check the Compass sensor function
Function is OK?
Yes
END
No
3XIS -OK?
Yes
No
Change the 3 AXIS Sensor IC(U8200))
3 AXIS Sensor function is OK?\
Yes
END
No
Change the main board
3 AXIS Sensor Function is OK?
Yes
Check the Accel sensor function
Function is OK?
Yes
END
No
6XIS -OK?
Yes
END
No
Change the 3 AXIS Sensor IC(U8100))
6 AXIS Sensor function is OK?
Yes
END
No
Change the main board
6 AXIS Sensor Function is OK?
Yes
END


U8200
U8100

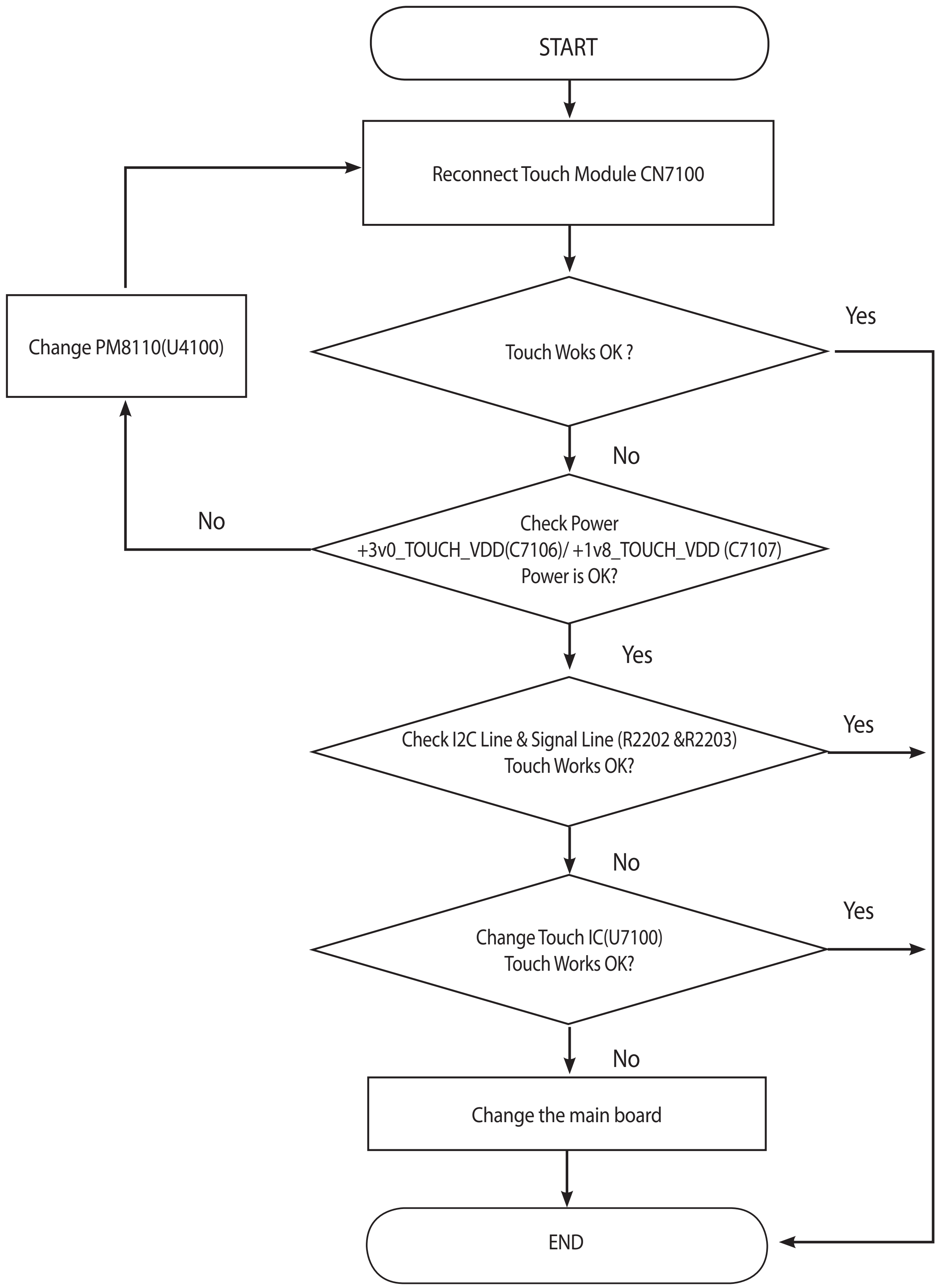
3.13 Proximity Sensor on/off TS

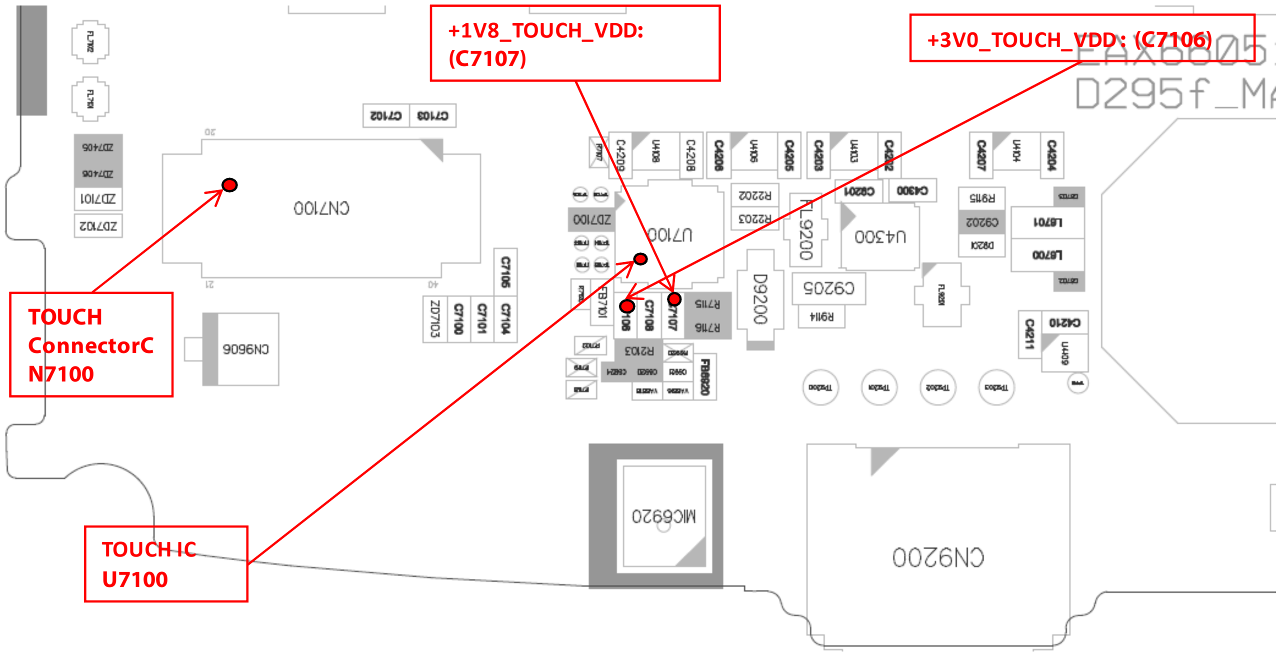
Proximity Sensor is worked as below:
Send Key click -> Phone number click -> Call connected -> Object moved at the sensor
-> Control the screen's on/off operation automatically




ANT1106
ANT1105
C7302
C7401
ZD7404
R2206
R2207
ZD7403
C8401
C8400
Bottom
Sensor Power: (C8401)
ZD7400
C7402
+1V8_VREG_L14
R8400
4.7K
SENSOR0_I2C_SDA
PROXIMITY_INT
U8400
APDS-9930-100R
SDA
INT
LDR
LED_K
LED_A
GND3
GND2
VDD
SCL
GND1
R8401
22
+2V85_VREG_L19
SENSOR0_I2C_SCL
C8401
1u
VA8402
C8400
1u
VA8400
VA8401

3.14. Touch Sensor Trouble Shooting

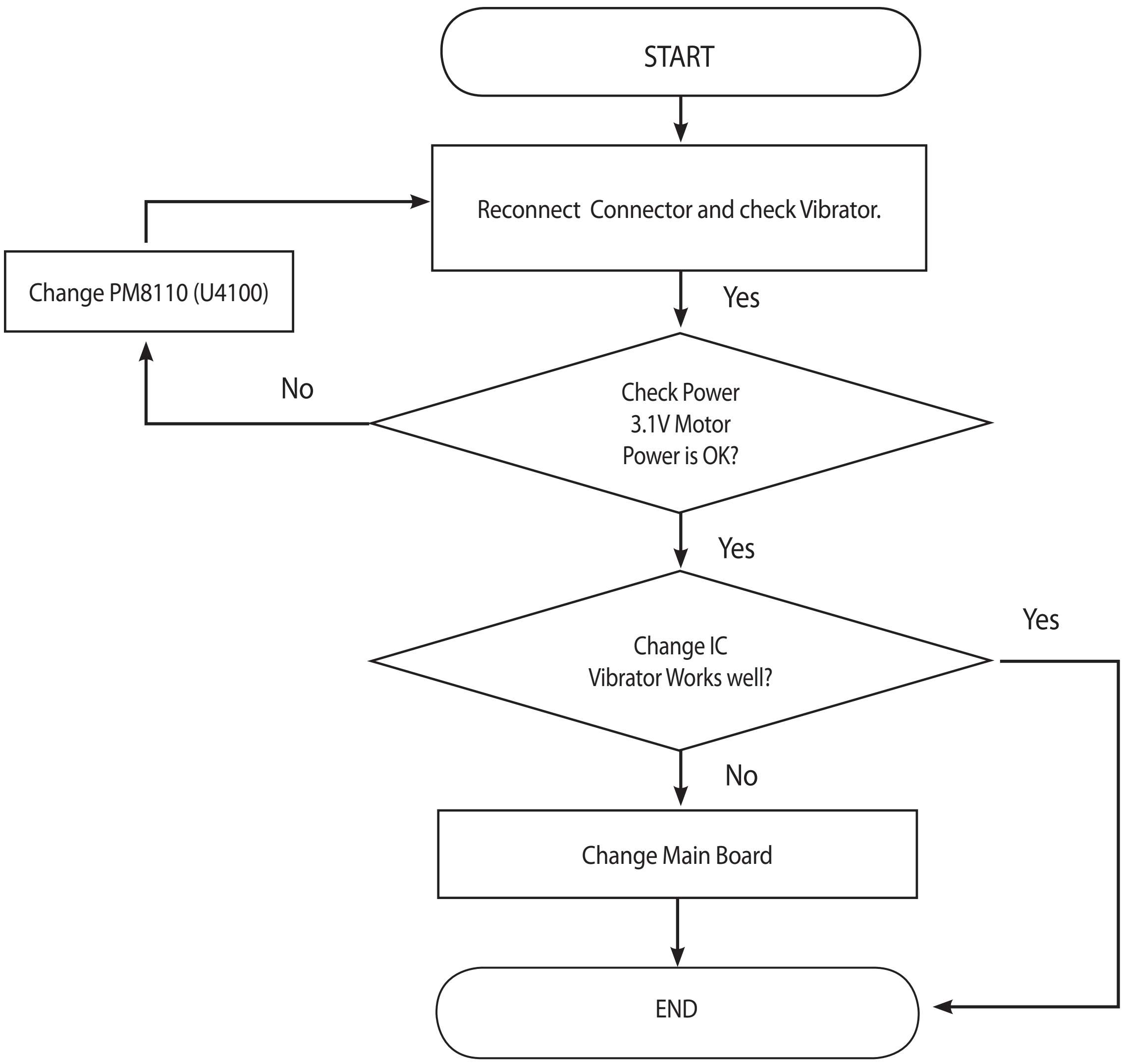


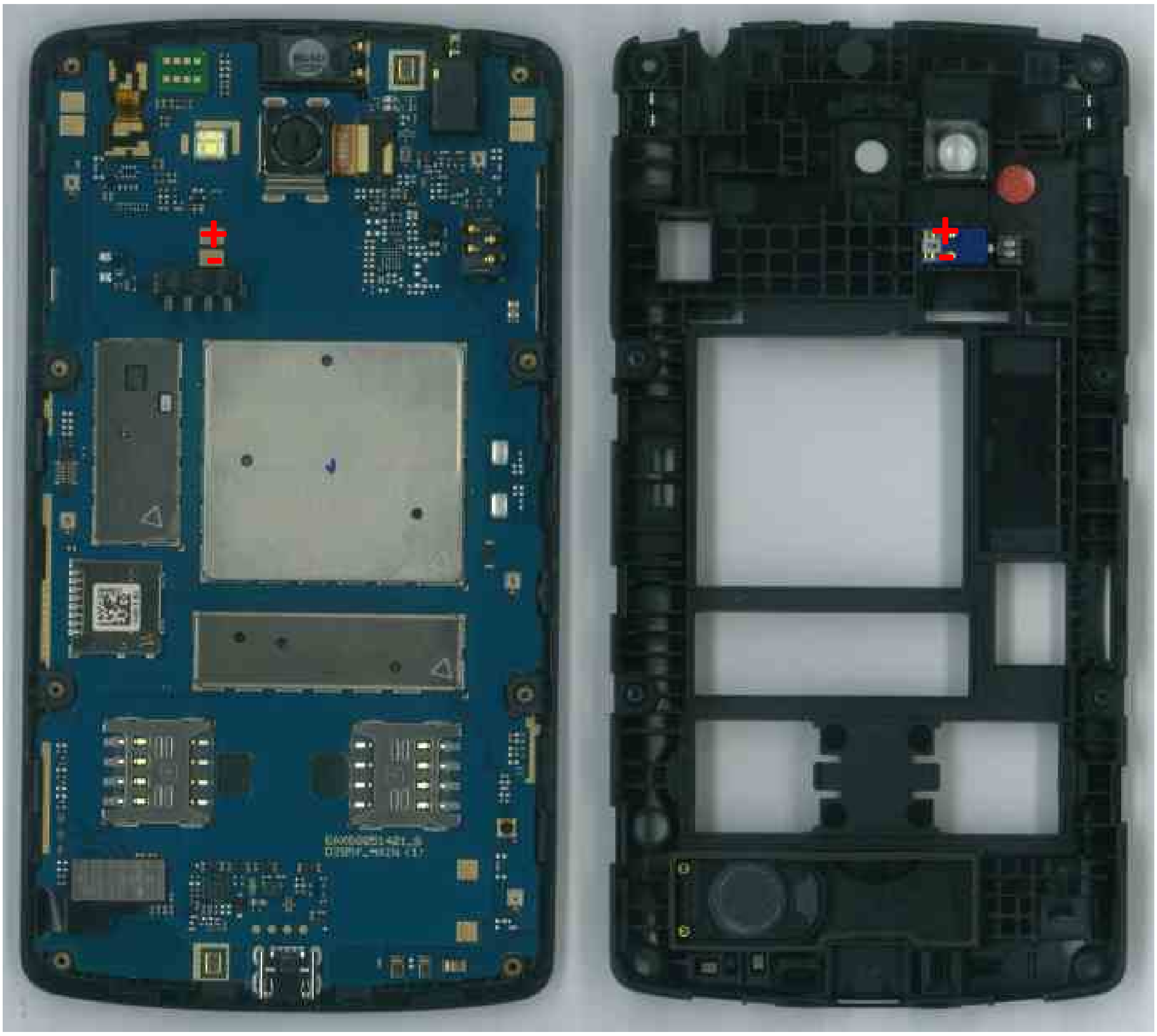


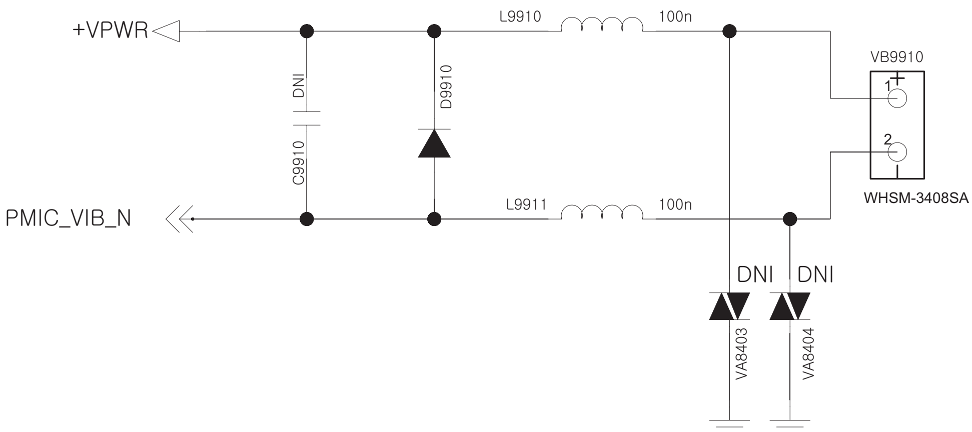
+1V8_TOUCH_VDD: (C7107)
+3V0_TOUCH_VDD: (C7106)
TOUCH ConnectorCN7100
TOUCH IC U7100

3.15. Vibrator Trouble Shooting

Vibrator signal path is PM8110(U4100) -> Vibrator. Motor Type is DC Motor



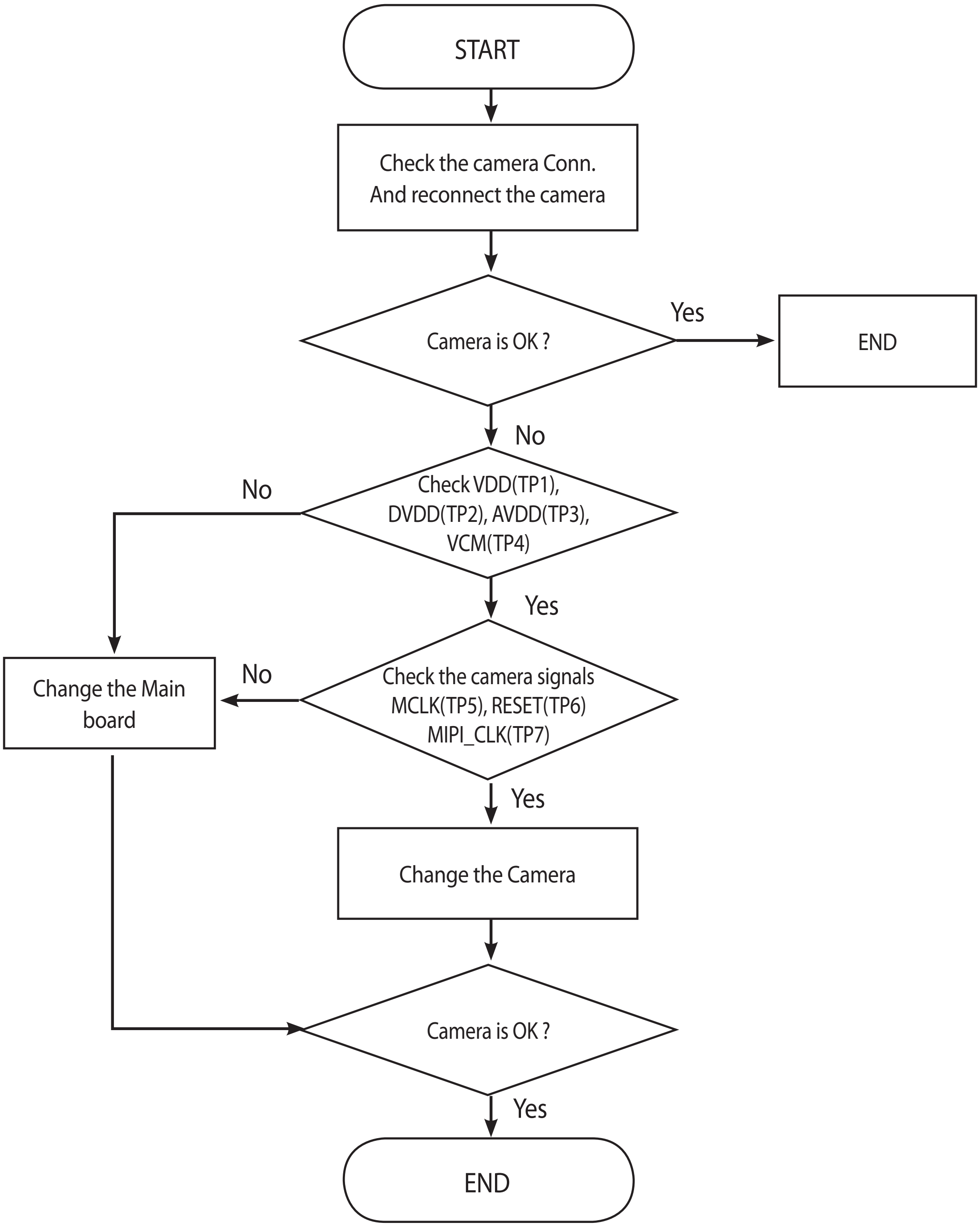

+
-
+
-

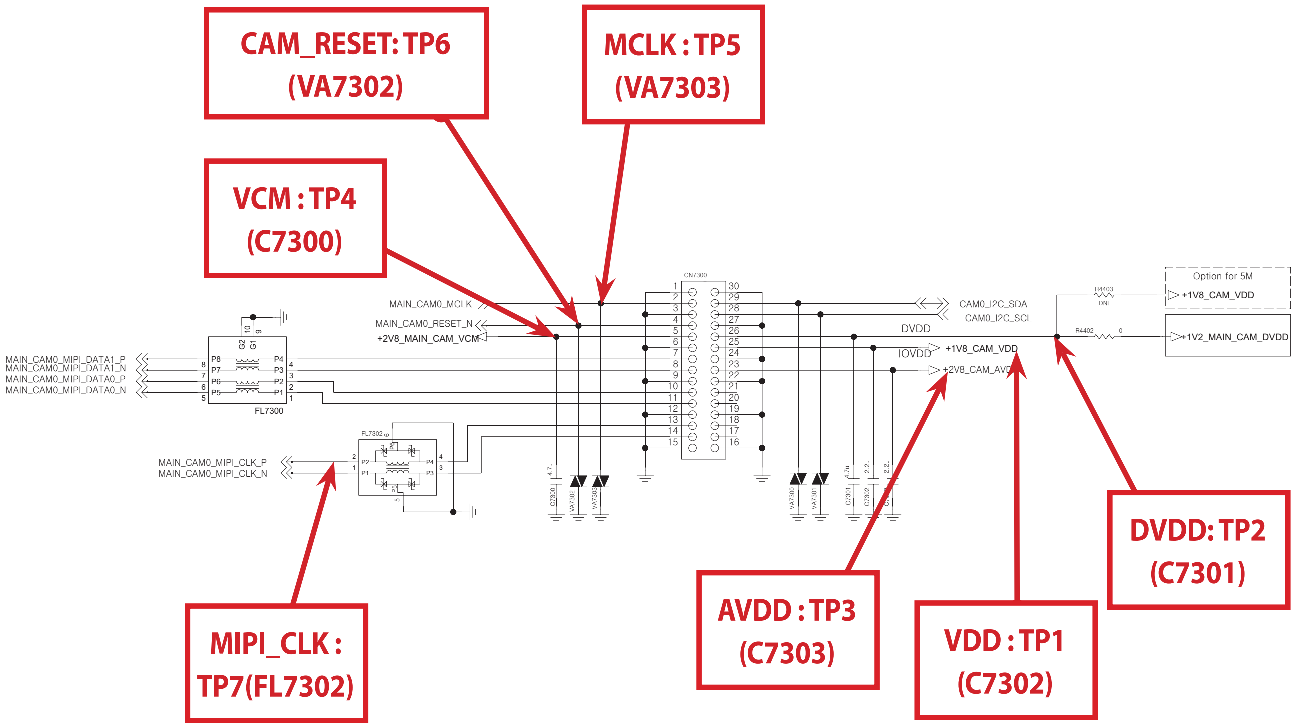

+VPWR
PMIC_VIB_N
L9910
100n
L9911
100n
DNI
C9910
D9910
VB9910
1
2
WHSM-3408SA
DNI
DNI
VA8403
VA8404

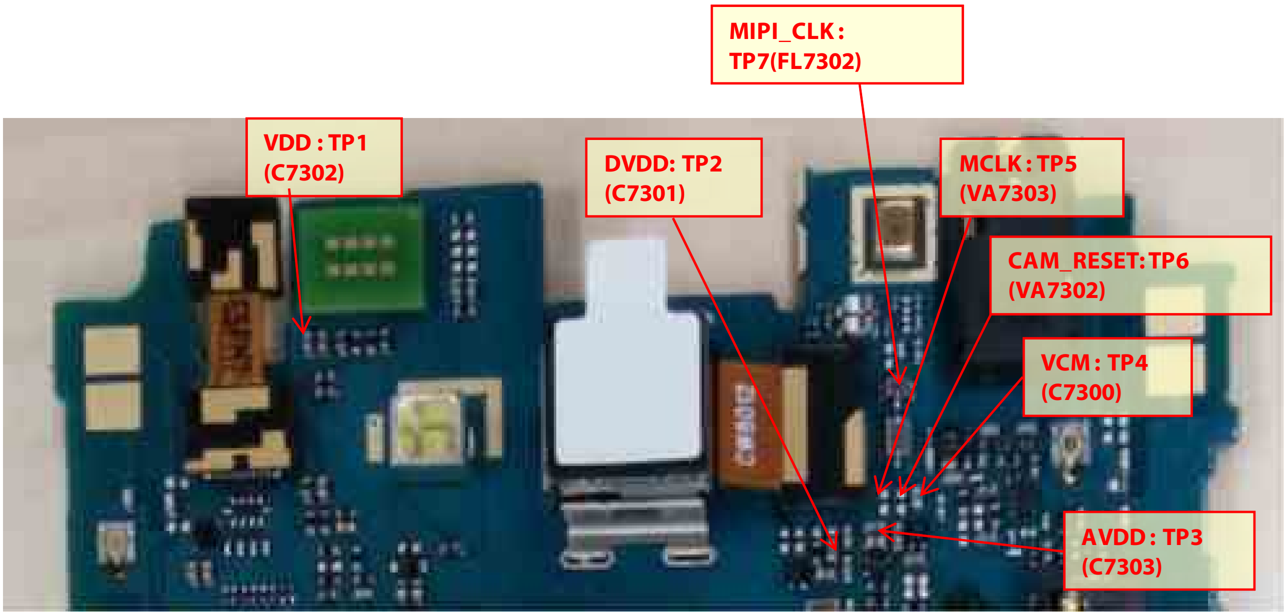
3.16 Camera TS

3.16.1 8M Main Camera TS

8M camera control signals are generated by MSM8212(U2100) and powered by voltage regulator (U4101, U4102, U4105 : LDO) , PM8110 (U4100 : PMIC).

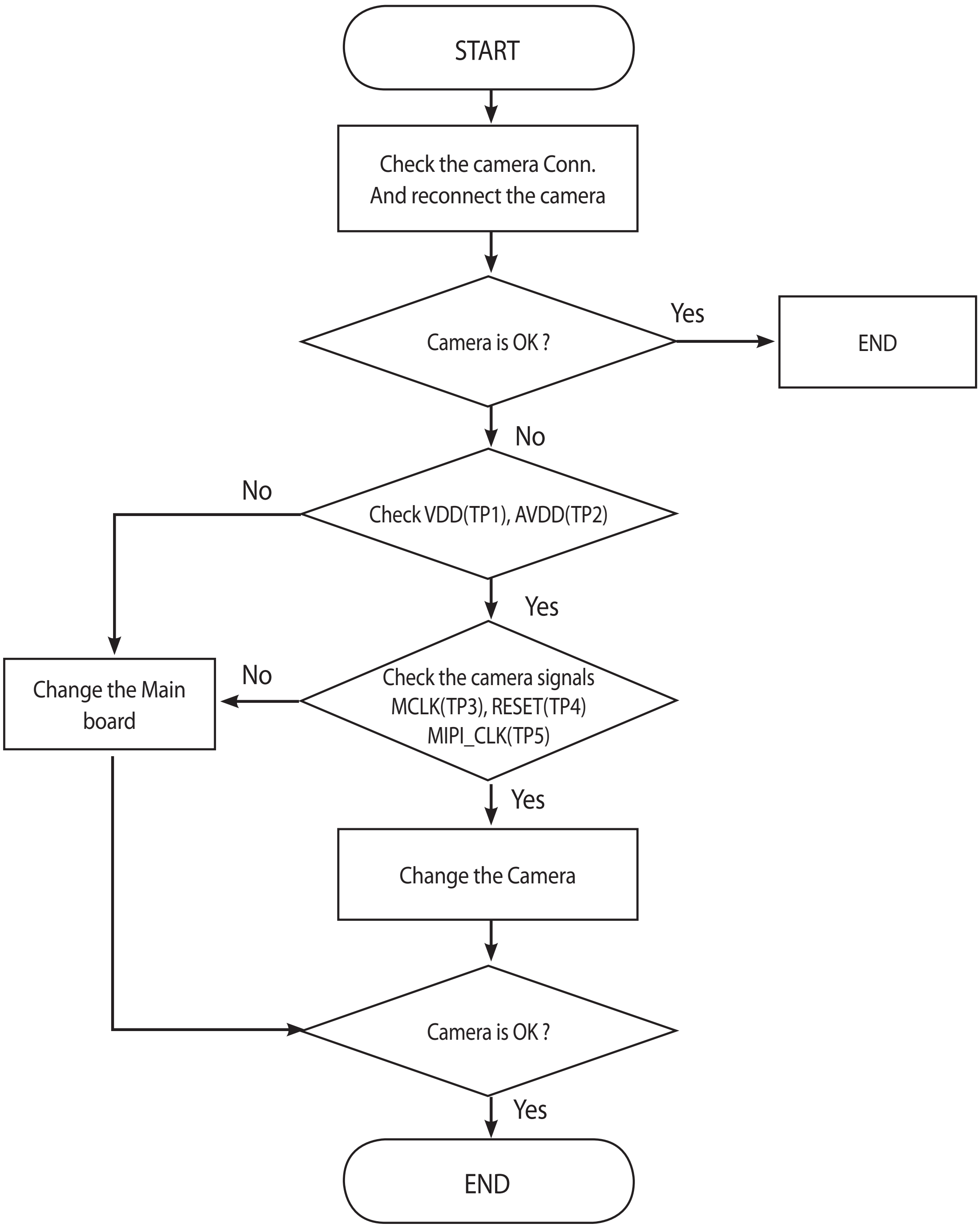


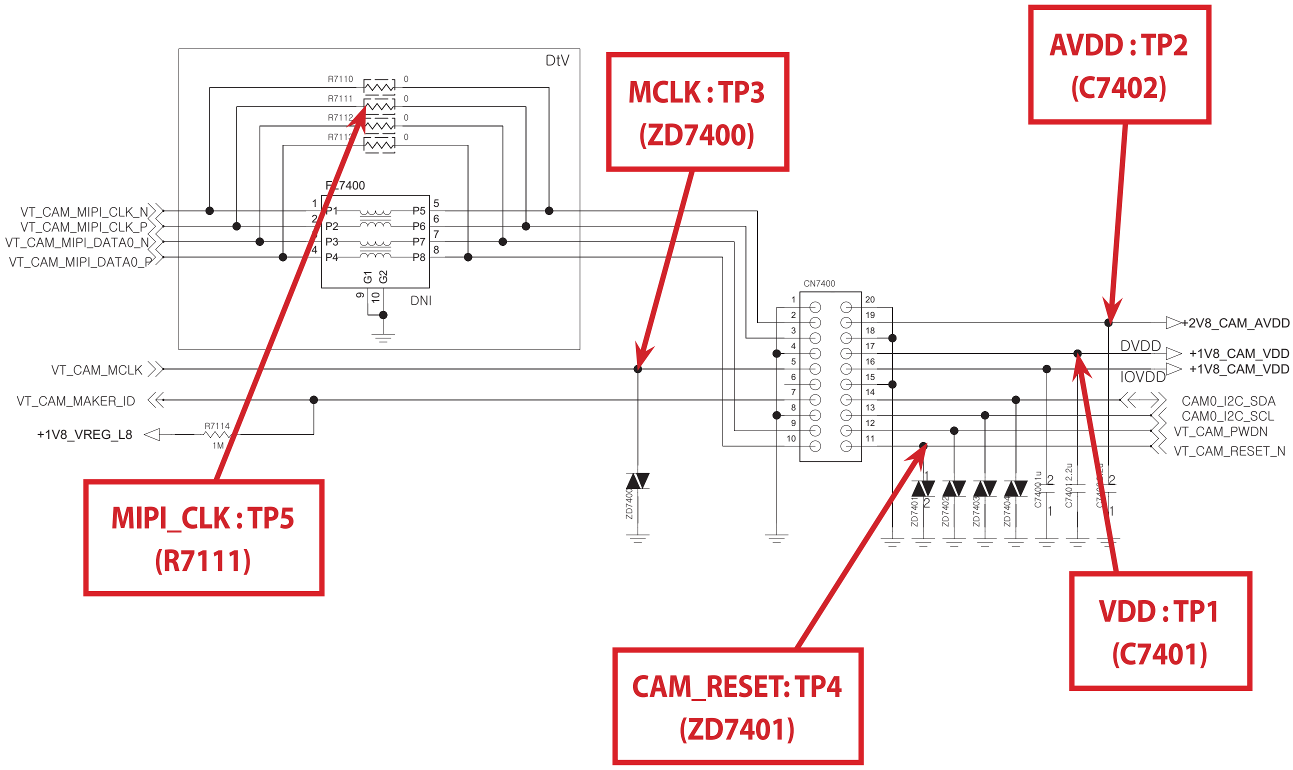

CAM_RESET: TP6
(VA7302)
MCLK : TP5
(VA7303)
VCM : TP4
(C7300)
MAIN_CAM0_MCLK
MAIN_CAM0_RESET_N
+2V8_MAIN_CAM_VCM
MAIN_CAM0_MIPI_DATA1_P
MAIN_CAM0_MIPI_DATA1_N
MAIN_CAM0_MIPI_DATA0_P
MAIN_CAM0_MIPI_DATA0_N
FL7300
MAIN_CAM0_MIPI_CLK_P
MAIN_CAM0_MIPI_CLK_N
CAM0_I2C_SDA
CAM0_I2C_SCL
DVDD
IOVDD
+1V8_CAM_VDD
+2V8_CAM_AVDD
Option for 5M
+1V8_CAM_VDD
+1V2_MAIN_CAM_DVDD
MIPI_CLK :
TP7(FL7302)
AVDD : TP3
(C7303)
VDD : TP1
(C7302)
DVDD: TP2
(C7301)


MIPI_CLK :
TP7(FL7302)
VDD : TP1
(C7302)
DVDD: TP2
(C7301)
MCLK : TP5
(VA7303)
CAM_RESET: TP6
(VA7302)
VCM : TP4
(C7300)
AVDD : TP3
(C7303)

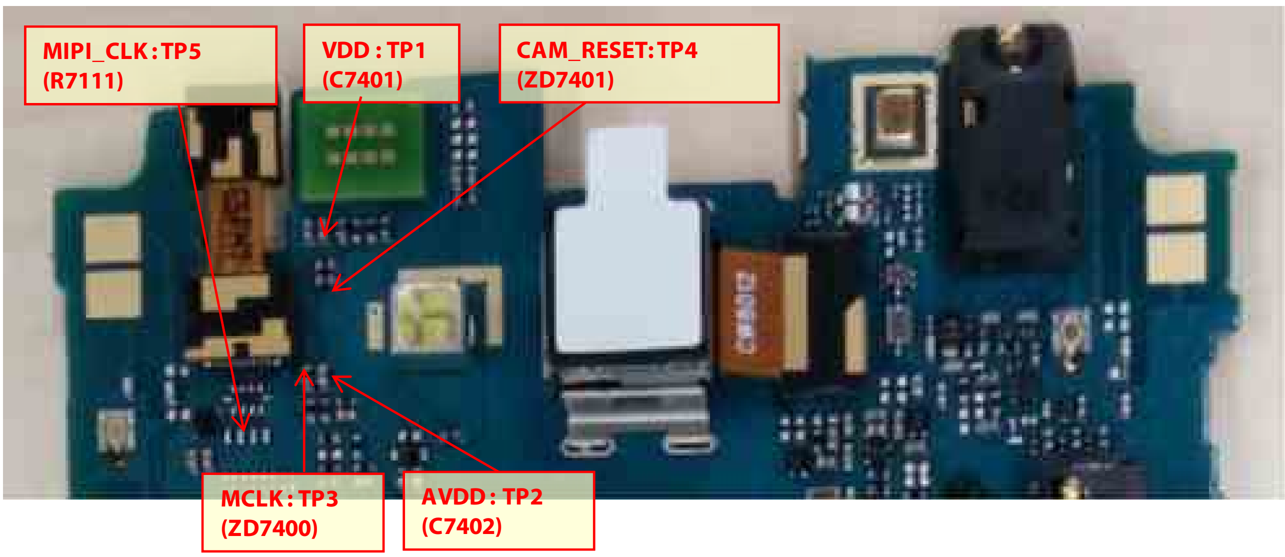
3.16.2 VGA Sub Camera TS

VGA camera control signals are generated by MSM8212(U2100) and powered by voltage regulato r (U4102 : LDO) , PM8110 (U4100 : PMIC).





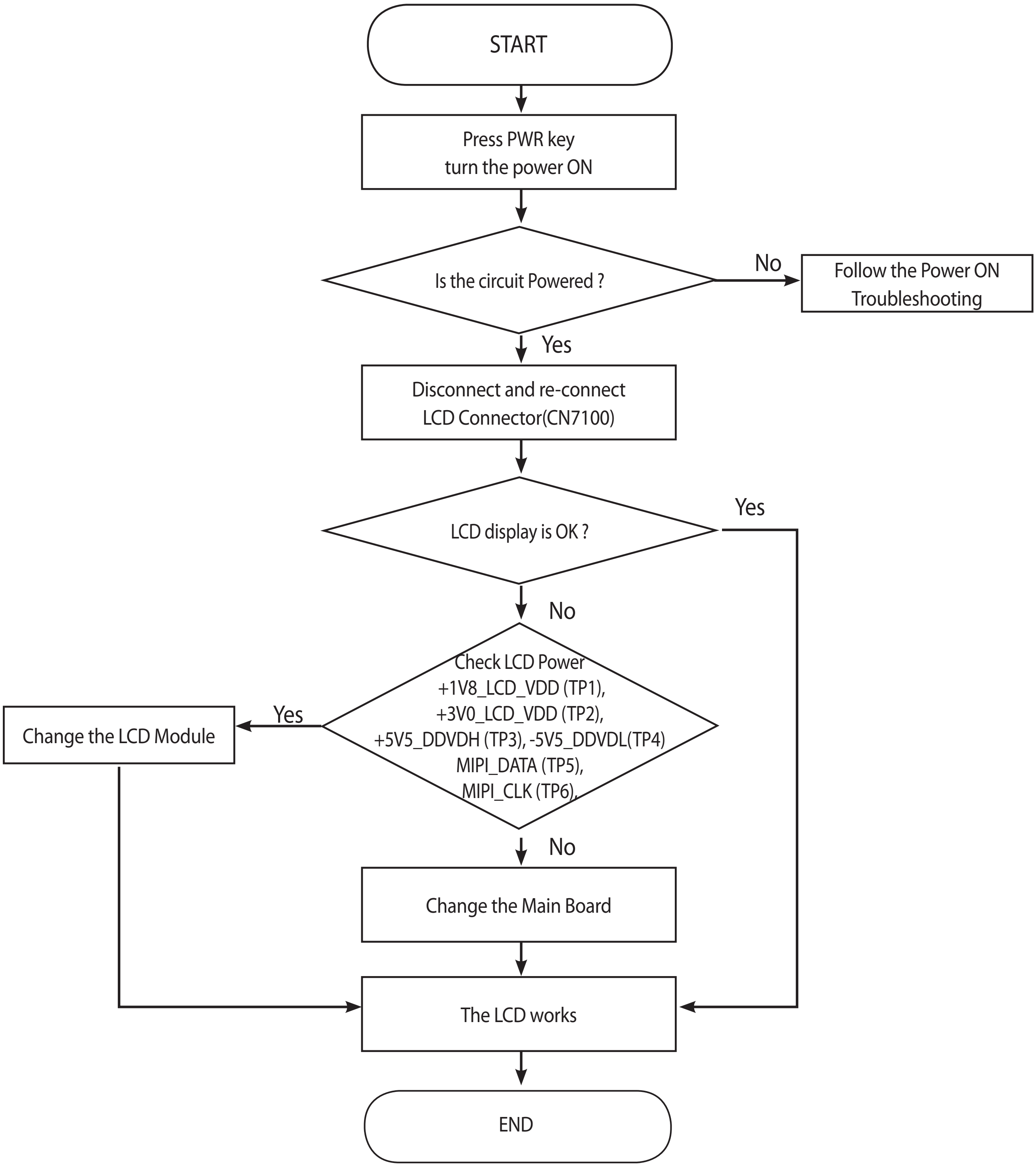
DtV
R7110
R7111
R7112
R7113
FL7400
VT_CAM_MIPI_CLK_N
VT_CAM_MIPI_CLK_P
VT_CAM_MIPI_DATA0_N
VT_CAM_MIPI_DATA0_P
DNI
CN7400
VT_CAM_MCLK
VT_CAM_MAKER_ID
+1V8_VREG_L8
R7114
1M
+2V8_CAM_AVDD
DVDD
+1V8_CAM_VDD
IOVDD
+1V8_CAM_VDD
CAM0_I2C_SDA
CAM0_I2C_SCL
VT_CAM_PWDN
VT_CAM_RESET_N
AVDD : TP2
(C7402)
MCLK : TP3
(ZD7400)
MIPI_CLK : TP5
(R7111)
CAM_RESET: TP4
(ZD7401)
VDD : TP1
(C7401)

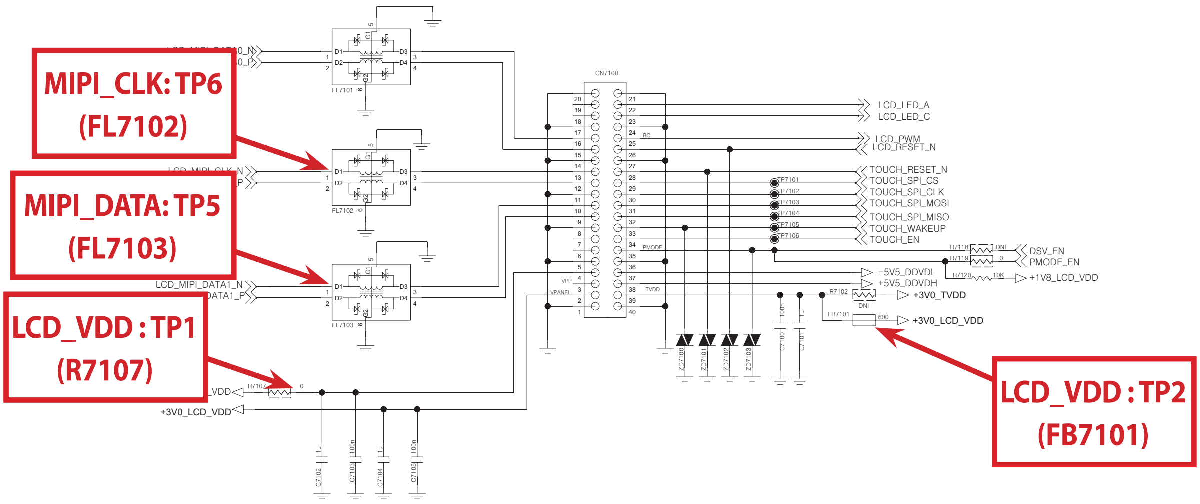


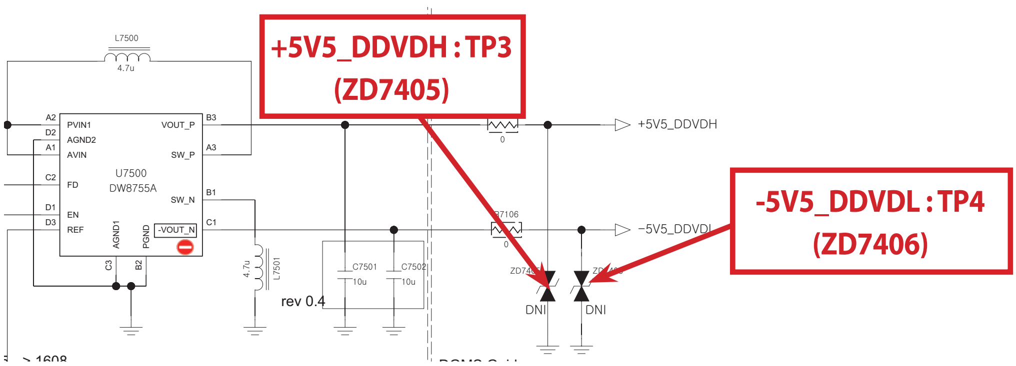
MIPI_CLK : TP5
(R7111)
VDD : TP1
(C7401)
CAM_RESET: TP4
(ZD7401)
MCLK : TP3
(ZD7400)
AVDD : TP2
(C7402)

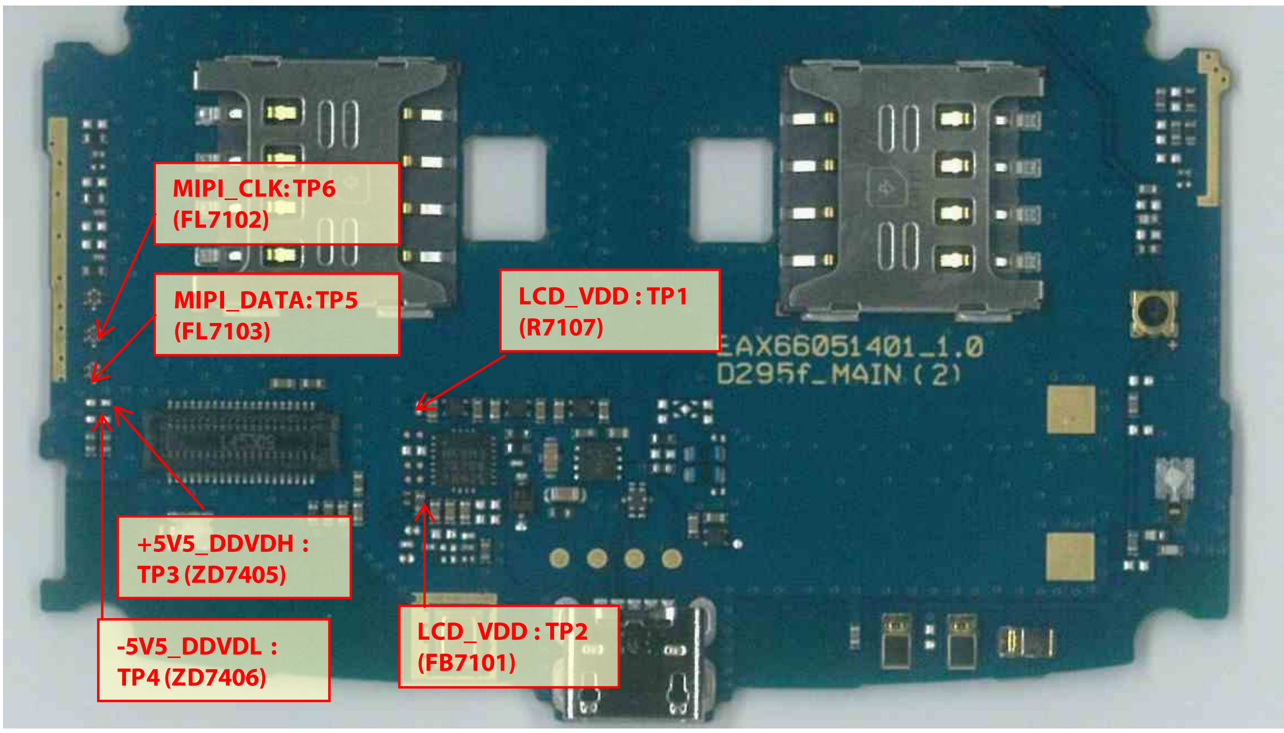
3.17. Main LCD TS

The LCD control signals are generated by MSM8212(U2100) and powered by PM8110 (U4100 : PMIC).

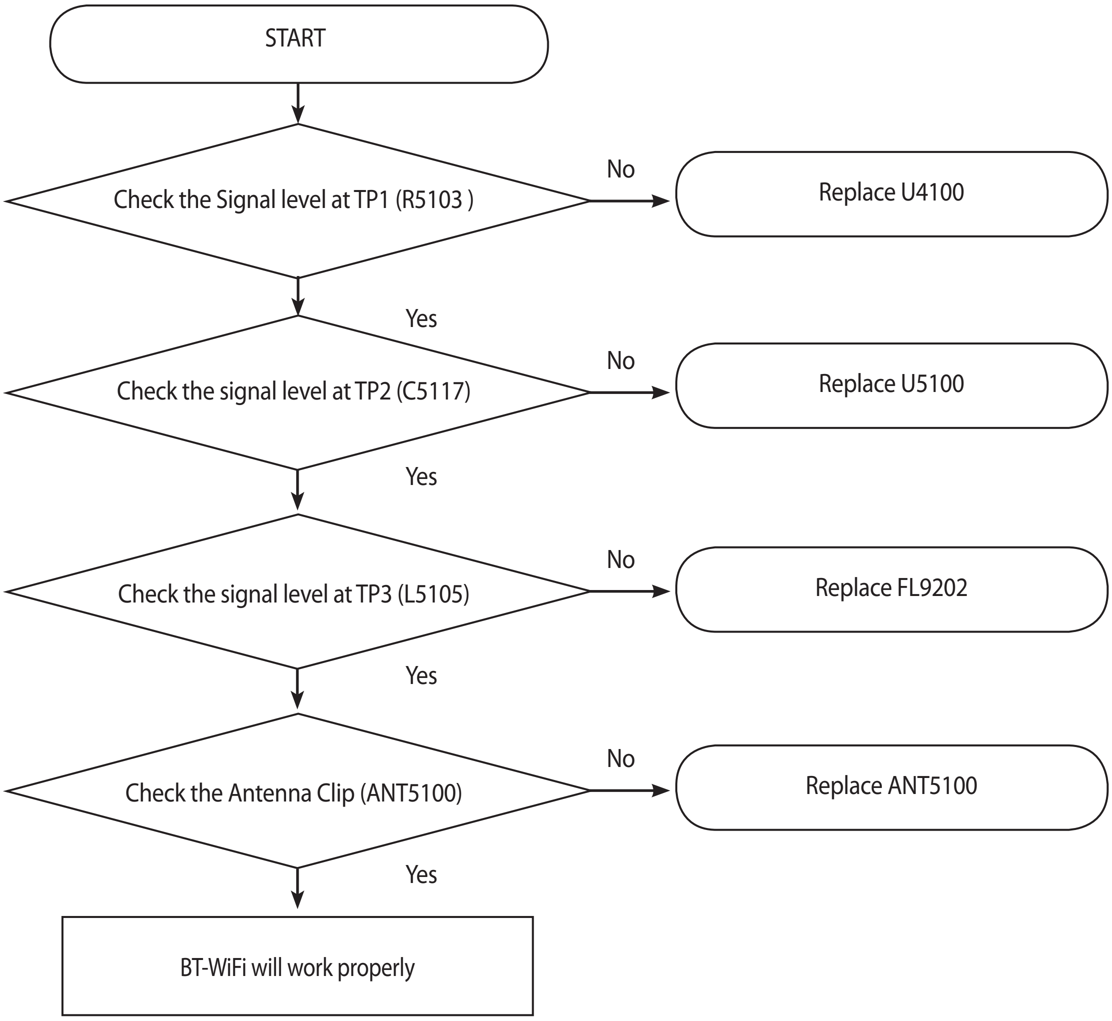


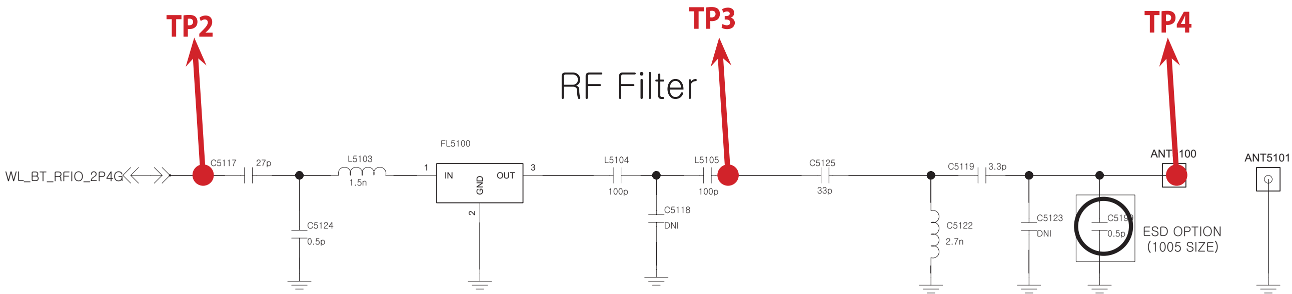

MIPI_CLK: TP6
(FL7102)
MIPI_DATA: TP5
(FL7103)
LCD_VDD : TP1
(R7107)
LCD_VDD : TP2
(FB7101)
CN7100
FL7101
FL7102
FL7103
LCD_LED_A
LCD_LED_C
LCD_PWM
LCD_RESET_N
TOUCH_RESET_N
TOUCH_SPI_CS
TOUCH_SPI_CLK
TOUCH_SPI_MOSI
TOUCH_SPI_MISO
TOUCH_WAKEUP
TOUCH_EN
DSV_EN
PMODE_EN
-5V5_DDVDL
+5V5_DDVDH
+1V8_LCD_VDD
+3V0_TVDD
+3V0_LCD_VDD

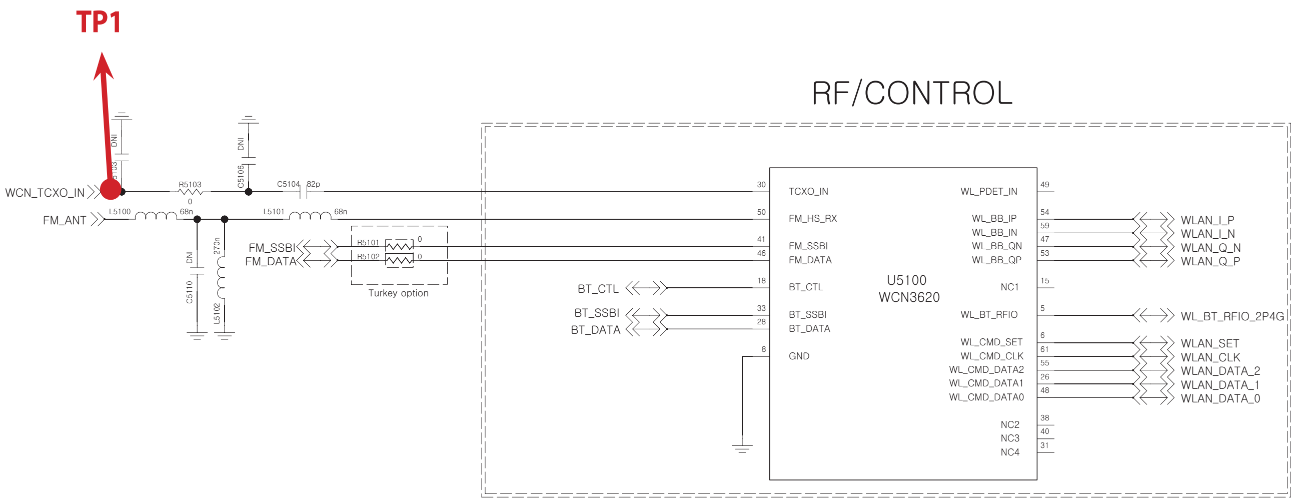

+5V5_DDVDH : TP3
(ZD7405)
-5V5_DDVDL : TP4
(ZD7406)
L7500
4.7u
PVIN1
AGND2
AVIN
VOUT_P
SW_P
U7500
DW8755A
FD
SW_N
EN
REF
AGND1
PGND
-VOUT_N
C7501
10u
C7502
10u
L7501
rev 0.4
+5V5_DDVDH
-5V5_DDVDL
R7106
DNI
DNI

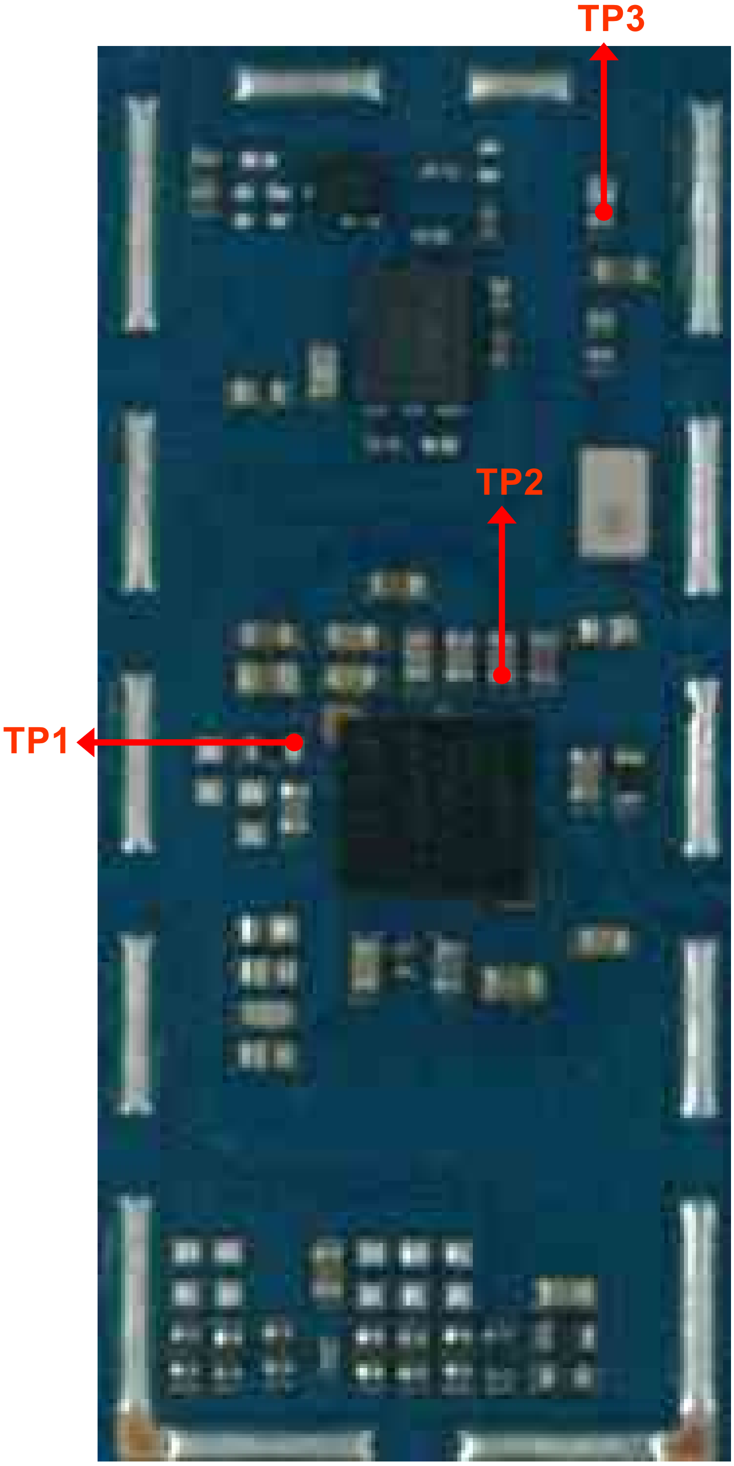

MIPI_CLK: TP6
(FL7102)
MIPI_DATA: TP5
(FL7103)
LCD_VDD : TP1
(R7107)
EAX66051401_1.0
D295f_MAIN (2)
+5V5_DDVDH :
TP3 (ZD7405)
-5V5_DDVDL :
TP4 (ZD7406)
LCD_VDD : TP2
(FB7101)

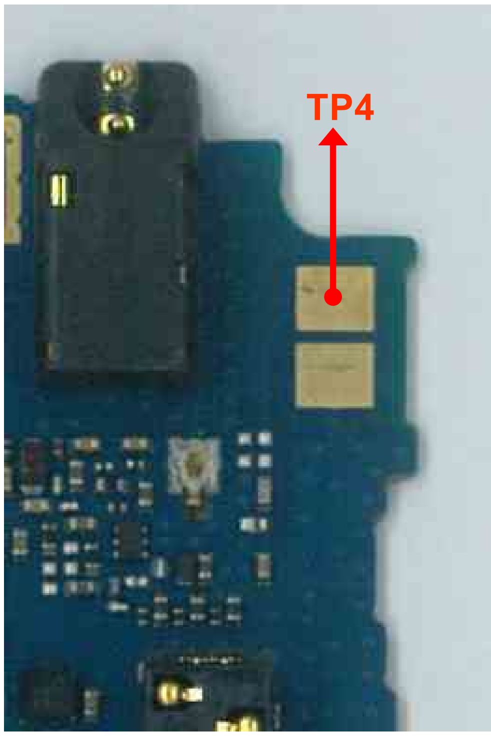
3.18. BT/Wifi TS



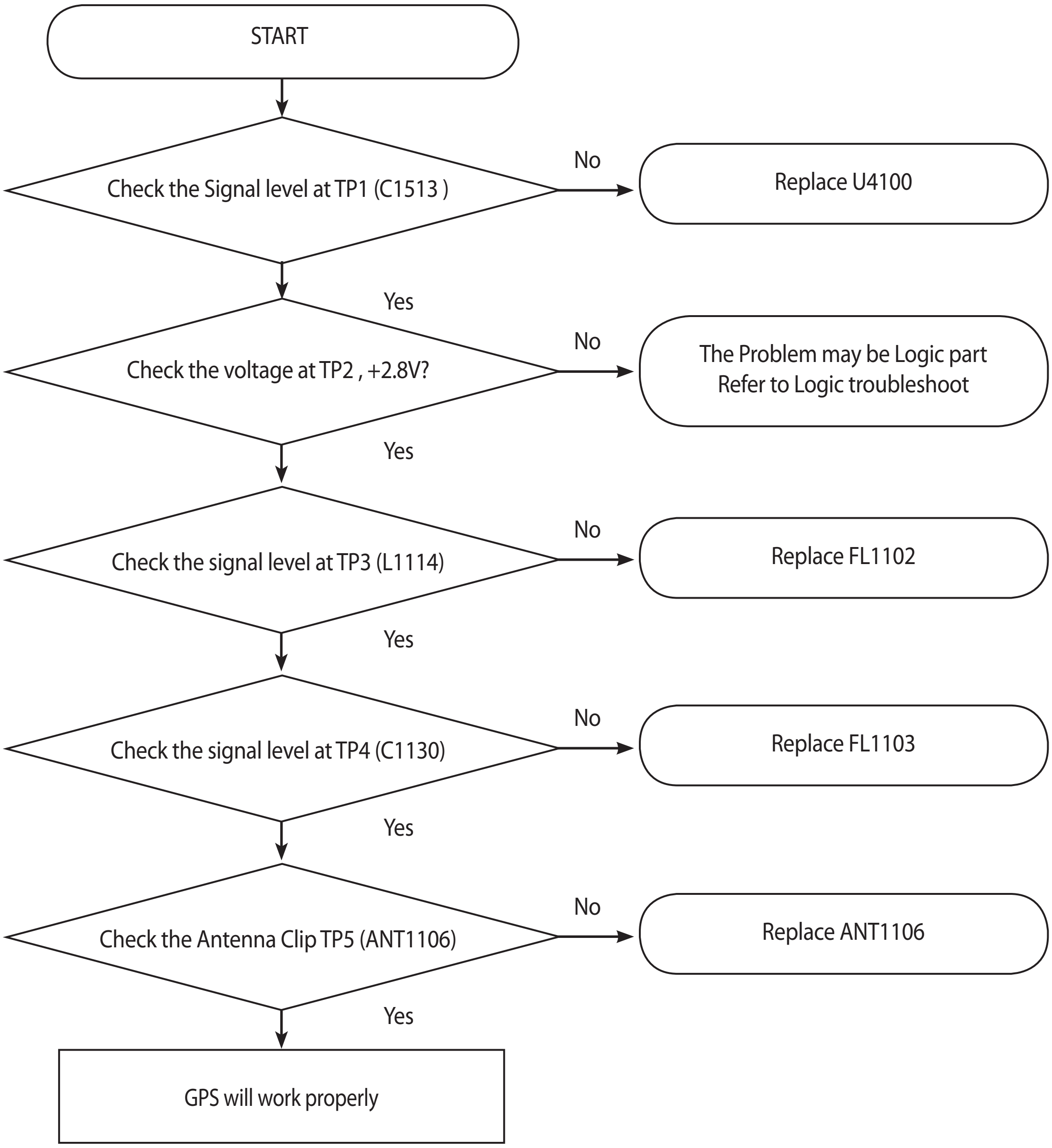

TP2
TP3
TP4
RF Filter
WL_BT_RFIO_2P4G
C5117
27p
C5124
0.5p
L5103
1.5n
FL5100
IN
GND
OUT
L5104
100p
C5118
DNI
L5105
100p
C5125
33p
C5119
3.3p
C5122
2.7n
C5123
DNI
0.5p
ESD OPTION
(1005 SIZE)
ANT5101

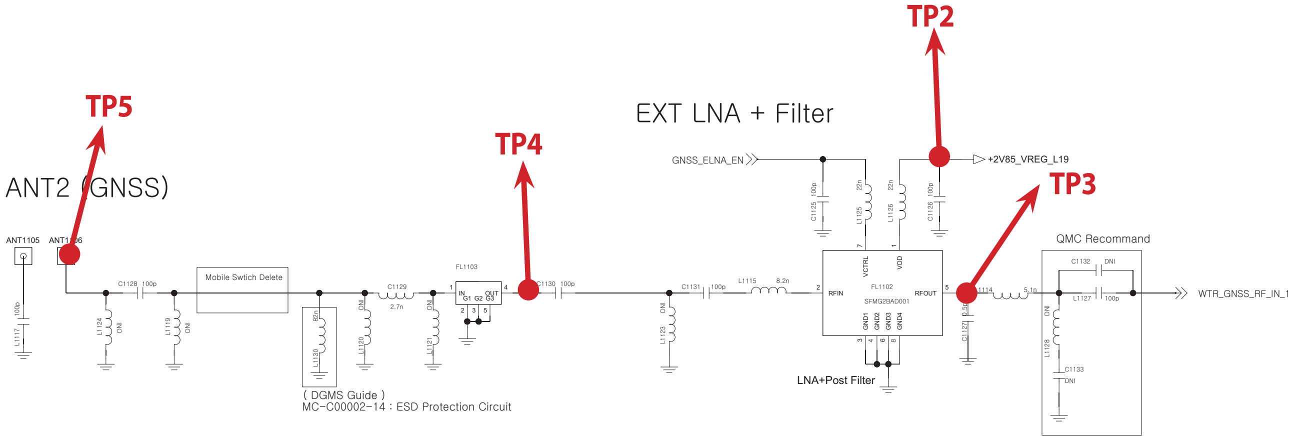

TP1
RF/CONTROL
WCN_TCXO_IN
FM_ANT
L5100
68n
R5103
C5104
82p
L5101
68n
FM_SSBI
FM_DATA
R5101
R5102
Turkey option
BT_CTL
BT_SSBI
BT_DATA
TCXO_IN
FM_HS_RX
FM_SSBI
FM_DATA
BT_CTL
BT_SSBI
BT_DATA
GND
U5100
WCN3620
WL_PDET_IN
WL_BB_IP
WL_BB_IN
WL_BB_QN
WL_BB_QP
NC1
WL_BT_RFIO
WL_CMD_SET
WL_CMD_CLK
WL_CMD_DATA2
WL_CMD_DATA1
WL_CMD_DATA0
NC2
NC3
NC4
WLAN_I_P
WLAN_I_N
WLAN_Q_N
WLAN_Q_P
WL_BT_RFIO_2P4G
WLAN_SET
WLAN_CLK
WLAN_DATA_2
WLAN_DATA_1
WLAN_DATA_0

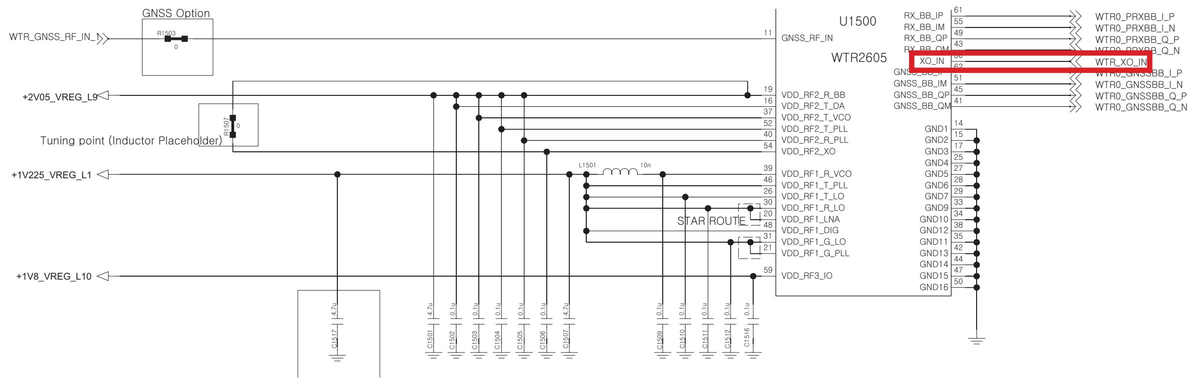

TP3
TP2
TP1

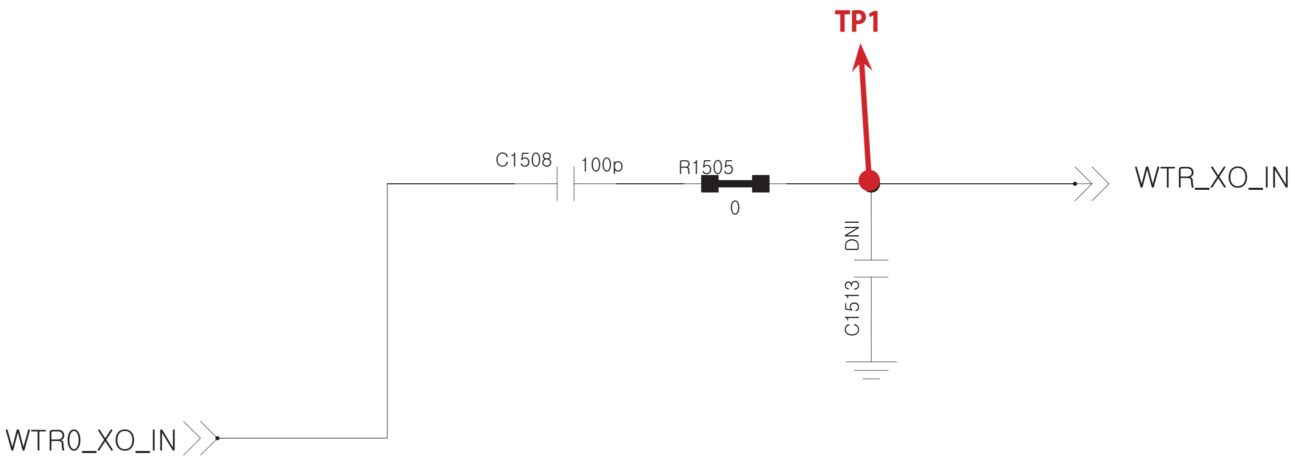

TP4

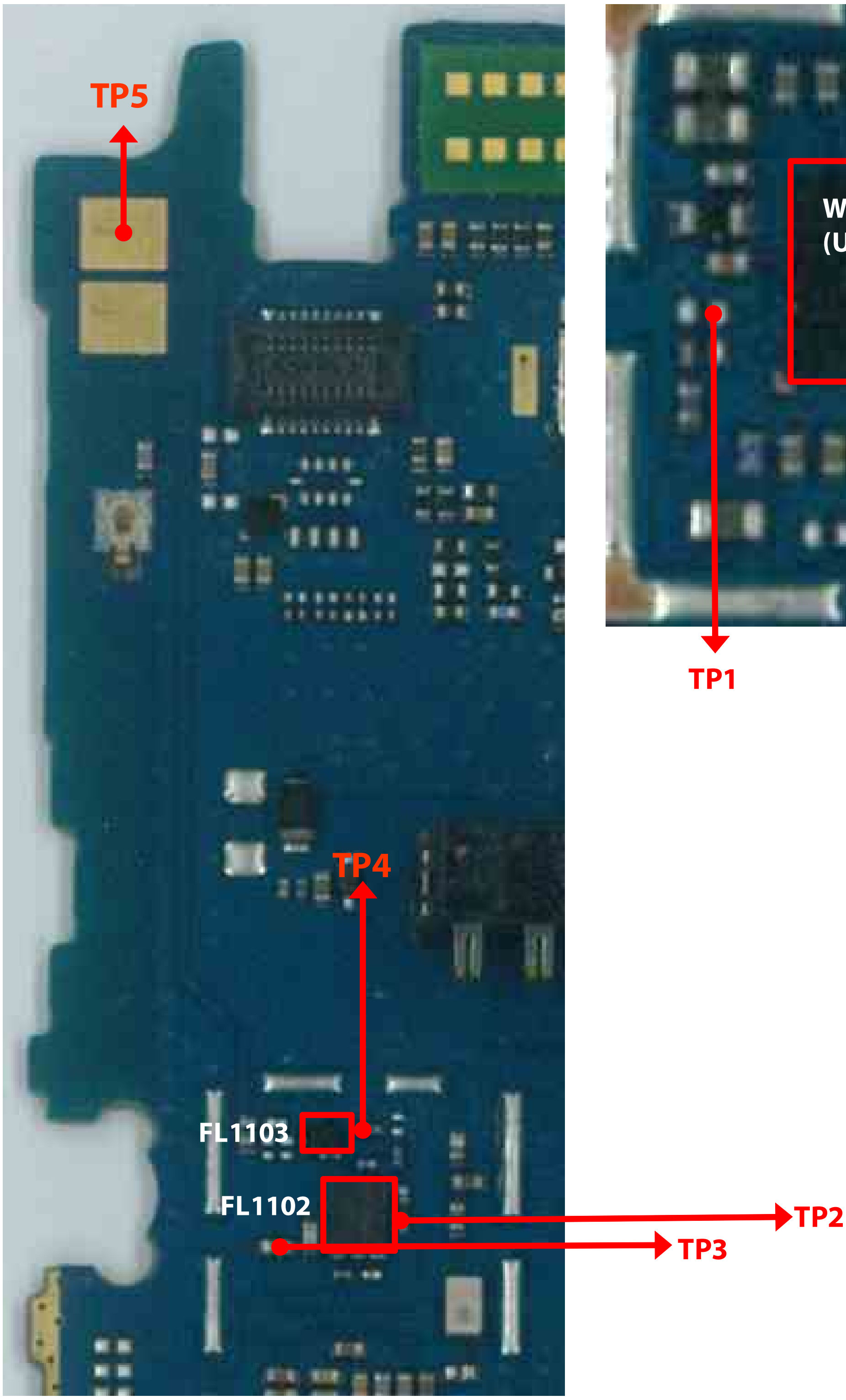
3.19. GPS TS

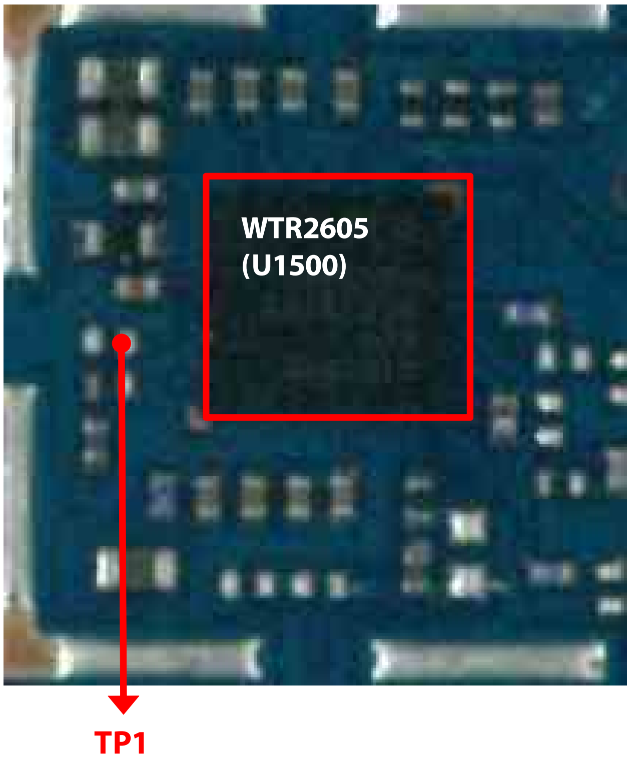

START
Check the Signal level at TP1 (C1513)
No
Replace U4100
Yes
Check the voltage at TP2 , +2.8V?
No
The Problem may be Logic part
Refer to Logic troubleshoot
Yes
Check the signal level at TP3 (L1114)
No
Replace FL1102
Yes
Check the signal level at TP4 (C1130)
No
Replace FL1103
Yes
Check the Antenna Clip TP5 (ANT1106)
No
Replace ANT1106
Yes
GPS will work properly


TP5
TP4
TP2
TP3
ANT2 (GNSS)
EXT LNA + Filter
GNSS_ELNA_EN
+2V85_VREG_L19
QMC Recommand
WTR_GNSS_RF_IN_1
Mobile Swtich Delete
LNA+Post Filter
(DGMS Guide)
MC-C00002-14 : ESD Protection Circuit

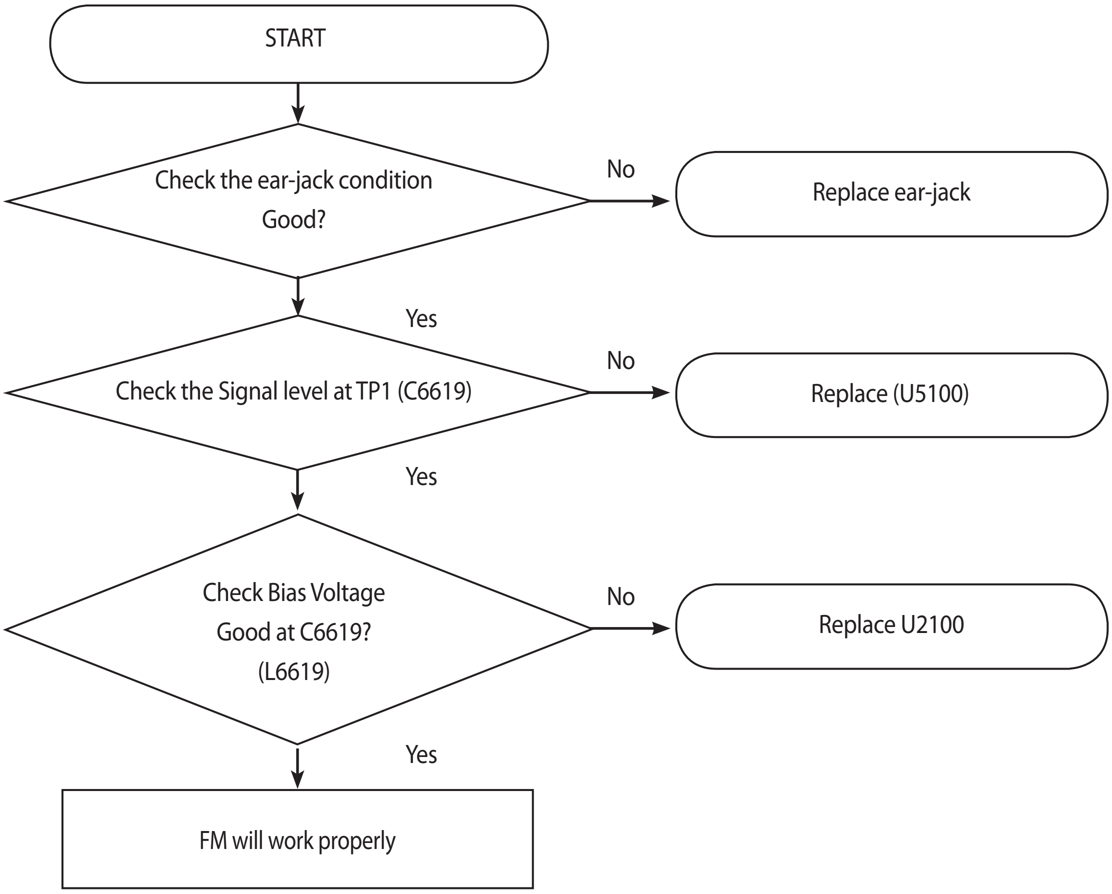

GNSS Option
WTR_GNSS_RF_IN_1
U1500
WTR2605
GNSS_RF_IN
+2V05_VREG_L9
Tuning point (Inductor Placeholder)
+1V225_VREG_L1
+1V8_VREG_L10
STAR ROUTE
XO_IN
WTR_XO_IN
WTR0_PRXBB_I_P
WTR0_PRXBB_I_N
WTR0_PRXBB_Q_P
WTR0_GNSSBB_I_N
WTR0_GNSSBB_Q_P
WTR0_GNSSBB_Q_N

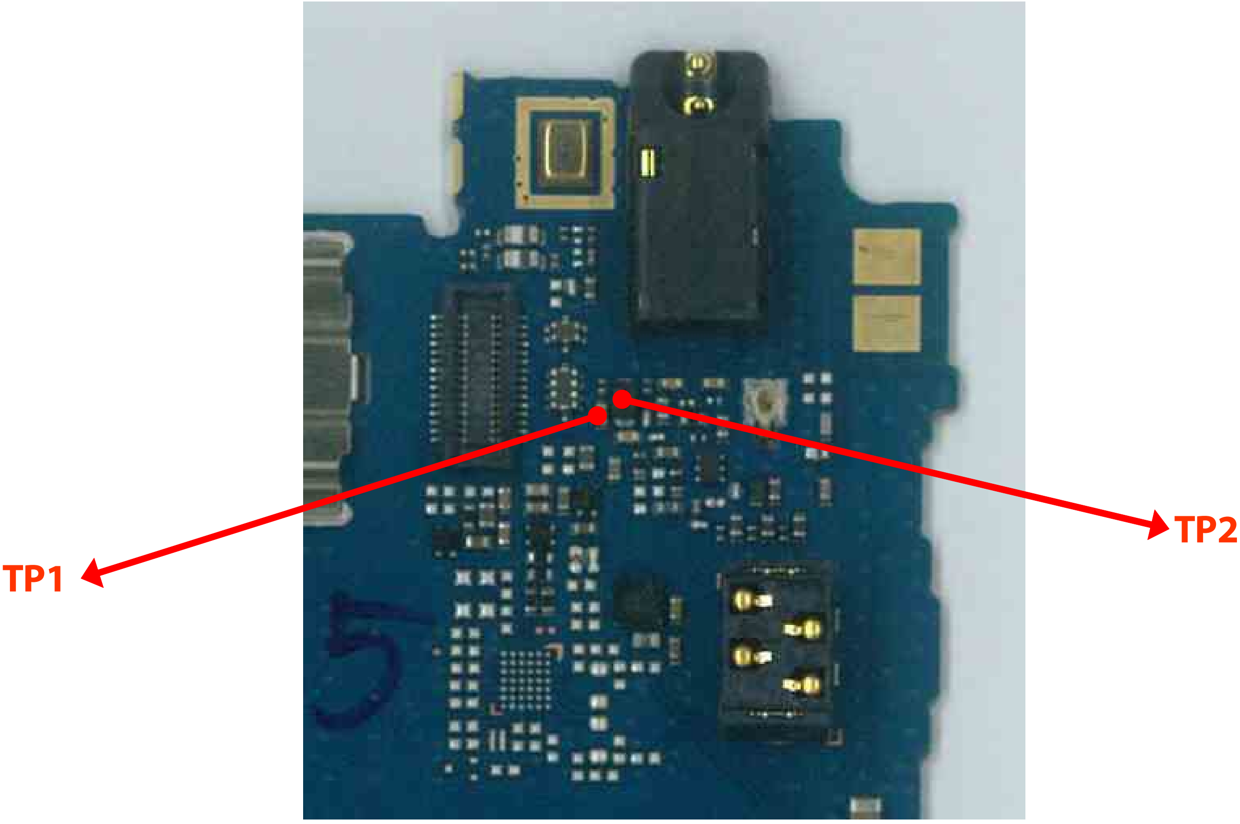

TP1
C1508
100p
R1505
0
DNI
C1513
WTR_XO_IN
WTR0_XO_IN

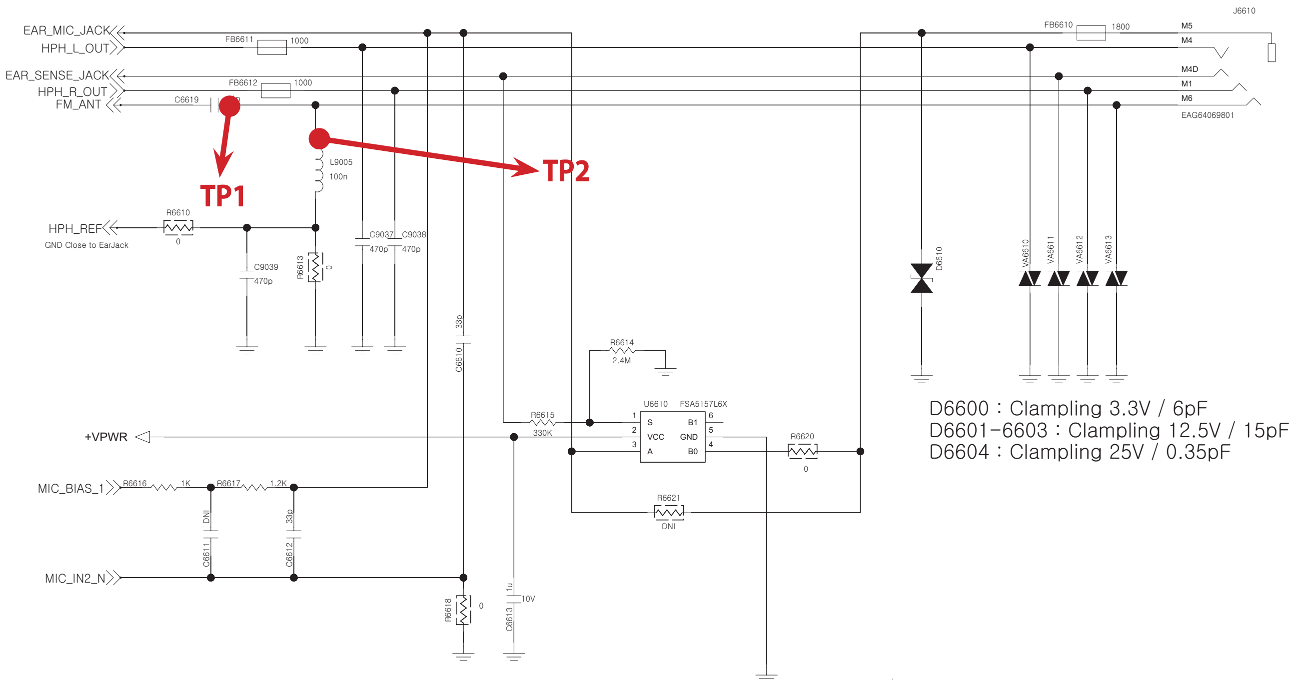

TP5
TP4
FL1103
FL1102
TP2
TP3


WTR2605
(U1500)
TP1

3.20. FM TS

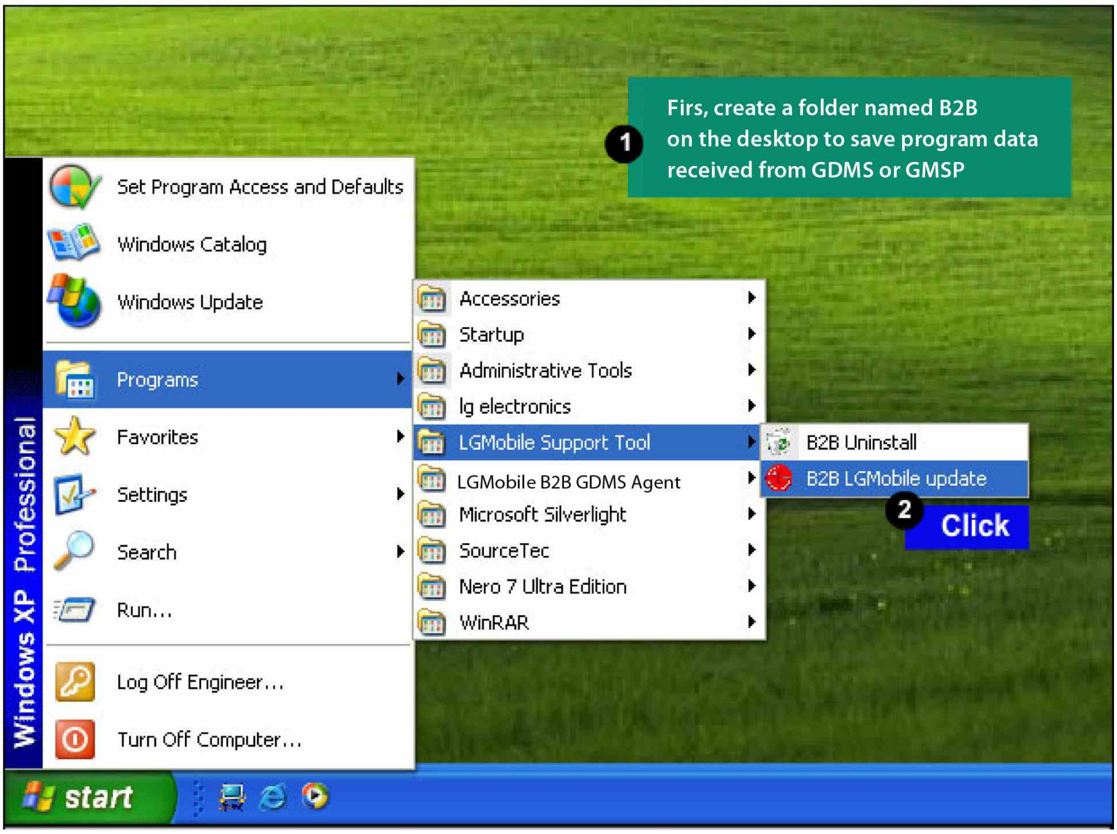

START
Check the ear-jack condition
Good?
No
Replace ear-jack
Yes
Check the Signal level at TP1 (C6619)
No
Replace (U5100)
Yes
Check Bias Voltage
Good at C6619?
(L6619)
No
Replace U2100
Yes
FM will work properly

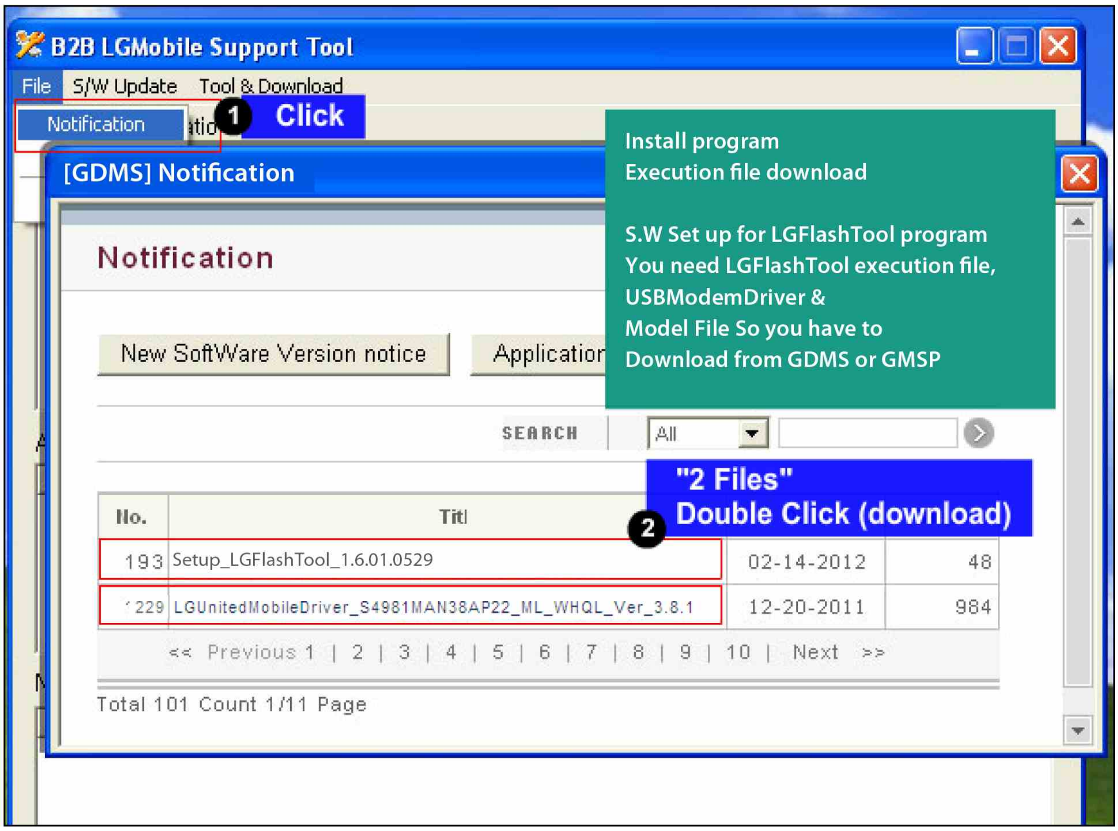

TP1
TP2

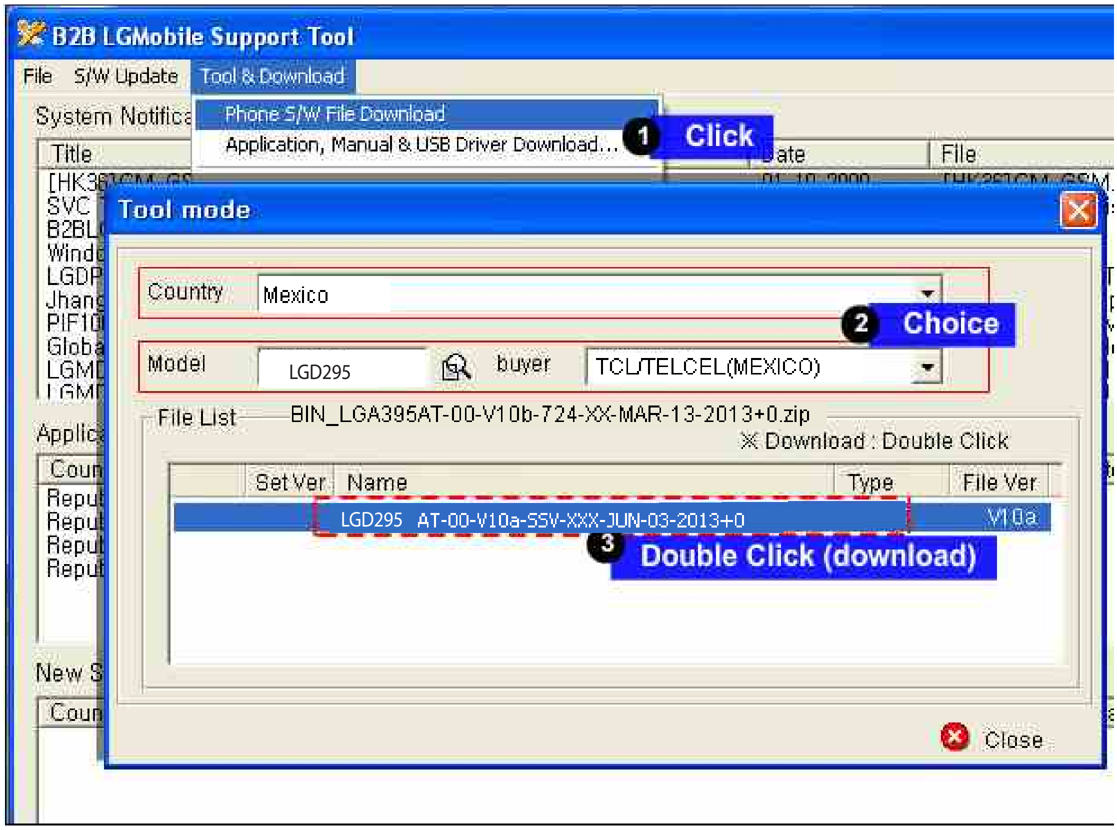

EAR_MIC_JACK
HPH_L_OUT
EAR_SENSE_JACK
HPH_R_OUT
FM_ANT
TP1
TP2
HPH_REF
GND Close to EarJack
+VPWR
MIC_BIAS_1
MIC_IN2_N
U6610 FSA5157L6X
J6610
EAG64069801
D6600 : Clampling 3.3V / 6pF
D6601~6603 : Clampling 12.5V / 15pF
D6604 : Clampling 25V / 0.35pF

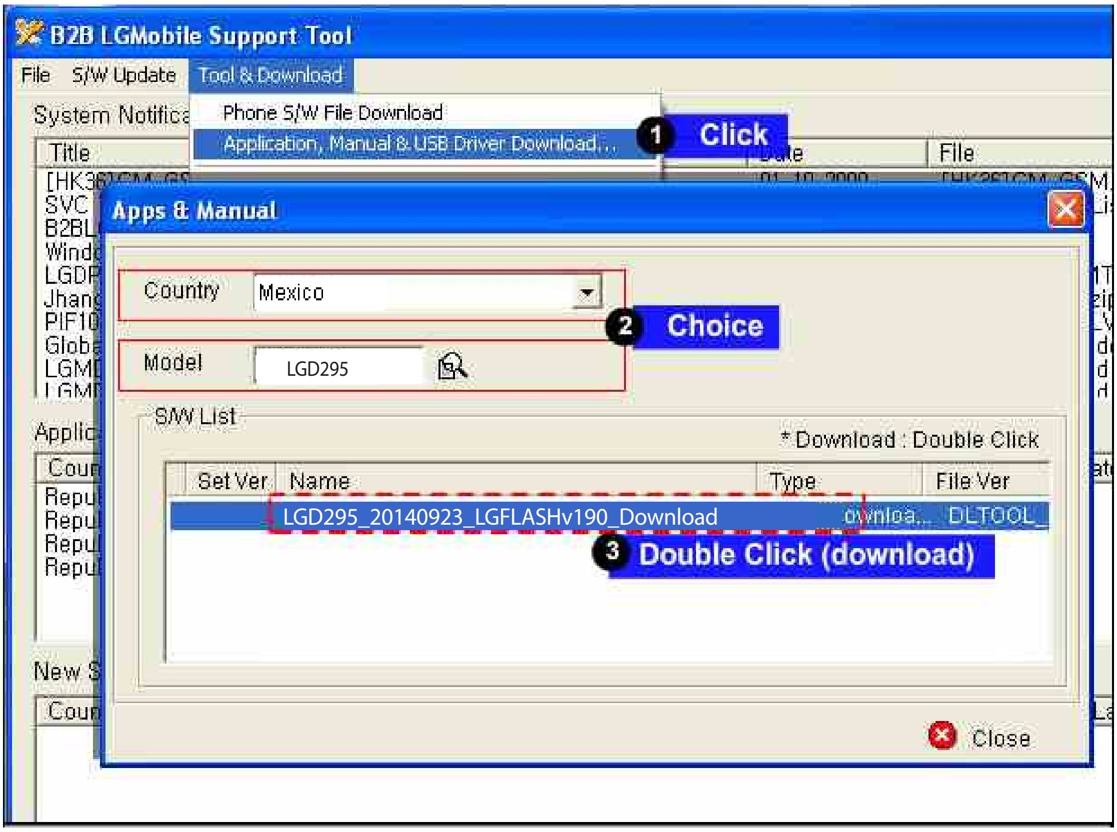
4. DOWNLOAD

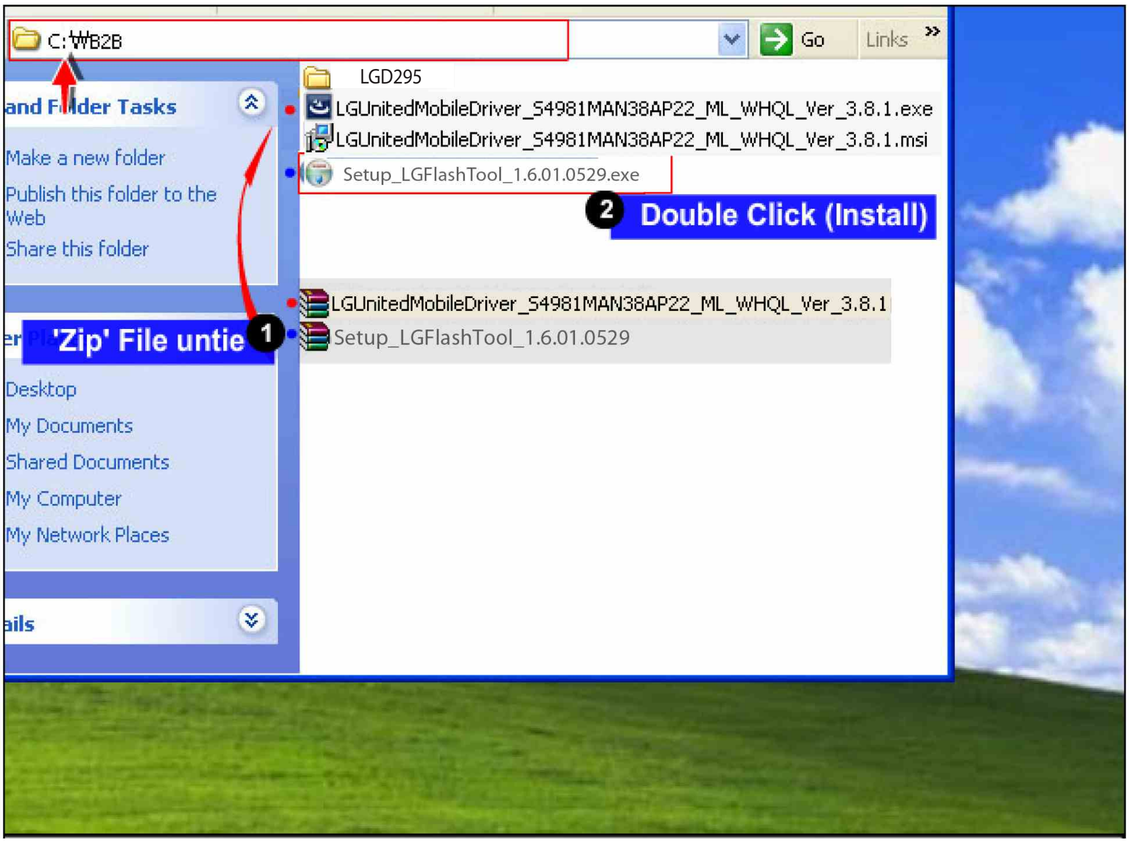
TOOL INFORMATION			
TOOL VERSION	**DLL NAME**	**USB DRIVER**	
LGFLASHv190	LGD295_20140923_LGFLASHv190_Download	LGUnitedMobileDriver_S50MAN311AP22_ML_WHQL_Ver_3.11.3	
Please Check the Version to "B2B"			
H/W			
	Name	**Part No.**	**SW**
D/L Cable	Micro 5P (56-open-910K) USB DLC	RAD32167835	TOT

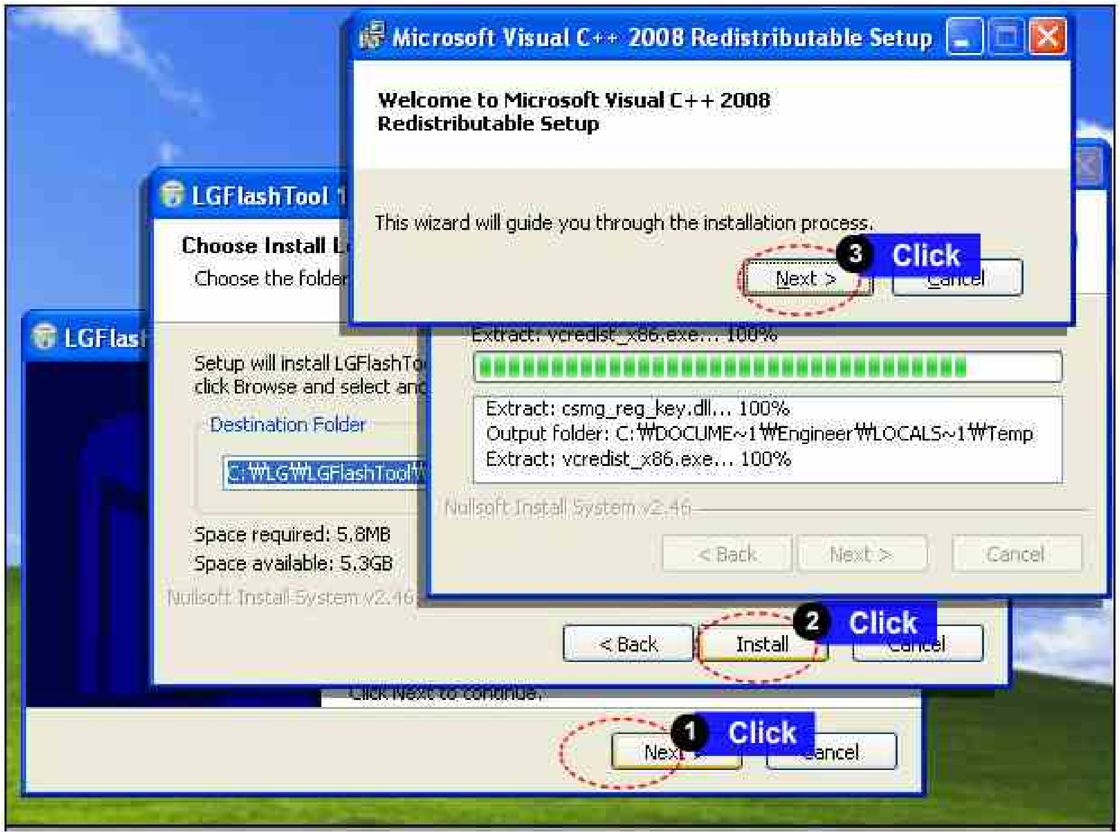



B2B LGMobile Support Tool
File
S/W Update
Tool & Download
Notification
1
Click
[GDMS] Notification
Install program
Execution file download
S.W Set up for LGFlashTool program
You need LGFlashTool execution file,
USBModemDriver &
Model File So you have to
Download from GDMS or GMSP
Notification
New SoftWare Version notice
Application
SEARCH
All
"2 Files"
Double Click (download)
2
No.
Titl
193
Setup_LGFlashTool_1.6.01.0529
02-14-2012
48
229
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1
12-20-2011
984
<< Previous 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Next >>
Total 101 Count 1/11 Page

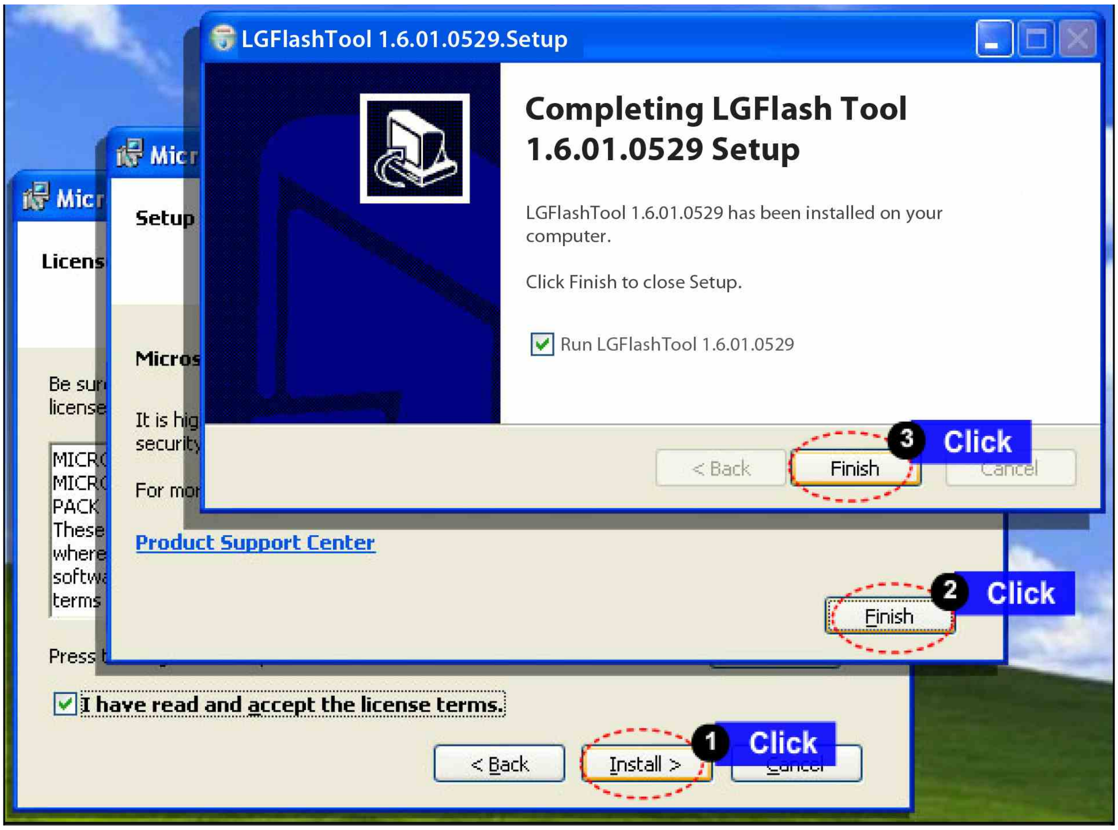

B2B LGMobile Support Tool
File
S/W Update
Tool & Download
Phone S/W File Download
Application, Manual & USB Driver Download...
1
Click
Tool mode
Country
Mexico
2
Choice
Model
LGD295
buyer
TCL/TELCEL(MEXICO)
File List
BIN_LGA395AT-00-V10b-724-XX-MAR-13-2013+0.zip
※ Download : Double Click
Set Ver
Name
Type
File Ver
LGD295 AT-00-V10a-SSV-XXX-JUN-03-2013+0
V10a
3
Double Click (download)
Close


B2B LGMobile Support Tool
File
S/W Update
Tool & Download
Phone S/W File Download
Application, Manual & USB Driver Download...
1 Click
Apps & Manual
Country
Mexico
2 Choice
Model
LGD295
S/W List
* Download : Double Click
Set Ver
Name
Type
File Ver
LGD295_20140923_LGFLASHv190_Download
DLTOOL_
3 Double Click (download)
Close

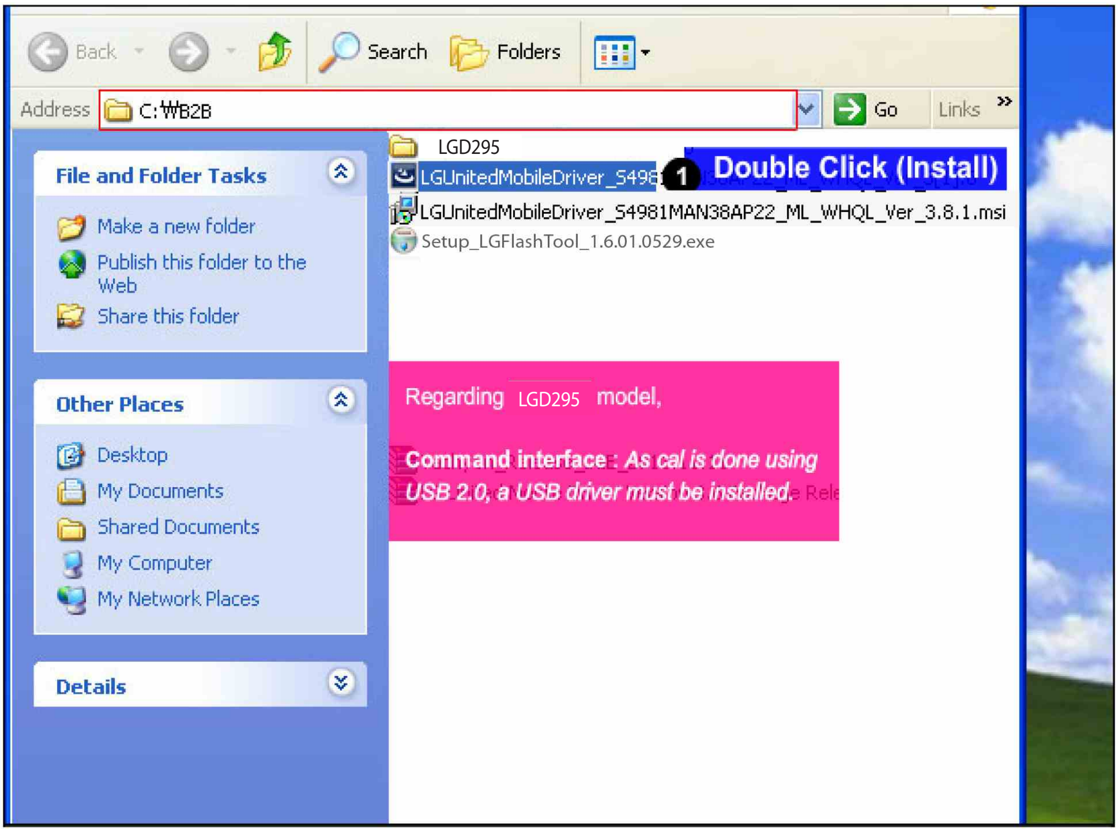

C:₩B2B
Go
Links
LGD295
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.exe
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.msi
Setup_LGFlashTool_1.6.01.0529.exe
2 Double Click (Install)
Make a new folder
Publish this folder to the Web
Share this folder
1 'Zip' File untie
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1
Setup_LGFlashTool_1.6.01.0529
Desktop
My Documents
Shared Documents
My Computer
My Network Places

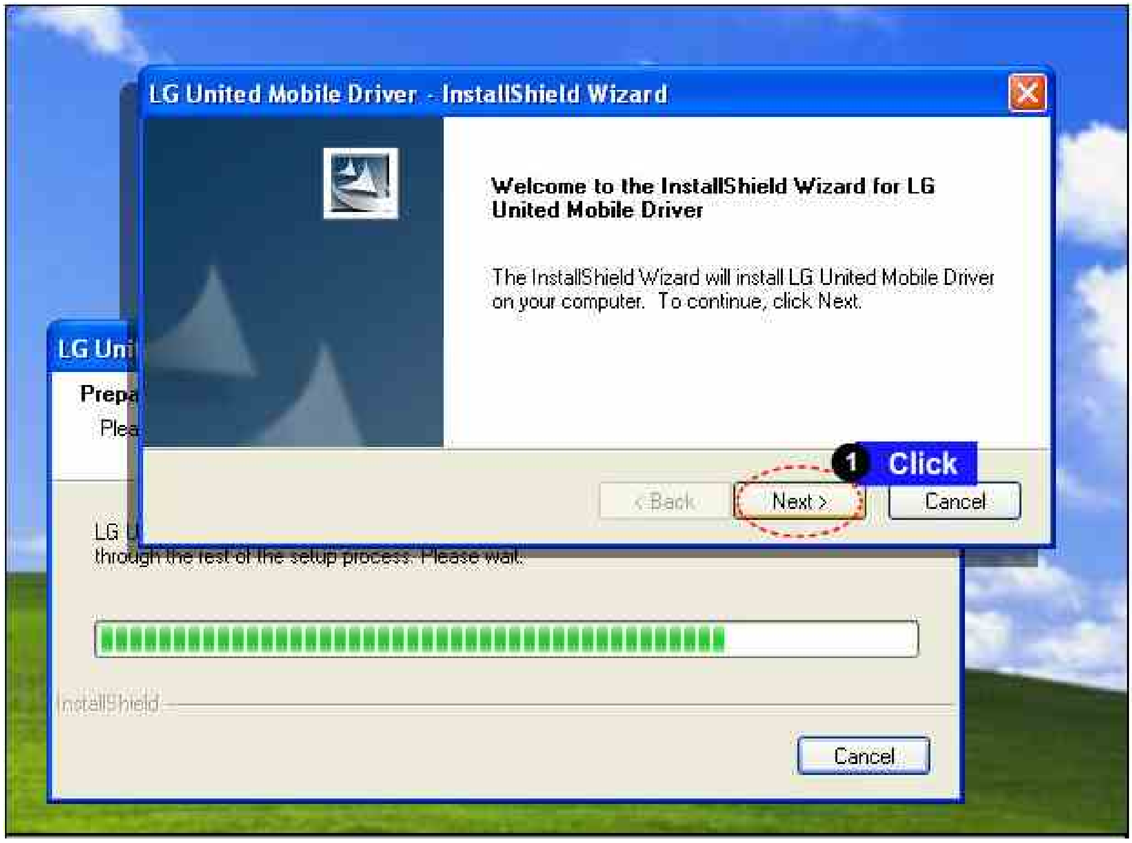


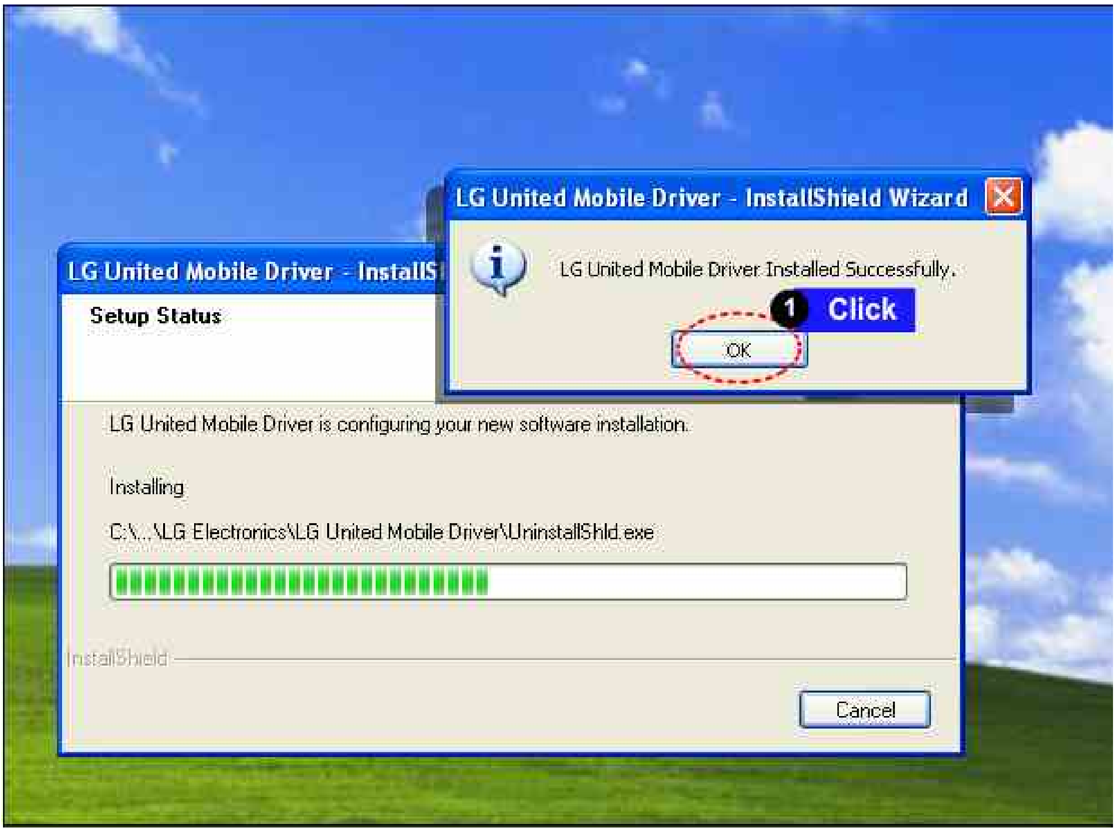
Microsoft Visual C++ 2008 Redistributable Setup
Welcome to Microsoft Visual C++ 2008 Redistributable Setup
This wizard will guide you through the installation process.
Next >
Cancel
3 Click
LGFlashTool
Choose Install L
Choose the folder
Setup will install LGFlashTo
click Browse and select an
Destination Folder
C:\LG\LGFlashTool\
Space required: 5.8MB
Space available: 5.3GB
Nullsoft Install System v2.46
Extract: vcredist_x86.exe... 100%
Extract: csmg_reg_key.dll... 100%
Output folder: C:\DOCUME~1\Engineer\LOCALS~1\Temp
Extract: vcredist_x86.exe... 100%
< Back
Next >
Cancel
< Back
Install
2 Click
Click Next to continue.
Next >
1 Click

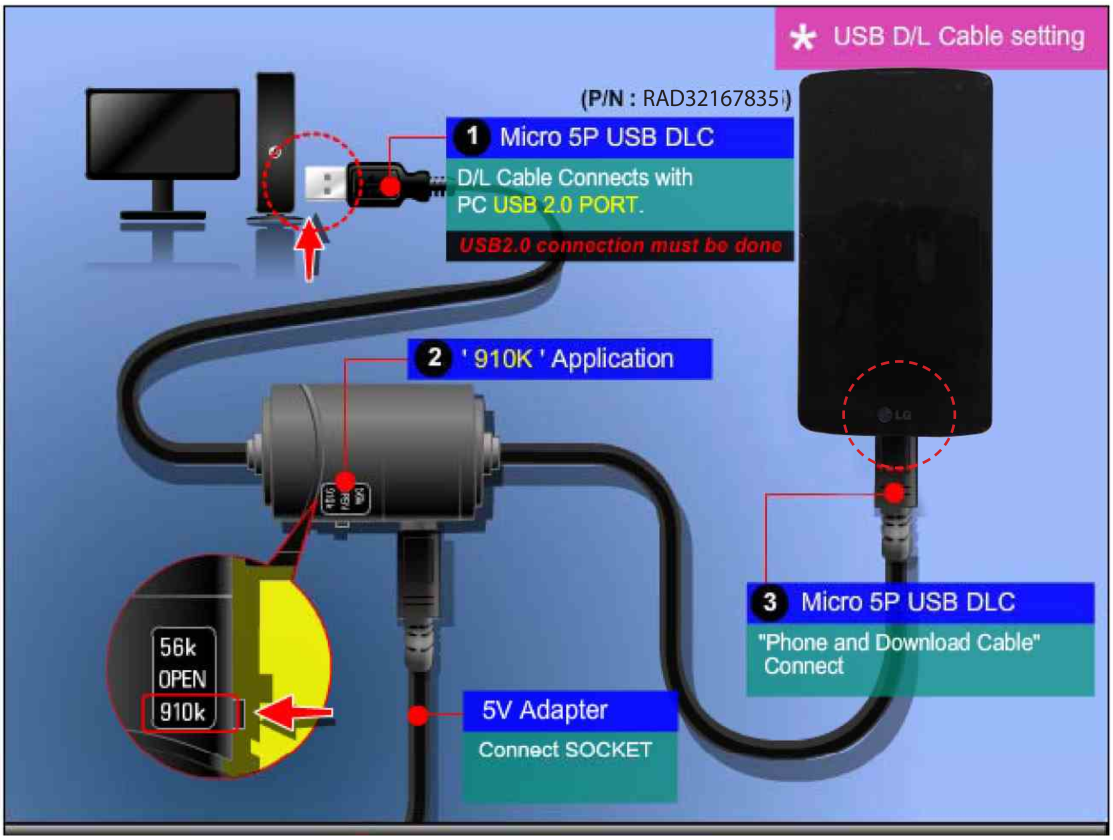


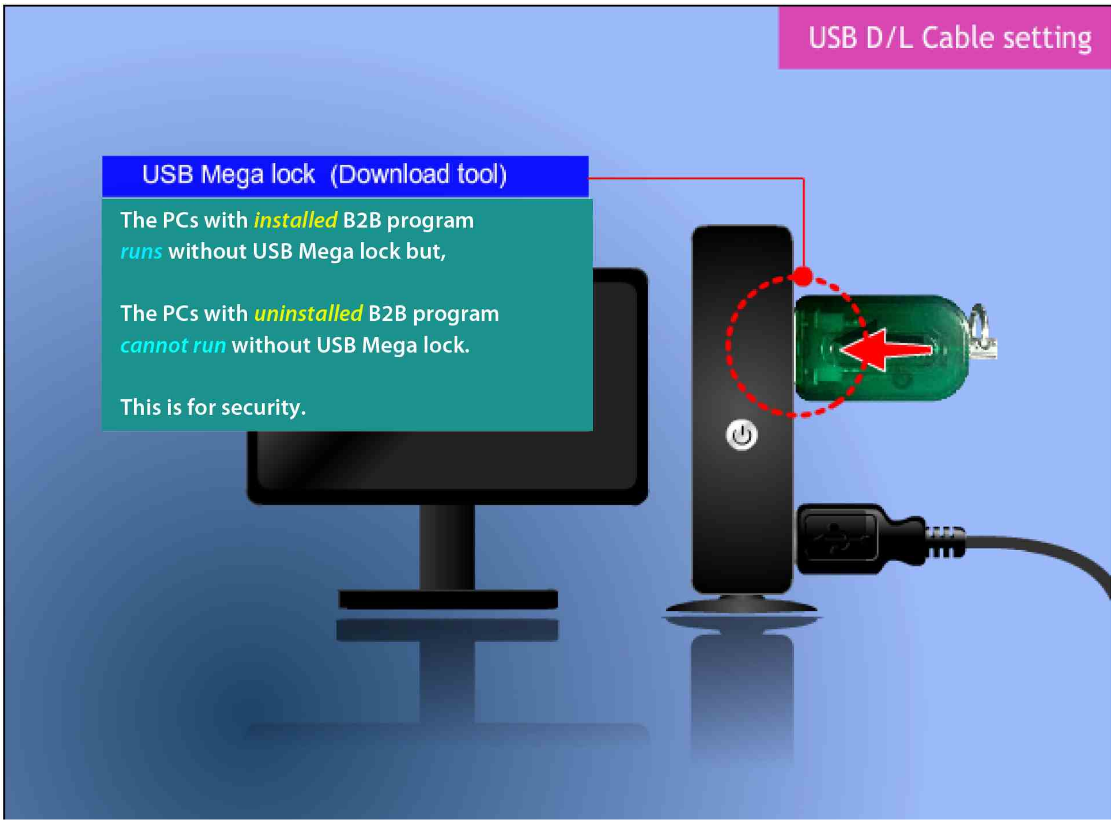
LGFlashTool 1.6.01.0529.Setup
Completing LGFlash Tool 1.6.01.0529 Setup
LGFlashTool 1.6.01.0529 has been installed on your computer.
Click Finish to close Setup.
Run LGFlashTool 1.6.01.0529
< Back
Finish
Cancel
3 Click
Product Support Center
Finish
2 Click
I have read and accept the license terms.
< Back
Install >
1 Click


Back
Search
Folders
Address
C:₩B2B
Go
Links
File and Folder Tasks
Make a new folder
Publish this folder to the Web
Share this folder
Other Places
Desktop
My Documents
Shared Documents
My Computer
My Network Places
Details
LGD295
LGUnitedMobileDriver_S498
1
Double Click (Install)
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.msi
Setup_LGFlashTool_1.6.01.0529.exe
Regarding LGD295 model,
Command interface : As cal is done using USB 2.0, a USB driver must be installed.

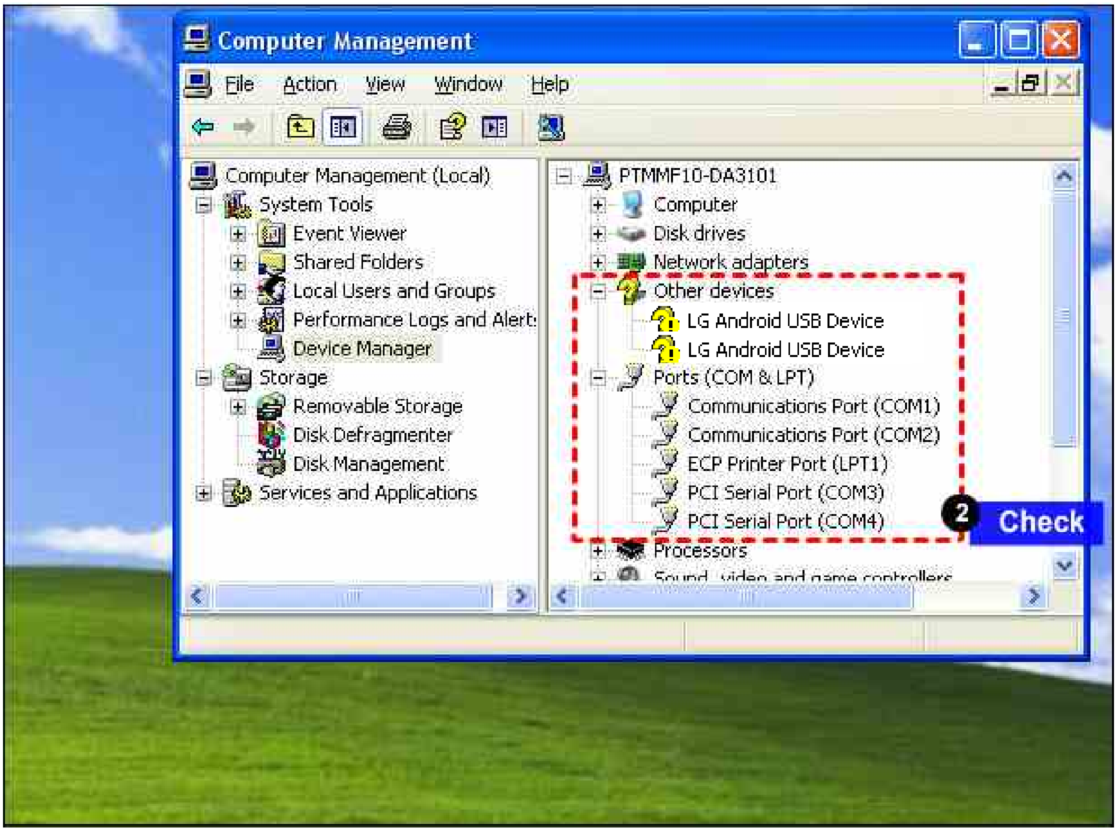

LG United Mobile Driver - InstallShield Wizard
Welcome to the InstallShield Wizard for LG United Mobile Driver
The InstallShield Wizard will install LG United Mobile Driver on your computer. To continue, click Next.
1
Click
< Back
Next >
Cancel
through the rest of the setup process. Please wait.
InstallShield
Cancel

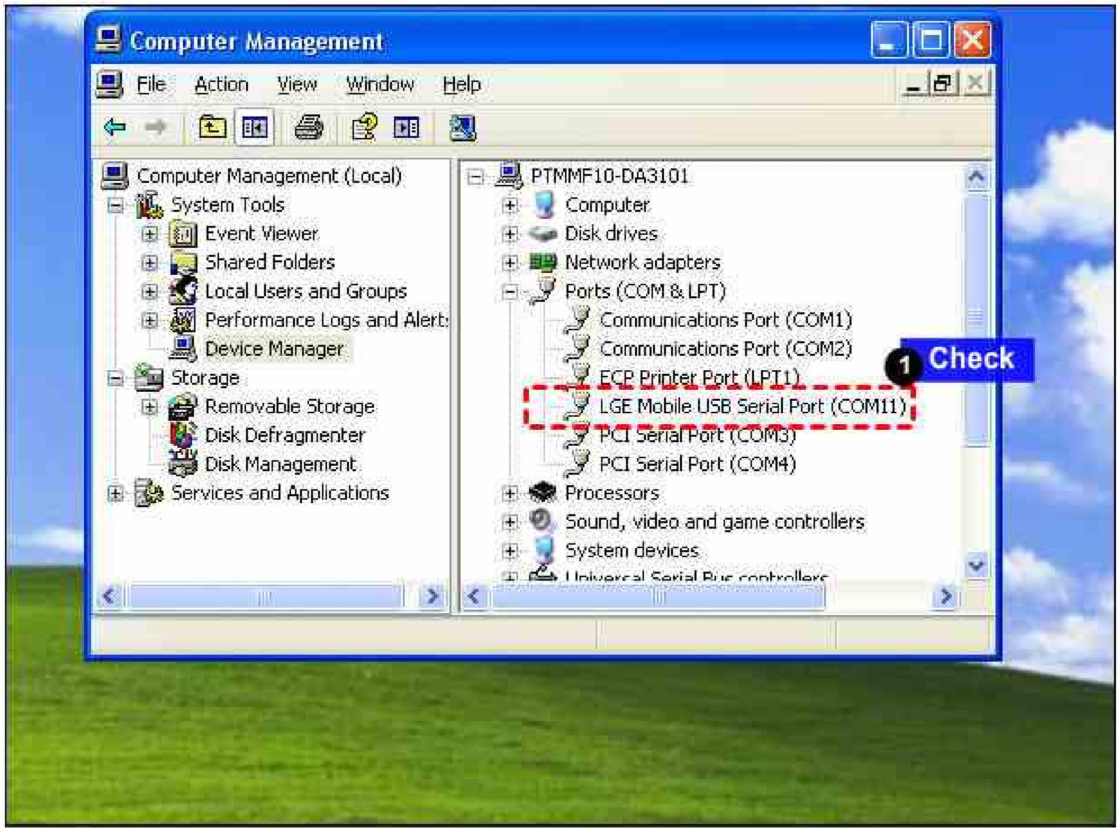

LG United Mobile Driver - InstallShield Wizard
LG United Mobile Driver Installed Successfully.
1 Click
OK
LG United Mobile Driver - InstallS
Setup Status
LG United Mobile Driver is configuring your new software installation.
Installing
C:\...\LG Electronics\LG United Mobile Driver\UninstallShld.exe
InstallShield
Cancel

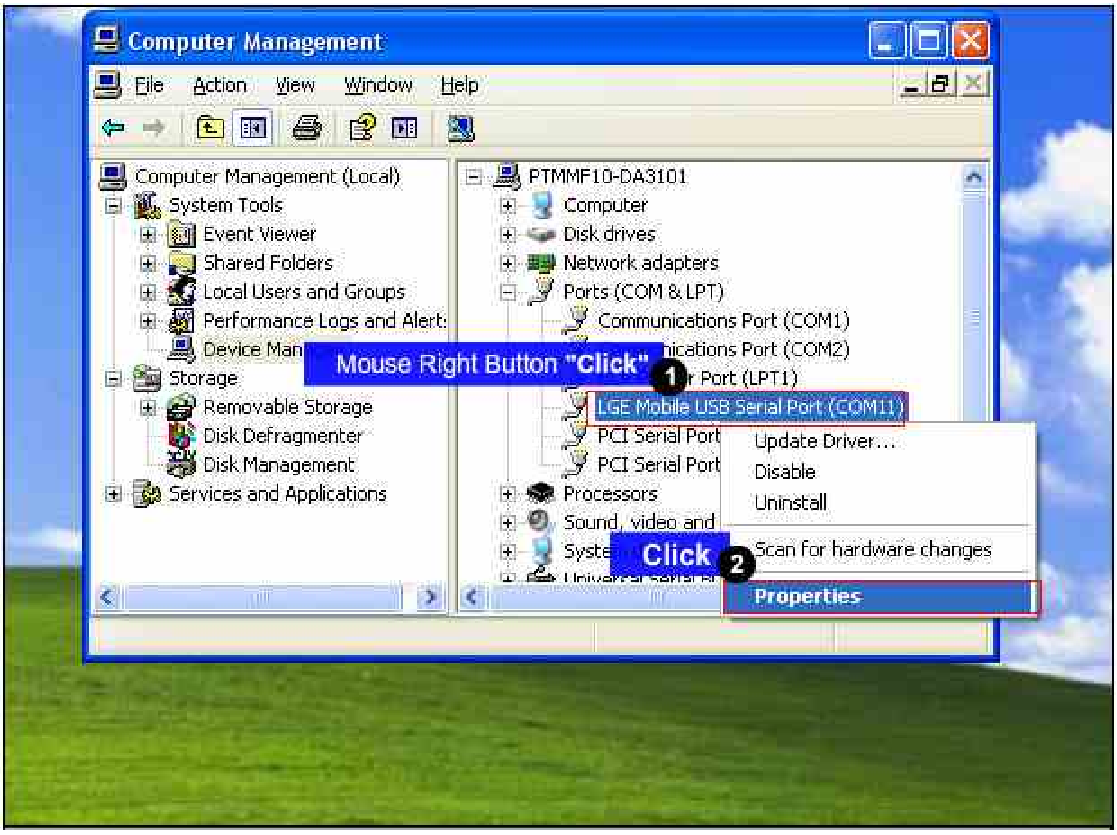

USB D/L Cable setting
(P/N : RAD32167835)
1 Micro 5P USB DLC
D/L Cable Connects with PC USB 2.0 PORT.
USB2.0 connection must be done
2 ' 910K ' Application
56k
OPEN
910k
3 Micro 5P USB DLC
"Phone and Download Cable" Connect
5V Adapter
Connect SOCKET


USB D/L Cable setting
USB Mega lock (Download tool)
The PCs with installed B2B program runs without USB Mega lock but,
The PCs with uninstalled B2B program cannot run without USB Mega lock.
This is for security.

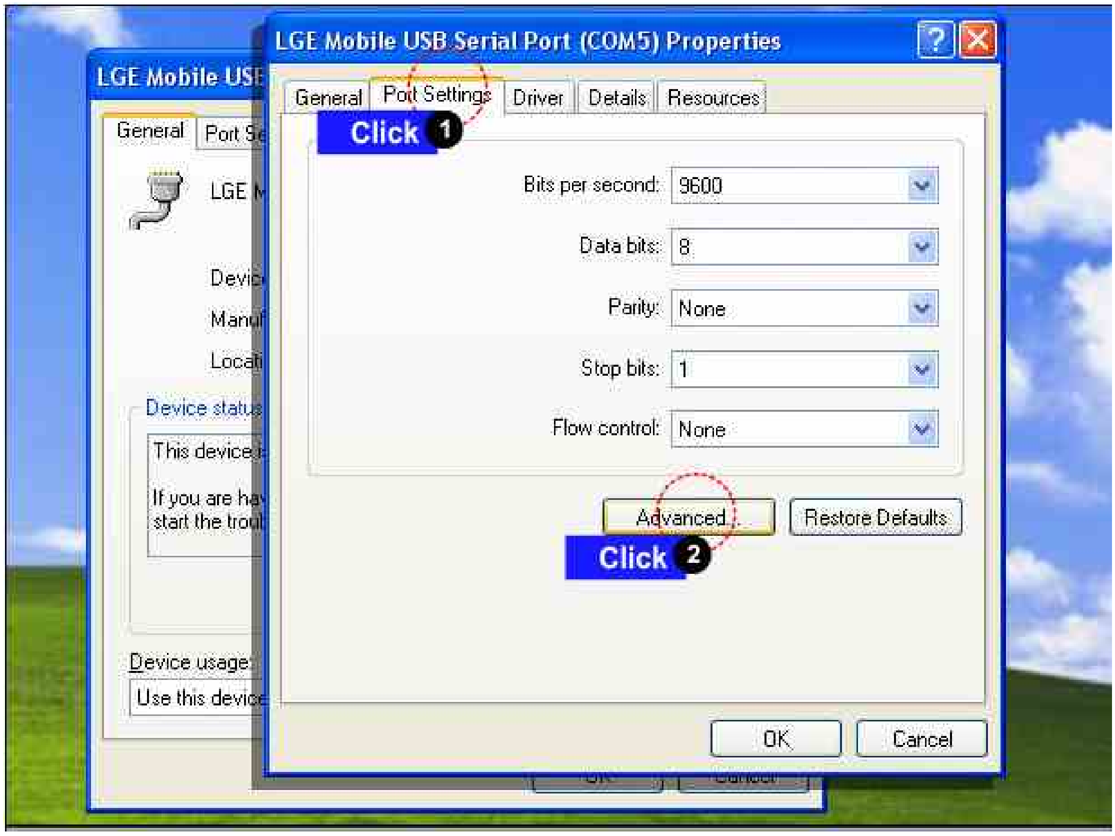

Open
Explore
Search...
Manage
Map Network Drive...
Disconnect Network Drive...
Create Shortcut
Delete
Rename
Properties
Internet Explorer
* If phone connected, Automatic Installation
1
Found New Hardware
LG Mobile Composite USB Device
EN
9:38 AM


Computer Management
File Action View Window Help
Computer Management (Local)
System Tools
Event Viewer
Shared Folders
Local Users and Groups
Performance Logs and Alert:
Device Manager
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
PTMMF10-DA3101
Computer
Disk drives
Network adapters
Other devices
LG Android USB Device
LG Android USB Device
Ports (COM & LPT)
Communications Port (COM1)
Communications Port (COM2)
ECP Printer Port (LPT1)
PCI Serial Port (COM3)
PCI Serial Port (COM4)
Processors
2 Check

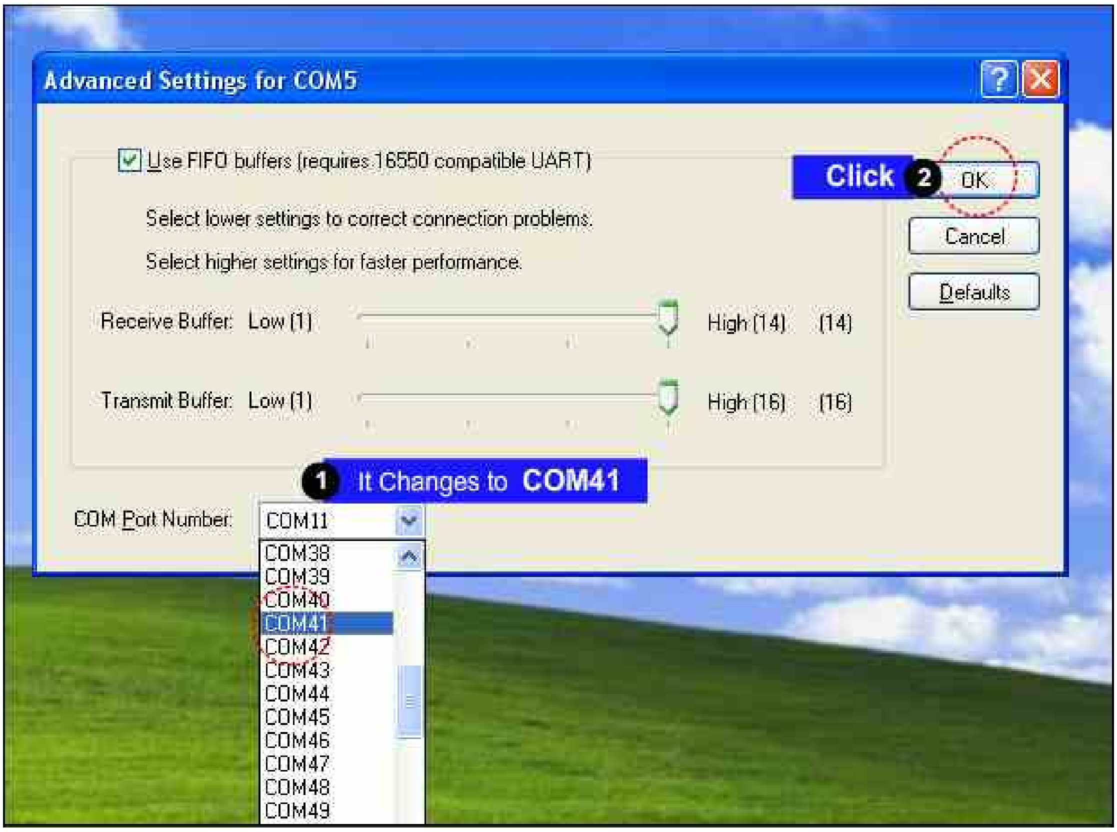

Computer Management
File Action View Window Help
Computer Management (Local)
System Tools
Event Viewer
Shared Folders
Local Users and Groups
Performance Logs and Alert:
Device Manager
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
PTMMF10-DA3101
Computer
Disk drives
Network adapters
Ports (COM & LPT)
Communications Port (COM1)
Communications Port (COM2)
ECP Printer Port (LPT1)
LGE Mobile USB Serial Port (COM11)
PCI Serial Port (COM3)
PCI Serial Port (COM4)
Processors
Sound, video and game controllers
System devices
1 Check



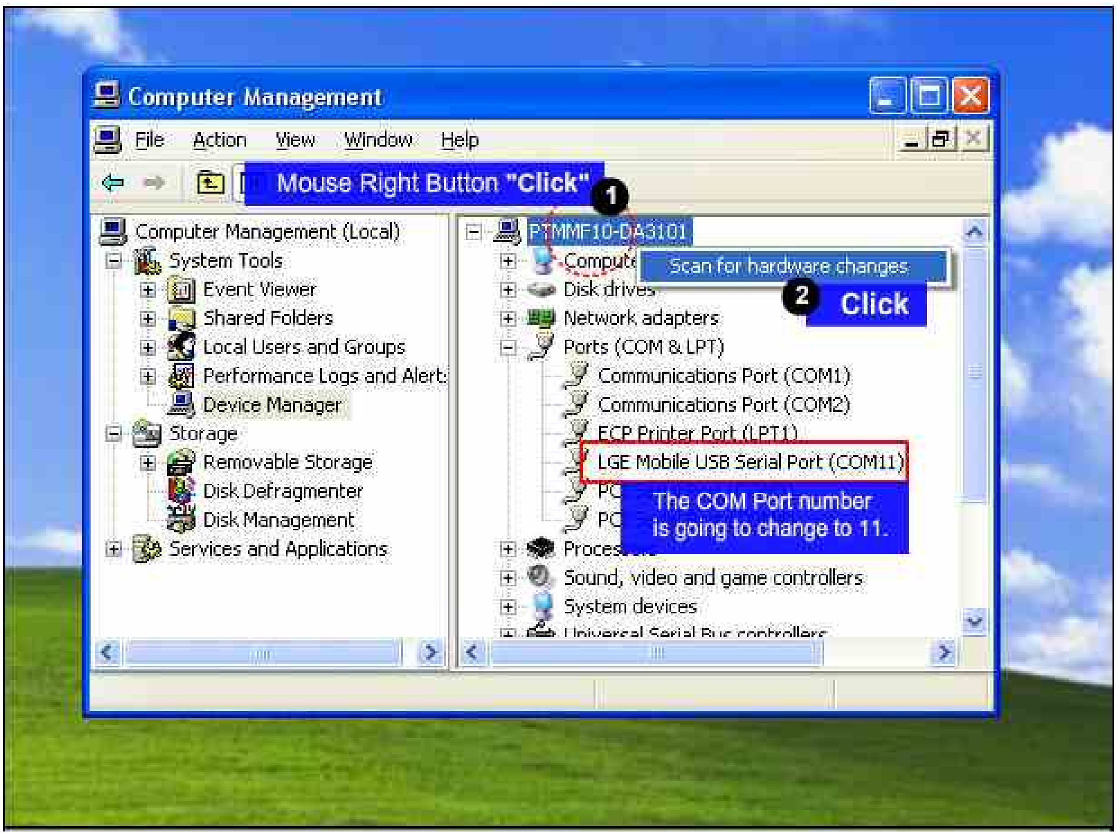
Computer Management
File Action View Window Help
Computer Management (Local)
System Tools
Event Viewer
Shared Folders
Local Users and Groups
Performance Logs and Alert:
Device Man
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
PTMMF10-DA3101
Computer
Disk drives
Network adapters
Ports (COM & LPT)
Communications Port (COM1)
nications Port (COM2)
r Port (LPT1)
LGE Mobile USB Serial Port (COM11)
PCI Serial Port
PCI Serial Port
Processors
Sound, video and
Mouse Right Button "Click" 1
Update Driver...
Disable
Uninstall
Scan for hardware changes
Click 2
Properties



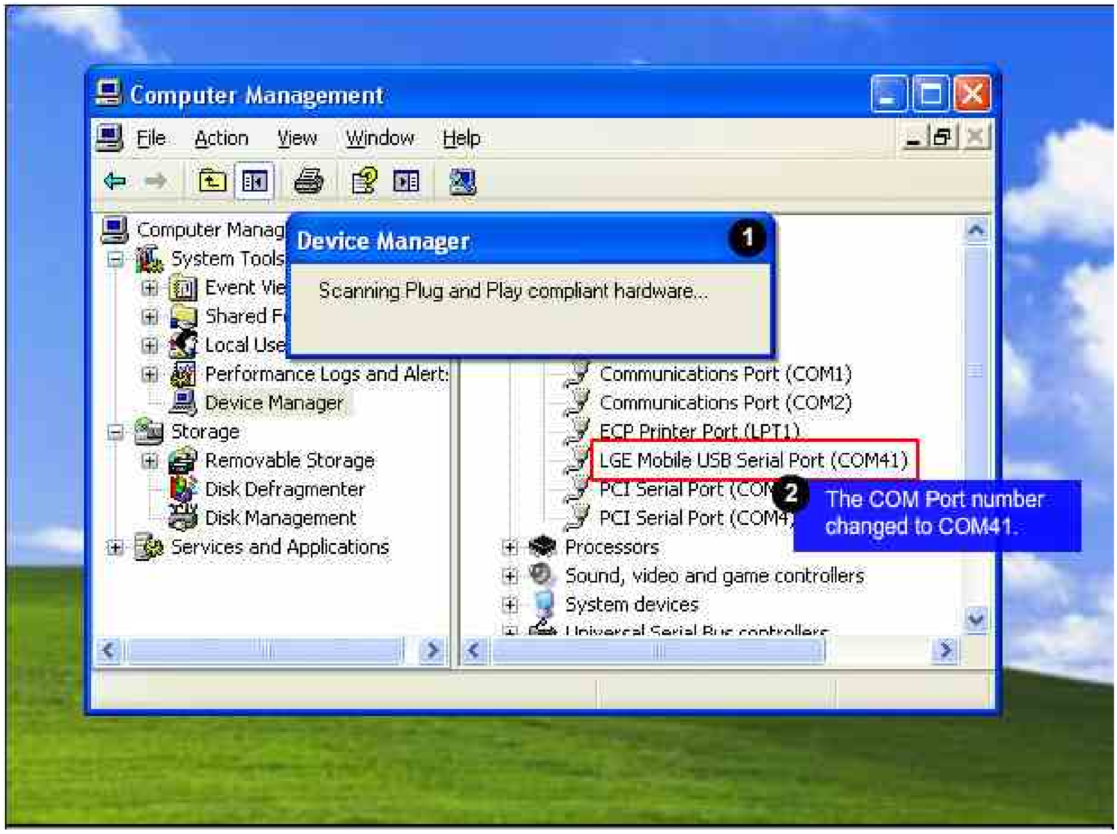
LGE Mobile USB Serial Port (COM5) Properties
General
Port Settings
Driver
Details
Resources
Click 1
Bits per second: 9600
Data bits: 8
Parity: None
Stop bits: 1
Flow control: None
Advanced...
Restore Defaults
Click 2
OK
Cancel

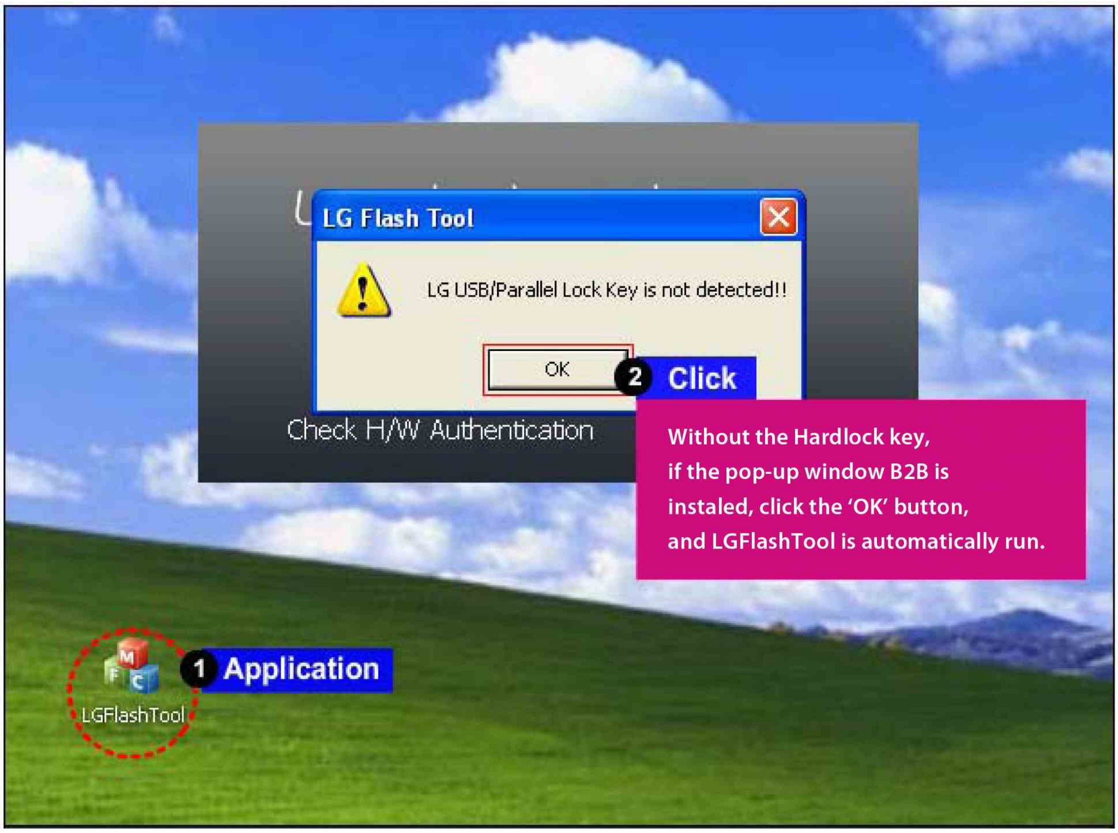


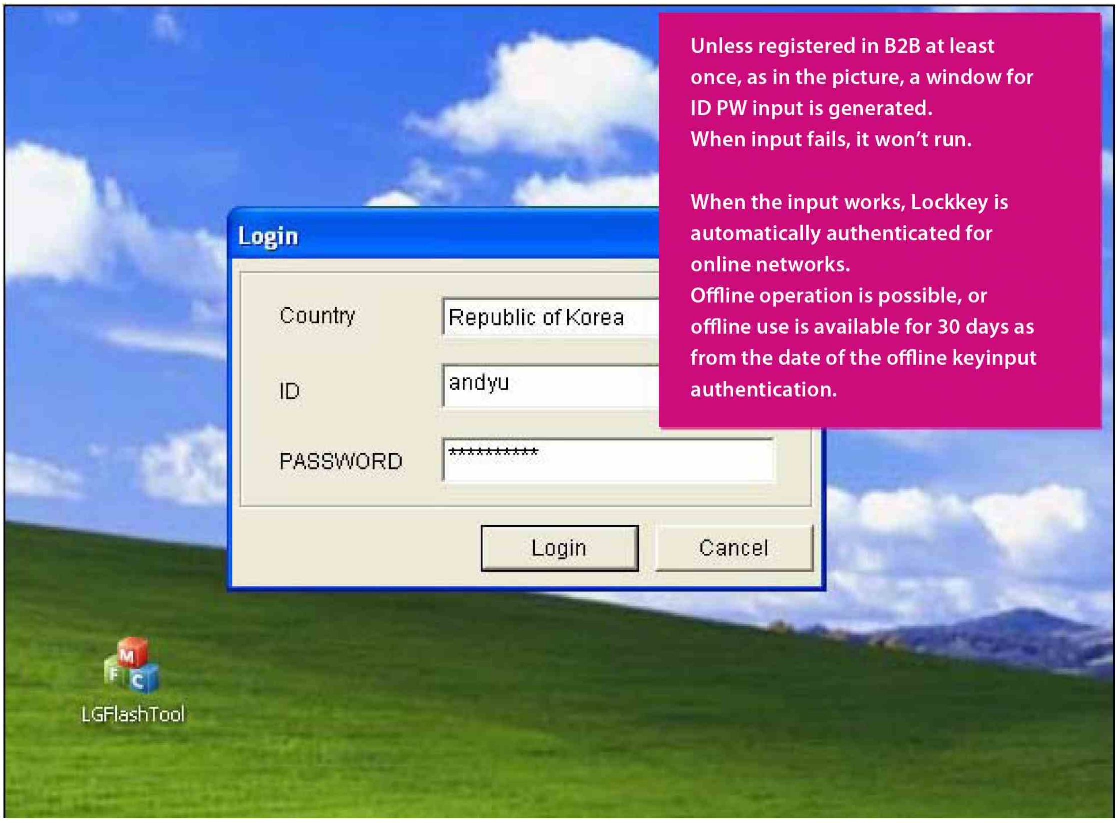
Advanced Settings for COM5
Use FIFO buffers (requires 16550 compatible UART)
Select lower settings to correct connection problems.
Select higher settings for faster performance.
Receive Buffer: Low (1)
High (14) (14)
Transmit Buffer: Low (1)
High (16) (16)
Click 2
OK
Cancel
Defaults
1 It Changes to COM41
COM Port Number:
COM11
COM38
COM39
COM40
COM41
COM42
COM43
COM44
COM45
COM46
COM47
COM48
COM49

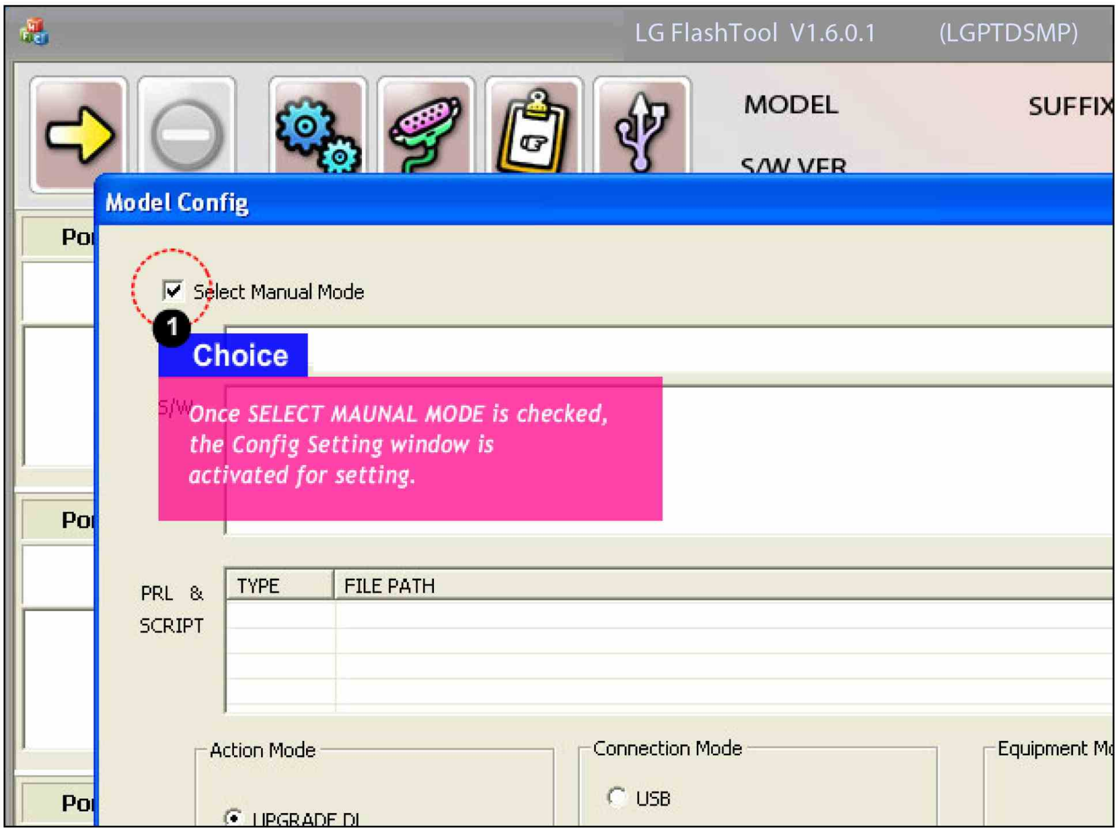


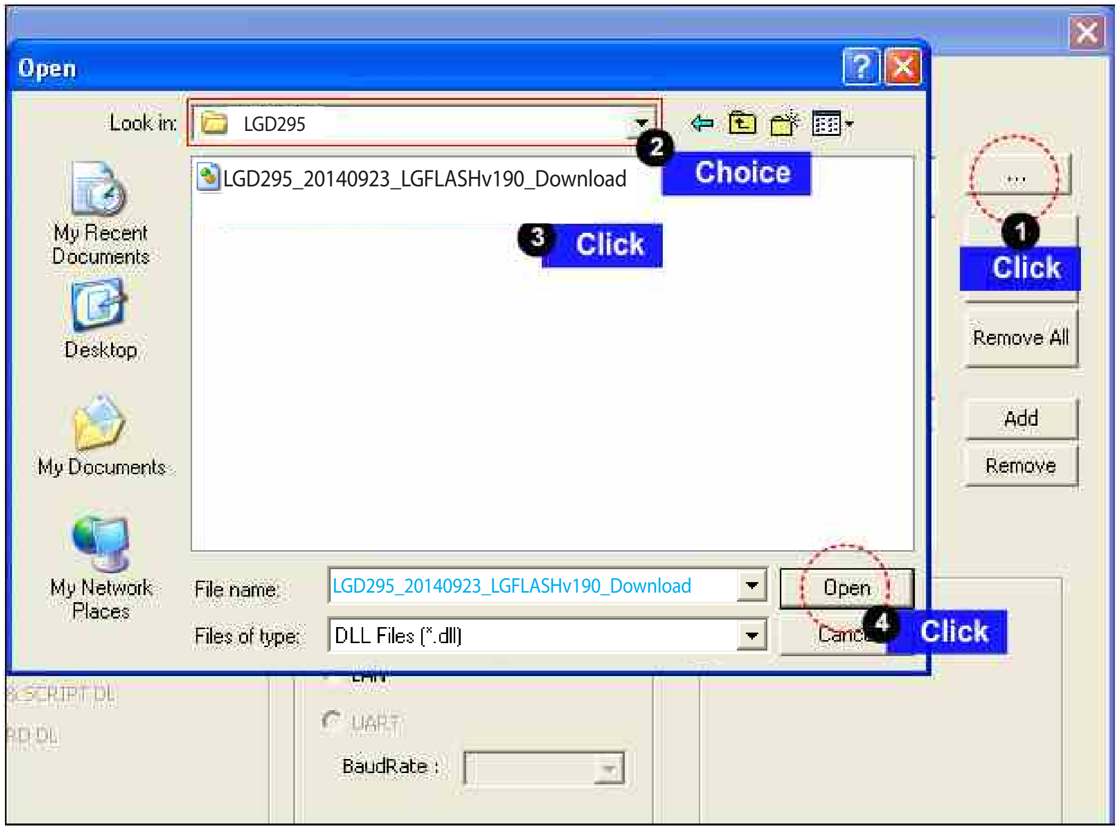
Computer Management
File Action View Window Help
Mouse Right Button "Click" 1
Computer Management (Local)
System Tools
Event Viewer
Shared Folders
Local Users and Groups
Performance Logs and Alert:
Device Manager
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
PTMMF10-DA3101
Scan for hardware changes
Disk drives
2 Click
Network adapters
Ports (COM & LPT)
Communications Port (COM1)
Communications Port (COM2)
ECP Printer Port (LPT1)
LGE Mobile USB Serial Port (COM11)
The COM Port number
is going to change to 11.
Sound, video and game controllers
System devices

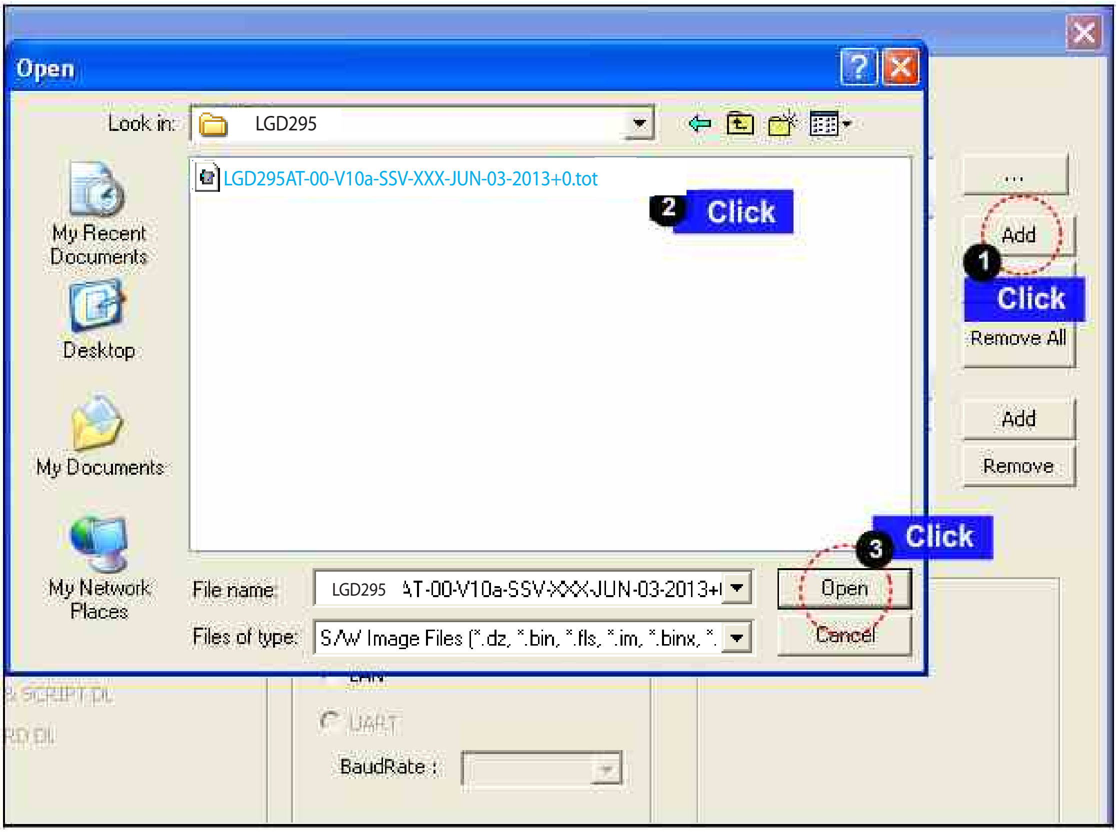

Computer Management
File Action View Window Help
Computer Manag
System Tools
Event Vie
Shared F
Local Use
Performance Logs and Alert:
Device Manager
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
Device Manager
1
Scanning Plug and Play compliant hardware...
Communications Port (COM1)
Communications Port (COM2)
ECP Printer Port (LPT1)
LGE Mobile USB Serial Port (COM41)
PCI Serial Port (COM
PCI Serial Port (COM4)
2
The COM Port number changed to COM41.
Processors
Sound, video and game controllers
System devices

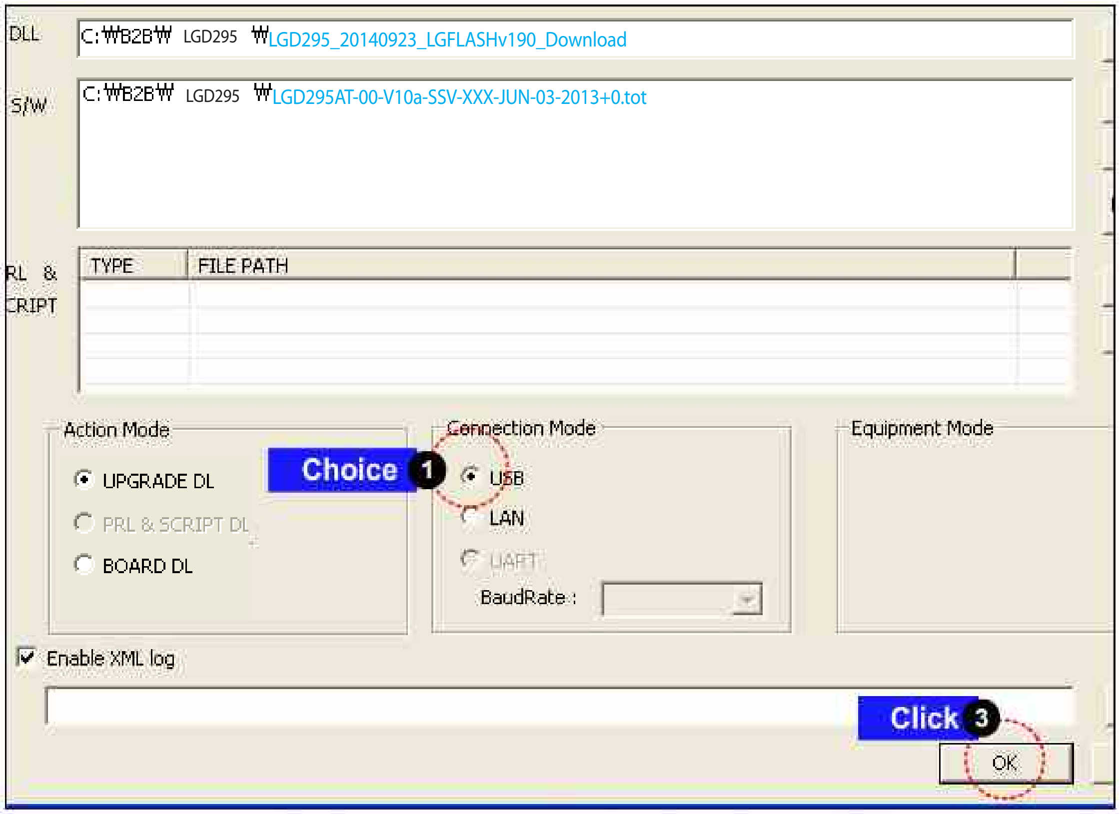

LG Flash Tool
LG USB/Parallel Lock Key is not detected!!
OK
2 Click
Check H/W Authentication
Without the Hardlock key,
if the pop-up window B2B is
instaled, click the 'OK' button,
and LGFlashTool is automatically run.
LGFlashTool
1 Application

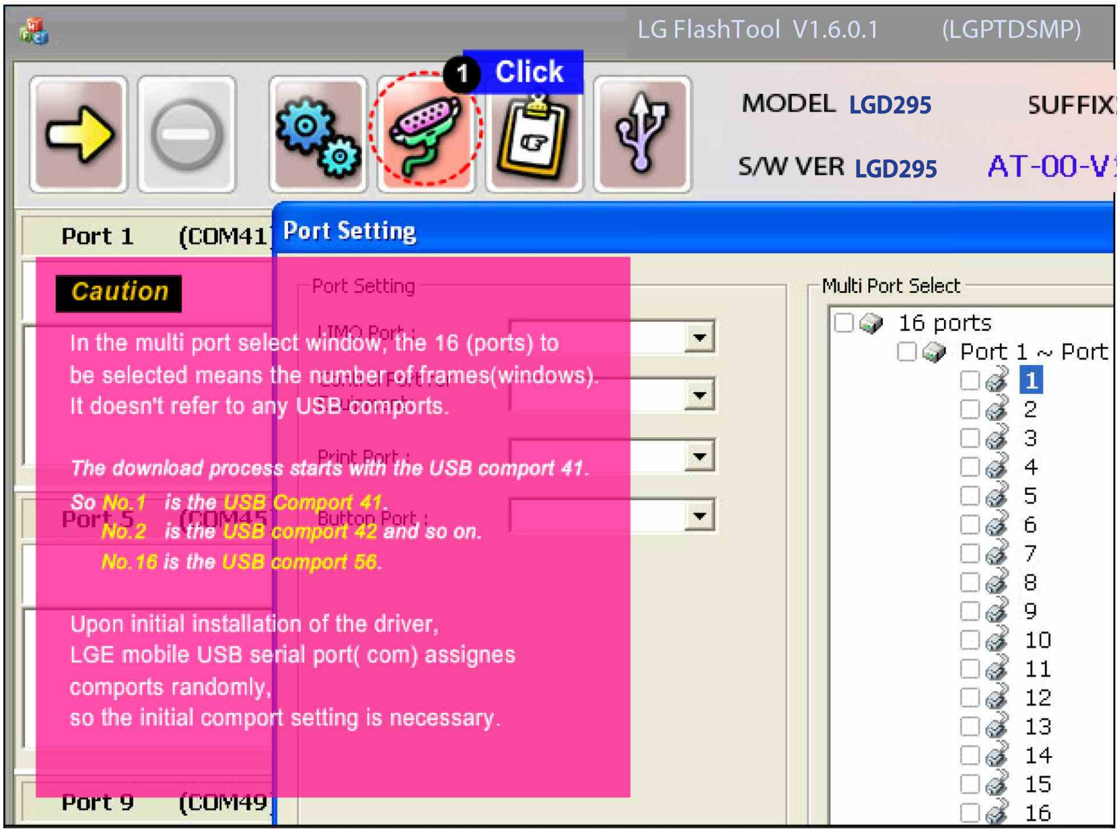

Unless registered in B2B at least once, as in the picture, a window for ID PW input is generated.
When input fails, it won't run.
When the input works, Lockkey is automatically authenticated for online networks.
Offline operation is possible, or offline use is available for 30 days as from the date of the offline keyinput authentication.
Login
Country
Republic of Korea
ID
andyu
PASSWORD
Login
Cancel
LGFlashTool

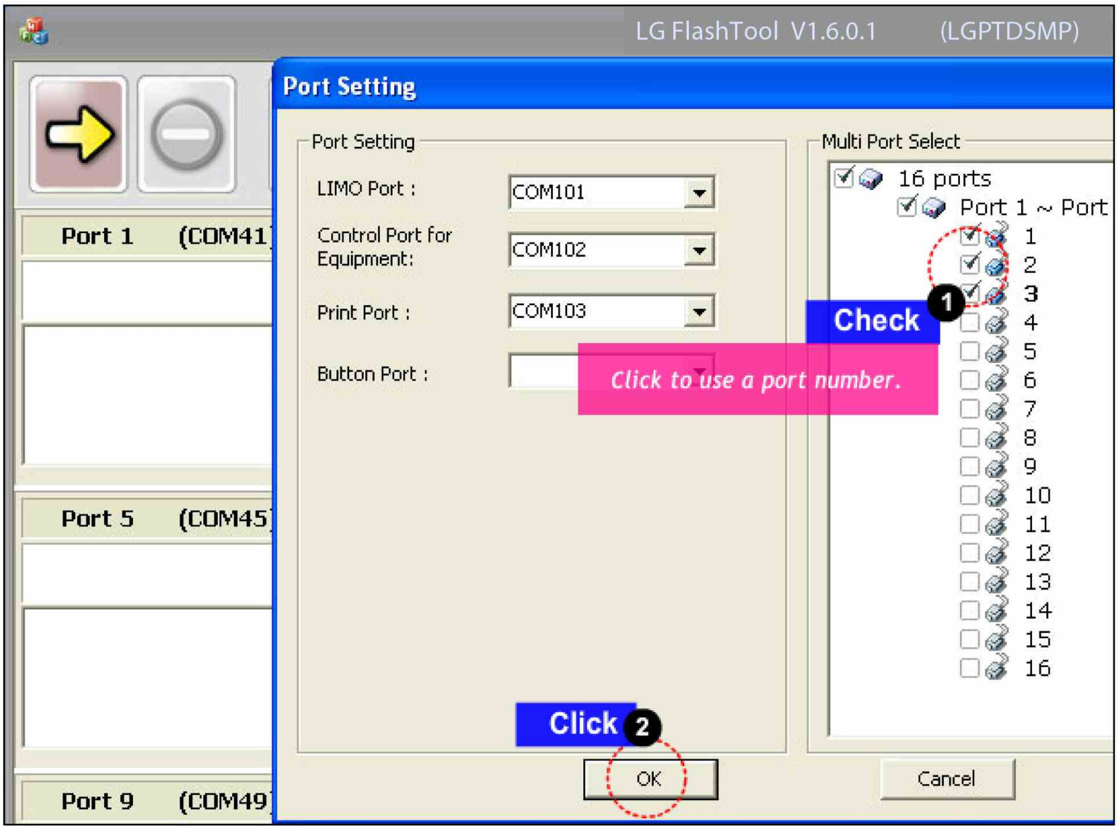

LG FlashTool V1.6.0.1 (LGPTDSMP)
MODEL
SUFFIX
Model Config
Select Manual Mode
1
Choice
Once SELECT MAUNAL MODE is checked, the Config Setting window is activated for setting.
PRL &
SCRIPT
TYPE
FILE PATH
Action Mode
Connection Mode
USB


Open
Look in:
LGD295
LGD295_20140923_LGFLASHv190_Download
2 Choice
3 Click
1 Click
My Recent Documents
Desktop
My Documents
My Network Places
Remove All
Add
Remove
File name:
LGD295_20140923_LGFLASHv190_Download
Open
Files of type:
DLL Files (*.dll)
4 Click
UART
BaudRate :

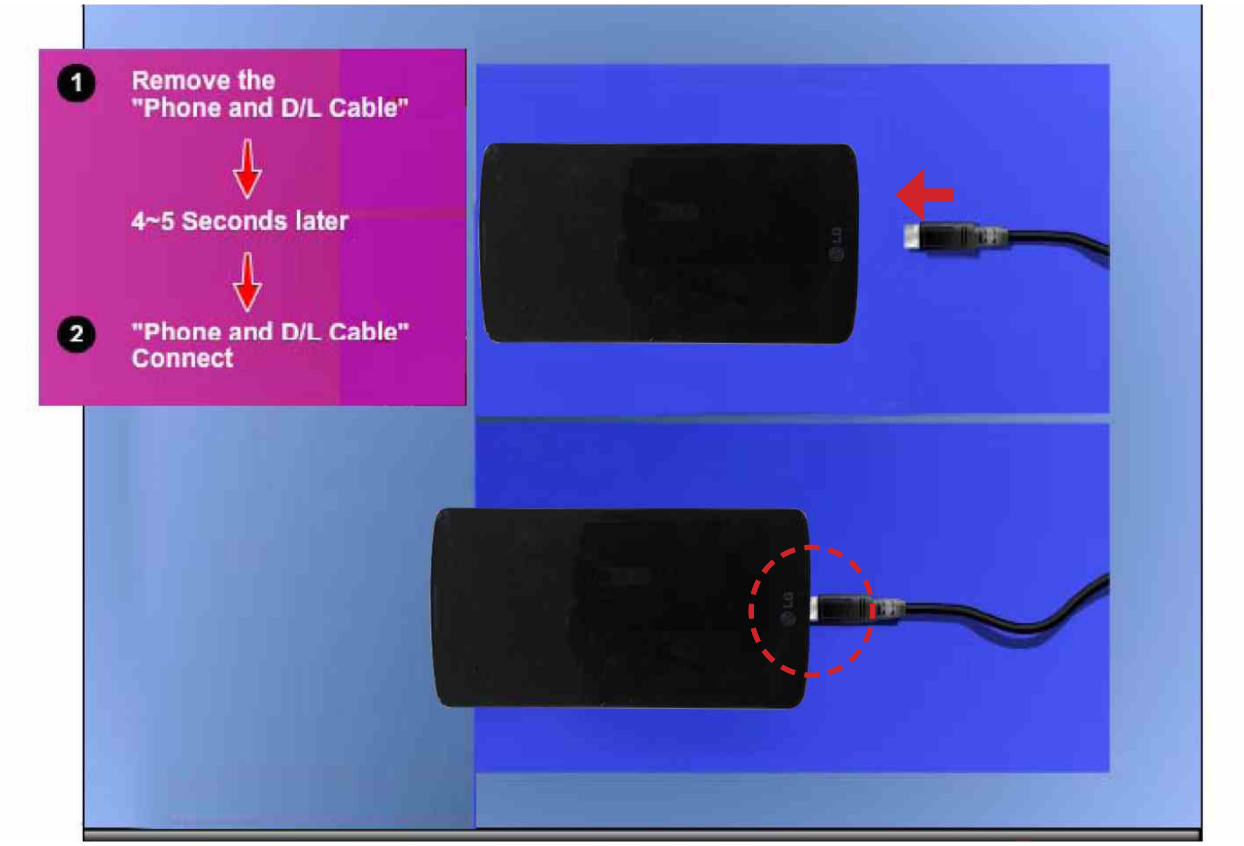

Open
Look in:
LGD295
LGD295AT-00-V10a-SSV-XXX-JUN-03-2013+0.tot
2 Click
Add
1 Click
Remove All
My Recent Documents
Desktop
My Documents
My Network Places
Add
Remove
3 Click
File name:
Open
Files of type:
S/W Image Files (*.dz, *.bin, *.fls, *.im, *.binx, *.
Cancel
UART
BaudRate :

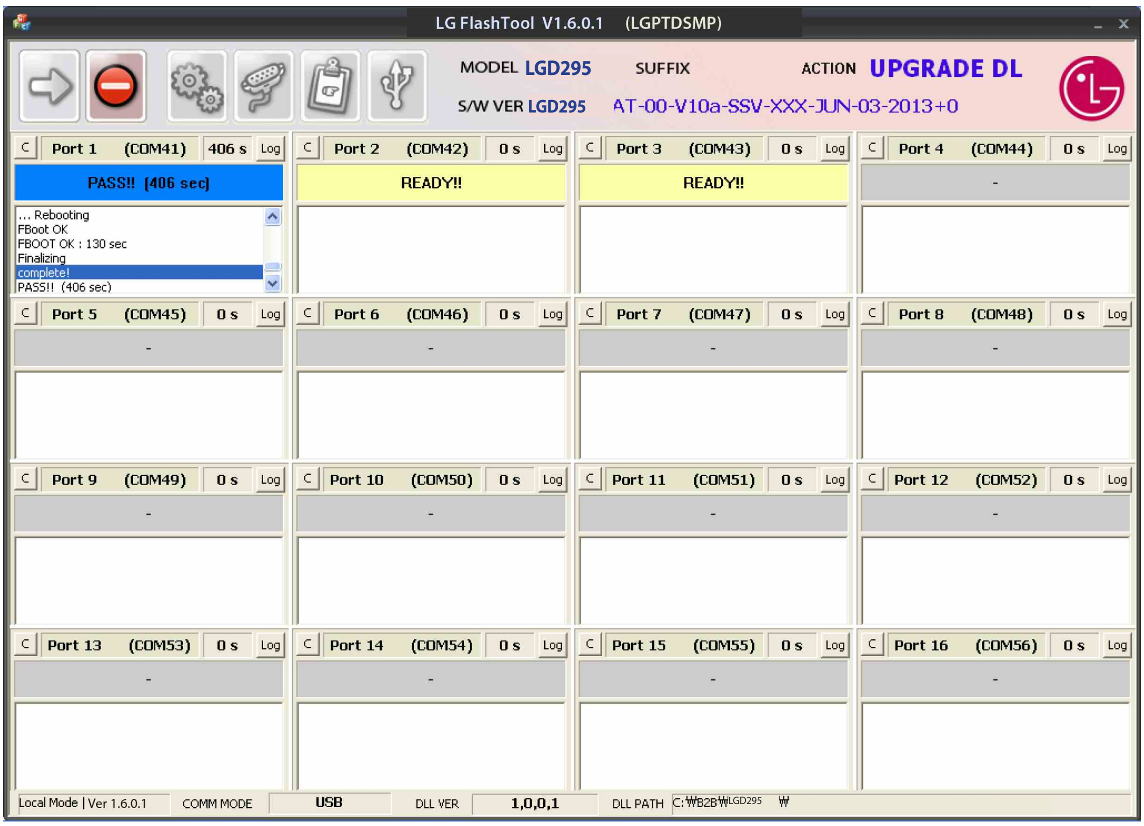

DLL
C:₩B2B₩ LGD295 ₩LGD295_20140923_LGFLASHv190_Download
S/W
C:₩B2B₩ LGD295 ₩LGD295AT-00-V10a-SSV-XXX-JUN-03-2013+0.tot
TYPE
FILE PATH
Action Mode
UPGRADE DL
PRL & SCRIPT DL
BOARD DL
Choice 1
Connection Mode
USB
LAN
UART
BaudRate :
Equipment Mode
Enable XML log
Click 3
OK

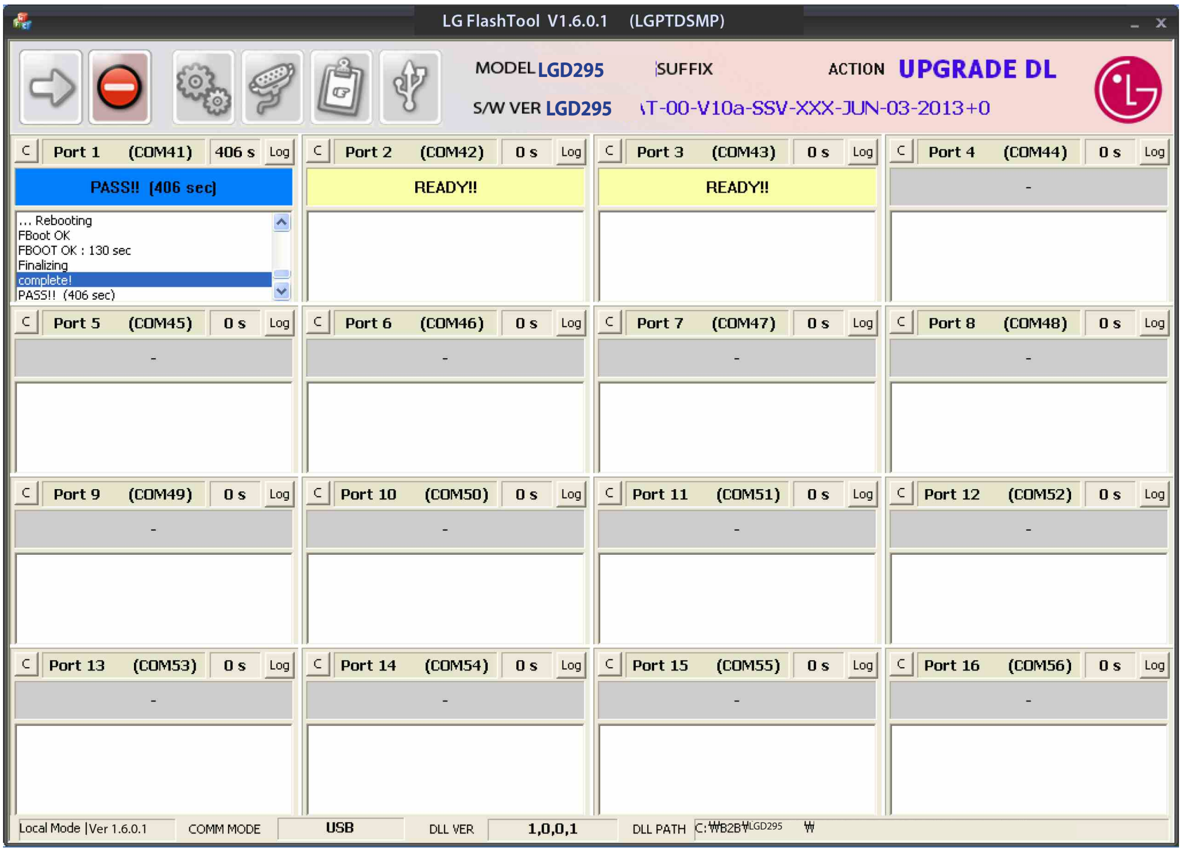

LG FlashTool V1.6.0.1 (LGPTDSMP)
1 Click
MODEL LGD295
SUFFIX
S/W VER LGD295
AT-00-V
Port 1 (COM41)
Port Setting
Caution
In the multi port select window, the 16 (ports) to be selected means the number of frames(windows). It doesn't refer to any USB comports.
The download process starts with the USB comport 41.
So No.1 is the USB Comport 41.
No.2 is the USB comport 42 and so on.
No.16 is the USB comport 56.
Upon initial installation of the driver, LGE mobile USB serial port(com) assignes comports randomly, so the initial comport setting is necessary.
Multi Port Select
16 ports
Port 1 ~ Port
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Port 9 (COM49)


LG FlashTool V1.6.0.1 (LGPTDSMP)
Port Setting
Port Setting
LIMO Port :
COM101
Control Port for Equipment:
COM102
Print Port :
COM103
Button Port :
Click to use a port number.
Multi Port Select
16 ports
Port 1 ~ Port
Check
Click
OK
Cancel
Port 1 (COM41)
Port 5 (COM45)
Port 9 (COM49)


LG FlashTool V1.6.0.1 (LGPTDSMP)
Click
MODEL LGD295
SUFFIX
S/W VER LGD295
Port 1 (COM41) 0 s Log
READY!!
Port 2 (COM42) 0 s Log
READY!!
Port 3
Port 5 (COM45) 0 s Log
Port 6 (COM46) 0 s Log
Port 7
Port 9 (COM49) 0 s Log
Port 10 (COM50) 0 s Log
Port 11

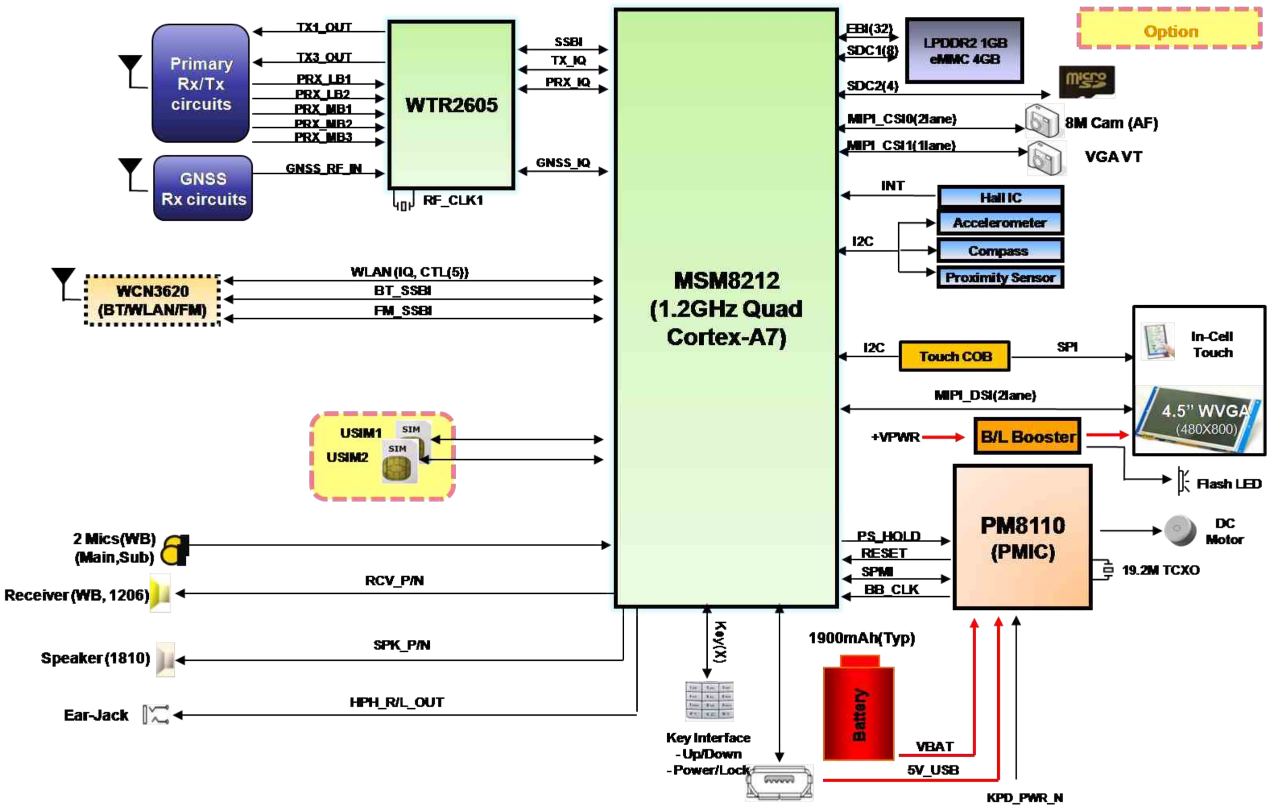

1
Remove the
"Phone and D/L Cable"
4~5 Seconds later
2
"Phone and D/L Cable"
Connect


LG FlashTool V1.6.0.1 (LGPTDSMP)
MODEL LGD295
SUFFIX
ACTION UPGRADE DL
S/W VER LGD295 AT-00-V10a-SSV-XXX-JUN-03-2013+0
Port 1 (COM41) 406 s Log
PASS!! [406 sec]
... Rebooting
FBoot OK
FBOOT OK : 130 sec
Finalizing
complete!
PASS!! (406 sec)
Port 2 (COM42) 0 s Log
READY!!
Port 3 (COM43) 0 s Log
READY!!
Port 4 (COM44) 0 s Log
Port 5 (COM45) 0 s Log
Port 6 (COM46) 0 s Log
Port 7 (COM47) 0 s Log
Port 8 (COM48) 0 s Log
Port 9 (COM49) 0 s Log
Port 10 (COM50) 0 s Log
Port 11 (COM51) 0 s Log
Port 12 (COM52) 0 s Log
Port 13 (COM53) 0 s Log
Port 14 (COM54) 0 s Log
Port 15 (COM55) 0 s Log
Port 16 (COM56) 0 s Log
Local Mode | Ver 1.6.0.1
COMM MODE
USB
DLL VER
1,0,0,1
DLL PATH

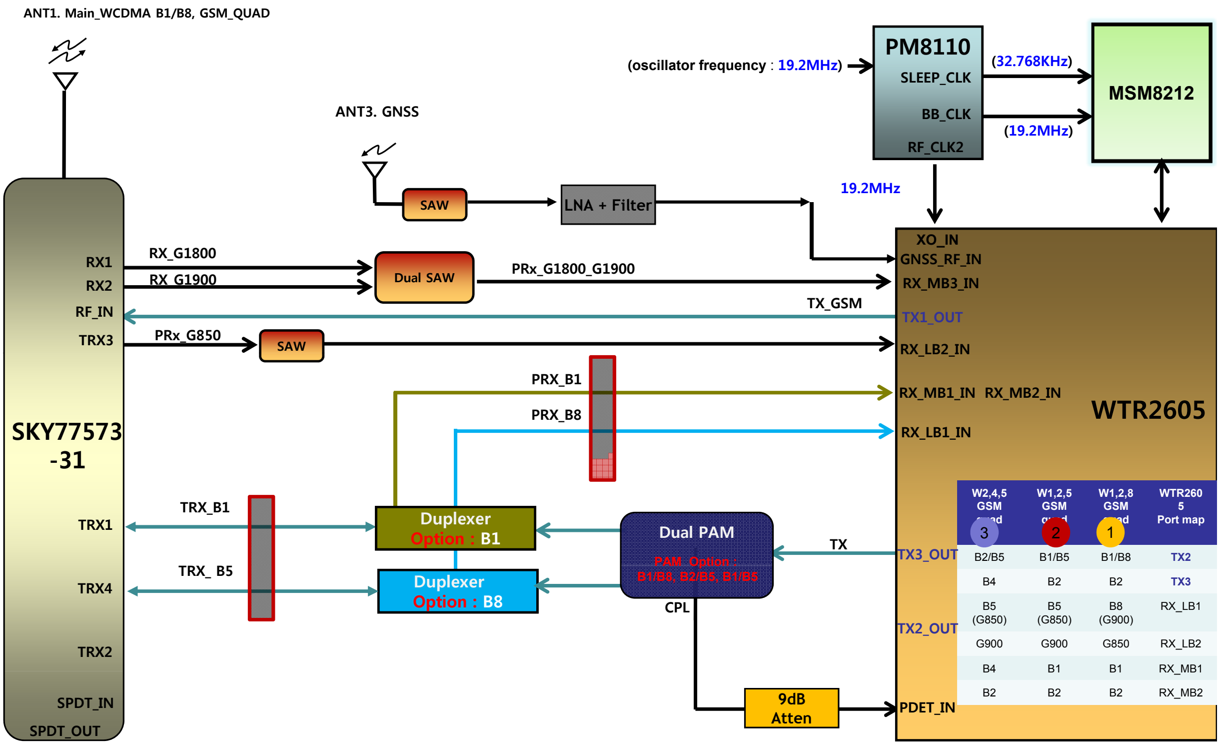

LG FlashTool V1.6.0.1 (LGPTDSMP)
MODEL LGD295
SUFFIX
ACTION UPGRADE DL
S/W VER LGD295
\T-00-V10a-SSV-XXX-JUN-03-2013+0
Port 1 (COM41) 406 s Log
PASS!! (406 sec)
... Rebooting
FBoot OK
FBOOT OK : 130 sec
Finalizing
complete!
PASS!! (406 sec)
Port 2 (COM42) 0 s Log
READY!!
Port 3 (COM43) 0 s Log
READY!!
Port 4 (COM44) 0 s Log
Port 5 (COM45) 0 s Log
Port 6 (COM46) 0 s Log
Port 7 (COM47) 0 s Log
Port 8 (COM48) 0 s Log
Port 9 (COM49) 0 s Log
Port 10 (COM50) 0 s Log
Port 11 (COM51) 0 s Log
Port 12 (COM52) 0 s Log
Port 13 (COM53) 0 s Log
Port 14 (COM54) 0 s Log
Port 15 (COM55) 0 s Log
Port 16 (COM56) 0 s Log
Local Mode |Ver 1.6.0.1
COMM MODE
USB
DLL VER
1,0,0,1
DLL PATH

5. BLOCK DIAGRAM

[D295] Total Block Diagram

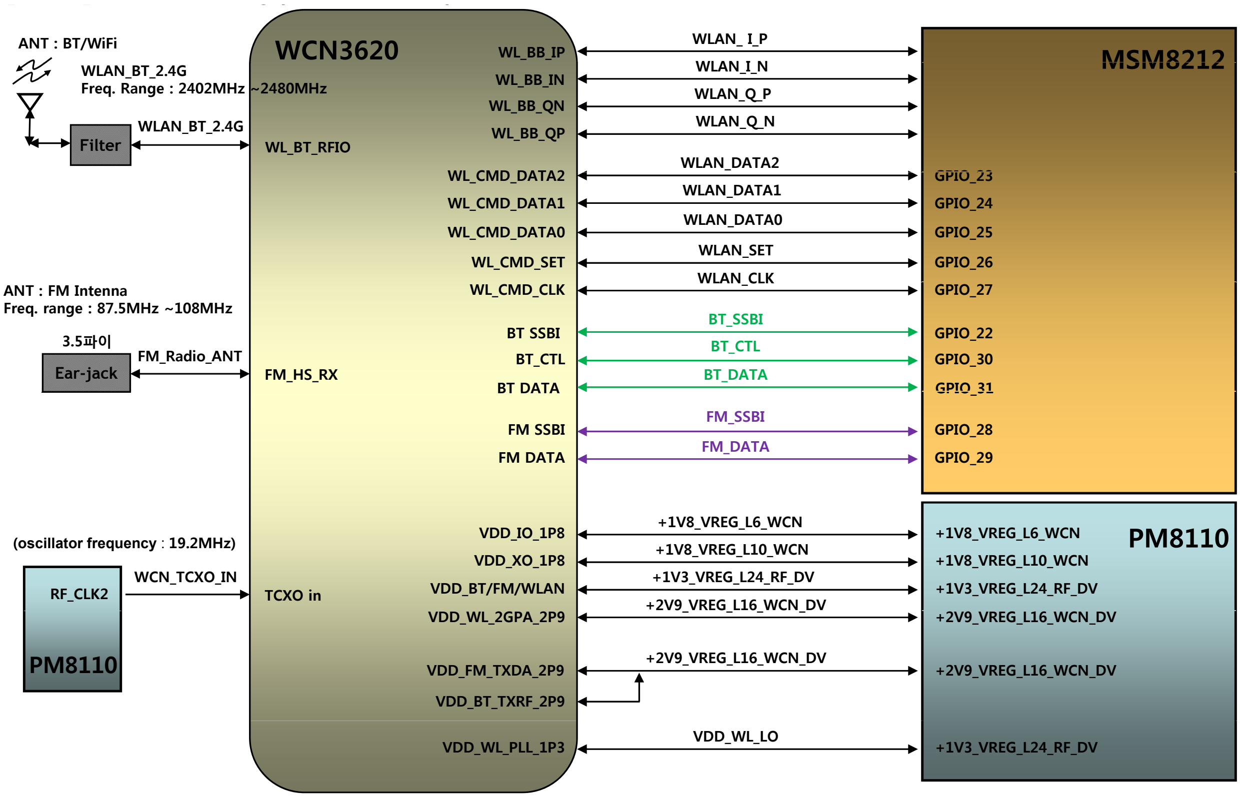


[D295] RF Main (3G & GSM)

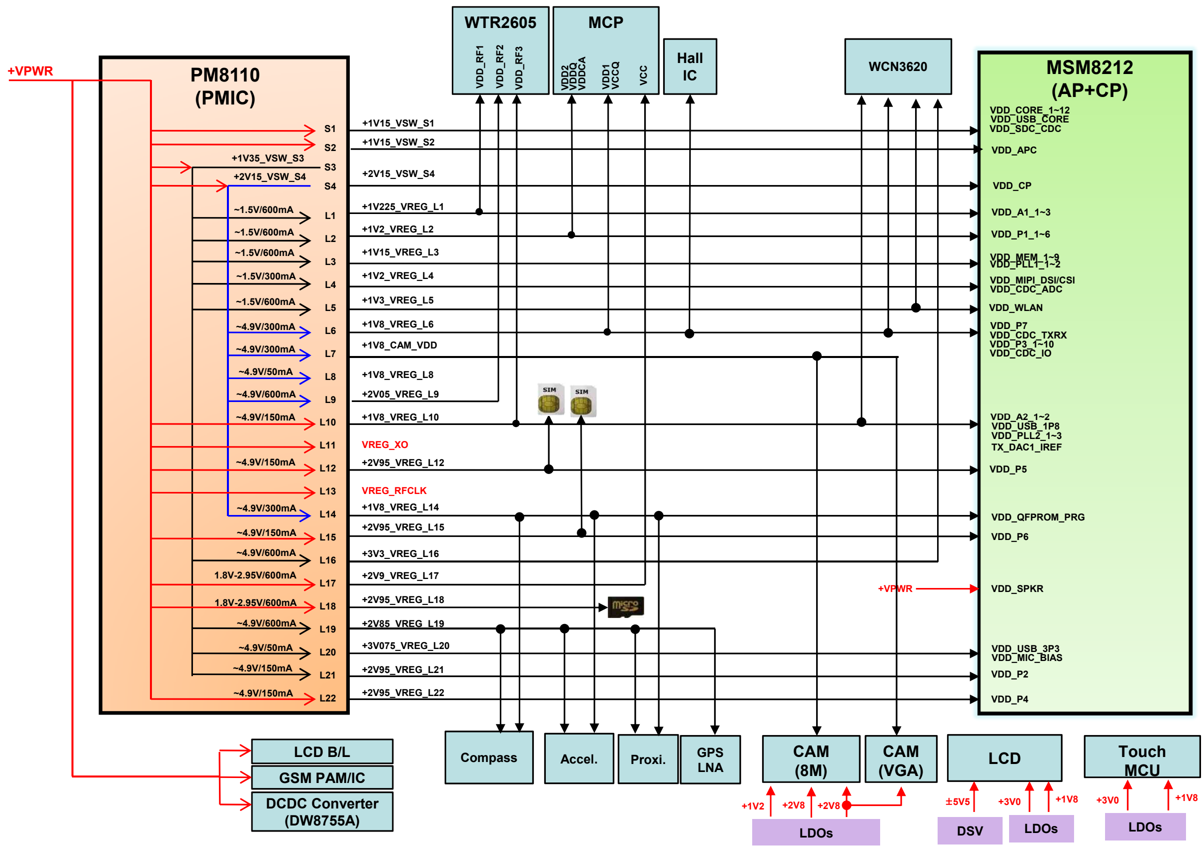


W2,4,5 GSM quad 3	W1,2,5 GSM quad 2	W1,2,8 GSM quad 1	WTR2605 Port map
B2/B5	B1/B5	B1/B8	TX2
B4	B2	B2	TX3
B5 (G850)	B5 (G850)	B8 (G900)	RX_LB1
G900	G900	G850	RX_LB2
B4	B1	B1	RX_MB1
B2	B2	B2	RX_MB2

[D295]Connectivity(BT/WiFi/FM)

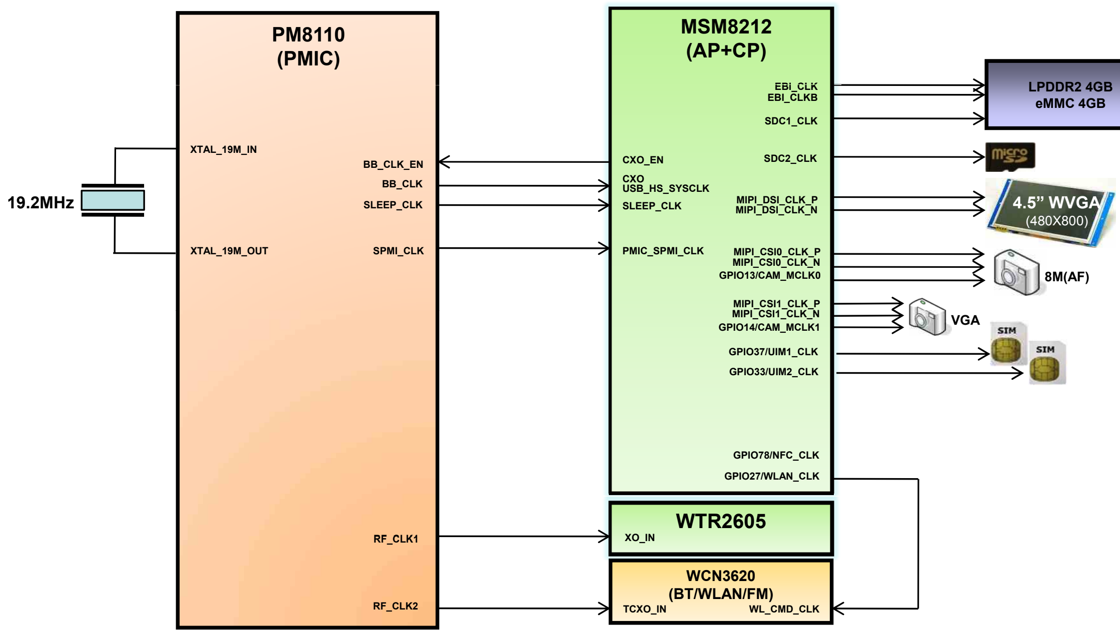


[D295] Power



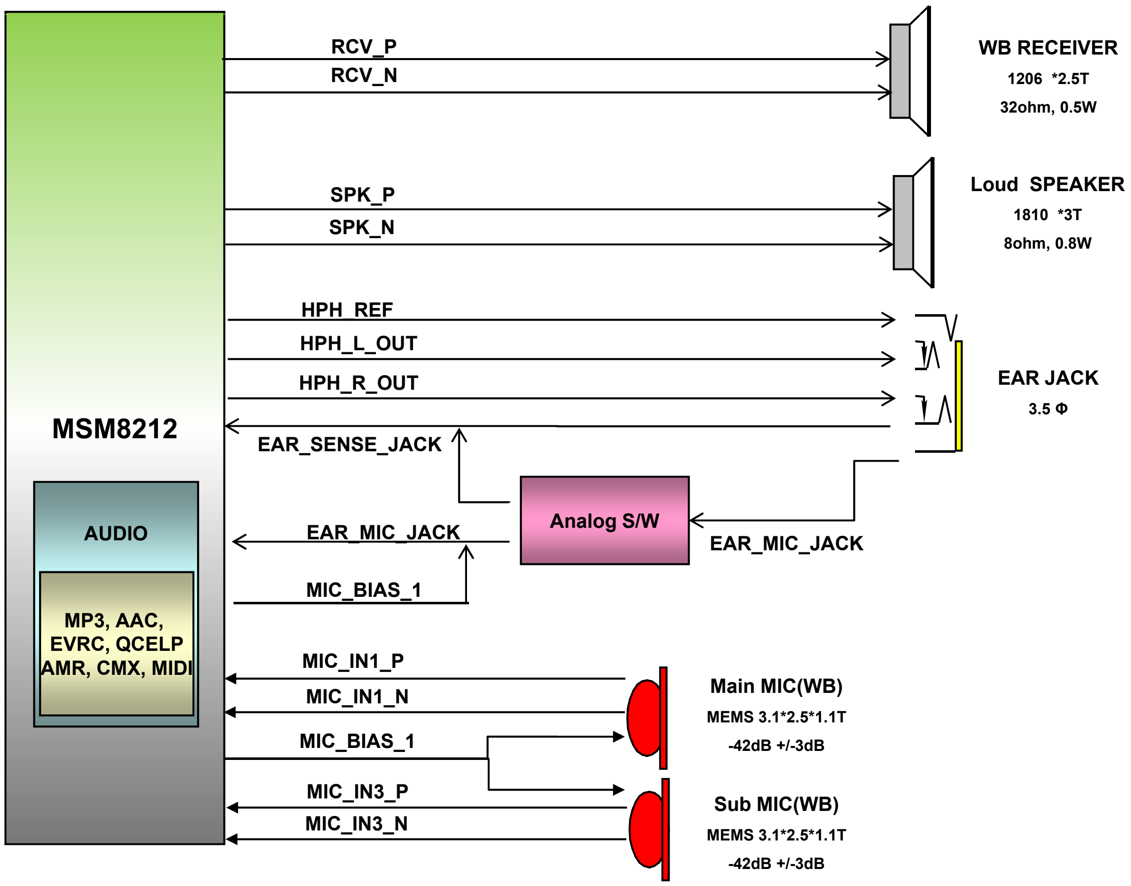
+VPWR
PM8110
(PMIC)
WTR2605
VDD_RF1
VDD_RF2
VDD_RF3
MCP
VCC
Hall
IC
WCN3620
MSM8212
(AP+CP)
+1V15_VSW_S1
+1V15_VSW_S2
+1V35_VSW_S3
+2V15_VSW_S4
+1V225_VREG_L1
+1V2_VREG_L2
+1V15_VREG_L3
+1V2_VREG_L4
+1V3_VREG_L5
+1V8_VREG_L6
+1V8_CAM_VDD
+1V8_VREG_L8
+2V05_VREG_L9
+1V8_VREG_L10
VREG_XO
+2V95_VREG_L12
VREG_RFCLK
+1V8_VREG_L14
+2V95_VREG_L15
+3V3_VREG_L16
+2V9_VREG_L17
+2V95_VREG_L18
+2V85_VREG_L19
+3V075_VREG_L20
+2V95_VREG_L21
+2V95_VREG_L22
VDD_APC
VDD_CP
VDD_A1_1~3
VDD_P1_1~6
VDD_WLAN
VDD_P5
VDD_QFPROM_PRG
VDD_P6
VDD_SPKR
VDD_P2
VDD_P4
LCD B/L
GSM PAM/IC
DCDC Converter
(DW8755A)
Compass
Accel.
Proxi.
GPS
LNA
CAM
(8M)
CAM
(VGA)
LCD
Touch
MCU
+1V2
+2V8
+3V0
+1V8
LDOs
DSV

[D295] Clock



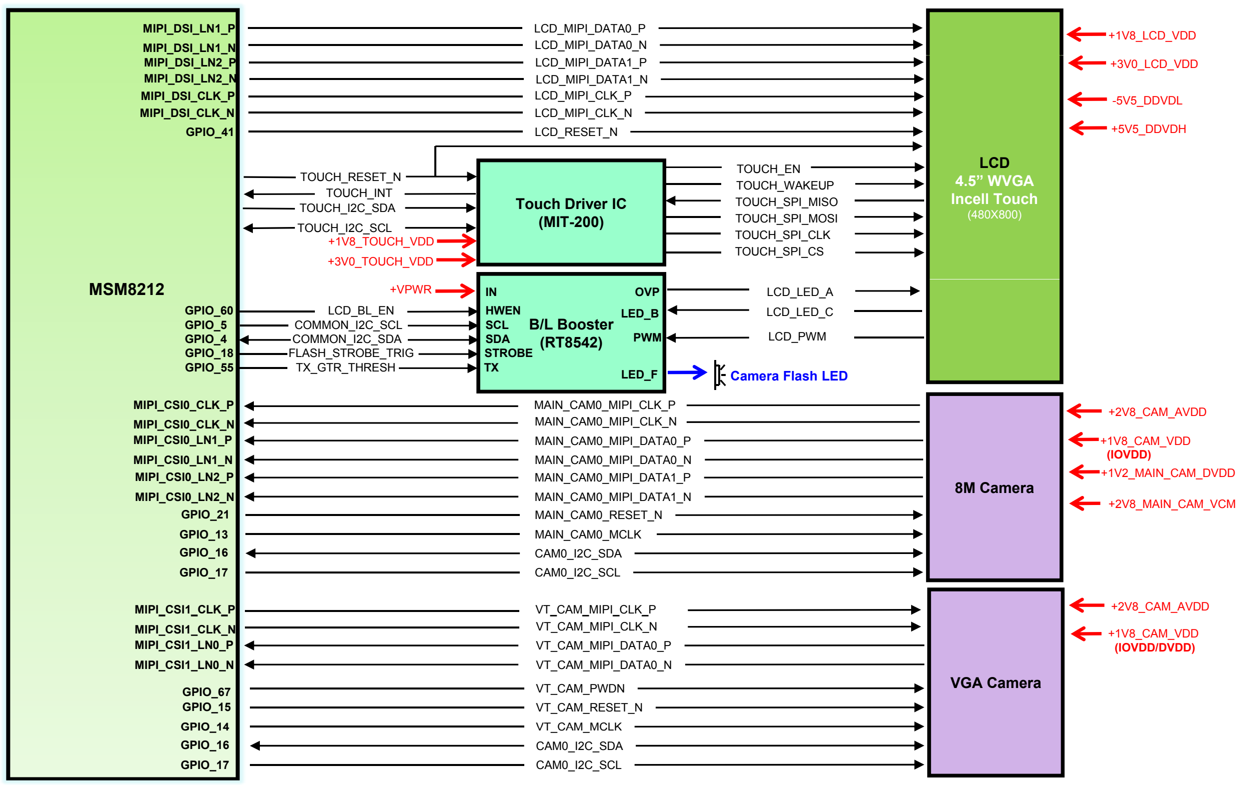
PM8110
(PMIC)
19.2MHz
XTAL_19M_IN
XTAL_19M_OUT
BB_CLK_EN
BB_CLK
SLEEP_CLK
SPMI_CLK
RF_CLK1
RF_CLK2
MSM8212
(AP+CP)
CXO_EN
CXO
USB_HS_SYSCLK
SLEEP_CLK
PMIC_SPMI_CLK
EBI_CLK
EBI_CLKB
SDC1_CLK
SDC2_CLK
MIPI_DSI_CLK_P
MIPI_DSI_CLK_N
MIPI_CSI0_CLK_P
MIPI_CSI0_CLK_N
GPIO13/CAM_MCLK0
MIPI_CSI1_CLK_P
MIPI_CSI1_CLK_N
GPIO14/CAM_MCLK1
GPIO37/UIM1_CLK
GPIO33/UIM2_CLK
GPIO78/NFC_CLK
GPIO27/WLAN_CLK
LPDDR2 4GB
eMMC 4GB
4.5" WVGA
(480X800)
8M(AF)
VGA
SIM
SIM
WTR2605
XO_IN
WCN3620
(BT/WLAN/FM)
TCXO_IN
WL_CMD_CLK

[D295] Audio

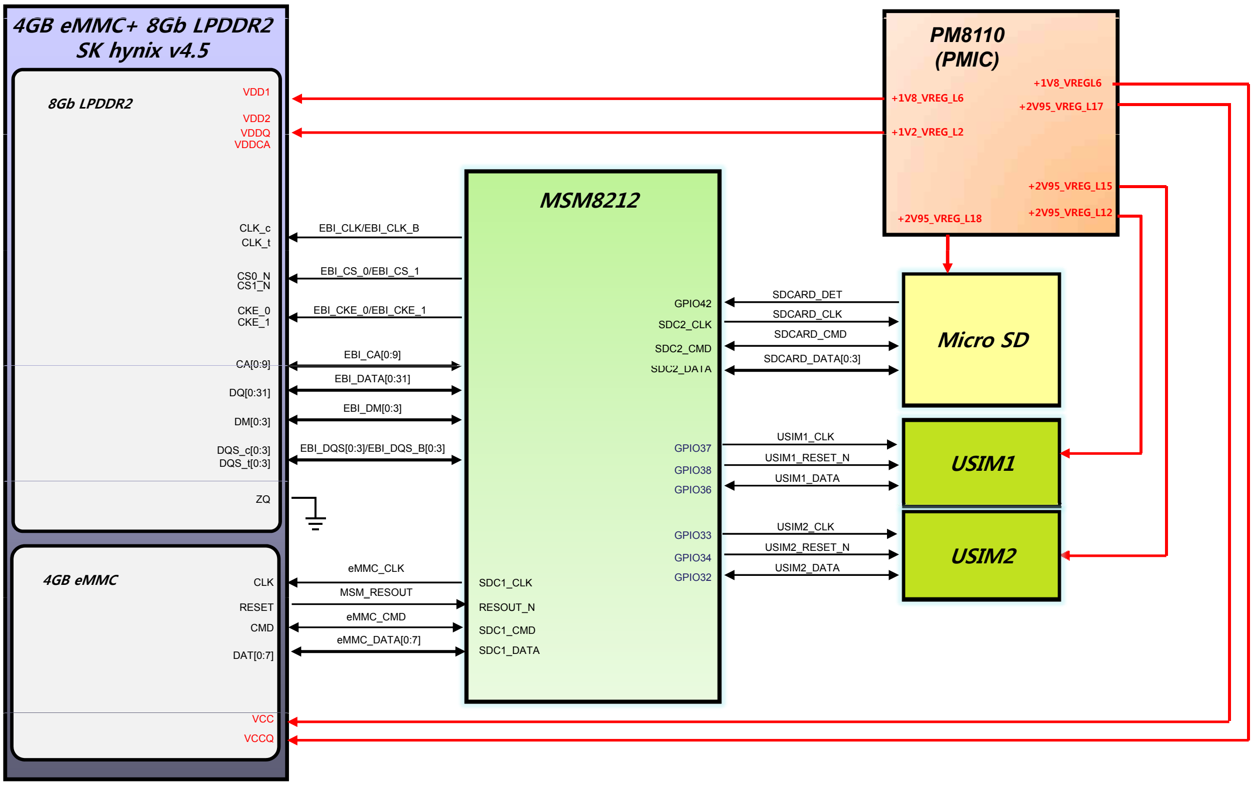


MSM8212
AUDIO
MP3, AAC, EVRC, QCELP AMR, CMX, MIDI
RCV_P
RCV_N
WB RECEIVER
1206 *2.5T
32ohm, 0.5W
SPK_P
SPK_N
Loud SPEAKER
1810 *3T
8ohm, 0.8W
HPH_REF
HPH_L_OUT
HPH_R_OUT
EAR JACK
3.5 Φ
EAR_SENSE_JACK
Analog S/W
EAR_MIC_JACK
EAR_MIC_JACK
MIC_BIAS_1
MIC_IN1_P
MIC_IN1_N
MIC_BIAS_1
Main MIC(WB)
MEMS 3.1*2.5*1.1T
-42dB +/-3dB
MIC_IN3_P
MIC_IN3_N
Sub MIC(WB)
MEMS 3.1*2.5*1.1T
-42dB +/-3dB

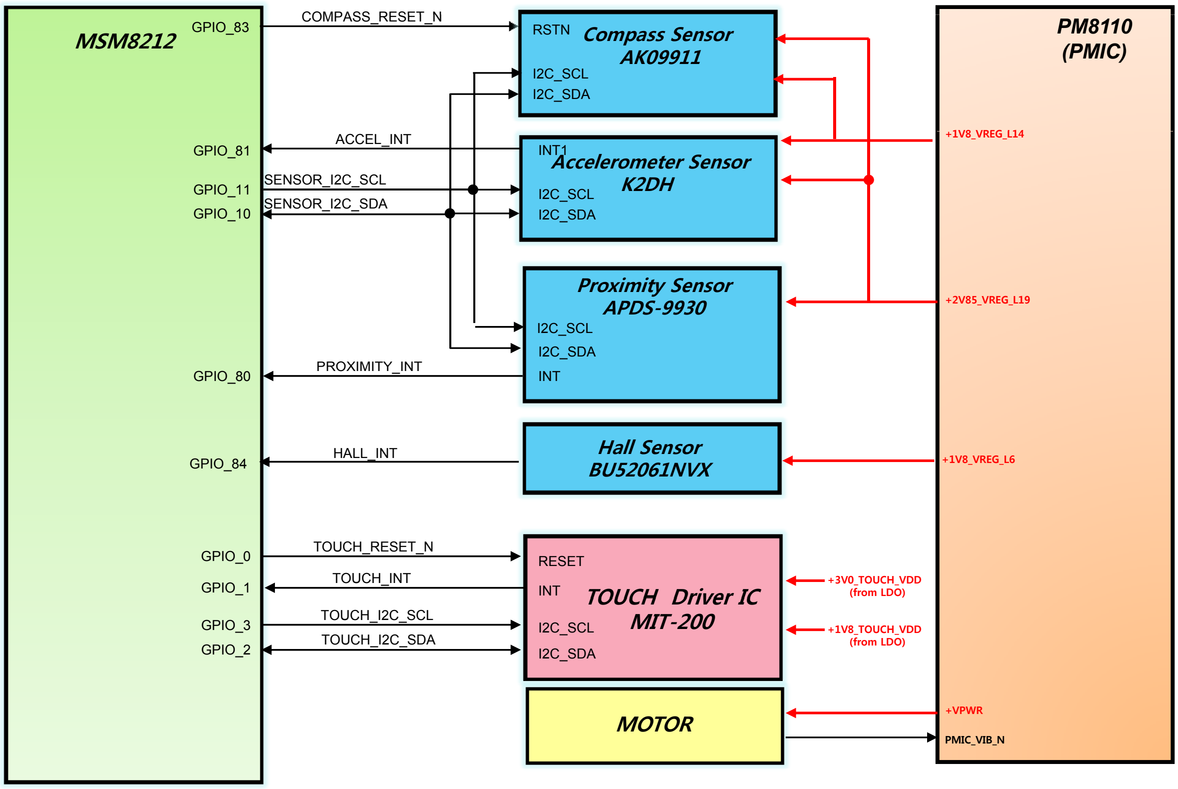
[D295] LCD/Touch/Camera



[D295] MCP/Peripherals



[D295] Sensors

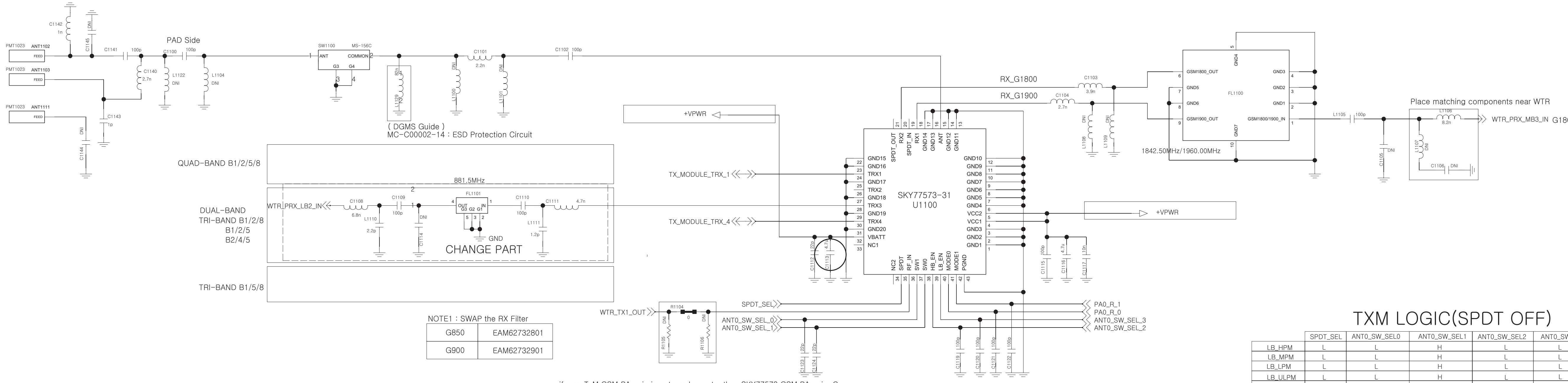


6. CIRCUIT DIAGRAM

< 1-1-1-4_ANT_WCDMA_NO_DRX >

Rev_0.6

ANT1 (GSM Quad / WCDMA B1, B2, B4, B5, B8)



NOTE1 : SWAP the RX Filter

G850	EAM62732801
G900	EAM62732901

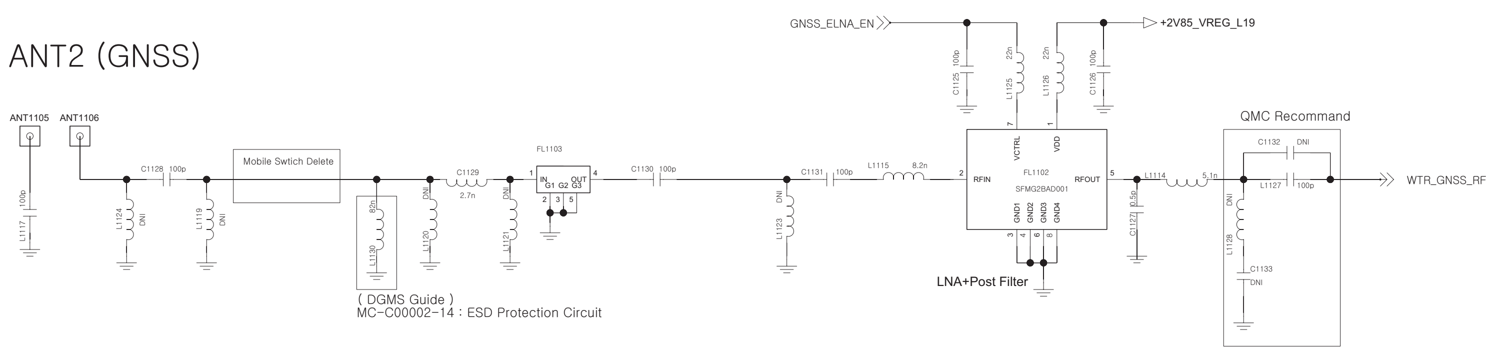
if your TxM GSM PA gain is not much greater than SKY77573 GSM PA gain. Can remove

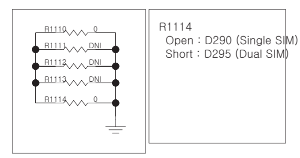
TXM LOGIC(SPDT OFF)

	SPDT_SEL	ANT0_SW_SEL0	ANT0_SW_SEL1	ANT0_SW_SEL2	ANT0_SW_SEL3	PA0_R0	PA0_R1
LB_HPM	L	L	H	L	L	L	L
LB_MPM	L	L	H	L	L	L	H
LB_LPM	L	L	H	L	L	H	L
LB_ULPM	L	L	H	L	L	H	H
HB_HPM	L	H	L	L	L	L	L
HB_MPM	L	H	L	L	L	L	H
HB_LPM	L	H	L	L	L	H	L
HB_ULPM	L	H	L	L	L	H	H
TRX1	H	L	L	L	L	-	-
TRX2	H	L	L	H	L	-	-
TRX3	H	L	H	L	L	-	-
TRX4	H	L	H	H	L	-	-
RX1	H	H	H	H	L	-	-
RX2	H	H	H	H	H	-	-

EXT LNA + Filter

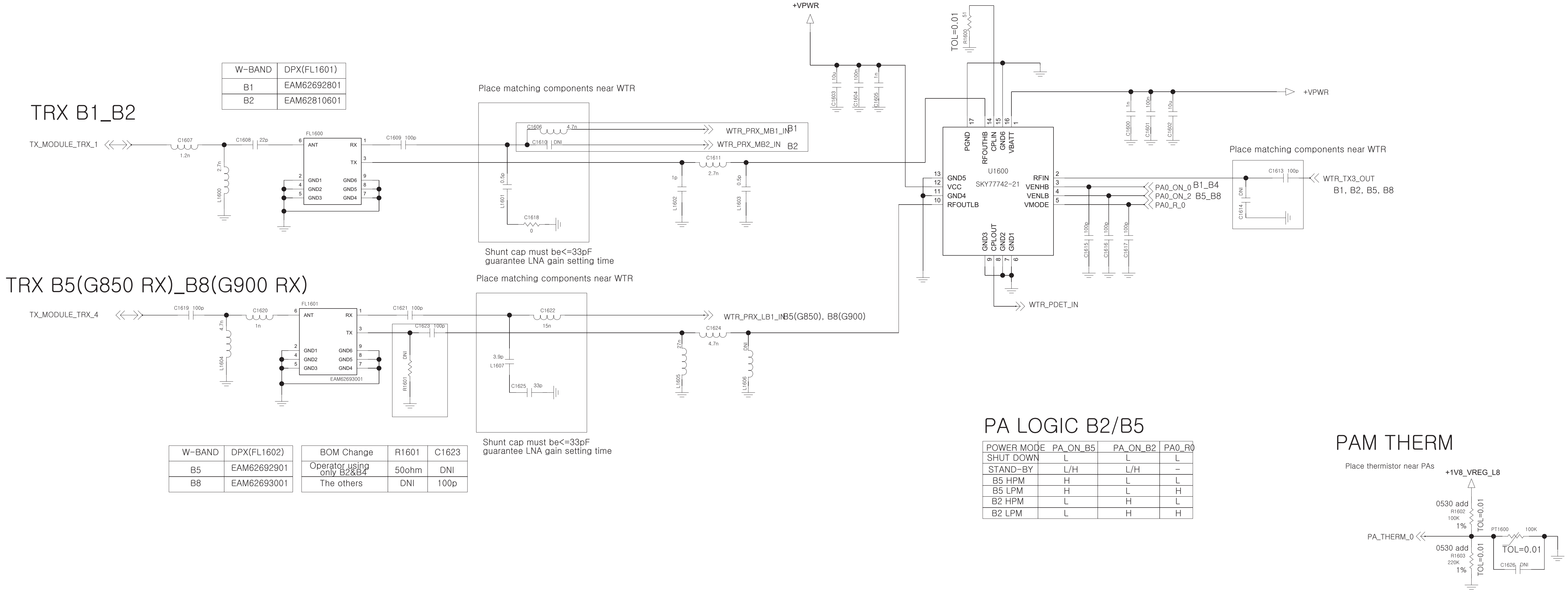
ANT2 (GNSS)





< 1-1-4-5 TRX_WCDMA_B1_B2_B5_B8 DUAL >

Rev 0.4



W-BAND	DPX(FL1601)
B1	EAM62692801
B2	EAM62810601

W-BAND	DPX(FL1602)
B5	EAM62692901
B8	EAM62693001

BOM Change	R1601	C1623
Operator using only B2&B4	50ohm	DNI
The others	DNI	100p

PA LOGIC B2/B5

POWER MODE	PA_ON_B5	PA_ON_B2	PA0_R0
SHUT DOWN	L	L	L
STAND-BY	L/H	L/H	-
B5 HPM	H	L	L
B5 LPM	H	L	H
B2 HPM	L	H	L
B2 LPM	L	H	H

< 1-1-3-1_RFIC_WTR2605 >

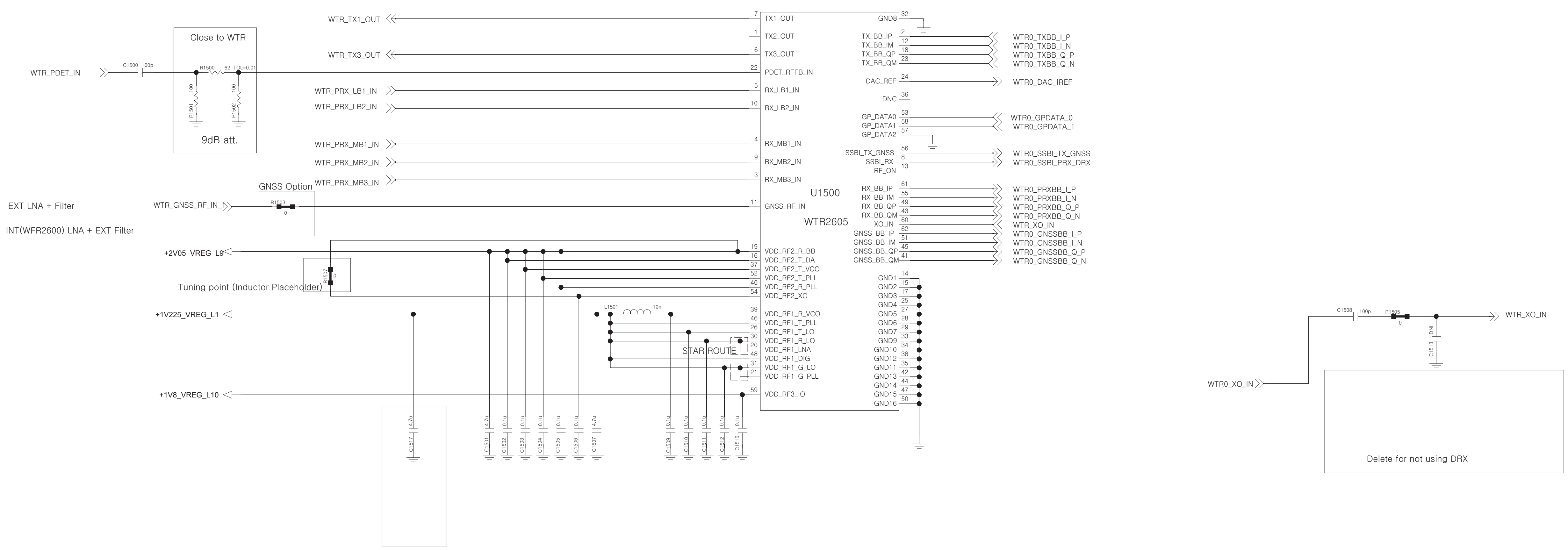
Rev_0.6

CAUTION: YOU MUST CHECK & MODIFY RF PORT AND GPIO ON YOUR SCHEMATICS.

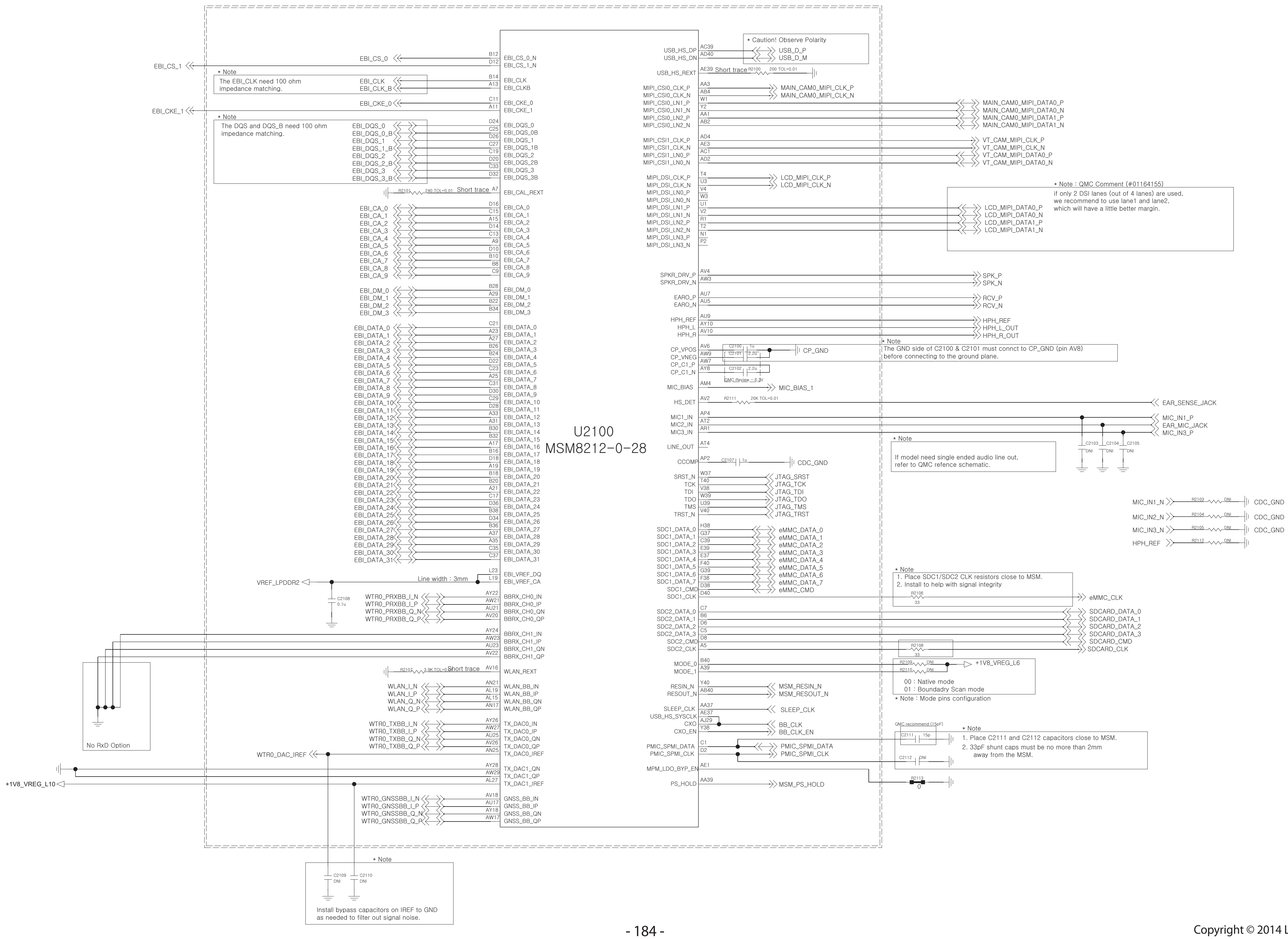
WCDMA							
WTR2605	B2, B4, B5	B1, B2, B5	B1, B2, B8	B1, B5, B8	B2, B5	B1, B5	B1, B8
TX1_OUT	GSM	GSM	GSM	GSM	GSM	GSM	GSM
TX2_OUT	B4	B1	B1	B5			
TX3_OUT	B2, B5	B2, B5	B2, B8	B1, B8	B2, B5	B1, B5	B1, B8
RX_LB1_IN	B5(G850)	B5(G850)	B8(G900)	B8(G900)	B5(G850)	B5(G850)	B8(G900)
RX_LB2_IN	G900	G900	G850	B5(G850)	G900	G900	G850
RX_MB1_IN	B4	B1	B1	B1			B1
RX_MB2_IN	B2	B2	B2		B2	B2	
RX_MB3_IN	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900

CDMA			
WTR2605	VZW	SPCS	Metro PCS
TX1_OUT			
TX2_OUT	BC0(BC10)	BC0(BC10)	BC0(BC10),BC15
TX3_OUT	BC1	BC1	BC1
RX_LB1_IN	BC0(BC10)	BC0(BC10)	BC0(BC10)
RX_LB2_IN			
RX_MB1_IN			BC15
RX_MB2_IN	BC1	BC1	BC1
RX_MB3_IN			

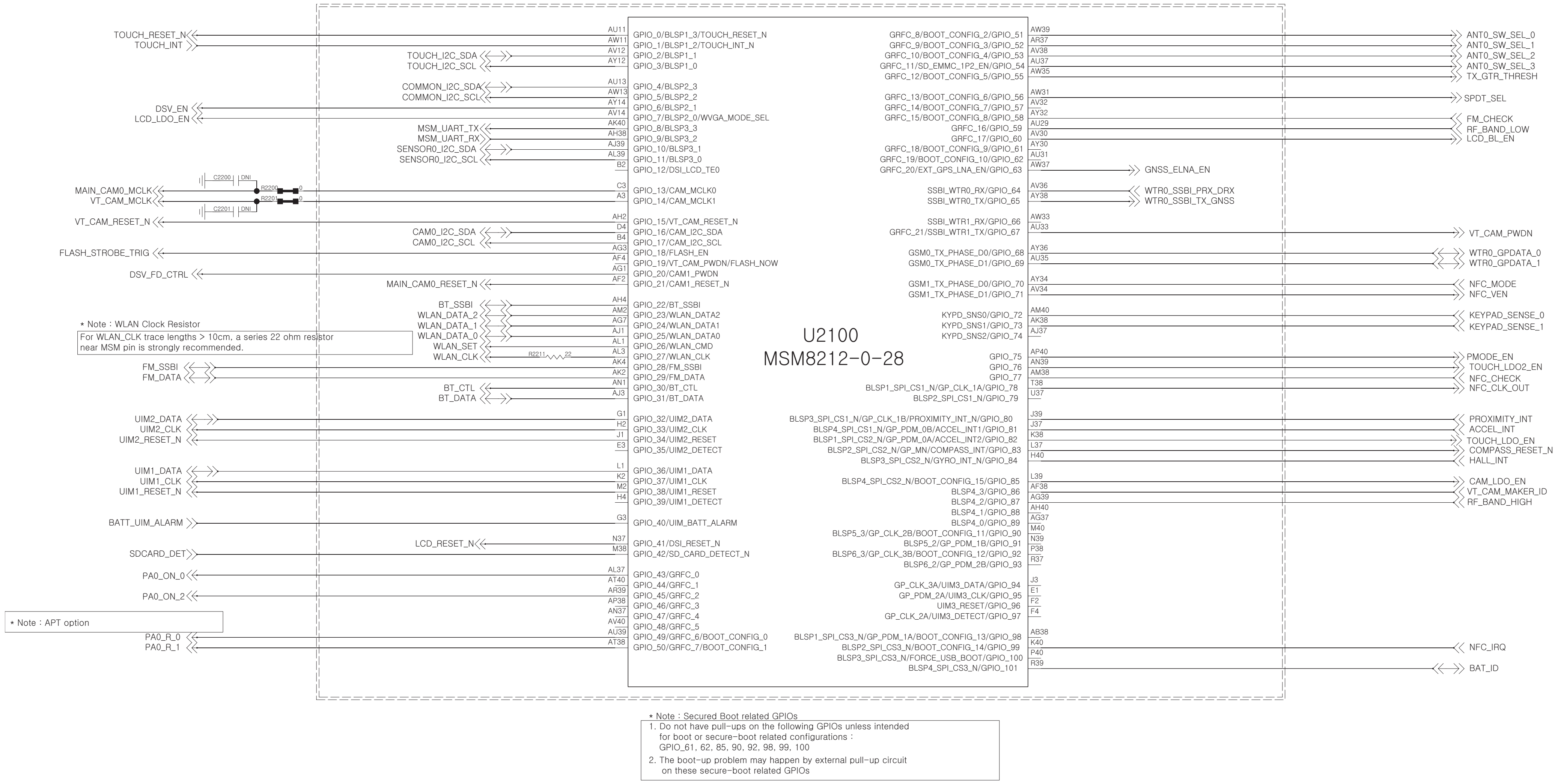
BOM CHANGE	R1606	R1607
WTR_GNSS_RF_IN1	0ohm	DNI
WTR_GNSS_RF_IN2	DNI	0ohm



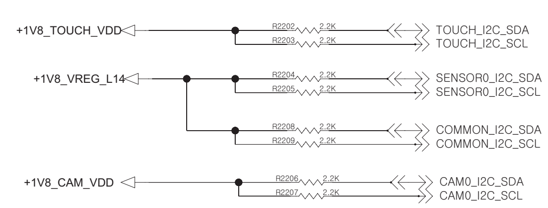
<2-1-6-1-1_MSM8212-0-28_CONTROL>REV.0.3



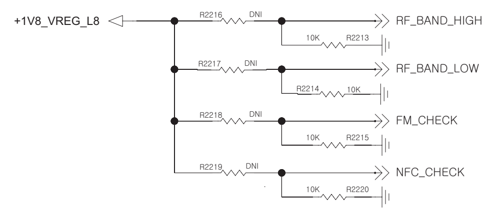
<2-1-6-1-2_MSM8212-0-28_GPIO> REV.0.3



I2C Pull-up

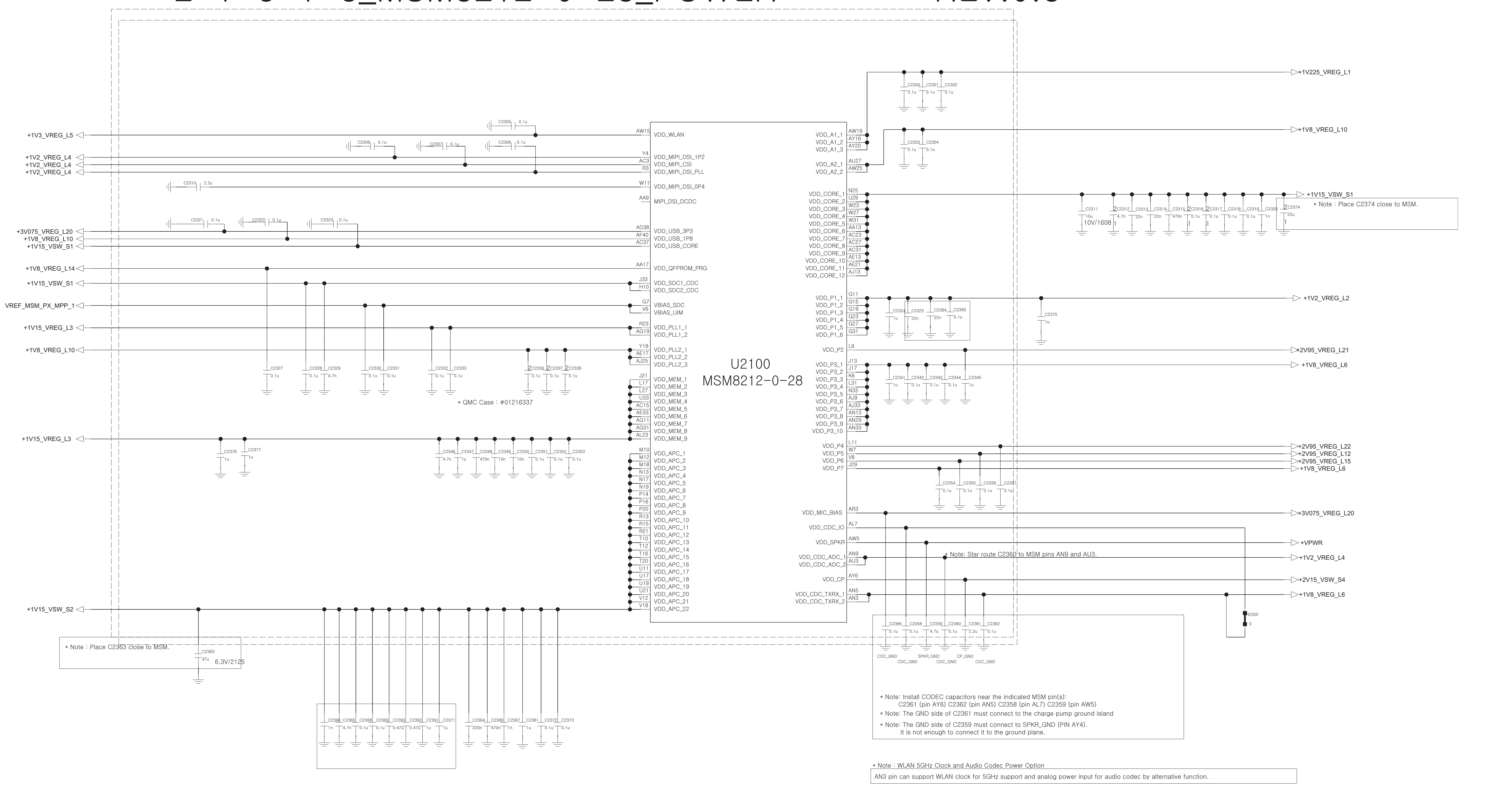


RF BAND Check

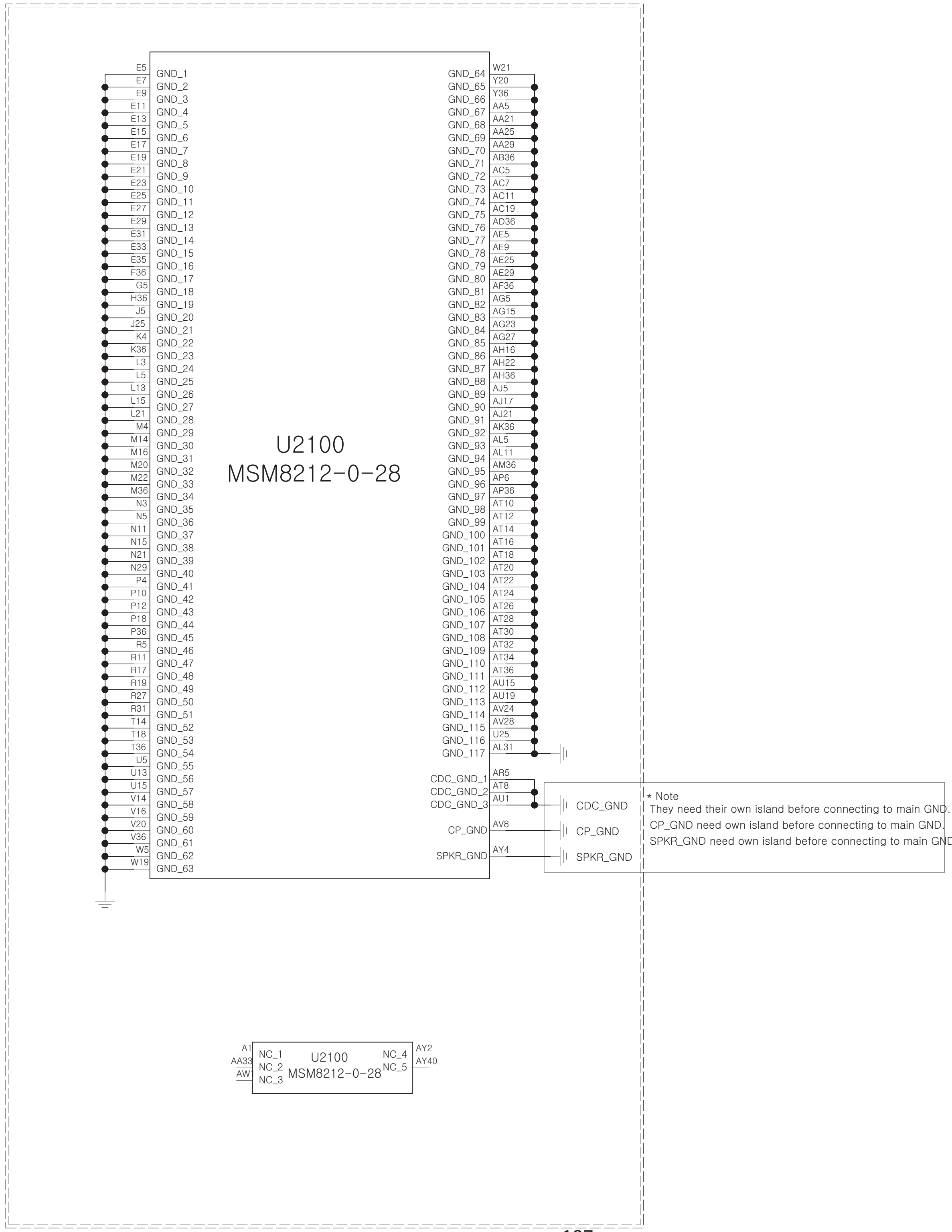


	LOW	HIGH	FM	NFC
GPIO	59	87	58	77
EU_1_8	L	L	L	L
F_1_5	H	L	L	L
G_2_5	H	H	L	L
J_2_8	L	H	L	L
FM	L	L	H	L
NFC	L	L	L	H

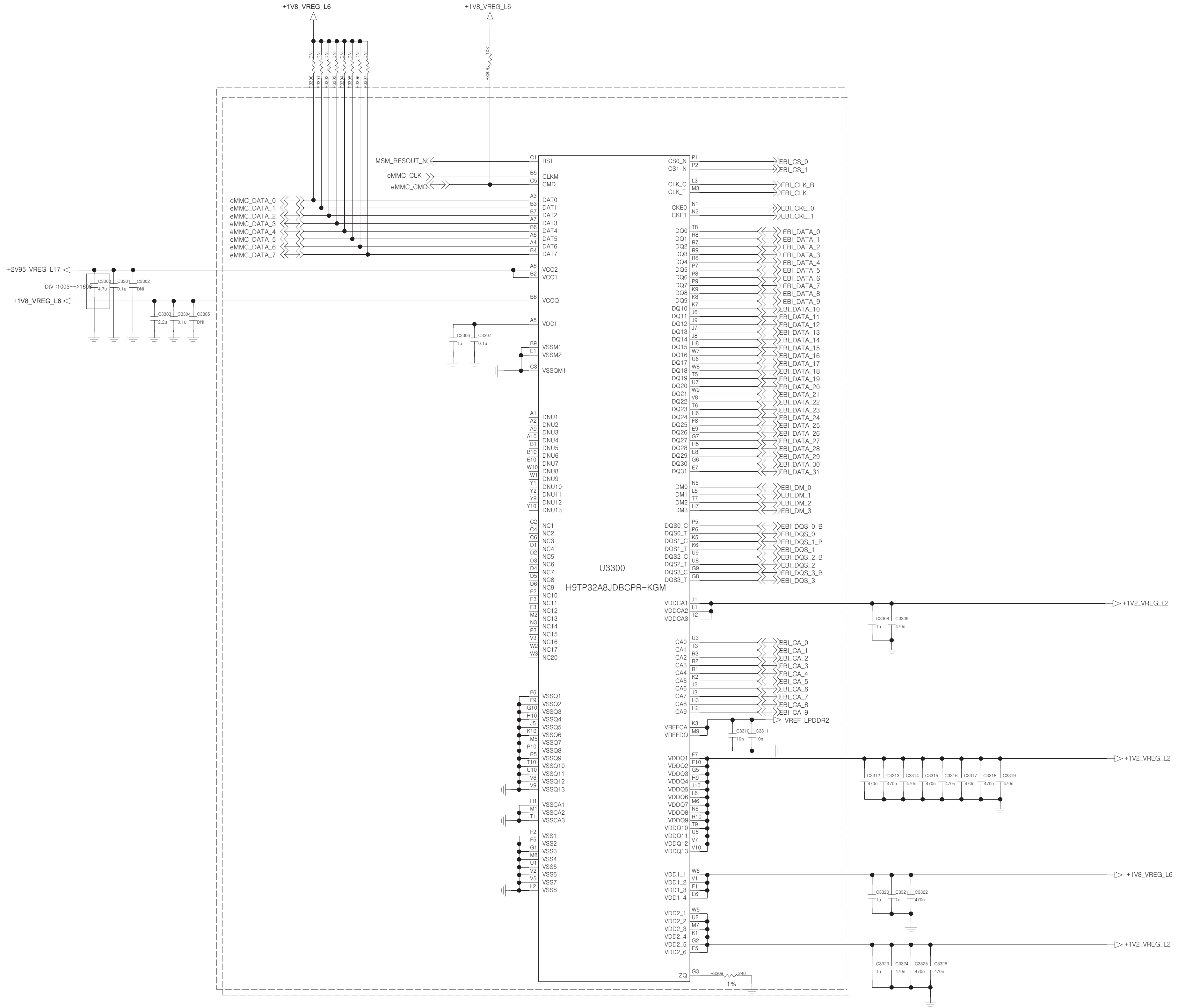
<2-1-6-1-3_MSM8212-0-28_POWER> REV.0.3



<2-1-6-1-4_MSM8212-0-28_GND/NC>
REV.0.3



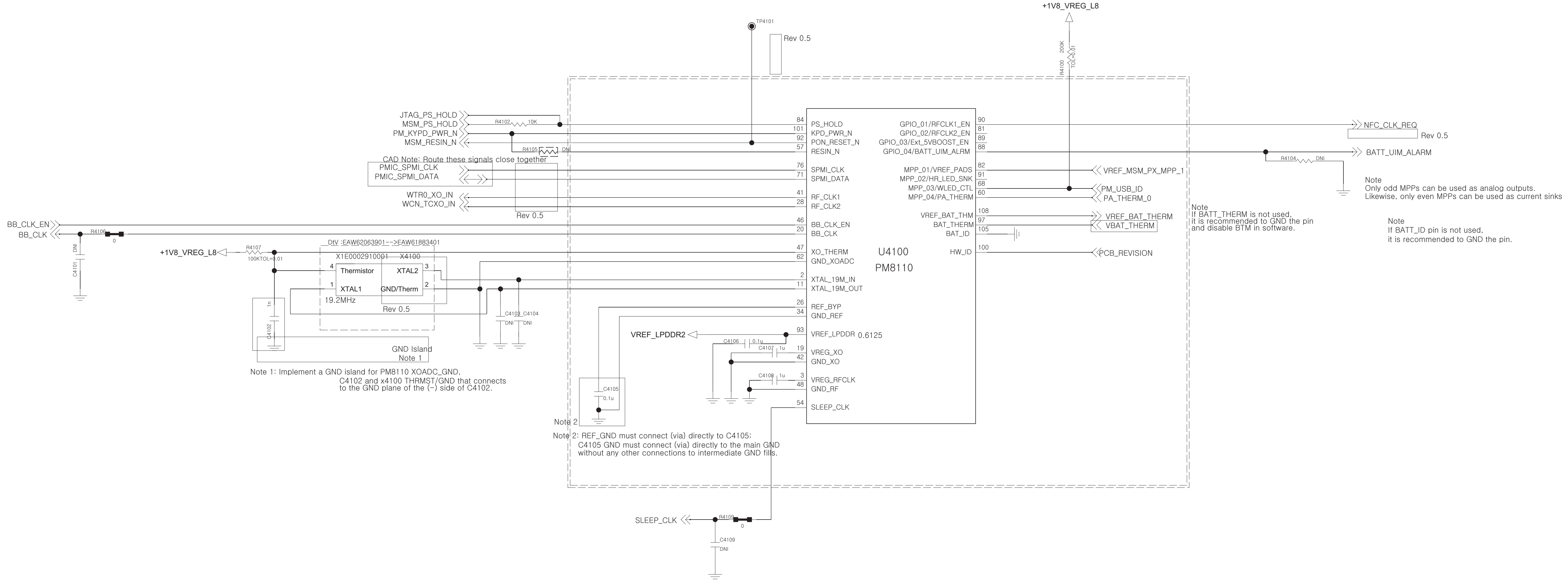
< 3_3_2_3_5_MCP_eMMC_v4_5_4G_DDR2_8G_SK_Hynix >
REV_1.0



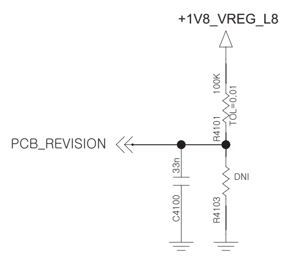
<4-1-4-1_PMIC_PM8110_DATA>

CCDS CARD Information	
Release Date	2013. 12. 2
Based on Reference Schematic	Rev.C
80-ND717-41 MSM8X10 + PM8110 REFERENCE SCHEMATIC	

REV.1.0



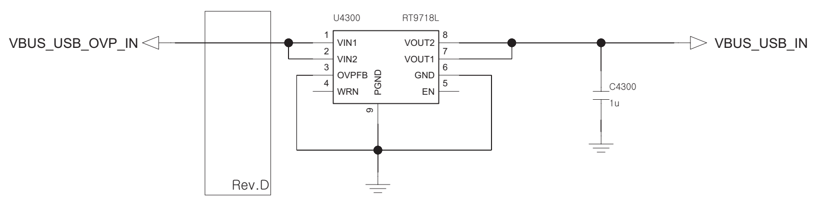
PCB REVISION



A	100K	20K	0.300
B	100K	27K	0.383
C	100K	39K	0.505
D	100K	51K	0.680
E	100K	75K	0.771
F	100K	100K	0.900
G	100K	130K	1.017
1.0	100K	DNI	1.800
1.1	DNI	100K	0.000

<4-3-3_OVP_RT9718L>

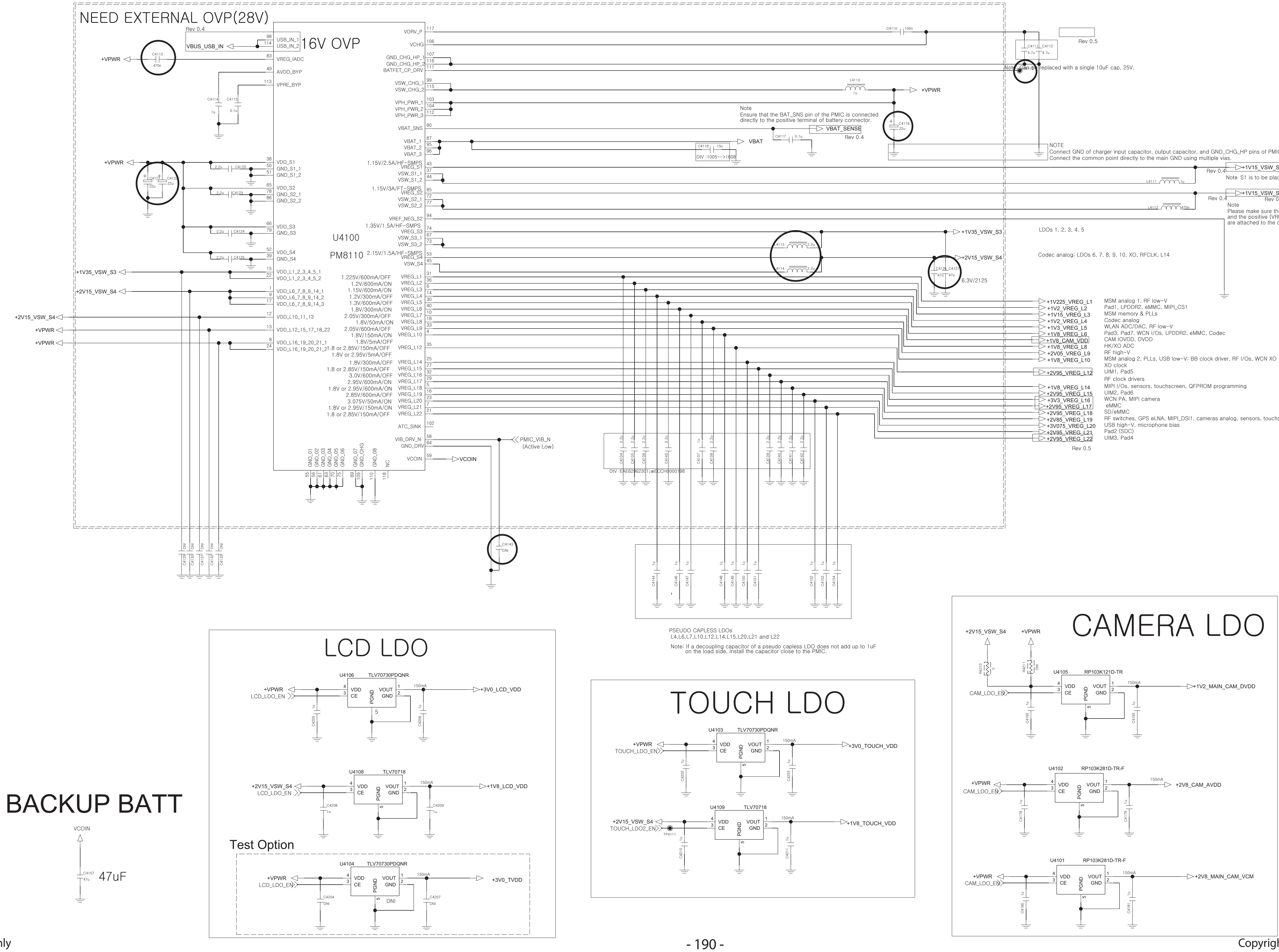
REV.0.3



<4-1-4-2_PMIC_PM8110_POWER>

REV.1.0

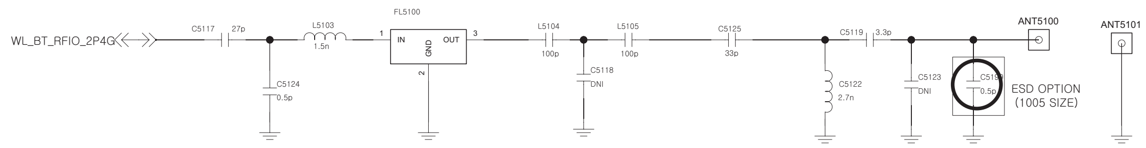
CCDS CARD Information	
Release Date	2013. 12. 2
Based on Reference Schematic	Rev.C
80-NO717-41 MSM8X10 + PM8110 REFERENCE SCHEMATIC	



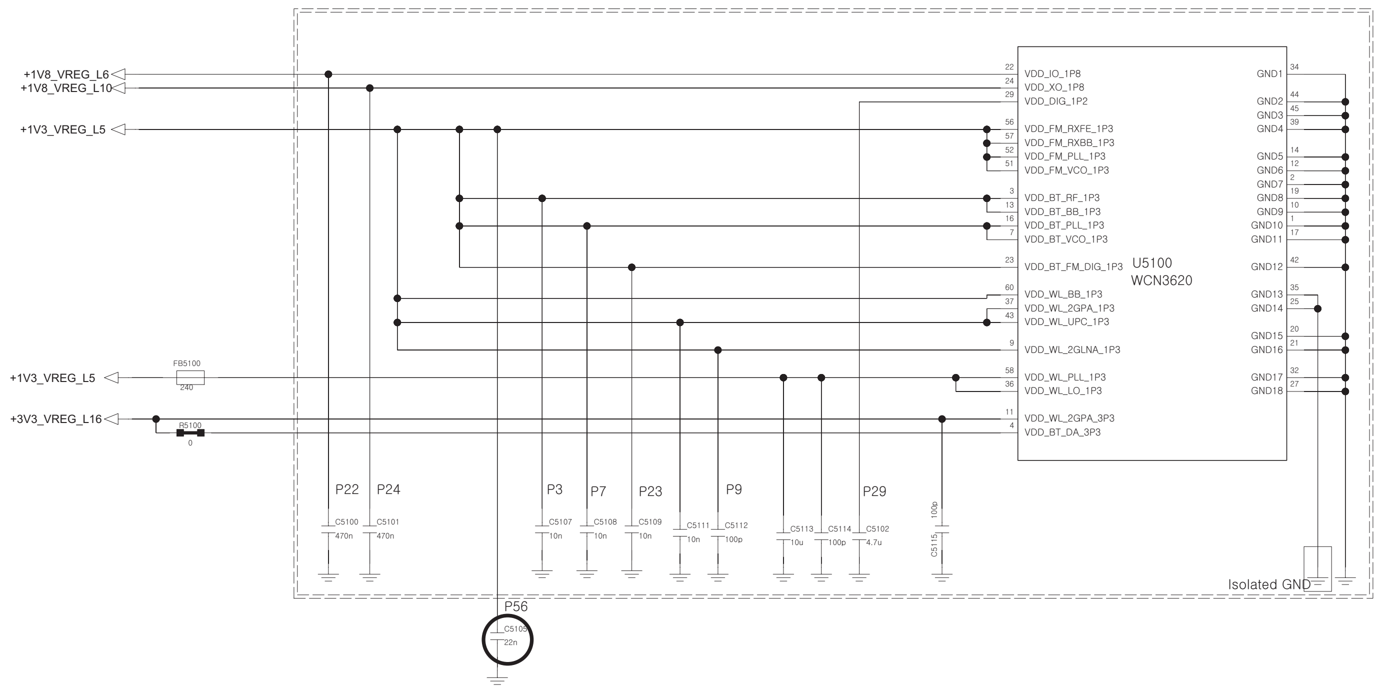
<5-1-1-4> BT_WiFi_Combo_WCN3620

REV 1.0

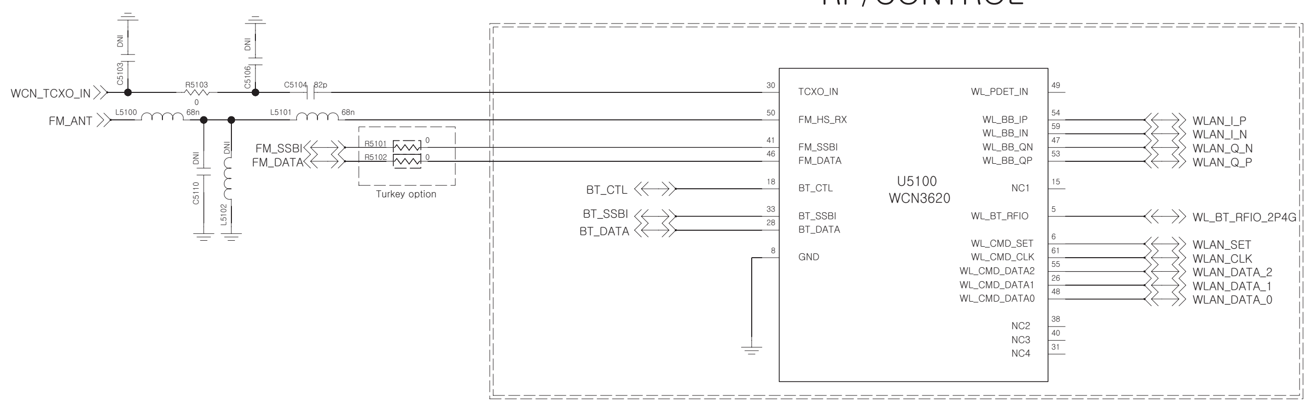
RF Filter



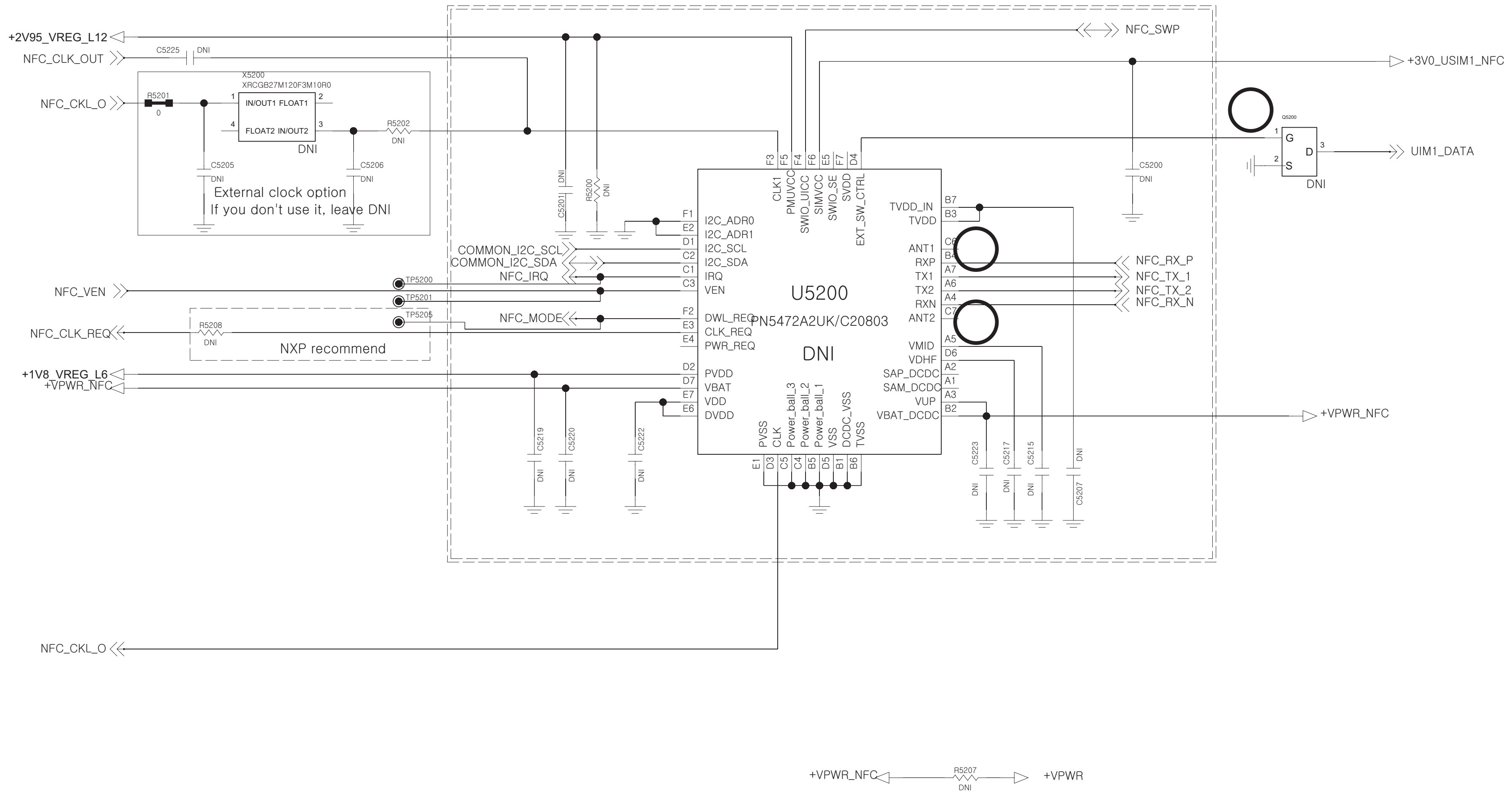
POWER/GND



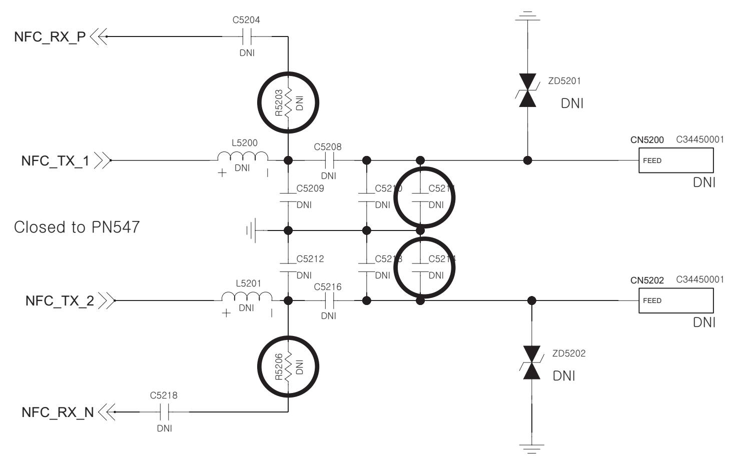
RF/CONTROL



<5-2-1-4_NFC_PN547> Rev_1.1

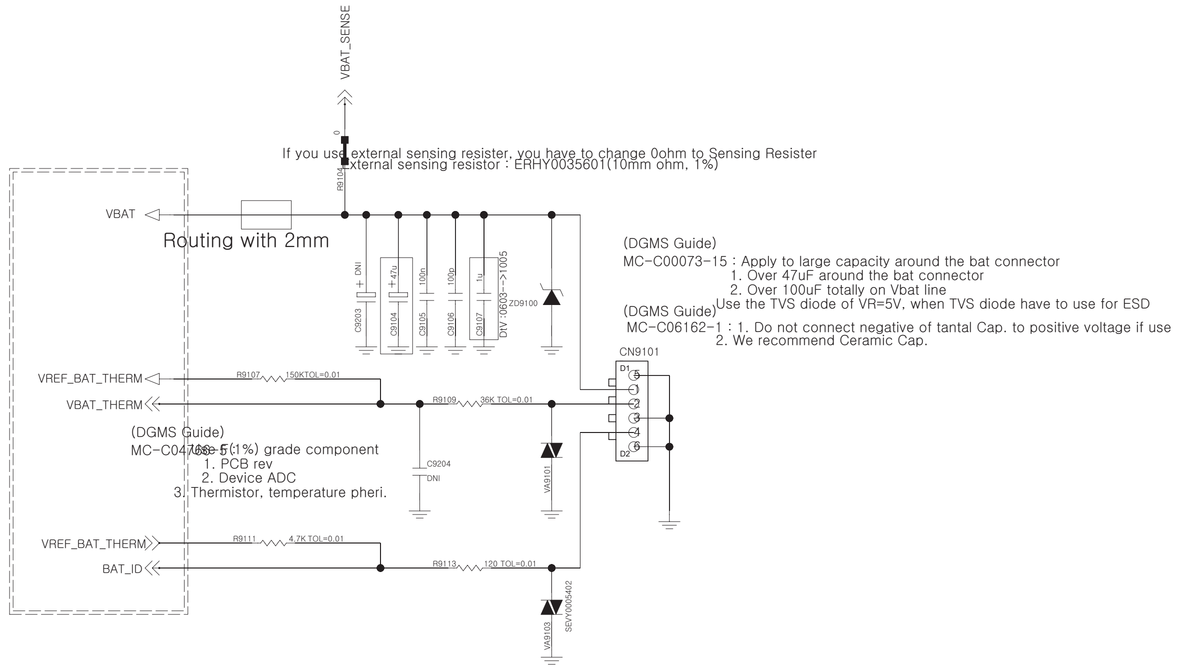


NFC Antenna

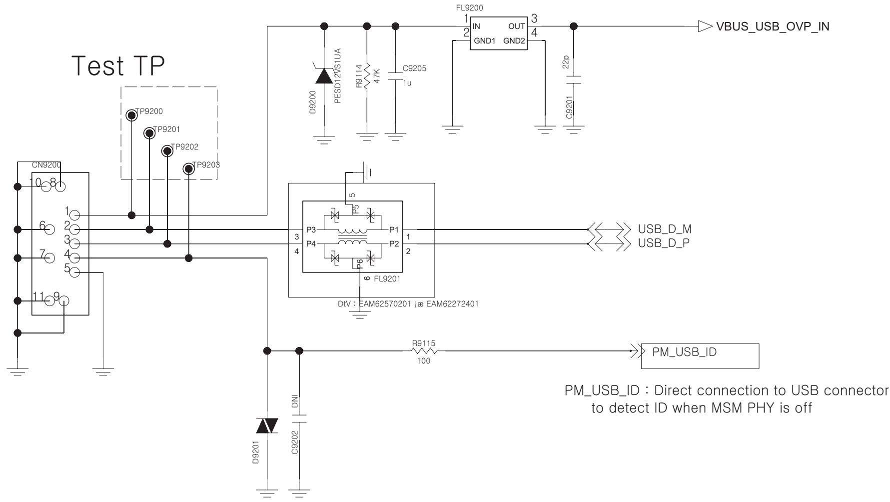


<9-1_Battery_Connector_4Pin>

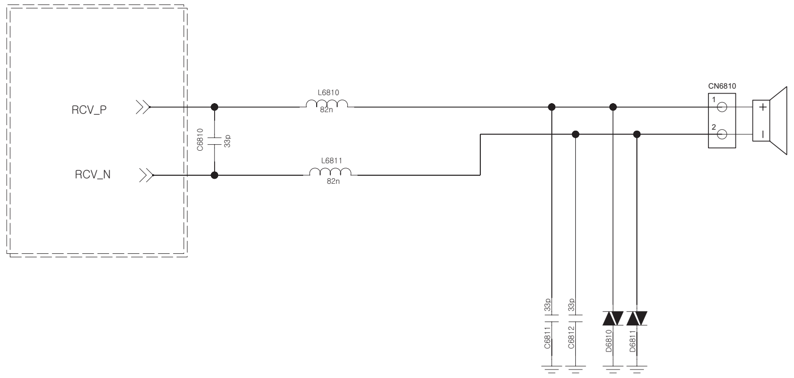
Circuit 1. Batt Conn. /wo EMI Filter



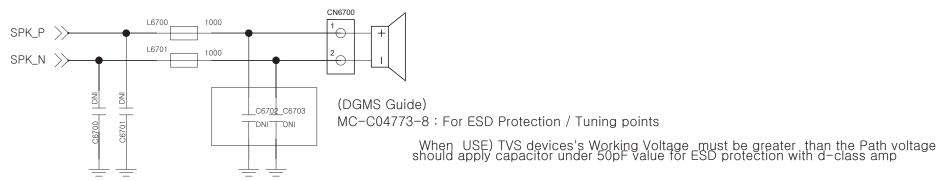
< 9-2-1_IO_USB2.0 > Rev_0.5



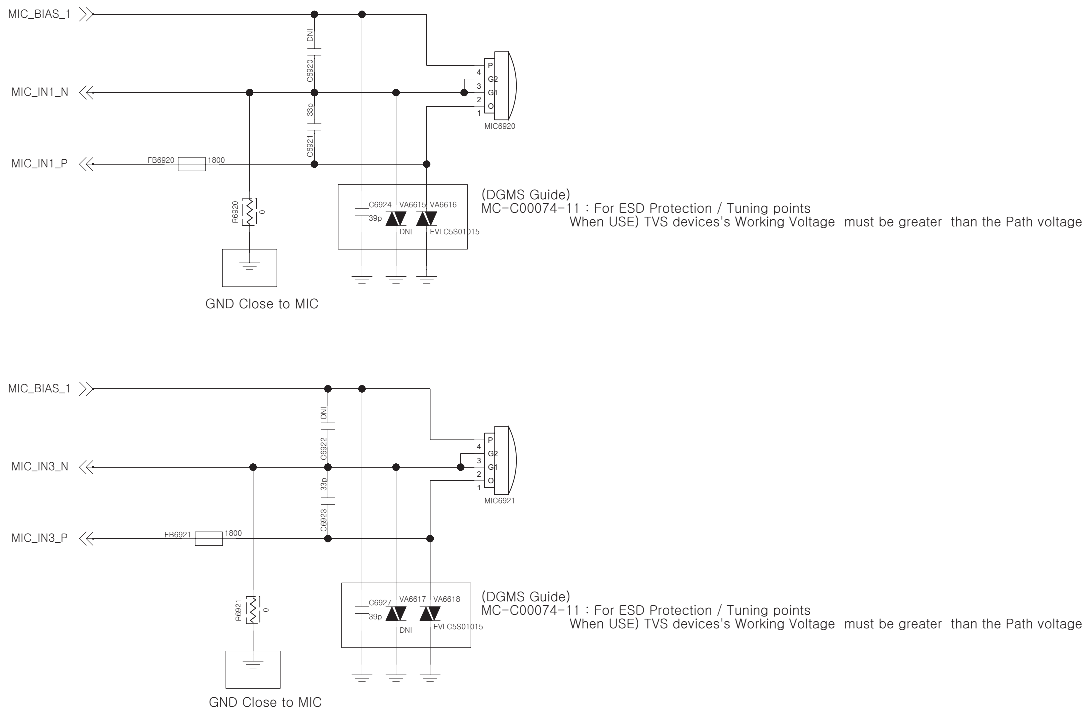
< 6-8-1_Receiver_Only >
Rev_0.3



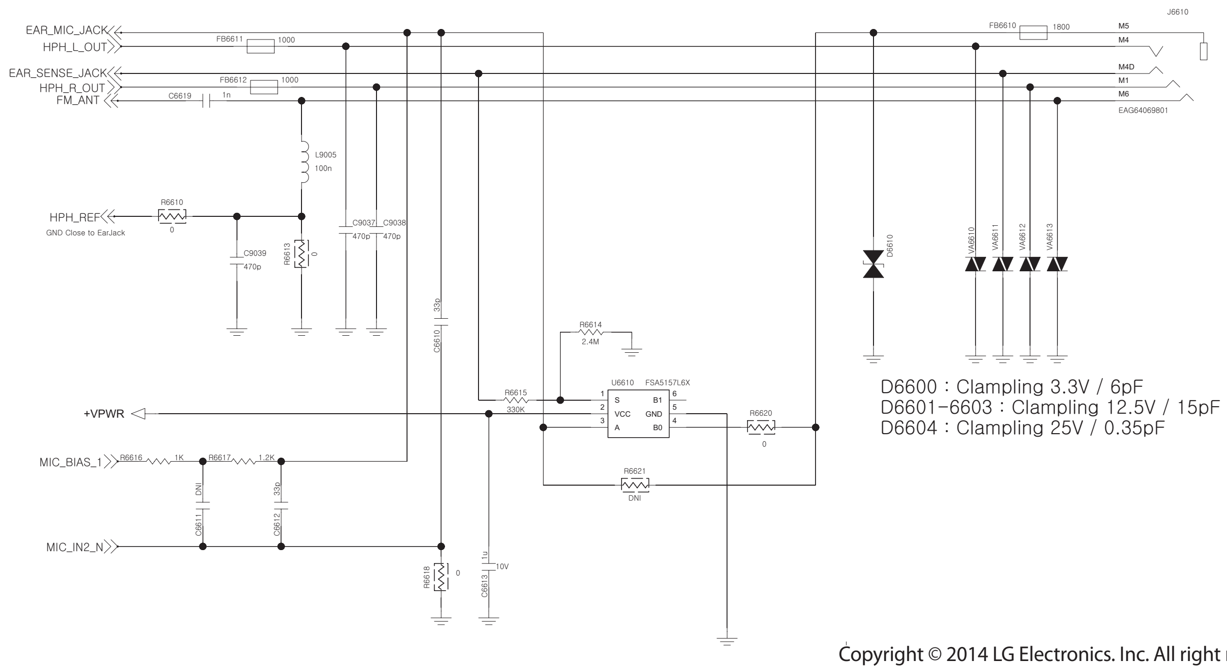
< 6-7-1_Speaker >
Rev_0.4



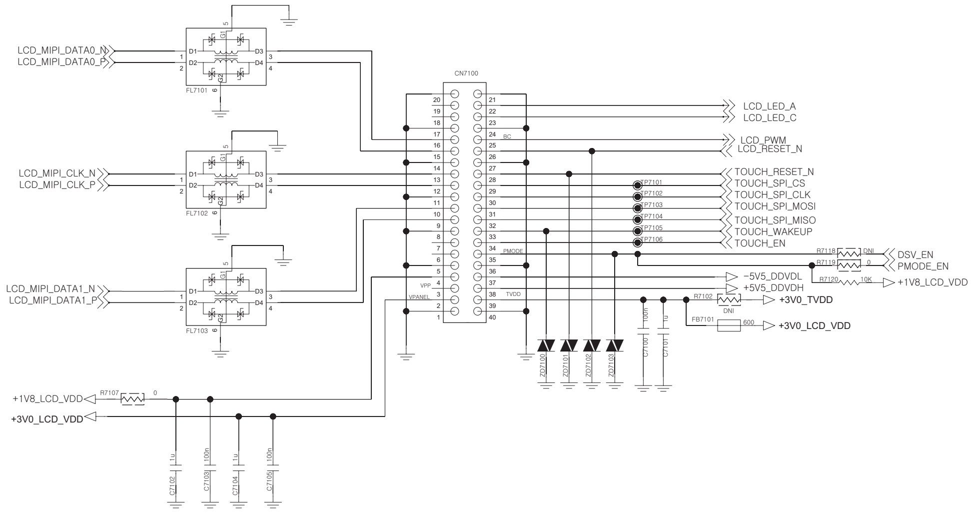
< 6-9-2_2MIC >
Rev_0.4



< 6-6-1_Earjack >



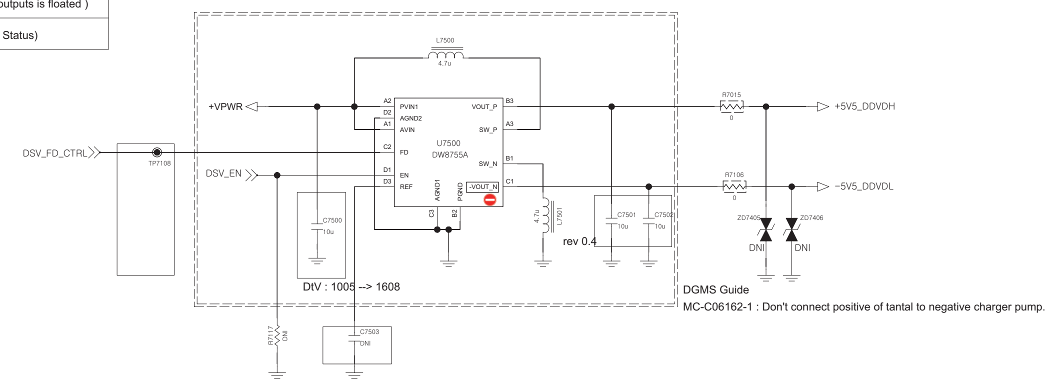
Phase 2 In-cell circuit check!!



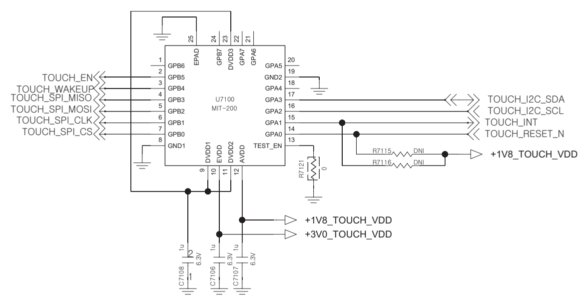
[FD operation Table]

EN	FD	Action
LOW	LOW	Device OFF (Fast discharge active(on) after IC disables)
LOW	High	Device OFF (The status of dual outputs is floated)
High	LOW/High	Device ON (Don't care about FD Status)

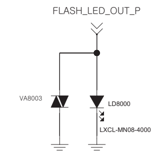
< 7-5-4_DSV_DW8755 >Rev_1.1



Touch MCU

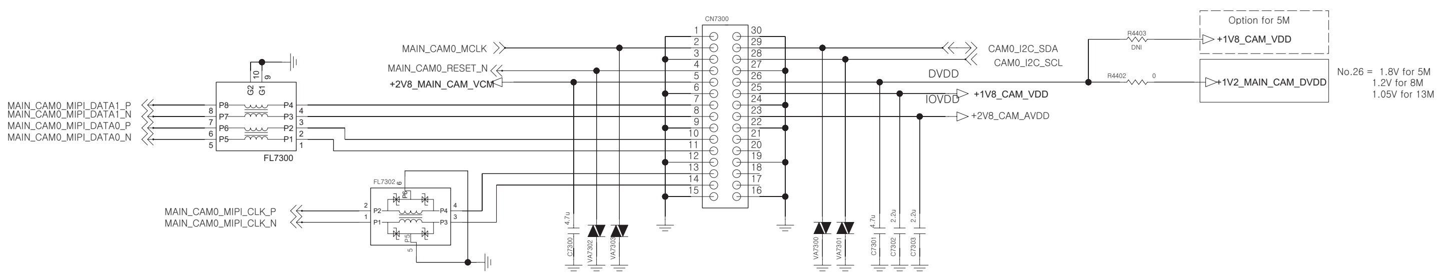


<9-8-3_Flash_LED>

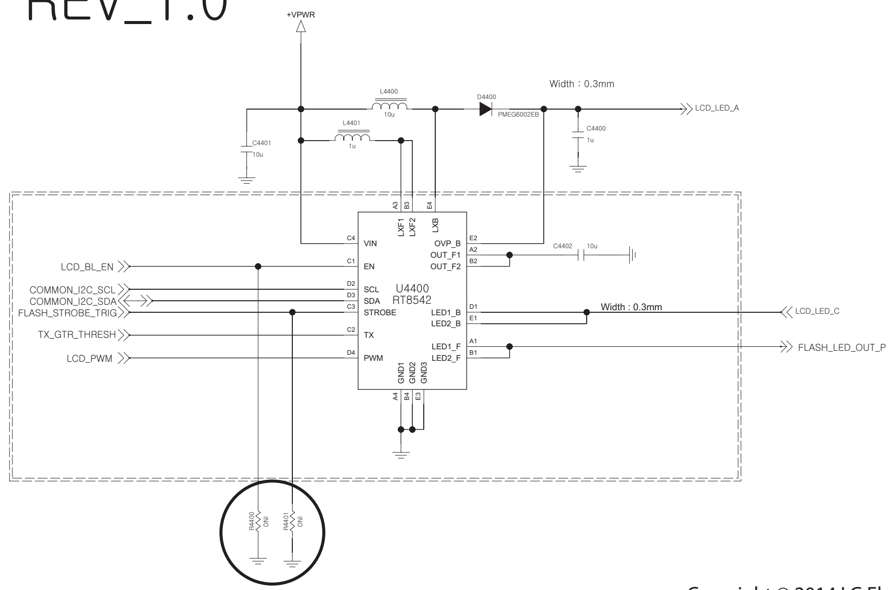


<7-3-5-1_30pin_5M_8M_13M_AF> Rev_0.3

Check the pin map

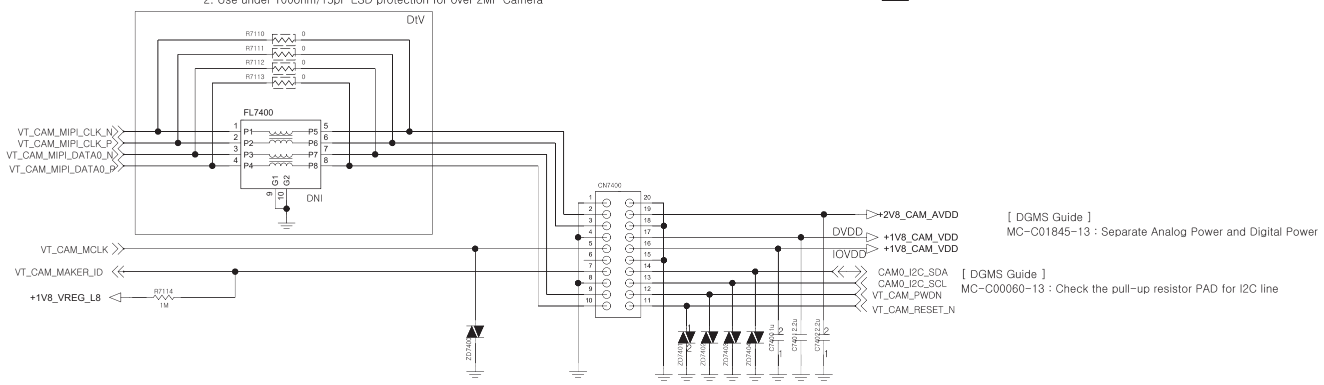


<4-4-3_LCD_Backlight_IC_RT8542> REV_1.0

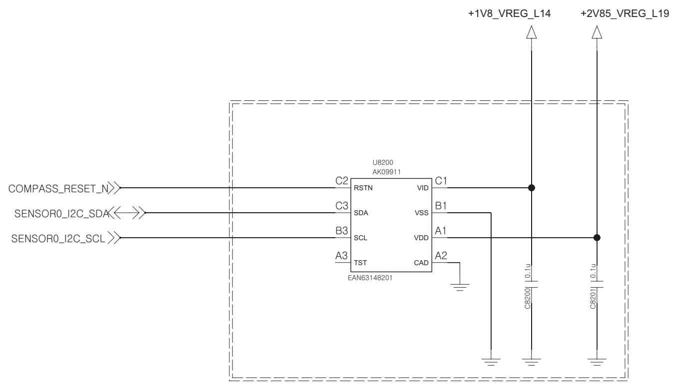


< 7-4-1_VT_Camera_20pin_VGA_1.26M_1.3M_2.4M> Rev_0.3

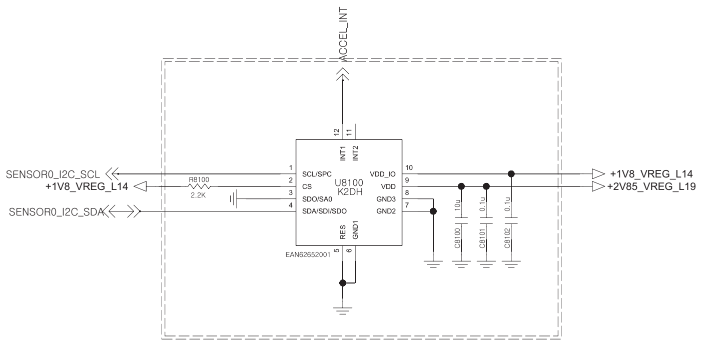
[DGMS Guide]
MC-C00081-17: 1. Add ESD Protection curcuit for Camera signal lines
2. Use under 100ohm/15pF ESD protection for over 2MP Camera



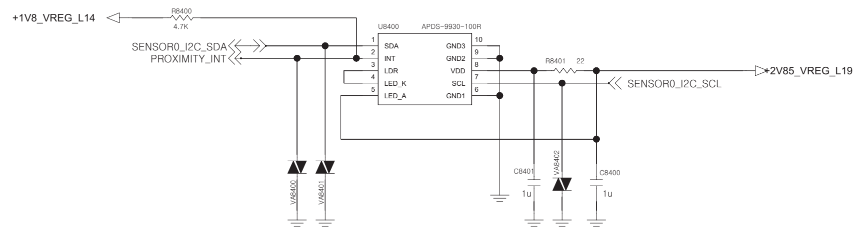
<8-2-1-3_Compass_AK09911 >
Rev_0.3



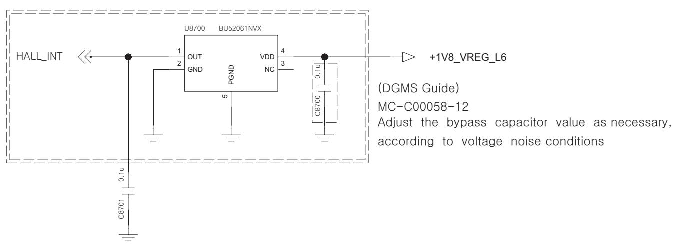
<8-1-1-3_Accelerometer_K2DH >
Rev_0.3



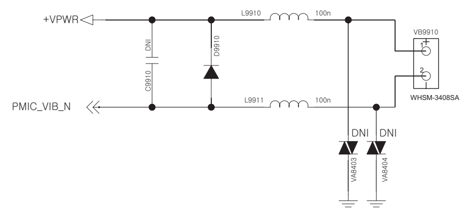
< 8-4-1-2_Proximity_APDS-9190 >
Rev_1.0



<8-7-1-6_Hall_IC_BU52033NVX>
Rev_1.0

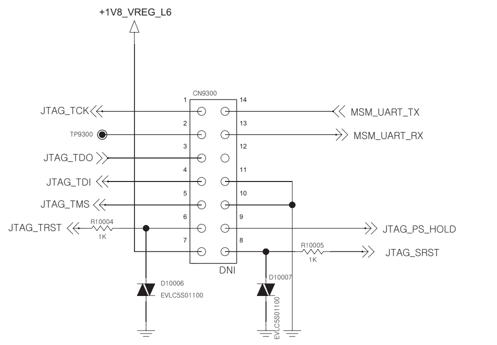


< 9-9-1-1_Motor >
Rev_0.3

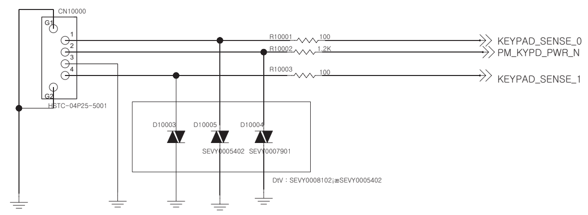


< 9-3-1_JTAG >

Rev_0.4

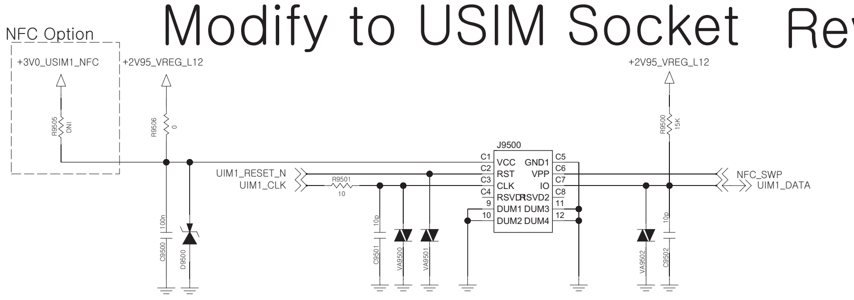


Back Volume Key



<9-5-2-2_Micro_SIM_8P_1_5T>

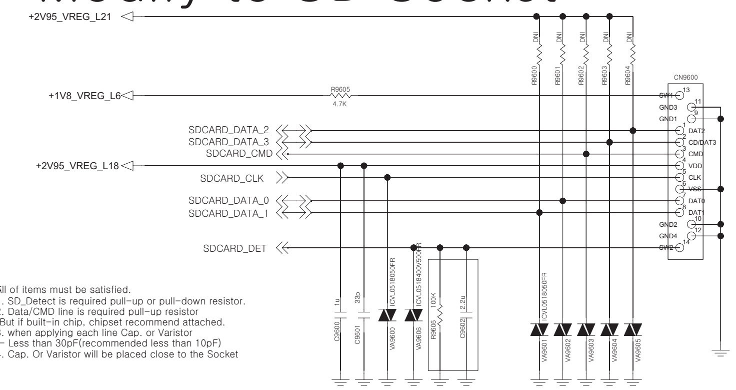
Modify to USIM Socket Rev_0.3



(DGMS Guide)
MC-C04581-13 : All of items must be satisfied.
1. SIM_DATA : It is required pull-up resistor
(But, if built-in chip, chipset recommend attached)
2. when applying each line cap. or Varistor
- Less than 33pF
- Do not use ESD suppressor type
3. when drawing Artwork, SIM_DATA and SIM_CLK line should be separated, and routing.
4. Cap. or Varistor will be placed close to the socket.

< 9-6-1_Socket_Micro_SD > Rev_0.5

Modify to SD Socket



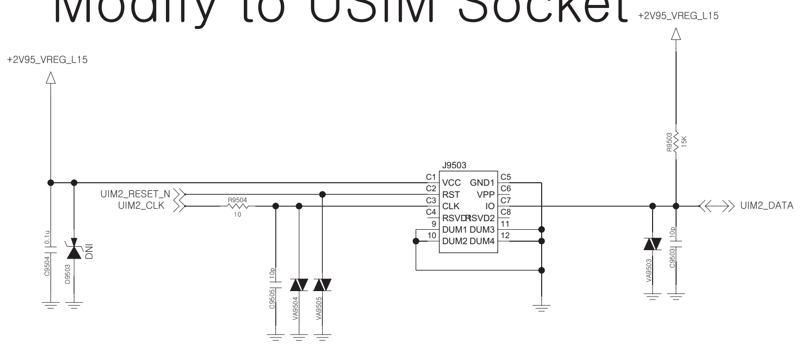
(DGMS Guide)
MC-C04582-13All of items must be satisfied.
1. SD_Detect is required pull-up or pull-down resistor.
2. Data/CMD line is required pull-up resistor
(But if built-in chip, chipset recommend attached.
3. when applying each line Cap. or Varistor
- Less than 30pF(recommended less than 10pF)
4. Cap. Or Varistor will be placed close to the Socket

(DGMS Guide)
MC-C05446-7All of items must be satisfied.
1. Power(VMMC) of Micr SD card circuit uses independent LDO/PMIC which can be ON/OFF
2. According to Chip type supported by SD card, the current capacity is managed.
: when 200mA High Speed Mode support, the MicroSD Card power is required Power PMIC Port which is obtained margin or LDO
3. SD Card power should not connect the element to serial due to voltage drop.

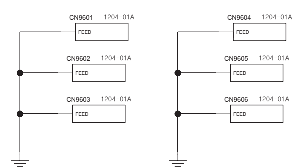
Card Detect	SW1 SW2	SD_CARD_DET
Card Installed		1.8V
Card Not Installed		GND

<9-5-2-2_Micro_SIM_8P_1_5T>

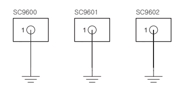
Modify to USIM Socket



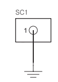
(DGMS Guide)
MC-C04581-13 : All of items must be satisfied.
1. SIM_DATA : It is required pull-up resistor
(But, if built-in chip, chipset recommend attached)
2. when applying each line cap. or Varistor
- Less than 33pF
- Do not use ESD suppressor type
3. when drawing Artwork, SIM_DATA and SIM_CLK line should be separated, and routing.
4. Cap. or Varistor will be placed close to the socket.

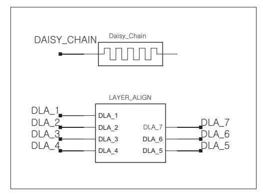


Shield CAN



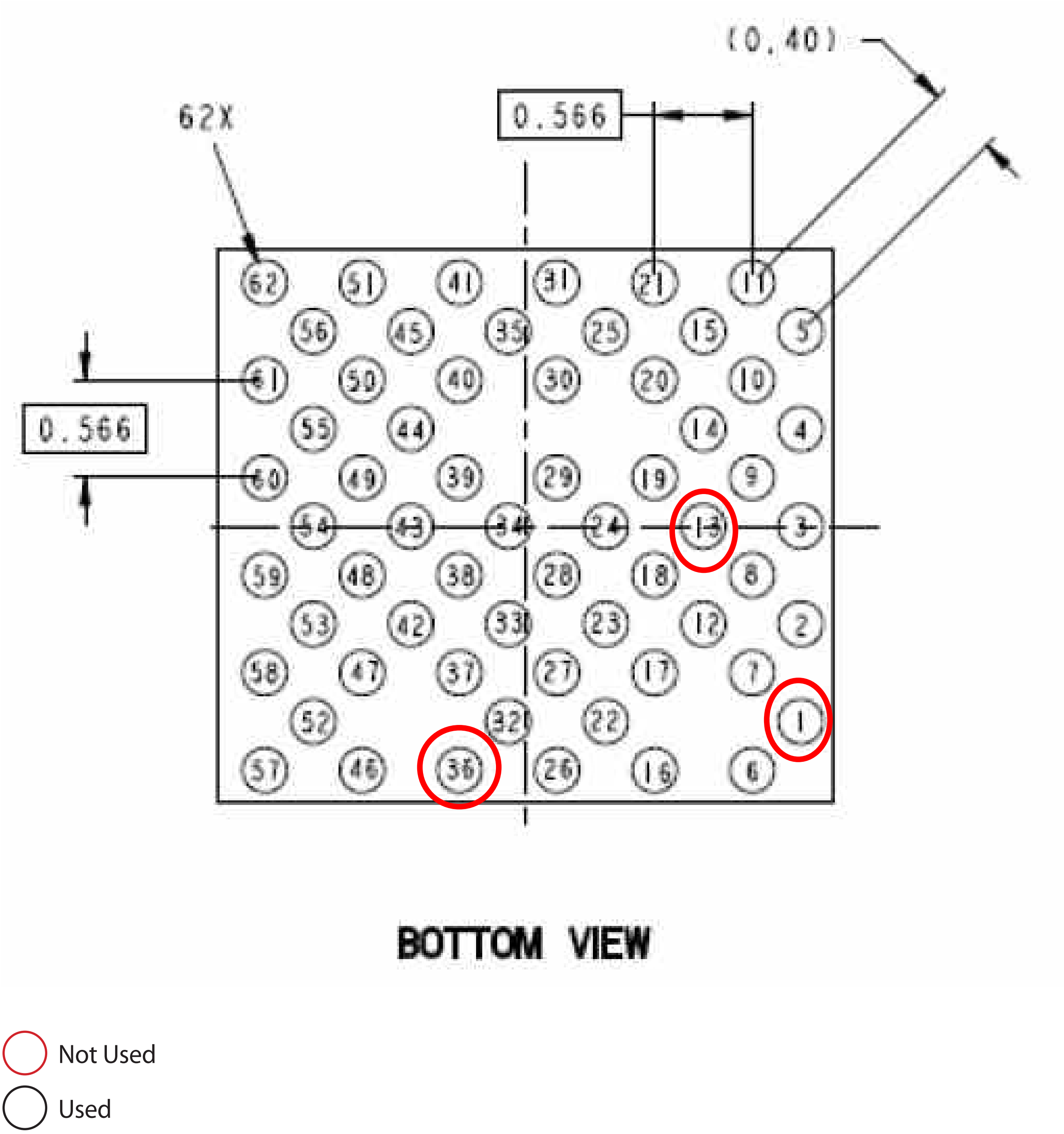
CAMERA BRACKET





7. BGA PIN MAP

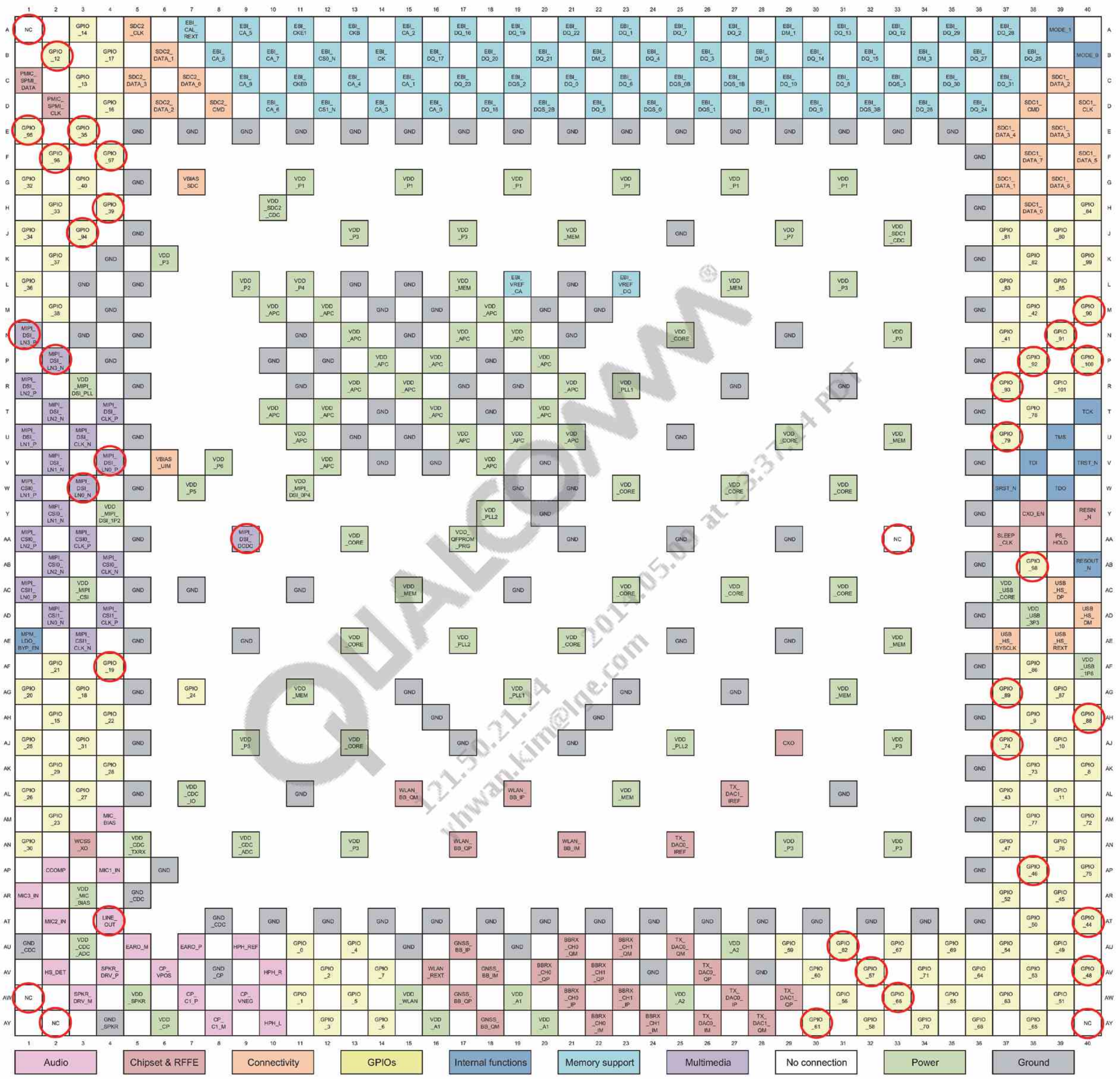
U1500 WTR2605



U4100 PM8110

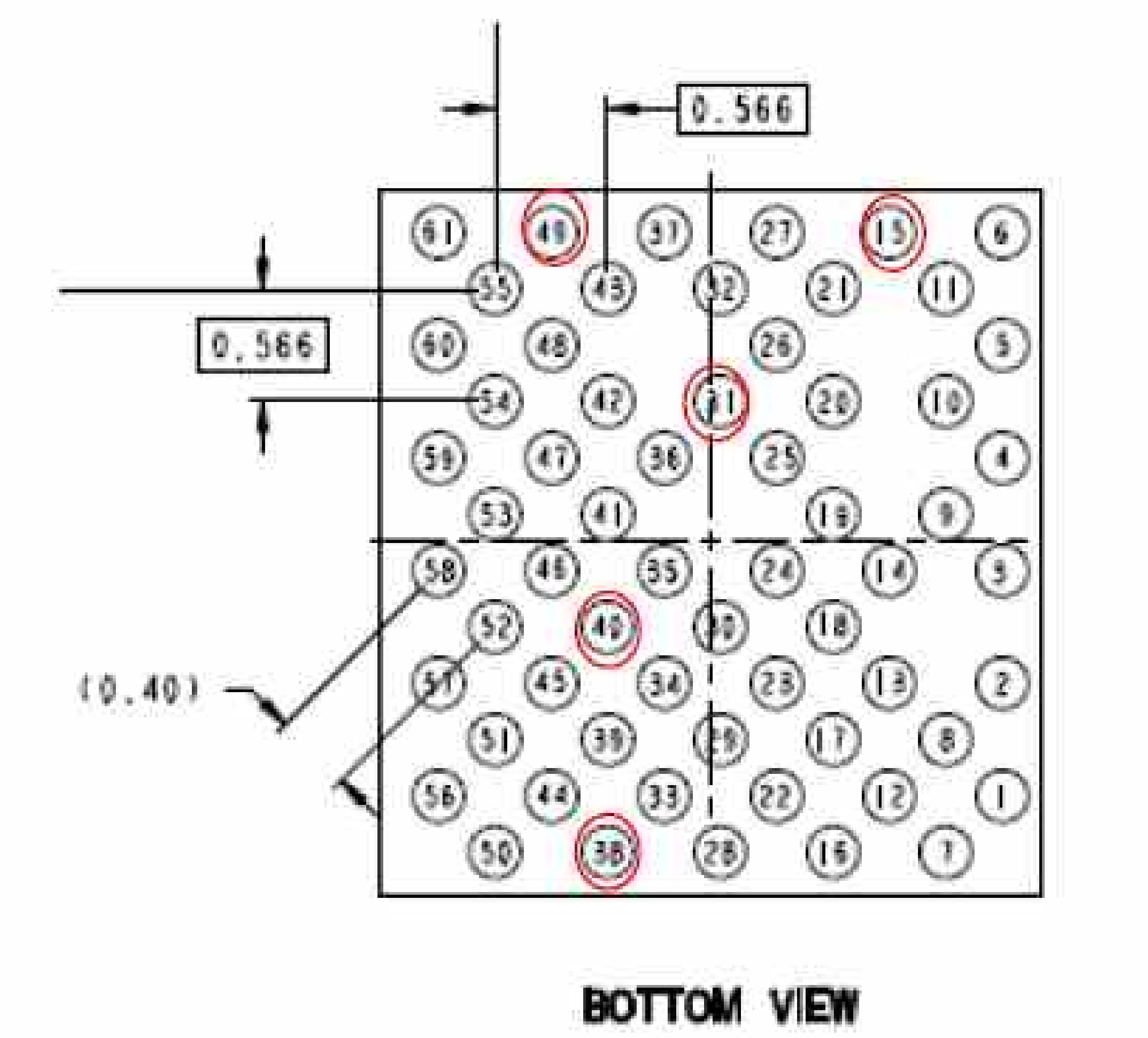
Not Used

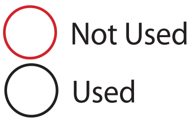
U2100 MSM8212



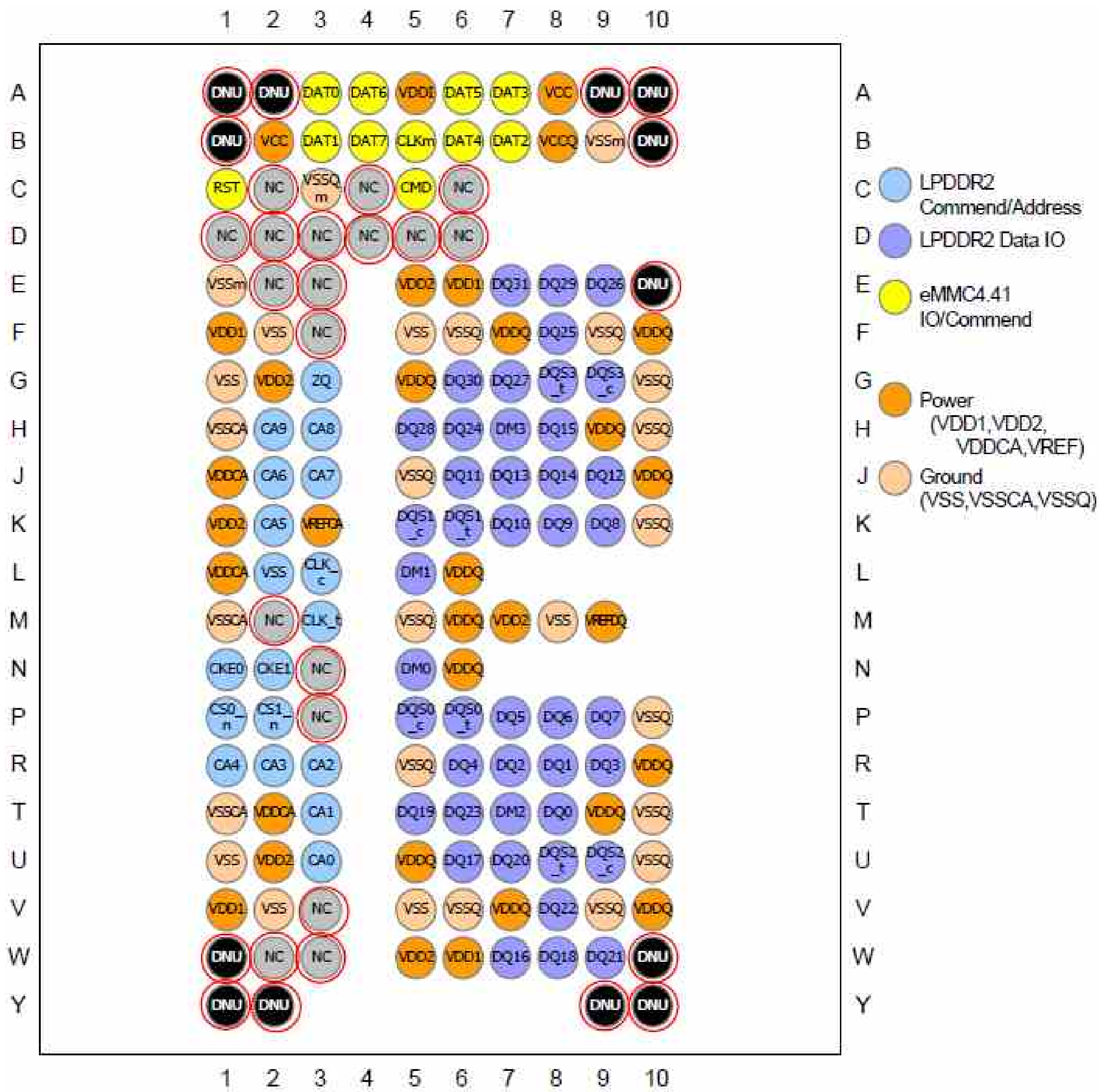
Not Used

U5100 WCN3620



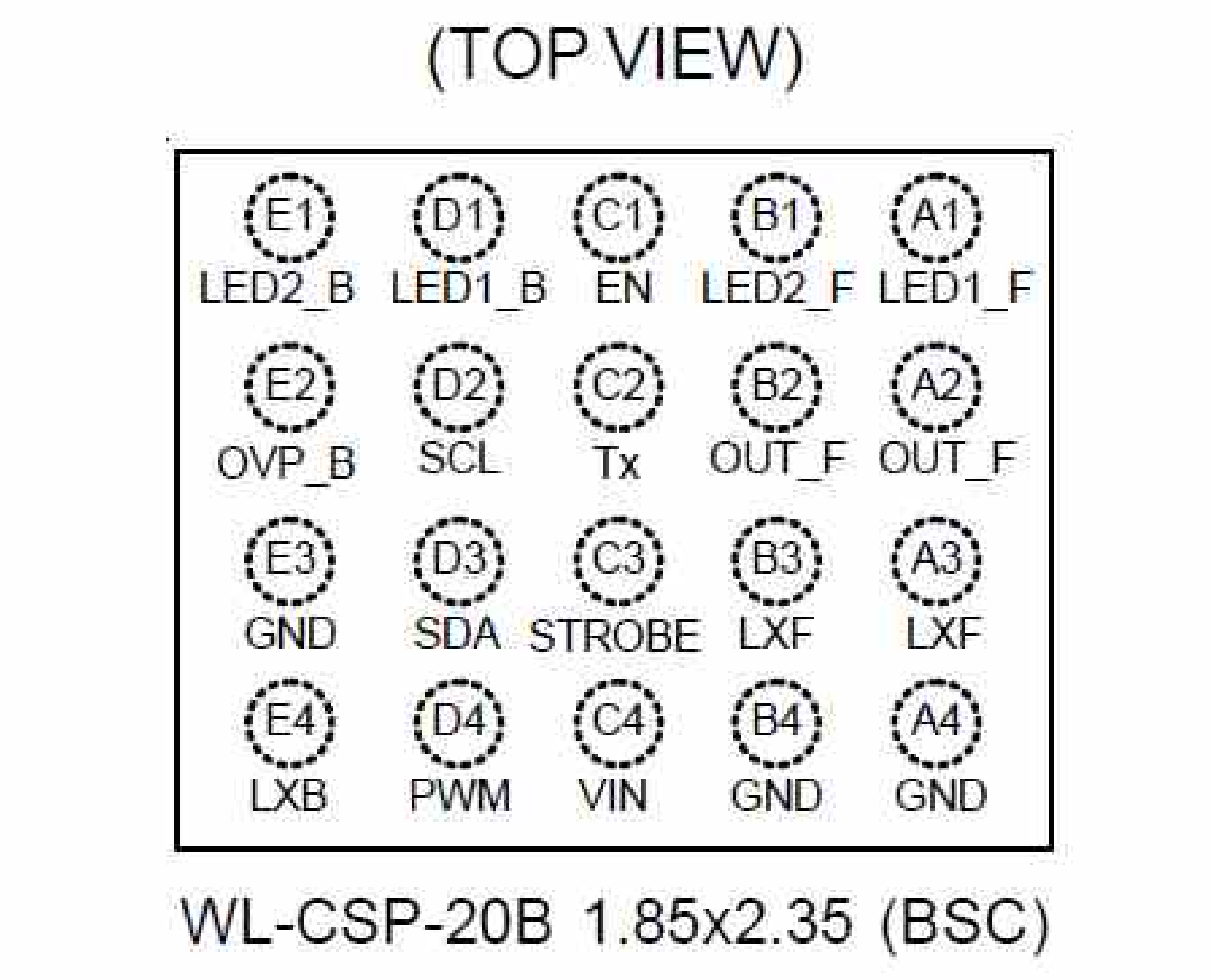


U3300 H9TP32A8JDBCPR-KGM



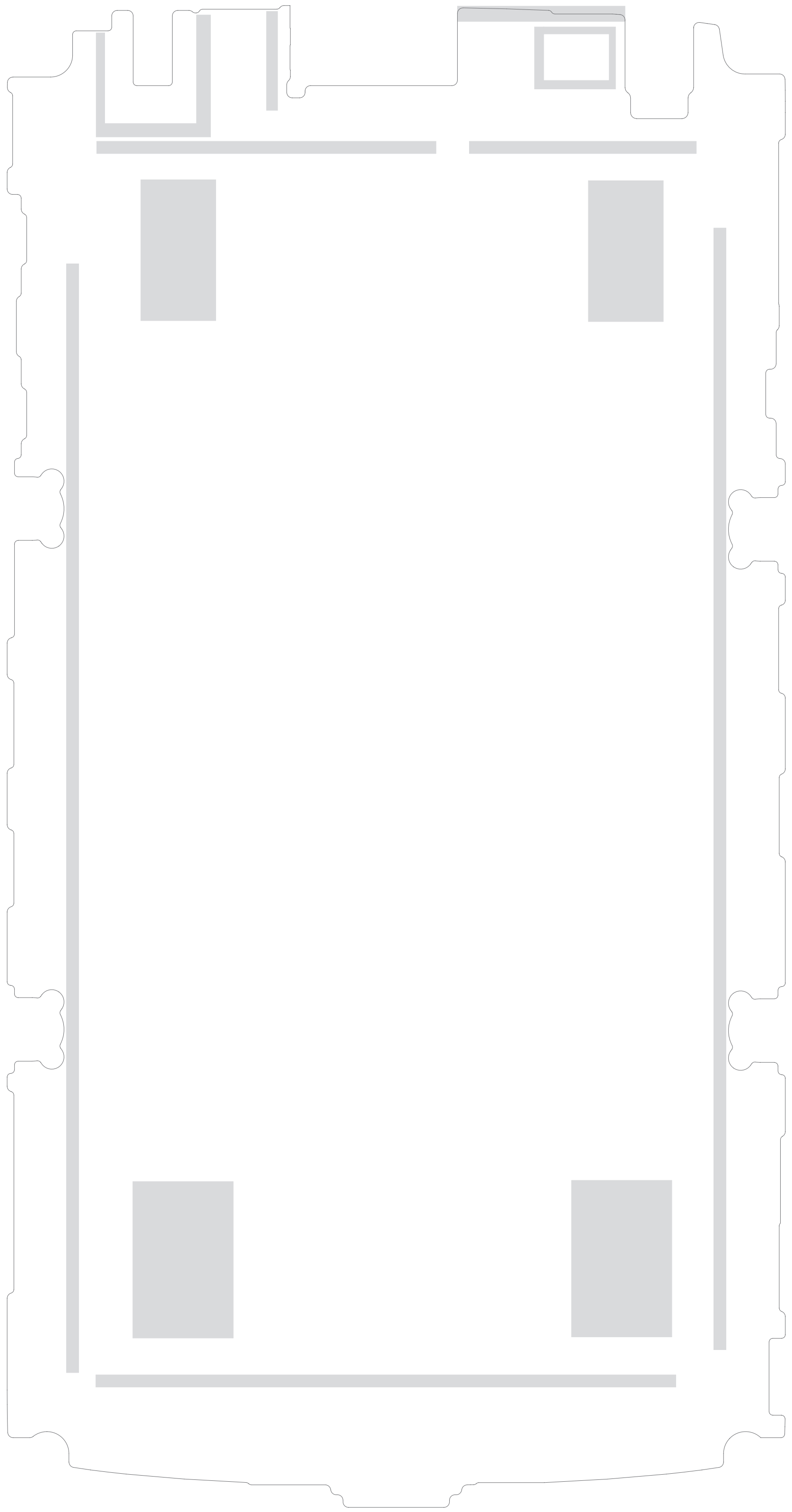
Not Used

U4400 RT8542



Not Used

8. PCB LAYOUT



LG-D295_MAIN_1.0_TOP

: Search parts by part number may not work on this layout. 206p to search parts.

U8400 Proximity Sensor
No Proximity Function

CN7300 Camera Connector No Camera Function

U3300 Memory (MCP)
No Boot / Reset, etc

CN9101 Battery Connector
No Power on / No Boot, No Charging

FL1102 GPS LNA filter
No GPS Function

U5100 BT/WIFI/FM Module
Can not use BT, WIFI, FM

U2100 MSM8212 (Main chipset)
No Boot / Reset, etc

U1500 RF Transceiver
No Network service

CN9600 SD SOCKET
No SD Card Detect

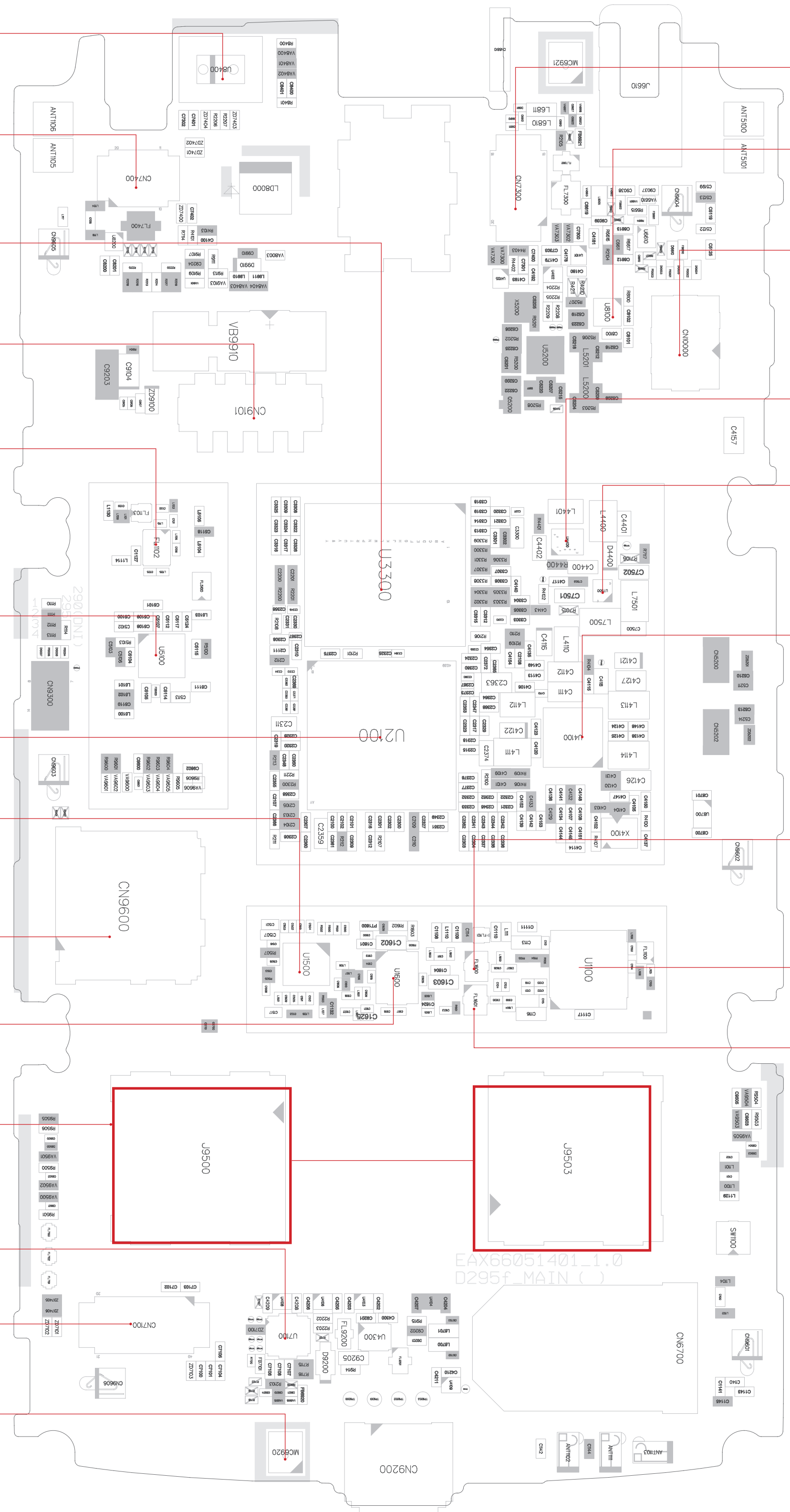
U1600 WCDMA PAM
W900, W2100 No Tx power

J9500/J9503 SIM SOCKET
No SIM Detect

U7100 Touch IC
No Touch Function

CN7100 LCD & Touch Connector
No LCD & Touch Function

MIC6920 Microphone
No Voice Sound



CN7300 VT Camera Connector
No VT Camera Function

U8100 Acceleration Sensor
Can not use Motion Sense

CN10000 Key Connector
Can not use Power & volume key

U4400 Backlight & Flash Driver IC
No LCD & Flash LED Function

U4400 DSV
No LCD

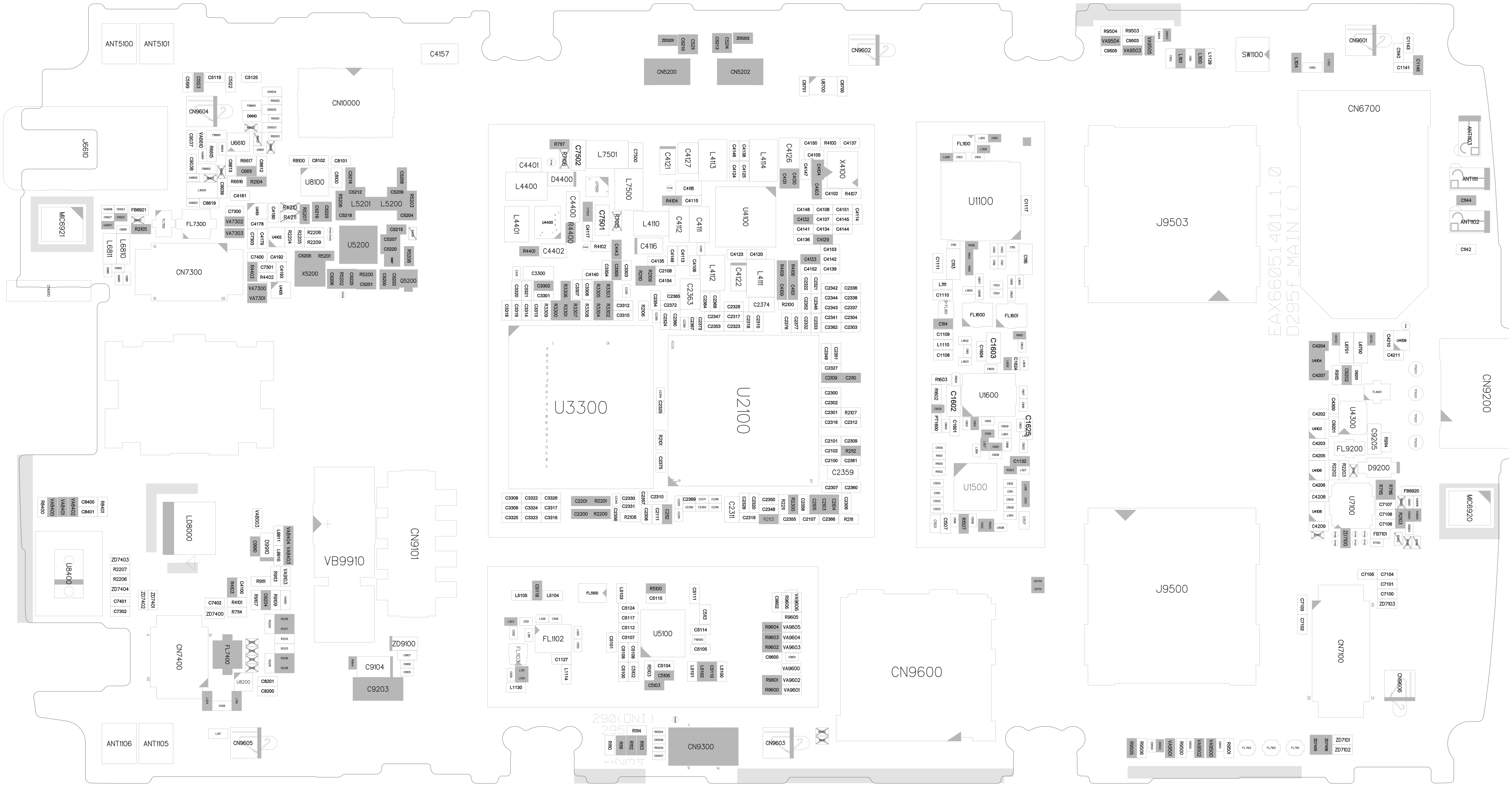
U4100 PMIC
No Power on, No Boot, No Charging

FL1600 Duplexer
W2100 Tx/Rx no service

U1100 Transmit & Tx Module
No Network service
GSM No Tx Power

FL1601 Duplexer
W900 Tx/Rx no service

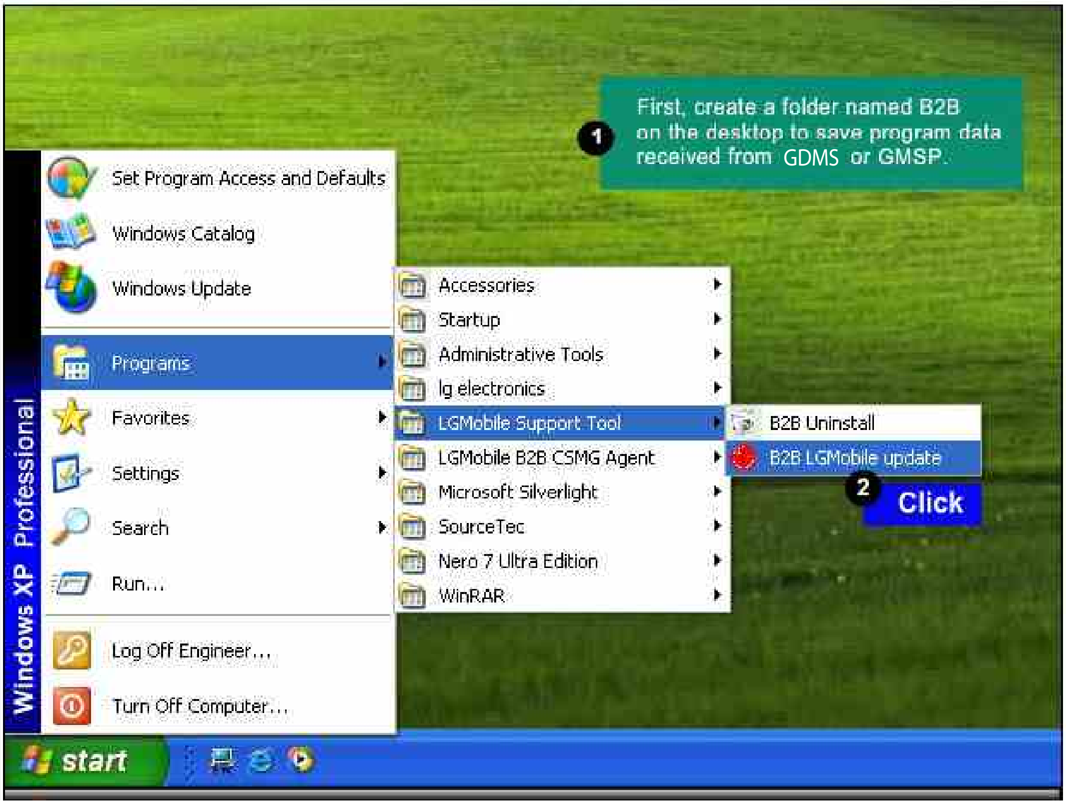
LG-D295_MAIN_1.0_BOT

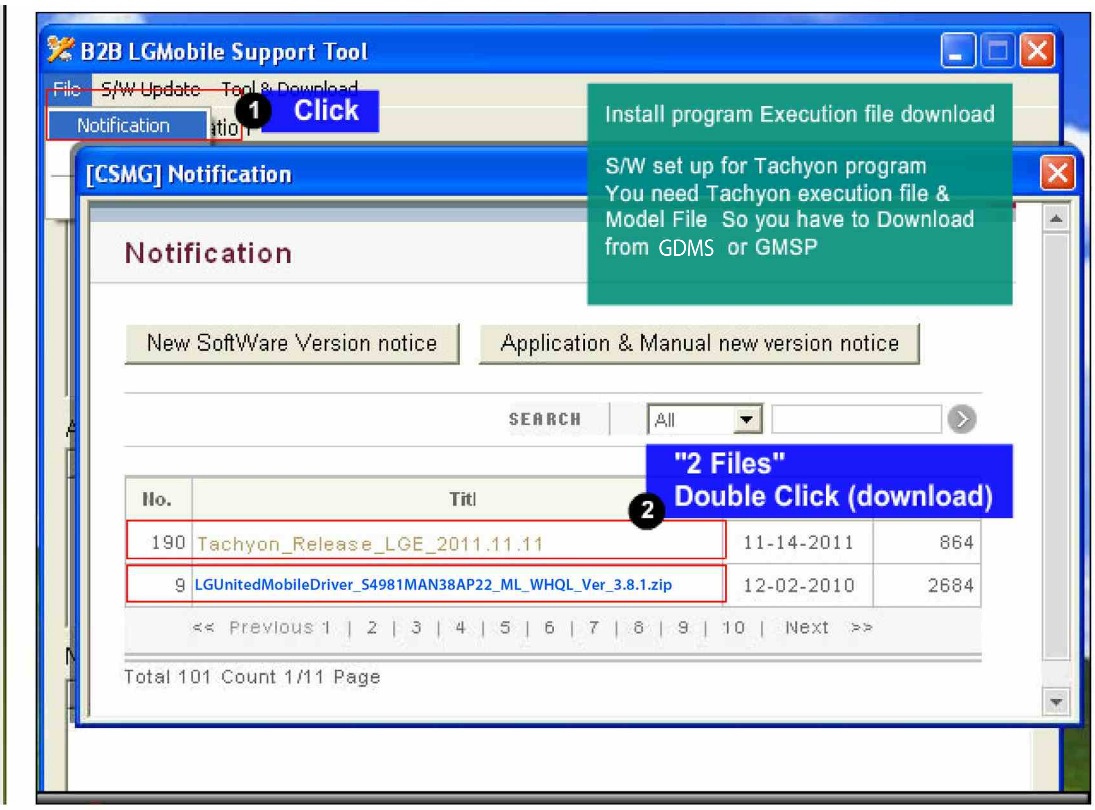


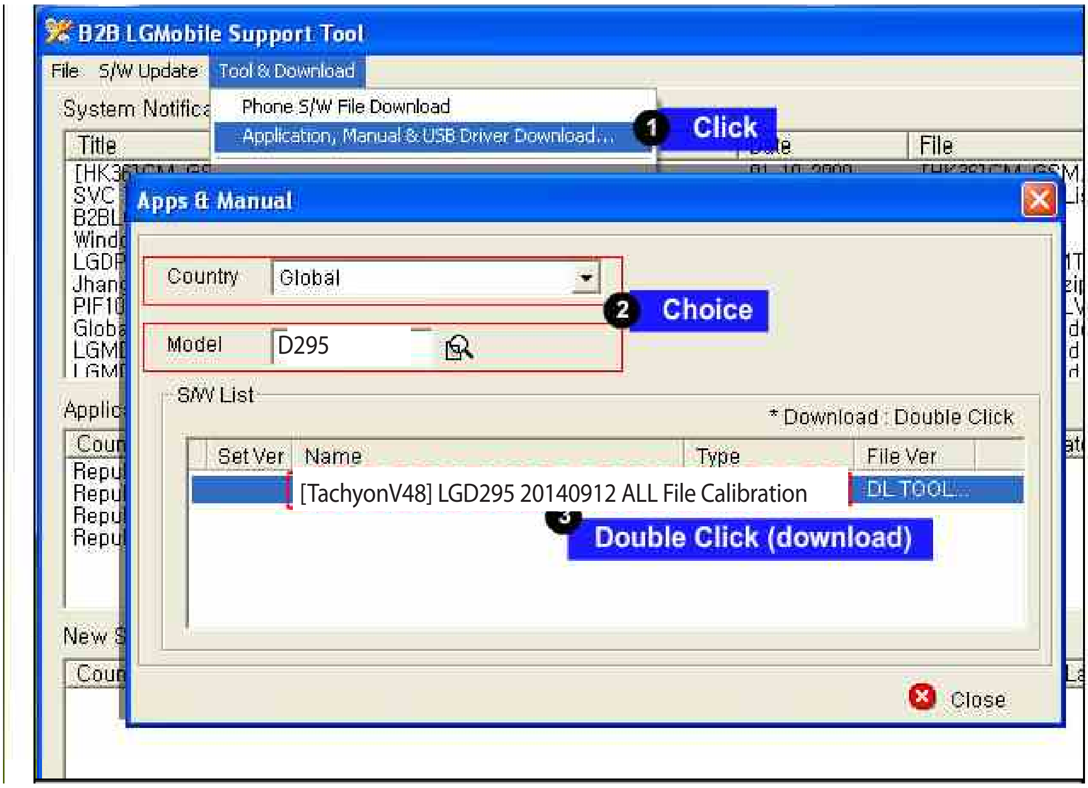
ANT5100
ANT5101
C4157
CN5200
CN5202
CN9602
SW1100
CN9601
CN10000
CN9604
J6610
U8100
L5201
L5200
U5200
X5200
MIC6921
CN7300
CN6700
J9503
EAX66051401_1.0
D295f_MAIN ( )
U1100
U3300
U2100
C2359
CN9200
U1600
U1500
FL1600
FL1601
U4100
X4100
L7501
L7500
D4400
L4400
C4401
C4402
L4110
C4116
L4112
L4111
C4122
C2363
C2374
FL9200
D9200
U7100
MIC6920
J9500
LD8000
U8400
VB9910
CN9101
CN7400
FL7400
C9104
C9203
ZD9100
FL1102
U5100
CN9600
CN7100
CN9606
ANT1106
ANT1105
CN9605
CN9300
CN9603
290(DNI)
295

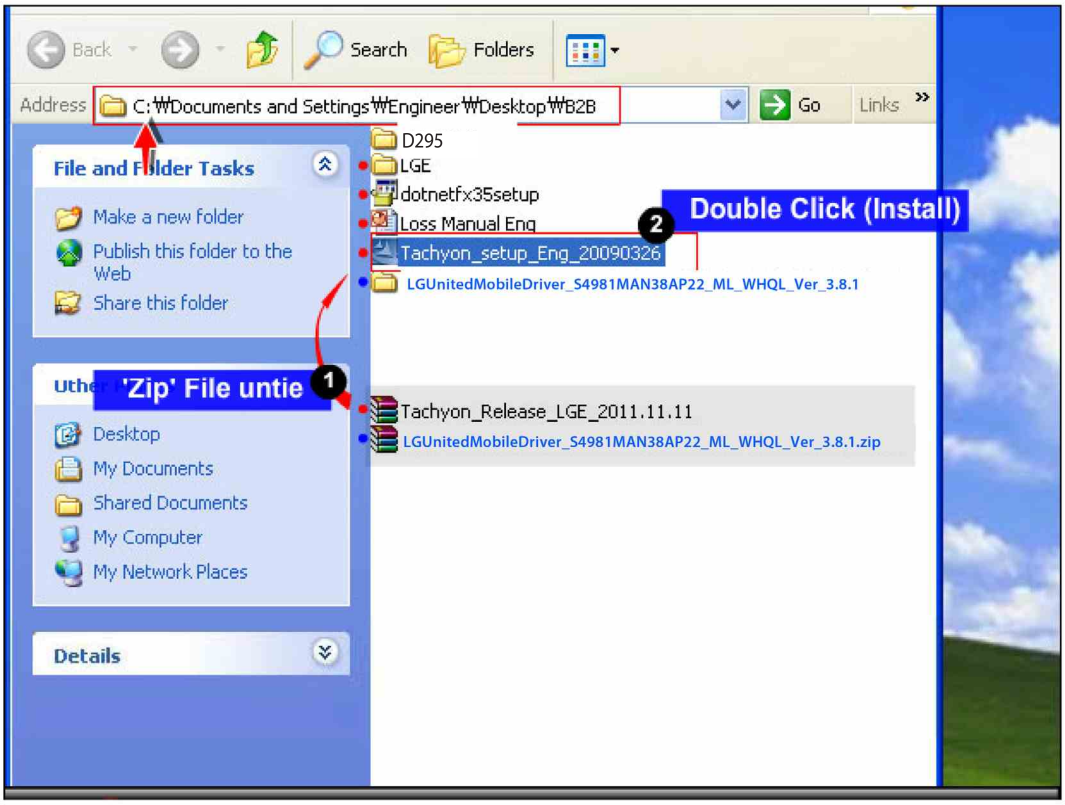
9. CALIBRATION

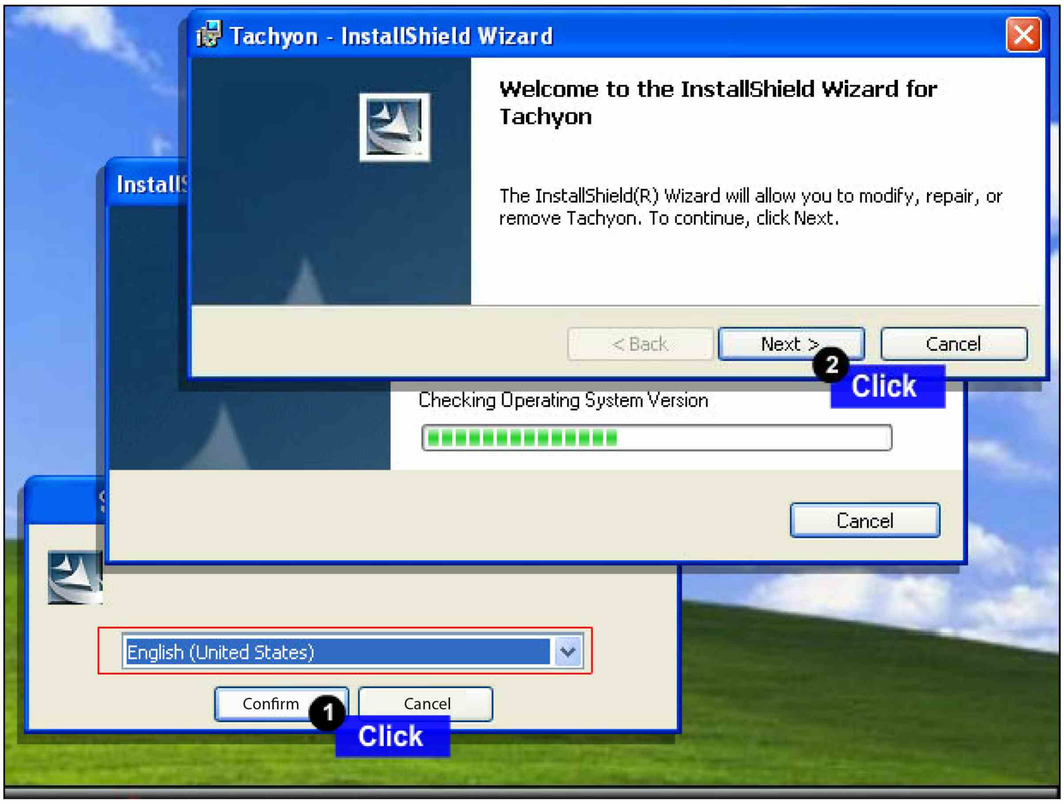
CAL INFORMATION		
S/W VERSION		
[TachyonV48] LGD295 20140912 ALL File Calibration		
Please Check the Version to "B2B"		
H/W		
	Name	**Part No.**
PIF	PIF200(All Type)	BJAY0024021
USB Cable	USB Cable	RAD32247898
Power Cable	DC Power Cable	RAD32247878
I/O Cable	5P E-SATA_DC_Plug	RAD32167861
RF Cable_Main	MS-156C	BJAY0024004
Power Supply_PIF	Power Supply 5.3V	
RF Test Equipment	Agilent/CMW500	
NOTICE	1. Use the Battery (Refer to Attached ppt) 1) Phone states: Power off 2) If do not use the battery, TX fails. 2. Port Setting (Refer to Attached ppt) 1) Uart Port1 : Use the "LGE Mobile USB Serial Port"	


First, create a folder named B2B on the desktop to save program data received from GDMS or GMSP.
1
Set Program Access and Defaults
Windows Catalog
Windows Update
Programs
Favorites
Settings
Search
Run...
Log Off Engineer...
Turn Off Computer...
Windows XP Professional
start
Accessories
Startup
Administrative Tools
lg electronics
LGMobile Support Tool
LGMobile B2B CSMG Agent
Microsoft Silverlight
SourceTec
Nero 7 Ultra Edition
WinRAR
B2B Uninstall
B2B LGMobile update
2
Click

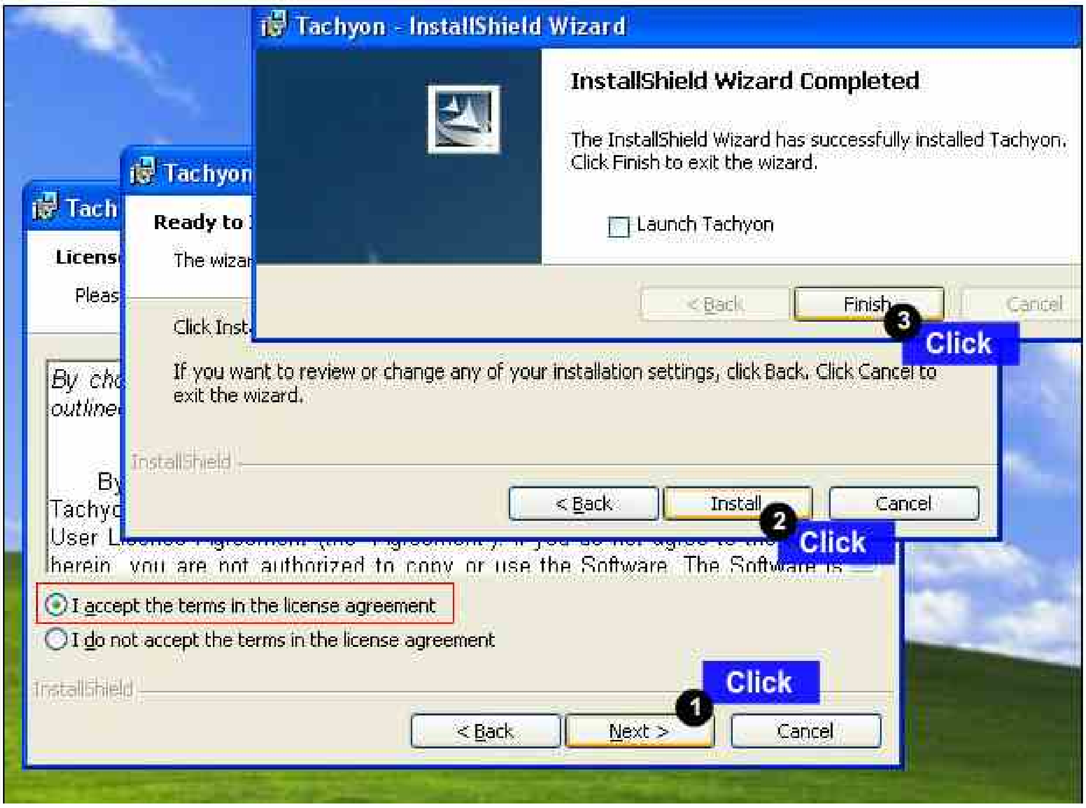

B2B LGMobile Support Tool
File S/W Update Tool & Download
Notification
1
Click
Install program Execution file download
S/W set up for Tachyon program
You need Tachyon execution file &
Model File So you have to Download
from GDMS or GMSP
[CSMG] Notification
Notification
New SoftWare Version notice
Application & Manual new version notice
SEARCH
All
"2 Files"
Double Click (download)
2
No.
Titl
190 Tachyon_Release_LGE_2011.11.11 11-14-2011 864
9 LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.zip 12-02-2010 2684
<< Previous 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Next >>
Total 101 Count 1/11 Page


B2B LGMobile Support Tool
File S/W Update Tool & Download
Phone S/W File Download
Application, Manual & USB Driver Download...
1 Click
Apps & Manual
Country Global
2 Choice
Model D295
S/W List
* Download : Double Click
Set Ver Name Type File Ver
[TachyonV48] LGD295 20140912 ALL File Calibration
DL TOOL...
Double Click (download)
Close

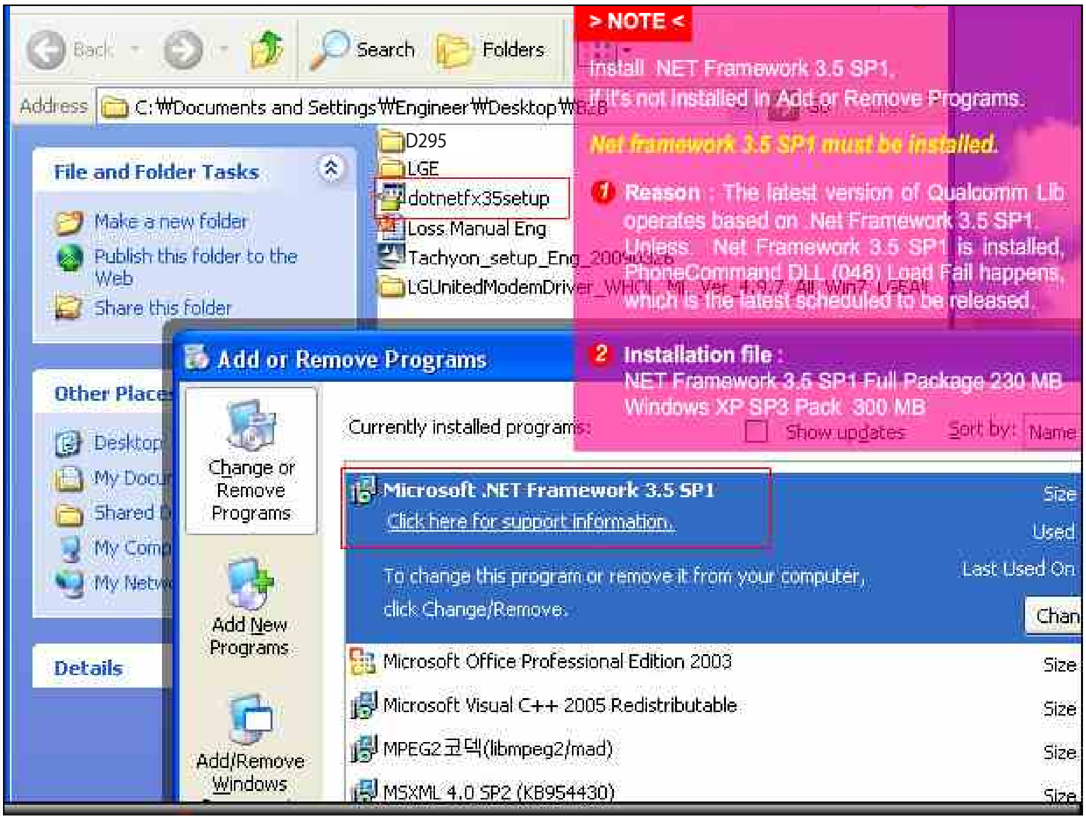

Back Search Folders
Address C:₩Documents and Settings₩Engineer₩Desktop₩B2B
Go Links
File and Folder Tasks
Make a new folder
Publish this folder to the Web
Share this folder
'Zip' File untie 1
Desktop
My Documents
Shared Documents
My Computer
My Network Places
Details
D295
LGE
dotnetfx35setup
Loss Manual Eng
Tachyon_setup_Eng_20090326
2 Double Click (Install)
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1
Tachyon_Release_LGE_2011.11.11
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.zip



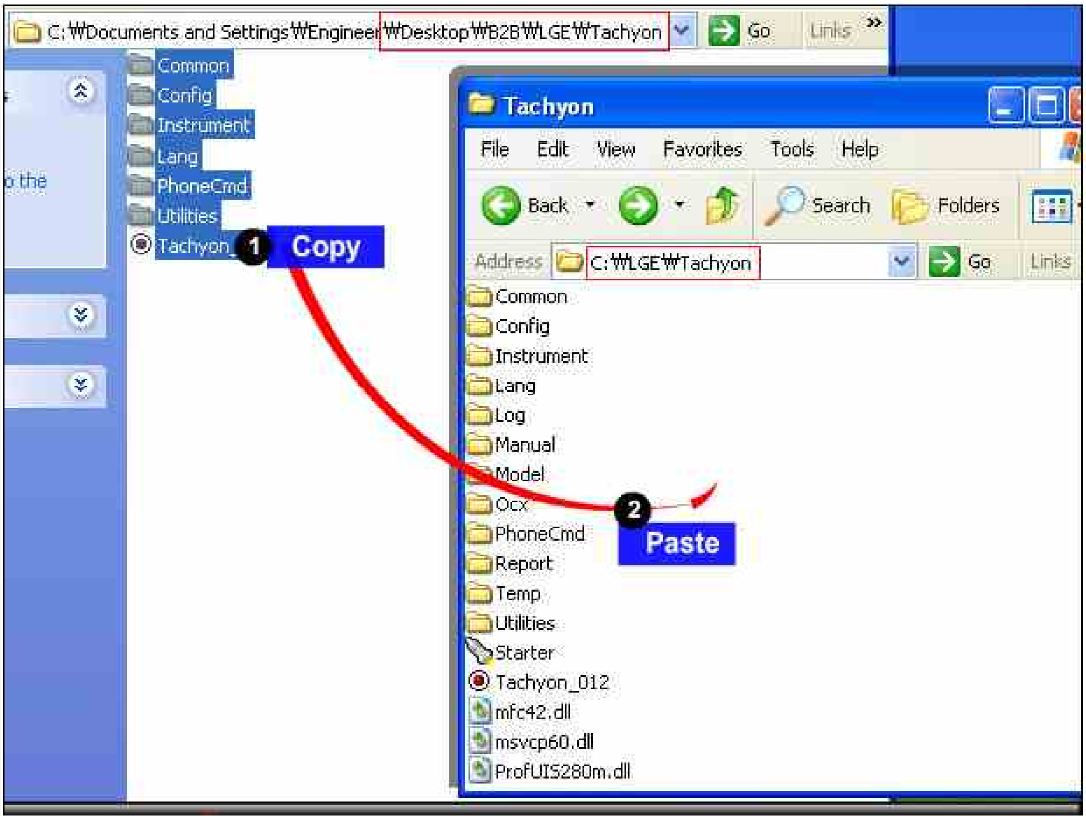
Tachyon - InstallShield Wizard
Welcome to the InstallShield Wizard for Tachyon
The InstallShield(R) Wizard will allow you to modify, repair, or remove Tachyon. To continue, click Next.
< Back
Next >
Cancel
2
Click
Checking Operating System Version
Cancel
English (United States)
Confirm
Cancel
1
Click



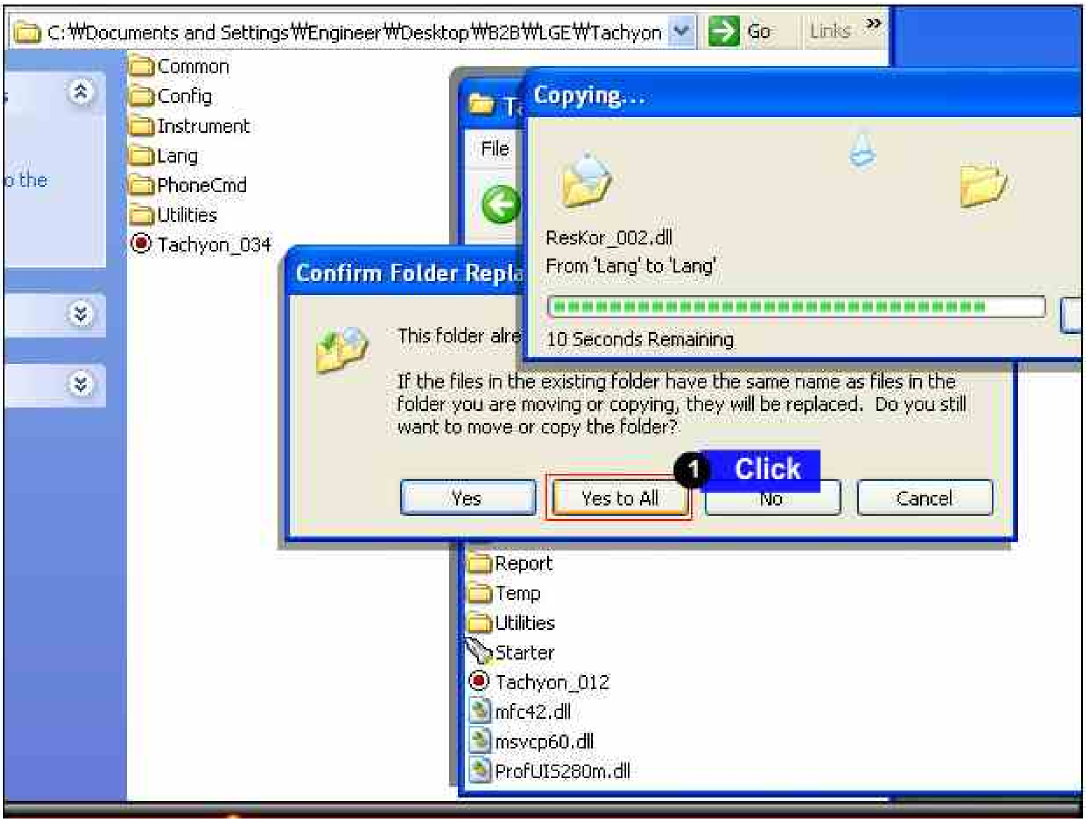
Tachyon - InstallShield Wizard
InstallShield Wizard Completed
The InstallShield Wizard has successfully installed Tachyon. Click Finish to exit the wizard.
Launch Tachyon
< Back
Finish
Cancel
3
Click
If you want to review or change any of your installation settings, click Back. Click Cancel to exit the wizard.
InstallShield
< Back
Install
Cancel
2
Click
I accept the terms in the license agreement
I do not accept the terms in the license agreement
InstallShield
Click
1
< Back
Next >
Cancel

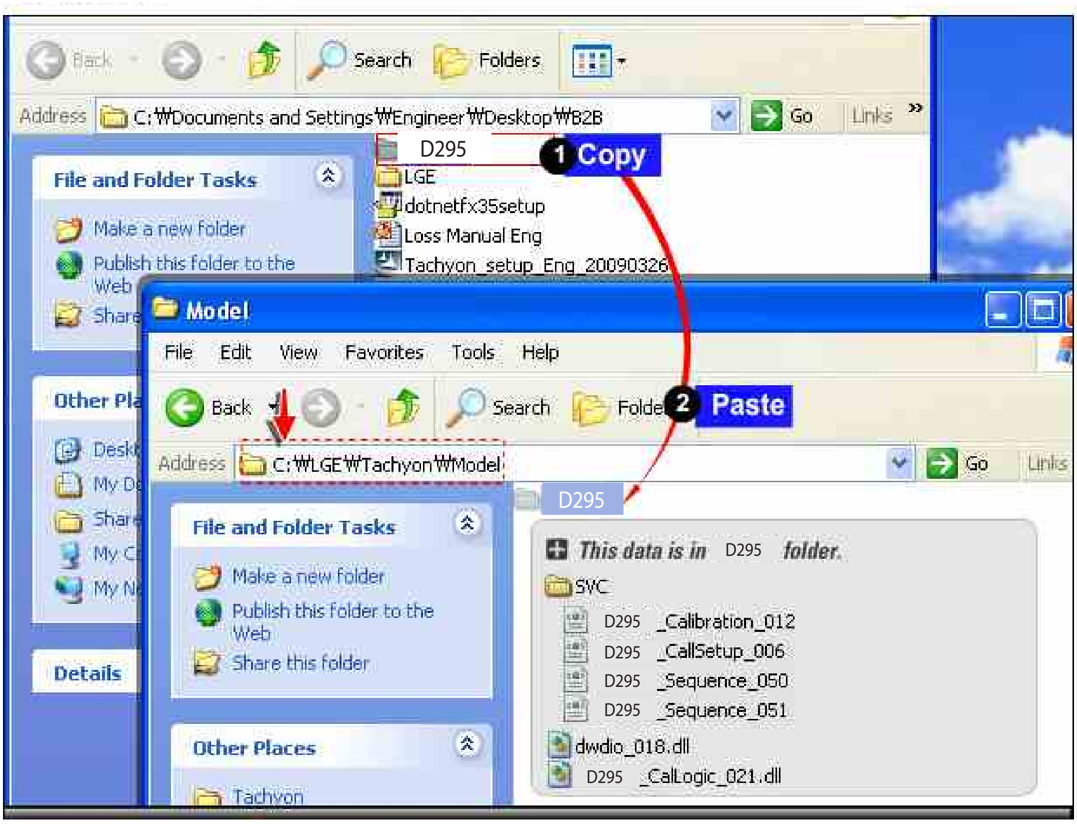


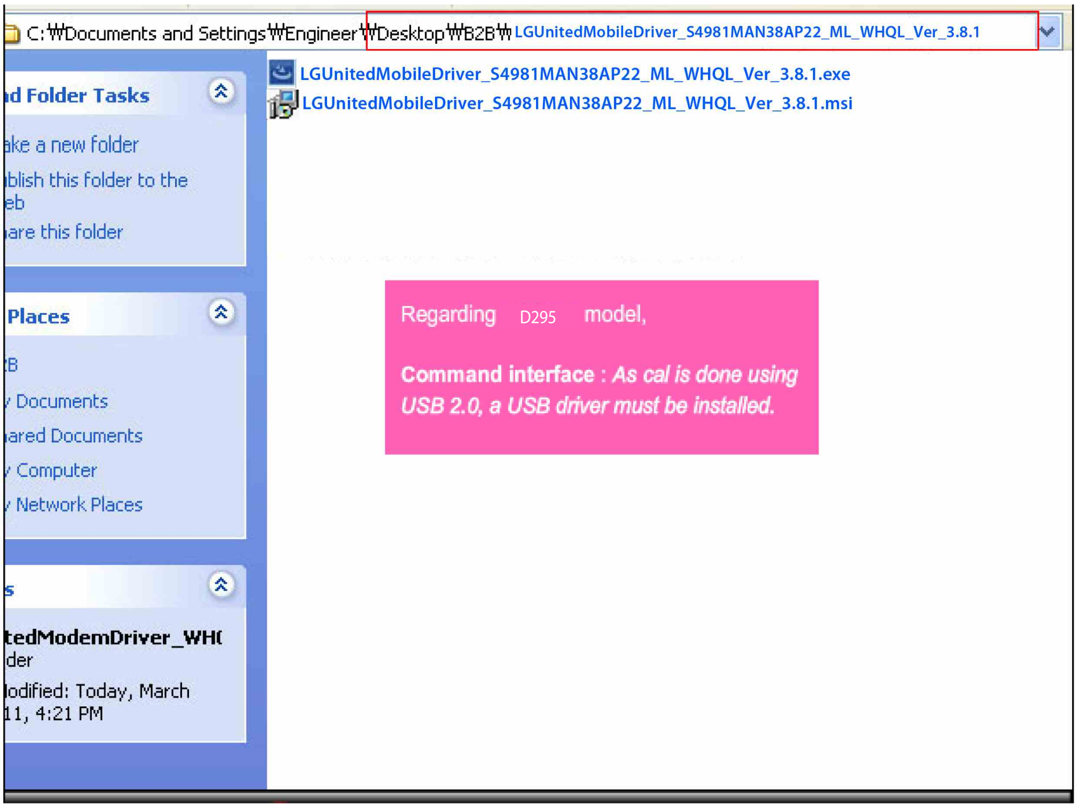
> NOTE <
Install .NET Framework 3.5 SP1,
if it's not installed in Add or Remove Programs.
Net framework 3.5 SP1 must be installed.
1 Reason : The latest version of Qualcomm Lib operates based on .Net Framework 3.5 SP1. Unless Net Framework 3.5 SP1 is installed, PhoneCommand DLL (048) Load Fail happens, which is the latest scheduled to be released.
2 Installation file :
NET Framework 3.5 SP1 Full Package 230 MB
Windows XP SP3 Pack 300 MB
Add or Remove Programs
Microsoft .NET Framework 3.5 SP1
dotnetfx35setup

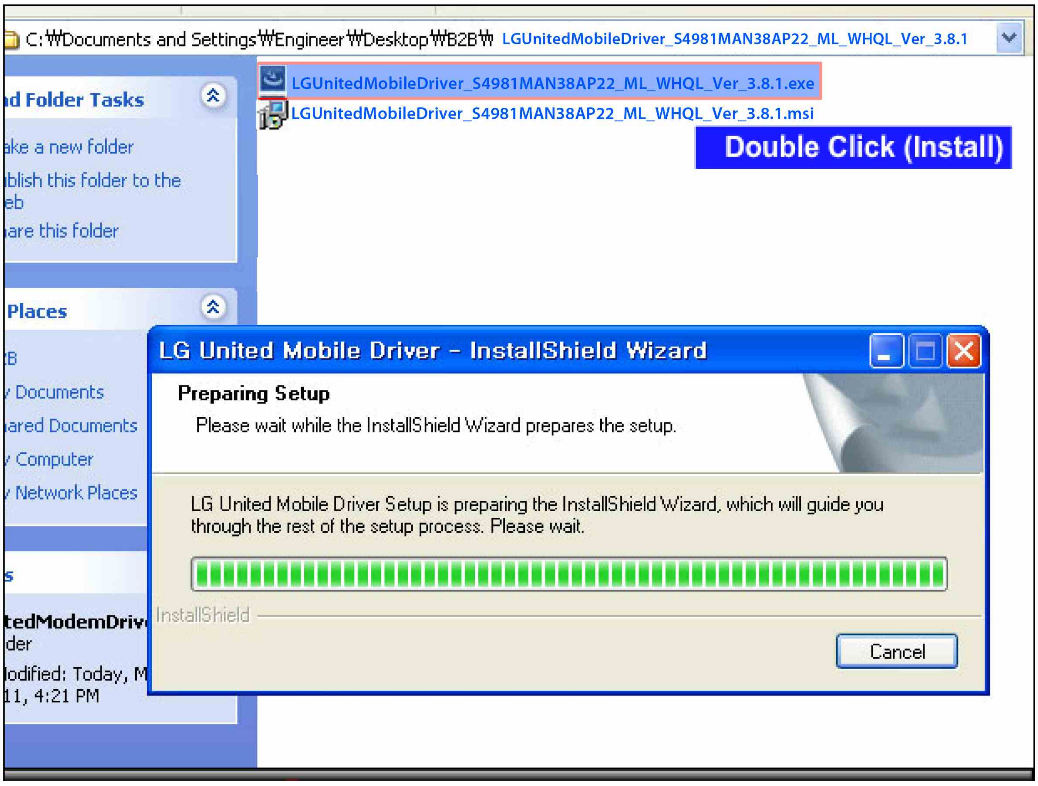


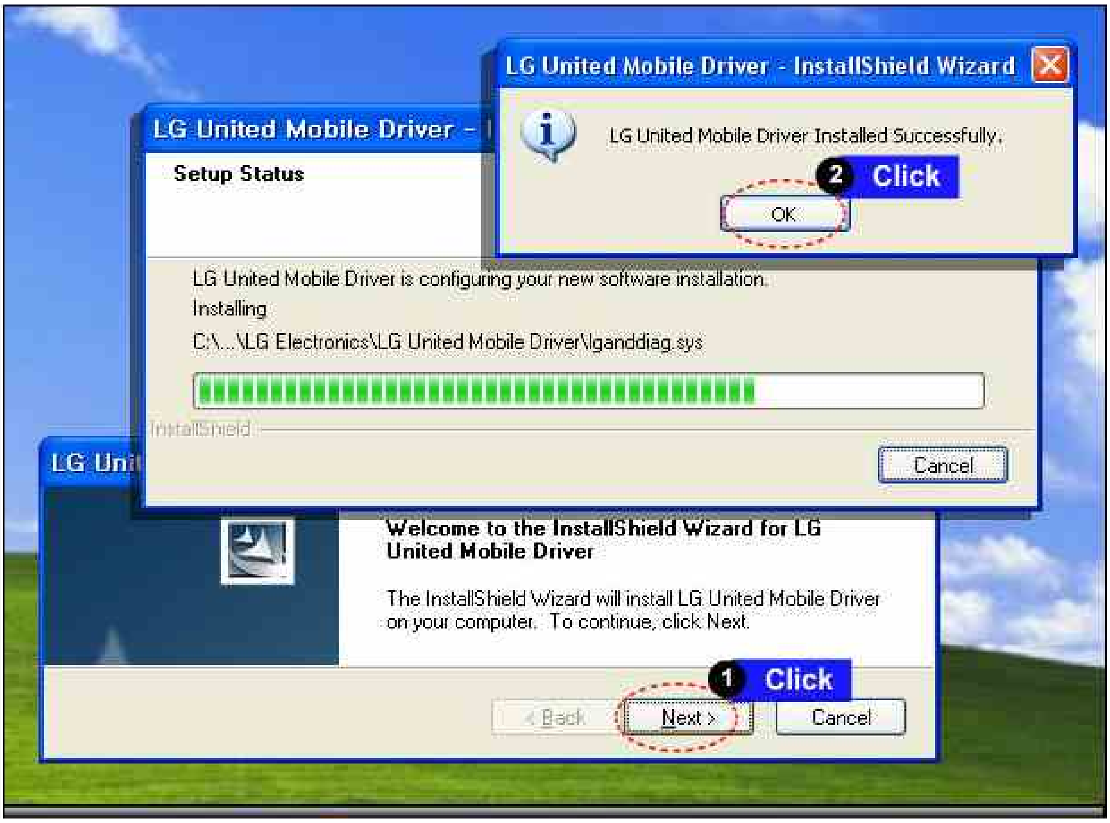
C:₩Documents and Settings₩Engineer₩Desktop₩B2B₩LGE₩Tachyon
1 Copy
2 Paste
Tachyon
C:₩LGE₩Tachyon

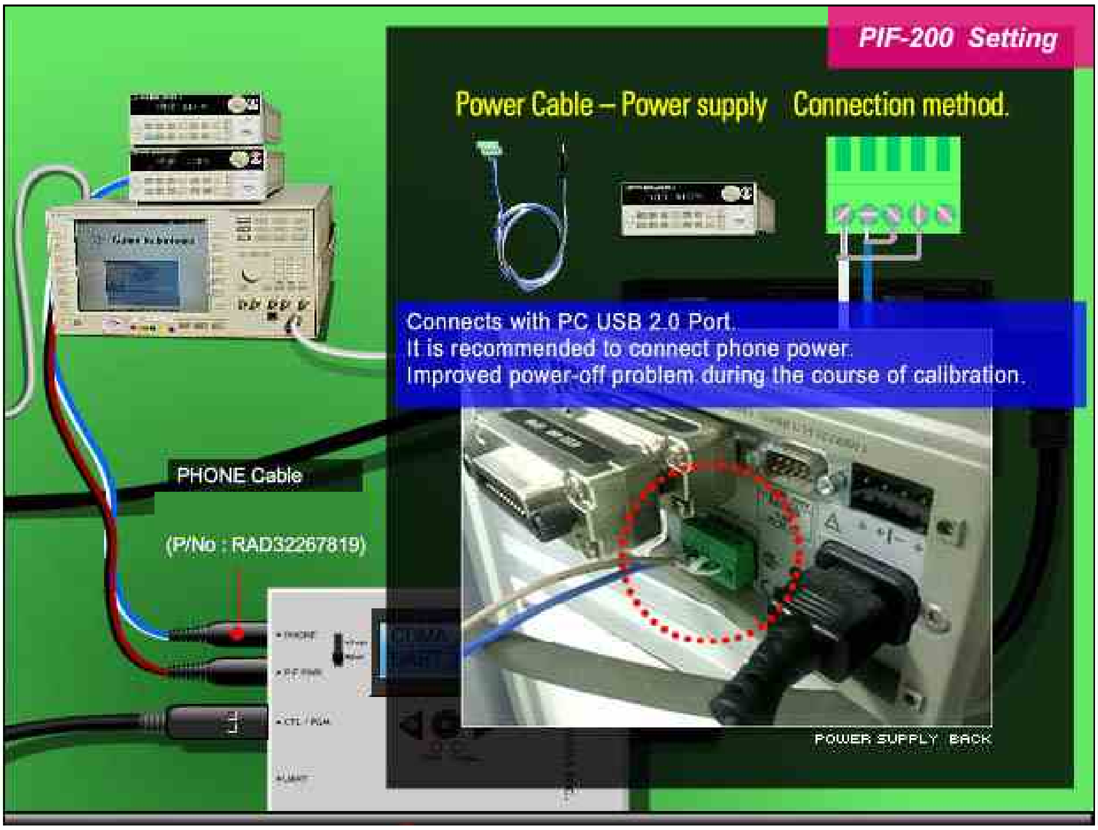

C:₩Documents and Settings₩Engineer₩Desktop₩B2B₩LGE₩Tachyon
Copying...
ResKor_002.dll
From 'Lang' to 'Lang'
10 Seconds Remaining
Confirm Folder Repla
If the files in the existing folder have the same name as files in the folder you are moving or copying, they will be replaced. Do you still want to move or copy the folder?
1 Click
Yes
Yes to All
No
Cancel

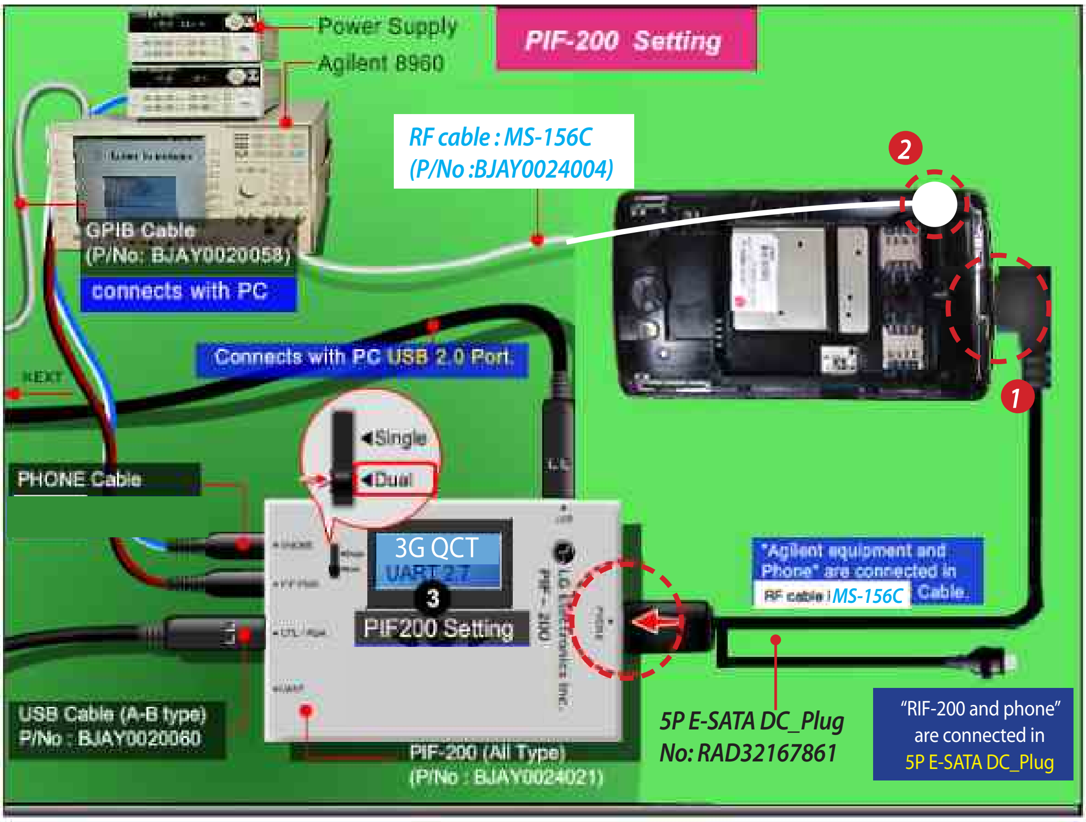

C:₩Documents and Settings₩Engineer₩Desktop₩B2B
D295
1 Copy
Model
C:₩LGE₩Tachyon₩Model
2 Paste
This data is in D295 folder.

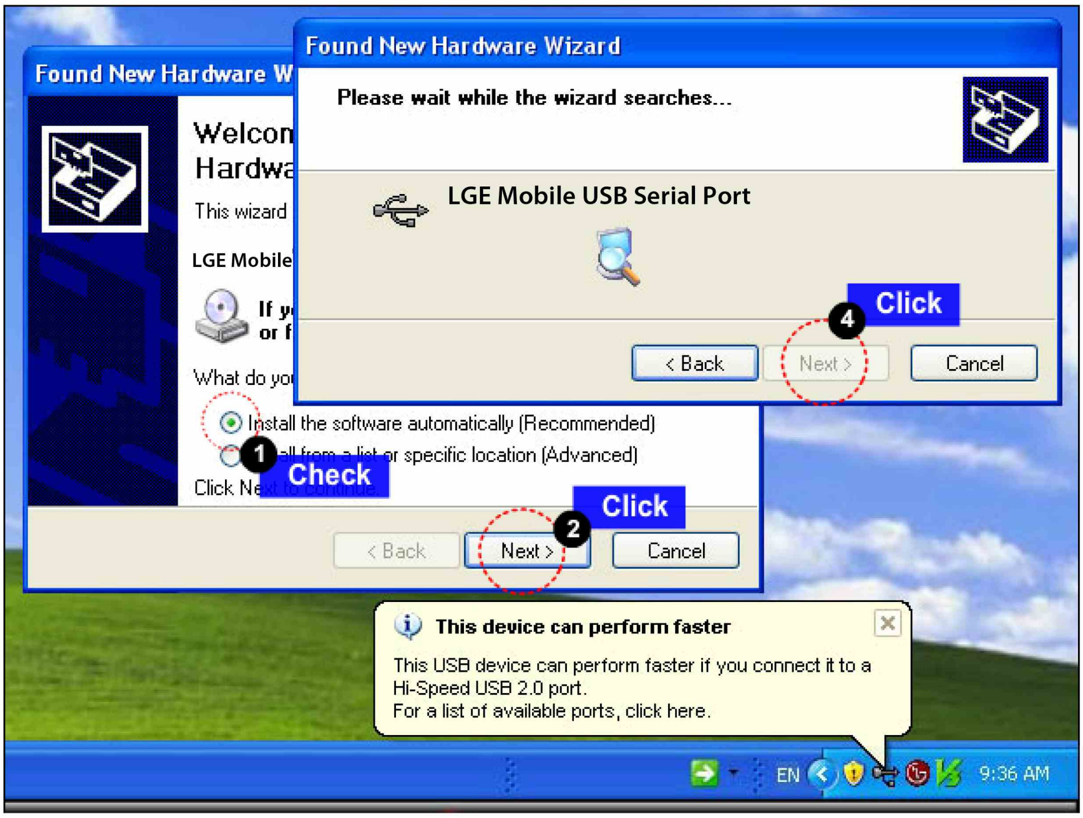

C:₩Documents and Settings₩Engineer₩Desktop₩B2B₩LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.exe
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.msi
Regarding D295 model,
Command interface : As cal is done using USB 2.0, a USB driver must be installed.

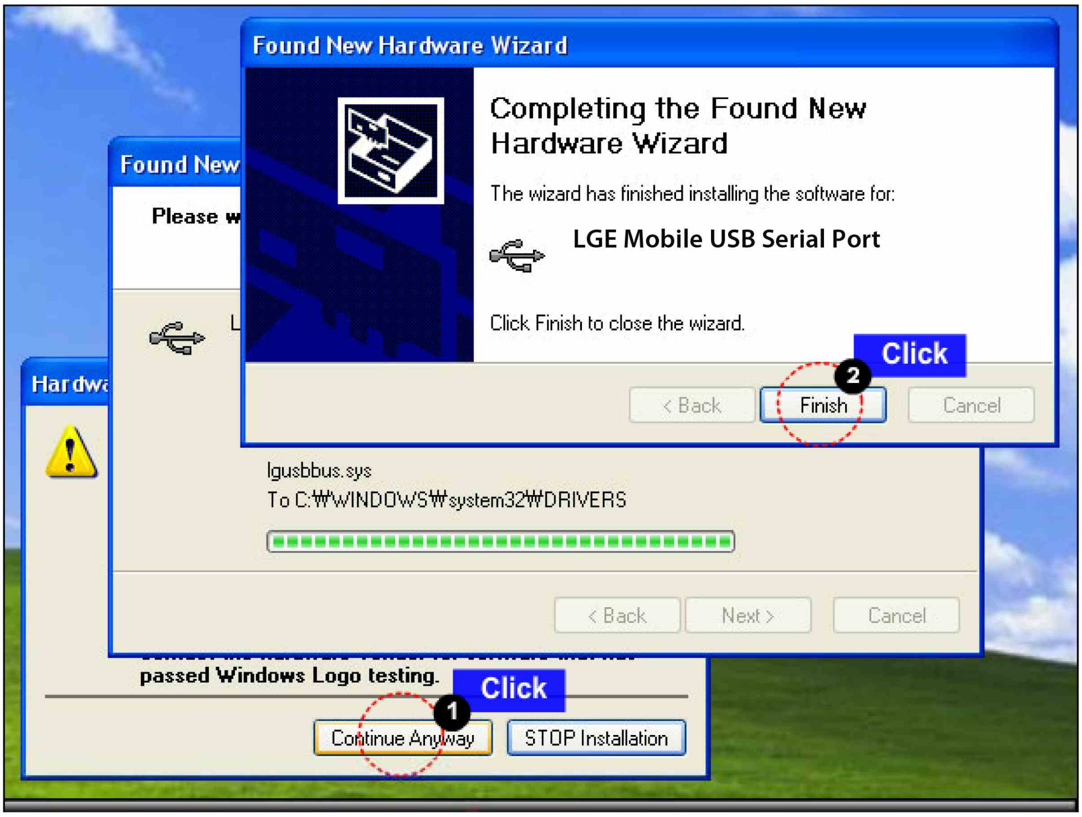

C:₩Documents and Settings₩Engineer₩Desktop₩B2B₩ LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.exe
LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_Ver_3.8.1.msi
Double Click (Install)
LG United Mobile Driver - InstallShield Wizard
Preparing Setup
Please wait while the InstallShield Wizard prepares the setup.
LG United Mobile Driver Setup is preparing the InstallShield Wizard, which will guide you through the rest of the setup process. Please wait.
InstallShield
Cancel


LG United Mobile Driver - InstallShield Wizard
LG United Mobile Driver Installed Successfully.
2 Click
OK
LG United Mobile Driver -
Setup Status
LG United Mobile Driver is configuring your new software installation.
Installing
C:\...\LG Electronics\LG United Mobile Driver\lganddiag.sys
InstallShield
Cancel
Welcome to the InstallShield Wizard for LG United Mobile Driver
The InstallShield Wizard will install LG United Mobile Driver on your computer. To continue, click Next.
1 Click
< Back
Next >
Cancel

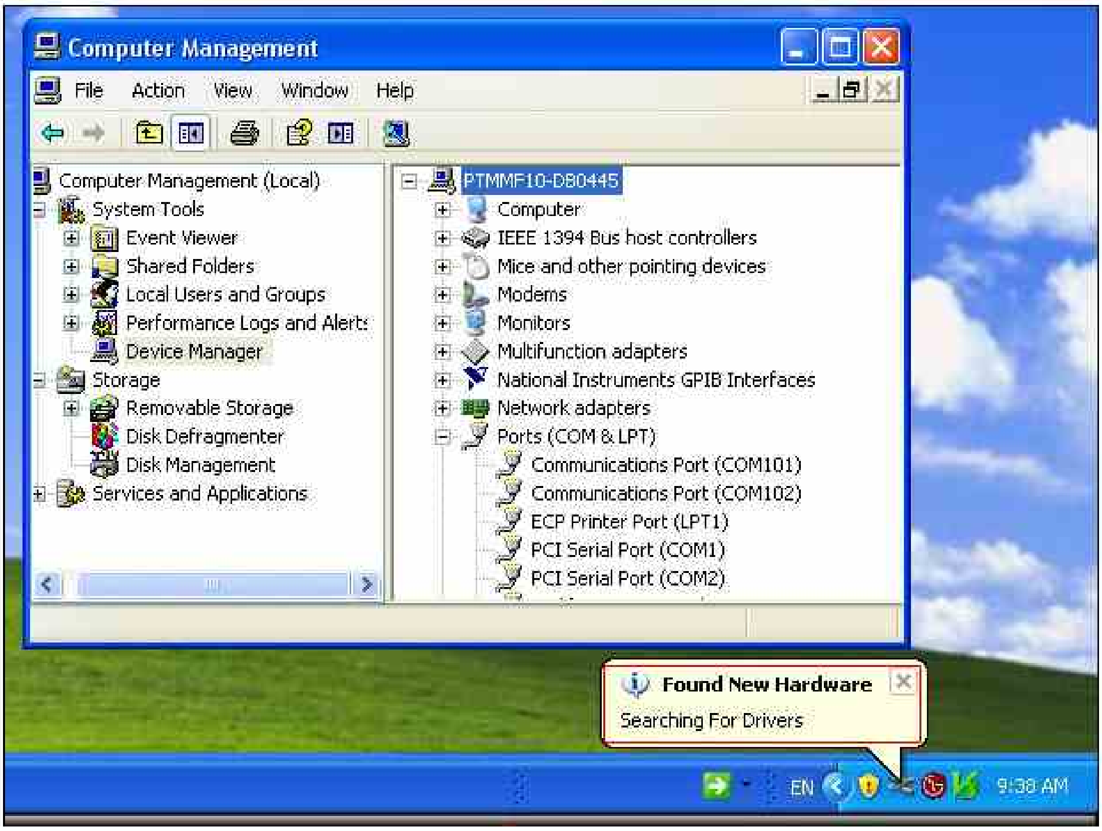

PIF-200 Setting
Power Cable – Power supply Connection method.
Connects with PC USB 2.0 Port.
It is recommended to connect phone power.
Improved power-off problem during the course of calibration.
PHONE Cable
(P/No : RAD32267819)
POWER SUPPLY BACK


Power Supply
Agilent 8960
PIF-200 Setting
RF cable : MS-156C
(P/No :BJAY0024004)
2
GPIB Cable
(P/No: BJAY0020058)
connects with PC
Connects with PC USB 2.0 Port
1
Single
Dual
PHONE Cable
3G QCT
UART 2.7
3
PIF200 Setting
"Agilent equipment and Phone" are connected in
RF cable : MS-156C Cable
USB Cable (A-B type)
P/No : BJAY0020060
PIF-200 (All Type)
(P/No : BJAY0024021)
5P E-SATA DC_Plug
No: RAD32167861
"RIF-200 and phone" are connected in
5P E-SATA DC_Plug

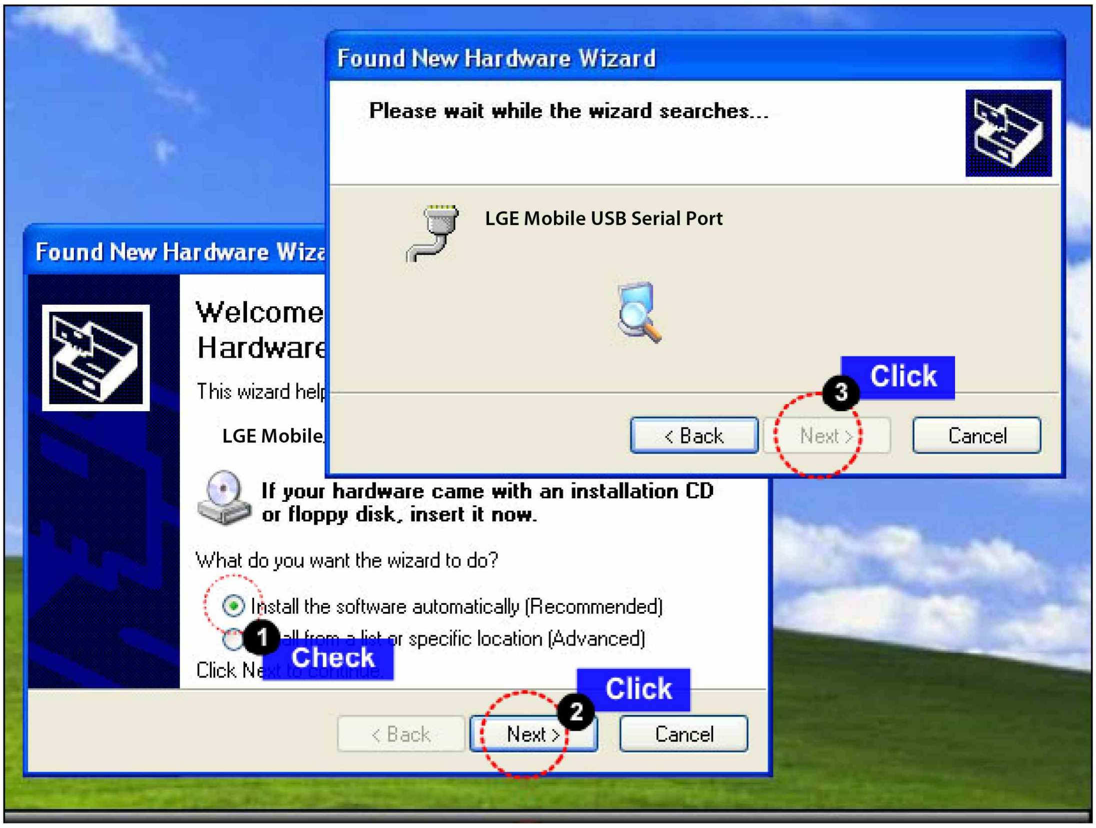

Found New Hardware Wizard
Please wait while the wizard searches...
LGE Mobile USB Serial Port
Click
4
< Back
Next >
Cancel
Found New Hardware W
Welcom
Hardwa
This wizard
LGE Mobile
What do yo
Install the software automatically (Recommended)
1
specific location (Advanced)
Check
Click
2
< Back
Next >
Cancel
This device can perform faster
This USB device can perform faster if you connect it to a
Hi-Speed USB 2.0 port.
For a list of available ports, click here.
EN
9:36 AM



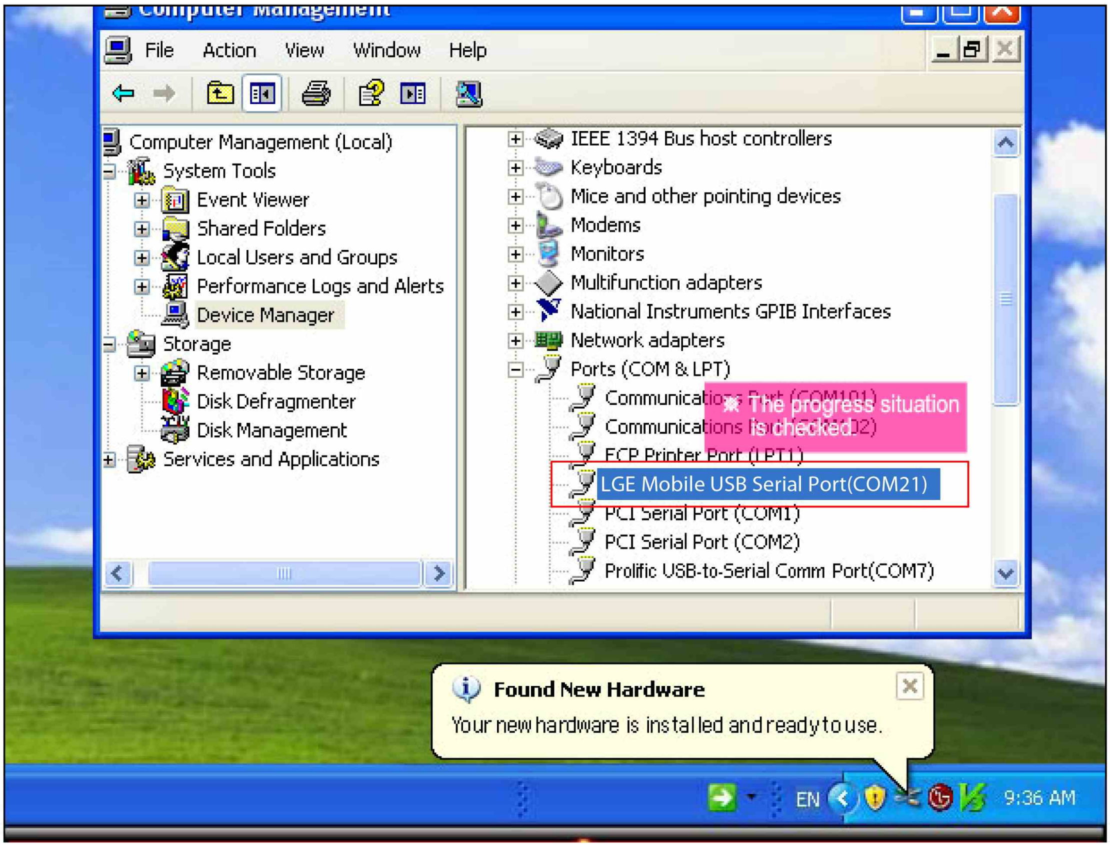
Found New Hardware Wizard
Completing the Found New Hardware Wizard
The wizard has finished installing the software for:
LGE Mobile USB Serial Port
Click Finish to close the wizard.
Click
< Back
Finish
Cancel
lgusbbus.sys
To C:₩WINDOWS₩system32₩DRIVERS
< Back
Next >
Cancel
passed Windows Logo testing.
Click
Continue Anyway
STOP Installation

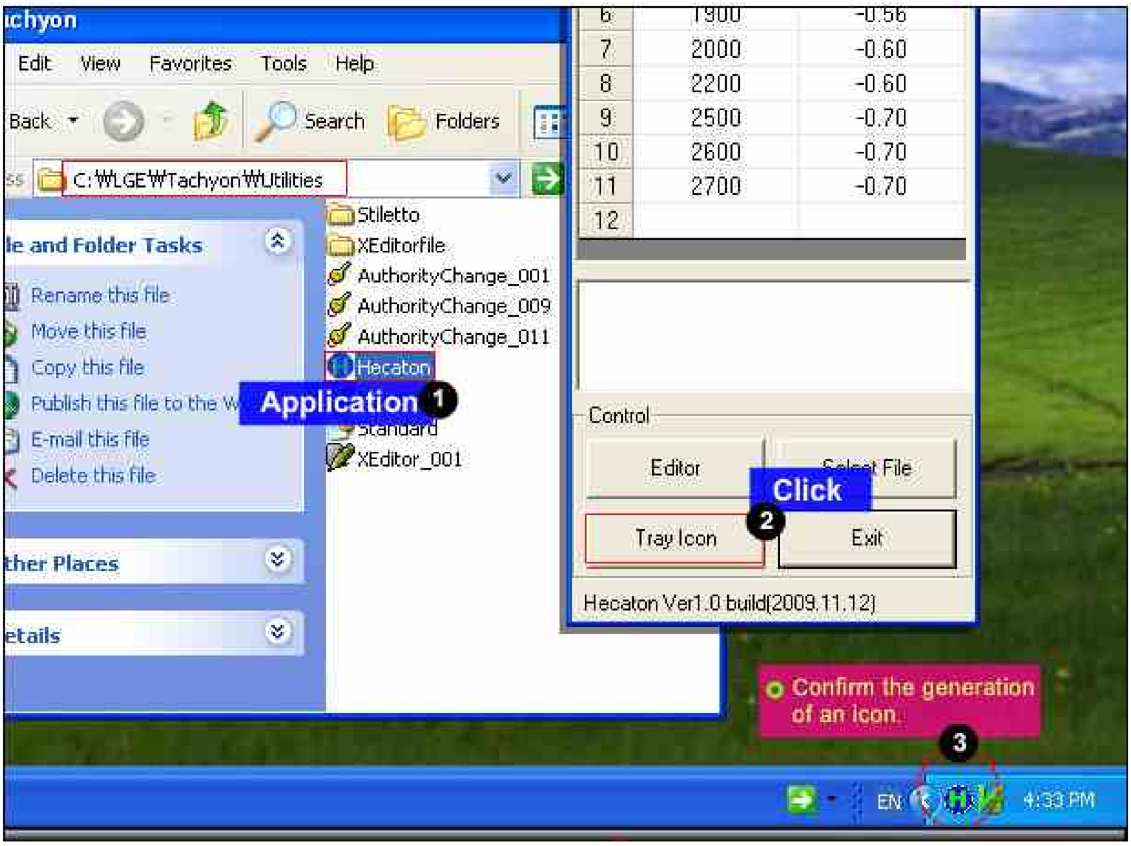


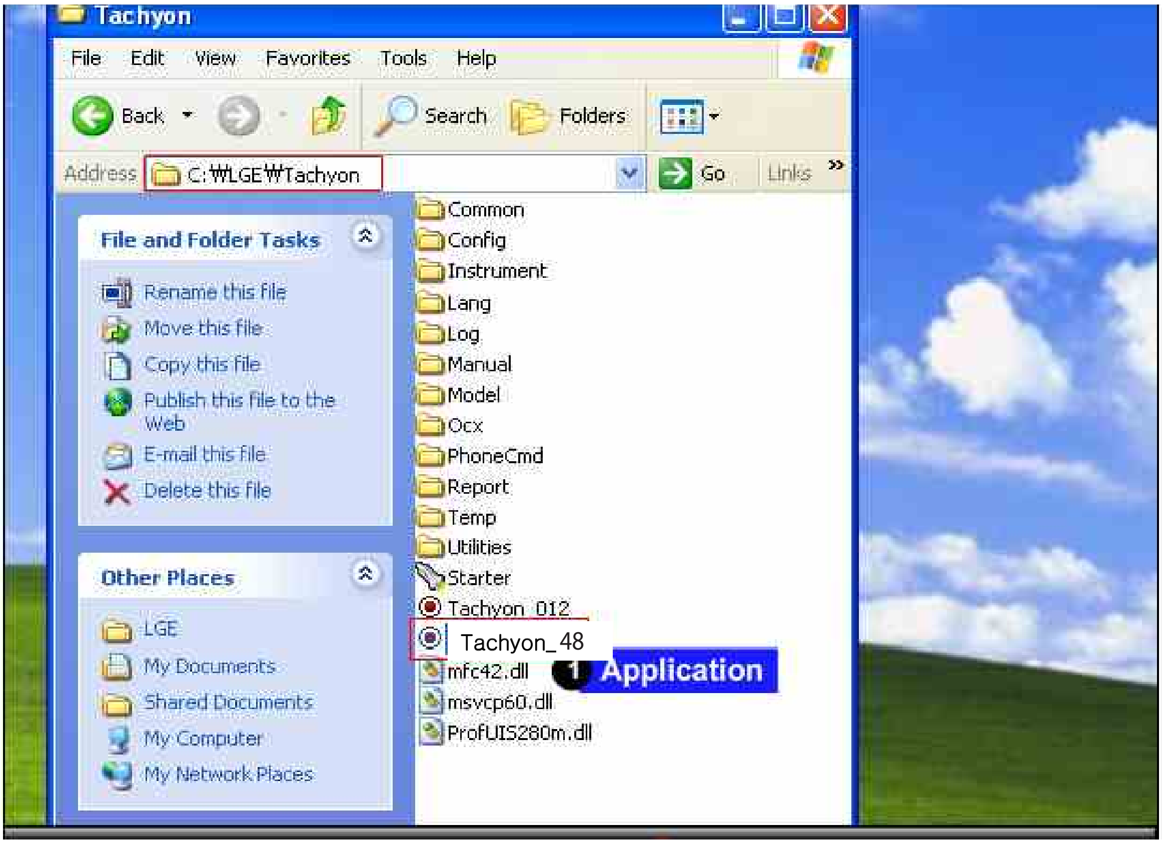
Computer Management
File
Action
View
Window
Help
Computer Management (Local)
System Tools
Event Viewer
Shared Folders
Local Users and Groups
Performance Logs and Alerts
Device Manager
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
PTMMF10-DB0445
Computer
IEEE 1394 Bus host controllers
Mice and other pointing devices
Modems
Monitors
Multifunction adapters
National Instruments GPIB Interfaces
Network adapters
Ports (COM & LPT)
Communications Port (COM101)
Communications Port (COM102)
ECP Printer Port (LPT1)
PCI Serial Port (COM1)
PCI Serial Port (COM2)
Found New Hardware
Searching For Drivers
EN
9:38 AM

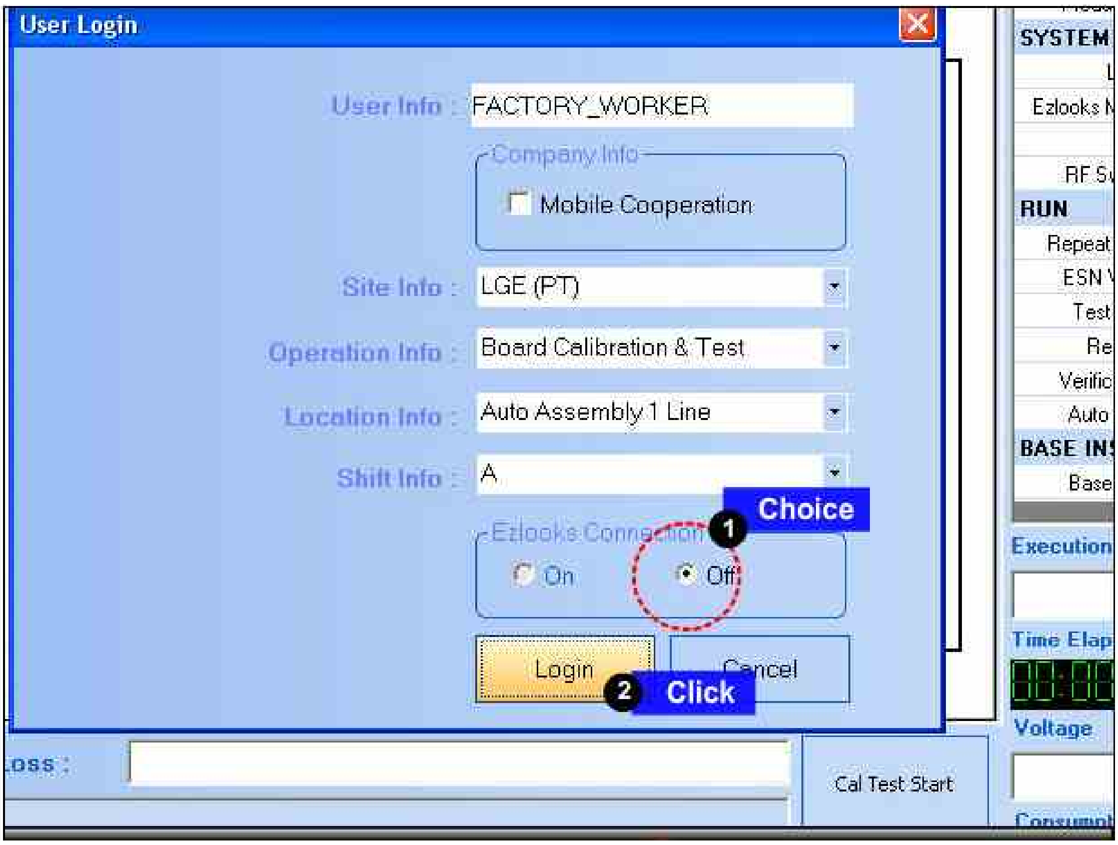

Found New Hardware Wizard
Please wait while the wizard searches...
LGE Mobile USB Serial Port
Click
3
< Back
Next >
Cancel
Found New Hardware Wiza
Welcome
Hardware
This wizard help
LGE Mobile
If your hardware came with an installation CD
or floppy disk, insert it now.
What do you want the wizard to do?
Install the software automatically (Recommended)
1
Check
all from a list or specific location (Advanced)
Click Ne
Click
2
< Back
Next >
Cancel

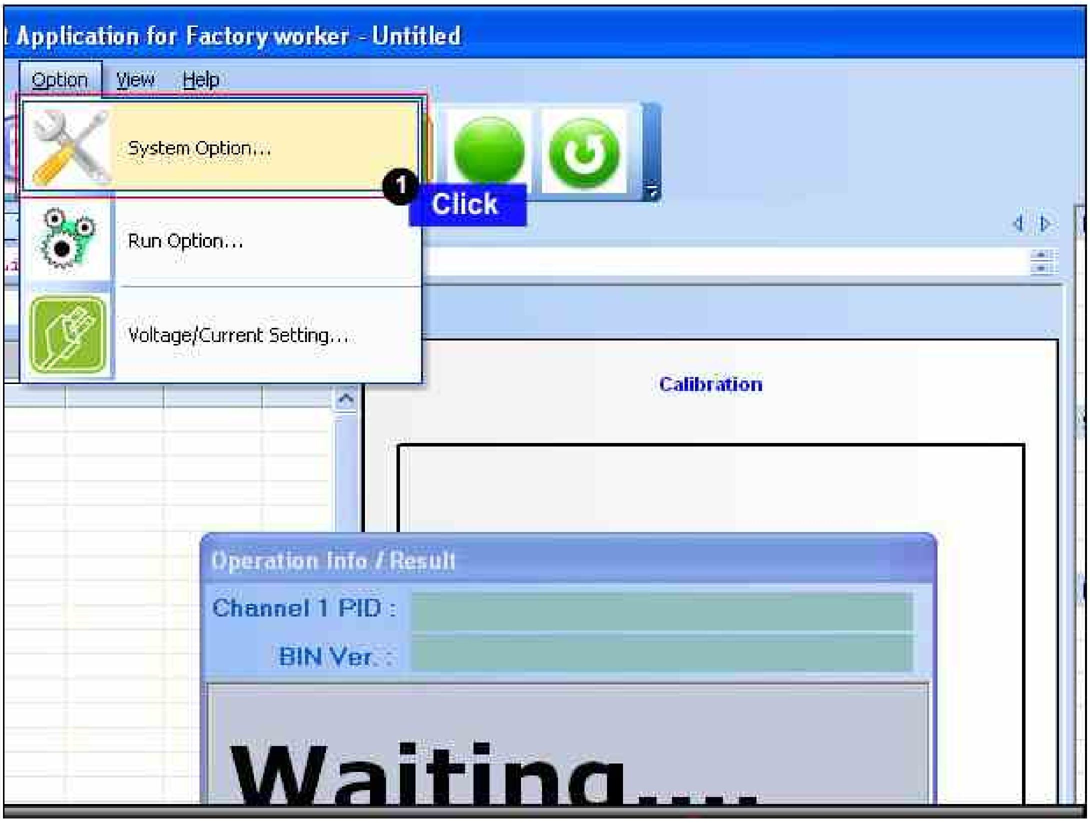

Found New Hardware Wizard
Completing the Found New
Hardware Wizard
The wizard has finished installing the software for:
LGE Mobile USB Serial Port
Click Finish to close the wizard.
Click
2
< Back
Finish
Cancel
Found New Har
Please wait w
LGE Mobile
lgusbdiag.sys
To C:₩WINDOWS₩system32₩DRIVERS
< Back
Next >
Cancel
Hardware Ins
The soft
LGE CDI
has not p
with Win
Continu
or dest
either i
recomm
contac
passed Windows Logo testing.
Click
1
Continue Anyway
STOP Installation

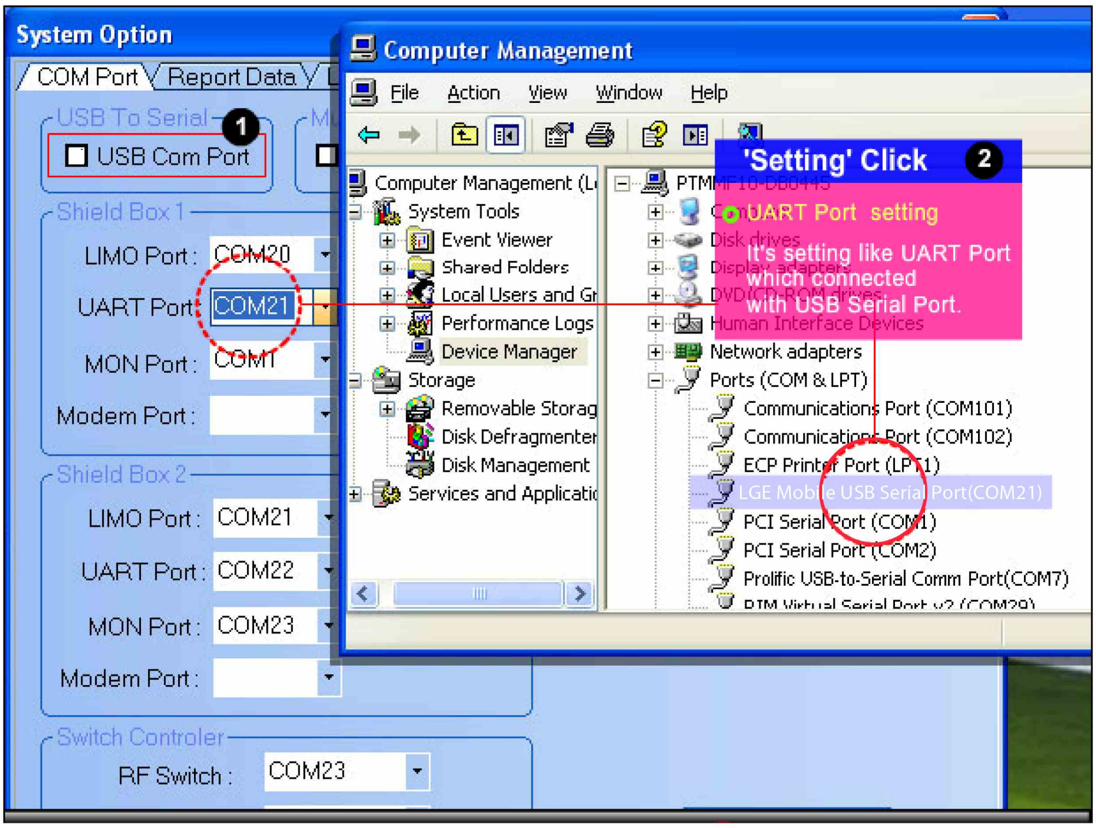

Computer Management
File Action View Window Help
Computer Management (Local)
System Tools
Event Viewer
Shared Folders
Local Users and Groups
Performance Logs and Alerts
Device Manager
Storage
Removable Storage
Disk Defragmenter
Disk Management
Services and Applications
IEEE 1394 Bus host controllers
Keyboards
Mice and other pointing devices
Modems
Monitors
Multifunction adapters
National Instruments GPIB Interfaces
Network adapters
Ports (COM & LPT)
※ The progress situation is checked
ECP Printer Port (LPT1)
LGE Mobile USB Serial Port(COM21)
PCI Serial Port (COM2)
Prolific USB-to-Serial Comm Port(COM7)
Found New Hardware
Your new hardware is installed and ready to use.
EN
9:36 AM

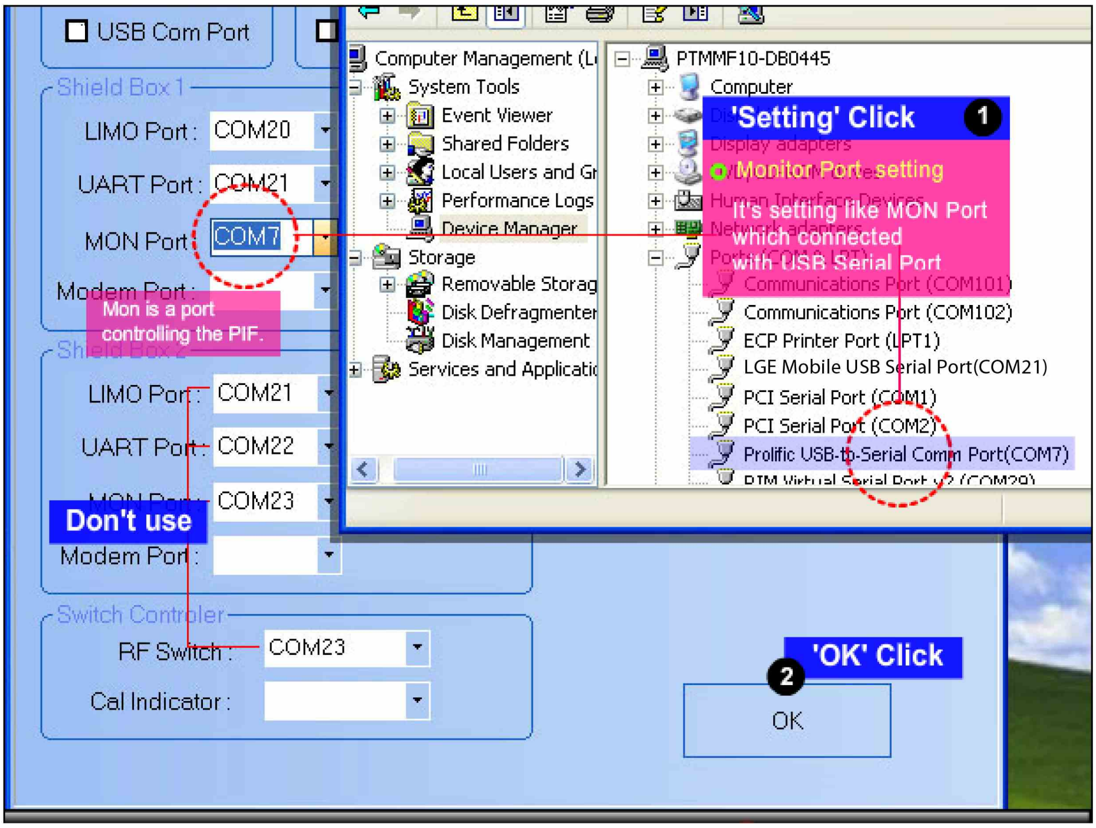

Edit View Favorites Tools Help
Back
Search
Folders
C:₩LGE₩Tachyon₩Utilities
Stiletto
XEditorfile
AuthorityChange_001
AuthorityChange_009
AuthorityChange_011
Hecaton
Application 1
XEditor_001
Rename this file
Move this file
Copy this file
E-mail this file
Delete this file
7 2000 -0.60
8 2200 -0.60
9 2500 -0.70
10 2600 -0.70
11 2700 -0.70
12
Control
Editor
Click
Tray Icon
2
Exit
Hecaton Ver1.0 build(2009.11.12)
Confirm the generation of an icon.
3
EN
4:33 PM

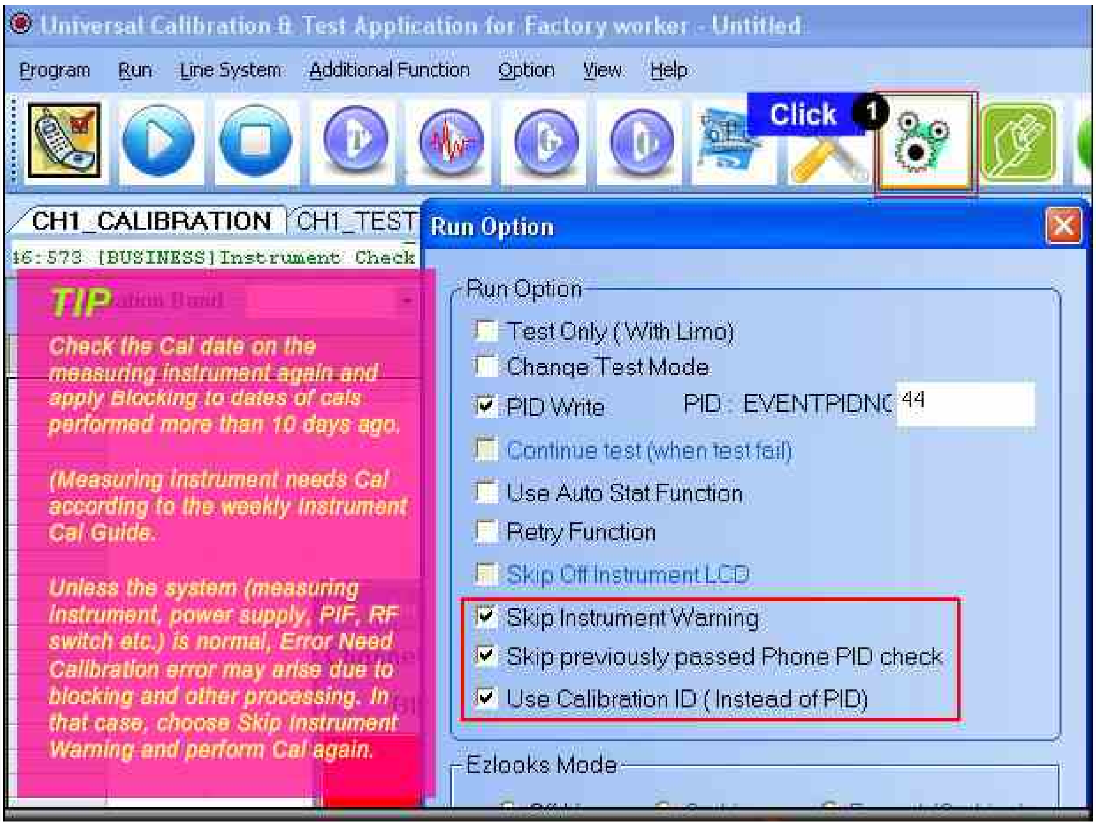

Tachyon
File Edit View Favorites Tools Help
Back Search Folders
Address C:₩LGE₩Tachyon
Go Links
File and Folder Tasks
Rename this file
Move this file
Copy this file
Publish this file to the Web
E-mail this file
Delete this file
Other Places
LGE
My Documents
Shared Documents
My Computer
My Network Places
Common
Config
Instrument
Lang
Log
Manual
Model
Ocx
PhoneCmd
Report
Temp
Utilities
Starter
Tachyon_012
Tachyon_48
mfc42.dll
msvcp60.dll
ProfUIS280m.dll
1 Application

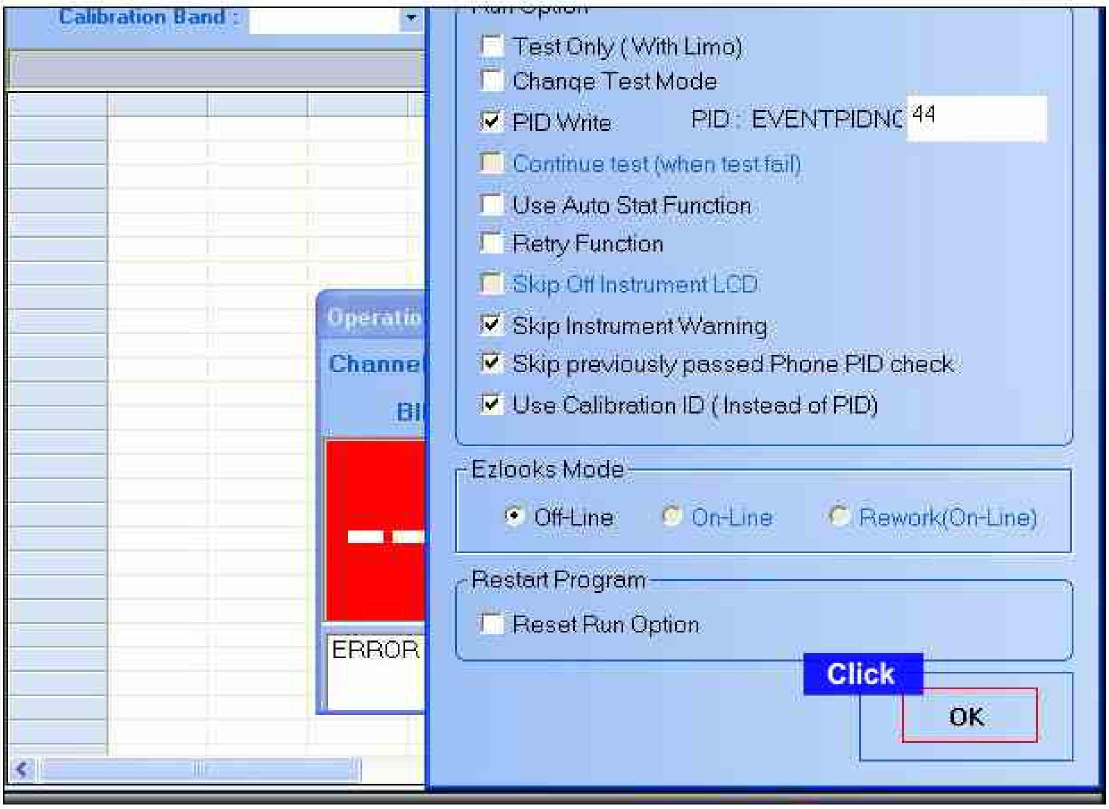

User Login
User Info : FACTORY_WORKER
Company Info
Mobile Cooperation
Site Info : LGE (PT)
Operation Info : Board Calibration & Test
Location Info : Auto Assembly 1 Line
Shift Info : A
Ezlooks Connection
On
Off
1 Choice
Login
Cancel
2 Click
Cal Test Start
SYSTEM
RUN
BASE INS
Execution
Voltage

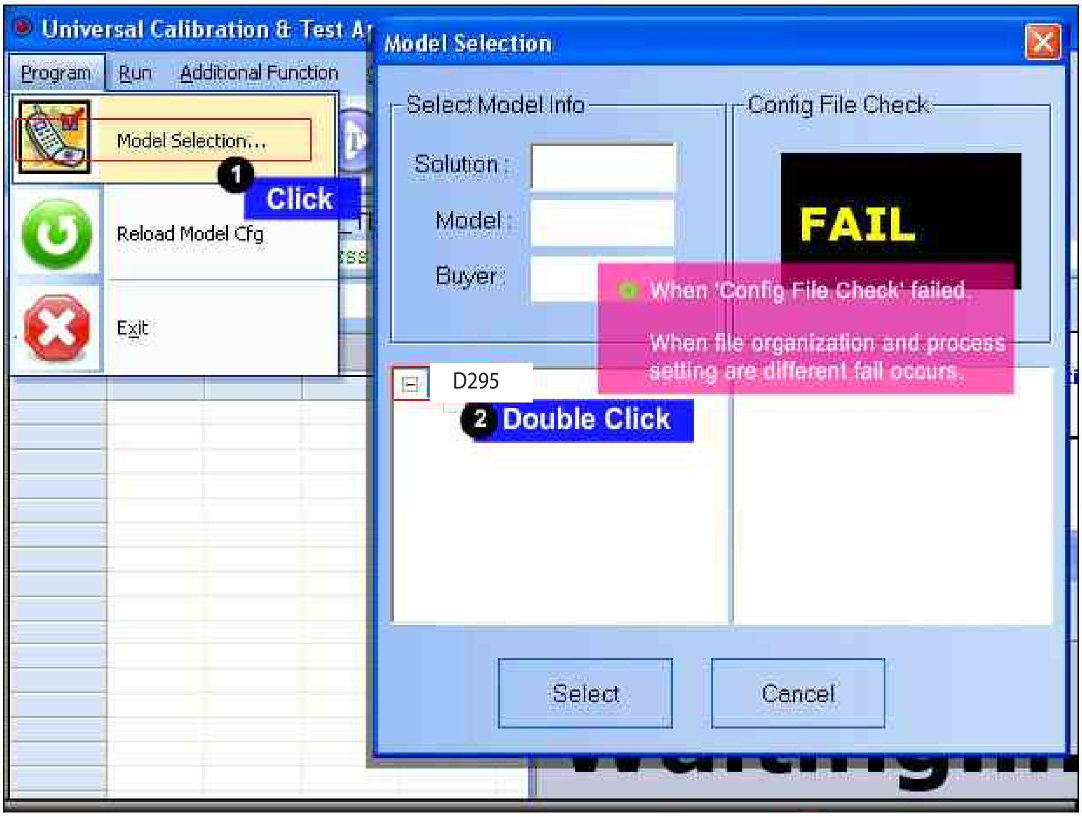

Application for Factory worker - Untitled
Option
View
Help
System Option...
Run Option...
Voltage/Current Setting...
1
Click
Calibration
Operation Info / Result
Channel 1 PID :
BIN Ver. :
Waiting....

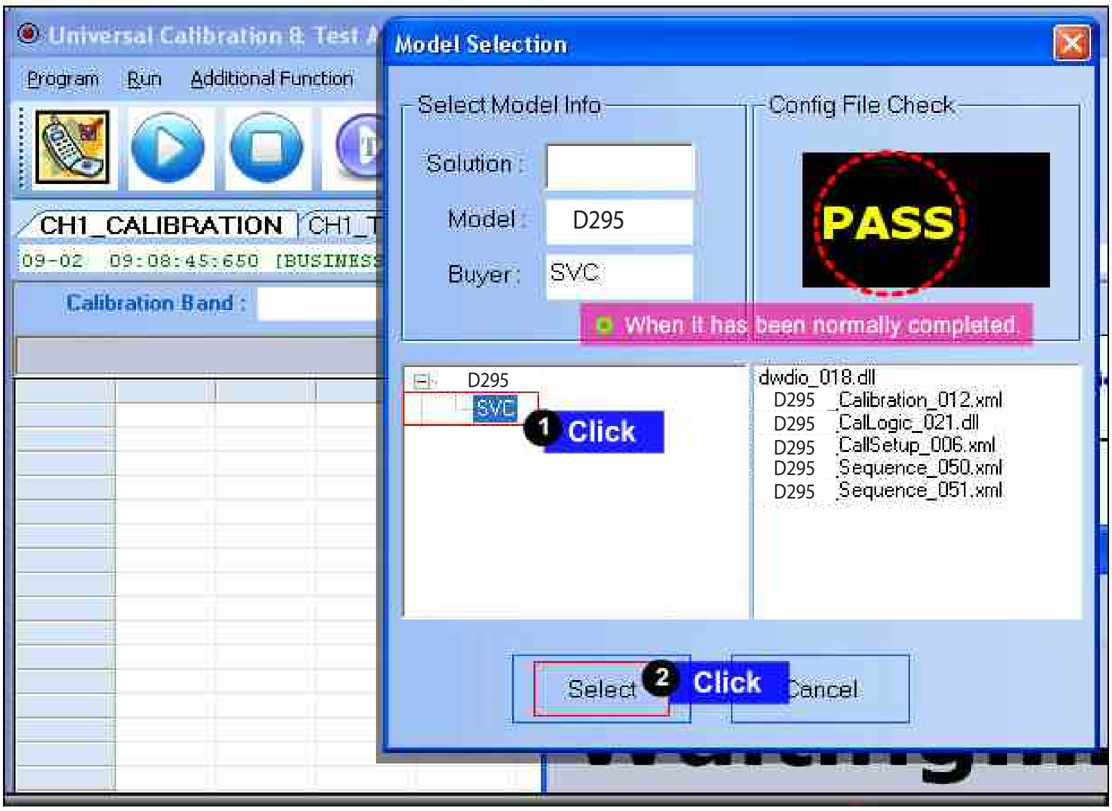

System Option
COM Port
Report Data
USB To Serial
1
USB Com Port
Shield Box 1
LIMO Port :
COM20
UART Port :
COM21
MON Port :
COM1
Modem Port :
Shield Box 2
LIMO Port :
COM21
UART Port :
COM22
MON Port :
COM23
Modem Port :
Switch Controler
RF Switch :
COM23
Computer Management
File
Action
View
Window
Help
'Setting' Click
2
UART Port setting
It's setting like UART Port which connected with USB Serial Port.
System Tools
Event Viewer
Shared Folders
Performance Logs
Device Manager
Storage
Disk Management
Network adapters
Ports (COM & LPT)
Communications Port (COM101)
Communications Port (COM102)
ECP Printer Port (LPT1)
LGE Mobile USB Serial Port(COM21)
PCI Serial Port (COM1)
PCI Serial Port (COM2)
Prolific USB-to-Serial Comm Port(COM7)

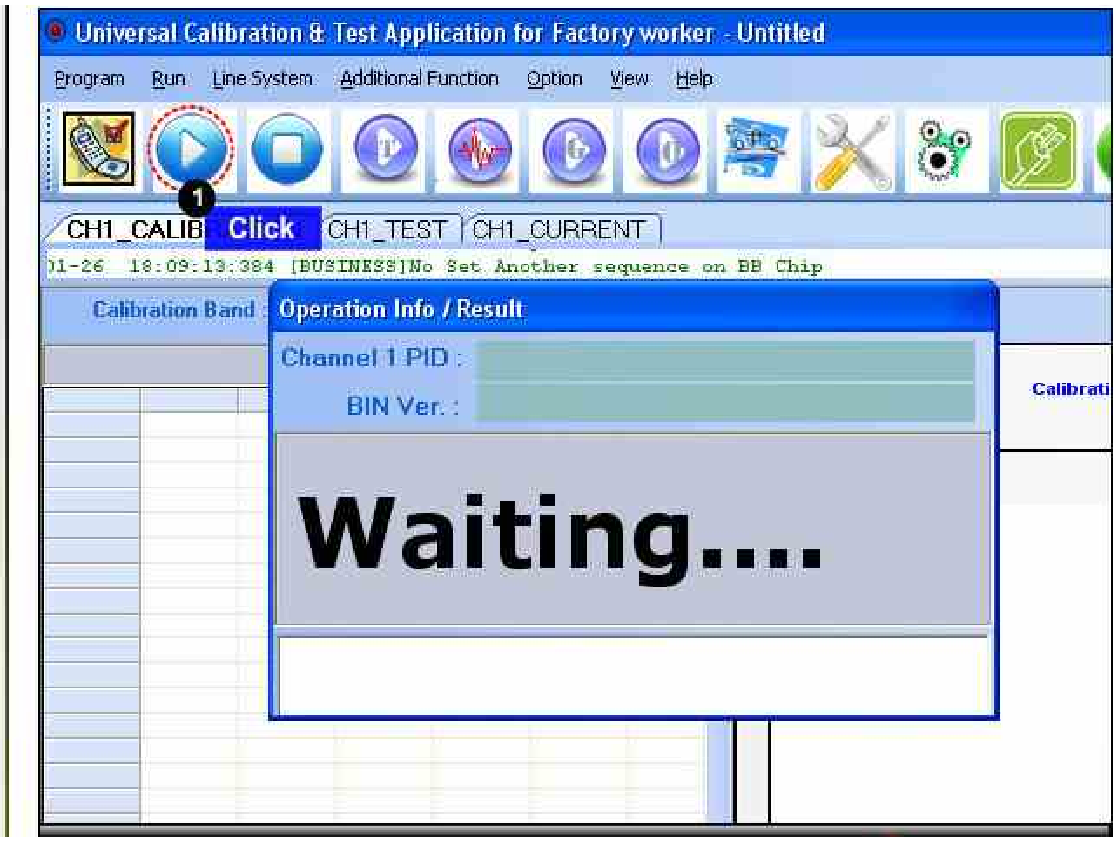

USB Com Port
Shield Box 1
LIMO Port : COM20
UART Port : COM21
MON Port : COM7
Modem Port :
Mon is a port controlling the PIF.
Shield Box 2
LIMO Port : COM21
UART Port : COM22
COM23
Don't use
Modem Port :
Switch Controller
RF Switch : COM23
Cal Indicator :
Computer Management
System Tools
Event Viewer
Shared Folders
Local Users and Gr
Performance Logs
Device Manager
Storage
Removable Storag
Disk Defragmenter
Disk Management
Services and Applicati
PTMMF10-DB0445
Computer
'Setting' Click 1
Display adapters
Monitor Port setting
It's setting like MON Port which connected with USB Serial Port
Communications Port (COM101)
Communications Port (COM102)
ECP Printer Port (LPT1)
LGE Mobile USB Serial Port(COM21)
PCI Serial Port (COM1)
PCI Serial Port (COM2)
Prolific USB-to-Serial Comm Port(COM7)
'OK' Click
2
OK

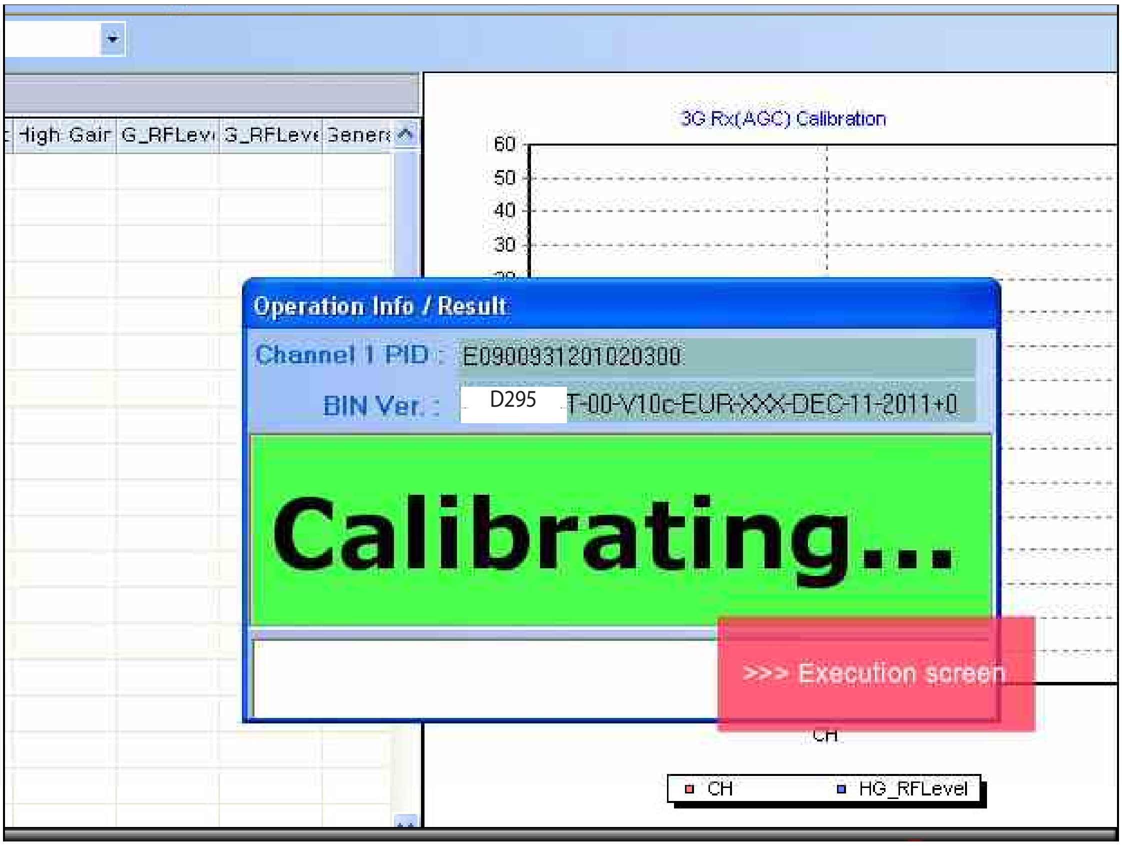

Universal Calibration & Test Application for Factory worker - Untitled
Program Run Line System Additional Function Option View Help
Click 1
CH1_CALIBRATION CH1_TEST
TIP
Check the Cal date on the measuring instrument again and apply Blocking to dates of cals performed more than 10 days ago.
(Measuring instrument needs Cal according to the weekly instrument Cal Guide.
Unless the system (measuring instrument, power supply, PIF, RF switch etc.) is normal, Error Need Calibration error may arise due to blocking and other processing. In that case, choose Skip Instrument Warning and perform Cal again.
Run Option
Run Option
Test Only (With Limo)
Change Test Mode
PID Write
PID : EVENTPIDNC 44
Continue test (when test fail)
Use Auto Stat Function
Retry Function
Skip Off Instrument LCD
Skip Instrument Warning
Skip previously passed Phone PID check
Use Calibration ID (Instead of PID)
Ezlooks Mode

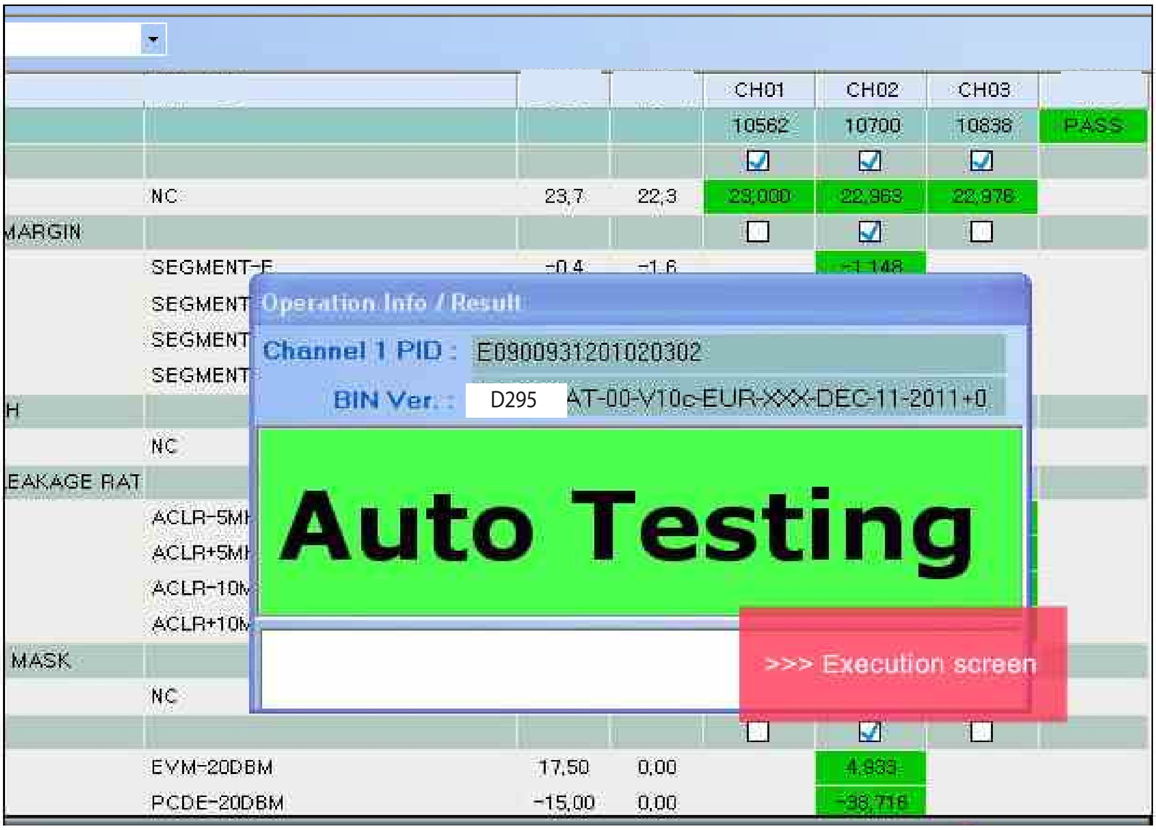

Calibration Band :
Test Only (With Limo)
Change Test Mode
PID Write
PID : EVENTPIDNC 44
Continue test (when test fail)
Use Auto Stat Function
Retry Function
Skip Off Instrument LCD
Skip Instrument Warning
Skip previously passed Phone PID check
Use Calibration ID (Instead of PID)
Ezlooks Mode
Off-Line
On-Line
Rework(On-Line)
Restart Program
Reset Run Option
ERROR
Click
OK

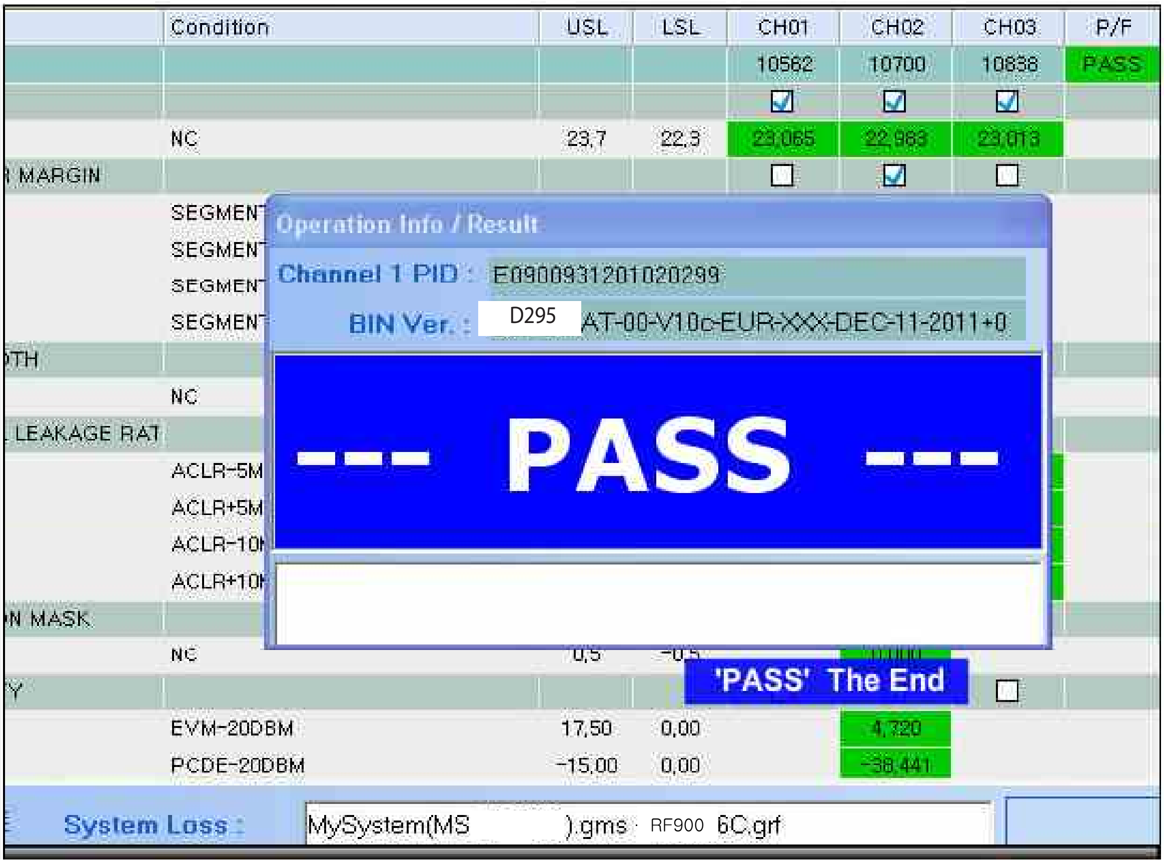

Universal Calibration & Test A
Program
Run
Additional Function
Model Selection...
1
Click
Reload Model Cfg
Exit
Model Selection
Select Model Info
Solution :
Model :
Buyer :
Config File Check
FAIL
When 'Config File Check' failed.
When file organization and process setting are different fail occurs.
D295
2 Double Click
Select
Cancel


Universal Calibration & Test A
Program Run Additional Function
CH1_CALIBRATION CH1_T
09-02 09:08:45:650 [BUSINESS
Calibration Band :
Model Selection
Select Model Info
Solution :
Model : D295
Buyer : SVC
Config File Check
PASS
When it has been normally completed.
D295
SVC
1 Click
dwdio_018.dll
D295 _Calibration_012.xml
D295 _CallLogic_021.dll
D295 _CallSetup_006.xml
D295 _Sequence_050.xml
D295 _Sequence_051.xml
Select
2 Click
Cancel


Universal Calibration & Test Application for Factory worker - Untitled
Program Run Line System Additional Function Option View Help
1 Click
CH1_CALIB CH1_TEST CH1_CURRENT
01-26 18:09:13:384 [BUSINESS]No Set Another sequence on BB Chip
Calibration Band
Operation Info / Result
Channel 1 PID :
BIN Ver. :
Waiting....

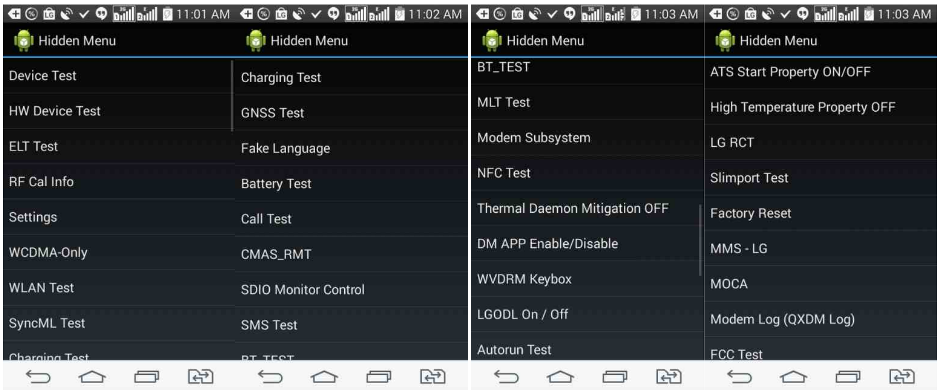

3G Rx(AGC) Calibration
Operation Info / Result
Channel 1 PID : E09009312010203000
BIN Ver. : D295 T-00-V10c-EUR-XXX-DEC-11-2011+0
Calibrating...
>>> Execution screen
CH
HG_RFLevel


CH01
CH02
CH03
10562
10700
10838
PASS
Operation Info / Result
Channel 1 PID : E09009312010203002
BIN Ver. : D295 AT-00-V10c-EUR-XXX-DEC-11-2011+0
Auto Testing
>>> Execution screen
EVM-20DBM
PCDE-20DBM



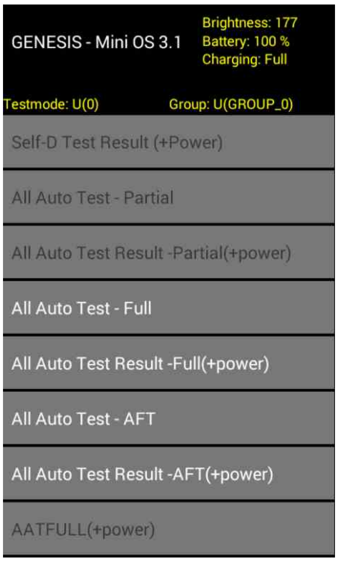
Condition
USL
LSL
CH01
CH02
CH03
P/F
10562
10700
10838
PASS
NC
23.7
22.3
23.065
22.963
23.013
MARGIN
SEGMENT
TH
LEAKAGE RAT
ACLR-5M
ACLR+5M
ACLR-10M
ACLR+10M
N MASK
NC
Y
EVM-20DBM
17.50
0.00
4.720
PCDE-20DBM
-15.00
0.00
-38.441
Operation Info / Result
Channel 1 PID : E0900931201020299
BIN Ver. : D295 AT-00-V10c-EUR-XXX-DEC-11-2011+0
--- PASS ---
'PASS' The End
System Loss : MySystem(MS).gms RF900 6C.grf

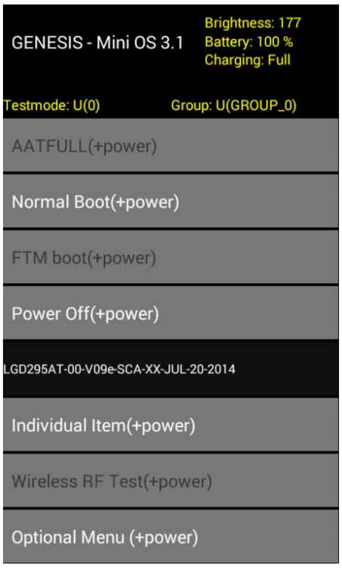
10. HIDDEN MENU



Hidden Menu Start

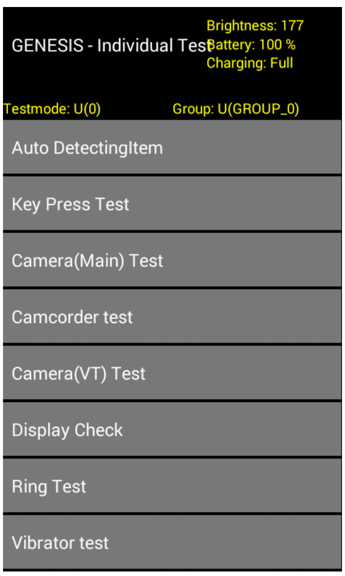
Start shortcut keys:3845#*295#

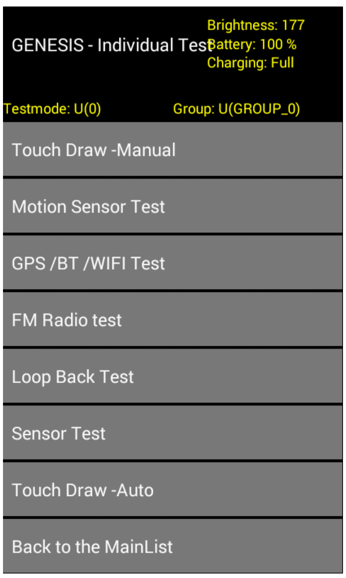




All Auto Test

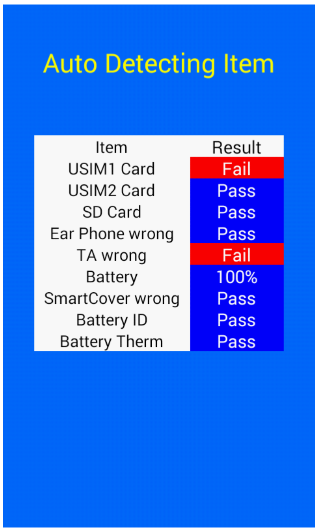
0. Self-D Test Result (+Power)
1. All Auto Test – Partial
2. All Auto Test Result– Partial(+power)
3. All Auto Test – Full
4. All Auto Test Result– Full(+power)
5. All Auto Test – AFT
6. All Auto Test Result– AFT(+power)
7. AATFULL (+power)
8. Normal Boot (+power key)
9. FTM Boot (+power key)
10. Power Off (+power key)
11. Version Name
12. Individual Item(+power)
13. Wireless RF Test(+power)
14. Optional Menu(+power)





Individual Item List:

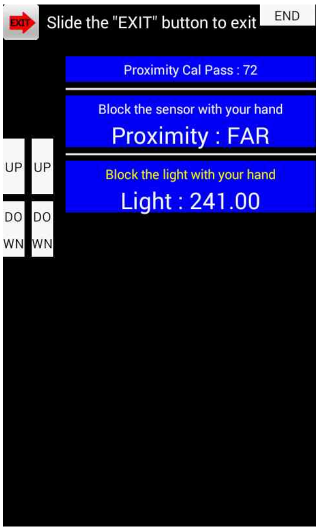
1. Auto DetectingItem
2. Key Press Test
3. Camera(Main) test
4. Camcorder Test
5. Camera(VT) test
6. LED / Display check Test
7. Ring Test
8. Vibrator Test
9. Touch Draw-Manual
10. Motion Sensor Test
11. GPS BT WIFI Test
12. NFC Test (not supported in D295)
13. FM Radio Test
14. Loopback Test
15. Sensor Test
16. Touch Draw-Auto



Auto Detecting Test

: USIM, SD card, Ear jack, TA, Battery

- Check the SIM card
- Check the SD card
- Check Ear phone(jack)
- Check TA
- Check Battery
  - Ejected TA is normal
  - Ejected ear jack is normal



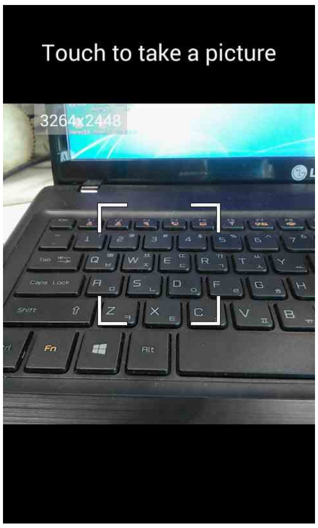
Key/Compass Test

Up/Down key : Check Recognition

Proximity Sensor : Phone contact with your fingers in the top of the sensor determine the sensor response

Far

Near



Camera Test 1

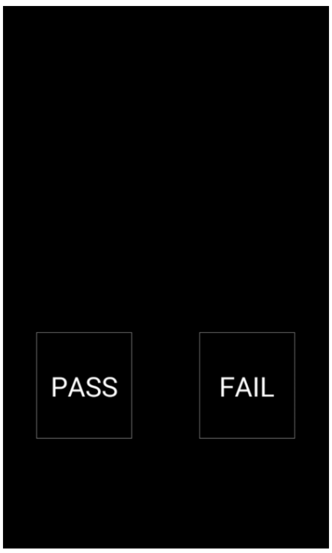
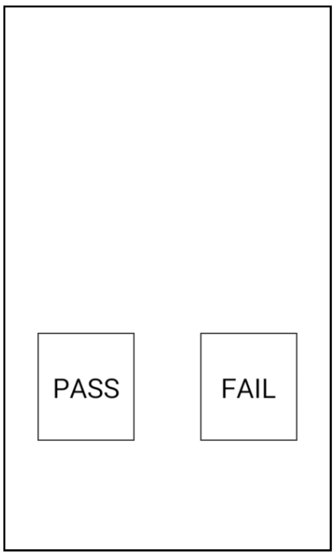
Main Camera

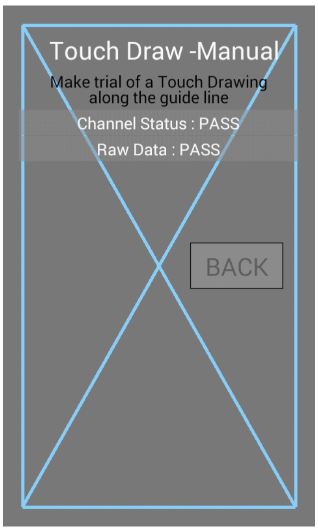
<test step>

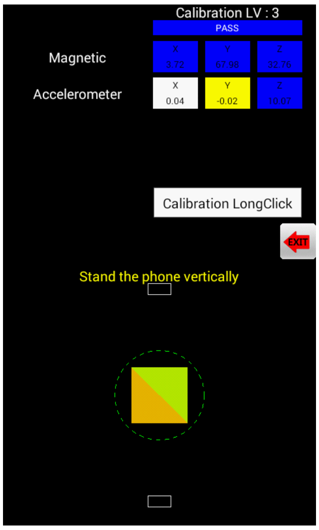
Touch to take a picture

Taking a picture

Verity the picture

	**Camera Test 2** Camcorder <test step> Taking a video Verity the video
	**Camera Test 1** VT Camera Preview
 	**LED / Display check Test** : LCD Black & White test (MURA 2.0 mm) • Check the Black display • Check the White display • Check MURA 2.0 mm

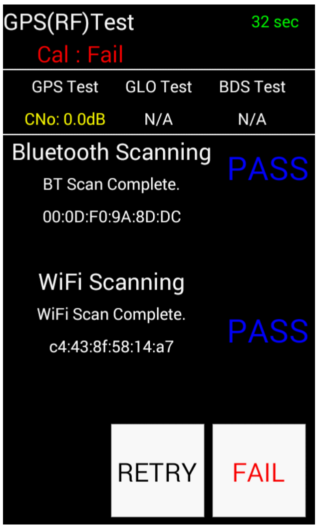
	Ring Test : Check sound • Check the sound
	**Vibrator test** A case-by-state vibration tests
	**Touch window test** Check the channel status & raw data And do touch drawing along the guide line



Accelerometer Test

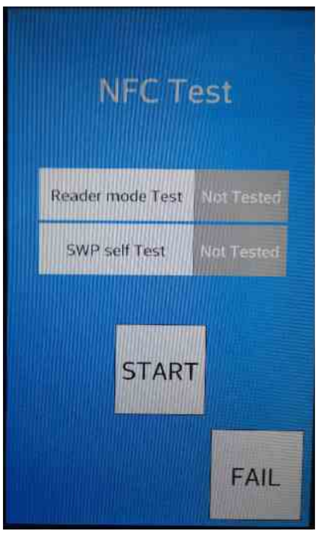
Magnetic sensor test, Accelerometer sensor test

- Check the magnetic sensor
- Check the accelerometer sensor



GPS BT WI-FI Test

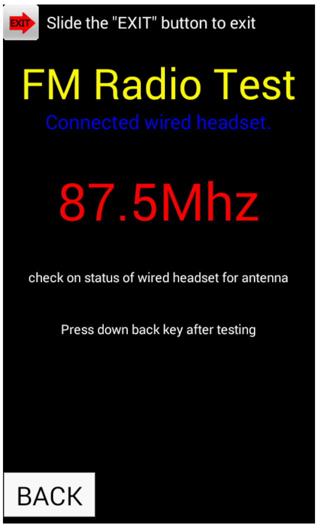
- Check the GPS scanning
- Check the WIFI scanning
- Check the BT scanning



NFC Test

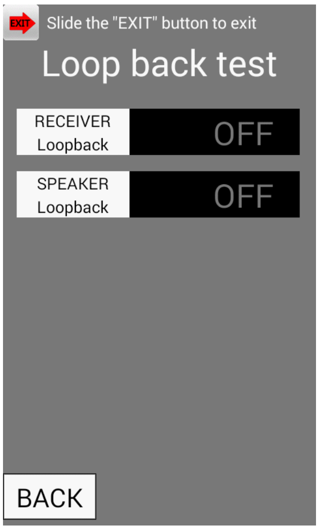
: Reader mode Test, SWP self Test

- Check the Reader with NFC tag
- Check the SWP (NFC Sim)



FM Radio Test

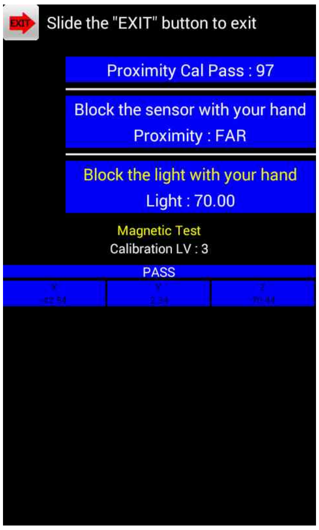
• Check the FM Radio



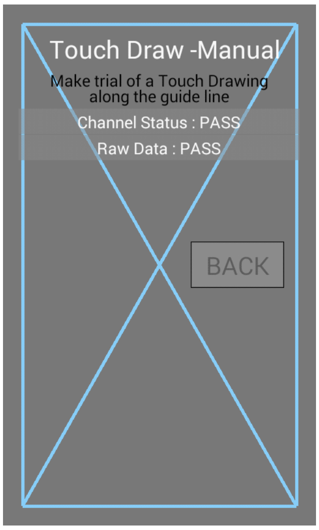
Audio Loopback Test

• Check the Receiver loopback

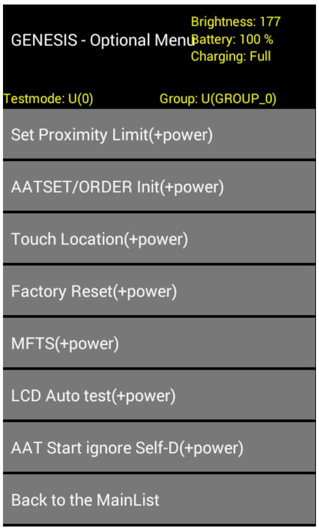
• Check the speaker loopback



Sensor Test

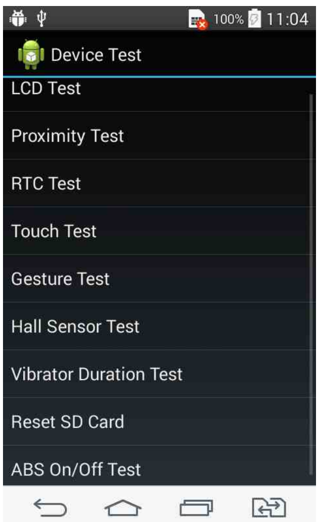


Touch Draw -Auto



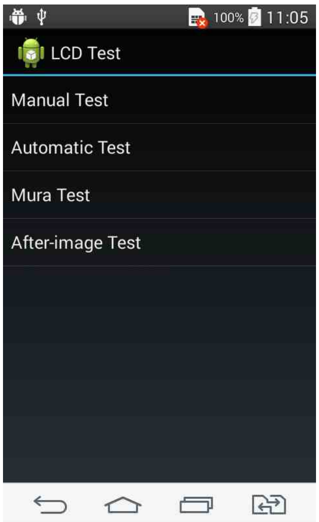
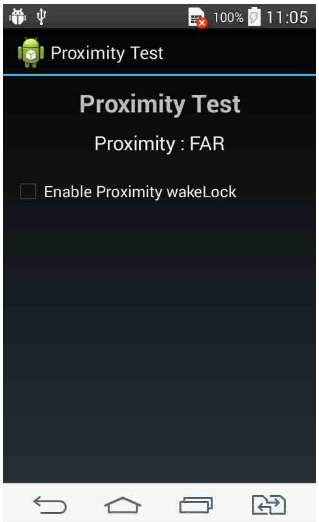
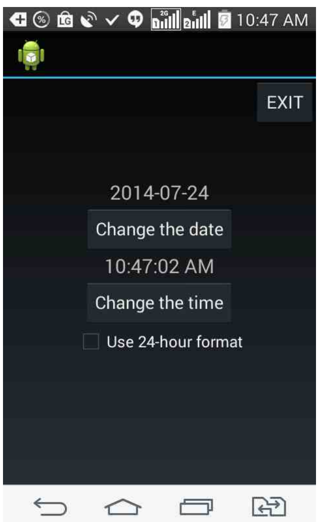
Optional Menu List

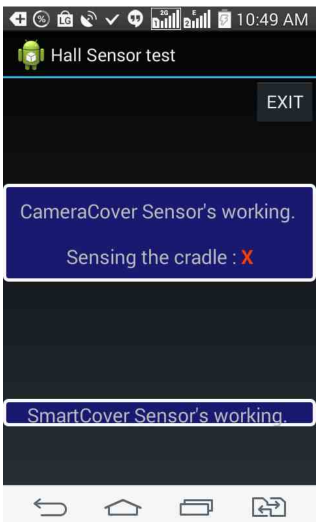
1. Set Proximity Limit(+power)
2. AATSET/ORDER Init (+power)
3. Touch Location (+power)
4. Factory Reset (+power)
5. MFTS (+power)
6. LCD Auto test (+power)
7. AAT Start ignore Self-D (+power)



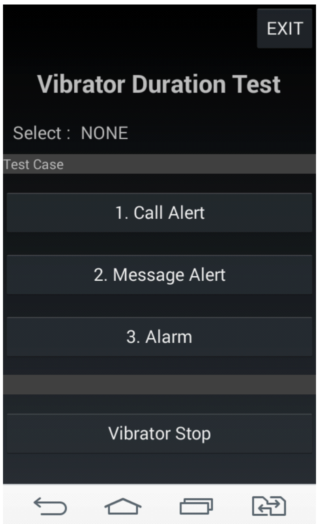
HW Device Test

1. LCD
2. Proximity
3. RTC
4. Touch
5. Gesture
6. Hall Sensor
7. Vibrator Duration
8. Reset SD Card
9. ABS On/Off Test

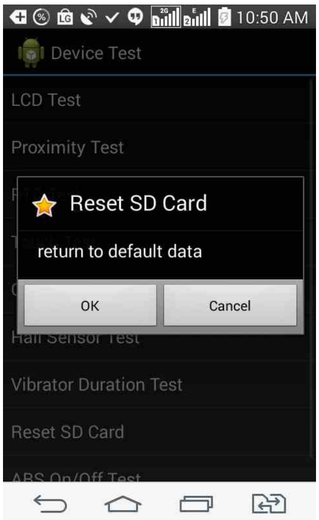
<table>
<tr>
<td>
</td>
<td>LCD list
- Manual Test
- Automatic Test
- Mura Test
- After-image Test</td>
</tr>
<tr>
<td>
</td>
<td>Proximity</td>
</tr>
<tr>
<td>
</td>
<td>RTC</td>
</tr>
</table>

	**Touch list** - Multi Touch Test - Finger Paint - Dot Finger Paint - Touch Sensing Test 5*9 - key Touch Sensing Test 5*7 - Grid Touch Test 5*7 - Touch Key Test - Grid Touch Color change - Block Touch Test - Touch Firmware Version - Touch Firmware Upgrade - Check Touch Panel
	**Gesture**
	**Hall Sensor**

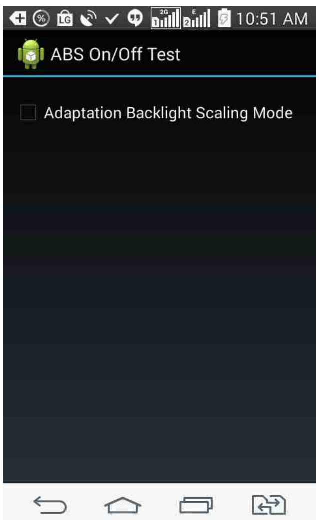
Vibrator Duration

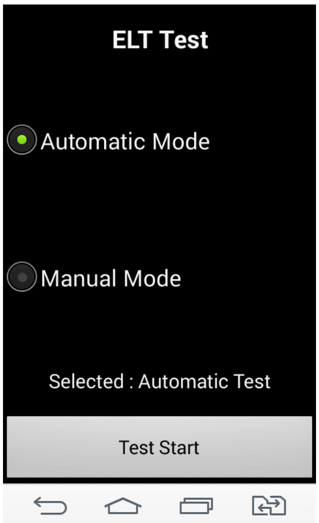
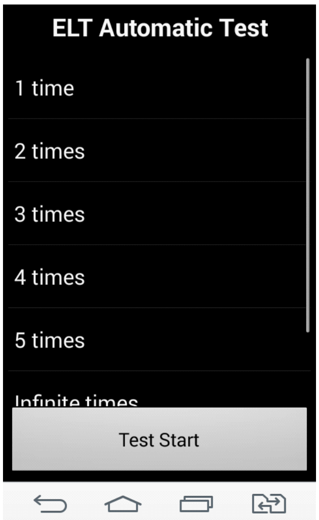
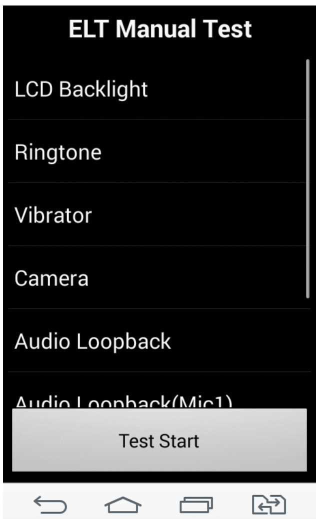


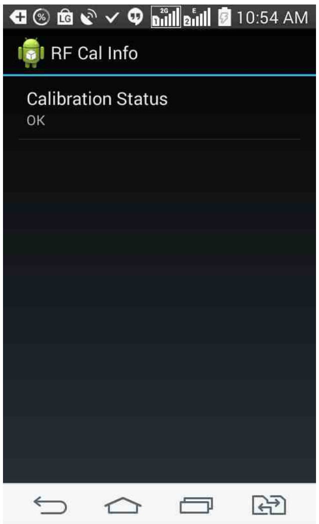
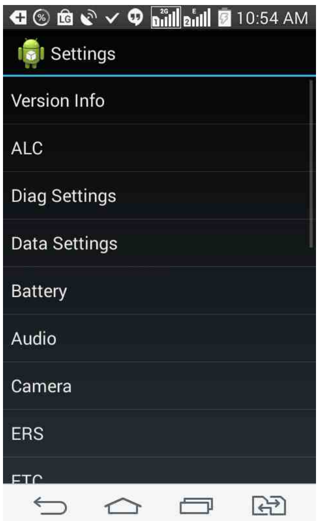
Reset SD Card

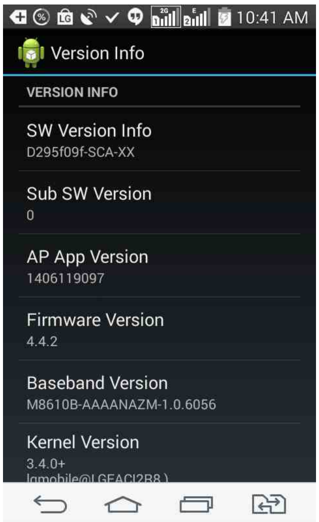
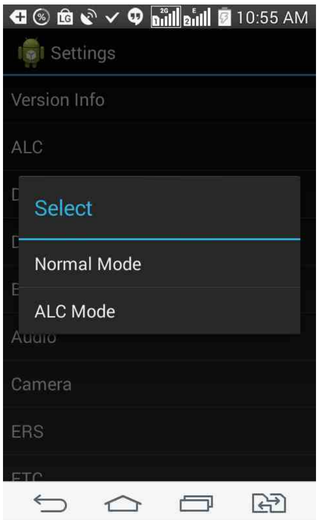
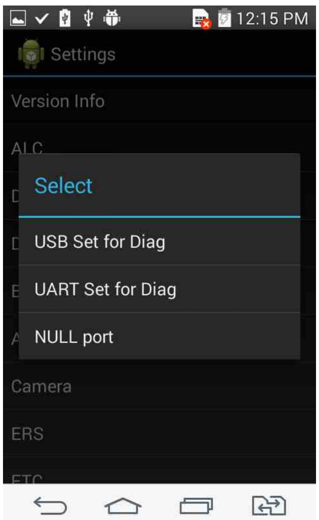


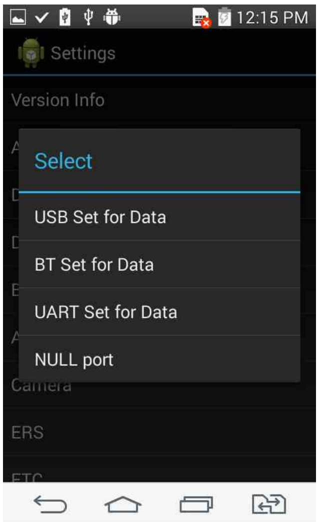
ABS On/Off Test



	**ELT Test** Automatic Mode : Test Automatically Manual Mode : Test selectivity
	**ELT Test** Automatic Mode : LCD Automatic on/off test -> time setting
	**ELT Manual Test** -LCD Backlight -Ringtone -Vibrator -Camera -Audio Loopback -Audio Loopback(Mic1) -Audio Loopback(Mic2) -Audio Loopback(Mic3) -> test on the device is working (The ability to use plant)

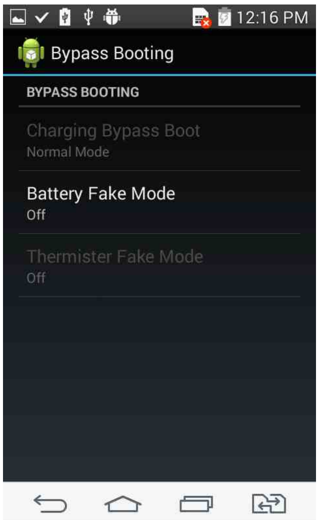
<table>
<tr>
<td>
</td>
<td>RF Cal Info</td>
</tr>
<tr>
<td>
</td>
<td>Settings
-Version Info
-ALC
-Diag Settings
-Data Settings
-Battery
-Audio
-Camera
-ERS (not implemented)
-ETC
-LCD Always On
-Auto Answer
-Upsell URL
-ATS Start Property On
-Update Touch Firmware (not implemented)
-APN Delete (not implemented)
-FastDormancy Mode
-Default IMS PDN Setting (not implemented)
-KnockOn On/Off Setting
-DataRecovery Mode
-CPJ Setting On (not implemented)</td>
</tr>
</table>

	**Version Info** (Classified Information representation) -SW Version Info -Sub SW Version -AP App Version -Firmware Version -Baseband Version -Kernel Version -HW Version -Factory Version -SW Original Version -Speed Bin -AVS Bin -NT code -SWFV -S/N
	**ALC** Normal Mode : Change to Normal Mode ALC Mode : Change to ALC Mode
	**Diag Settings** - USB Set for Diag - UART Set for Diag - Null port



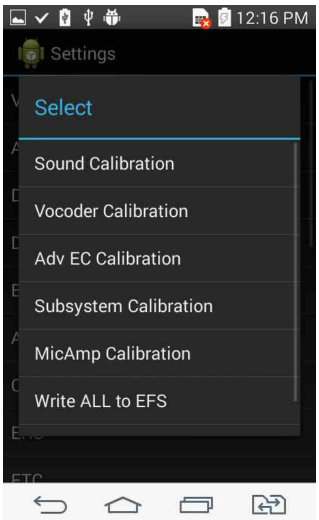
Data Settings

- USB Set for Data
- BT Set for Data
- UART Set for Data
- Null port



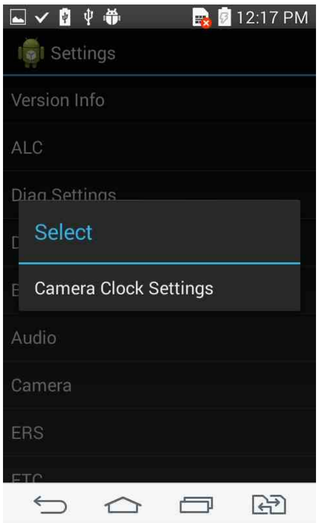
Battery

-Battery Fake Mode(On/Off)



Audio

- Sound Calibration
- Vocoder Calibration
- Adv EC Calibration
- Subsystem Calibration
- MicAmp Calibration
- Write All to EFS
- Write All to MEM

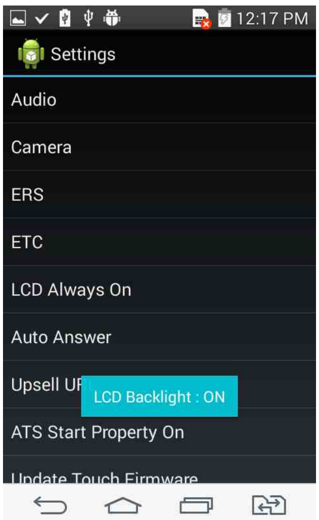


Camera

- Camera Clock Settings

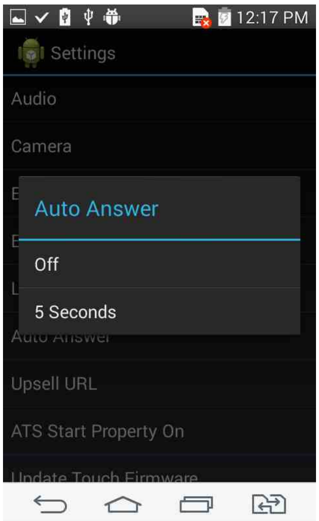
ETC

ADC Info

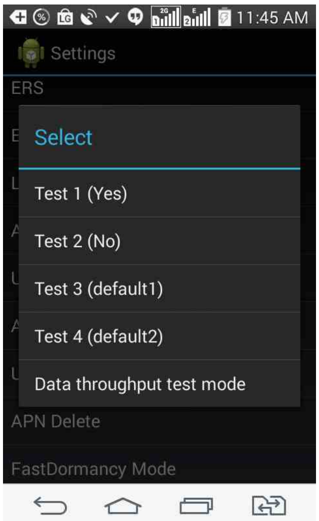


LCD Always On

Switch to LCD Backlight ON/OFF

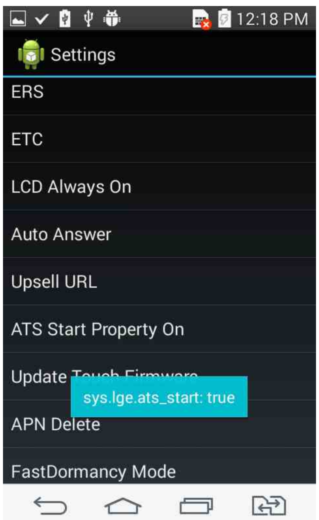


Auto Answer



Upsell URL

- Test 1 (Yes)
- Test 2 (No)
- Test 3 (default1)
- Test 4 (default2)
- Data throughput test mode



ATS Start Property On

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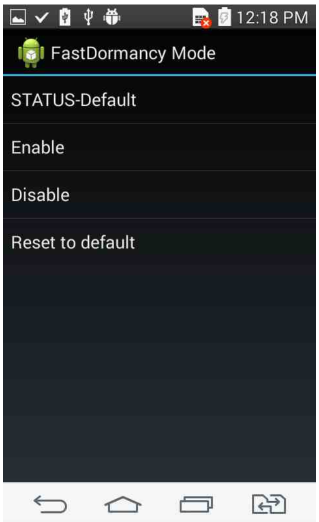
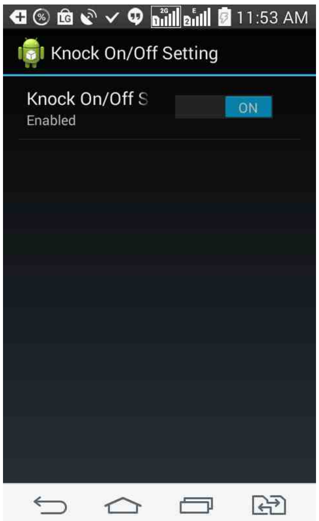
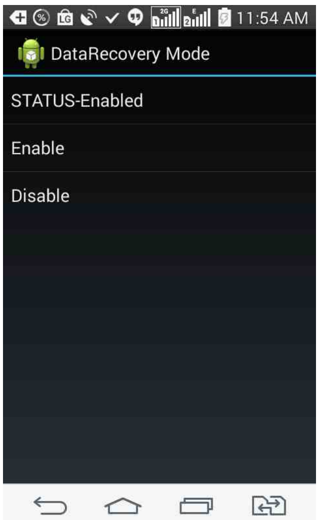
<tr><td>
</td><td>FastDormancy</td></tr>
<tr><td>
</td><td>Knock On/Off Setting</td></tr>
<tr><td>
</td><td>DataRecovery Mode</td></tr>
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<table>
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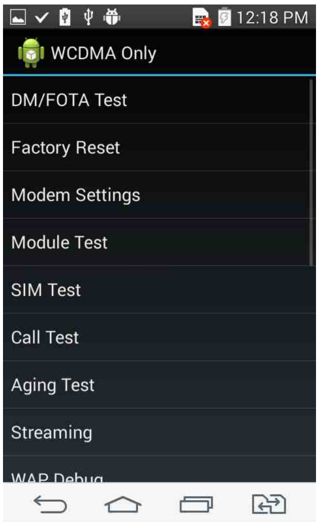
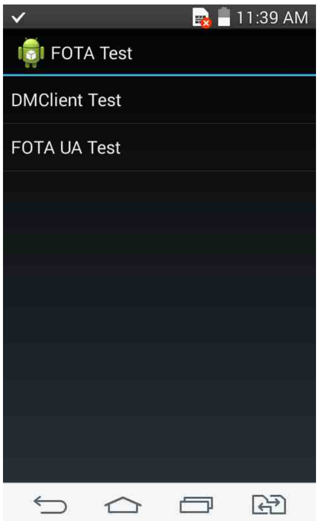
<td>
</td>
<td>WCDMA Only test list
-DM/FOTA Test
-Factory Reset
-Modem Settings
-Module Test
-SIM Test(Not Implemented)
-Call Test
-Aging Test
-Streaming
-WAP Debug
-DRM Test
-Contacts Test
-Log service
-Fake Language
-Mms Test
-Flex test
-FastDormancy Setting
-Data Performance
-FCC Test</td>
</tr>
<tr>
<td>
</td>
<td>DM/FOTA Test:
-DMClient Test
-FOTA UA Test</td>
</tr>
</table>

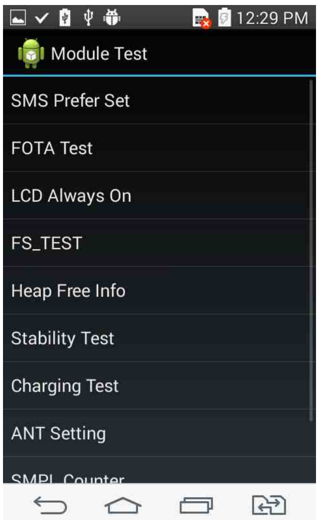
Factory Reset:

-Factory Reset :Changing the Factory

-Factory Reset (Power Off) : Reset and Power off

Modem Settings

- Engineering Mode
- RRC Version Setting
- QCRIL Log On/Off
- PDP Setting
- HSDPA Category
- GSM A5 Algorithm
- Protocol Test
- GCF Flag
- Arm9 Log On/Off
- Network Mode
- RAT Selection



Module Test list:

-SMS Prefer Set

-FOTA Test

-LCD Always On

-FS_TEST

-Heap Free Info

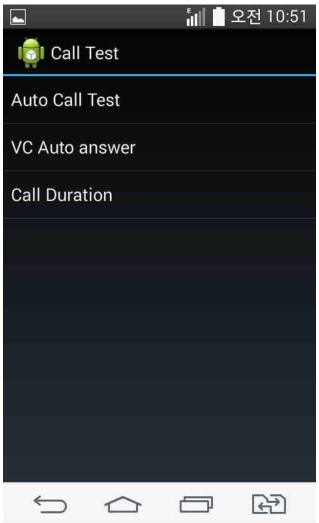
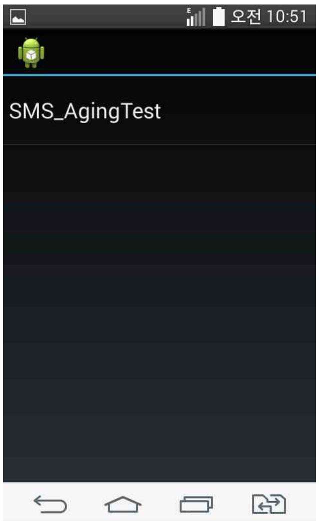
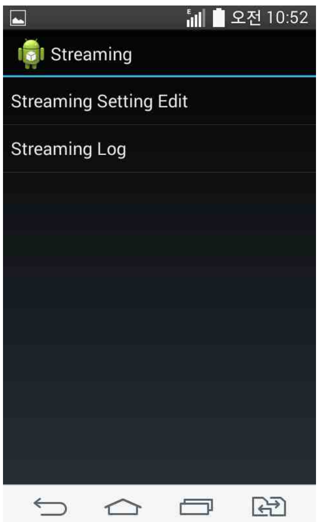
-Stability Test

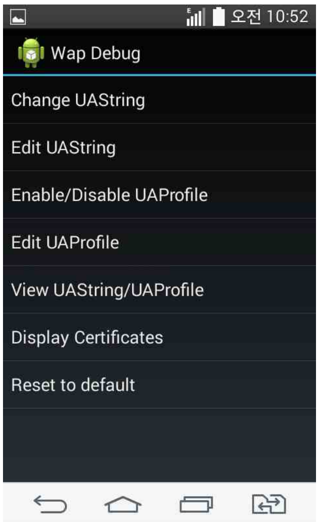
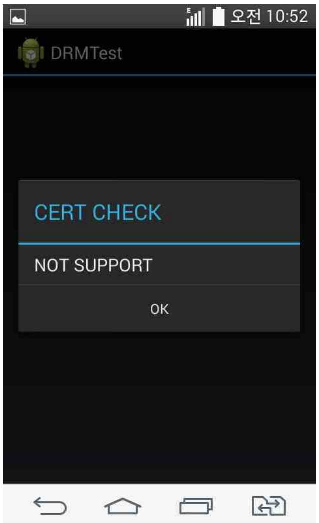
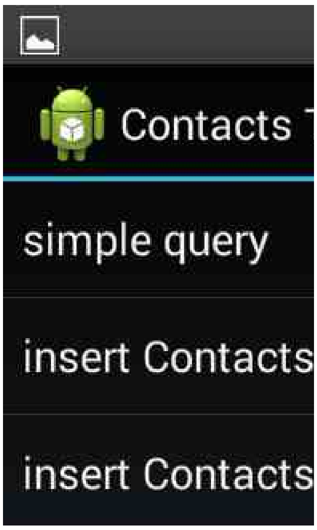
-Charging Test

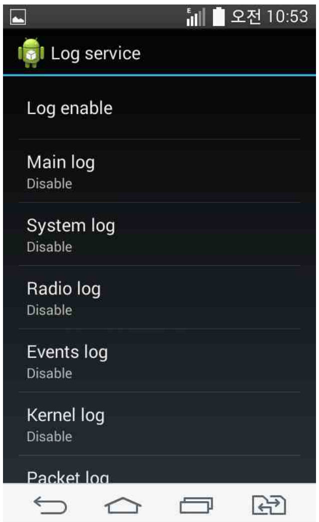
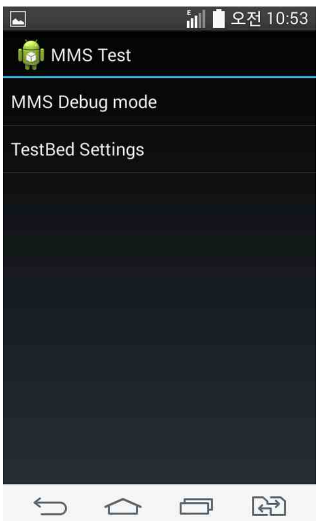
-ANT Setting

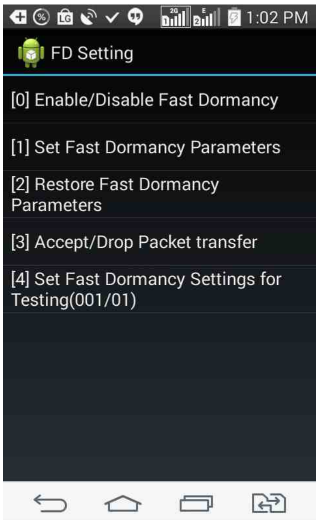
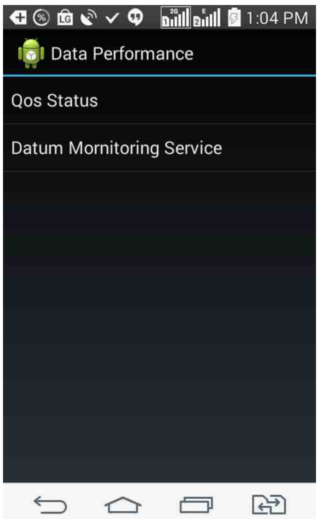
-SMPL Counter

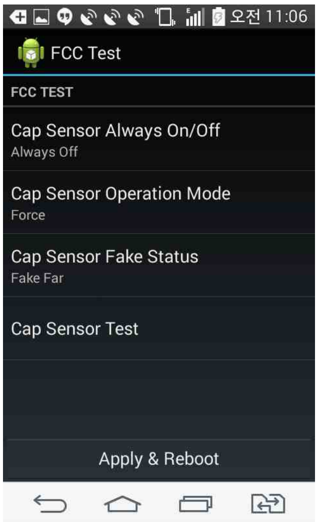
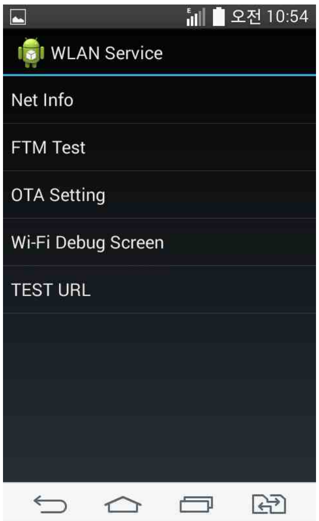
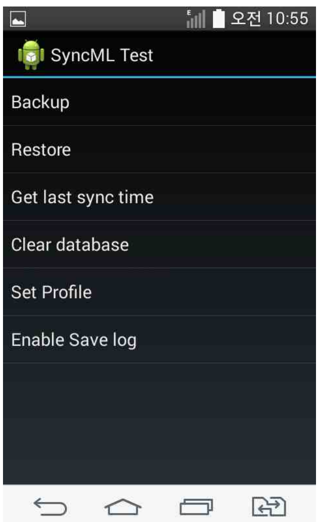
-BT DUT Setting

	**Call Test** -Auto Call Test -VC Auto answer -Call Duration
	**Aging Test**
	**Streaming** Streaming Setting Edit ->Edit Enable or not Streaming Log ->Log Enable or not

	**Wap Debug** -Change UAString -Edit UAString -Enable/Disable UAProfile -Edit UAProfile -View UASting/UAProfile -Display Certificates -Reset to default
	**DRM Test**
	**Contacts Test** Insert specific number of contact info

	**Log service** **Enable of disable creating below log files.** - Main log - System log - Radio log - Events log - Kernel log - Packet log
	**Fake Language**
	**MMS test**

	**Flex Test**
	**FastDormancy Setting**
	**Data Performance**

<table>
<tr><td>
</td><td>FCC Test</td></tr>
<tr><td>
</td><td>WLAN Test
WLAN performance on SW</td></tr>
<tr><td>
</td><td>SyncML Test</td></tr>
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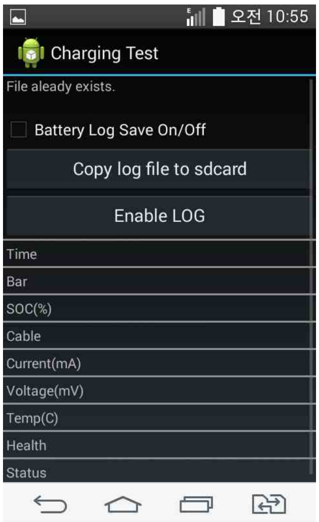
<td>
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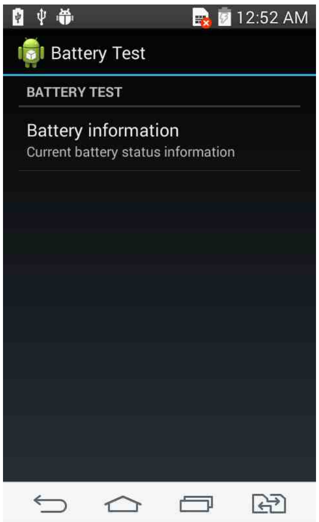
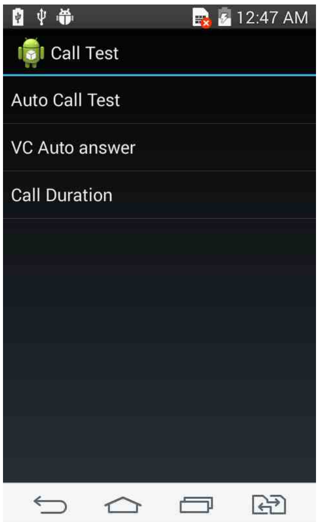
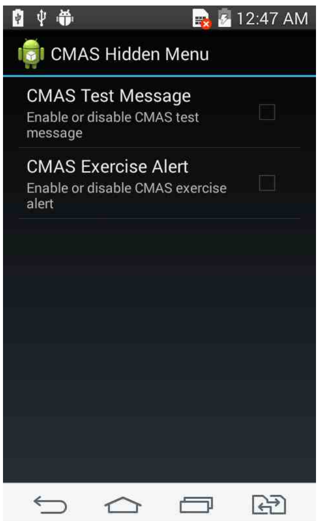
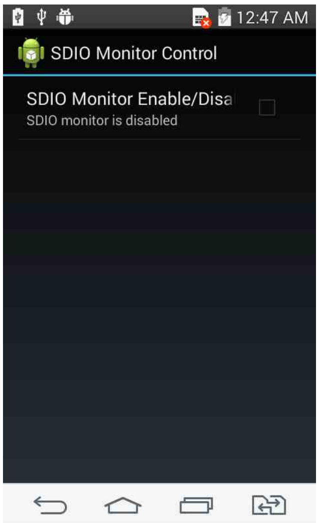
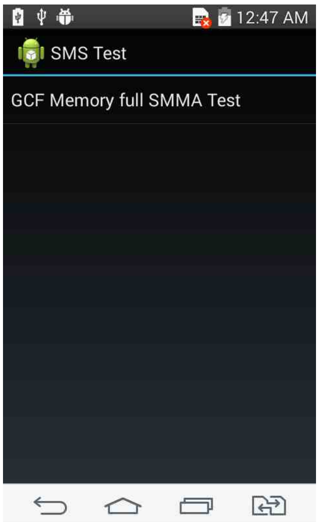
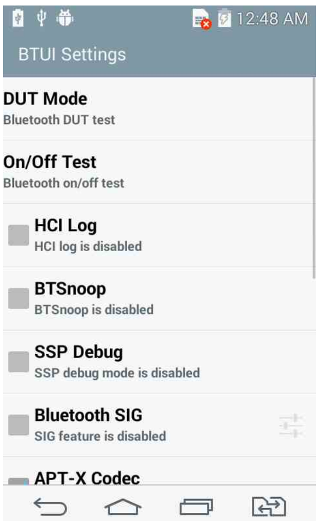
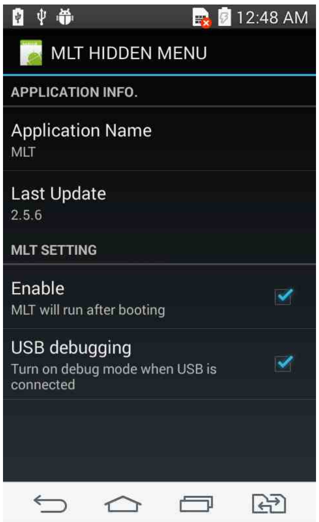
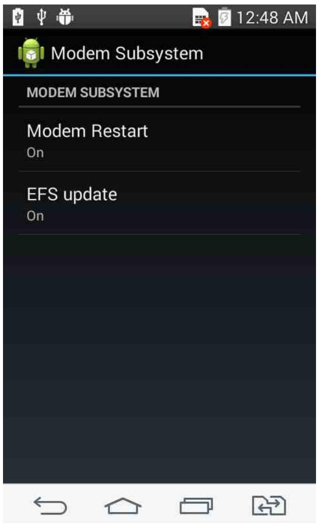
<td>Charging Test</td>
</tr>
<tr>
<td>
</td>
<td>Gnss Test</td>
</tr>
<tr>
<td>
</td>
<td>Same as
WCDMA Only – Fake Language</td>
</tr>
</table>

	**Battery Test**
	**Call Test**
	**CMAS_RMT**

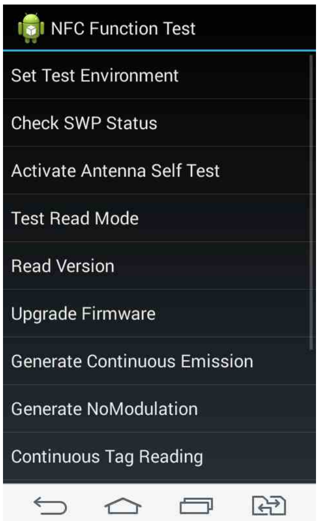
	**SDIO Monitor Contorl**
	**SMS Test**
	**BT_TEST**



MLT Test



Modem Subsystem

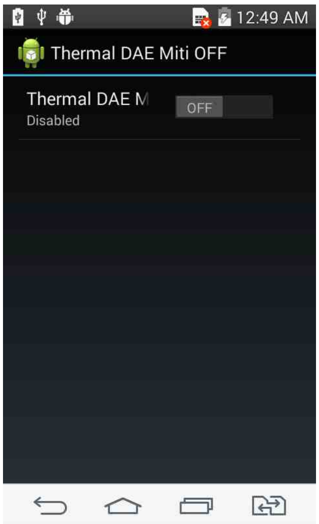
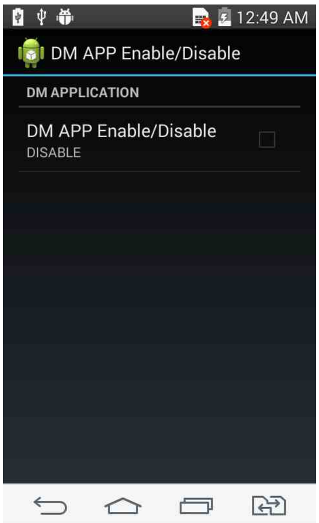
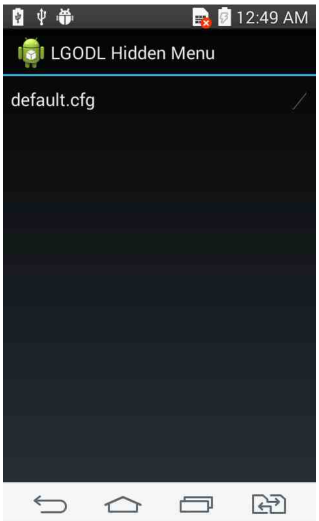


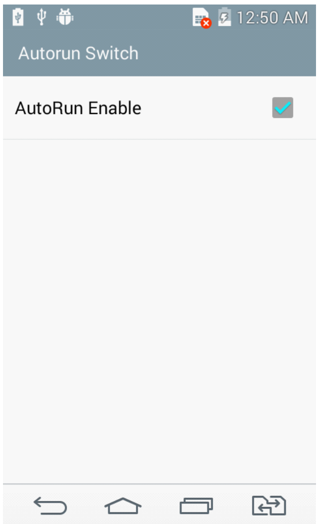
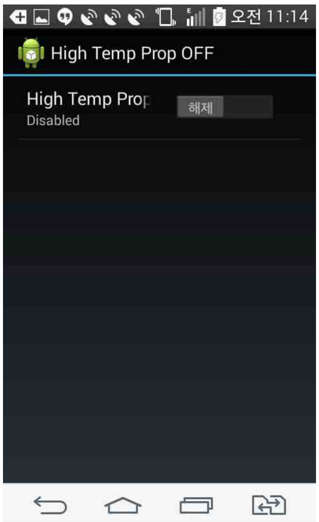
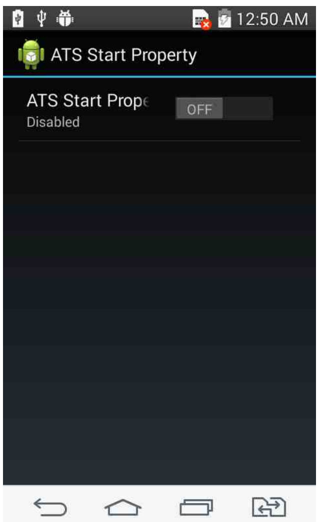
NFC Test

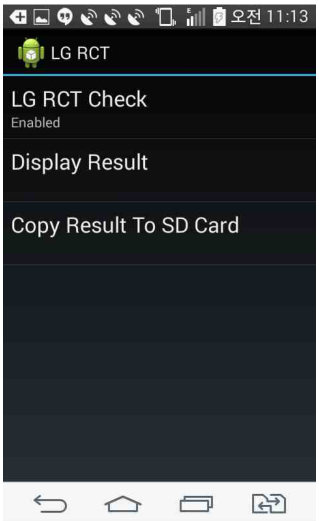
** Touch 'Set Test Environment'

If you want to Exit, touch Exit menu

(do not touch back key)

	**Thermal Daemon Mitigation OFF**
	**DM APP Enable/Disable**
	**First Execute, default.cfg is made**

	**Autorun Test**
	**Hiht Temperature Perperty OFF**
	**ATS Start Property ON/OFF**

<table>
<tr>
<td>


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<td>LG RCT Test</td>
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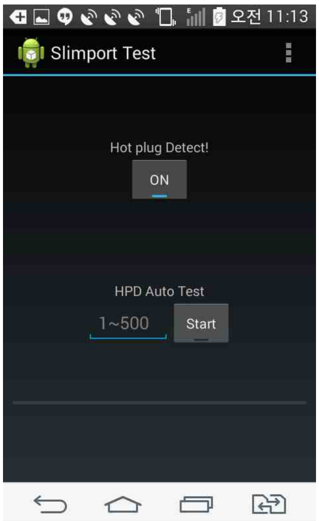

</td>
<td>Slimport Test</td>
</tr>
<tr>
<td>


</td>
<td>Facroty Reset</td>
</tr>
</table>

<table>
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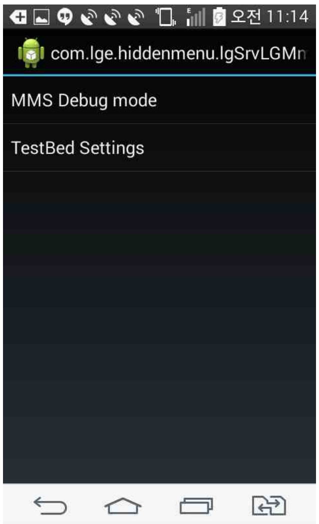
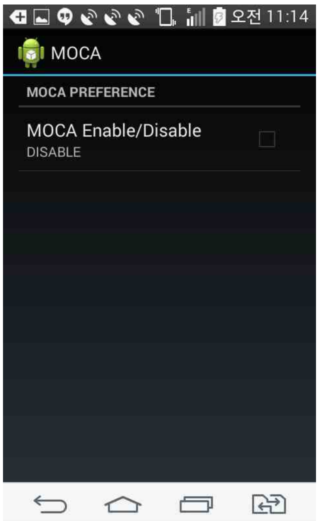
<td>
</td>
<td>MMS LG</td>
</tr>
<tr>
<td>
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<td>MOCA</td>
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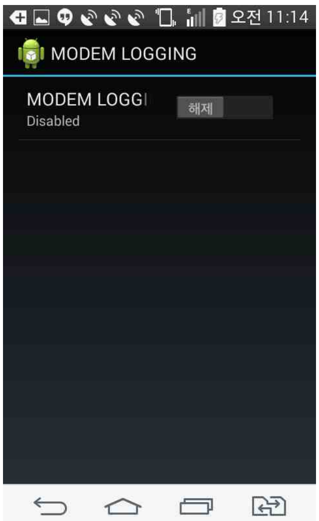
<td>
</td>
<td>Modem Log (QXDM Log)</td>
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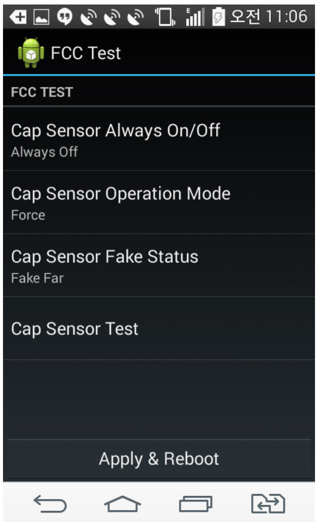
오전 11:06 FCC Test FCC TEST Cap Sensor Always On/Off Always Off Cap Sensor Operation Mode Force Cap Sensor Fake Status Fake Far Cap Sensor Test Apply & Reboot	**FCC Test**

11. DISASSEMBLE GUIDE

11.1 DISASSEMBLE GUIDE

1. Disassemble Battery Cover , Rear	
Disassemble sequence : 1 -> 2 -> 3 -> 4 	1 
2 	3 

4

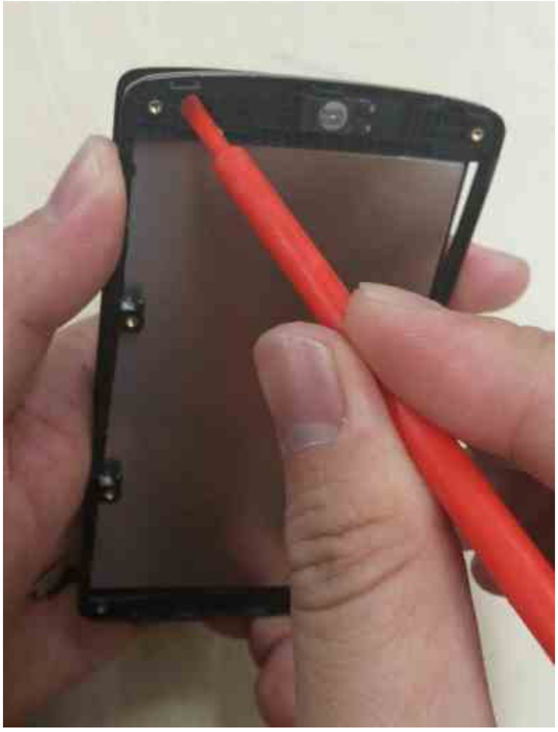
1. Remove Battery Cover
2. Remove 8 Screws from Rear
3. Remove Rear Cover

2. Disassemble PCB

1. Remove LCD Connector from PCB
2. Remove PCB

3. Disassemble LCD

1. User the Hair drier to LCD window front for 15 seconds



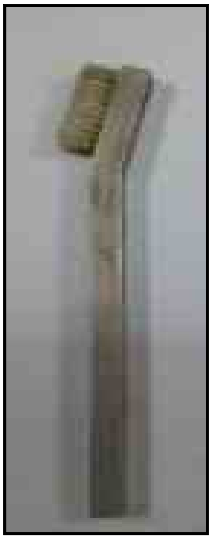
2. Press the 1 holes of below images with Tool.
and disassemble the window.

3. Complete as below.

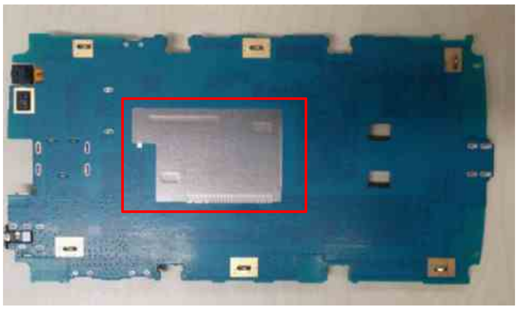
Remove LCD from Front

11.2 SHIELD CAN REWORK GUIDE

<table>
<tr><th colspan="6">HW Environment</th></tr>
<tr><td colspan="3">
</td><td colspan="3">
</td></tr>
<tr><td colspan="3">BE/T 853A : Hot Plate</td><td colspan="3">WHA3000P : Heat Gun</td></tr>
<tr><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td colspan="2">Soldering Iron</td><td colspan="2">PINSET</td><td colspan="2">H/F Solder</td></tr>
</table>

	
NEOZOL	H/F Flux
	
Bracket	Brush

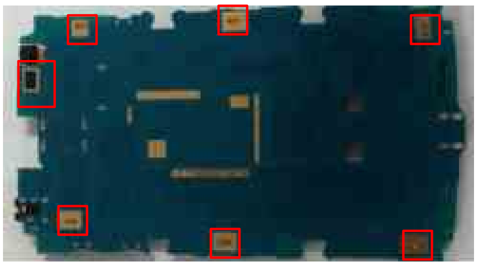
Setting Guide



1. Remove the gasket & cameras from the PCB.

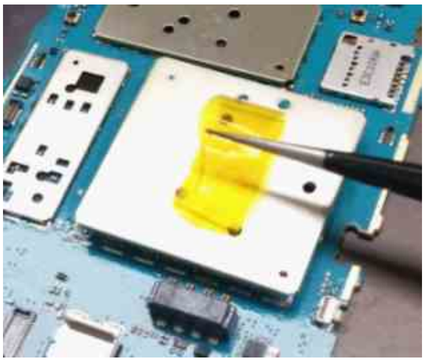
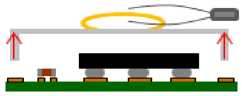
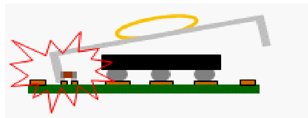
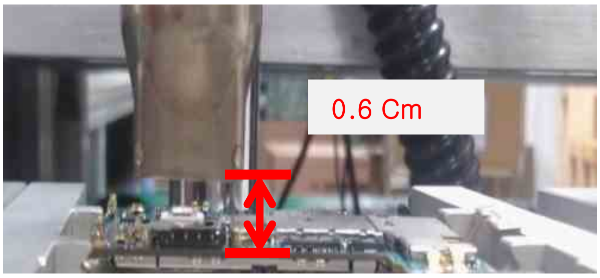
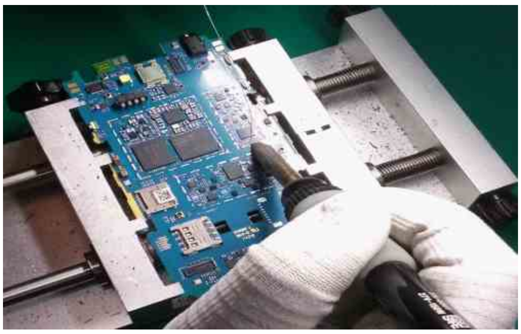
2. Set the BE/T 853A (Hot Plate) temperature to 200˚C
3. Pre-Heat the PCB for 1 minute on the Board support guide as image

*Caution : C-Clip & Sensor Damage

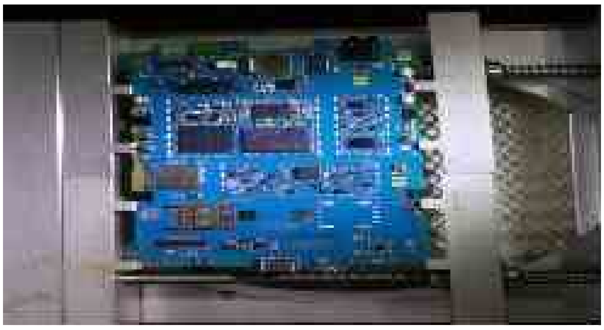


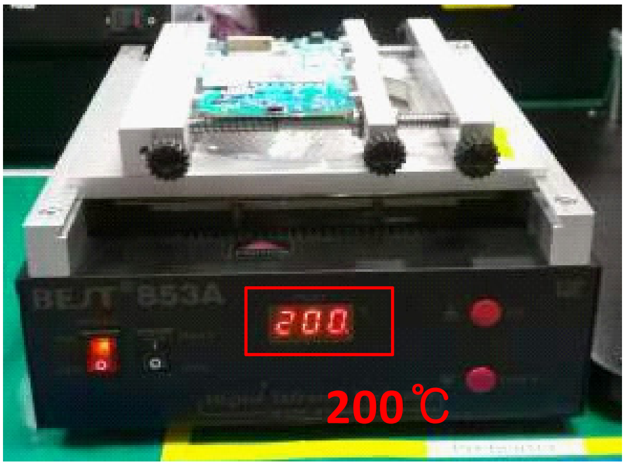
4. Set the WHA3000P (Heat Gun) as below

TEMP	AIR	PREHEAT
400˚C	15L	OFF

Disassembly Rework Guide	
 	1. Remove the shield can with pin set * Caution 1 : Lift the 'shield can' for vertical direction. If not, that is cause of defect.   * Caution 2 : Protect the near ICs with 'bracket' from HEAT DAMAGE * Caution 3 : Keep distance 0.6Cm PCB and Heat Gun nozzle 
	2. Apply additional solder on the soldering spot on PCB 3. Clean the PCB with Neozol.

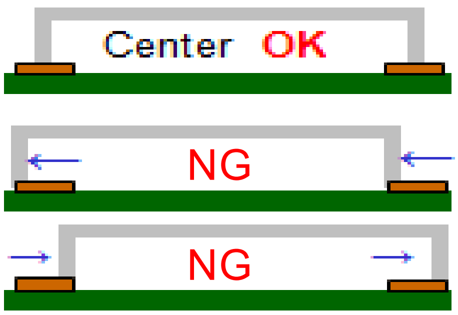
Assembly Rework Guide





1. Set the PCB as "Setting Guide"

2. Assemble the "shield can" on right place of PCB.
   *Caution : Shield can location



11.3 REWORK CAUTION POINTS

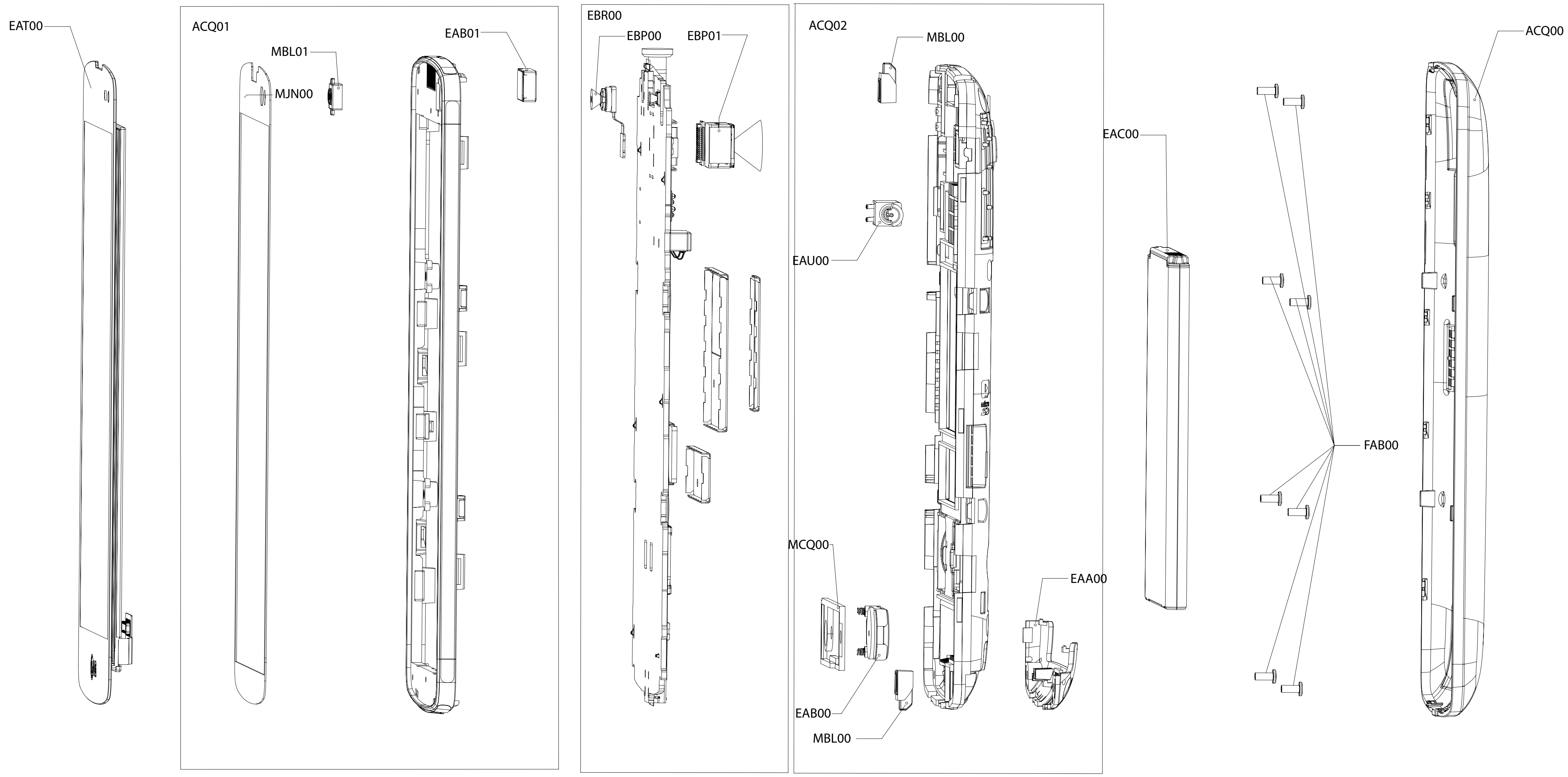
1. Issue : Use new part & rework short time, When change shieldcan
2. Cause : Shieldcan can be yellowish
    * It's appeared to customer for Phone Ass'y
3. Recommend rework time (200˚C) : Finish rework within 2min 30sec
   -> It's base on the below S/C yellowish test on Hot Plate

Time	New	1min	2min	3min	4min	5min
1Piece Shield can						
Enlarge Image						

Not Good

12. EXPLODED VIEW & REPLACEMENT PART LIST

12.1 EXPLODED VIEW(SBOM)



Location	Description	Location	Description
FAB00	Screw,Machine	MBL01	Cap
ACQ02	Cover Assembly,Rear	EAB01	Receiver
EAU00	Motor,DC	EBR00	PCB Assembly,Main
EAB00	Speaker,Dual Mode	EBP00	Camera Module
EAA00	PIFA Antenna,Multiple	EBP01	Camera Module
MCQ00	Damper	EAC00	Rechargeable Battery,Lithium Ion
MBL00	Cap	ACQ00	Cover Assembly,Battery
EAT00	Module,Hybrid Touch LCD		
ACQ01	Cover Assembly,Front		
MJN00	Tape,Window		

12.2 Replacement Parts
<Mechanic component>

Note: This Chapterisused for reference,Part order is ordered by SBOM standard on GCSC

Level	Location no	Description	P/N	Specification	Remark
1	AGQ000000	Phone Assembly	AGQ88014501	LGD295.ACISKT KT:BLACK TITAN -	
2	FAB00	Screw,Machine	GMEY0009201	GMEY0009201 BH + 2.7mM 3.5mM MSWR3 FZB N N LG ELECTRONICS INC.	
2	MEZ002100	Label,Approval	MLAA0062305	COMPLEX KB770 DEUBK ZZ:Without Color -	
2	ACQ100400	Cover Assembly, EMS	ACQ87682701	LGD295.ACISKT KT:BLACK TITAN -	
3	ACQ02	Cover Assembly, Rear	ACQ87738701	LGD295.ACISKW BK:Black -	
4	MCQ00	Damper	MCQ68104801	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
4	ACQ105800	Cover Assembly, Rear(SVC)	ACQ87611601	LGD295F.ABRAWH BK:Black -	
5	MCK063300	Cover,Rear	MCK68349901	MOLD PC LGD295F.ABRAWH BK:Black -	
5	MBL00	Cap	MBL66119801	MOLD TRU LGD295F.ABRAWH BK:Black -	
5	MCQ009400	Damper,Camera	MCQ68025901	COMPLEX LGD295F.ABRAWH ZY:Color Unfixed -	
5	MCQ000000	Damper	MCQ68125201	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
5	MJN009400	Tape,Camera	MJN69169801	COMPLEX LGD295F.ABRAWH ZY:Color Unfixed -	
5	MEZ000901	Label, After Service	MEZ64319901	COMPLEX LGF120L.ALGTWA ZZ:Without Color F120I AS Label	
5	MFB029600	Lens,Flash	MFB63753001	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
5	MCQ000001	Damper	MCQ68224701	COMPLEX LGD295F.ABRAKW ZZ:Without Color -	
5	MKC009400	Window,Camera	MKC65239101	CASTING GLASS LGD295F.ABRAWH WA:White -	
5	MJN000001	Tape	MJN69287401	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
5	MJN000000	Tape	MJN69267501	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
5	MCR000000	Decor	MCR65866801	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
3	ACQ003400	Cover Assembly, Bar	ACQ87611803	LGD295.AKAZKW BK:Black -	
4	MJN107400	Tape,USP Film	MJN69288002	COMPLEX LGD295.AKAZKW ZZ:Without Color -	

Level	Location no	Description	P/N	Specification	Remark
4	MEZ000000	Label	MLAZ0038301	COMPLEX LG-VX6000 ZZ:Without Color PID Label 4 Array PRINTING,	
4	MJN061100	Tape,Protect	MJN69308501	COMPLEX LGD392.AINDWH ZZ:Without Color -	
4	ACQ01	Cover Assembly, Front	ACQ87617501	LGD295F.ABRABK BK:Black -	
5	MJN000001	Tape	MJN69427701	COMPLEX LGD295F.KBRAKW ZZ:Without Color -	
5	MJN061101	Tape,Protect	MJN69347201	COMPLEX LGD390N.ADEUBK ZZ:Without Color -	
5	MJN061100	Tape,Protect	MJN69311201	COMPLEX LGD390N.ADEUBK ZZ:Without Color -	
5	MJN00	Tape,Window	MJN69290501	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
5	MJN000000	Tape	MJN69147801	COMPLEX LGD390.ADEUBK ZZ:Without Color TAPE, RCV	
5	MEV000000	Insulator	MEV65131101	COMPLEX LGD392.AINDWH ZZ:Without Color -	
5	MCR000000	Decor	MCR65747101	COMPLEX LGD390.ADEUBK ZZ:Without Color -	
5	MBL01	Cap	MBL66157501	MOLD TPU LGD392.AINDWH BK:Black -	
5	ACQ033200	Cover Assembly, Front(Sub)	ACQ87678401	LGD295F.ABRABK BK:Black -	
6	MCK032700	Cover,Front	MCK68527601	MOLD PC LGD295F.ABRABK BK:BLACK BLACK -	
6	MET099500	Insert,Nut	MICE0016905	COMPLEX MECH_COMMON ZZ:Without Color -	
5	MEZ000000	Label	MEZ65049701	COMPLEX LGLS720.ASPRTS ZZ:Without Color PID label	
6	SC1	Bracket	MAZ64486501	PRESS SUS 304 0.2 LGD295F.ABRAWH ZZ:Without Color -	
6	SC9600	Can,Shield	MBK64013301	PRESS SUS 304 0.2 LGD295F.ABRAWH ZZ:Without Color -	
6	SC9602	Can,Shield	MBK64033101	PRESS SUS 304 0.2 LGD295F.ABRAWH ZZ:Without Color -	
6	SC9601	Can,Shield	MBK64053001	PRESS SUS 304 0.2 LGD295F.ABRAWH ZZ:Without Color -	
1	AGF000000	Package Assembly	AGF77606008	LGD295.ACISKT ZZ:Without Color L70+/S6-1H32/CIS UB/CIS UL/CIS SEAL L/AL_STD_H-BODY/480ea	

Level	Location no	Description	P/N	Specification	Remark
2	MEZ047200	Label,Master Box	MLAJ0004402	PRINTING CG300 CGR DG ZZ:Without Color LABEL MASTER BOX(for CGR TDR 2VER. mbox_label) GSM standard_master box label	
2	AGJ000000	Pallet Assembly	AGJ73978404	LGD618.ACISBK ZZ:Without Color S6-1H/AL_STD_H-BODY/PLB_4ea/480ea	
3	MCQ000000	Damper	MCQ66486911	COMPLEX LGE730.ACISBK ZZ:Without Color S6T-CIS Dead Space Sleeve(480EA/1200*800)	
3	MBL007000	Cap,Box	MCCL0002604	COMPLEX LGE730.ANLDKT ZZ:Without Color S6T-STD Cap(800EA/1200*800)	
3	MAY010800	Box,Carton	MBEC0004405	COMPLEX LGE730.ACISBK ZZ:Without Color S6T-CIS Body(480EA/h:800 under/1200*800)	
3	MEZ047200	Label,Master Box	MLAJ0004402	PRINTING CG300 CGR DG ZZ:Without Color LABEL MASTER BOX(for CGR TDR 2VER. mbox_label) GSM standard_master box label	
3	MGA000000	Pallet	MPCY0012403	COMPLEX KG800 FRABK DB:DARK BLUE -	
3	MEZ000000	Label	MLAZ0050901	COMPLEX KU990.AGBRBK ZZ:Without Color Battery Warning Label (Lithium ion Battery Label)	
2	MLAC0000001	Label,Barcode	MLAC0004541	PRINTING HB620 KPNBK ZZ:Without Color GSM standard_unit box label_90*40	
2	MEZ084101	Label,Unit Box	MEZ64571201	COMPLEX LGP725.ACISBK WW:SNOW WHITE Seal Label_CIS For Only	
2	MAY047100	Box,Master	MBEE0061004	COMPLEX LGE730.ANLDKT ZZ:Without Color S6T_STD Master Box(10EA)	
2	MAY084000	Box,Unit	MAY66813303	BOX Paper 146 95 56 MULTI COLOR LGD295.ACISKT ZZ:Without Color L70+/S6-1H-32/CIS UB	
2	MAF086500	Bag,Vinyl	MAF63267402	COMPLEX LGE615.ACISBK ZZ:Without Color HDPE Phone Bag_add CIS Mark_85X200	
2	MEZ084100	Label,Unit Box	MEZ64686921	COMPLEX LGP880.ACISBK ZZ:Without Color Unit Box Label_CIS Peel Cutting Label_40X20, 4EA	
1	AAD000000	Addition Assembly	AAD87160601	LGD295.ACISKT KT:BLACK TITAN -	
2	MEZ002101	Label	MEZ65889901	COMPLEX LGD855.A6ISKG ZZ:Without Color -	
2	AFN053800	Manual Assembly, Operation	AFN76693510	LGD295.ACISKT ZZ:Without Color -	
2	ACQ00	Cover Assembly, Battery	ACQ87651703	LGD295.ACISKT WH:WHITE WHITE -	

Level	Location no	Description	P/N	Specification	Remark
3	MCK004100	Cover,Battery	MCK68350002	MOLD PC LGD295F.ABRAWH KT:BLACK TITAN -	
3	ABH000000	Button Assembly	ABH75199702	LGD295F.ABRAWH TK:TITAN BLACK -	
4	MGJ000000	Plate	MGJ64270901	PRESS SUS 304 0.3 LGD295F.ABRAWH ZZ:Without Color -	
4	MJN000001	Tape	MJN69247901	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
4	MJN000000	Tape	MJN69267701	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	
4	AEX046800	Keypad Assembly,Main	AEX74297412	LGD295F.ABRAWH TK:TITAN BLACK -	
5	MEX046800	Keypad,Main	MEX62388201	MOLD RUBBER LGD295F.ABRAWH ZZ:Without Color -	
5	MBG000000	Button	MBG65303902	MOLD PC LGD295F.ABRAWH TK:TITAN BLACK -	
4	ADB048600	Dome Assembly, Metal	ADB74378101	LGD295F.ABRAWH ZZ:Without Color -	
3	MHK000000	Sheet	MHK64805701	COMPLEX LGD295F.ABRAWH ZZ:Without Color -	

12.2 Replacement Parts
<Main component>

Note: This Chapterisused for reference, Part order is ordered by SBOM standard on GCSC

Level	Location no	Description	P/N	Specification	Remark
4	EAU00	Motor,DC	EAU62127101	DM-YK423-19 3V 80mA 0A 11KRPM 0RPM 0SEC 0GF.CM 0OHM DONGYANG CHENGJI CO.	
4	EAB00	Speaker, Dual Mode	EAB63468801	SB101830CS02N Nd-Fe-B 700mW 8OHM 90DB 850HZ 18x10x3.0 SPRING BSE CO., LTD.	
4	EAA00	PIFA Antenna, Multiple	EAA63646201	BMI14A032 MULTI -4DB 4 SIMA - Main Carrier AT&C CO.,LTD	
5	EAA030100	PIFA Antenna, GPS	EAA63666201	ACA-00328 SINGLE -3DB 5.0 SUS - GPS Carrier ACE TECHNOLOGIES CORPORATION	
5	EAA030101	PIFA Antenna, WiFi	EAA63607901	ACA-00329 SINGLE -3DB 5.0 SUS - WLAN+BT Carrier ACE TECHNOLOGIES CORPORATION	
4	EAT00	Module,Hybrid Touch LCD	EAT62493701	LH450WV5-SD01 CAPACITIVE TOUCH Incell Touch (AIT) Dongbu Hitek DB7436 LCD Driver IC Interpolation 120Hz OCR 0.15mm 4.49" WVGA (480*800) B to B - LG Display Co. Ltd.	
5	EAB01	Receiver	EAB63288401	WS061225HS01N 50mW 32OHM 109DB 100HZTO7KHZ PIN - BSE CO., LTD.	
3	EBR00	PCB Assembly, Main	EBR79712701	LGD295.ACISKT 1.0 Main	
4	EBR071500	PCB Assembly, Main,Insert	EBR79730501	LGD295.ACISKT 1.0 Main	
5	EBP00	Camera Module	EBP62201701	CW8012-AX219 CW8012-AX219 8M AF, Sony(1/4") IMX219 BSI, 8.5x8.5x4.95t, MIPI4, FPCB 3.5mm, 90 degree, 30pin COWELL ELECTRONICS CO.,LTD	
5	RAA050100	Resin,PC	BRAH0001304	: EFC-3A03-20-MH(Sn99%/Ag0.3%/Cu0.7%)) . . NONE	
5	EBP01	Camera Module	EBP62081901	CVFA-H498A CVFA-H498A VGA FF 1/10" Hi707 2P 0degree Front LG INNOTEK CO., LTD	
4	EBR071800	PCB Assembly, Main,SMT	EBR79673601	LGD295.ACISKT 1.0 Main	
5	SAD010000	Software,Mobile	SAD34945601	Android KK Base D29510a - EUROPE QCT -	
5	EBR071700	PCB Assembly, Module,SMT Top	EBR79691101	LGD295.ACISKT 1.0 Module	

Level	Location no	Description	P/N	Specification	Remark
6	EAX010000	PCB,Main	EAX66051401	LGD295F.ABRAWH 1.0 FR-4 Staggered via 6 1.0 Main	
5	EBR071600	PCB Assembly, Main,SMT Bottom	EBR79691001	LGD295.ACISKT 1.0 Main	
6	L1111	Capacitor, Ceramic,Chip	EAE62743101	C1005NPO129CGTQ 1.2pF 0.25PF 50V C0G -55TO+125C 1005 R/TP 0.55T max. DARFON ELECTRONICS CORP.	
6	L1110	Capacitor(High Frequency), Ceramic,Chip	EAE62882801	GRM1555C1H2R2C_H 2.2pF 0.25PF 50V C0G -55TO+125C 1005 R/TP 0.5+/-0.05 L:1.0+-0.05 W:0.5+-0.05 T:0.5+-0.05 MURATA MANUFACTURING CO.,LTD.	
6	VA6613	Varistor	EAF61891401	IECS0305C040 5V 0% 0.5pF 0.6x0.3 IEC61000-4-2 (ESD) level #4 SMD R/TP INNOCHIPS TECHNOLOGY	
6	FL1601	Filter,Duplexer	EAM62693001	B8000 942.5MHz 925.0to960.0MHz 897.5MHz 880.0to915.0MHz 2.1typ/3.2max 2.3typ/3.8max 2.0x1.6x0.45t DUAL SMD R/TP 9P EPCOS PTE LTD.	
6	C1112 C1123 C1124 C1608	Capacitor, Ceramic,Chip	ECCH0009216	GRM0335C1E220J 22pF 5% 25V X7R -55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	C6924 C6927	Capacitor, Ceramic,Chip	ECCH0009226	GRM0335C1E390J 39pF 5% 25V X7R -55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	C5125 C6612	Capacitor, Ceramic,Chip	ECZH0000830	C1005C0G1H330JT000F 33pF 5% 50V NP0 -55TO+125C 1005 R/TP - TDK KOREA COOPERATION	
6	C1140 C5122	Inductor, Multilayer,Chip	ELCH0001056	1005GC2T2N7SLF 2.7NH 0.3NH - 300mA 0.17OHM 5.5GHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP PILKOR ELECTRONICS LTD.	
6	D6810 D6811 VA6610 VA6611 VA6612 VA6616 VA6618	Varistor	SEVY0008102	EVLC5S01015 5.5V 0% 15pF 0.6*0.3*0.33 NONE SMD R/TP AMOTECH CO., LTD.	
6	C4400 C9205	Capacitor, Ceramic,Chip	EAE62505701	CL10A105KB8NNNC 1uF 10% 50V X5R -55TO+85C 1608 R/TP 0.9T max. - SAMSUNG ELECTRO-MECHANICS CO., LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C5102	Capacitor, Ceramic,Chip	EAE62506501	CL05A475MP5NRNC 4.7uF 20% 10V X5R -55TO+85C 1005 R/TP - - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2324 C2341 C2347 C2375 C2376 C2377 C4300	Capacitor, Ceramic,Chip	EAE62685301	CL05A105KA5NQNC 1uF 10% 25V X5R -55TO+85C 1005 R/TP 0.6T max. Samsung(1.0+-0.1 0.5+-0.1 0.5+-0.1) Murata (1.0+-0.05 0.5+-0.05 0.5+-0.05) SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C4116 C4121 C4122	Capacitor,TA, Conformal	EAE62748301	F981A226MMA 22uF 20% 10V 11UA -55TO+125C 8OHM 1.6X0.85X0.8MM NONE SMD R/TP 0.9T max. UNITEL ELECTRONICS CO. LTD.	
6	C9104	Capacitor,TA, Conformal	EAE62748401	F981A476MSAFZH 47uF 20% 10V 9.4UA -55TO+125C 5OHM 2.2X1.25X0.9MM NONE SMD R/TP 0.9T max. UNITEL ELECTRONICS CO. LTD.	
6	C2371 C2381 C2391	Capacitor, Ceramic,Chip	EAE62762301	CL03A105MP3NSNC 1uF 20% 10V X5R -55TO+85C 0603 R/TP 0.33 MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2363 C4126 C4127	Capacitor,Ceramic, Chip	EAE62767801	JDK212BBJ476MD 47uF -20TO20% 6.3V X5R -55TO+85C 2012 R/TP 0.85T - TAIYO YUDEN CO.,LTD	
6	C1602 C1603 C4401 C4402 C7501 C7502	Capacitor, Ceramic,Chip	EAE62843001	CL10A106MO8NQNC 10uF -20TO20% 16V X5R -55TO+85C 1608 R/TP 0.8+-0.15 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C1143	Capacitor(High Frequency), Ceramic,Chip	EAE62882601	GRM1555C1H1R0C_H 1pF 0.25PF 50V C0G -55TO+125C 1005 R/TP 0.5+/-0.05 L:1.0+-0.05 W:0.5+-0.05 T:0.5+-0.05 MURATA MANUFACTURING CO.,LTD.	
6	C1127	Capacitor(High Frequency), Ceramic,Chip	EAE62945301	GRM1555C1HR50B_H 0.5pF 0.1PF 50V C0G -55TO+125C 1005 R/TP 0.5+/-0.05 L:1.0+-0.05 W:0.5+-0.05 T:0.5+-0.05 MURATA MANUFACTURING CO.,LTD.	
6	C5119	Capacitor(High Frequency), Ceramic,Chip	EAE62945601	GRM1555C1H3R3C_H 3.3pF 0.25PF 50V C0G -55TO+125C 1005 R/TP 0.5+/-0.05 L:1.0+-0.05 W:0.5+-0.05 T:0.5+-0.05 MURATA MANUFACTURING CO.,LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C2361 C3303 C4120 C4123 C4124 C4125	Capacitor, Ceramic,Chip	EAE62962301	GRM155R61A225KE95 2.2uF -10TO+10% 10V X5R -55TO+85C 1005 R/TP 0.5 L:1.0+-0.05 W:0.5+-0.05 T:0.5+-0.05 MURATA MANUFACTURING CO.,LTD.	
6	C9503 C9505	Capacitor(High Frequency), Ceramic,Chip	EAE63023501	RMUMK105CG100JV-F_H 10pF 5% 50V C0G -55TO+125C 1005 R/TP 0.5+-0.05 L:1.0+-0.05, W:0.5+-0.05, T:0.5+-0.05 KOREA TAIYO YUDEN.CO., LTD.	
6	C2313 C2314	Capacitor, Ceramic,Chip	EAE63070001	GRM033R60J223JE01 22nF -5 to +5% 6.3V X5R -55TO+85C 0603 R/TP Max 0.33T L:0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	D10006 D10007	Varistor	EAF61450601	EVLC 5S 01 100 5.5V 0% 100pF 0.6X0.3X0.3MM IEC61000-4-2 (ESD) level #4 SMD R/TP AMOTECH CO., LTD.	
6	CN9101	Connector, Terminal Block	EAG62911101	KQ03LV-4R 4P 2.50MM ANGLE DIP R/TP Twin contact Block 5.4mm Contact 4.1mm HIROSE KOREA CO.,LTD	
6	CN9200	Connector,I/O	EAG63090001	04-5161-005-100-868 7P 0.90MM ANGLE RECEPTACLE DIP R/TP Normal New IO Connector KYOCERA ELCO KOREA SALES CO.,LTD.	
6	CN9600	Card Socket	EAG63292001	3.2.04.0048 Micro-SD 8P STRAIGHT SMD T/REEL - SHENZHEN EVERWIN PRECISION TECHNOLOGY CO.,LTD.	
6	J9500 J9503	Socket,DIMM/SIM M	EAG63351601	5000-8P-1.5ML2 8P STRAIGHT STANDARD SMD T/REEL - HYUPJIN I&C CO.,LTD.	
6	CN10000	Connector, Terminal Block	EAG64091201	HSTC-04P25-5001 4P 2.50MM STRAIGHT SMD REEL - HANSHIN TERMINAL CO., LTD	
6	ANT1102 ANT1103 ANT1111	C-Clip	EAG63670501	PMT1023 1P ANT TERMINAL STRAIGHT SMD T/REEL AU 3.0X1.5X1.3(LXWXH) SERVEONE CO., LTD.	
6	J6610	Jack,Phone	EAG64069801	JAM3335-F53-7H 5P 4P STRAIGHT R/TP 3.5M BLACK 5P - FOXCONN INTERCONNECT TECHNOLGY LIMITED(FIT)	
6	CN9601 CN9602 CN9603 CN9604 CN9605 CN9606	C-Clip	EAG64069901	1204-01A 1P ANT TERMINAL STRAIGHT SMD REEL AU 2.8X2.0X1.7 (L X W X H)1.7T 2x2.8 SERVEONE CO., LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	D9910	Diode,Switching	EAH61532901	BA891_ 1V 35V - - 0SEC 715mW SOD523 R/TP 2P 1 NXP Semiconductors	
6	D9200	Diode,TVS	EAH61872601	PESD12VS1UA 12V 13.3V min. 19V 22.5A 360mW SOD323 R/TP 2P 1 NXP Semiconductors	
6	D4400	Diode,Schottky	EAH61992701	PMEG6002EB 600mV 60V 200mA 0SEC 20pF 0W SOD523 R/TP 2P 1 NXP Semiconductors	
6	D6610	Diode,TVS	EAH62335501	ESD105-B1-02EL 5.5V 6.1V min / 8V typ 14V 5A 70W TSLP-2-20 R/TP 2P 1 INFINEON TECHNOLOGIES (ASIA PACIFIC) PTE LTD.	
6	FL5100	Filter,Ceramic	EAM62250401	LFB212G45CG7D227 BPF 2.45KHZ 100Hz SMD R/TP 3P MURATA MANUFACTURING CO.,LTD.	
6	FL9201	Filter,EMI/Power	EAM62272401	ICMEF112P500MFR COMMON MODE NOISE FILTER 0HZ 0.0000000000001F 0H SMD R/TP - INNOCHIPS TECHNOLOGY	
6	FB5100	Filter,Bead	EAM62471001	BLM03AX241SN1D 240 ohm 0.6X0.3X0.33 25% 0.38 ohm 0.35A SMD R/TP 2P 0 MURATA MANUFACTURING CO.,LTD.	
6	FL7101 FL7102 FL7103	Filter,EMI/Power	EAM62570201	ICMEF062P900MFR COMMON MODE NOISE FILTER - 0.0000000000017F 0H SMD R/TP - INNOCHIPS TECHNOLOGY	
6	FB6611 FB6612	Filter,Bead	EAM62630901	BLM15AX102SN1D_ 1000 1.0x0.5x0.5 25% 0.49 0.35 SMD R/TP 0P - MURATA MANUFACTURING CO.,LTD.	
6	FL1103	Filter,Saw	EAM62633001	SFHG89DQ102 1575.42@GPS/1601.72@GLONASS 1.4x1.1x0.5t SMD R/TP 5P WISOL.CO.,LTD	
6	FL1600	Filter,Duplexer	EAM62692801	B8597 2140.0MHz 2112.4to2167.6MHz 1950.0MHz 1922.4to1977.6MHz 2.1typ/2.6max 1.6typ/2.0max 2.0x1.6x0.47t DUAL SMD R/TP 9P EPCOS PTE LTD.	
6	C5113	Capacitor, Ceramic,Chip	EAE62502901	CL05A106MP5NUNC 10uF 20% 10V X5R -55TO+85C 1005 R/TP 0.55T max. SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	MIC6920 MIC6921	Microphone, Condenser	EAB62950801	SPW4430HR5H -42DB 450OHM OMNI 1.5 TO 3.6V 3.10 * 2.5 * 1.1T SMD KNOWLES ACOUSTICS	
6	C2367 C2390	Capacitor, Ceramic,Chip	EAE62282201	GRM033R71E102KA01D 0.000000001F 10% 25V X7R -55TO+125C 0603 R/TP 0.3+/-0.03 MM - MURATA MANUFACTURING CO.,LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C4110 C9105 C9500	Capacitor, Ceramic,Chip	EAE62286801	CL03A104KP3NNNC 0.0000001F 10% 10V X5R -55TO+85C 0603 R/TP 0.3 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	U4108 U4109	IC,LDO Voltage Regulator	EAN62496301	TLV70718 V2 to V5.5 V1.8 0W QFN R/TP 4P EUSY0355501 RP103K181D ,Richo Pin to Pin TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	FL1101	Filter,Saw	EAM62732801	B8808 881.5MHz 1.1x0.9x0.45t SMD R/TP 5P EPCOS PTE LTD.	
6	FL1100	Filter,Saw,Dual	EAM62792801	SFRG42DB002 1842.50MHz/1960.00MHz 1.5x1.1x0.60t SMD R/TP 10P WISOL.CO.,LTD	
6	U4103 U4106	IC,LDO Voltage Regulator	EAN62512901	TLV70730PDQNR V2 to V5.5 V3.0 0W QFN R/TP 4P EUSY0373901 RP103K301D-TR. RICHO PIN TO PIN TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	U6610	IC,Analog Switch	EAN62578201	FSA5157L6X 1.65 to 5.5V 60NSEC 20NSEC 180mW MICROPAK R/TP 6P - FAIRCHILD SEMICONDUCTOR HONG KONG LTD.	
6	U8100	IC,Acceleration Sensor	EAN62652001	K2DH 3-Axis Accelerometer 2X2X1 LGA R/TP 12P - ST MICROELECTRONICS ASIA PACIFIC PTE LTD.	
6	U8700	IC,Hall Effect Switch	EAN62682901	BU52061NVX Hall IC 1.2X1.6 SSON R/TP 4P - ROHM Semiconductor KOREA CORPORATION	
6	U1500	IC Assembly	EAN62686601	WTR2605 2G/3G, WCDMA, GSM, EDGE, CDMA and TDSCDMA Multi Mode QUALCOMM INCORPORATED.	
6	U5100	IC,WiFi	EAN62698901	WCN-3620-0-61WLNSP WiFi(11bgn)+BT4.0+FM IC for QMC 28nm BB, 3.32*3.55*0.63, 0.4pitch WLCSP R/TP 61P QUALCOMM INCORPORATED.	
6	U4100	IC,PMIC	EAN62737301	PM8110 to 5.5V adj 0W CSP R/TP 118P SMBC, PWM, 19.2MHz XO QUALCOMM INCORPORATED.	
6	U4400	IC,DC,DC Converter	EAN62766201	RT8542 2.7 to 5.5V adj 0W CSP R/TP 20P - RICHTEK TECHNOLOGY CORP.	
6	U3300	IC,MCP,eMMC	EAN62825901	H9TP32A8JDBCPR-KGM NAND/4G SDRAM/4G(2G*2/ 32bit) 2.7VTO3.6V,1.7VTO1.95V,1.14VTO1.3V 11.5x13.0x1.0 TR 162P NAND+DDR SDRAM FBGA 4GB eMMC v4.5+8Gb LPDDR2 MCP (20nm 32Gb MLC+29nm 4Gb 533MHz LPDDR2) HYNIX SEMICONDUCTOR INC.	

Level	Location no	Description	P/N	Specification	Remark
6	FL1102	IC,RF Amplifier	EAN62826801	SFMG2BAD001 2.85 0 0 0W 0W 0 0 MOLD R/TP 8P - WISOL.CO.,LTD	
6	U1600	IC,Power Amplifier	EAN62885901	SKY77742-21 3.2V to 4.35V 0 0 0W 0W 0 2 SMD R/TP 16P 3*4size, for WTR2605, single input dual output, Band1/2/5/8, with CPL, GPIO SKYWORKS SOLUTIONS INC.	
6	U8200	IC,Geomagnetic Sensor	EAN63148201	AK09911 3 Axis Geomagnetic Sensor,measurable range : +-49G, ADC: 14bit, sensitivity : 0.6uT 1.2X1.2X0.5 WLCSP R/TP 8P - ASAHI KASEI CORPORATION	
6	U2100	IC,Digital Baseband Processor,3G	EAN63187901	MSM8212-0-28 Quad 1.2GHz A7, 1x333 LPDDR2, DC-HSPA+,DSDS, RxD, HD display@ 60fps, Adreno 302, 12x12mm NSP R/TP 488P QUALCOMM INCORPORATED.	
6	U7100	IC,Capacitive Touch Sensor	EAN63266301	MIT-200 Cpacitive Type Touch IC 3X3X0.38 UQFN R/TP 24P - Melfas	
6	U8400	IC,Proximity	EAN63386101	APDS-9930-100R Bottom SMT Type Proximity Sensor 6.6X5.16X2.45 DFN R/TP 8P - AVAGO TECHNOLOGIES INTERNATIONAL SALES PTE. LIMITED	
6	U7500	IC,DC,DC Converter	EAN63409101	DW8755A 2.7V to 5.3V Pos. Vout: 4.6V to 6.0 V (0.1V Step) / Neg. Vout: -4.6V to -6.0V (0.1V Step) 0W WLCSP R/TP 12P 1.30x1.53x0.45 mm DONGWOON ANATECH CO.,LTD.	
6	L4400	Inductor,Wire Wound,Chip	EAP62106301	1239AS-H-100N=P2 10UH 30% 30V 700mA 1 0.7 0.46OHM - - SHIELD 2.5X2.0X1.2MM NONE R/TP TOKO, INC.	
6	L7500 L7501	Inductor,Wire Wound,Chip	EAP62107801	1269AS-H-4R7M=P2 4.7UH 20% - 1A 1.3 1.0 0.3OHM - - SHIELD 2.5X2.0X1.0 MM - R/TP TOKO, INC.	
6	C1103	Inductor, Multilayer,Chip	EAP62226101	LQP03TG3N9C02D 3.9NH 0.2NH - 350mA - - 0.35OHM 6GHZ 13 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	L1106 L1115	Inductor, Multilayer,Chip	EAP62108301	LQP03TG8N2J02D 8.2NH 5% - 200mA - - 1.4OHM 4.8GHZ 12 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	C1620	Inductor, Multilayer,Chip	EAP62225901	LQP03TG1N0C02D 1NH 0.2NH - 600mA - - 0.15OHM 6GHZ 12 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C1607	Inductor, Multilayer,Chip	EAP62226001	LQP03TG1N2C02D 1.2NH 0.2NH - 600mA - - 0.15OHM 6GHZ 13 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	C1622	Inductor, Multilayer,Chip	EAP62226601	LQP03TG15NJ02D 15NH 5% - 170mA - - 1.9OHM 3.1GHZ 11 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	L1605	Inductor, Multilayer,Chip	EAP62228001	LG HK 0603 27NJ-T 27NH 5% - 120mA - - 1.35OHM 1.8GHZ 4 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	C1129	Inductor, Multilayer,Chip	EAP62245901	LG HK 0603 2N7S-T 2.7NH 0.3NH - 340mA - - 0.21OHM 7.7GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	C1606 C1624 L1604	Inductor, Multilayer,Chip	EAP62246101	LG HK 0603 4N7S-T 4.7NH 0.3NH - 280mA - - 0.3OHM 5.3GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	L1125 L1126	Inductor, Multilayer,Chip	EAP62246401	LG HK 0603 22NJ-T 22NH 5% - 150mA - - 1OHM 1.8GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	C1104 C1611 L1600	Inductor, Multilayer,Chip	EAP62266401	LQP03TN2N7B02D 2.7NH 0.1NH - 500mA - - 0.2OHM 6GHZ 14 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	L4111	Inductor,Wire Wound,Chip	EAP62266501	MAMK2520T1R0M 1UH 20% - 2.7A 3.1 2.7 0.059OHM - - SHIELD 2.5X2.0X1.2MM - R/TP TAIYO YUDEN CO.,LTD	
6	L4112	Inductor,Wire Wound,Chip	EAP62327801	PIFE20161B-R47MS-39 470NH 20% - 3.5A 3.5 3.5 0.036OHM - - SHIELD 2.0X1.6X1.2 MM - R/TP CYNTEC CO., LTD.	
6	L1501	Inductor, Multilayer,Chip	EAP62526201	LQP03TN10NH02D 10NH 3% - 250mA - - 0.7OHM 3.2GHZ 14 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	L4110 L4401	Inductor,Wire Wound,Chip	EAP62526501	TFM201610GHM-1R0MTAA 1UH 20% - 3A 3.6 3 0.06OHM - - SHIELD 2.0X1.6X1.0 MM - R/TP TDK KOREA COOPERATION	
6	U1100	Module, Tx Module	EAT62073401	SKY77573-31 0DBM 0DB 0% 0A 0A 0DB 0DBM 0DBM 42P 6.0x6.0x0.9MM WEDGE TxM for WTR2605, SP10T SKYWORKS SOLUTIONS INC.	
6	LD8000	Module, Assembly	EAT62473701	LXCL-MN08-4000 Low Cost RO Type Flash Module_3.0T LUMILEDS LIGHTING U.S.,LLC	
6	X4100	Crystal	EAW61883401	X1E0002910001 19.2MHZ 10PPM 7F - SMD R/TP EPSON TOYOCOM CORP	

Level	Location no	Description	P/N	Specification	Remark
6	R6614	Resistor,Chip	EBC61997701	RC0201JR-072M4L 2.4MOHM 5% 1/20W 0603 R/TP - YAGEO CORPORATION	
6	R1500	Resistor,Chip	EBC62497901	RC0201FR-0762RL 62OHM 1% 1/20W 0603 R/TP - YAGEO CORPORATION	
6	PT1600	Thermistor,NTC	EBG61306601	NTCG104EF104FT 100KOHM 1% 35V 35A 4.25MK SMD R/TP - TDK CORPORATION	
6	C2111	Capacitor, Ceramic,Chip	ECCH0000112	MCH155C150J 15pF 5% 50V NP0 -55TO+125C 1005 R/TP - - ROHM Semiconductor KOREA CORPORATION	
6	C5104	Capacitor, Ceramic,Chip	ECCH0000127	MCH155A820J 82pF 5% 50V NP0 -55TO+125C 1005 R/TP - ROHM Semiconductor KOREA CORPORATION	
6	C9201	Capacitor, Ceramic,Chip	ECCH0000115	MCH155A220JK 22pF 5% 50V NP0 -55TO+125C 1005 R/TP - ROHM Semiconductor KOREA CORPORATION	
6	C5117	Capacitor, Ceramic,Chip	ECCH0000117	CL05C270JB5NNNC 27pF 5% 50V NP0 -55TO+125C 1005 R/TP 0.5 SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2320 C4102 C6619	Capacitor, Ceramic,Chip	ECCH0000143	MCH155CN102KK 1nF 10% 50V X7R -55TO+125C 1005 R/TP - ROHM Semiconductor KOREA CORPORATION	
6	C1117 C2349 C2350 C5107 C5108 C5109 C5111	Capacitor, Ceramic,Chip	ECCH0000155	MCH153CN103KK 10nF 10% 16V X7R -55TO+125C 1005 R/TP - ROHM Semiconductor KOREA CORPORATION	
6	C5105	Capacitor, Ceramic,Chip	ECCH0000179	GRM155R71C223K 22nF 10% 16V X7R -55TO+85C 1005 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	C2108 C4117 C8101 C8102	Capacitor, Ceramic,Chip	ECCH0000182	GRM155R61A104K 0.1uF 10% 10V X5R -55TO+85C 1005 R/TP - - MURATA MANUFACTURING CO.,LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C2101 C2102 C2310 C4134 C4135 C4136 C4138 C4139 C4140 C4141 C4142 C4145 C7302 C7303 C7401 C7402 C9602	Capacitor, Ceramic,Chip	ECCH0000198	CL05A225MQ5NSNC 2.2uF 20% 6.3V X5R -55TO+85C 1005 R/TP . SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2300 C2301 C2302 C2303 C2304 C2306 C2307 C2308 C2309 C2316 C2317 C2318 C2319 C2321 C2322 C2323 C2327 C2328 C2332 C2333 C2336 C2337 C2338 C2342 C2343	Capacitor, Ceramic,Chip	ECCH0002001	C1005JB0J104KT000F 0.1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - - TDK CORPORATION	

Level	Location no	Description	P/N	Specification	Remark
6	C2344 C2351 C2352 C2353 C2354 C2355 C2358 C2360 C2362 C2366 C2368 C2372 C3301 C3304 C3307 C4105 C4106 C4115 C8200 C8201 C8700 C8701	Capacitor, Ceramic,Chip	ECCH0002001	C1005JB0J104KT000F 0.1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - - TDK CORPORATION	
6	C4100	Capacitor, Ceramic,Chip	ECCH0002003	C1005JB1C333KT000F 33nF -10TO+10% 16V X7R -55TO+125C 1005 R/TP - - TDK CORPORATION	
6	C2100 C2107 C3306 C3308 C3320 C3321 C3323 C4107 C4108 C4114 C4137 C4144 C4146 C4147 C4148 C4149 C4150 C4151 C4152 C4153 C4154	Capacitor, Ceramic,Chip	ECCH0004904	GRM155R60J105K 1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - MURATA MANUFACTURING CO.,LTD.	

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6	C4179 C4181 C4193 C4203 C4206 C4209 C4211 C7101 C7102 C7104 C7106 C7107 C7108 C7400 C8400 C8401 C9600	Capacitor, Ceramic,Chip	ECCH0004904	GRM155R60J105K 1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	C1113 C1116 C2359 C3300	Capacitor, Ceramic,Chip	ECCH0007802	CL10A475KP8NNNC 4.7uF 10% 10V X5R -55TO+85C 1608 R/TP - - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2311 C4118 C7500	Capacitor, Ceramic,Chip	ECCH0007803	CL10A106MP8NNNC 10uF 20% 10V X5R -55TO+85C 1608 R/TP 0.8MM SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C8100	Capacitor, Ceramic,Chip	ECCH0007805	CL05A106MQ5NUNC 10uF 20% 6.3V X5R -55TO+85C 1005 R/TP 0.5+-0.05 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C1502 C1503 C1504 C1505 C1506 C1509 C1510 C1511 C1512 C1516 C2330 C2331 C2356 C2357 C2369 C2373 C2395 C9504	Capacitor, Ceramic,Chip	ECCH0009101	C0603X5R0J104KT00NN 0.1uF 10% 6.3V X5R -55TO+85C 0603 R/TP - TDK CORPORATION	

Level	Location no	Description	P/N	Specification	Remark
6	C6610 C6921 C6923 C9601	Capacitor, Ceramic,Chip	ECCH0009104	C0603C0G1H330JT00NN 33pF 5% 50V C0G -55TO+125C 0603 R/TP - - TDK CORPORATION	
6	C3310 C3311	Capacitor, Ceramic,Chip	ECCH0009106	C0603X7R1C103KT 10nF 10% 10V X7R -55TO+125C 0603 R/TP - TDK CORPORATION	
6	C2325 C2394	Capacitor, Ceramic,Chip	ECCH0009110	C0603X7R0J223KT 22nF 10% 6.3V X7R -55TO+125C 0603 R/TP - - TDK CORPORATION	
6	L1601 L1603	Capacitor, Ceramic,Chip	ECCH0009208	GRM0335C1ER50C 0.5pF 0.25PF 25V X7R -55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	L1602	Capacitor, Ceramic,Chip	ECCH0009209	GRM0335C1E1R0C 1pF 0.25PF 25V X7R -55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	C4157	Capacitor, Ceramic,Chip	ECCH0009603	CE JMK212 BJ476MG-T 47uF 20% 6.3V X5R -55TO+85C 2012 R/TP 1.25mm - TAIYO YUDEN CO.,LTD	
6	C2365	Capacitor, Ceramic,Chip	ECCH0009228	GRM033R61A472K 4700pF 10% 10V X5R -55TO+85C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	
6	C9501 C9502	Capacitor, Ceramic,Chip	ECCH0009514	MCH032A(AN)100DK 10pF 0.5PF 25V X7R -55TO+125C 0603 R/TP - ROHM.	
6	C2345	Capacitor, Ceramic,Chip	ECCH0017301	CL03A105MQ3CSNH 0.000001F 20% 6.3V X5R -45TO+85C 0603 R/TP - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2374	Capacitor, Ceramic,Chip	ECCH0017501	CL10A226MQ8NRNE 22uF 20% 6.3V X5R -55TO+85C 1608 R/TP 0.8MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C1501 C1507 C1517 C7300 C7301	Capacitor, Ceramic,Chip	ECCH0017601	CL05A475MQ5NRNC 4.7uF 20% 6.3V X5R -55TO+85C 1005 R/TP 0.5MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C2392 C2393	Capacitor, Ceramic,Chip	ECCH0034801	CL03A474MQ3NNNH 0.47uF -20TO20% 6.3V X5R -55TO+85C 0603 R/TP 0.3+-0.03 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C4111 C4112	Capacitor, Ceramic,Chip	ECCH0042302	CL21A475KACLRNC 4.7uF -10TO+10% 25V X5R -55TO+85C 2012 R/TP 0.85+-0.1 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C2312 C2329 C2346	Capacitor, Ceramic,Chip	ECZH0001106	C1005X7R1E472KT000F 4.7nF 10% 25V X7R -55TO+125C 1005 R/TP - - TDK Electronics KOREA CORPORATION	
6	C9037 C9038 C9039	Capacitor, Ceramic,Chip	ECZH0001121	C1005X7R1H471KT000F 470pF 10% 50V X7R -55TO+125C 1005 R/TP - TDK KOREA COOPERATION	
6	C2380 C3309 C3312 C3313 C3314 C3315 C3316 C3317 C3318 C3319 C3322 C3324 C3325 C3326 C4113	Capacitor, Ceramic,Chip	ECZH0001210	C1005Y5V1A474ZT000F 470nF -20TO+80% 10V Y5V -30TO+85C 1005 R/TP - TDK KOREA COOPERATION	
6	C4178 C4180 C4192 C4202 C4205 C4208 C4210 C6613 C9107	Capacitor, Ceramic,Chip	ECZH0001215	C1005X5R1A105KT000F 1uF 10% 10V X5R -55TO+85C 1005 R/TP - - TDK Electronics KOREA CORPORATION	
6	C2364	Capacitor, Ceramic,Chip	ECZH0001216	C1005X5R1A224KT000E 220nF 10% 10V X5R -55TO+85C 1005 R/TP - TDK KOREA COOPERATION	
6	C2315 C2348 C5100 C5101	Capacitor, Ceramic,Chip	ECZH0001217	GRM155R60J474K 470nF 10% 6.3V X5R -25TO+70C 1005 BK-DUP - MURATA MANUFACTURING CO.,LTD.	
6	C1601 C1604 C7100 C7103 C7105	Capacitor, Ceramic,Chip	ECZH0004402	CL05F104ZO5NNNC 0.1uF -20TO+80% 16V Y5V -30TO+85C 1005 R/TP - - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C1625 C6810 C6811 C6812	Capacitor, Ceramic,Chip	ECZH0025916	GRM0335C1E330J 33pF 5% 25V NP0 -55TO+125C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	C1600 C1605	Capacitor, Ceramic,Chip	ECZH0025920	GRM033R71C102K 1nF 10% 16V X7R -55TO+125C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	
6	ZD9100	Diode,Zener	EDNY0013602	EDZVT2R5.1B 4.98V 4.98TO5.2V 80OHM 150mW EMD2 R/TP 2P 1 ROHM Semiconductor KOREA CORPORATION	
6	C1111	Inductor,Multilayer, Chip	ELCH0001035	HK1005 4N7S-T 4.7NH 0.3NH - 300mA 0.21OHM 4GHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP TAIYO YUDEN CO.,LTD	
6	L5103	Inductor,Multilayer, Chip	ELCH0001404	LL1005-FHL1N5S 1.5NH 0.3NH - 400mA 0.13OHM 15GHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP TOKO, INC.	
6	L9910 L9911	Inductor,Multilayer, Chip	ELCH0001430	LL1005-FHLR10J 100NH 5% - 150mA 2.2OHM 1.03GHZ 10 SHIELD NONE 1.0X0.5X0.5MM R/TP TOKO, INC.	
6	L5100	Inductor,Multilayer, Chip	ELCH0001431	LL1005-FHL68NJ 68NH 5% - 180mA 1.7OHM 1.3GHZ 10 SHIELD NONE 1.0X0.5X0.5MM R/TP TOKO, INC.	
6	C1142	Inductor,Multilayer, Chip	ELCH0003831	LQG15HS1N0S02D 1NH 0.3NH - 300mA - - 0.07OHM 10GHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	L1129 L1130	Inductor,Multilayer, Chip	ELCH0003841	LQG15HS82NJ02D 82NH 5% - 150mA 1.2OHM 700MHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	C1108	Inductor,Multilayer, Chip	ELCH0004713	1005GC2T6N8JLF 6.8NH 5% - 250mA 0.32OHM 3GHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP PILKOR ELECTRONICS LTD.	
6	L6810 L6811	Inductor,Multilayer, Chip	ELCH0004903	1608GC2T82NJLF 82NH 5% - 300mA - - 0.95OHM 600MHZ 12 SHIELD NONE 1.6X0.8X0.8MM R/TP PILKOR ELECTRONICS LTD.	
6	C1101	Inductor,Multilayer, Chip	ELCH0005001	HK1005 2N2S 2.2NH 0.3NH - 300mA - - 0.13OHM 6GHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP TAIYO YUDEN CO.,LTD	
6	L1114	Inductor,Multilayer, Chip	ELCH0009110	0402CS-5N1XJEW 5.1NH 5% - 800mA 0.083OHM 4.8GHZ 20 NON SHIELD NONE 1.19X0.64X0.66MM R/TP COILCRAFT SINGAPORE PTE LTD.	
6	L9005	Inductor,Multilayer, Chip	ELCH0010302	0603LS-101XJEC 100NH 5% - 1.4A 0.13OHM 1.15GHZ 12 NON SHIELD NONE 1.80X1.12X1.12MM R/TP COILCRAFT SINGAPORE PTE LTD.	

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6	L4113 L4114	Inductor,Wire Wound,Chip	ELCP0014201	1239AS-H-2R2N=P2 2.2UH 30% - 1.6A 2.2 1.6 0.108OHM - - SHIELD 2.5X2.0X1.2MM NONE R/TP TOKO, INC.	
6	CN7100	Connector,BtoB	ENBY0036001	GB042-40S-H10-E3000 40P 0.4MM STRAIGHT SOCKET SMD R/TP 1M ENGINEERING PLASTIC UL94V-0 AU OVER NI LS Mtron Ltd.	
6	CN7400	Connector,BtoB	ENBY0039601	GB042-20S-H10-E3000 20P 0.4MM STRAIGHT SOCKET SMD R/TP 1M - LS Mtron Ltd.	
6	CN7300	Connector,BtoB	ENBY0040701	GB042-30S-H10-E3000 30P 0.4MM STRAIGHT FEMALE SMD R/TP 1M - LS Mtron Ltd.	
6	R8400 R9605	Resistor,Chip	ERHY0000254	MCR01MZP5J472 4.7KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	SW1100	Connector,RF	ENWY0008701	MS-156C NONE STRAIGHT SOCKET SMD T/REEL AU 50OHM 400mDB HIROSE KOREA CO.,LTD	
6	R9109	Resistor,Chip	ERHY0000140	MCR01MZP5F3602 36KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R4100	Resistor,Chip	ERHY0000161	MCR01MZP5F2003 200KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R9115	Resistor,Chip	ERHY0003301	MCR01MZP5J101 100OHM 5% 1/16W 1005 R/TP - ROHM.	
6	R1600	Resistor,Chip	ERHY0009311	MCR006YZPF51R0 51OHM 1% 1/20W 0603 R/TP - ROHM.	
6	C1618	Resistor,Chip	ERHY0009501	MCR006YZPJ000 0OHM 5% 1/20W 0603 R/TP - ROHM.	
6	R1501 R1502	Resistor,Chip	ERHY0009503	MCR006YZPJ101 100OHM 5% 1/20W 0603 R/TP - ROHM.	
6	R10004 R10005	Resistor,Chip	ERHY0009504	MCR006YZPJ102 1KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	R7120	Resistor,Chip	ERHY0009505	MCR006YZPJ103 10KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	R1602 R4101 R4107	Resistor,Chip	ERHZ0000204	MCR01MZP5F1003 100KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R9113	Resistor,Chip	ERHZ0000210	MCR01MZP5F1200 120OHM 1% 1/16W 1005 R/TP - ROHM.	
6	R9107	Resistor,Chip	ERHZ0000222	MCR01MZP5F1503 150KOHM 1% 1/16W 1005 R/TP - ROHM.	

Level	Location no	Description	P/N	Specification	Remark
6	R2100	Resistor,Chip	ERHZ0000235	MCR01MZP5F2000 200OHM 1% 1/16W 1005 R/TP - ROHM.	
6	R2111	Resistor,Chip	ERHZ0000237	MCR01MZP5F2002 20KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R1603	Resistor,Chip	ERHZ0000245	MCR01MZP5F2203 220KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R2101 R3309	Resistor,Chip	ERHZ0000250	MCR01MZP5F2400 240OHM 1% 1/16W 1005 R/TP - ROHM.	
6	R2107	Resistor,Chip	ERHZ0000278	MCR01MZP5F3901 3.9KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R9111	Resistor,Chip	ERHZ0000286	MCR01MZP5F4701 4.7KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R9501 R9504	Resistor,Chip	ERHZ0000402	MCR01MZP5J100 10OHM 5% 1/16W 1005 R/TP - ROHM.	
6	R7114	Resistor,Chip	ERHZ0000407	MCR01MZP5J105 1MOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R6616	Resistor,Chip	ERHZ0000404	MCR01MZP5J102 1KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R2213 R2214 R2215 R2220 R3308 R4102	Resistor,Chip	ERHZ0000405	MCR01MZP5J103 10KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R9606	Resistor,Chip	ERHZ0000406	MCR01MZP5J104 100KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R6617	Resistor,Chip	ERHZ0000412	MCR01MZP5J122 1.2KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R9500 R9503	Resistor,Chip	ERHZ0000422	MCR01MZP5J153 15KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R2211 R8401	Resistor,Chip	ERHZ0000441	MCR01MZP5J220 22OHM 5% 1/16W 1005 R/TP - ROHM.	

Level	Location no	Description	P/N	Specification	Remark
6	R2202 R2203 R2204 R2205 R2206 R2207 R2208 R2209 R8100	Resistor,Chip	ERHZ0000443	MCR01MZP5J222 2.2KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R2106 R2108	Resistor,Chip	ERHZ0000463	MCR01MZP5J330 33OHM 5% 1/16W 1005 R/TP - ROHM.	
6	R6615	Resistor,Chip	ERHZ0000467	MCR01MZP5J334 330KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	R9114	Resistor,Chip	ERHZ0000486	MCR01MZP5J473 47KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	U4101 U4102	IC,LDO Voltage Regulator	EUSY0355701	RP103K281D-TR-F 1.7V TO 5.25V 2.8V 400mW DFN R/TP 4P - RICOH COMPANY, LTD.	
6	U4105	IC,LDO Voltage Regulator	EUSY0374201	RP103K121D-TR RP103K121D-TR RP103K121D-TR,PLP1010 ,4 ,R/TP ,1.2V 150mA Single LDO High PSRR ver RICOH COMPANY, LTD. RICOH COMPANY, LTD.	
6	U4300	IC,Over Voltage Protection	EUSY0407301	RT9718L WDFN8,8,R/TP,Programmable OVP,IC,ChargerIC,Charger 4~7V 0W WDFN R/TP 8P - RICHTEK TECHNOLOGY CORP.	
6	VA8003	Varistor	SEVY0003601	ICVL0505101V150FR 5.6V 0% 100pF 1.0*0.5*0.55 NONE SMD R/TP INNOCHIPS TECHNOLOGY	
6	D9201 VA9606 ZD7101 ZD7102 ZD7103	Varistor	SEVY0004401	ICVL0518400V500FR 18V 0% 40pF 1.0*0.5*0.55 NONE SMD R/TP INNOCHIPS TECHNOLOGY	
6	VA9600 VA9601 VA9602 VA9603 VA9604 VA9605	Varistor	SEVY0005101	ICVL0518050FR 18V 0% 5F 1.0*0.5*0.55 NONE SMD R/TP INNOCHIPS TECHNOLOGY	
6	FB7101	Filter,Bead	SFBH0008101	BLM15AG601SN1D 600 ohm 1.0X0.5X0.5 25% 0.6 ohm 0.3A SMD R/TP 2P 0 MURATA MANUFACTURING CO.,LTD.	

Level	Location no	Description	P/N	Specification	Remark
6	FB6610 FB6920 FB6921	Filter,Bead	SFBH0008102	BLM15HD182SN1D 1800 ohm 1.0X0.5X0.5 25% 2.2 ohm 0.2A SMD R/TP 2P 0 MURATA MANUFACTURING CO.,LTD.	
6	FL9200	Filter,EMI/Power	SFEY0015301	NFM18PC104R1C3 ESD/EMI 0HZ 0.1uF 0H SMD R/TP - MURATA MANUFACTURING CO.,LTD.	
6	FL7300	Filter,EMI/Power	SFEY0015901	ICMEF214P101MFR ICMEF214P101MFR ICMEF214P101MFR,SMD ,ESD Common mode Filter INNOCHIPS TECHNOLOGY INNOCHIPS TECHNOLOGY	
6	FL7302	Filter,EMI/Power	SFEY0016301	ICMEF112P900M COMMON MODE NOISE FILTER 0HZ 0F 0H SMD R/TP - INNOCHIPS TECHNOLOGY	
6	R4105 R6621 R7102 R7118	Wire Pad,Open	SAFO0000401	AX3100 ATL SV_SHIPBACK,MAIN,A,0OHM DNI	
6	R5101 R5102 R6610 R6613 R6618 R6620 R6920 R6921 R7107 R7110 R7111 R7112 R7113 R7119 R7121	Wire Pad,Short	SAFP0000401	AX3100 ATL SV_SHIPBACK,MAIN,A	
6	R4211	Wire Pad,Open	SAFO0000501	AX3100 ATL SV_SHIPBACK,MAIN,A,0OHM_1005_DNI	
6	R4210 R7015 R7106	Wire Pad,Short	SAFP0000501	LG-VS760 VRZ	

Level	Location no	Description	P/N	Specification	Remark
6	C1115 C1119 C1120 C1121 C1122 C1125 C1126 C1130 C1131 C1500 C1508 C1609 C1613 C1615 C1616 C1617 C1619 C1621 C1623 C9106 L1105 L1127	Capacitor, Ceramic,Chip	ECCH0009103	C0603C0G1H101JT00NN 100pF 5% 50V C0G -55TO+125C 0603 R/TP - - TDK CORPORATION	
6	C1100 C1102 C1109 C1110 C1128 C1141 C5112 C5114 C5115 L1117 L5104 L5105	Capacitor, Ceramic,Chip	ECZH0000813	C1005C0G1H101JT 100pF 5% 50V C0G -55TO+125C 1005 R/TP - TDK KOREA COOPERATION	
6	D10003 D10005 VA9101 VA9103 ZD7400 ZD7401 ZD7402 ZD7403 ZD7404	Varistor	SEVY0005402	ICVS0505500FR 5.6V 0% 50F 1.0*0.5*0.55 - SMD R/TP INNOCHIPS TECHNOLOGY	
6	C5124 C5199	Capacitor(High Frequency),Ceramic,Chip	EAE63042301	RMUMK105CG0R5BV-F_H 0.5pF 0.1PF 50V C0G -55TO+125C 1005 R/TP 0.5+-0.05 L:1.0+-0.05, W:0.5+-0.05, T:0.5+-0.05 KOREA TAIYO YUDEN.CO., LTD.	
6	R10001 R10003	Resistor,Chip	ERHY0009301	MCR006YZPF1000 100OHM 1% 1/20W 0603 R/TP - ROHM.	

Level	Location no	Description	P/N	Specification	Remark
6	R1110 R1114 R4402 R5103 R9506	Resistor,Chip	ERHZ0000401	MCR01MZSJ000 0OHM 5% 1/16W 1005 R/TP - ROHM.	
6	L6700 L6701	Filter,Bead	EAM62870101	MPZ1608S102A 1000 1.6x0.8x0.8 25% 0.3 0.8 SMD R/TP 2P - TDK CORPORATION	
6	D10004	Varistor	SEVY0007901	ICVS0505481FR 5.6V 0% 480F 1.0*0.5*0.55 NONE SMD R/TP INNOCHIPS TECHNOLOGY	
6	R10002	Resistor,Chip	ERHY0009508	MCR006MZPJ122 1.2KOHM 5% 1/20W 0603 R/TP - ROHM Semiconductor KOREA CORPORATION	
6	L1607	Capacitor(High Frequency), Ceramic,Chip	EAE62946501	GRM0335C1E3R9B_H 3.9pF 0.1PF 25V C0G -55TO+125C 0603 R/TP 0.3+-0.03 L:0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	L5101	Inductor, Multilayer,Chip	ELCH0005019	HK1005 68NJ 68NH 5% - 180mA - - 1.2OHM 750MHZ 8 SHIELD NONE 1.0X0.5X0.5MM R/TP TAIYO YUDEN CO.,LTD	
1	SAF010100	Software Assembly, Common	SAF30560101	LGD295.ACISKT KK -	
2	SAD010500	Software,USB Driver	SAD34932301	Windows -	
2	SAD010600	Software,OSP	SAD34991501	Windows Windows KK - -	
2	SAD010100	Software, Application	SAD35024401	Android KK - PCSuiteV - - EUROPE PC SYNC -	
4	EAX010700	PCB,Flexible	EAX66092601	LGD295F.ABRAWH 1.0 POLYI Multi 2 0.15mm Flexible	

12.3 Accessory

Note: This Chapterisused for reference, Part order is ordered by SBOM standard on GCSC

Level	Location no	Description	P/N	Specification	Remark
2	MFL053800	Manual, Operation	MFL68623712	PRINTING LGD295.ACISKT ZZ:Without Color Russian Federation ENGLISH+RUSSIAN+KAZAKH+UKRAINIAN -	
2	EAC00	Rechargeable Battery, Lithium Ion	EAC62378401	BL-41ZH_WW_TOCAD 3.8V 1.82AH 450mAH 66 x 48 x 4.2 69 x 48 x 4.3 GRAY Innerpack Bar-Type BL-41ZH_Tocad_WW_424866_Innerpack TOCAD DONG-HWA CO., LTD	
3	MBM062600	Card,Quick Reference	MBM64678012	PRINTING LGD295.ACISKT ZZ:Without Color -	
3	MBM087200	Card,Warranty	MCDF0011303	COMPLEX GD350 CISBK ZZ:Without Color -	
2	EAY060000	Adapters	EAY62709907	MCS-02ER_EAC 90Vac~264Vac 4.75Vdc~5.25Vdv 850mA 50/60Hz CE WALL 2P USB - SUNLIN ELECTRONICS CO.,LTD	
2	EAD010000	Cable,Assembly	EAD62377902	LG0108 USB USB 1.0M 4 BLACK UL N ningbo broad telecommunication co.,ltd	